Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
9A30P2	CONNECTOR
9A35	YAW RATE GYRO ASSY (1)
9A35P1	CONNECTOR (1)
9A35P2	CONNECTOR (1)
9B1	PITCH LINEAR ACTUATOR NO. 1
9B1J1	CONNECTOR
9B2	ROLL LINEAR ACTUATOR NO. 1
9B2J1	CONNECTOR
9B3	YAW LINEAR ACTUATOR NO. 1
9B3J1	CONNECTOR
9B4	PITCH LINEAR ACTUATOR NO. 2
9B4J1	CONNECTOR
9B5	ROLL LINEAR ACTUATOR NO. 2
9B5J1	CONNECTOR
9B6	PITCH TRIM ACTUATOR
9B6P1	CONNECTOR
9B7	ROLL TRIM ACTUATOR
9B7P1	CONNECTOR
9B8	NO. 2 HYD PRESSURE XMTR
9B8P1	CONNECTOR
9B9	NO. 1 HYD PRESSURE XMTR
9B9P1	CONNECTOR
9B32	YAW 2 LINEAR ACTUATOR
9B32P1	CONNECTOR
9DS1	CLOGGED HYD FLTR INDICATOR
9J/P3	RIGHT AFT DISCONNECT
9J/P4	RIGHT BEAM DISCONNECT
9J/P30	RIGHT BEAM DISCONNECT
9J/P31	LEFT BEAM DISCONNECT
9L1	NO. 1 HYD BYPASS SOLENOID
9L1P1	CONNECTOR
9L2	NO. 2 HYD BYPASS SOLENOID
9L2P1	CONNECTOR
9L8	YAW TRIM MAG BRAKE
9L8P1	CONNECTOR
9M1	NO. 1 HYD P&T INDICATOR
9M1P1	CONNECTOR
9M2	NO. 1 HYD P&T INDICATOR
9M2P1	CONNECTOR
9MT1	PITCH CMT
9MT2	ROLL CMT
9MT3	YAW CMT
9MT4	PITCH CMT

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
9MT5	ROLL CMT
9MT30	COLL CMT
9MT31	YAW CMT
9R1	CYCLIC F.T. PULL-UP RESISTOR
9S1	NO. 1 HYD FLTR SWITCH
9S1P1	CONNECTOR
9S2	NO. 2 HYD FLTR SWITCH
9S2P1	CONNECTOR
9S51	PILOT AFCS TRIM SWITCH
9S52	PILOT FRC TRM REL SWITCH
9S53	COPILOT AFCS TRIM SWITCH
9S54	COPILOT FRC TRM REL SWITCH
9TB1	TERM BOARD
9TB1P1	CONNECTOR
9TB3	TERM MODULE
9TB4	TERM MODULE
9TB4P1	CONNECTOR
9TB6	TERM MODULE
9TB30	TERM MODULE
9TB30P1	CONNECTOR
9TB31	TERM MODULE
9TB31P1	CONNECTOR
9TB32	TERM MODULE
9TB32P1	CONNECTOR
9TB33	TERM MODULE
9TB33P1	CONNECTOR
9TB34	TERM MODULE
9TB34P1	CONNECTOR
9WT1	GROUND MODULE
9WT1P1	CONNECTOR
9WT2	GROUND MODULE
9WT2P1	CONNECTOR
9WT3	GROUND MODULE
9WT3P1	CONNECTOR
9WT4	GROUND MODULE
9WT4P1	CONNECTOR
9Z1	NO. 1 HYD TEMPERATURE BULB
9Z1P1	CONNECTOR
9Z2	NO. 2 HYD TEMPERATURE BULB
9Z2P1	CONNECTOR
10A1	RESISTOR PANEL
10A1P1	CONNECTOR
10A2	RESISTOR PANEL

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
10A2P1	CONNECTOR
10B1	COPILOT WNDSHLD WIPER MOTOR
10B1P1	CONNECTOR
10B2	PILOT WNDSHLD WIPER MOTOR
10B2P1	CONNECTOR
10B5	CABIN TEMPERATURE CONTROL MOTOR
10B5P1	CONNECTOR
10HR1	COPILOT PITOT HEATER
10HR1P1	CONNECTOR
10HR2	PILOT PITOT HEATER
10HR2P1	CONNECTOR
10K1	HEATER RELAY
10K1P1	CONNECTOR
10L2	DOOR POST OUTLET VALVE SOLENOID
10L2P1	CONNECTOR
10L3	NO. 1 BLEED AIR VALVE SOLENOID
10L3P1	CONNECTOR
10L4	NO. 2 BLEED AIR VALVE SOLENOID
10L4P1	CONNECTOR
10L5	AFT OUTLET VALVE SOLENOID
10L5P1	CONNECTOR
10S1	AFT OUTLET LIMIT SWITCH
10S2	HEATER AIR LINE OVERHEAT SWITCH
10S3	HEATER OVERHEAT SWITCH
10Z1	CABIN TEMPERATURE CONTROL
22J/P1	LEFT NOSE DISCONNECT
22J/P2	RIGHT NOSE DISCONNECT
22J/P3	LEFT NOSE DISCONNECT
22J/P4	RIGHT NOSE DISCONNECT
22J/P5	LEFT NOSE DISCONNECT
22J/P6	RIGHT NOSE DISCONNECT
22J/P7	LEFT BEAM DISCONNECT
22J/P8	RIGHT BEAM DISCONNECT
22J/P9	LEFT BEAM DISCONNECT
22J/P10	RIGHT BEAM DISCONNECT
22J/P11	LEFT DOOR DISCONNECT
22J/P12	RIGHT WNSHLD POST DISCONNECT
22J/P13	LEFT DOOR DISCONNECT
22J/P14	LEFT WNSHLD POST DISCONNECT
22J/P15	LEFT DOOR DISCONNECT
22J/P16	RIGHT INSTR PANEL DISCONNECT
22J/P17	LEFT INSTR PANEL DISCONNECT
22J/P19	LEFT BEAM DISCONNECT

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
22WT1	GROUND MODULE
22WT1P1	CONNECTOR
22WT2	GROUND MODULE
22WT2	GROUND MODULE
22WT3	GROUND MODULE
22WT3P1	CONNECTOR
22WT4	GROUND MODULE
22WT4P1	CONNECTOR
22WT5	GROUND MODULE
22WT6	GROUND MODULE
22WT7	GROUND MODULE
22WT8	GROUND MODULE
22WT9	GROUND MODULE
22WT10	GROUND MODULE
22WT11	GROUND MODULE
22WT12	GROUND MODULE
22WT13	GROUND MODULE
22TW14	GROUND MODULE
24A1	MISC CONTROL PANEL
24A1P1	CONNECTOR
24A1P2	CONNECTOR
24S10	PILOT WIPER SWITCH
27AR1	FLIGHT DATA ACQUISITION UNIT
27AR1P1A	CONNECTOR
27AR1P1B	CONNECTOR
3400TB1	TERMINAL BOARD
3400TB1P1	CONNECTOR
3400TB2	TERMINAL BOARD
3400TB2P1	CONNECTOR
3407L1	REMOTE COMPENSATOR (1)
3407MP1	COPILOT THREE-AXIS REFERENCE SENSOR NO. 1 (1)
3407MP1	COPILOT ATTITUDE HEADING REFERENCE UNIT (AHRU) NO. 1 (2)
3407MP1P1	CONNECTOR (1) (2)
3407MP1P2	CONNECTOR (1) (2)
3407MP1P3	CONNECTOR (2)
3407MP1P4	CONNECTOR (2)
3407MP1P5	CONNECTOR (2)
3407MP2	PILOT THREE-AXIS REFERENCE SENSOR NO. 2 (1)
3407MP2	ATTITUDE HEADING REFERENCE UNIT (AHRU) PILOT (2)
3407MP2P1	CONNECTOR (1) (2)
3407MP2P2	CONNECTOR (1) (2)
3407MP2P3	CONNECTOR (2)
3407MP2P4	CONNECTOR (2)

Table 4. Reference Designators (S/N 36087 through 36999) (continued)

REF DES	NOMENCLATURE
3407MP2P5	CONNECTOR (2)

NOTES:
1
2

Table 5. 4-Axis Configuration Electrical Equipment List

REF DES	NOMENCLATURE
10DS1	LIGHT, WINDSHIELD HEATER
10DS1P1	CONN, WINDSHIELD HEATER LIGHT
22WT12	TERMINAL BOARD, GROUND
22WT14	TERMINAL BOARD, GROUND
22WT8	TERMINAL BOARD, GROUND
2300J1	CONN
24A1	CONTROL PANEL, MISCELLANEOUS
24A1P1	CONN, MISCELLANEOUS CONTROL PANEL
24A3	CONTROL UNIT, HOIST PENDANT
24A3J2	CONN, HOIST PENDANT CONTROL
24A3J3	CONN, HOIST PENDANT CONTROL
24A3P2	CONN, HOIST PENDANT CONTROL
24A3P3	CONN, HOIST PENDANT CONTROL
24A3S1	SWITCH, BOOM
24A3S3	SWITCH, HOIST UP-DOWN
24P1	CONN, HOIST REMOTE BEEP - CABIN ROOF DISCONNECT
27A2	COMPARATOR, AIR DATA
27A2P1	CONNECTOR, AIR DATA COMPARATOR
27TB1	TERMINAL BOARD, FLIGHT DATA RECORDER
3400CB3	CIRCUIT BREAKER, HSI
3400CP30	NAV CONTROL
3400CP30P1	CONN, NAV CONTROL
3400P13B	CONN
3400TB30	TERMINAL BOARD
3400WT2	TERMINAL BOARD, GROUND
3400WT8	TERMINAL BOARD, GROUND
3407CP1	PANEL, COPILOT COMPASS CONTROL
3407CP1P1	CONN, COPILOT COMPASS CONTROL PANEL
3407CP2	PANEL, PILOT COMPASS CONTROL
3407CP2P1	CONN, PILOT COMPASS CONTROL PANEL
3440A10	FAN, EFIS
3440A10	TERMINATOR, RESISTIVE LOAD
344A10J1	CONN, RESISTIVE LOAD TERMINATOR
3440A11	FAN, EFIS
3440A11	TERMINATOR, RESISTIVE LOAD
3440A11J1	CONN, RESISTIVE LOAD TERMINATOR

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
3440A12	MOD, COMM
3440A12	TERMINATOR, RESISTIVE LOAD
3440A12J1	CONN, RESISTIVE LOAD TERMINATOR
3440A13	MOD, COMM
3440A1P1A	CONN, COPILOT SYMBOL GENERATOR
3440A1P1B	CONN, COPILOT SYMBOL GENERATOR
3440A1P2A	CONN, COPILOT SYMBOL GENERATOR
3440A1P2B	CONN, COPILOT SYMBOL GENERATOR
3440A2	SYMBOL GENERATOR, PILOT
3440A2P1A	CONN, PILOT SYMBOL GENERATOR
3440A2P1B	CONN, PILOT SYMBOL GENERATOR
3440A2P2A	CONN, PILOT SYMBOL GENERATOR
3440A2P2B	CONN, PILOT SYMBOL GENERATOR
3440A3	SWITCH PANEL, COPILOT EFIS AUXILIARY
3440A3P1	CONN, COPILOT EFIS AUXILIARY SWITCH PANEL
3440A4	SWITCH PANEL, PILOT EFIS AUXILIARY
3440A4P1	CONN, PILOT EFIS AUXILIARY SWITCH PANEL
3440A5	TERMINATOR, RESISTIVE LOAD
3440A5J1	CONN, RESISTIVE LOAD TERMINATOR
3440A6	TERMINATOR, RESISTIVE LOAD
3440A6J1	CONN, RESISTIVE LOAD TERMINATOR
3440A7	TERMINAL, RESISTIVE LOAD
3440A7J1	CONN, RESISTIVE LOAD TERMINATOR
3440A8	TERMINATOR, RESISTIVE LOAD
3440A8J1	CONN, RESISTIVE LOAD TERMINATOR
3440A9	TERMINATOR, RESISTIVE LOAD
3440A9J1	CONN, RESISTIVE LOAD TERMINAL
3440CB1	CIRCUIT BREAKER, EFIS NO. 1 ADI
3440CB10	CIRCUIT BREAKER, EFIS BLOWER
3440CB2	CIRCUIT BREAKER, EFIS NO. 2 ADI
3440CB3	CIRCUIT BREAKER, EFIS NO. 1 HSI
3440CB4	CIRCUIT BREAKER, EFIS NO. 2 HSI
3440CB5	CIRCUIT BREAKER, EFIS NO. 1 SG/DC
3440CB6	CIRCUIT BREAKER, EFIS NO. 2 SG/DC
3440CB7	CIRCUIT BREAKER, EFIS 26V NO. 1
3440CB8	CIRCUIT BREAKER, EFIS 26V NO. 2
3440CB9	CIRCUIT BREAKER, EFIS BLOWER
3440CP1	INSTRUMENT CONTROLLER
3440CP1P1	CONN, INSTRUMENT CONTROLLER
3440CP1P2	CONN, INSTRUMENT CONTROLLER
3440CP3	DISPLAY CONTROLLER, COPILOT
3440CP3P1	CONN, COPILOT DISPLAY CONTROLLER
3440CP4	DISPLAY CONTROLLER, PILOT

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
3440CP4P1	CONN, PILOT DISPLAY CONTROLLER
3440CR1	DIODE
3440DS1	EADI, COPILOT
3440DS1P1	CONN, COPILOT EADI
3440DS10	SWITCH, EFIS MASTER
3440DS2	PILOT, EADI
3440DS2P1	CONN, PILOT EADI
3440DS3	EHSI, COPILOT
3440DS3P1	CONN, COPILOT EHSI
3440DS4	EHSI, PILOT
3440DS4P1	CONN, PILOT EHSI
3440J1	CONN
3440J10	CONN, TEE
3440J10C	CONN
3440J11	CONN, TEE
3440J11C	CONN
3440J12	CONN, TEE
3440J12C	CONN
3440J13	CONN, TEE
3440J13C	CONN
3440J14	CONN, TEE
3440J14C	CONN
3440J15	CONN, TEE
3440J15C	CONN
3440J16	CONN, TEE
3440J16C	CONN
3440J17	CONN, TEE
3440J17C	CONN
3440J2	CONN
3440J3	CONN
3440J4	CONN
3440K1	RELAY
3440K2	RELAY
3440K3	RELAY
3440K4	RELAY
3440K5	RELAY
3440K6	RELAY
3440P1	CONN
3440P10A	CONN
3440P10B	CONN
3440P11A	CONN
3440P11B	CONN
3440P12A	CONN

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
3440P12B	CONN
3440P13A	CONN
3440P14A	CONN
3440P14B	CONN
3440P15A	CONN
3440P15B	CONN
3440P16A	CONN
3440P16B	CONN
3440P17A	CONN
3440P17B	CONN
3440P2	CONN
3440P3	CONN
3440P4	CONN
3440TB1	TERMINAL BOARD
3440TB2	TERMINAL BOARD
3440TB3	TERMINAL BOARD
3440WT1	TERMINAL BOARD, GROUND
3440WT2	TERMINAL BOARD, GROUND
4A2	COLLECTIVE STICK, PILOT
4A2J1	CONN, PILOT COLLECTIVE STICK
4M5	CLOCK, PILOT DIGITAL
4M5P1	CONN, PILOT DIGITAL CLOCK
5A2	COLLECTIVE STICK, COPILOT
5A2J1	CONN, COPILOT COLLECTIVE STICK
5M5	CLOCK, COPILOT DIGITAL
5M5P1	CONN, COPILOT DIGITAL CLOCK
8CR52	DIODE
8CR53	DIODE
8A1	PANEL, CAUTION
8A1P1	CONN
8A1P2	CONN
8K20	RELAY, SWITCHING
8Q20	TRANSISTOR
8Q21	TRANSISTOR
8TB1	TERMINAL BOARD, LIGHTING
8TB2	TERMINAL BOARD, RPM WARNING
8TB3	TERMINAL BOARD, 5 VOLT LIGHT POWER
9A1	COMPUTER, COPILOT DFC NO. 1
9A10	SWITCH, PITCH DETENT
9A10J1	CONN, PITCH DETENT SWITCH
9A10P1	CONN, PITCH DETENT SWITCH
9A11	INDICATOR PANEL AFCS ACTUATOR
9A11P1	CONN, AFCS ACTUATOR PANEL

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9A1P1A	CONN, COPILOT DFC COMPUTER NO. 1
9A1P1B	CONN, COPILOT DFC COMPUTER NO. 1
9A1P1C	CONN, COPILOT DFC COMPUTER NO. 1
9A2	COMPUTER, PILOT DFC NO. 2
9A2P1A	CONN, PILOT DFC COMPUTER NO. 2
9A2P1B	CONN, PILOT DFC COMPUTER NO. 2
9A2P1C	CONN, PILOT DFC COMPUTER NO. 2
9A3	MODE SELECTOR, COPILOT
9A3P1	CONN, COPILOT MODE SELECTOR
9A30	CONTROL BOX, AFCS RELAY
9A30P1	CONN, AFCS RELAY CONTROL BOX
9A30P1	CONN, AFCS RELAY CONTROL BOX
9A31	GYRO NO. 1 YAW RATE
9A31P1	CONN, YAW RATE GYRO NO. 1
9A32	GYRO NO. 2, YAW RATE
9A32P1	CONN, YAW RATE GYRO NO. 2
9A33	SWITCH, YAW DETENT
9A33J1	CONN, YAW DETENT SWITCH
9A33P1	CONN, YAW DETENT SWITCH
9A34	SWITCH, COLLECTIVE DETENT
9A34J1	CONN, COLLECTIVE DETENT SWITCH
9A34P1	CONN, COLLECTIVE DETENT SWITCH
9A35	GYRO ASSEMBLY, YAW RATE
9A35J1	CONN, YAW RATE GYRO ASSEMBLY
9A35J2	CONN, YAW RATE GYRO ASSEMBLY
9A35P1	CONN, YAW RATE GYRO ASSEMBLY
9A35P2	CONN, YAW RATE GYRO ASSEMBLY
9A3P1	CONN, COPILOT MODE SELECTOR
9A4	MODE SELECTOR, PILOT
9A4P1	CONN, PILOT MODE SELECTOR
9A5	CONTROLLER, AUTOPILOT
9A5P1	CONN, AUTOPILOT CONTROLLER
9A5P2	CONN, AUTOPILOT CONTROLLER
9A6	PRESELECT, ALTITUDE
9A6P1	CONN, ALTITUDE PRESELECT
9A7	SENSOR, AIR DATA NO. 1
9A7P1	CONN, AIR DATA SENSOR NO. 1
9A8	SENSOR, AIR DATA NO. 2
9A8P1	CONN, AIR DATA SENSOR NO. 2
9A9	SWITCH, ROLL DETENT
9A9J1	CONN, ROLL DETENT SWITCH
9A9P1	CONN, ROLL DETENT SWITCH
9B1	ACTUATOR, PILOT LINEAR NO. 1

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9B1J1	CONN, PITCH LINEAR ACTUATOR NO. 1
9B2	ACTUATOR, ROLL LINEAR NO. 1
9B2J1	CONN, ROLL LINEAR ACTUATOR NO. 1
9B3	ACTUATOR, YAW LINEAR NO. 1
9B30	ACTUATOR, YAW TRIM
9B30P1	CONN, YAW TRIM ACTUATOR
9B31	ACTUATOR, COLLECTIVE TRIM
9B31P1	CONN, COLLECTIVE TRIM ACTUATOR
9B32	LINEAR ACTUATOR, YAW NO. 2
9B32J1	CONN, YAW LINEAR ACTUATOR NO. 2
9B3J1	CONN, YAW LINEAR ACTUATOR NO. 1
9B4	LINEAR ACTUATOR, PITCH NO. 2
9B4J1	CONN, PITCH LINEAR ACTUATOR NO. 2
9B5	LINEAR ACTUATOR, ROLL NO. 2
9B5J1	CONN, ROLL LINEAR ACTUATOR NO. 2
9B6	ACTUATOR, PITCH TRIM
9B6P1	CONN, PITCH TRIM ACTUATOR
9B7	ACTUATOR, YAW TRIM
9B7P1	CONN, YAW TRIM ACTUATOR
9CB2	CIRCUIT BREAKER, AFCS NO. 2
9CB3	CIRCUIT BREAKER, FORCE TRIM
9CB30	CIRCUIT BREAKER, RATE GYRO NO. 2
9CB31	CIRCUIT BREAKER, RATE GYRO NO. 1
9CB32	CIRCUIT BREAKER, AFCS NO. 1 ADD/CAUTION
9CB33	CIRCUIT BREAKER, AFCS NO. 2 CAUT
9CB4	CIRCUIT BREAKER, AFCS 26V NO. 1
9CB7	CIRCUIT BREAKER, AFCS NO. 2
9CR30	DIODE
9CR31	DIODE
9CR32	DIODE
9CR33	DIODE
9CR34	DIODE
9CR35	DIODE
9CR36	DIODE
9CR37	DIODE
9CR38	DIODE
9CR39	DIODE
9CR40	DIODE
9CR41	DIODE
9CR42	DIODE
9CR43	DIODE
9CR44	DIODE
9CR45	DIODE

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9CR46	DIODE
9CR47	DIODE
9CR48	DIODE
9CR49	DIODE
9CR50	DIODE
9CR51	DIODE
9CR52	DIODE
9CR53	DIODE
9CR54	DIODE
9CR55	DIODE
9CR56	DIODE
9CR57	DIODE
9CR58	DIODE
9CR59	DIODE
9DS30	LIGHT, COPILOT AUTOPILOT
9DS30P1	CONN, COPILOT AUTOPILOT LIGHT
9DS31	LIGHT COPILOT PITCH, ROLL, YAW
9DS31P1	CONN, COPILOT PITCH, ROLL, YAW LIGHT
9DS32	LIGHT, COPILOT FORCE TRIM DECOUPLER
9DS32P1	CONN, COPILOT FORCE TRIM DECOUPLER LIGHT
9DS33	LIGHT, PILOT AUTOPILOT
9DS33P1	CONN, PILOT AUTOPILOT LIGHT
9DS34	LIGHT, PILOT PITCH, ROLL, YAW
9DS34P1	CONN, PILOT PITCH, ROLL, YAW LIGHT
9DS35	LIGHT, PILOT FORCE TRIM DECOUPLER
9DS35P1	CONN, PILOT FORCE TRIM DECOUPLER LIGHT
9DS36	LIGHT, MOT/ON
9DS36P1	CONN, MOT/ON LIGHT
9DS37	LIGHT, CLIMB/ON
9DS37P1	CONN, CLIMB/ON LIGHT
9DS38	LIGHT, APR/1/2
9DS38P1	CONN, APR/1/2 LIGHT
9DS39	LIGHT, HOIST/EN
9DS39P1	CONN, HOIST/EN LIGHT
9J30	CONN
9J31	CONN
9K2	RELAY, TRIM FAIL ANNUN
9K20	RELAY
9K21	RELAY
9K22	RELAY
9K23	RELAY
9K24	RELAY
9K25	RELAY

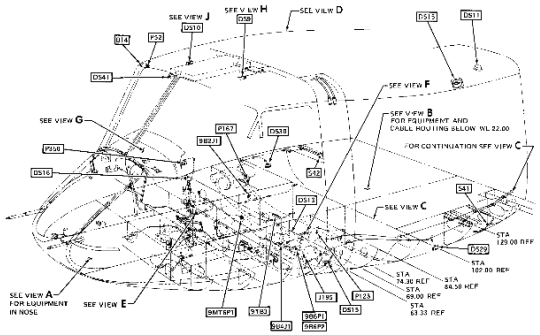
Table 5. 4-Axis Configuration Electrical Equipment List (continued)

REF DES	NOMENCLATURE
9K3	RELAY, CLTV WARN ANNUN
9MT1	DUAL CMT, COPILOT CYCLIC PITCH
9MT1J1	CONN, COPILOT CYCLIC PITCH DUAL CMT
9MT2	DUAL CMT, COPILOT CYCLIC ROLL
9MT2J1	CONN, COPILOT CYCLIC ROLL DUAL CMT
9MT3	DUAL CMT, COPILOT PEDALS YAW
9MT30	DUAL CMT, PILOT COLLECTIVE
9MT30J1	CONN, PILOT COLLECTIVE DUAL CMT
9MT31	DUAL CMT, PILOT PEDALS YAW
9MT31J1	CONN, PILOT PEDALS YAW DUAL CMT
9MT32	ACCELEROMETER, LONGITUDE NO. 1
9MT32P1	CONN, LONGITUDE NO. 1 ACCELEROMETER
9MT33	ACCELEROMETER, LONGITUDE
9MT33P1	CONN, LONGITUDE ACCELEROMETER
9MT34	ACCELEROMETER, LATITUDE NO. 1
9MT34P1	CONN, LATITUDE ACCELEROMETER
9MT35	ACCELEROMETER, LATITUDE
9MT35P1	CONN, LATITUDE ACCELEROMETER
9MT36	ACCELEROMETER, VERTICAL NO. 1
9MT36P1	CONN, VERTICAL NO. 1 ACCELEROMETER
9MT37	ACCELEROMETER, VERTICAL
9MT37P1	CONN, VERTICAL ACCELEROMETER
9MT3J1	CONN, COPILOT PEDALS YAW DUAL CMT
9MT4	DUAL CMT, PILOT PITCH CYCLIC
9MT4J1	CONN, PILOT PITCH CYCLIC DUAL CMT
9MT5	DUAL CMT, PILOT ROLL CYCLIC
9MT5J1	CONN, PILOT ROLL CYCLIC DUAL CMT
9MT6	DUAL CMT, COPILOT COLLECTIVE
9MT6J1	CONN, COPILOT COLLECTIVE DUAL CMT
9P30	CONN
9P31	CONN
9S34	SWITCH, READ EEPROM
9S35	SWITCH, ERASE EEPROM
9S36	SWITCH, HOIST REMOTE BEEP
9TB1	TERMINAL BOARD
9TB3	TERMINAL BOARD
9TB30	TERMINAL BOARD
9TB31	TERMINAL BLOCK
9TB32	TERMINAL BOARD
9TB33	TERMINAL BOARD
9TB34	TERMINAL BOARD
9TB4	TERMINAL BOARD
9TB6	TERMINAL BOARD

Table 5. 4-Axis Configuration Electrical Equipment List (continued)

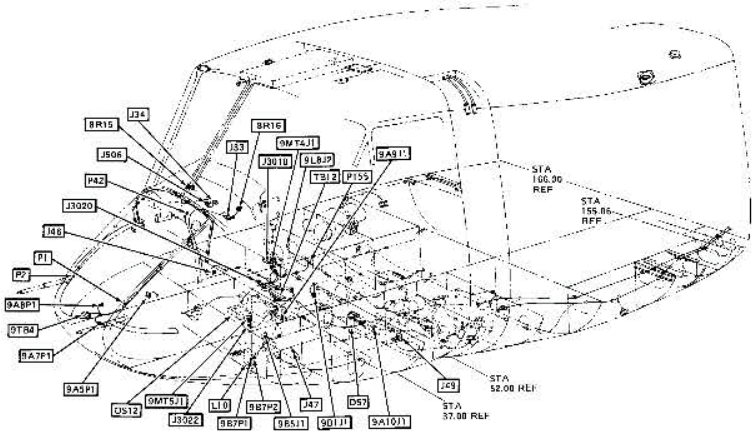
REF DES	NOMENCLATURE
9WT30	TERMINAL BOARD, GROUND
9WT31	TERMINAL BOARD, GROUND
9WT32	TERMINAL BOARD, GROUND
9WT33	TERMINAL BOARD, GROUND
9WT34	TERMINAL BOARD, GROUND
9WT35	TERMINAL BOARD, GROUND
J14	CONN, GLARE SHIELD FLOOD LIGHT ASSEMBLY
P14	CONN, GLARE SHIELD FLOOD LIGHT ASSEMBLY

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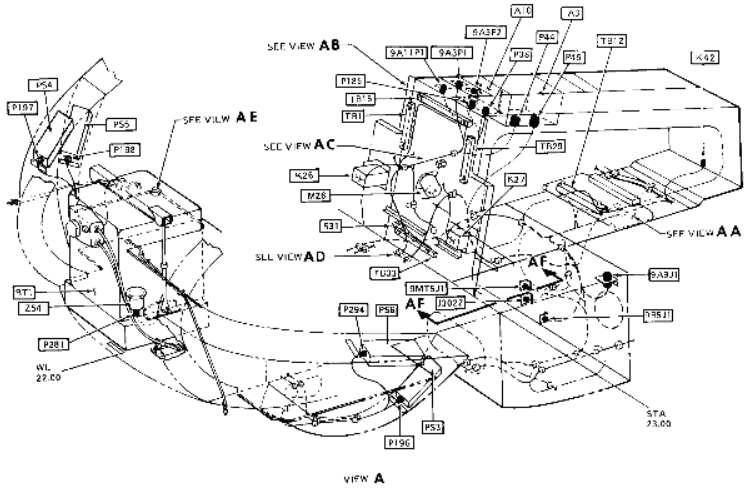
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 1 of 11)



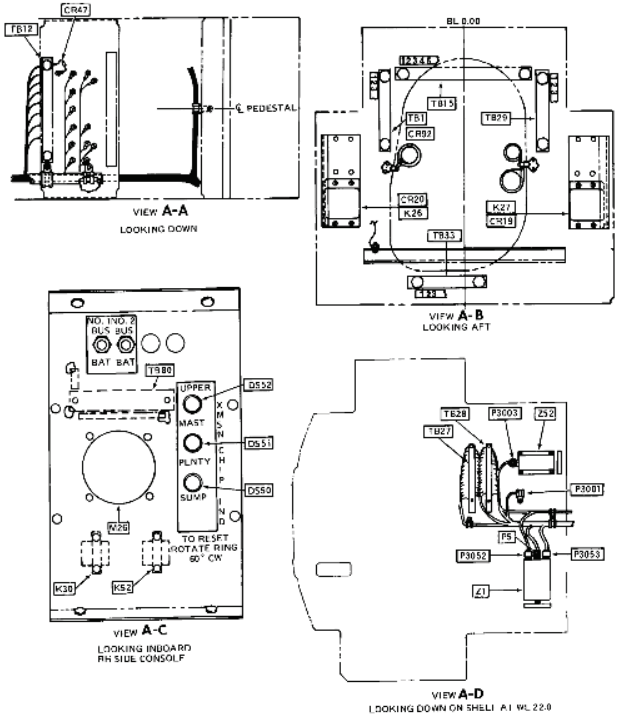
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 2 of 11)



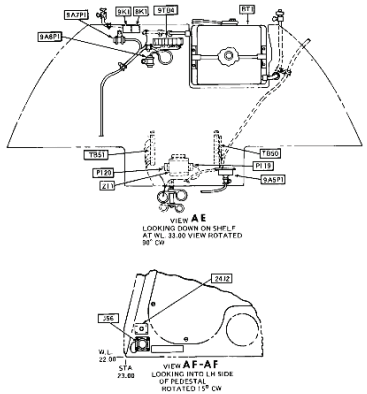
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 3 of 11)



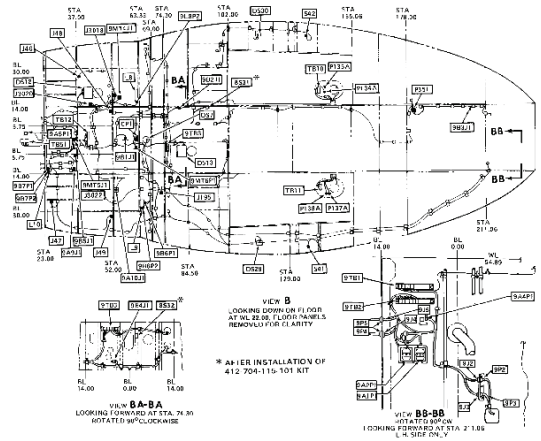
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 4 of 11)



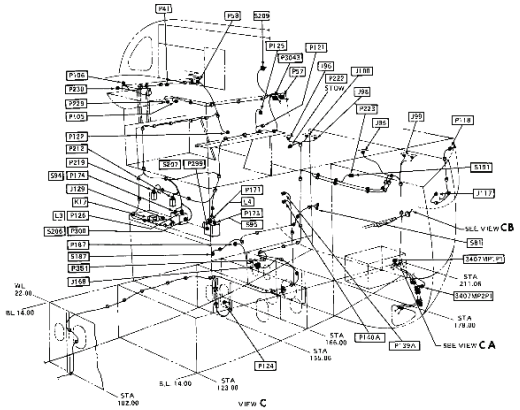
ICN-412-A-960002-B-97499-00006-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 5 of 11)



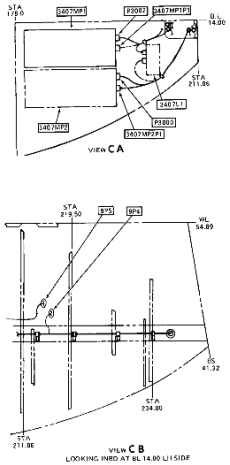
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 6 of 11)



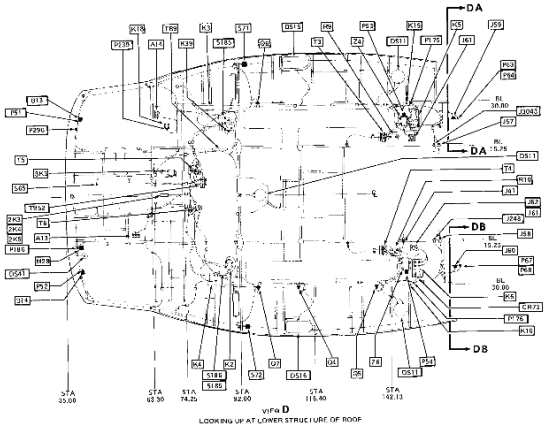
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 7 of 11)



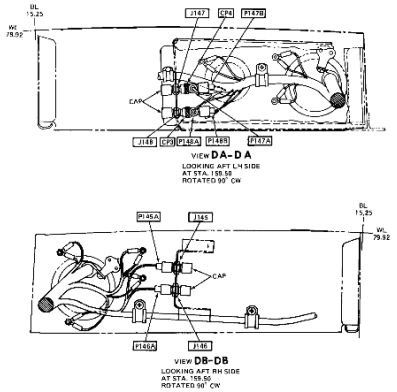
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 8 of 11)



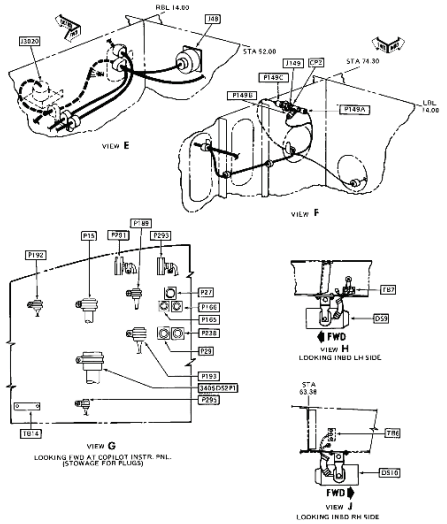
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Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 9 of 11)



ICN-412-A-960002-B-97499-00011-A-001-01

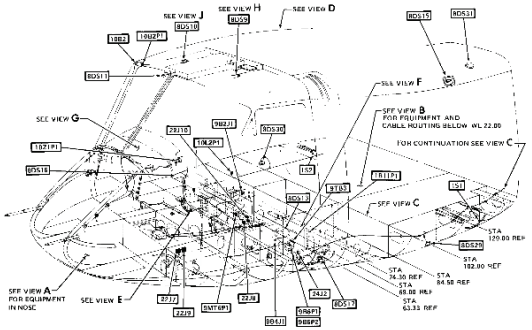
Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 10 of 11)



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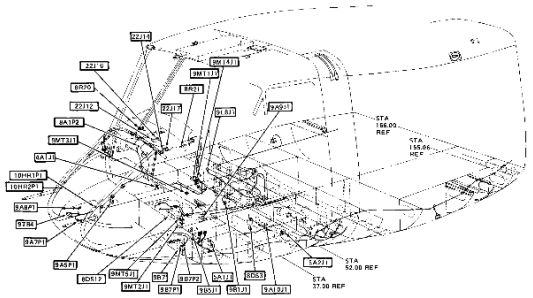
Figure 1. Fuselage Equipment Location (S/N 33001 through 33100) (Sheet 11 of 11)

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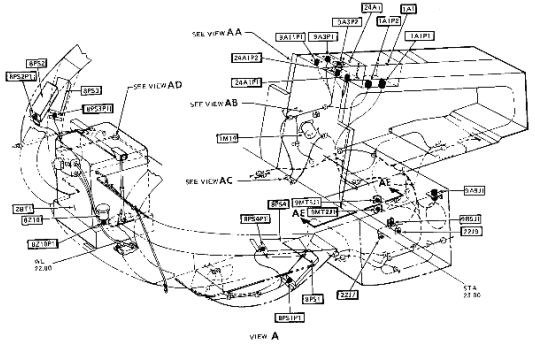
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 1 of 11)



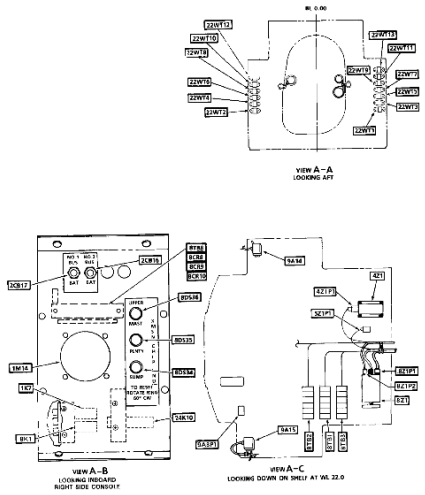
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 2 of 11)



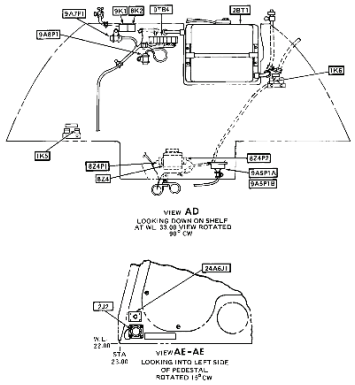
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 3 of 11)



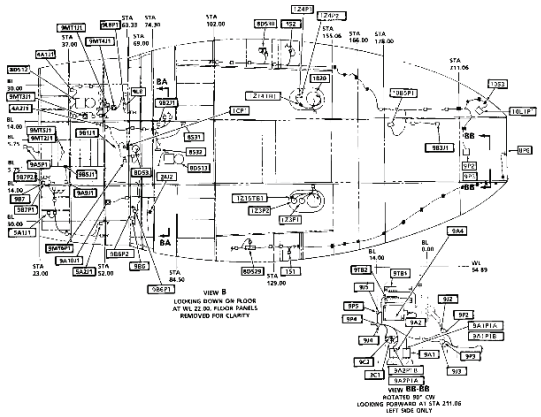
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 4 of 11)



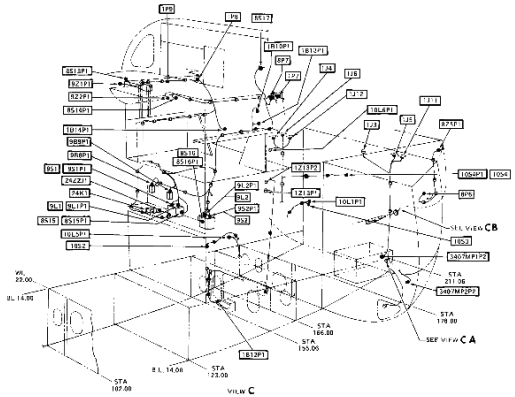
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 5 of 11)



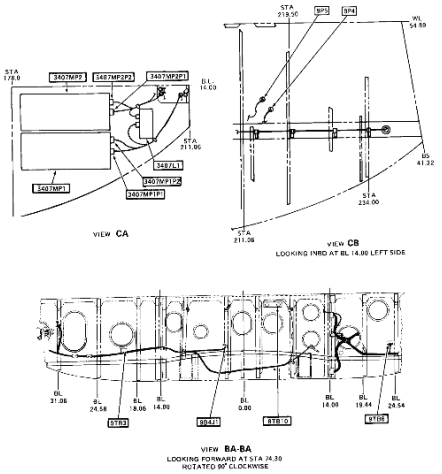
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 6 of 11)



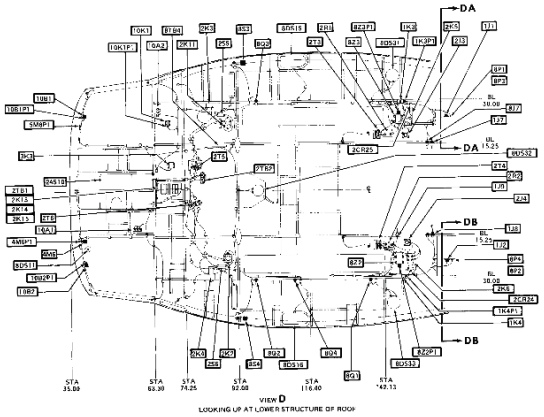
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 7 of 11)



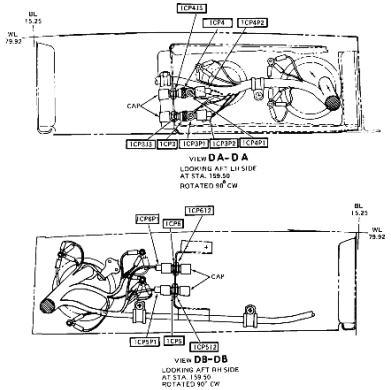
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 8 of 11)



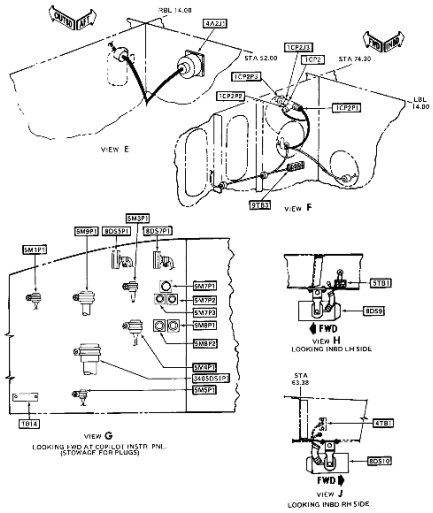
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Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 9 of 11)



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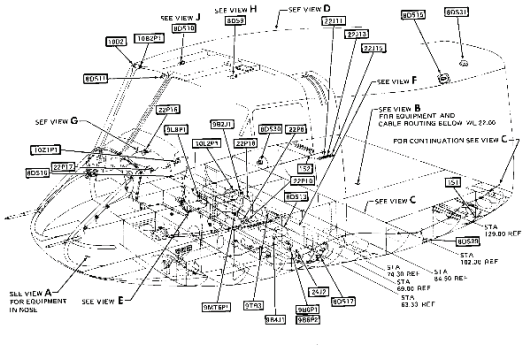
Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 10 of 11)



ICN-412-A-960002-B-97499-00023-A-001-01

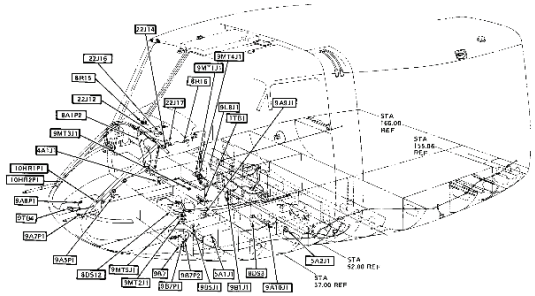
Figure 2. Fuselage Equipment Location (S/N 33101 through 33107) (Sheet 11 of 11)

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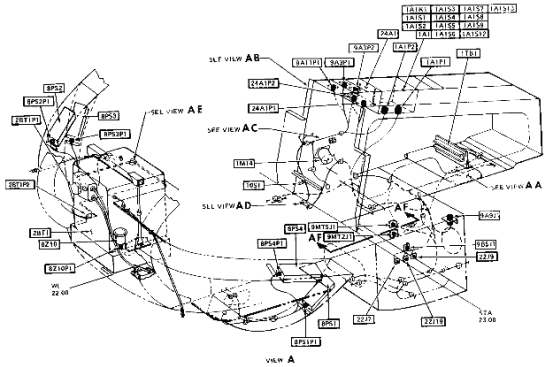
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 1 of 11)



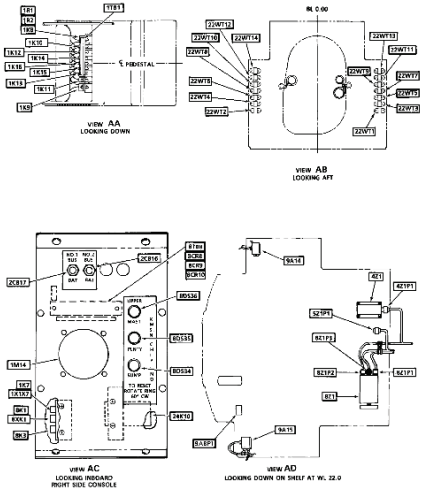
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 2 of 11)



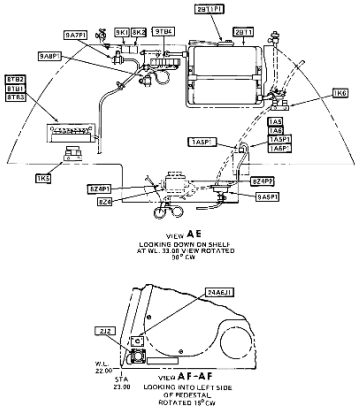
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 3 of 11)



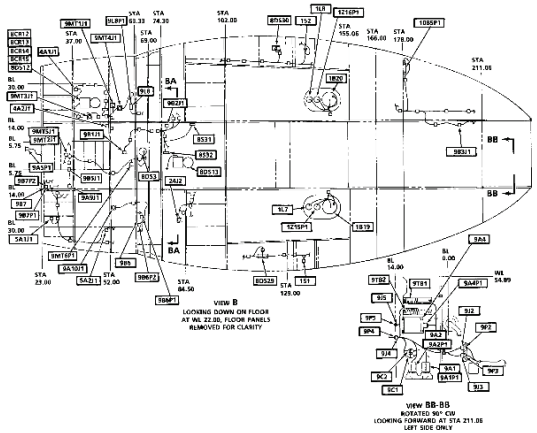
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 4 of 11)



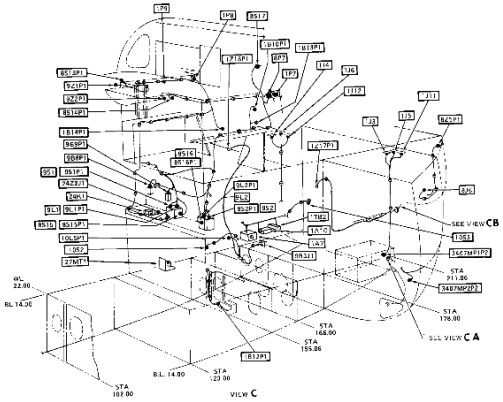
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 5 of 11)



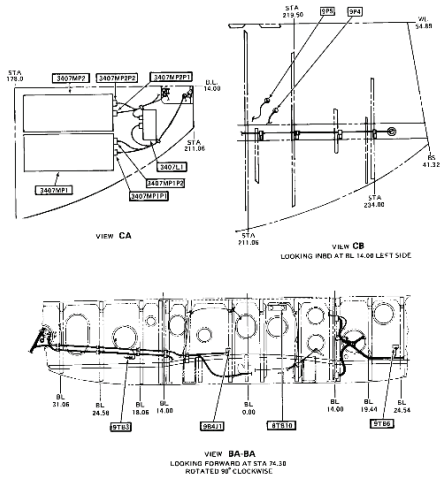
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 6 of 11)



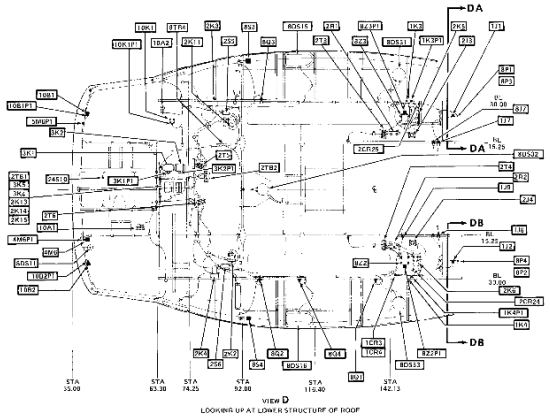
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 7 of 11)



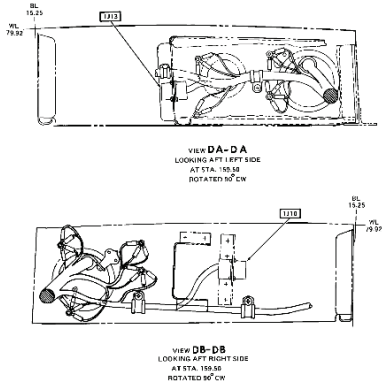
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 8 of 11)



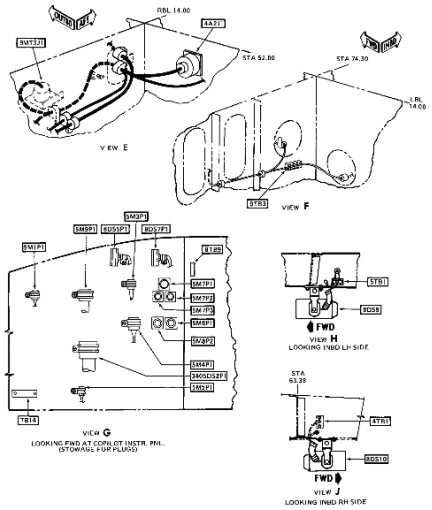
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Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 9 of 11)



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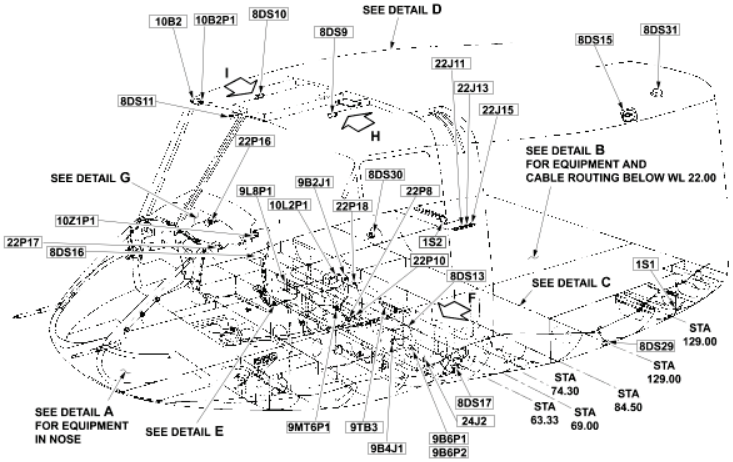
Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 10 of 11)



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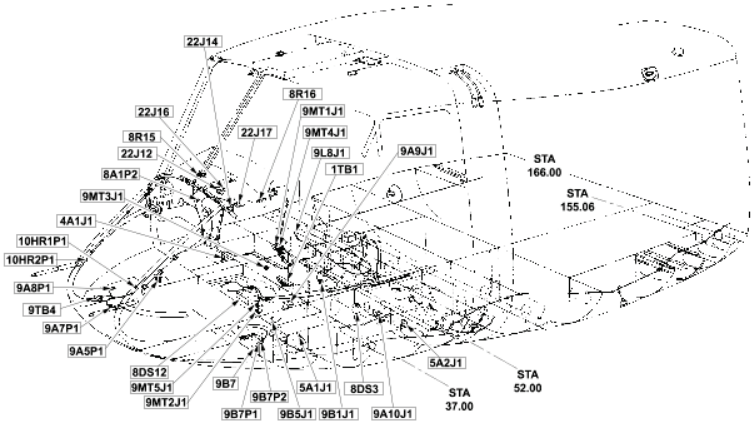
Figure 3. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 11 of 11)

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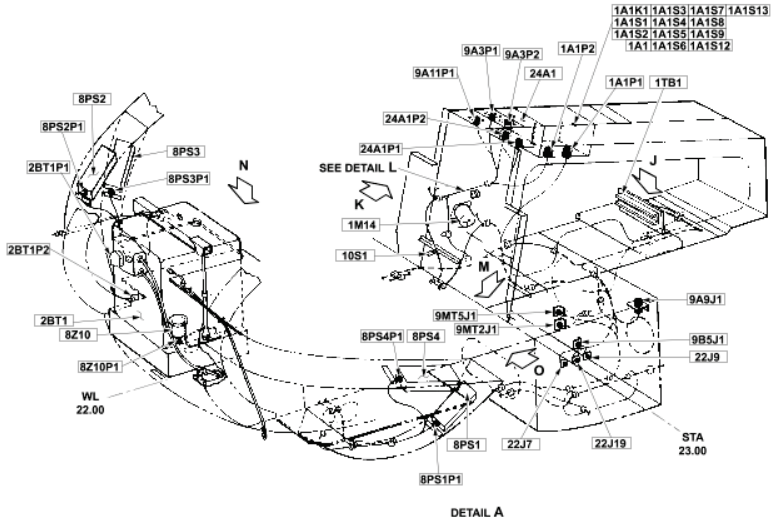
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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 1 of 15)



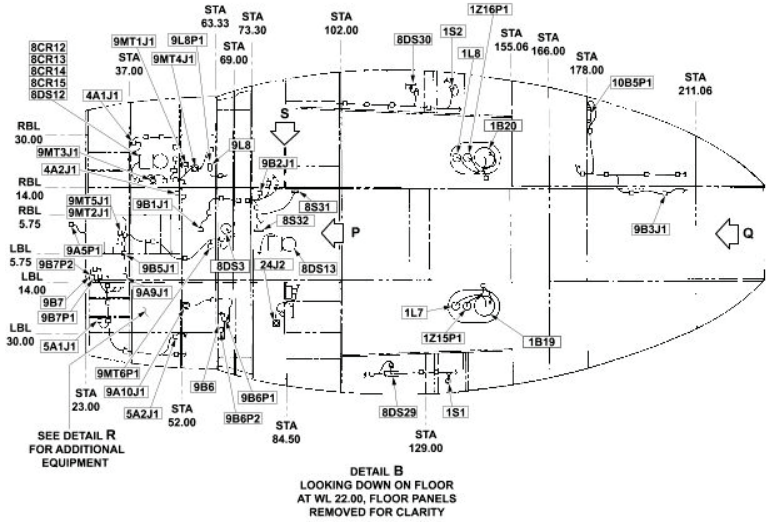
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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 2 of 15)



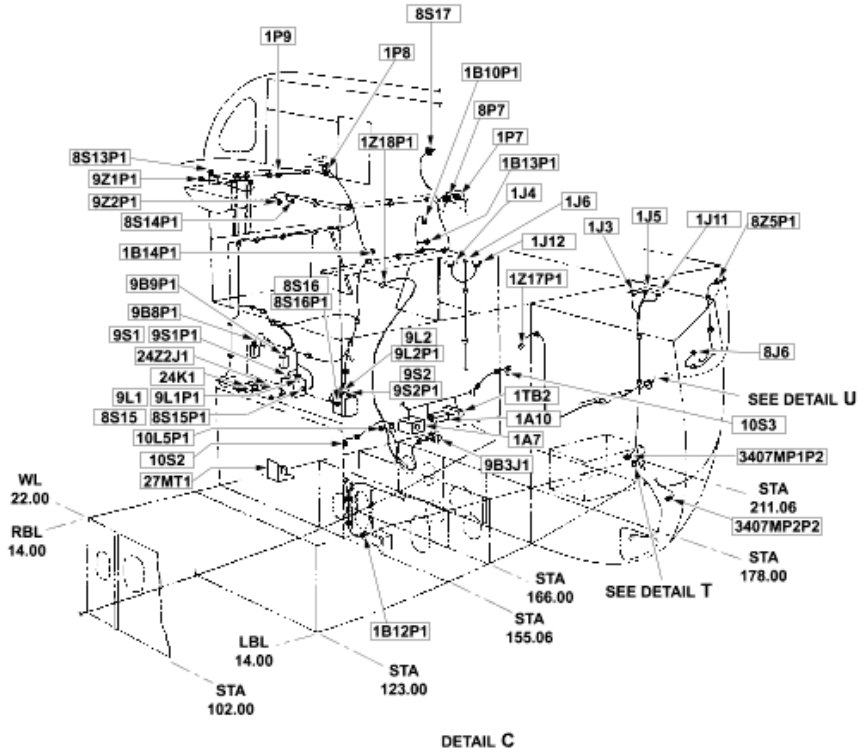
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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 3 of 15)



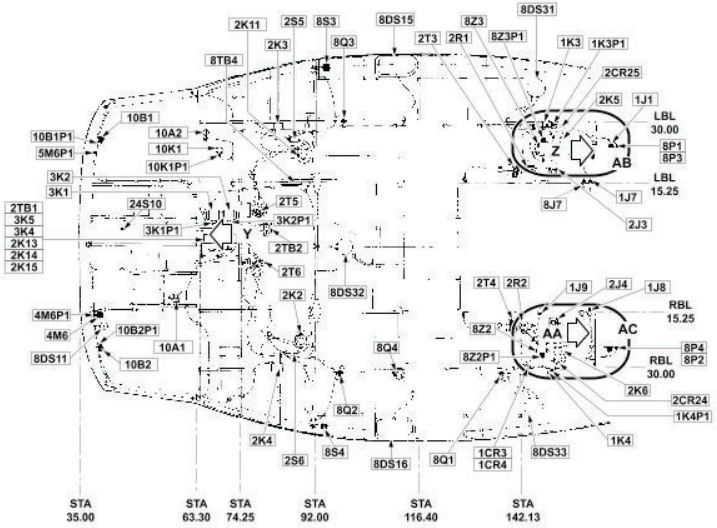
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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 4 of 15)



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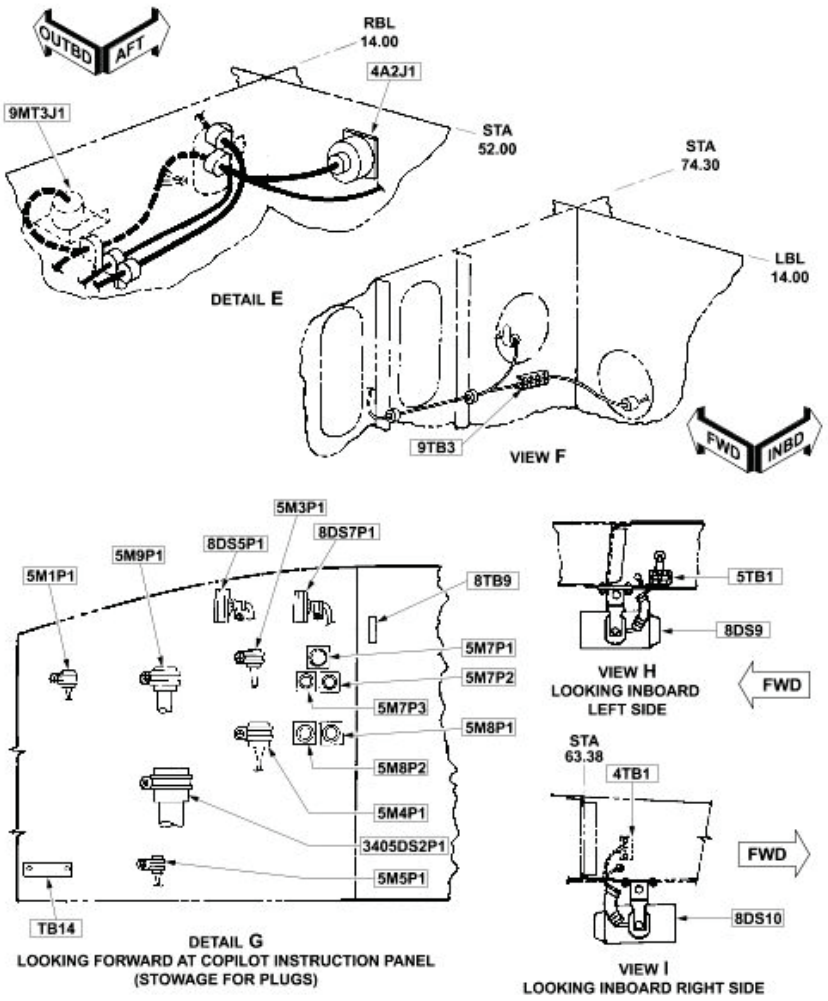
Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 5 of 15)



DETAIL D
LOOKING UP AT LOWER STRUCTURE OF ROOF

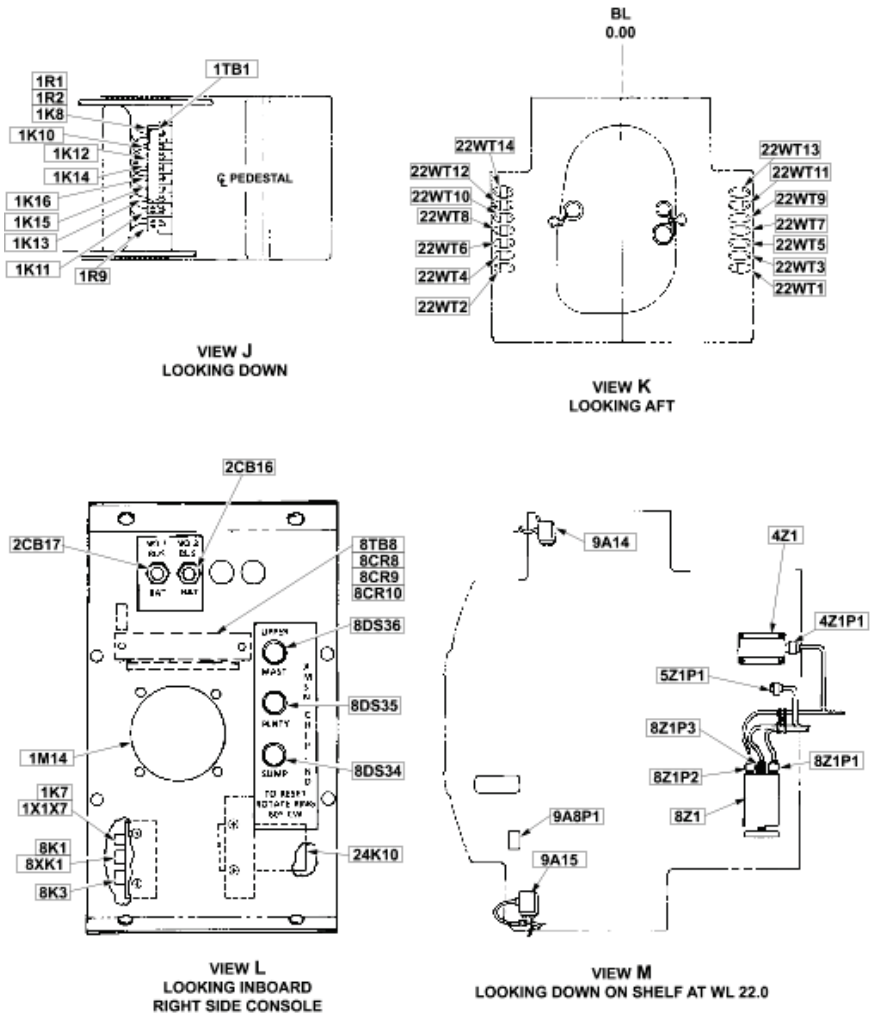
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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 6 of 15)



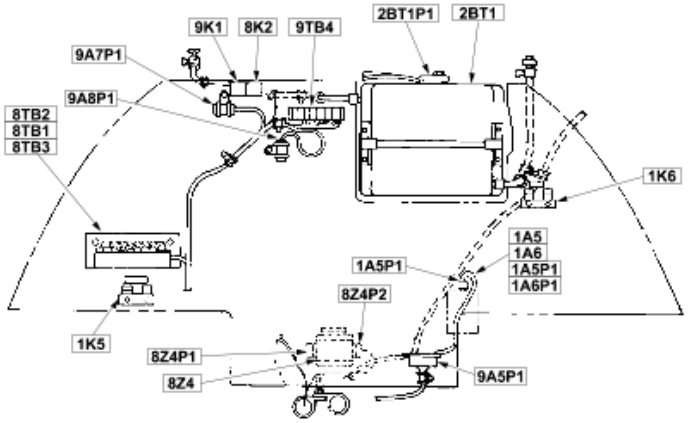
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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 7 of 15)

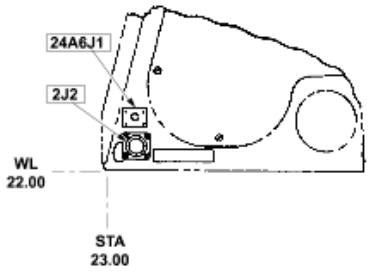


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Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 8 of 15)



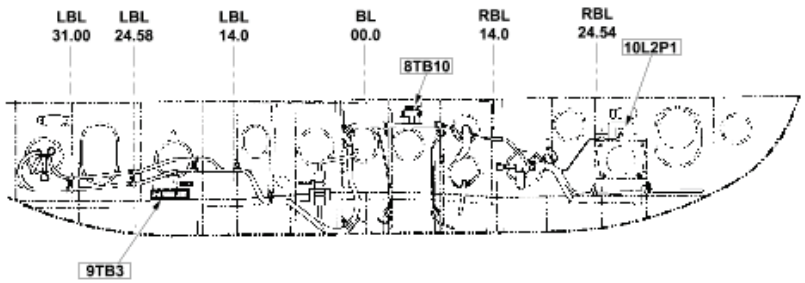
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LOOKING DOWN ON SHELF
AT WL 33.00 VIEW ROTATED 90° CLOCKWISE



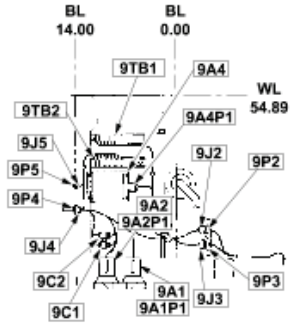
VIEW O
LOOKING INTO LEFT SIDE OF PEDESTAL
ROTATED 15° CLOCKWISE

ICN-412-A-960002-B-97499-00043-A-001-01

Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 9 of 15)



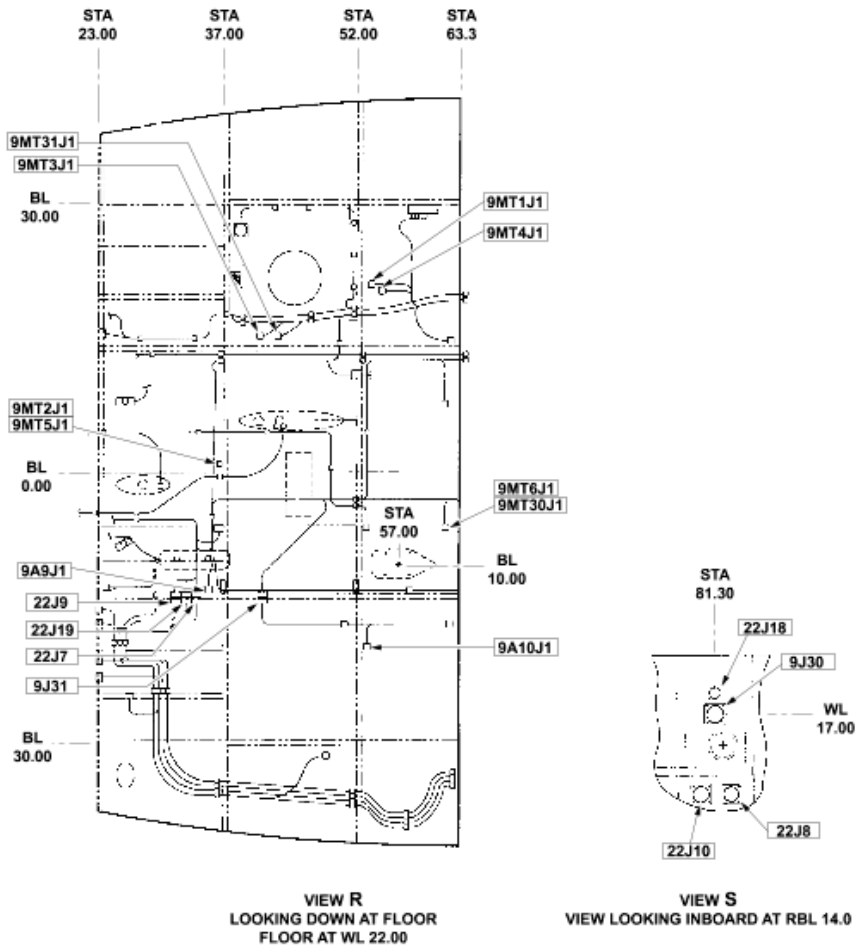
VIEW P
LOOKING FORWARD AT STA 74.3



VIEW Q
ROTATED 90° CLOCKWISE
LOOKING FORWARD AT STA 211.06
LEFT SIDE ONLY

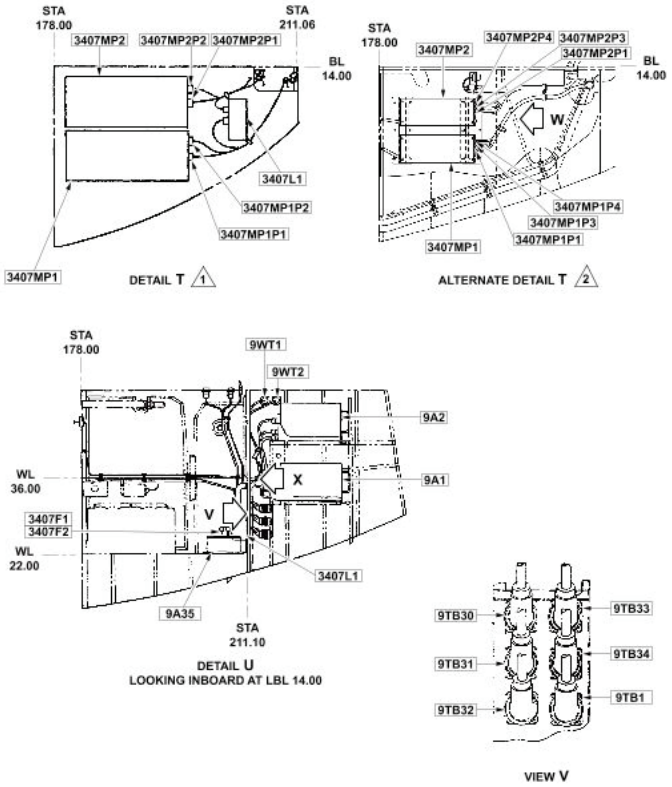
ICN-412-A-960002-B-97499-00044-A-001-01

Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 10 of 15)



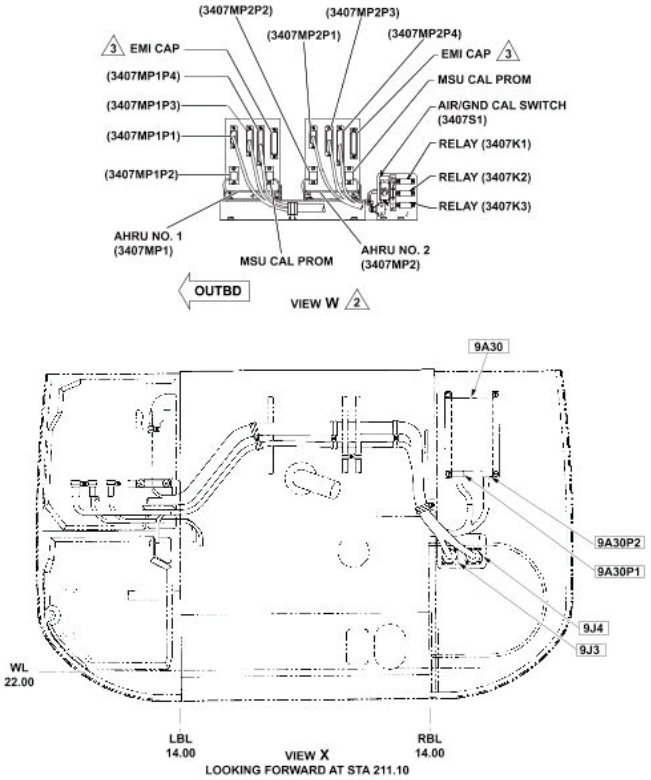
ICN-412-A-960002-B-97499-00045-A-001-01

Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 11 of 15)



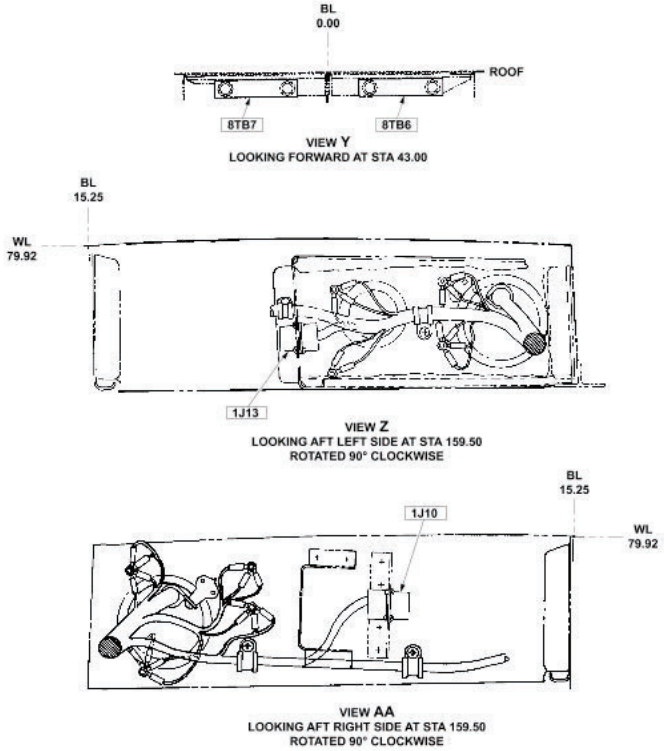
ICN-412-A-960002-B-97499-00046-A-001-01

Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 12 of 15)



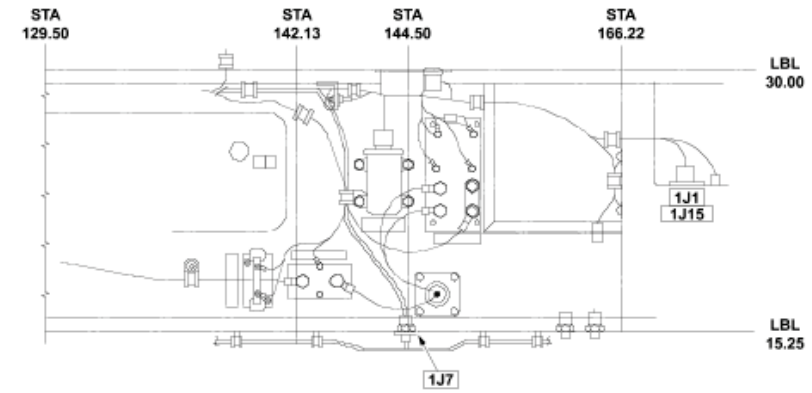
ICN-412-A-960002-B-97499-00047-A-001-01

Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 13 of 15)

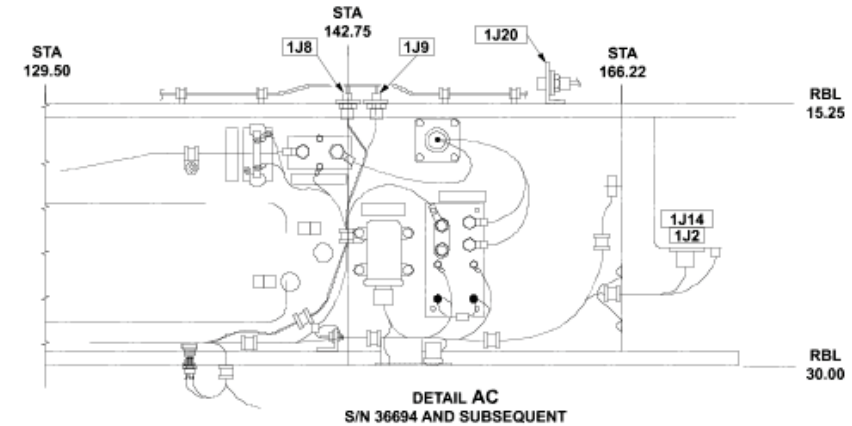


ICN-412-A-960002-B-97499-00048-A-001-01

Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 14 of 15)



DETAIL AB
S/N 36694 AND SUBSEQUENT



DETAIL AC
S/N 36694 AND SUBSEQUENT

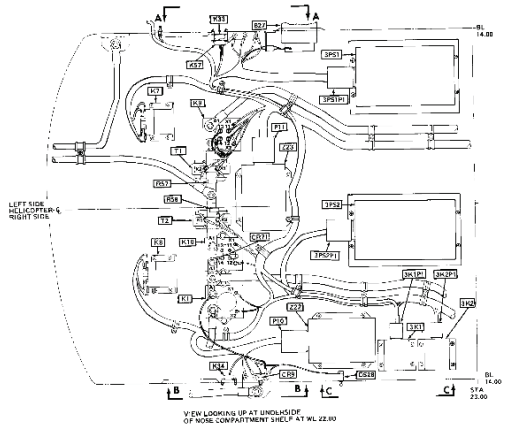
NOTES

1. S/N 36087 through 36247.
2. S/N 36248 through 36999.
3. S/N 36248 through 36704 Post ASB 412-15-169 and 36705 through 36999.

ICN-412-A-960002-B-97499-00049-A-001-01

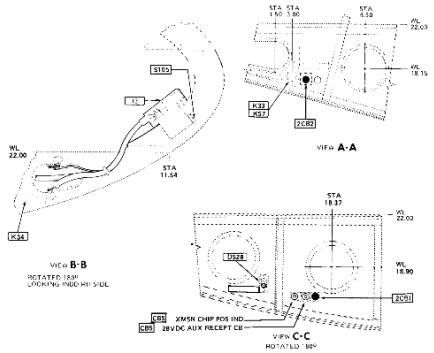
Figure 4. Fuselage Equipment Location (S/N 36087 through 36999) (Sheet 15 of 15)

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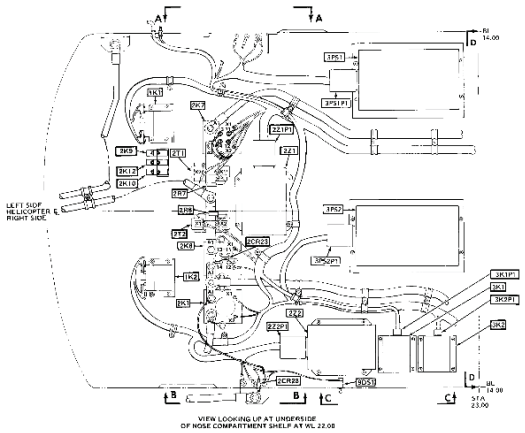
ICN-412-A-960002-B-97499-00050-A-001-01

Figure 5. Nose Compartment Equipment Location (S/N 33001 through 33100) (Sheet 1 of 2)



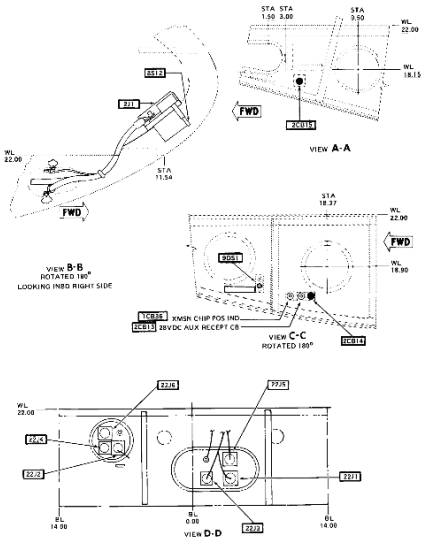
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Figure 5. Nose Compartment Equipment Location (S/N 33001 through 33100) (Sheet 2 of 2)



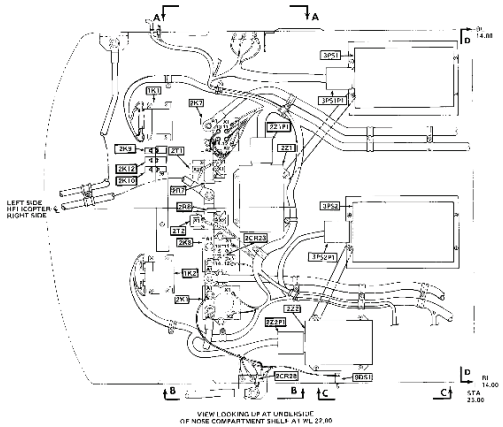
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Figure 6. Nose Compartment Equipment Location (S/N 33101 through 33107) (Sheet 1 of 2)



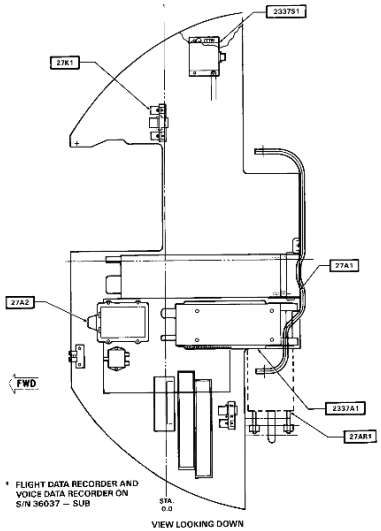
ICN-412-A-960002-B-97499-00053-A-001-01

Figure 6. Nose Compartment Equipment Location (S/N 33101 through 33107) (Sheet 2 of 2)



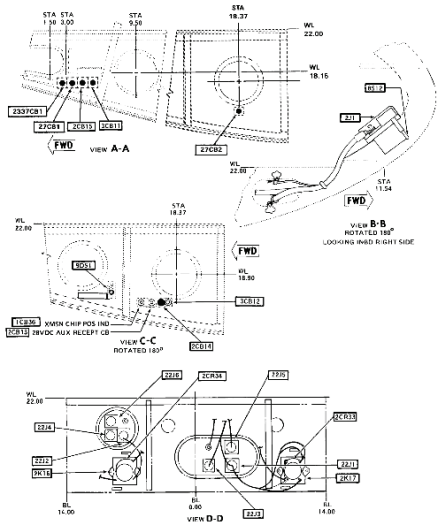
ICN-412-A-960002-B-97499-00054-A-001-01

Figure 7. Nose Compartment Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 1 of 3)



ICN-412-A-960002-B-97499-00055-A-001-01

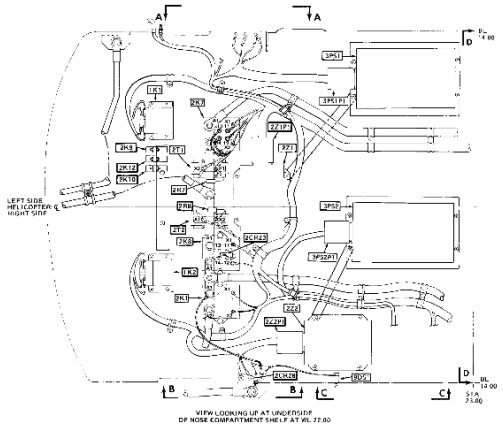
Figure 7. Nose Compartment Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 2 of 3)



ICN-412-A-960002-B-97499-00056-A-001-01

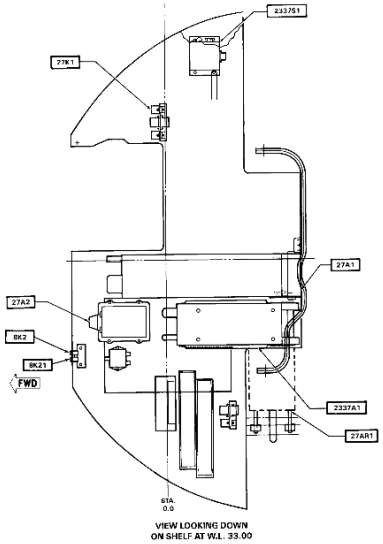
Figure 7. Nose Compartment Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 3 of 3)

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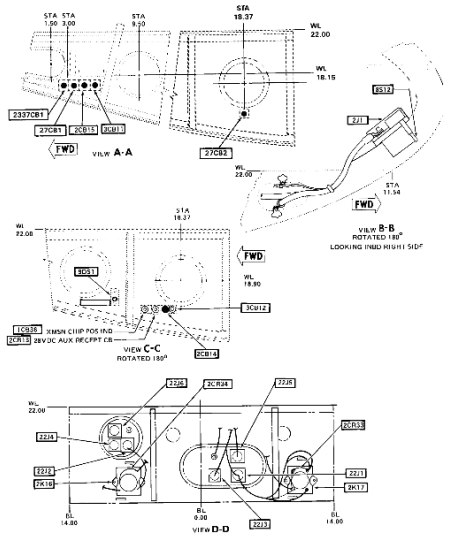
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Figure 8. Nose Compartment Equipment Location (S/N 36087 through 36999) (Sheet 1 of 4)



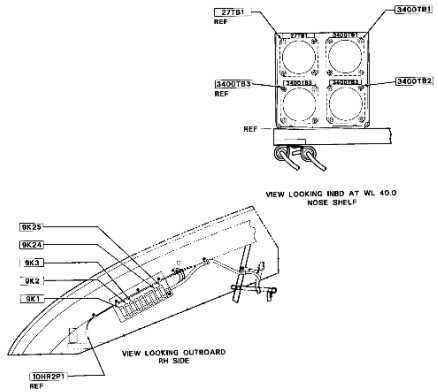
ICN-412-A-960002-B-97499-00058-A-001-01

Figure 8. Nose Compartment Equipment Location (S/N 36087 through 36999) (Sheet 2 of 4)



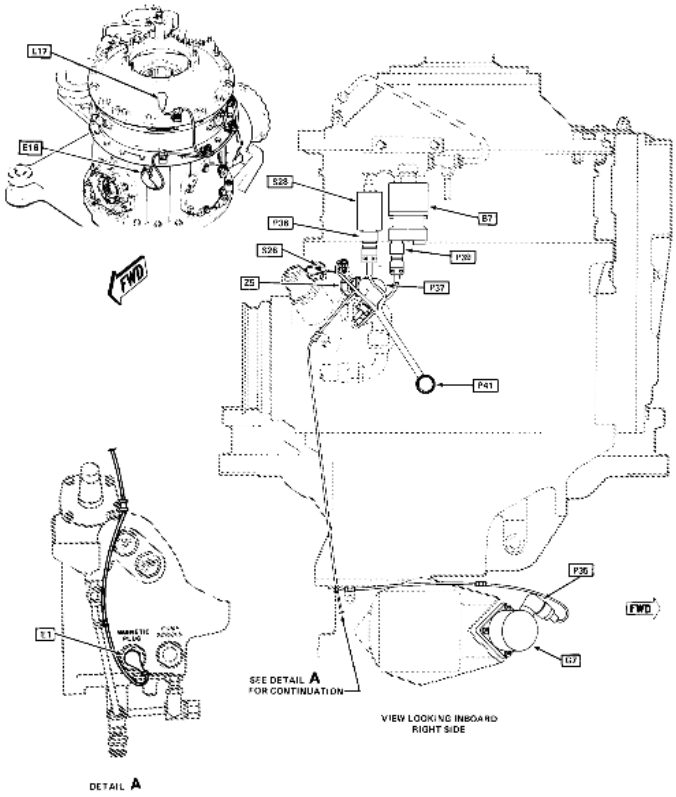
ICN-412-A-960002-B-97499-00059-A-001-01

Figure 8. Nose Compartment Equipment Location (S/N 36087 through 36999) (Sheet 3 of 4)



ICN-412-A-960002-B-97499-00060-A-001-01

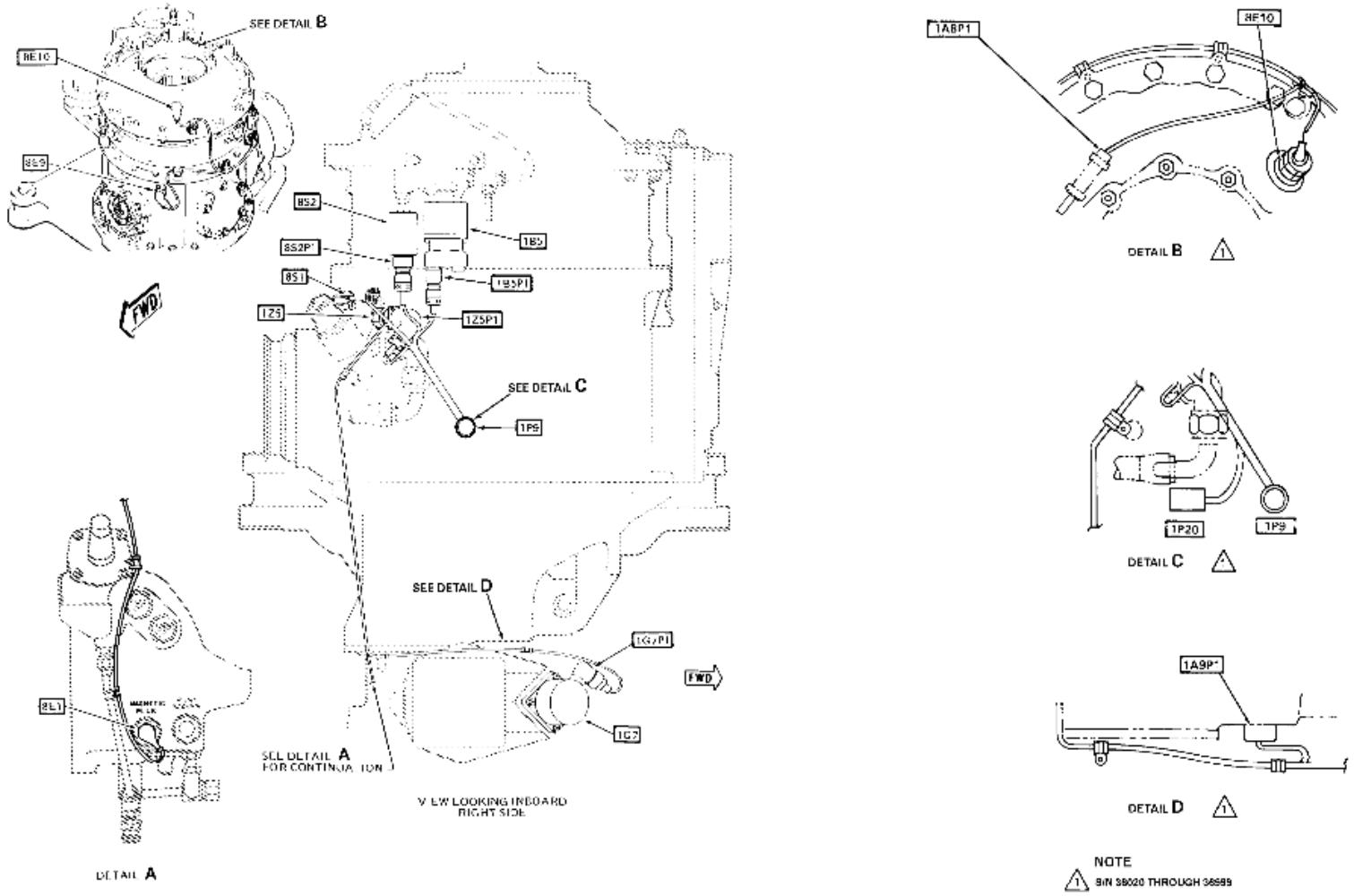
Figure 8. Nose Compartment Equipment Location (S/N 36087 through 36999) (Sheet 4 of 4)



ICN-412-A-960002-B-97499-00061-A-001-01

Figure 9. Transmission Equipment Location (S/N 33001 through 33100) (Sheet 1 of 1)

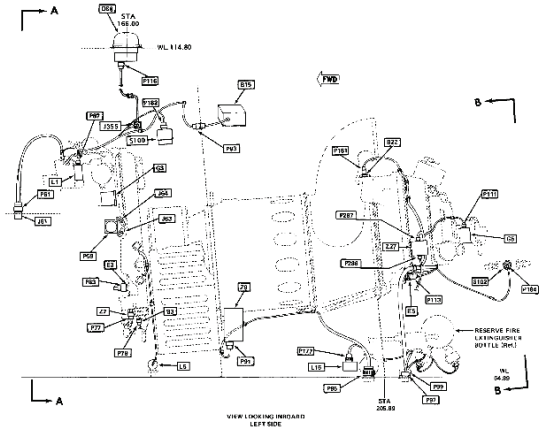
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ICN-412-A-960002-B-97499-00062-A-001-01

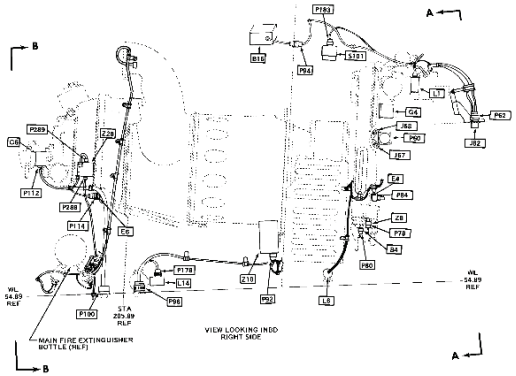
Figure 10. Transmission Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 1 of 1)

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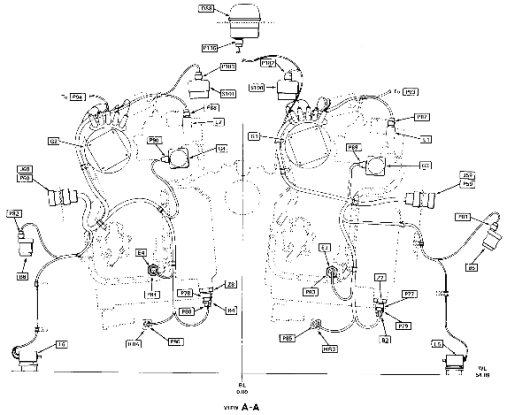
ICN-412-A-960002-B-97499-00063-A-001-01

Figure 11. Engine Equipment Location (S/N 33001 through 33100) (Sheet 1 of 4)



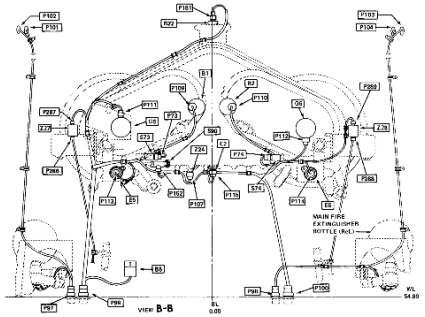
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Figure 11. Engine Equipment Location (S/N 33001 through 33100) (Sheet 2 of 4)



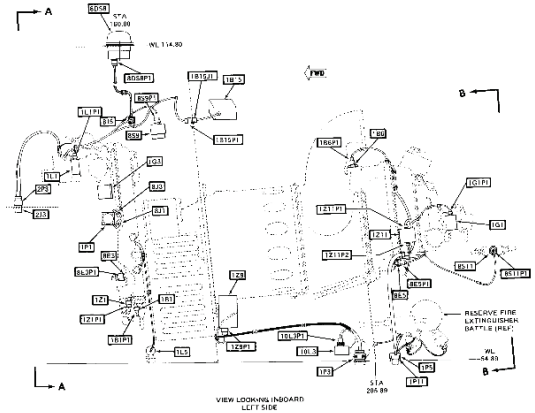
ICN-412-A-960002-B-97499-00065-A-001-01

Figure 11. Engine Equipment Location (S/N 33001 through 33100) (Sheet 3 of 4)



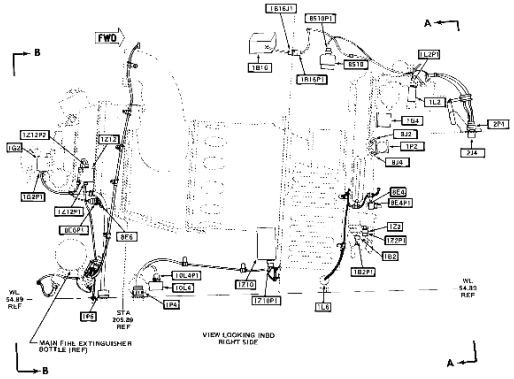
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Figure 11. Engine Equipment Location (S/N 33001 through 33100) (Sheet 4 of 4)



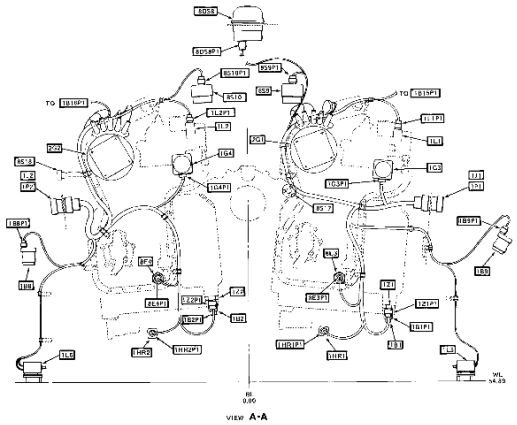
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Figure 12. Engine Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 1 of 5)



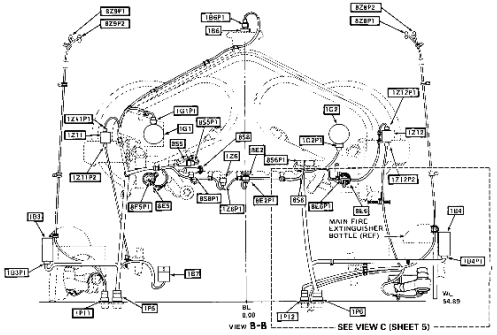
ICN-412-A-960002-B-97499-00068-A-001-01

Figure 12. Engine Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 2 of 5)



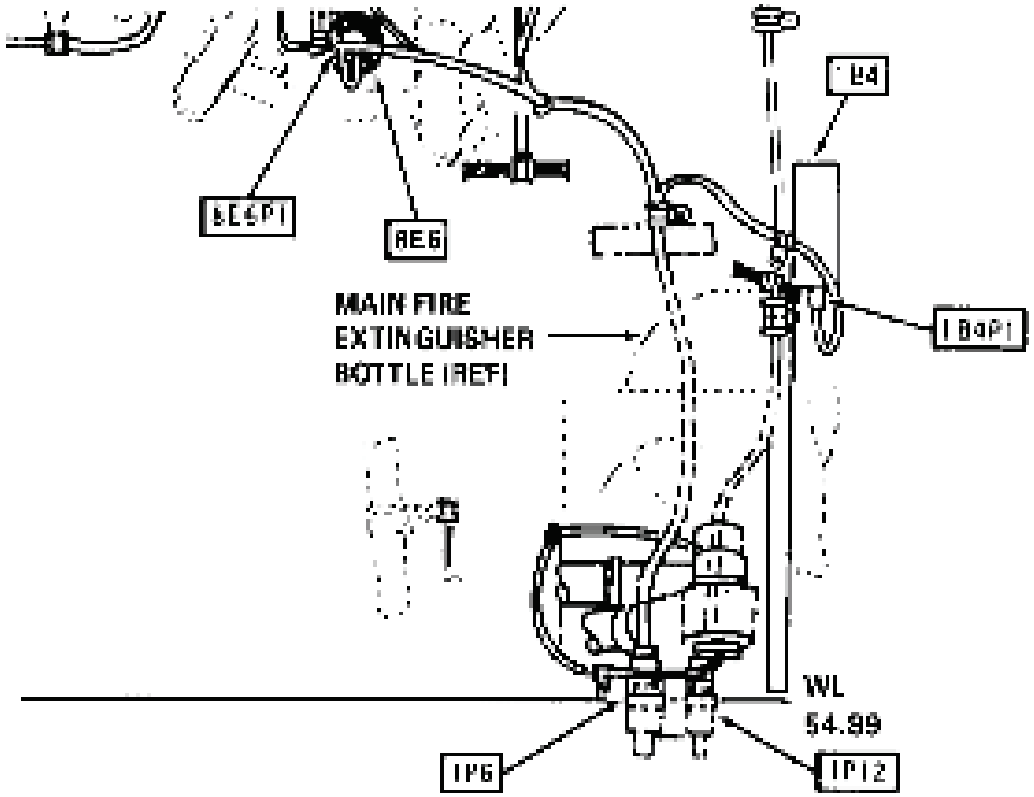
ICN-412-A-960002-B-97499-00069-A-001-01

Figure 12. Engine Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 3 of 5)



ICN-412-A-960002-B-97499-00070-A-001-01

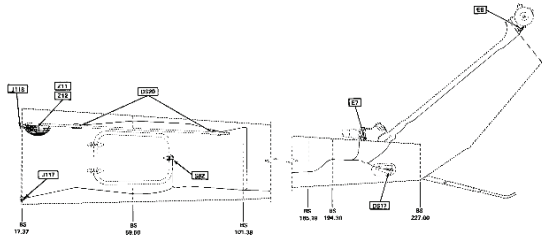
Figure 12. Engine Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 4 of 5)



ICN-412-A-960002-B-97499-00071-A-001-01

Figure 12. Engine Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 5 of 5)

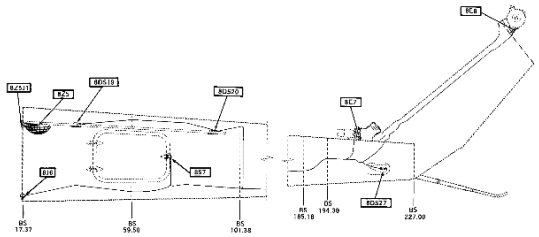
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Figure 13. Tailboom Equipment Location (S/N 33001 through 33100) (Sheet 1 of 1)

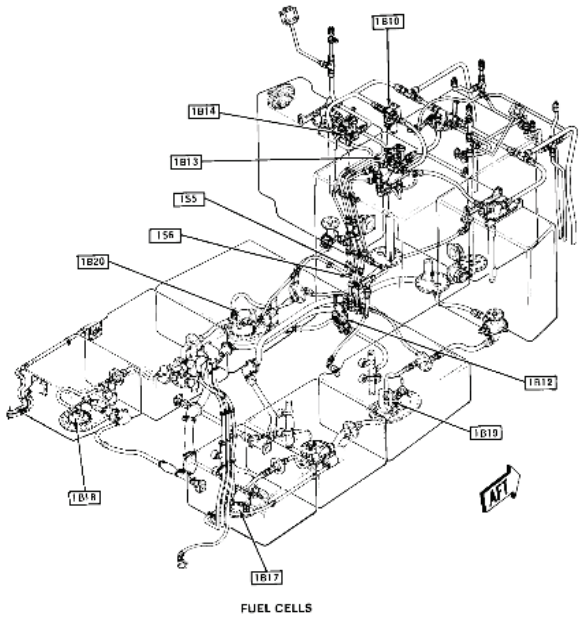
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ICN-412-A-960002-B-97499-00073-A-001-01

Figure 14. Tailboom Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 1 of 1)

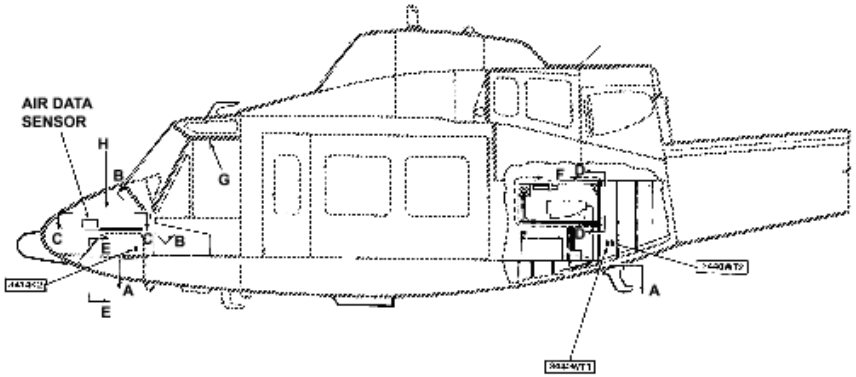
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ICN-412-A-960002-B-97499-00074-A-001-01

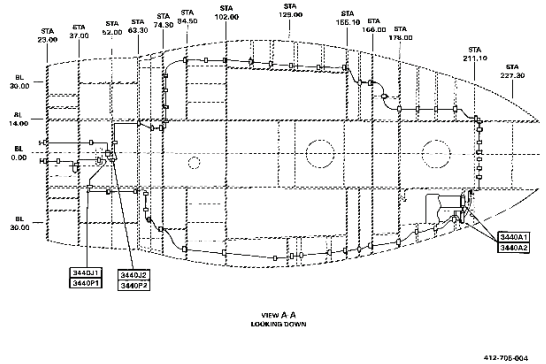
Figure 15. Fuel Cell Equipment Location (S/N 33101 through 33213 and 36001 through 36999) (Sheet 1 of 1)

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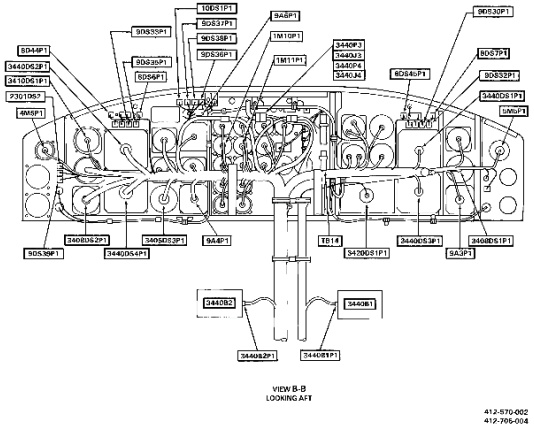
ICN-412-A-960002-B-97499-00075-A-001-01

Figure 16. 4-Axis Configuration Related Component Location (Sheet 1 of 7)



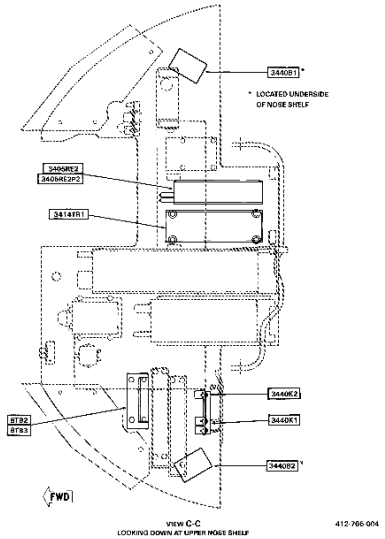
ICN-412-A-960002-B-97499-00076-A-001-01

Figure 16. 4-Axis Configuration Related Component Location (Sheet 2 of 7)



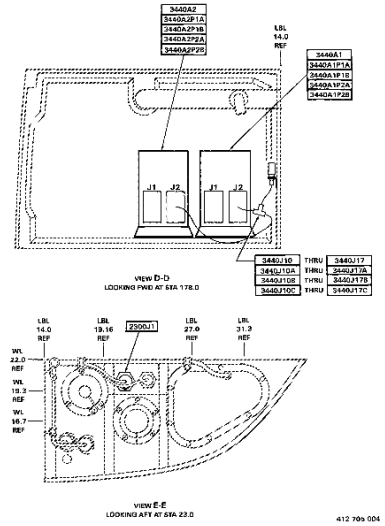
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Figure 16. 4-Axis Configuration Related Component Location (Sheet 3 of 7)



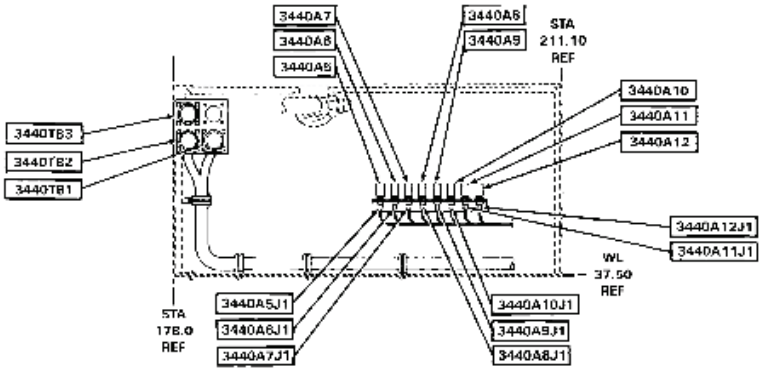
ICN-412-A-960002-B-97499-00078-A-001-01

Figure 16. 4-Axis Configuration Related Component Location (Sheet 4 of 7)



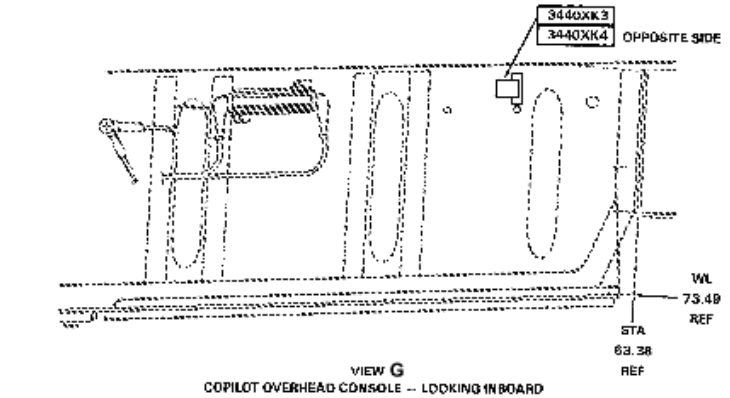
ICN-412-A-960002-B-97499-00079-A-001-01

Figure 16. 4-Axis Configuration Related Component Location (Sheet 5 of 7)



ICN-412-A-960002-B-97499-00080-A-001-01

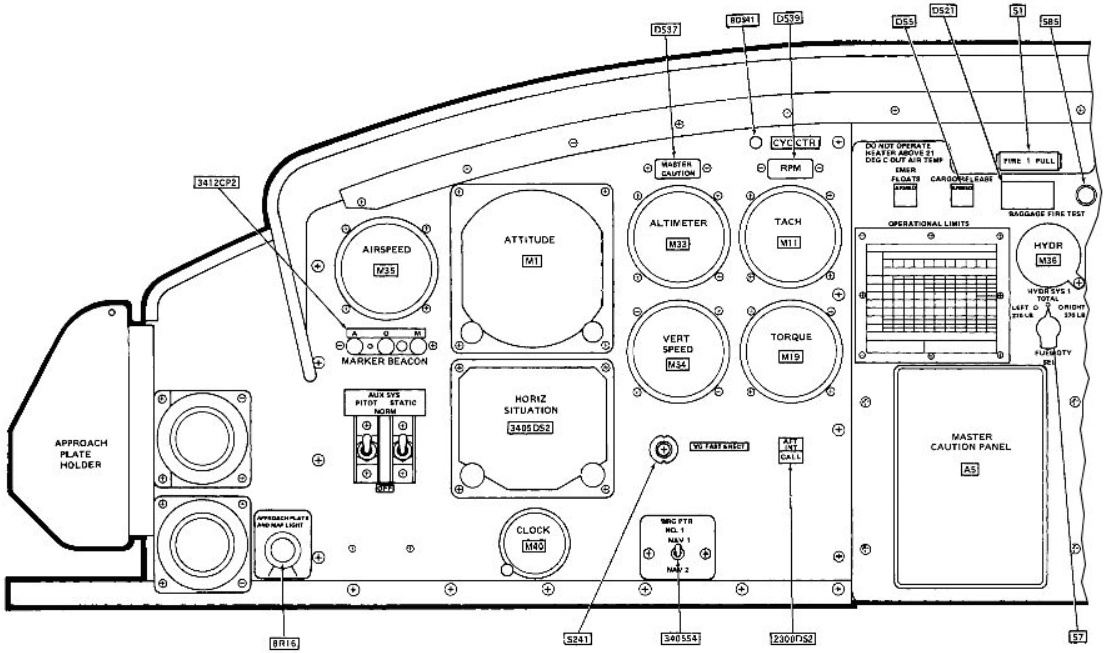
Figure 16. 4-Axis Configuration Related Component Location (Sheet 6 of 7)



ICN-412-A-960002-B-97499-00081-A-001-01

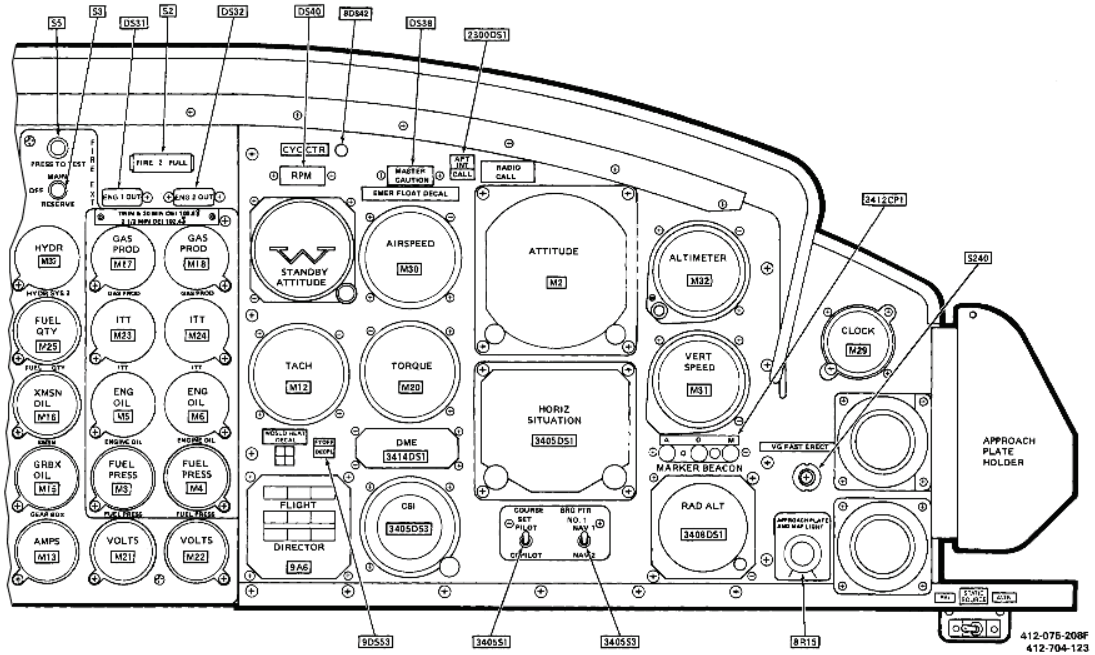
Figure 16. 4-Axis Configuration Related Component Location (Sheet 7 of 7)

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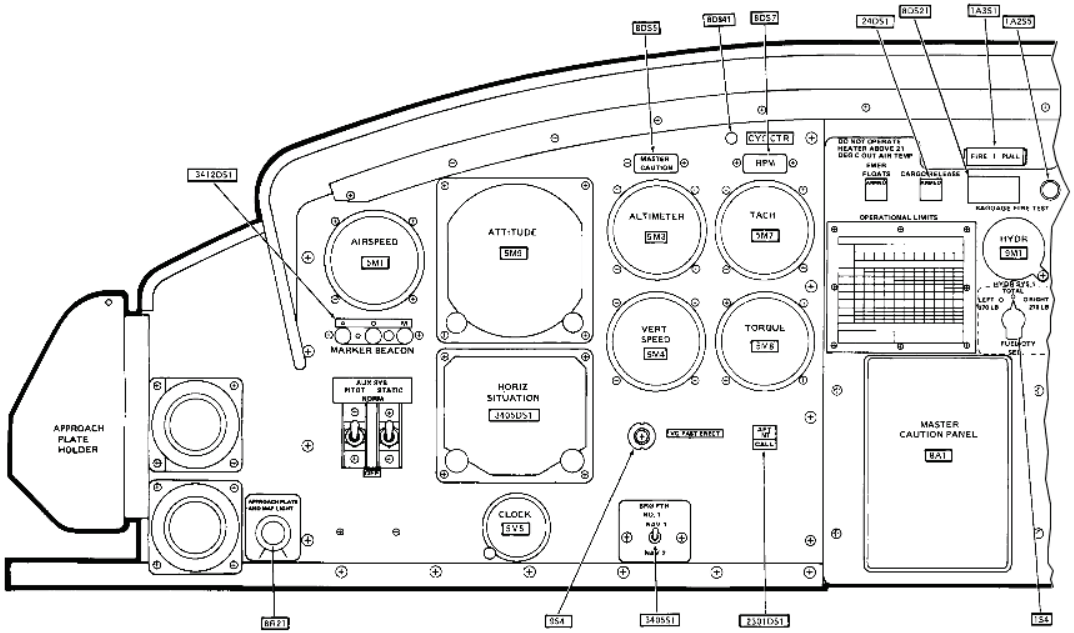
ICN-412-A-960002-B-97499-00082-A-001-01

Figure 17. Instrument Panel (S/N 33001 through 33100) (Post 412-704-115-101) (Sheet 1 of 2)



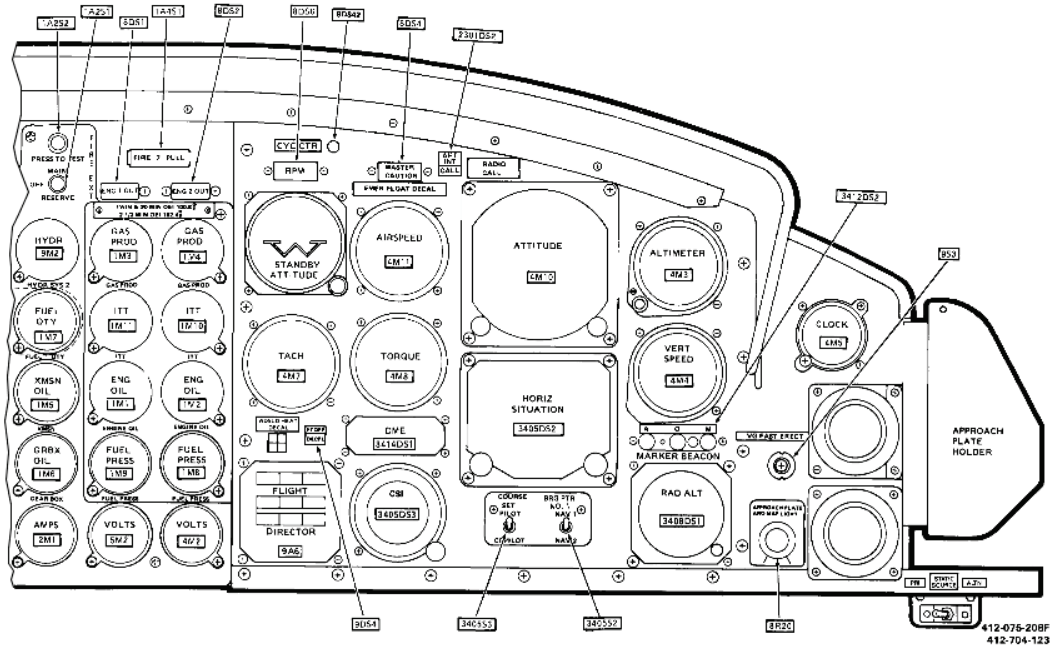
ICN-412-A-960002-B-97499-00083-A-001-01

Figure 17. Instrument Panel (S/N 33001 through 33100) (Post 412-704-115-101) (Sheet 2 of 2)



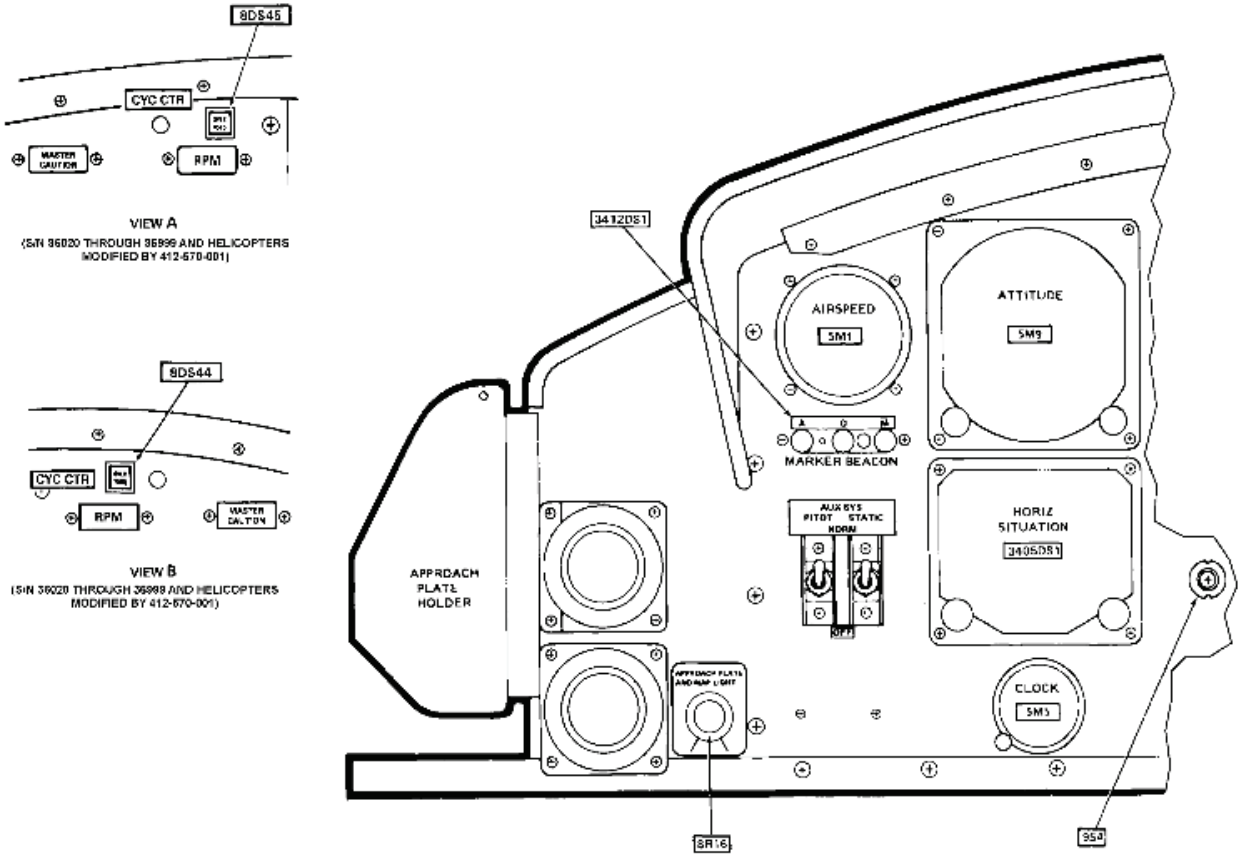
ICN-412-A-960002-B-97499-00084-A-001-01

Figure 18. Instrument Panel (S/N 33101 through 33107) (Sheet 1 of 2)



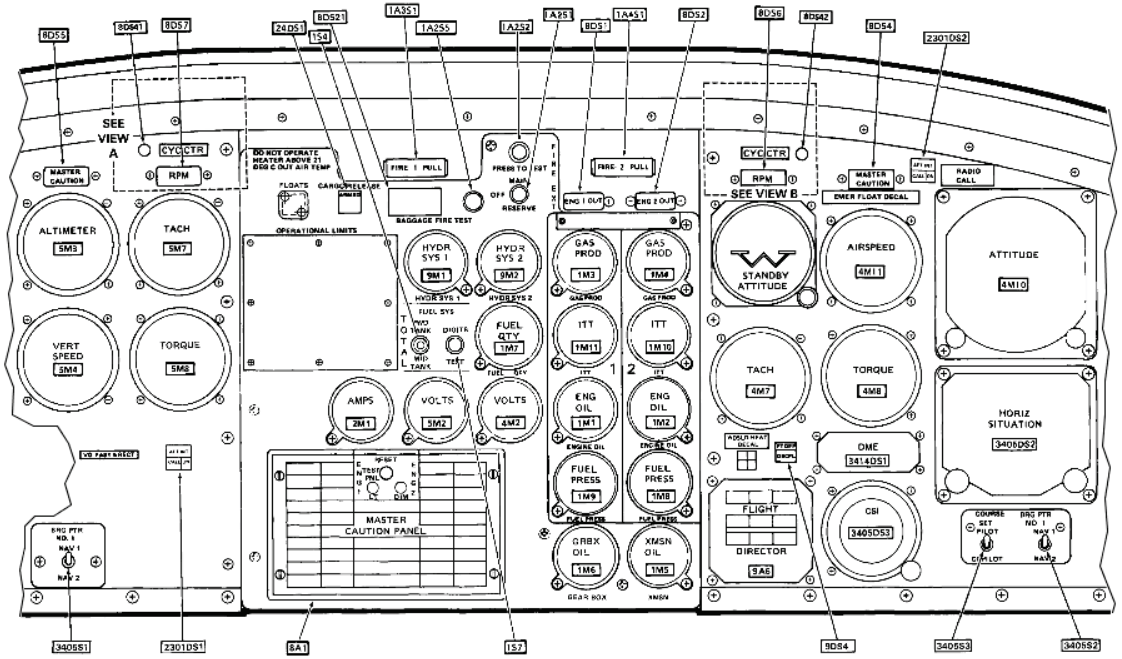
ICN-412-A-960002-B-97499-00085-A-001-01

Figure 18. Instrument Panel (S/N 33101 through 33107) (Sheet 2 of 2)



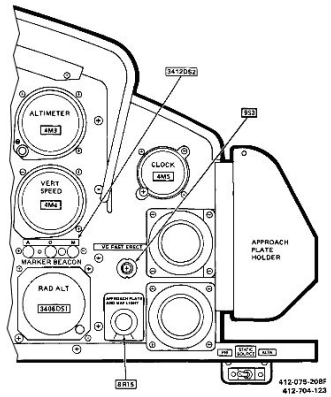
ICN-412-A-960002-B-97499-00086-A-001-01

Figure 19. Instrument Panel (S/N 33108 through 33213 and 36001 through 36086) (Sheet 1 of 3)



ICN-412-A-960002-B-97499-00087-A-001-01

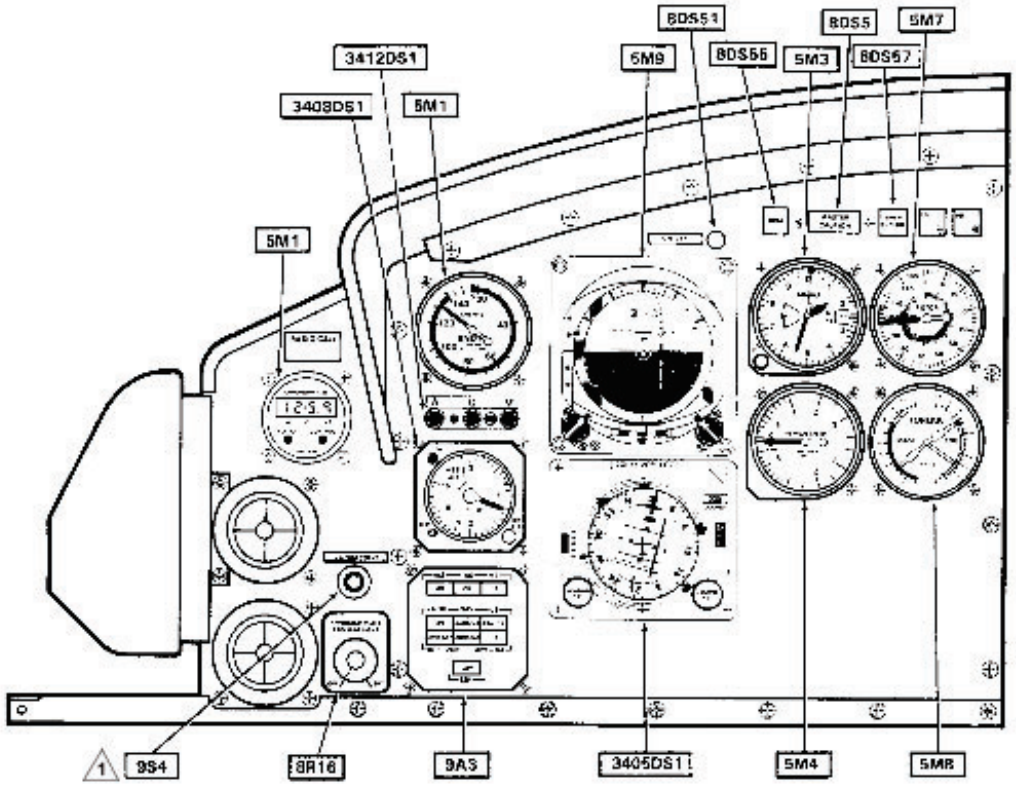
Figure 19. Instrument Panel (S/N 33108 through 33213 and 36001 through 36086) (Sheet 2 of 3)



ICN-412-A-960002-B-97499-00088-A-001-01

Figure 19. Instrument Panel (S/N 33108 through 33213 and 36001 through 36086) (Sheet 3 of 3)

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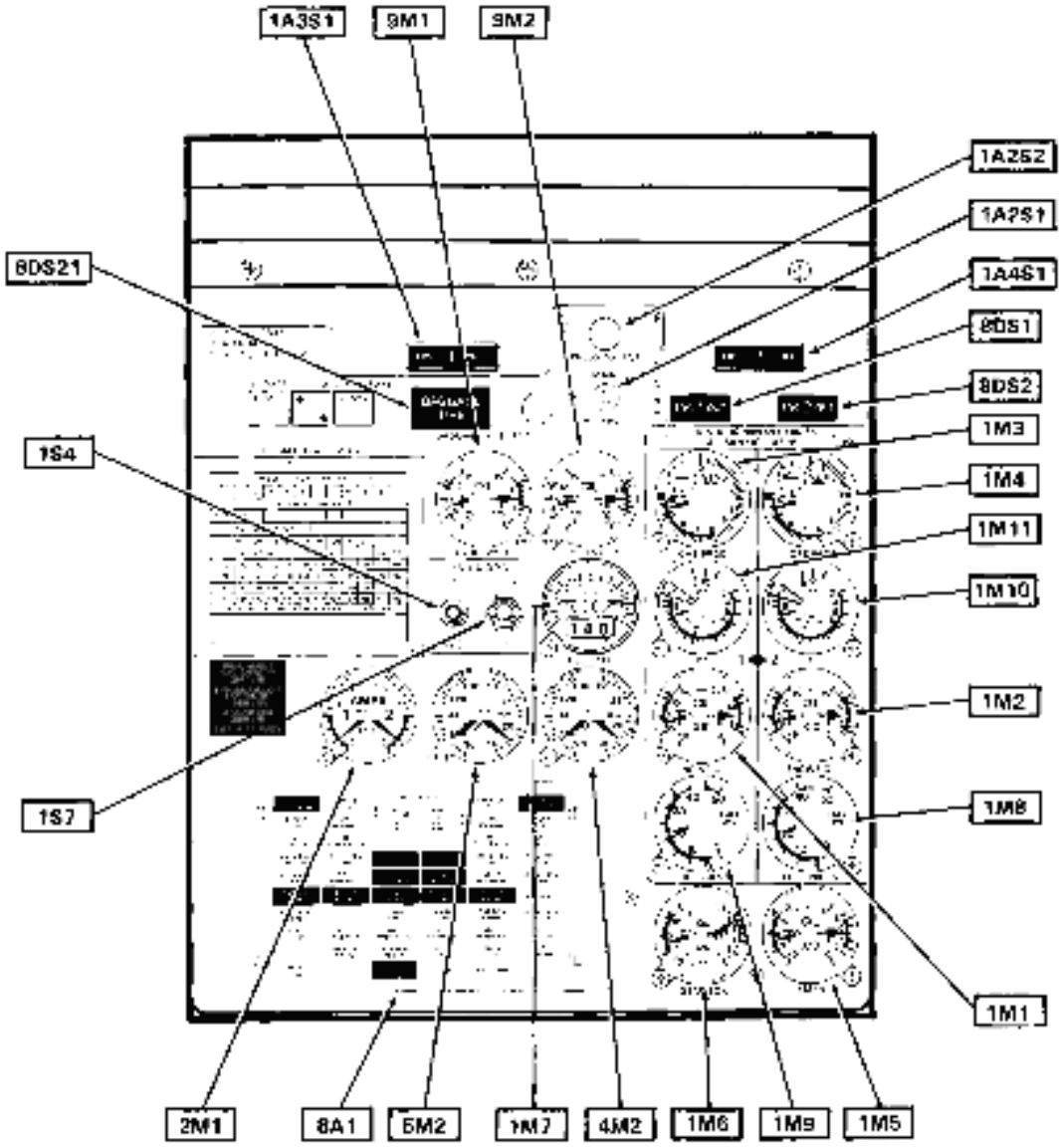


NOTE

1 Applicable to S/N 36087 through 36247.

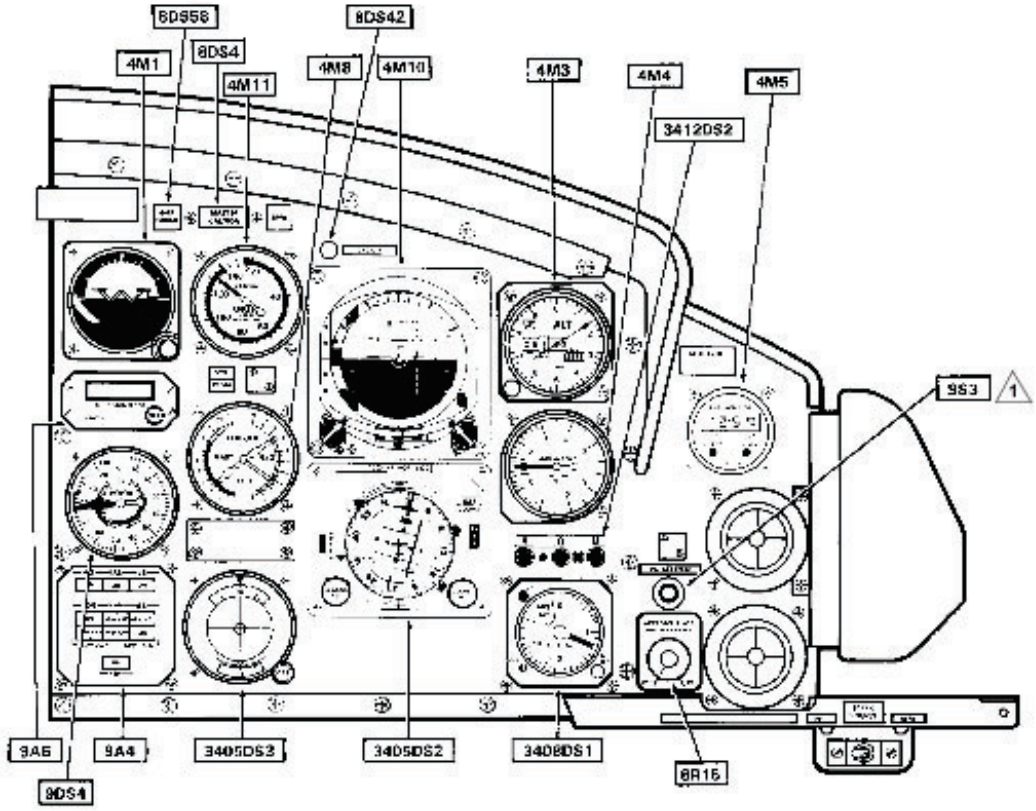
ICN-412-A-960002-B-97499-00089-A-001-01

Figure 20. Instrument Panel (S/N 36087 through 36999) (Sheet 1 of 3)



ICN-412-A-960002-B-97499-00090-A-001-01

Figure 20. Instrument Panel (S/N 36087 through 36999) (Sheet 2 of 3)

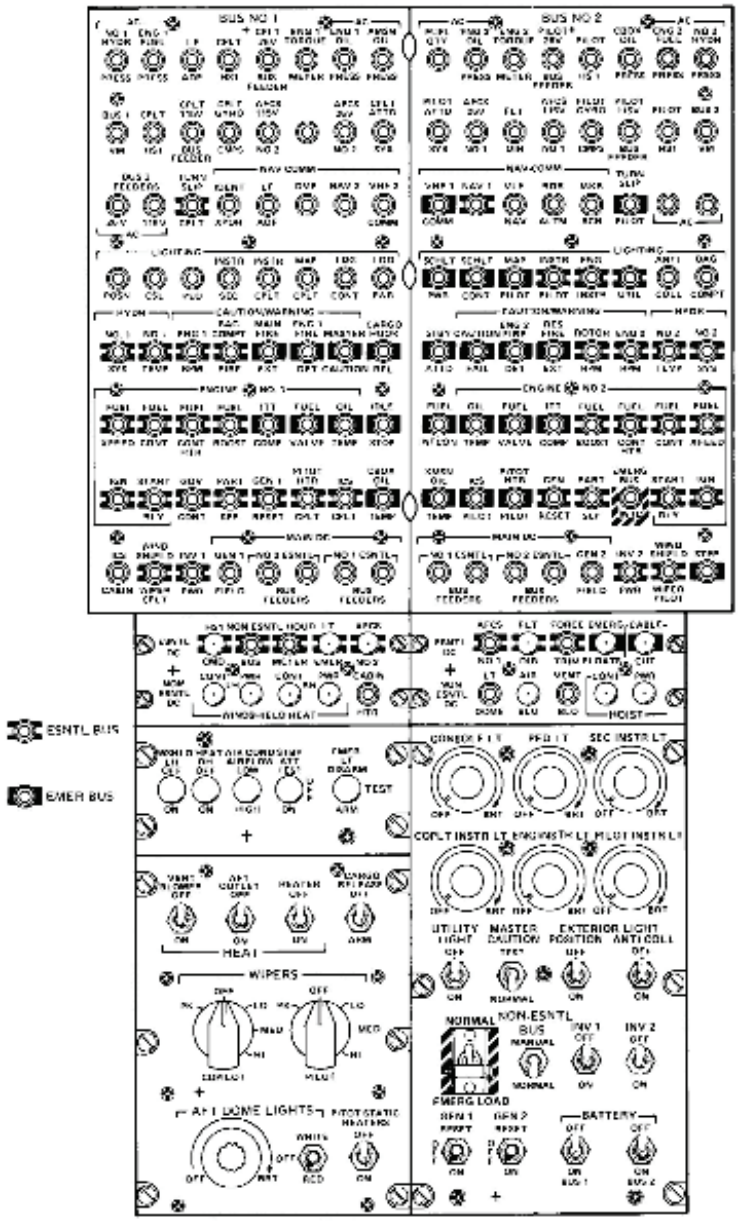


NOTE

1 Applicable to 36087 through 36247.

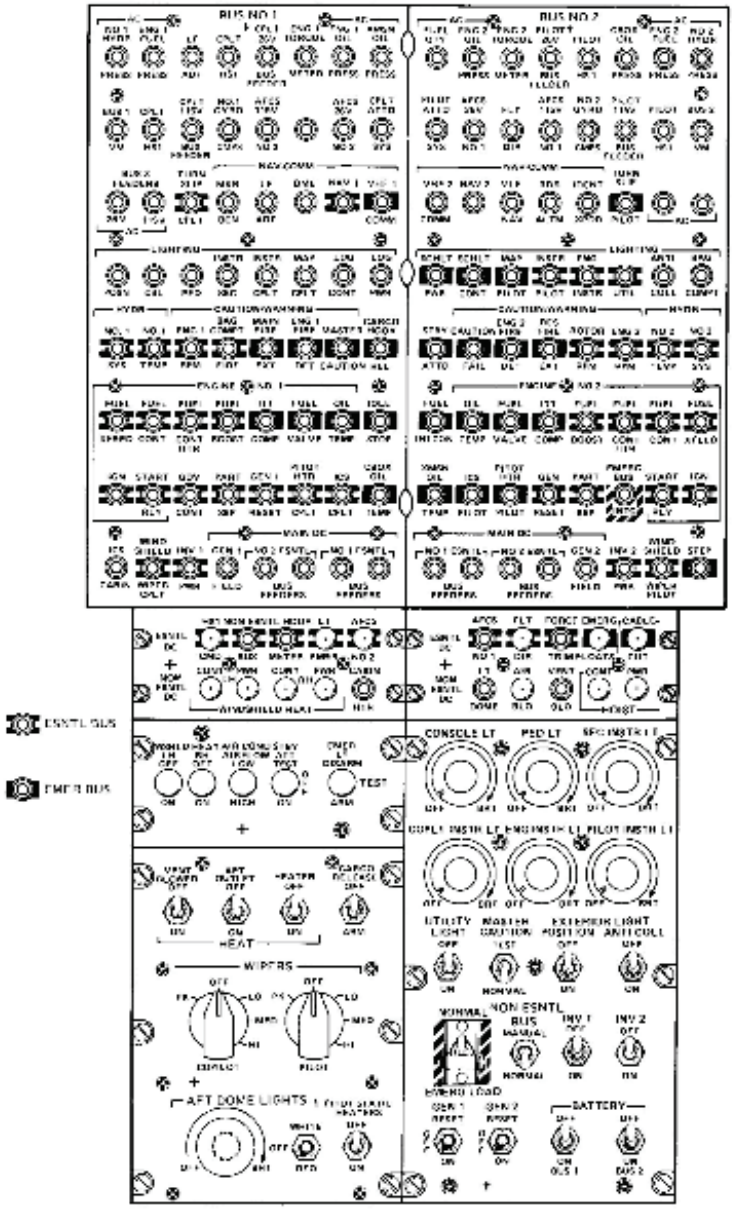
ICN-412-A-960002-B-97499-00091-A-001-01

Figure 20. Instrument Panel (S/N 36087 through 36999) (Sheet 3 of 3)



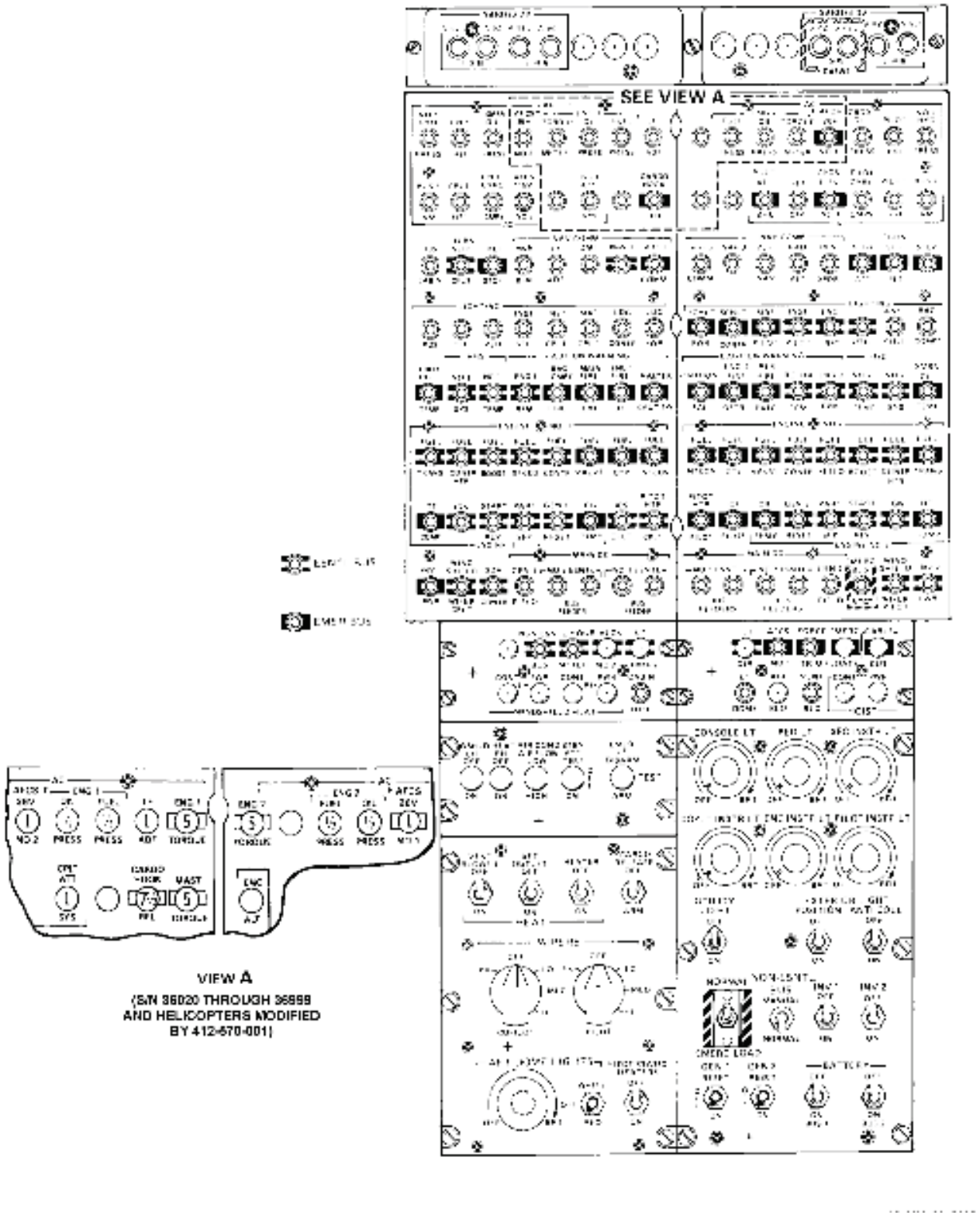
ICN-412-A-960002-B-97499-00092-A-001-01

Figure 21. Overhead Console (S/N 33001 through 33100) (Sheet 1 of 1)



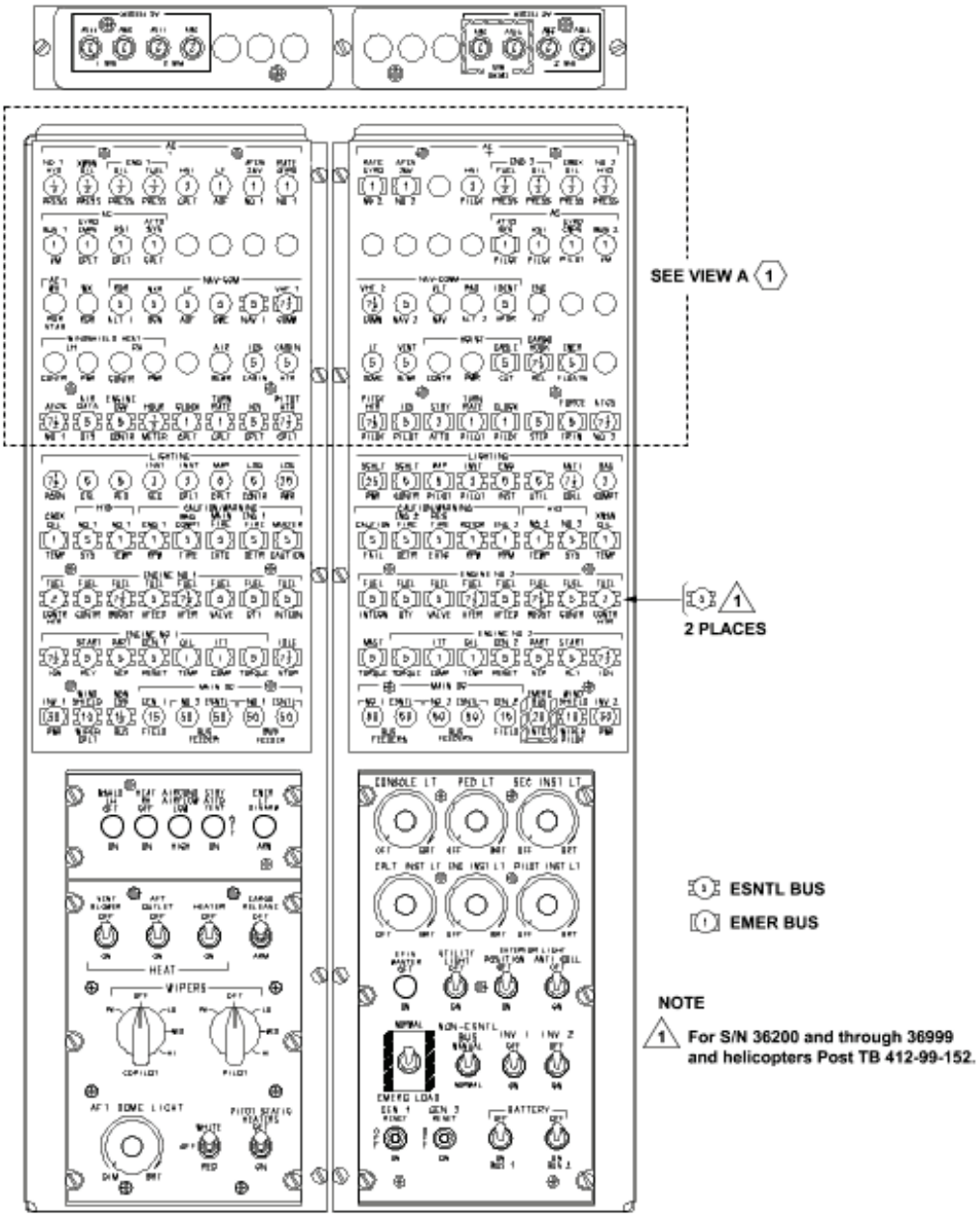
ICN-412-A-960002-B-97499-00093-A-001-01

Figure 22. Overhead Console (S/N 33101 through 33107) (Sheet 1 of 1)



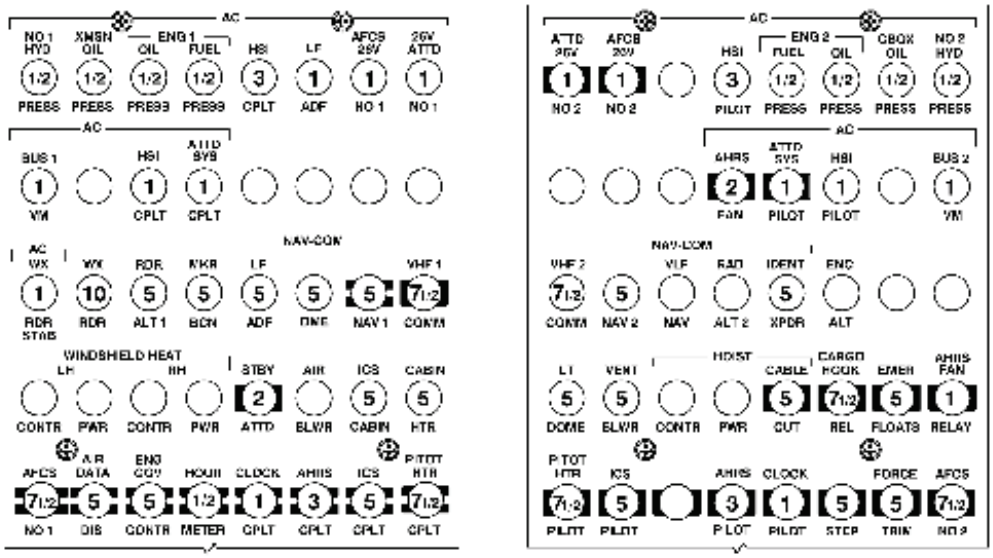
ICN-412-A-960002-B-97499-00094-A-001-01

Figure 23. Overhead Console (S/N 33108 through 36086) (Sheet 1 of 1)



ICN-412-A-960002-B-97499-00095-A-001-01

Figure 24. Overhead Console (S/N 36087 through 36999) (Sheet 1 of 2)



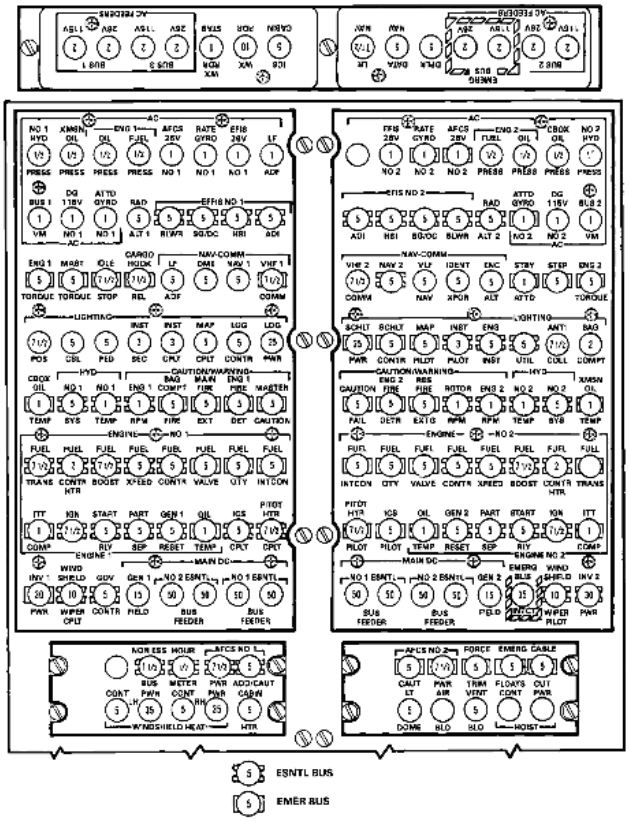
VIEW A (1)

EFFECTIVITY

(1) S/N 36248 THROUGH 36999.

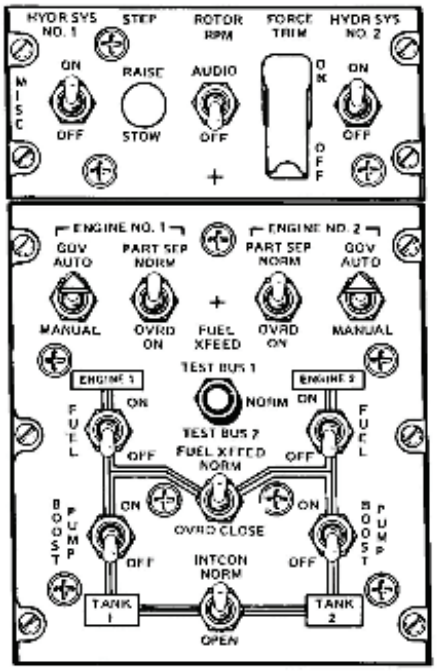
ICN-412-A-960002-B-97499-00096-A-001-01

Figure 24. Overhead Console (S/N 36087 through 36999) (Sheet 2 of 2)



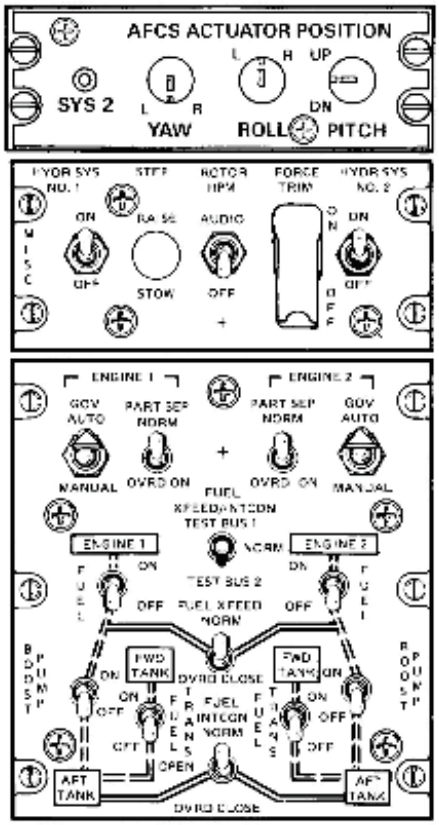
ICN-412-A-960002-B-97499-00097-A-001-01

Figure 25. Overhead Console (Typical 4-Axis Configuration) (Sheet 1 of 1)



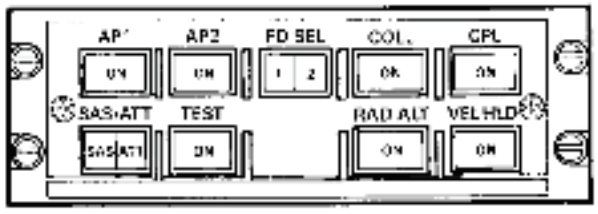
ICN-412-A-960002-B-97499-00098-A-001-01

Figure 26. Pedestal Panels (S/N 33001 through 33107) (Sheet 1 of 1)

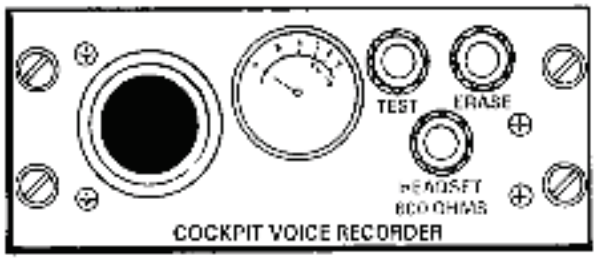


ICN-412-A-960002-B-97499-00099-A-001-01

Figure 27. Pedestal Panels (S/N 33108 through 33213 and 36001 through 36999) (Sheet 1 of 2)



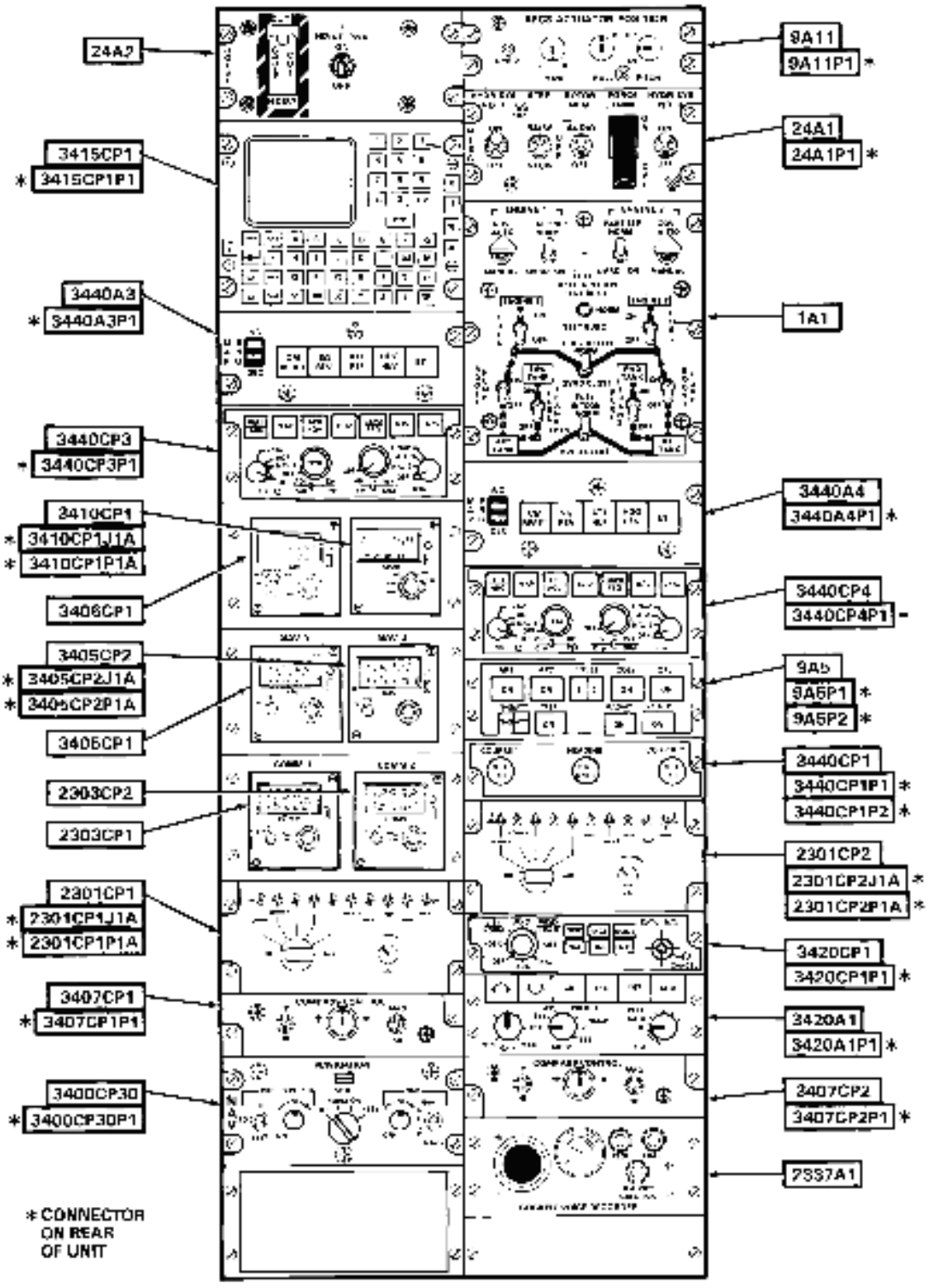
(HELICOPTERS S/N 36087 THROUGH 36999)



(HELICOPTERS S/N 36037 THROUGH 36999)

ICN-412-A-960002-B-97499-00100-A-001-01

Figure 27. Pedestal Panels (S/N 33108 through 33213 and 36001 through 36999) (Sheet 2 of 2)



ICN-412-A-960002-B-97499-00101-A-001-01

Figure 28. Pedestal (Typical) - 4-Axis Configuration (Sheet 1 of 1)

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DC Power Systems - Description

The DC power systems include the battery, external power, starter-generator, and igniter systems. Refer to (412-A-98-00-00-07A-051A-A / 02467, or 412-A-98-00-00-08A-051A-A / 02468, or 412-A-98-00-00-09A-051A-A / 02469, or 412-A-98-00-00-11A-051A-A / 02471, or 412-A-98-00-00-12A-051A-A / 02472, or 412-A-98-00-00-13A-051A-A / 02473) for the wiring diagrams.

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Battery System - Description

The battery is provided for engine start and as a backup source of emergency power. The battery circuit is activated by any of the following procedures.

Verify that the No. 1 BUS BAT and No. 2 BUS BAT circuit breakers are closed.

Close both the No. 1 ESNTL BUS FEEDER and No. 2 ESNTL BUS FEEDER circuit breakers on the left overhead console and set the BATTERY BUS switch to ON.

Close both the No. 1 ESNTL BUS FEEDER and No. 2 ESNTL BUS FEEDER circuit breakers on the right overhead console and set the BATTERY BUS switch to ON.

The SYS 1 and SYS 2 dual voltmeter indicate battery voltage when the GEN 1 RESET and GEN 2 RESET circuit breakers are set. After the battery has been used for engine start and the starter-generators are on line, the dual ammeter will indicate a high current. As the state of the battery charge increases the ammeter reading will decrease until the battery is completely charged. With the battery completely charged there will be no perceptible difference in the dual ammeter reading when the BATTERY switch is moved from ON to OFF.

A battery vent installation is connected to the battery to ventilate the gas fumes and heat generated by the battery.

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Battery System - Operational Check

Procedure

1. Before connecting power to the battery circuitry, do as follows:

1.1. Open all the circuit breakers.

1.2. Set all the switches to the off or normal position.

1.3. Verify that an open circuit exists between the positive terminals of the quick-disconnect and ground.

1.4. Check all the battery circuitry connections for tightness and correct polarity.

NOTE

The following tests of the battery circuitry shall be done using the installed 24 VDC battery as a power source, or alternatively, an external 24 VDC power source connected to the battery connector.

2. Connect the battery.

3. Close the following circuit breakers in the nose compartment:
- BUS 1 BATT
- EMERG BUS 1

4. Set the BATTERY BUS 1 switch to ON.

NOTE

The No. 1 MAIN DC BUS and No. 2 MAIN DC BUS are located in the overhead console.

5. Verify the battery voltage at the No. 1 MAIN DC BUS.

6. Verify that no voltage is present at the No. 2 MAIN DC BUS.

7. Close the following circuit breaker switches in the nose compartment:
- BUS 2 BATT
- EMERG BUS 2

8. Set the BATTERY BUS 2 switch to ON.

9. Verify that battery voltage is present at the No.1 and No. 2 MAIN DC BUS.

10. Verify that the SYS 1 and SYS 2 dual voltmeters indicate 0 (zero) volts.

11. Close the following circuit breakers on the left side overhead console:
- No. 1 ESNTL BUS FEEDER
- GEN 1 RESET

12. Verify that the SYS 1 dual voltmeter indicates battery voltage.

13. Verify that the SYS 2 dual voltmeter indicates 0 (zero) voltage.

14. Repeat Step 11. through Step 13. with the No. 1 ESNTL BUS FEEDER circuit breaker open on the left side overhead console and closed on the right side overhead console.

15. Open the No. 1 ESNTL BUS FEEDER circuit breaker.

16. Close the following circuit breakers on the left side overhead console:
- No. 2 ESNTL BUS FEEDER
- GEN 2 RESET

17. Verify that the SYS 2 dual voltmeter indicates the battery voltage.

18. Verify that the SYS 1 dual voltmeter indicates 0 (zero) voltage.

19. Repeat Step 16. through Step 18. with the No. 2 ESNTL BUS FEEDER circuit breaker open on the left side overhead console and closed on the right side overhead console.

20. Place the NON-ESNTL BUS switch to NORMAL and close the NON ESS BUS circuit breaker.

21. Close all the No. 1 ESNTL BUS FEEDER circuit breakers.

22. Verify that no voltage is present on the NON-ESSENTIAL BUS.

23. Set the NON-ESNTL BUS switch to MANUAL.

24. Verify that battery voltage is present on the NON-ESSENTIAL BUS.

25. Set the BATTERY BUS 1 switch to OFF.

26. Verify that battery voltage is present on the NON-ESSENTIAL BUS.

27. Set the BATTERY BUS 1 switch to ON.

28. Set the BATTERY BUS 2 switch to OFF.

29. Verify that battery voltage is present on the NON-ESSENTIAL BUS.

30. Close the MASTER CAUTION circuit breaker.

31. Set the EMERG LOAD switch to EMERG LOAD.

32. Verify that the BATTERY caution light is illuminated.

33. Switch the BATTERY BUS 2 switch to ON then to OFF.

34. Verify that the BATTERY caution light remains illuminated.

35. Set the BATTERY BUS 1 switch to OFF.

36. Verify that the BATTERY caution light remains illuminated.

37. Set the EMERG LOAD switch to NORMAL.

38. Set the BATTERY BUS 1 switch to ON.

39. Set the BATTERY BUS 2 switch to ON.

40. Verify that the BATTERY caution light is illuminated.

41. Set the BATTERY BUS 1 switch to OFF.

42. Verify that the BATTERY caution light is extinguished.

43. Set the BATTERY BUS 1 switch to ON.

44. Verify that the BATTERY caution light is illuminated.

45. Set the BATTERY BUS 2 switch to OFF.

46. Verify that the BATTERY caution light is extinguished.

Battery - Description

NOTE

Information pertaining to specific battery maintenance should be obtained from the battery manufacturer maintenance manual (BHT-412-CR&O-V, Chapter 96) and the BHT-ELEC-SPM, Electrical Standard Practices Manual.

The battery (2BT1) is installed in the nose compartment of the nose cabin section. The battery is a source of power for engine start and it also supplies emergency DC power, if required. The basic battery is a 24 VDC Nickel-Cadmium (Ni-Cad) battery with a rated capacity of 40 amp/hours. The optional battery is a 24 VDC Lead-Acid battery (VRLA) with a rated capacity of 53 amp/hours. The Ni-Cad battery contains 20 cells. The Lead-Acid battery contains 12 cells.

The Ni-Cad battery contains negative plate cells of cadmium oxide and positive plate cells of nickel oxide. The plates are submerged in a solution of approximately 30% potassium hydroxide as the electrolyte. The cells are ventilated through the helicopter fuselage by the battery vent installation.

The battery temperature is monitored by thermal switches installed between the cells of the battery. The thermal switches supply a ground path to the caution/warning/advisory panel or Crew Alerting System (CAS) when the battery overheats. This causes the BATTERY HOT warning light or CAS warning message to come on. The overheat temperature for the 28 amp/hour battery is 160° ±5°F (71.1° ±2.8°C).

The Valve-Regulated Lead-Acid (VRLA) battery has optimum lead alloy with tin and copper to provide the best electrode. The plates are submerged in a solution of sulfuric acid that is contained in glass mat separators, with no free electrolyte. The battery is considered a sealed or recombinant-gas battery. The battery has vent caps (with valves included) that are sealed in place and that cannot be accessed for maintenance. The vent caps shall not be removed.

The cells are ventilated through the helicopter fuselage by the battery vent installation. With the Lead-Acid battery, there is no BATTERY HOT warning CAS message.

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FOLLOW THE WARNINGS THAT FOLLOW WHEN YOU HANDLE AND WORK ON A BATTERY. FAILURE TO OBEY THESE INSTRUCTIONS CAN CAUSE SERIOUS INJURY TO PERSONNEL.

DO NOT WEAR JEWELRY WHEN YOU WORK ON OR HANDLE A BATTERY. ACCIDENTAL CONTACT OF JEWELRY WITH A BATTERY PART THAT CARRIES CURRENT CAN CAUSE A SHORT THAT COULD CAUSE SERIOUS BURNS.

USE ONLY TOOLS THAT ARE COVERED WITH A NON-CONDUCTIVE INSULATION MATERIAL. IF YOU DROP A METAL TOOL IN THE BATTERY, A SHORT-CIRCUIT BETWEEN THE CELL CONNECTION BARS COULD RESULT. THE SHORT-CIRCUIT COULD CAUSE ARCS THAT COULD DAMAGE THE BATTERY AND CAUSE SERIOUS INJURY TO PERSONNEL.

WEAR PROTECTIVE GOGGLES, A RUBBER APRON, AND RUBBER GLOVES WHEN YOU WORK ON OR HANDLE A BATTERY. FOR A NICKEL-CADMIUM BATTERY, THE POTASSIUM HYDROXIDE AND WATER SOLUTION THAT FORMS ELECTROLYTE IS ALKALINE AND CORROSIVE. FOR THE LEAD-ACID BATTERY, THE ELECTROLYTE IS A SULFURIC ACID AND HIGHLY CORROSIVE. THE ELECTROLYTE CAUSES SERIOUS BURNS WHEN IT COMES IN CONTACT WITH THE SKIN. IF THE ELECTROLYTE COMES IN CONTACT WITH THE SKIN, IMMEDIATELY WASH THE AFFECTED AREA WITH A LARGE QUANTITY OF WATER OR TAKE A SHOWER.

NEUTRALIZE THE ELECTROLYTE WITH A 3% SOLUTION OF ACETIC ACID, VINEGAR, LEMON JUICE, OR WITH A 10% SOLUTION OF BORIC ACID. WASH THE AFFECTED AREA WITH WATER.

WEAR SAFETY GLASSES TO AVOID CONTACT OF THE ELECTROLYTE WITH THE EYES. IF THE ELECTROLYTE GETS IN CONTACT WITH THE EYES, IMMEDIATELY WASH THE EYES WITH A LARGE QUANTITY OF WATER. GET IMMEDIATE MEDICAL ATTENTION.

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Battery Vent - Description

The battery vent installation has an inlet tube connected between a ram air inlet and the battery case. An exhaust tube connects the opposite side of the battery case to an outlet installed on the lower fuselage. In flight, the ram air inlet collects the ram air and distributes it to the inside of the battery case. The ram air exits the battery case through the vent tube and is evacuated overboard.

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Battery - Maintenance Practices

Procedure

NOTE

Batteries shall not be serviced while installed in the helicopter. The battery shall be serviced by authorized battery shop personnel only.

1. Specific battery maintenance information should be obtained from the battery manufacturer maintenance manual (BHT-412-CR&O-V, Chapter 96). Information is also available in the BHT-ELEC-SPM, Electrical Standard Practices Manual.

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Battery Case and Battery Compartment - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-156	AR	
Varnish	C-237	AR	
Boric acid	C-489	AR	

Procedure



ONLY AUTHORIZED PERSONNEL INSTRUCTED IN THE MAINTENANCE PRECAUTIONS AND ASSOCIATED HAZARDS RELATED TO THE HANDLING OF BATTERIES SHOULD DO THIS PROCEDURE. SERIOUS INJURY TO PERSONNEL MAY OCCUR IF THE PROPER PRECAUTIONS ARE NOT TAKEN.

DO NOT USE A WIRE BRUSH OR STEEL WOOL TO CLEAN THE BATTERY TERMINALS. THIS COULD RESULT IN SERIOUS INJURY TO PERSONNEL AND/OR DAMAGE TO THE BATTERY IF AN ELECTRICAL SHORT OCCURS. USE ONLY A STIFF FIBER BRUSH. MAKE SURE THE FIBER BRUSH IS CLEAN AND HAS NOT BEEN USED ON A LEAD-ACID BATTERY.



DO NOT CLEAN THE BATTERY IN THE BATTERY COMPARTMENT. CORROSION IN THE BATTERY COMPARTMENT CAN RESULT. MAKE SURE THE BATTERY VENTS ARE NOT OPEN TO AVOID CONTAMINATION INSIDE THE BATTERY CASE DURING CLEANING. CLOSE THE BATTERY VENTS WITH caps or plugs (C-156).

MAKE SURE THE EXTERIOR OF THE BATTERY CASE IS DRY BEFORE IT IS INSTALLED IN THE BATTERY COMPARTMENT. FAILURE TO OBEY THIS PRECAUTION CAN RESULT IN CORROSION IN THE BATTERY COMPARTMENT.

DO NOT USE THE SAME TOOLS THAT WERE USED ON LEAD-ACID BATTERIES TO CLEAN NICKEL-CADMIUM BATTERIES. THIS INCLUDES GLOVES, GLASSES, AND PROTECTIVE CLOTHING. A STRONG CHEMICAL REACTION WILL RESULT IF THE TWO ELECTROLYTES COME INTO CONTACT WITH EACH OTHER.

1. Clean the exterior of the battery case and terminals with a dry stiff fiber brush or wash with water.
2. Make sure that the battery compartment is free of alkaline deposits. If deposits are found, clean them with a 10% boric acid (C-489) solution.
3. For nickel-cadmium batteries, if necessary, retouch the varnished surfaces of the battery case with a coat of alkaline resistant varnish (C-237) on the surfaces that have been contaminated with alkaline deposits.

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Battery - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-156	AR	

Safety conditions

WARNING

OBEY ALL OF THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.

Procedure

NOTE

The following procedure applies to the standard 40 amp/hour Ni-Cad battery and the optional Lead-Acid battery kit (412-706-131). The procedure is the same except where noted.

1. Open the upper hinge nose door (412-A-52-40-00-01A-520A-A / 00832).

2. Remove the lower hinged nose door (412-A-52-40-00-01A-520A-A / 00832).

3. Remove the external power.

NOTE

The connector (2BT1P2) (8, Figure 1) is bagged and stowed for the 53 amp/hour Lead-Acid battery (2BT1) (18).

4. Disconnect the connector (2BT1P2) (8) from the battery Ni-Cad (2BT1) (1).

5. Disconnect the battery power connector (2BT1P1) (9) from the battery (2BT1) (1 or 18), as applicable.

6. Install caps or plugs (C-156) on the receptacle of the battery (2BT1) (1 or 18), as applicable, connector (2BT1P2) (8), as applicable, and battery power connector (2BT1P1) (9).

7. On the each side of the battery (2BT1) (1 or 18), as applicable, remove the clamps (3 and 4) and the vent tube assemblies (2 and 5).

8. Lift the clamps (11) to unlock the rod assemblies (10) from the battery (2BT1) (1 or 18), as applicable.

9. Move the rod assemblies (10) to clear the battery (2BT1) (1 or 18), as applicable.

WARNING

HANDLE THE BATTERY WITH CARE. SERIOUS INJURY TO PERSONNEL CAN OCCUR IF THE CORRECT TECHNIQUES TO LIFT AND HANDLE THE BATTERY ARE NOT FOLLOWED. THE WEIGHT OF THE 40 AMP/HOUR NI-CAD BATTERY IS APPROXIMATELY 80 POUNDS (36.3 KG). THE 53 AMP/HOUR LEAD-ACID BATTERY IS APPROXIMATELY 93 POUNDS (42.2 KG).

10. Remove the battery (2BT1) (1 or 18), as applicable, from the pan (6).

11. If required, remove the rod assembly (10) as follows:

11.1. Remove the nut (17), washers (16), and bolt (15).

11.2. Remove the rod assembly (10) from the eyebolt (7).

11.3. Do Step 11. again for the remaining rod assembly (10).

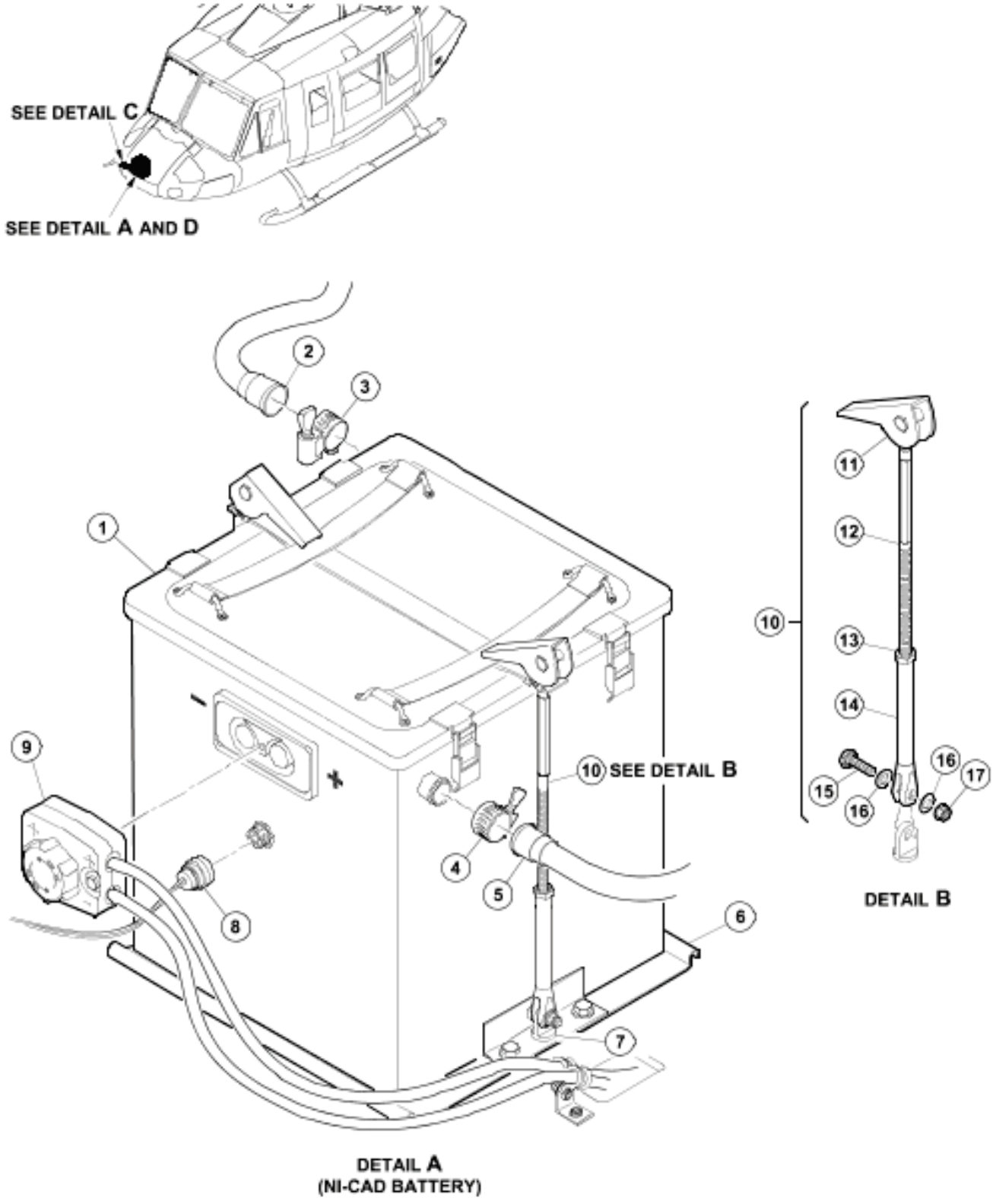
12. Disassemble the rod assembly (10) as follows:

12.1. Loosen the nut (13).

12.2. Remove the rod (12) and the nut (13) from the rod end (14).

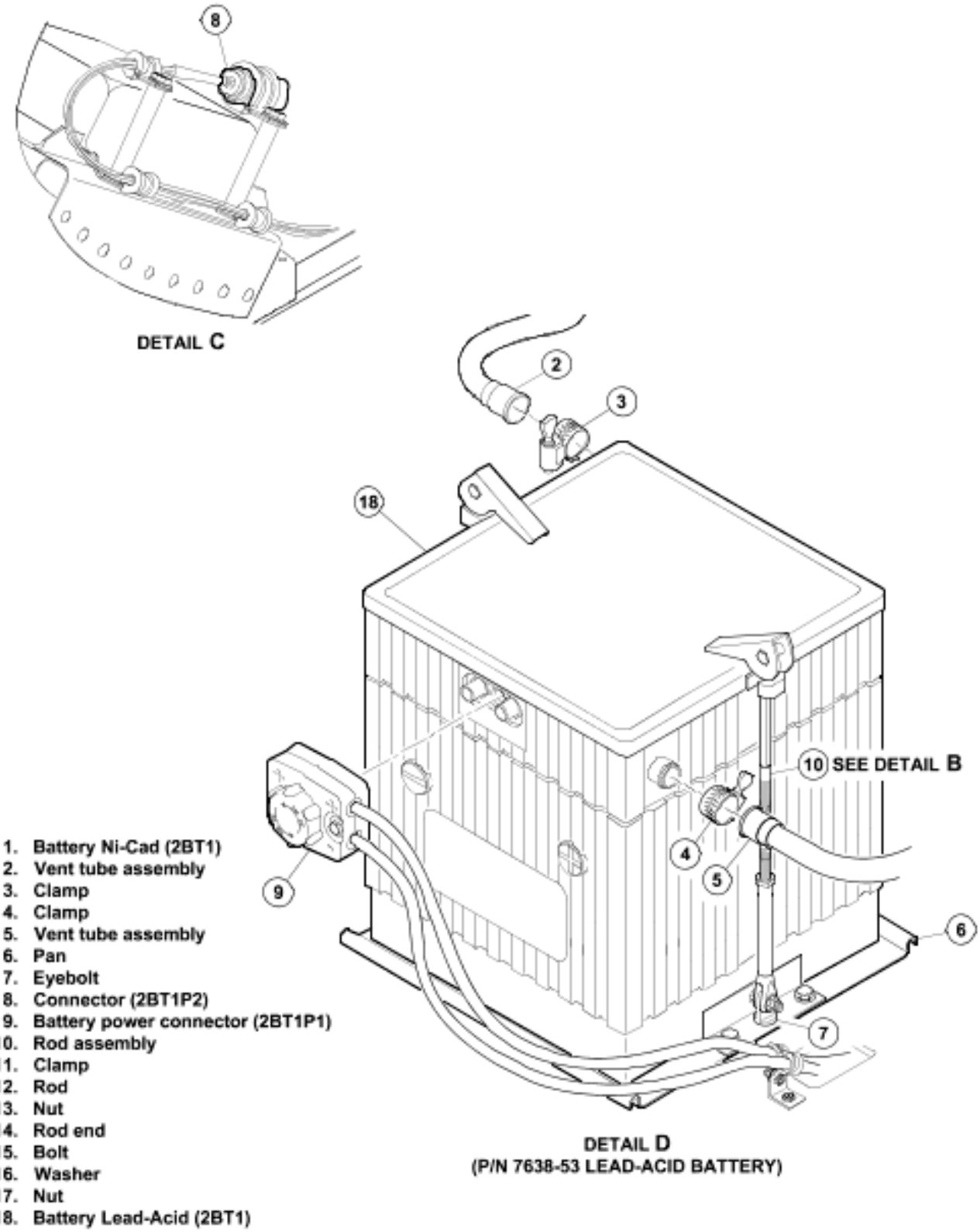
12.3. Remove the nut (13) from the rod (12).

12.4. Do Step 12. again to disassemble the remaining rod assembly (10).



ICN-412-A-966100-B-97499-00001-A-001-01

Figure 1. Battery - Removal and Installation (Sheet 1 of 2)



ICN-412-A-966100-B-97499-00002-A-001-01

Figure 1. Battery - Removal and Installation (Sheet 2 of 2)

Battery - Inspection

Procedure

NOTE

The following inspection criteria are primarily intended when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

NOTE

Specific information pertaining to battery inspection can be obtained from the battery manufacturer maintenance manual (BHT-412-CR&O-V, Chapter 96).

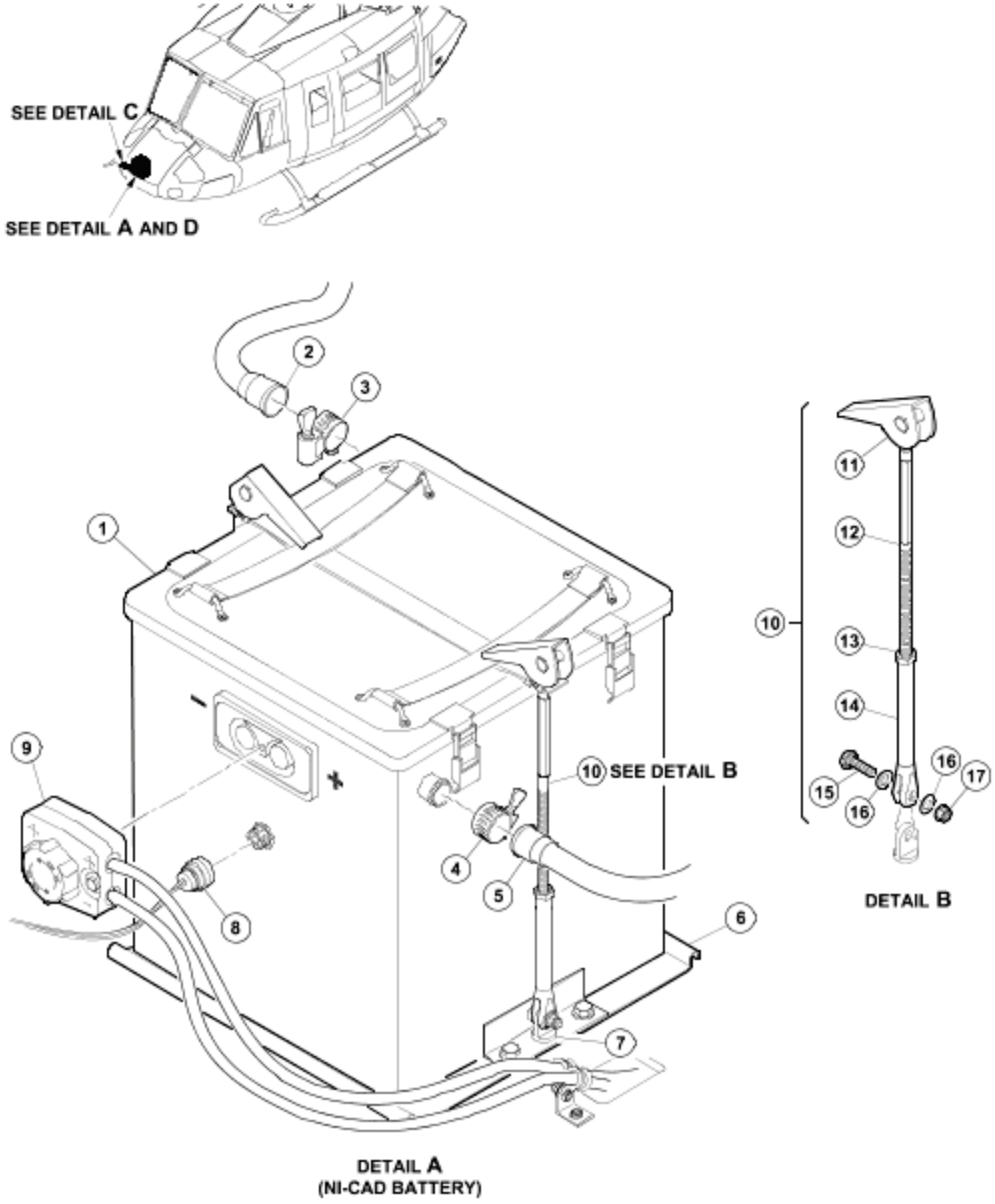
1. Visually examine the external case of the battery (2BT1) (1 or 18, Figure 1), as applicable, for cracks, mechanical damage, deformation, and leakage. Visually examine the terminal pins for wear, mechanical damage, and corrosion. Replace the battery, as required.

2. Visually examine the battery power connector (2BT1P1) (9) for wear to the contact pins, cracks, mechanical damage, and corrosion. Repair or replace the connector, as required.

3. Visually examine the connector (2BT1P2) (8) for wear to the contact pins, cracks, mechanical damage, and corrosion. Repair or replace the connector, as required.

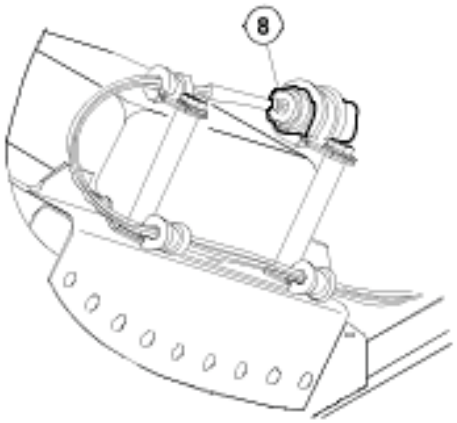
4. Visually examine the vent tube assemblies (2 and 5) for chafing, kinks, cracks, and general condition. Make sure that there is no internal blockage. Replace the vent tube assemblies, as required.

5. Visually examine all attaching hardware for cracks, wear, mechanical damage, and corrosion. Replace the hardware, as required.

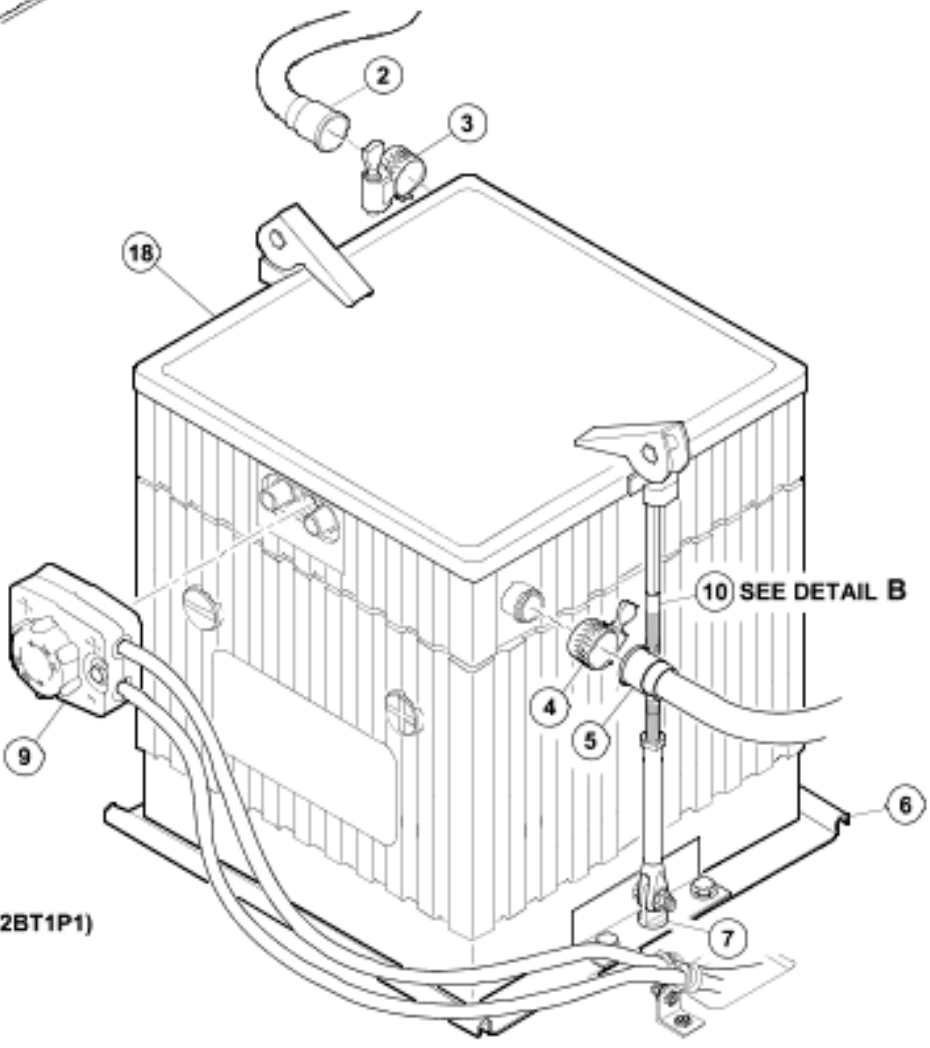


ICN-412-A-966100-B-97499-00001-A-001-01

Figure 1. Battery - Removal and Installation (Sheet 1 of 2)



DETAIL C



DETAIL D
(P/N 7638-53 LEAD-ACID BATTERY)

1. Battery Ni-Cad (2BT1)
2. Vent tube assembly
3. Clamp
4. Clamp
5. Vent tube assembly
6. Pan
7. Eyebolt
8. Connector (2BT1P2)
9. Battery power connector (2BT1P1)
10. Rod assembly
11. Clamp
12. Rod
13. Nut
14. Rod end
15. Bolt
16. Washer
17. Nut
18. Battery Lead-Acid (2BT1)

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Figure 1. Battery - Removal and Installation (Sheet 2 of 2)

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Battery - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-156	AR	

Safety conditions



OBEY ALL OF THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT.



HANDLE THE BATTERY WITH CARE. SERIOUS INJURY TO PERSONNEL COULD OCCUR IF THE CORRECT TECHNIQUES TO LIFT AND HANDLE THE BATTERY ARE NOT FOLLOWED. THE WEIGHT OF THE 40 AMP/HOUR NI-CAD BATTERY IS APPROXIMATELY 80 POUNDS (36.3 KG). THE 53 AMP/HOUR LEAD-ACID BATTERY IS APPROXIMATELY 93 POUNDS (42.2 KG).



IF THE 53 AMP/HOUR BATTERY P/N 7638-53 IS REPLACED, WEIGH THE BATTERY BEFORE INSTALLATION. MAKE SURE THAT THE WEIGHT OF THE BATTERY AND THE INSTALLED BALLAST DOES NOT EXCEEDS 155 LBS (52.2 KG).

Procedure

NOTE

The following procedure applies to the standard 40 amp/hour Ni-Cad battery and the optional Lead-Acid battery kit (412-706-131). The procedure is the same except where noted.

1. Remove the external power.

2. Make sure that the battery being installed is fully charged and caps or plugs (C-156) are installed on the main power receptacle.

3. Install the battery (2BT1) (1 or 18, Figure 1), as applicable, on the pan (6) in the nose cabin compartment.

4. If previously disassembled, assemble the rod assembly (10) as follows:

4.1. Thread the nut (13) on the rod (12).

4.2. Install the rod (12) and the nut (13) on the rod end (14). Do not tighten the nut at this time.

4.3. Do Step 4. again to assemble the remaining rod assembly (10).

5. If previously removed, install the rod assembly (10) as follows:

5.1. Put the rod assembly (10) in position on the eye bolt (7) and secure with the bolt (15), washer (16), and nut (17). Tighten the nut.

5.2. Adjust the length of the rod assembly (10) and tighten the nut (13).

5.3. Do Step 12. again to install the remaining rod assembly (10).

6. Move the rod assemblies (10) to engage the battery (2BT1) (1 or 18), as applicable.

7. Push down the clamps (11) to lock the rod assemblies (10) on the battery (2BT1) (1 or 18), as applicable.

8. On each sides of the battery (2BT1) (1 or 18), as applicable, connect the vent tube assemblies (2 and 5) and secure with the clamps (3 and 4), respectively. Tighten the clamps.

9. Remove the caps and/or plugs from the receptacle of the battery (2BT1) (1 or 18), as applicable, connector (2BT1P2) (8), as applicable, and battery power connector (2BT1P1) (9).

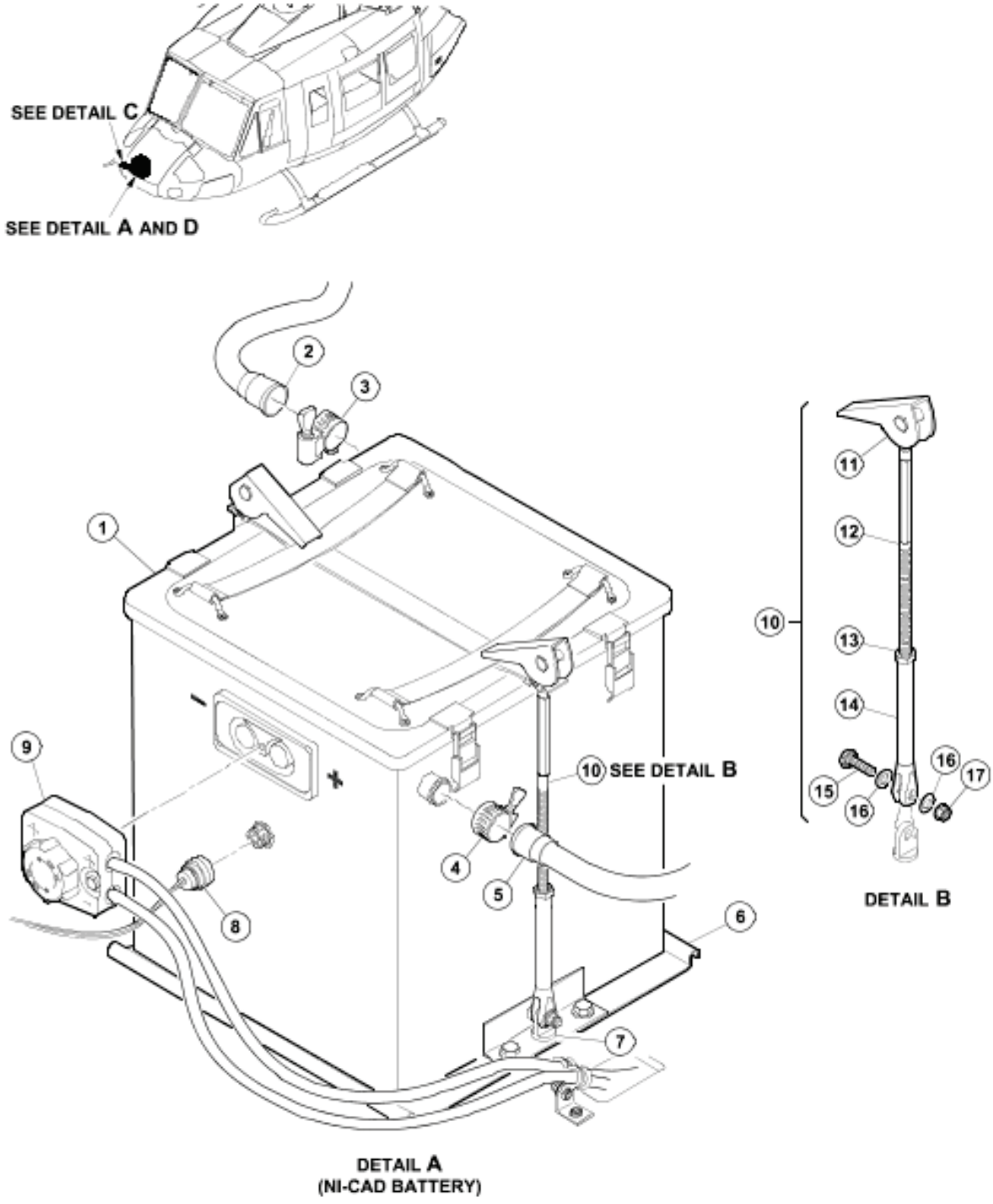
NOTE

The connector (2BT1P2) (8) is bag and stowed for the 53 amp/hour lead acid battery (2BT1) (18).

10. Connect the connector (2BT1P2) (8) to the Ni-Cad battery (2BT1) (1).

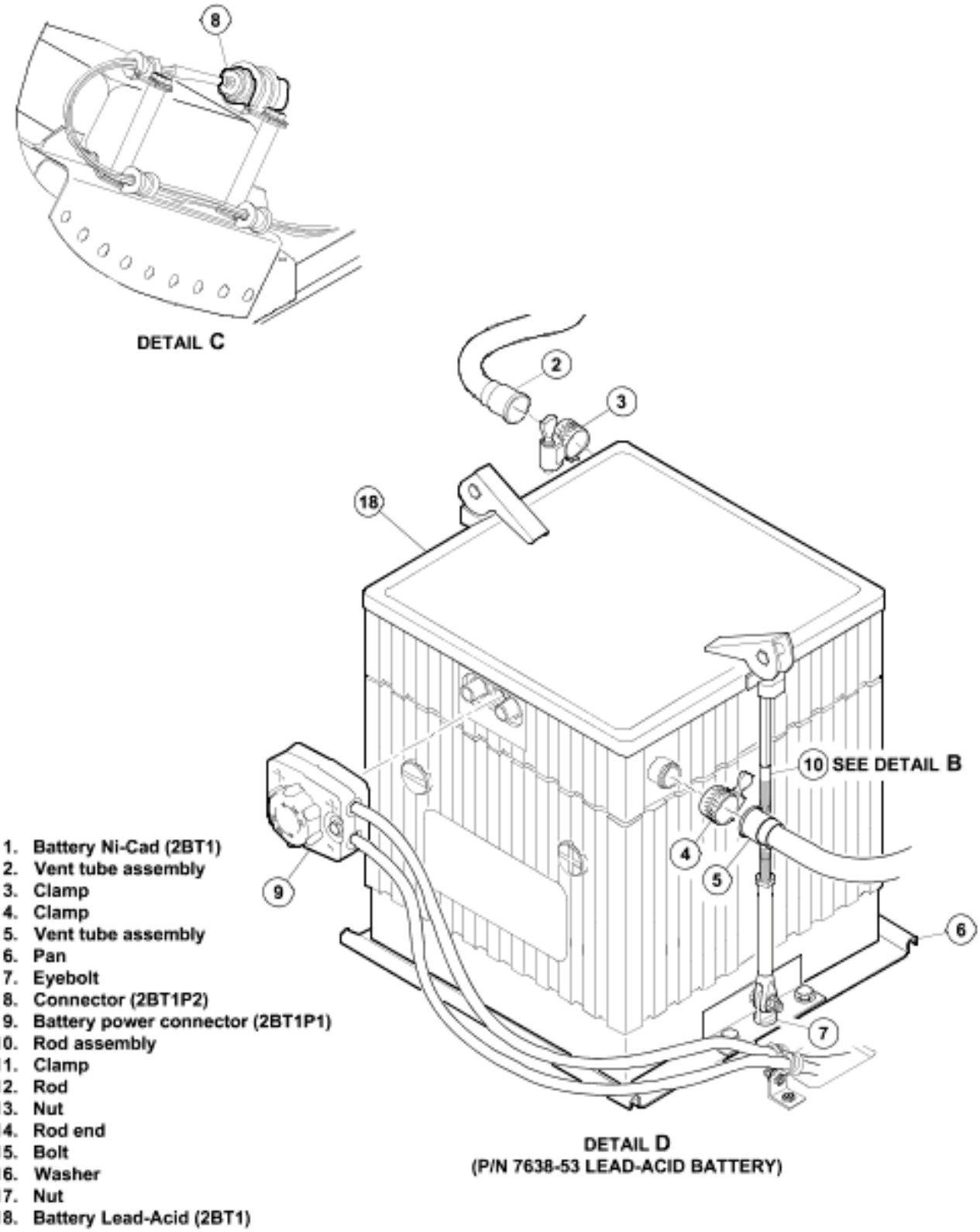
11. Connect the battery power connector (2BT1P1) (9) to the battery (2BT1) (1 or 18), as applicable.

12. If the Ni-Cad battery is installed, do the battery sensor temperature caution light system operational check (412-A-96-03-01-24A-320A-A / 02078 or 412-A-96-03-01-24B-320A-A / 02080), as applicable.



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Figure 1. Battery - Removal and Installation (Sheet 1 of 2)



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Figure 1. Battery - Removal and Installation (Sheet 2 of 2)

External Power System - Description



ELECTRICAL/ELECTRONIC SYSTEMS ARE NOT PROTECTED FROM AN OVERVOLTAGE CONDITION WHEN OPERATING ON EXTERNAL POWER. TO PREVENT DAMAGE TO ELECTRICAL/ELECTRONIC SYSTEMS, MAKE SURE THE EXTERNAL POWER SOURCE IS EQUIPPED WITH AN OVERVOLTAGE PROTECTION CIRCUIT.

External power is connected to the helicopter electrical systems through the external power receptacle. No special action or switching is required to connect external power. If external power connections are of proper polarity, the external power relay closes automatically and connects the ground unit to the main DC bus 2. If the connections are not of proper polarity, no action occurs.

Applicable: SN: 33001-33107:

No. 1 and No. 2 DC essential buses may be energized by closing the GEN 1 RESET, GEN 2 RESET, and all ESNTL BUS FEEDER circuit breakers. The non-essential bus may be energized by setting the NON-ESNTL BUS switch to MANUAL. This allows complete checks of all equipment or other ground operations to be done.

Applicable: SN: 33108-36999:

No. 1 and No. 2 DC essential buses and No. 1 and No. 2 emergency buses are energized, when all ESNTL BUS FEEDER circuit breakers are closed. With battery installed, non-essential bus No. 1 and No. 2 may be energized by placing the NON-ESNTL BUS switch to MANUAL. This allows accomplishment of operational checks and other ground operations.

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External Power System - Operational Check

Procedure

1. Before you connect the external power do the following:

1.1. Make sure that all the switches are set to the off or the normal position.

1.2. Visually check the external power receptacle wiring for correct polarity and terminations.

1.3. Apply 28 VDC reverse polarity power between the small pin in the external power receptacle and the helicopter structure (positive to the helicopter structure and negative to the small pin in the external power receptacle).

1.4. Verify that the external power relay (K1/2K1) does not close.

1.5. Remove the 28 VDC reverse polarity power.

1.6. Connect the DC power source to the external power receptacle and energize.

1.7. Adjust the external power source to 28 ±0.5 VDC.

1.8. Verify that 28 ±0.5 VDC is present on the No. 2 MAIN DC BUS and 0 VDC is present on the No. 1 MAIN DC BUS.

1.9. Close the following circuit breakers:
- GEN 1 RESET
- GEN 2 RESET
- NO. 1 ESNTL BUS FEEDER
- NO. 2 ESNTL BUS FEEDER

1.10. Verify that both dual volt meters indicate 28 ±0.5 VDC.

1.11. Verify that 0 VDC is present on the No.1 MAIN DC BUS.

1.12. De-energize the external power source.

1.13. Verify that the dual voltmeter indicates 0 VDC.

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External Power Receptacle - Removal

Procedure

1. Make sure that the electrical power is off.
2. Remove the nuts and the washers from the terminal post of the receptacle.
3. Identify and remove the conductors from the receptacle.
4. Protect the conductor ends with electrical tape.
5. Remove the mounting screws and lift the receptacle from the bracket.

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External Power Receptacle - Installation

Procedure

1. Position the receptacle on the bracket and secure with the mounting screws.
2. Remove the electrical tape from the conductors and position on the terminals.
3. Secure the conductors to the terminals with the existing hardware.

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External Power Receptacle - Inspection

Procedure

1. Visually check the contact pins in the receptacle for corrosion and looseness.
2. Check the wiring and the terminals for evidence of overheating.

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External Power Receptacle - Replacement

Procedure

1. Replace the external power receptacle if the inspection requirements are not met.

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Generator System - Description

The primary 28 VDC power system consists of two starter-generator systems, designed to work independently of each other, or in conjunction with each other to produce a composite dual generator system. The two generating systems normally operate in parallel and equally share total load on DC buses. In event of failure of one generator, the remaining generator is capable of supplying essential loads on the helicopter bus. For maintenance of the starter-generators refer to 412-A-71-01-00-11A-042A-A / 01693.

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Generator No. 1 System - Operational Check

Procedure

1. Remove the lower nose access panel located under the nose of the helicopter.
2. Remove the voltage setting protective cover from the No. 1 DC CONTROL PANEL.
3. Set the GEN 1, GEN 2, BATTERY BUS 1, and BATTERY BUS 2 switches to the OFF positions.
4. Remove all the electrical power from the helicopter buses.
5. Connect one end of a 3-foot (91-cm) length of helicopter wire to the load side of a 5 amp circuit breaker.
6. Connect the bus side of the circuit breaker to terminal A1 (generator side) of the bus No. 1 BUS/BAT CONTROL RELAY.
7. Leave the other end of the wire disconnected, but protect it from an accidental short to ground or to other circuitry.
8. Make sure that the 5 amp circuit breaker is open.

NOTE

Either the helicopter battery or an external 28 VDC power source may be used for starting power.

9. Do a normal start of the No. 1 engine.
10. Bring the engine No. 1 starter-generator to normal operating speed but do not close the GEN 1 switch.
11. Energize only the minimum electrical equipment necessary for normal starting and ground run.
12. Insert the voltmeter probes into the test jacks on the No. 1 DC CONTROL PANEL and observe the correct polarity.
13. Verify that the voltmeter indicates normal generator voltage (27 to 28 VDC).
14. Close the 5 amp circuit breaker and place the GEN 1 switch to ON.
15. Momentarily short the loose end of the wire on the 5 amp circuit breaker to ground.
16. Remove the short and check that the voltmeter reading is approximately 0 volts.
17. Set the GEN 1 switch to RESET and then to OFF.
18. Verify that the voltmeter indicates normal generator voltage.
19. Set the GEN 1 switch to ON.
20. Verify that the ENG 1 DC GENERATOR caution light on the master caution panel extinguishes and both dual voltmeters indicate normal generator voltage.
21. Momentarily energize a load supplied by the essential DC bus No. 1, such as the landing light.
22. Verify that the No. 1 ammeter reads upscale.
23. Set the GEN 1 switch to OFF.
24. Verify that the ENG 1 DC GENERATOR caution light on the master caution panel illuminates.
25. Shut down the No. 1 engine.
26. Remove the electrical power from the helicopter buses.
27. Remove the voltmeter probes from the test jacks of the No. 1 DC CONTROL PANEL.
28. Install the voltage setting protective cover on the No. 1 DC CONTROL PANEL.
29. Remove the wire and the 5 amp circuit breaker.
30. Install the lower nose access panel.

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Generator No. 2 System - Operational Check

Procedure

1. Remove the lower nose access panel, located under the nose of the helicopter.

2. Remove the voltage setting protective cover from the No. 2 DC CONTROL PANEL.

3. Set the GEN 1, GEN 2, BATTERY BUS 1, and BATTERY BUS 2 switches to OFF.

4. Connect one end of a 3-foot (91-cm) length of helicopter wire to the load side of a 5 amp circuit breaker.

5. Connect the bus side of the circuit breaker to terminal A2 (generator side) of the No. 2 BUS/BAT CONTROL RELAY.

6. Leave the other end of the wire disconnected, but protect it from an accidental short to ground or to other circuitry.

7. Verify that the 5 amp circuit breaker is open.

NOTE

Either the helicopter battery or an external 28 VDC power source may be used for starting power.

8. Do a normal start of the No. 2 engine.

9. Bring the engine No. 2 starter-generator to normal operating speed but do not close the GEN 2 switch.

10. Energize only the minimum electrical equipment necessary for normal starting and ground run.

11. Insert the voltmeter probes into the test jacks on the No. 2 DC CONTROL PANEL and observe the correct polarity.

12. Verify that the voltmeter indicates normal generator voltage (27 to 28 VDC).

13. Close the 5 amp circuit breaker and set the GEN 2 switch to ON.

14. Momentarily short to ground the loose end of the wire on the 5 amp circuit breaker.

15. Remove the short and check that the voltmeter reading is approximately 0 volts.

16. Set the GEN 2 switch to RESET and then to OFF.

17. Verify that the voltmeter indicates generator output voltage.

18. Set the GEN 2 switch to ON.

19. Verify that the ENG 2 DC GENERATOR caution light on the master caution panel extinguishes and both dual voltmeters indicate generator voltage.

20. Momentarily energize a load supplied by the essential DC bus No. 2, such as the landing light.

21. Verify that the No. 2 ammeter reads upscale.

22. Set the GEN 2 switch to OFF.

23. Verify that the ENG 2 DC GENERATOR caution light on the master caution panel illuminates.

24. Shut down the No. 2 engine.

25. Remove the electrical power from the helicopter buses.

26. Remove the voltmeter probes from the test jacks of the No. 2 DC CONTROL PANEL.

27. Install the voltage setting protective cover on the No. 2 DC CONTROL PANEL.

28. Remove the wire and the 5 amp circuit breaker.

29. Install the lower nose access panel.

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Generator No. 1 and Generator No. 2 System - Operational Check

Procedure

1. Remove the lower nose access panel located under the nose of the helicopter.

2. Operate both engines at flight idle speed in accordance with normal ground run operations.

3. Make sure that both BATTERY switches are ON.

4. Verify that the battery voltage is present on both buses. This indicates that the two primary DC buses are connected.



IF RESULTS DESCRIBED IN THE FOLLOWING Step 5. OR Step 6. ARE NOT OBTAINED, IMMEDIATELY SET BATTERY SWITCHES AND BOTH GEN SWITCHES TO OFF. THIS WILL PREVENT DAMAGE TO THE STARTER-GENERATORS IF EITHER OR BOTH DC CONTROL UNITS SHOULD BECOME DEFECTIVE.

5. Set the GEN 1 and GEN 2 switches to ON. Verify that both dual voltmeters indicate approximately 27 to 28 VDC.

6. Connect the voltmeter to the generator circuitry terminal of the No. 2 START-GEN RELAY. Verify that the voltmeter indicates the generator voltage.

7. To simulate an engine failure, shut off the No. 2 engine fuel supply.

8. Monitor the helicopter dual voltmeter and verify the following:
- ENG 2 DC GENERATOR caution light on the master caution panel illuminates
- Engine No. 2 starter-generator loses speed
- No. 2 generator voltage decreases
- Both dual voltmeters indicate previous values

9. Disconnect the voltmeter from the No. 2 START-GEN RELAY.

10. Connect the voltmeter to the generator circuitry terminal of the engine No. 1 starter-generator.

11. To simulate a second engine failure, shut off the No. 1 engine fuel supply.

12. Monitor both dual voltmeters and verify the following:
- ENG 1 DC GENERATOR caution light on the master caution panel illuminates
- Engine No. 1 starter-generator loses speed
- No. 1 generator voltage decreases
- Both dual voltmeter indications drop to and remain at battery potential

13. Set the BATTERY switches to OFF.

14. Verify that both dual voltmeters indicate 0 VDC.

15. Set all switches to the off or normal position.

16. Remove the voltmeter.

17. Close the lower nose access panel.

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Voltage Control Paralleling Adjustment - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Digital Multimeter	Fluke Model 85	1	or Equivalent

Procedure

NOTE

Do the equalization adjustment only after the operational check of the generator No. 1 and No. 2 systems is complete (412-A-96-61-00-04A-320C-A / 01925).

1. Remove the lower nose access panel.



DO NOT DISCONNECT OR RECONNECT CABLE TO CONTROL UNIT WHEN GENERATORS ARE ROTATING, AS THIS MAY CAUSE DAMAGE TO CONTROL UNIT.

2. Close the circuit breakers required for engine start and ground run.

3. Start both engines.

NOTE

Do not set the generator switches to ON until specified in the following steps.

4. Increase and maintain the engine speed to indicate 71% N_1 or greater.

5. Set the GEN 1 switch to RESET and then to OFF.

6. Set the GEN 2 switch to RESET and then to OFF.

7. Measure the voltage at the No. 1 DC CONTROL UNIT test jacks with a digital voltmeter (Figure 1).

Applicable: SN: 33001-33213 or 36001-36086:

7.1. Equipped with a 19 cell battery, if necessary, adjust the No. 1 DC CONTROL UNIT for the required voltage as follows:

SEASONAL TEMPERATURE	VOLTAGE SETTING
Maximum daily ambient temperature does not exceed 58°F (14.4°C)	28.0 ±0.1 VDC
Maximum daily ambient temperature exceeds 58°F (14.4°C)	27.0 ±0.1 VDC

Applicable: SN: 36087-36999 or 37002-37999 or 38001-38999 or 39101-39999:

7.2. *Equipped with a 20 cell battery, if necessary, adjust the No. 1 DC CONTROL UNIT for the required voltage as follows:*

SEASONAL TEMPERATURE	*VOLTAGE SETTING*
Maximum daily ambient temperature does not exceed 58°F (14.4°C)	*28.5 ±0.1 VDC*
Maximum daily ambient temperature exceeds 58°F (14.4°C)	*28.0 ±0.1 VDC*

8. Measure the voltage at the No. 2 DC CONTROL UNIT test jacks with a digital voltmeter (Figure 1). If necessary, adjust the No. 2 DC CONTROL UNIT until the voltage is ±0.1 VDC of that voltage set for the No. 1 GENERATOR.

9. Connect the digital voltmeter positive lead to the positive test jack of the No. 1 DC CONTROL UNIT and connect the negative lead to the positive test jack of the No. 2 DC CONTROL UNIT (Figure 1).

10. Verify that the meter reads 0.00 ±0.1 VDC.

11. If necessary, adjust only one of the DC CONTROL UNITS until the voltmeter indicates 0.1 VDC or less.

NOTE

When both GEN switches are set to ON, the BATTERY BUS 1 switch automatically switches to OFF.

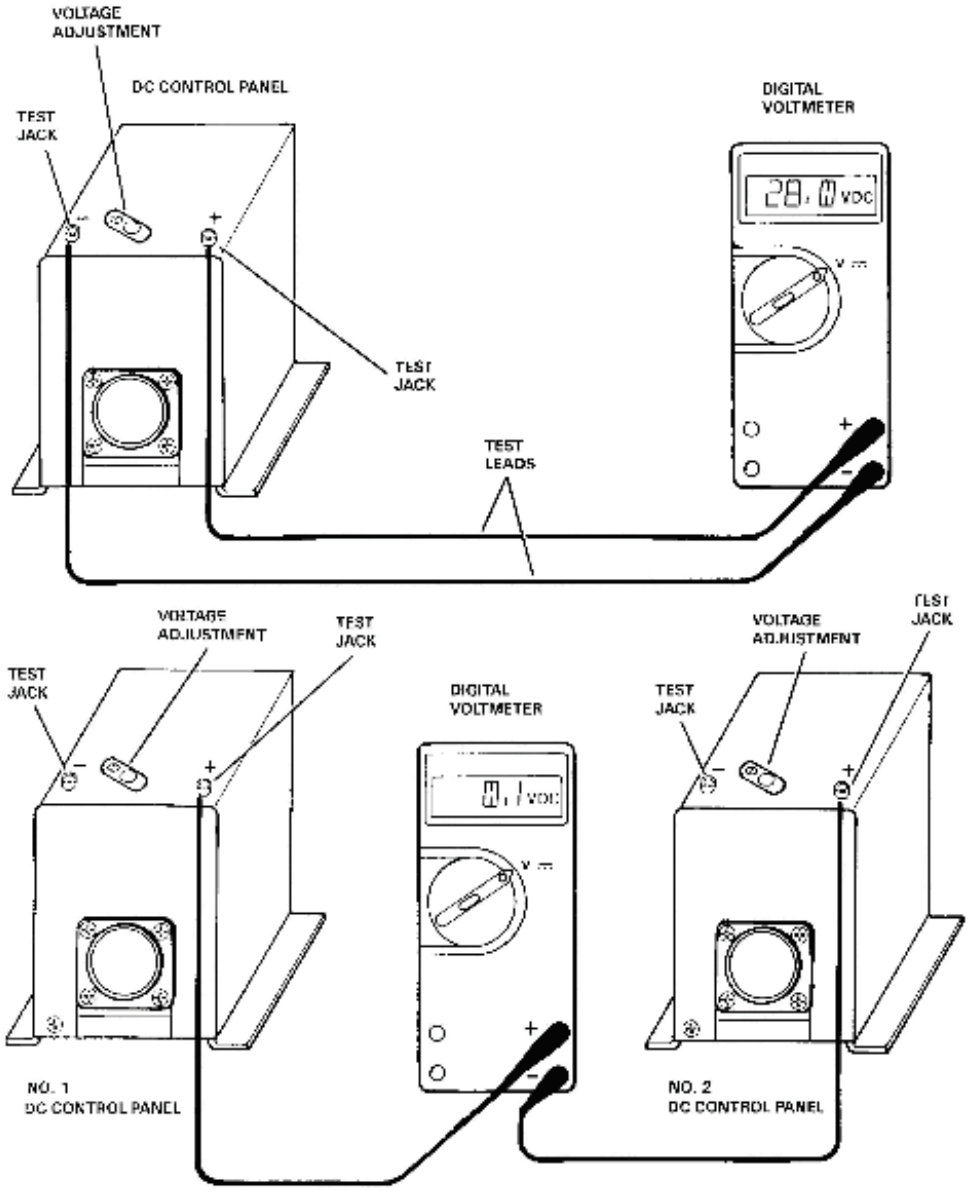
12. Set the GEN 1 and GEN 2 switches to ON.

13. Set the BATTERY BUS 2 switch to OFF.

14. Continue the parallel operation (both generators operating) for 5 minutes. During this time, observe the dual ammeter and verify that the differential between the two generators does not exceed 20 amps.

15. Shut down both engines.

16. Install the lower nose access panel.



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Figure 1. Voltage Control Paralleling Adjustment (Sheet 1 of 1)

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Non-essential Bus System - Operational Check

Procedure

NOTE

The operational check of the non-essential bus system should be done immediately after you complete the equalization adjustment with both generators operating and parallel.

1. Close the NON-ESNTL BUS circuit breaker.

2. Place the NON-ESNTL BUS switch to MANUAL.

3. Check for 28 VDC on the non-essential bus in the overhead console.

4. Set the NON-ESNTL BUS switch to NORMAL. Check for 28 VDC on the non-essential bus in the overhead console.

5. Set the GEN 2 switch to OFF.

6. Verify that the ENG 2 DC GENERATOR caution light on the MASTER CAUTION PANEL is illuminated.

7. Check for 0 volts on the non-essential bus.

8. Set the NON-ESNTL BUS switch to MANUAL.

9. Check for 28 VDC on the non-essential bus.

10. Set the GEN 2 switch to ON and the GEN 1 switch to OFF.

11. Check for 28 VDC on the non-essential bus.

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Starter and Igniter Systems - Description

One starter-generator is mounted on each of the two engines. If electrical power is supplied to the series (or starter) winding, the starter-generator will produce a rotational torque to start the engine. Electrical power may be from the battery or from an external source of 28 VDC power. To start engine No. 1, close the ENGINE NO. 1 IGN circuit breaker and the ENGINE NO. 1 START RLY circuit breaker. Set the ENGINE 1 FUEL switch to ON and the START switch (on pilot collective switch) to the ENG 1 position. This energizes the No. 1 start control relay and the No. 1 starter-generator relay, which energizes the No. 1 generator relay, and also energizes the engine No. 1 ignition unit. The current now flows through the contacts of the No. 1 generator relay and the No. 1 starter-generator relay to the series (starter) winding of the starter-generator No. 1. To start the engine No. 2, repeat the procedure with the control items of the engine No. 2 system. For the maintenance of the starter-generators, refer to (412-A-71-01-00-11A-042A-A / 01693).

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Starter No. 1 Circuitry - Operational Check

Procedure

NOTE

Disconnect the wire in the following step only if the starter fails to rotate.

1. Disconnect wire P95A1 from terminal C on the engine No. 1 starter-generator and isolate from the helicopter structure.
2. Close the engine No. 1 START RLY, No. 1 ESNTL BUS FEEDER(S), GEN 1 RESET, and No. 1 BUS BAT switch circuit breakers.
3. Set the BATTERY BUS 1 switch to ON.
4. Position the START switch on the pilot collective stick to ENG 1.
5. Verify that the battery voltage is present on the disconnected end of wire P95A1 and 0 voltage is present at terminal B (wire P72A1) of the starter-generator.
6. Verify that the battery voltage is present on the No. 1 dual voltmeter.



DUE TO HIGH ELECTRICAL SHOCK HAZARD, DO NOT RECONNECT STARTER-GENERATOR WIRE IF AN OPERATIONAL CHECK OF THE ENGINE IGNITION IS TO BE DONE.

7. Set the START switch to OFF.
8. Verify that 0 VDC is present on the disconnected wire P95A1.
9. Turn off the electrical power and connect wire P95A1.

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Starter No. 2 Circuitry - Operational Check

Procedure

NOTE

Disconnect the wire in the following step only if the starter fails to rotate.

1. Set the BATTERY BUS 1 switch to OFF.

2. Disconnect wire P98A1 from terminal C on the engine No. 2 starter-generator and isolate from the helicopter structure.

3. Connect the DC external power source to the external power receptacle and energize.

4. Close the engine No. 2 START RLY, GEN 2 RESET, and No. 2 ESNTL BUS FEEDER circuit breakers.

5. Set the START switch on the pilot collective stick to ENG 2.

6. Verify that the external power source voltage is present on the disconnected wire P98A1 and 0 VDC is present at terminal B (wire P85A1) of the engine No. 2 starter-generator.



DUE TO HIGH ELECTRICAL SHOCK HAZARD, DO NOT RECONNECT STARTER-GENERATOR WIRE IF AN OPERATIONAL CHECK OF THE ENGINE IGNITION IS TO BE DONE.

7. Set the START switch to OFF.

8. Verify that 0 VDC is present on the disconnected wire P98A1.

9. Turn off the electrical power and connect wire P98A1.

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Engine Ignition Unit No. 1 - Operational Check

NOTE

This check must be done on battery power.

Procedure

1. Close the engine No. 1 IGN and the engine No. 1 START RLY circuit breakers.

2. Set the ENGINE 1 FUEL switch to ON.

3. Set the BATTERY BUS 1 switch to OFF.

4. Set the BATTERY BUS 2 switch to ON.

5. Position the START switch on the pilot collective stick to ENG 1.

6. Verify that the igniters can be heard arcing.

7. Release the START switch.

8. Open the engine No. 1 IGN and the engine No. 1 START RLY circuit breakers.

9. Set the ENGINE 1 FUEL switch to OFF.

10. If the igniters were heard, proceed to engine No. 2 (412-A-96-61-00-06A-320C-A / 01932). If not, do the following steps.

11. Disconnect the engine No. 1 ignition unit connector (1Z9P1) from the engine No. 1 ignition unit.

12. Close the engine No. 1 IGN and the engine No. 1 START RLY circuit breakers.

13. Position the START switch on the pilot collective stick to ENG 1.

14. Verify that 0 VDC is present between pin A and pin B (ground) of the engine No. 1 ignition unit connector.

15. Set the ENGINE 1 FUEL switch to ON.

16. Verify that the battery voltage is present between pin A and pin B of the engine No. 1 ignition unit connector.

17. Set the BATTERY BUS 2 switch to OFF.

18. Set the ENGINE 1 FUEL switch to OFF.

19. Reconnect the engine No. 1 ignition unit connector to the engine No. 1 ignition unit.

20. Verify that the connector is properly mated, tight, and secure.

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Engine Ignition Unit No. 2 - Operational Check

NOTE

This check must be done on battery power.

Procedure

1. Close the engine No. 2 IGN and the engine No. 2 START RLY circuit breakers.

2. Set the ENGINE 2 FUEL switch to ON.

3. Set the BATTERY BUS 1 switch to ON.

4. Set the BATTERY BUS 2 switch to OFF.

5. Position the START switch on the pilot collective stick to ENG 2.

6. Verify that the igniters can be heard arcing.

7. Release the START switch.

8. Open the engine No. 2 IGN and the engine No. 2 START RLY circuit breakers.

9. Set the ENGINE 2 FUEL switch to OFF.

10. If the igniters were heard, the check is complete. Secure the helicopter. If not, do the following steps.

11. Disconnect the connector (1Z10P1) from the engine No. 2 ignition unit.

12. Close the engine No. 2 IGN and the engine No. 2 START RLY circuit breakers.

13. Set the START switch on the pilot collective stick to ENG 2.

14. Check that 0 VDC is present between pin A and pin B (ground) of the connector.

15. Set the ENGINE 2 FUEL switch to ON.

16. Check that the battery voltage is present between pin A and pin B of the engine No. 2 ignition unit connector.

17. Set the BATTERY BUS 1 switch to OFF.

18. Set the ENGINE 2 FUEL switch to OFF.

19. Reconnect the connector to the ignition unit.

20. Check that the connector is properly mated, tight, and secure.

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Power Diodes - Functional Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Digital Multimeter	Fluke Model 85	1	or Equivalent

Procedure

1. Make sure that all electrical power is OFF.

2. Open the copilot No. 1 ESNTL BUS FEEDERS circuit breakers and the No. 2 ESNTL BUS FEEDERS circuit breakers.

3. Open the pilot No. 1 ESNTL BUS FEEDERS circuit breakers and the No. 2 ESNTL BUS FEEDERS circuit breakers.

4. Loosen the fasteners and lower the cockpit overhead circuit breaker panels.

NOTE

SN: 33001-33100:

Refer to Figure 1 for simplified electrical bus wiring.

NOTE

SN: 33101-33213 or 36001-36999:

Refer to Figure 2 for simplified electrical bus wiring.

NOTE

SN: 33001-33100:

Refer to Figure 3 as a guide to power source, circuit breaker, and switch positions for testing various diodes.

NOTE

SN: 33101-33213 or 36001-36999 or 37002-37999:

Refer to Figure 4 as a guide to power source, circuit breaker, and switch positions for testing various diodes.

4.1. Remove the access panels for the non-essential power diode bus feeder, located on the roof, left and right sides.

4.2. Remove the power diode assemblies from the roof.

4.3. Disconnect the bus bar from the power diodes (anode side).

5. Set the digital multimeter (Fluke Model 85) (DMM) to the DIODE TEST setting.

6. Connect the positive lead of the DMM to the wire terminal (anode) of the diode and the negative lead to the heat sink (cathode) to forward bias the diode.

NOTE

Replace the diodes that fail either Step 7., Step 9., or Step 11..

7. Verify a DMM reading of 0.25 to 1.15 V.

8. Connect the negative lead of the DMM to the wire terminal (anode) of the diode and the positive lead to the heat sink (cathode) to reverse bias the diode.

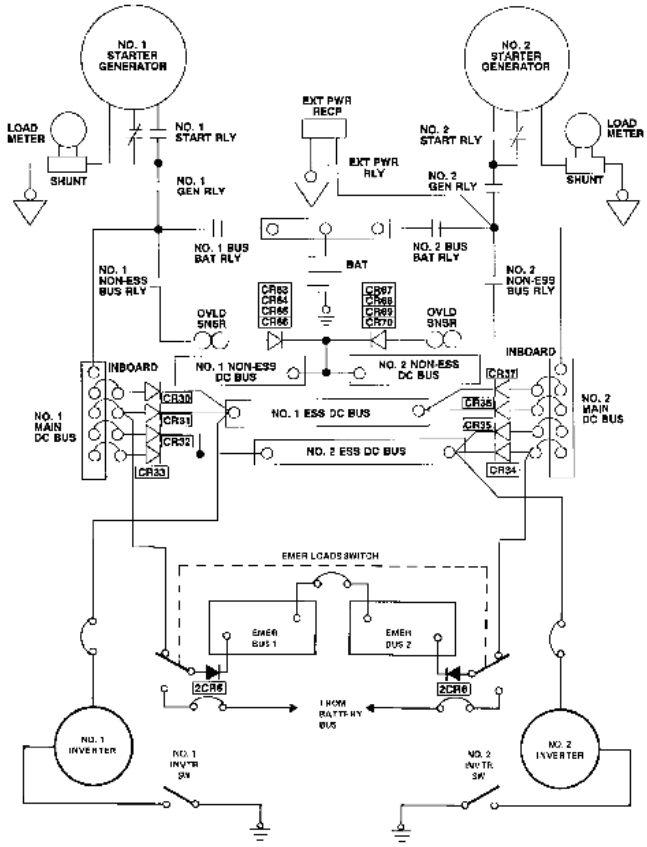
9. Verify that the DMM reads OPEN (high resistance).

10. Apply sideway pressure with the multimeter leads and monitor the meter.

11. Check for an intermittent short circuit.

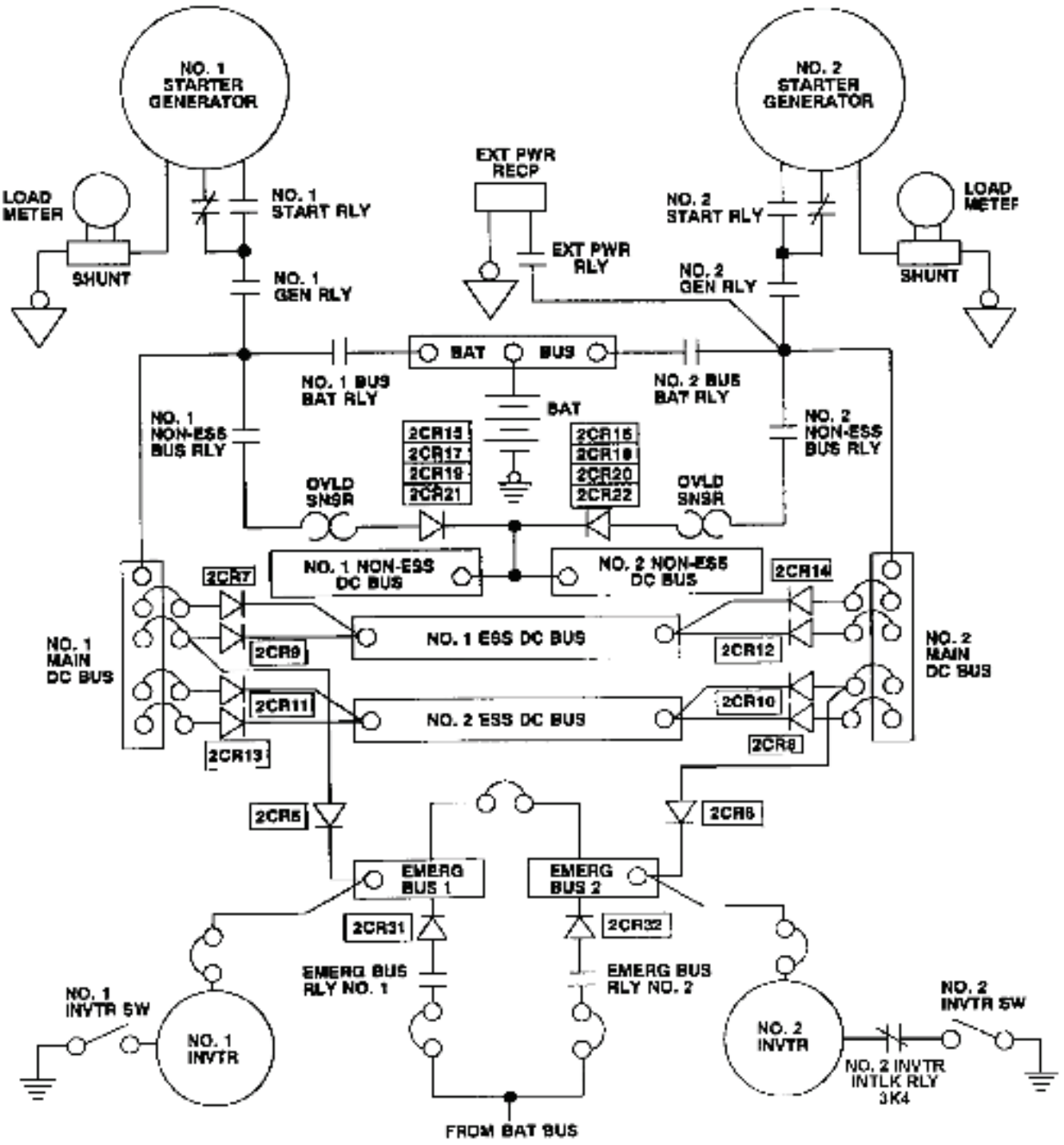
12. Close the copilot No. 1 ESNTL BUS FEEDERS circuit breakers and the No. 2 ESNTL BUS FEEDERS circuit breakers.

13. Close the pilot No. 1 ESNTL BUS FEEDERS circuit breakers and the No. 2 ESNTL BUS FEEDERS circuit breakers.



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Figure 1. Simplified Electrical Bus Wiring (S/N 33001 through 33100) (Sheet 1 of 1)



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Figure 2. Simplified Electrical Bus Wiring (S/N 33101 through 33213 and 36001 through 36999) (Sheet 1 of 1)

INSTRUCTIONS:
1. POSITION ALL SWITCHES NOT LISTED BELOW IN OFF OR NORMAL POSITION AND CLOSE ALL CIRCUITS NOT LISTED BELOW.
2. PERFORM TEST A, B, C, ETC.
3. REPLACE ALL DIODES FOUND TO BE OPEN OR SHORTED.

CIRCUIT BREAKER, SWITCH & POWER SOURCE TEST SETTINGS

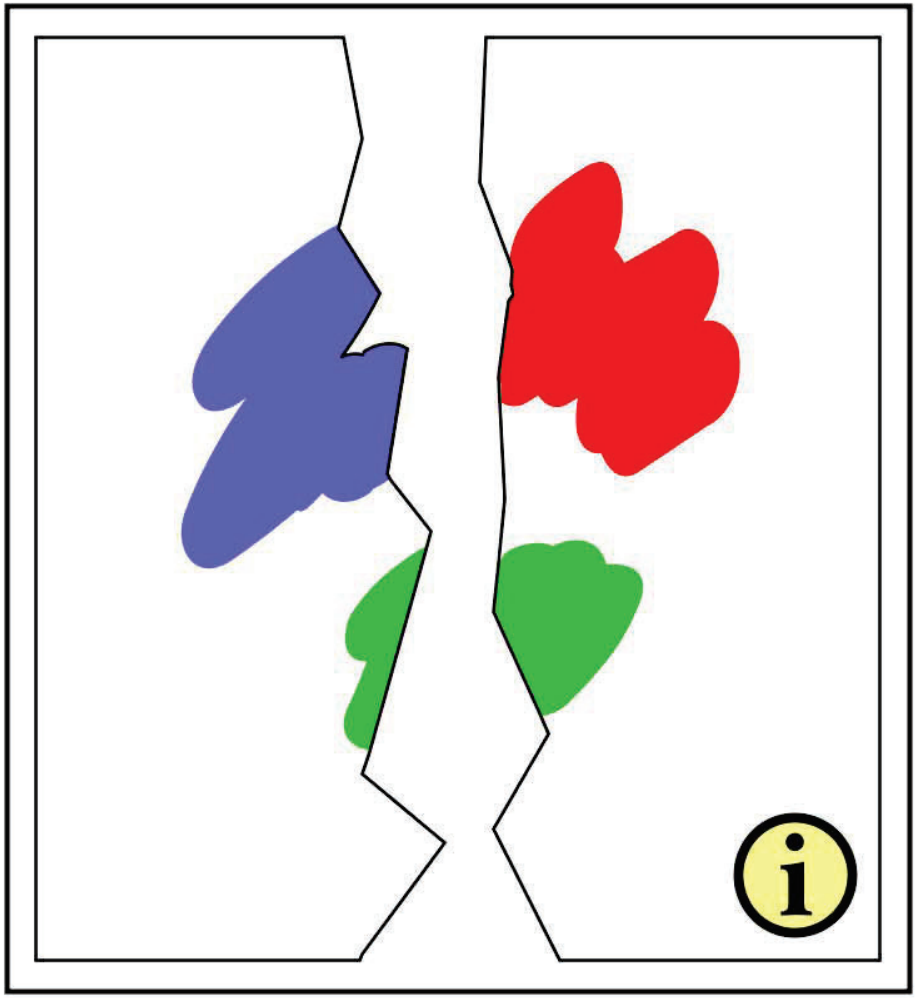
	CIRCUIT BREAKERS, SWITCHES & POWER SOURCES TO BE SET	1	A	B	C	D	E	F	G	H	I	J	K
1	DC External Power (If available)	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
2	BATTERY BUS 1 switch	1	OFF	ON	ON	ON	ON	ON	ON	ON	OFF	OFF	OFF
3	BATTERY BUS 2 switch	1	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
4	NON-ESNTL BUS switch (OFF=NORMAL, ON=MANUAL)	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
5	INV 1 switch	1	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON
6	INV 2 switch	1	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON
7	NORMAL/EMERG LOAD switch (OFF – NORMAL, ON – EMERG LOAD)	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
8	Copilot side NO. 1 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	CLOSE	OPEN	CLOSE	CLOSE	OPEN	OPEN	OPEN	CLOSE	CLOSE
9	Copilot side NO. 1 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	OPEN	CLOSE	CLOSE
10	Copilot side NO. 2 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	OPEN	OPEN	CLOSE	CLOSE
11	Copilot side NO. 2 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	OPEN	CLOSE	CLOSE
12	Pilot side NO. 1 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	CLOSE	OPEN
13	Pilot side NO. 1 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	CLOSE
14	Pilot side NO. 2 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	OPEN
15	Pilot side NO. 2 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	OPEN
16	EMERG BUS INTC circuit breaker	1	OPEN	OPEN	OPEN	OPEN	CLOSE	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN

	OBSERVATIONS, MEASUREMENTS AND TEST POINTS	1	RESULTS NECESSARY FOR PROPER OPERATION										
17	DC Volt Meter NO. 1	1	0 VDC	0 VDC	+ VDC	+ VDC	+ VDC	+ VDC	0 VDC	0 VDC	0 VDC	– VDC	+ VDC
18	DC Volt Meter NO. 2	1	0 VDC	0 VDC	0 VDC	0 VDC	0 VDC	0 VDC	+ VDC	+ VDC	0 VDC	0 VDC	0 VDC
19	Caution Panel Segment INVERTER NO. 1	1	OFF	OFF	–	–	OFF	OFF	–	–	–		–
20	Caution Panel Segment INVERTER NO. 2	1	OFF	OFF	–	–	ON	ON	–	–	–	–	–
21	Deleted	1	–	–	–	–		–	–	–	–	–	–
22	Deleted	1	–			–	–	–	–	–	–	–	–
23	Four NO. 1 Non-Esntl Bus Diodes	1	–	–	–	–	–	–	–	–	–	–	–
24	Four NO. 2 Non-Esntl Bus Diodes	1	–	–	–	–	–	–	–	–	–	–	–

TEST	PURPOSE
A	INITIAL SETTING WITHOUT POWER APPLIED
B	INITIAL SETTING WITH POWER APPLIED
C	**INSURE NO OPEN CIRCUIT AT CR30 DIODE**
D	**INSURE NO OPEN CIRCUIT AT CR31 DIODE**
E	INSURE NO OPEN CIRCUIT AT 2CR6 DIODE
F	**INSURE PROPER REVERSE BIAS ISOLATION FROM CR36, CR37, AND 2CR6 DIODES (CR36 AND CR37)**
G	**INSURE NO OPEN CIRCUIT AT CR32 DIODE AND PROPER REVERSE BIAS ISOLATION OF CR34 AND CR35 DIODES**
H	**INSURE NO OPEN CIRCUIT AT CR33 DIODE AND PROPER REVERSE BIAS ISOLATION OF CR34 AND CR35 DIODES**

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Figure 3. Power Diode Functional Check (S/N 33001 through 33100) (Sheet 1 of 2)



ICN-412-A-966100-B-97499-00007-A-001-01

Figure 3. Power Diode Functional Check (S/N 33001 through 33100) (Sheet 2 of 2)

INSTRUCTIONS:
1. POSITION ALL SWITCHES NOT LISTED BELOW IN OFF OR NORMAL POSITION AND CLOSE ALL CIRCUITS NOT LISTED BELOW.
2. PERFORM TEST A, B, C, ETC.
3. REPLACE ALL DIODES FOUND TO BE OPEN OR SHORTED.

CIRCUIT BREAKER, SWITCH & POWER SOURCE TEST SETTINGS

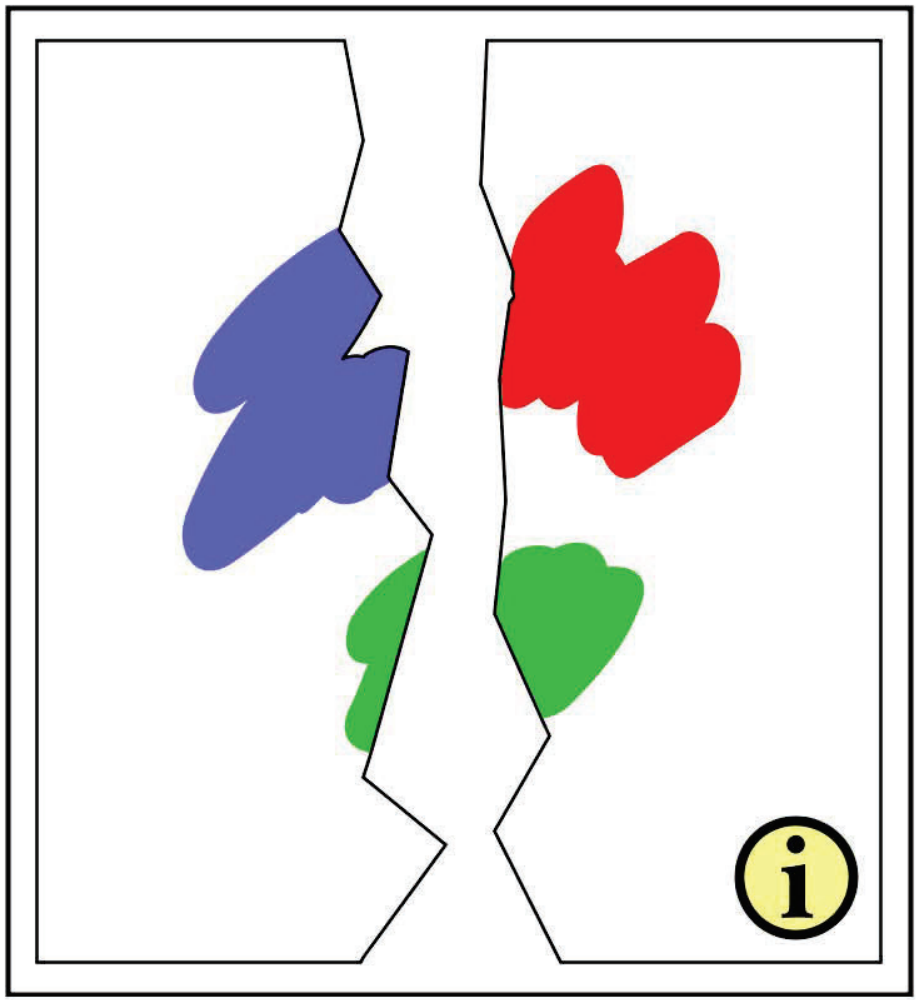
	CIRCUIT BREAKERS, SWITCHES & POWER SOURCES TO BE SET	1	A	B	C	D	E	F	G	H	I	J	K
1	DC External Power (If available)	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
2	BATTERY BUS 1 switch	1	OFF	ON	ON	ON	ON	ON	ON	ON	OFF	OFF	OFF
3	BATTERY BUS 2 switch	1	OFF	ON	OFF	OFF	OFF	OFF	OFF	OFF	ON	ON	ON
4	NON-ESNTL BUS switch (OFF=NORMAL, ON=MANUAL)	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
5	INV 1 switch	1	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON
6	INV 2 switch	1	OFF	OFF	ON	ON	ON	ON	ON	ON	ON	ON	ON
7	NORMAL/EMERG LOAD switch (OFF = NORMAL, ON = EMERG LOAD)	1	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF	OFF
8	Copilot side NO. 1 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	CLOSE	OPEN	CLOSE	CLOSE	OPEN	OPEN	OPEN	CLOSE	CLOSE
9	Copilot side NO. 1 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	OPEN	CLOSE	CLOSE
10	Copilot side NO. 2 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	OPEN	OPEN	CLOSE	CLOSE
11	Copilot side NO. 2 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	OPEN	CLOSE	CLOSE
12	Pilot side NO. 1 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	CLOSE	OPEN
13	Pilot side NO. 1 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	CLOSE
14	Pilot side NO. 2 ESNTL BUS FEEDER circuit breaker (inboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	OPEN
15	Pilot side NO. 2 ESNTL BUS FEEDER circuit breaker (outboard)	1	OPEN	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	CLOSE	OPEN	OPEN	OPEN
16	EMERG BUS INTC circuit breaker	1	OPEN	OPEN	OPEN	OPEN	CLOSE	CLOSE	OPEN	OPEN	OPEN	OPEN	OPEN

	OBSERVATIONS, MEASUREMENTS AND TEST POINTS	1	RESULTS NECESSARY FOR PROPER OPERATION										
17	DC Volt Meter NO. 1	1	0 VDC	0 VDC	+ VDC	− VDC	+ VDC	+ VDC	0 VDC	0 VDC	0 VDC	− VDC	+ VDC
18	DC Volt Meter NO. 2	1	0 VDC	0 VDC	0 VDC	0 VDC	0 VDC	0 VDC	+ VDC	+ VDC	0 VDC	0 VDC	0 VDC
19	Caution Panel Segment INVERTER NO. 1	1	OFF	OFF	—	—	OFF	OFF	—	—	—	—	—
20	Caution Panel Segment INVERTER NO. 2	1	OFF	OFF	—	—	OFF	OFF	—	—	—	—	—
21	**Emergency Bus No. 1 Power Relay Terminal A2 (2K17-A2)**	1	—	—	—	—	—	—	—	—	—	—	—
22	**Emergency Bus No. 2 Power Relay Terminal A2 (2K16-A2)**	1	—	—	—	—	—	—	—	—	—	—	—
23	Four NO. 1 Non-Esntl Bus Diodes	1	—	—	—	—			—		—		
24	Four NO. 2 Non-Esntl Bus Diodes	1	—	—	—	—	—	—	—	—	—		

TEST	PURPOSE
A	INITIAL SETTING WITHOUT POWER APPLIED
B	INITIAL SETTING WITH POWER APPLIED
C	INSURE NO OPEN CIRCUIT AT 2CR7 DIODE
D	INSURE NO OPEN CIRCUIT AT 2CR9 DIODE
E	INSURE NO OPEN CIRCUIT AT 2CR5 DIODE
F	INSURE PROPER REVERSE BIAS ISOLATION FROM 2CR12, 2CR14 AND 2CR6 DIODES
G	INSURE NO OPEN CIRCUIT AT 2CR11 DIODE AND PROPER REVERSE BIAS ISOLATION OF 2CR8 AND 2CR10 DIODES
H	INSURE NO OPEN CIRCUIT AT 2CR13 DIODE AND PROPER REVERSE BIAS ISOLATION OF 2CR8 AND 2CR10 DIODES

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Figure 4. Power Diode Functional Check (S/N 33101 through 33213, 36001 through 36999, and 37002 through 37999) (Sheet 1 of 2)



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Figure 4. Power Diode Functional Check (S/N 33101 through 33213, 36001 through 36999, and 37002 through 37999) (Sheet 2 of 2)

Power Diodes - Removal

Procedure

1. Disconnect the battery.
2. Make sure that all electrical power is OFF.
3. Remove the attaching hardware from the defective diode.
4. Remove the diode.

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Power Diodes - Inspection

Procedure

1. Visually inspect the diodes for the following:

- Charred/discolored insulation
- Proper bonding and security of mounting hardware
- Cleanliness and corrosion

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Power Diodes - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Heatsink compound	C-041	AR	

Procedure

1. Apply silicone heatsink compound (C-041) between the diode and the heatsink.

2. Install the mounting hardware.

NOTE

The threads of the diode stud should be dry. Make sure that no silicone heatsink compound is on the threads of the diode stud.

3. Torque the diode stud fixing nut 20 to 30 inch-pounds (2.3 to 3.4 Nm).

3.1. Connect the bus bar on the non-essential power diodes.

3.2. Secure the power diode assembly to the roof.

3.3. Install the access panels.

4. Do a functional check of the replaced diode (412-A-96-61-00-07A-340A-A / 01933).

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AC Power Systems - Description

This system includes components of the inverter systems. Refer to (412-A-98-00-00-18A-051A-A / 02478, or 412-A-98-00-00-19A-051A-A / 02479, or 412-A-98-00-00-20A-051A-A / 02480) for the applicable wiring diagrams.

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Inverter Systems - Description

The Model 412 helicopter has two different alternating current power inverters. These are 26/115 volt, 400 Hz, single-phase, solid-state inverters.

Applicable: SN: 33001-33213 or 36001-36247:

Helicopters have two identical 450VA inverters.

Applicable: SN: 36248-36705:

Have two identical 250VA or 450VA inverters.

Applicable: SN: 36706-36999:

Have two identical 250VA inverters.

Applicable: SN: 33001-33107:

One inverter is energized by each of the two essential DC buses.

Applicable: SN: 33108-33213 or 36001-36999:

One inverter is energized by each of the two DC emergency buses. Inverter No. 1 supplies power to the 115 and 26 VAC buses 1 and 2, and inverter No. 2 supplies power to the 115 and 26 VAC bus 3. The two inverters are completely independent of each other.

Applicable: SN: 33001-33107:

AC power is distributed through the respective AC voltage sensor relays and common AC bus control relay.

Applicable: SN: 33108-33213 or 36001-36999:

AC power is distributed through the respective AC voltage sensor relays, inverter No. 2 interlock relay, and inverter No. 1 emergency bus relay. Caution light segments are provided for each inverter system.

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Inverter Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Digital Multimeter	Fluke Model 85	1	
Frequency Meter	Commercial	1	

Procedure

1. Preliminary Setup

1.1. Connect the external power.

1.2. Make sure that the following circuit breakers, in the nose compartment, are closed:
- INVERTER 1 EMERG INTLK
- INVERTER 2 EMERG INTLK

1.3. Close the following circuit breakers:
- NO. 1 ESNTL FEEDER BUS
- NO. 2 ESNTL FEEDER BUS
- GEN 1 RESET
- GEN 2 RESET
- MASTER CAUTION
- BUS 1 VM
- BUS 2 VM

1.4. Set the BATTERY BUS 1 and the BATTERY BUS 2 switches ON.

1.5. Verify that the NO. 1 INVERTER and NO. 2 INVERTER caution lights on the master caution panel are illuminated.

2. Inverter No. 1 Circuitry

2.1. Close the following circuit breakers:
- INV 1 PWR
- 26/115 AC FEEDER BUS 1
- 26/115 AC FEEDER BUS 2
- 26/115 AC FEEDER BUS 3
- **SN: 33108-33213 or 36001-36999:**
  26/115 EMERG BUS

2.2. Set the INV 1 switch to ON.

2.3. Verify that the NO. 1 INVERTER caution light is extinguished.

2.4. Verify that both dual voltmeters indicate approximately 115 VAC.

2.5. Verify that the NO. 2 INVERTER caution light is illuminated.

2.6. Check for 110 to 120 VAC on the 115 VAC buses No. 1, No. 2, and No. 3.

Applicable: SN: 33108-33213 or 36001-36999:

2.7. Check for 110 to 120 VAC on the 115 VAC EMERG BUS.

2.8. Check for 24.5 to 27.5 VAC on the 26 VAC buses No. 1, No. 2, and No. 3.

Applicable: SN: 33108-33213 or 36001-36999:

2.9. Check for 24.5 to 27.5 VAC on the 26 VAC EMERG BUS.

2.10. Verify that the inverter No. 1 output frequency is 400 ±4 Hz with a frequency meter.

2.11. Set the INV 1 switch to OFF.

2.12. Verify that the NO. 1 INVERTER caution light illuminates.

2.13. Verify that both dual voltmeters indicate zero.

3. Inverter No. 2 Circuitry

3.1. Close the INV 2 PWR circuit breaker.

3.2. Set the INV 2 switch to ON.

3.3. Verify that the NO. 2 INVERTER caution light is extinguished.

3.4. Verify that both dual voltmeters indicate approximately 115 VAC.

3.5. Verify that the NO. 1 INVERTER caution light illuminates.

3.6. Check for 110 to 120 VAC on the 115 VAC bus No. 1, No. 2, and No. 3.

Applicable: SN: 33108-33213 or 36001-36999:

3.7. Check for 110 to 120 VAC on the 115 VAC EMERG BUS.

3.8. Check for 24.5 to 27.5 VAC on the 26 VAC bus No. 1, No. 2, and No. 3.

Applicable: SN: 33108-33213 or 36001-36999:

3.9. Check for 24.5 to 27.5 VAC on the 26 VAC EMERG BUS.

3.10. Verify that the inverter No. 2 output frequency is 400 ±4 Hz with a frequency meter.

3.11. Set the INV 2 switch to OFF.

3.12. Verify that the NO. 2 INVERTER caution light illuminates.

Inverter No. 1 and Inverter No. 2 Circuitry - Operational Check

Procedure

1. Set the INV 1 and INV 2 switch to ON.
2. Verify that the NO. 1 INVERTER and the NO. 2 INVERTER caution lights are extinguished.
3. Verify that the No. 1 and No. 2 bus voltmeters indicate approximately 115 VAC.
4. Set the EMERG LOAD switch to EMERG LOAD.
5. Verify that the NO. 1 INVERTER caution light remains extinguished.
6. Verify that the NO. 2 INVERTER caution light illuminates.
7. Verify that voltage is present on the 115 VAC EMERG BUS and the 26 VAC EMERG BUS.
8. Verify that no voltage is present on any other 115 VAC or 26 VAC BUS.

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Inverters - Removal

Procedure



TO PREVENT ELECTRICAL SHOCK OR DAMAGE TO EQUIPMENT, MAKE SURE EXTERNAL POWER AND BATTERY ARE DISCONNECTED.

1. Disconnect the battery.
2. Make sure that all electrical power is off.
3. Disengage the connectors and protect with a cap or tape.
4. While supporting the inverter, remove the mounting hardware.
5. Carefully lower and remove the inverter from the compartment.

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Inverters - Inspection

Procedure



TO PREVENT ELECTRICAL SHOCK OR DAMAGE TO EQUIPMENT, MAKE SURE EXTERNAL POWER AND BATTERY ARE DISCONNECTED.

1. Visually inspect the inverters for the following conditions:

- Cracked or damaged cases that could impair proper operation of the inverter
- Proper bonding and security of the mounting hardware
- Broken connector pins or cracked connector inserts

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Inverters - Repair

Procedure



TO PREVENT ELECTRICAL SHOCK OR DAMAGE TO EQUIPMENT, MAKE SURE EXTERNAL POWER AND BATTERY ARE DISCONNECTED.

1. Repair or replace damaged airframe connectors only.

2. Replace the inverter if other inspection criteria is not met.

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Inverters - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-508	AR	

Procedure

1. While supporting the inverter on its mounting pad, install and secure the mounting hardware.



DO NOT USE ANY TYPE OF TOOL TO INSTALL PLUG. USE MODERATE HAND PRESSURE ONLY.

2. Remove the protective covers from the connectors and engage and secure the connectorslockwire (C-508).

3. Restore the electrical power and do an operational check.

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AC Indicator Systems - Description

The AC indicator systems include the torque pressure, engine oil pressure, transmission oil pressure, hydraulic pressure, combining gearbox oil pressure, fuel pressure, attitude director indicator, triple tachometer, and gas producer tachometer indicator systems. Refer to (412-A-98-00-00-18A-051A-A / 02478, or 412-A-98-00-00-19A-051A-A / 02479, or 412-A-98-00-00-20A-051A-A / 02480) for the applicable wiring diagrams.

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Torque Pressure Indicator Systems - Description

The torque pressure indicator system simultaneously displays torque output of both engines on the inner dial transmission torque on the outer dial. Torque output of engine No. 1 is indicated by pointer No. 1, engine No. 2 is indicated by pointer No. 2, and the resultant sum of these torques is displayed by the cursor (bug). All indications are in terms of present allowable to transmission. AC power is supplied through the TORQUE METER circuit breakers. The pilot and copilot systems are totally independent of each other but function identically.

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Torque Pressure Indicator Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Deadweight Tester	2311F	1	or Equivalent
Pressure Gauge	302-00102	1	

Procedure

NOTE

Refer to (412-A-76-10-00-00A-271A-A / 01727) for the engine torque transmitter adjustment and functional check.

1. Set the BATTERY BUS 1 and the BATTERY BUS 2 switches to ON.
2. Set the INV 1 and INV 2 switches to ON.
3. Close the ENG 1 TORQUE METER circuit breaker.
4. Disconnect the engine 1 torque pressure transmitter from the engine No. 1.



MAKE SURE THAT THE PRESSURE TESTER IS SERVICED WITH THE SAME OIL AS USED IN THE COMBINING GEARBOX.

5. Connect the pressure tester to the input port.
6. Apply pressure to the input port not to exceed 75 PSIG (517 kPa).

NOTE

When you do the following, the pressure tester and gauge must be at the same height as the reduction gearbox torque meter (approximate level of the oil cooler blower center line).

7. Adjust the pressure to 55 ±1.5 PSIG (379 ±10 kPa).
8. Verify that the pilot and copilot (if installed) ENG 1 TORQUE pointer reads approximately 70% on the inner ENG dial of the indicator.
9. Verify that the pilot and copilot (if installed) ENG 2 TORQUE pointer remains at the low end of the dial.
10. Verify that the TRANSMISSION cursor indicates the sum of the ENG 1 and ENG 2 pointers.
11. Remove the applied pressure.
12. Verify that the pilot and copilot (if installed) ENG 1 and ENG 2 pointers are at the low end of the engine dial.
13. Verify that the pilot and copilot (if installed) TRANSMISSION cursor is at the low end dial.
14. Disconnect the pressure from the engine 1 torque pressure transmitter.
15. Connect the torque transmitter to the engine No. 1.
16. Verify that the torque transmitter is properly mated, tight, and secure.
17. Repeat Step 3. through Step 16. for the engine No. 2 (references to engine 1 shall be engine 2 and vise versa).
18. Set the INV 1 and INV 2 switches to OFF.
19. Set the BATTERY BUS 1 and BATTERY BUS 2 switches to OFF.

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Engine Oil Pressure Indicator Systems - Description

The oil pressure indicators for engine No. 1 and engine No. 2 receive signals from their respective engine oil pressure transmitters.

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Engine Oil Pressure Indicator Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

1. Close the ENG 1 OIL PRESS circuit breaker. Check that the engine No. 1 ENGINE OIL temperature and pressure indicator reads 0 ±7.5 PSI.

NOTE

A pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 150 PSI (1034 kPa) must be used when you do the following checks.

2. Disconnect the pressure input line from the engine 1 oil pressure transmitter on the engine No. 1.

3. Attach the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) and apply pressure to the transmitter input line, but do not exceed 110 PSIG (758 kPa).

4. Check that the ENGINE OIL indicator reads 100 PSI when the input pressure is 100 ±10 PSIG (689 ±69 kPa).

5. Disconnect the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F).

6. Connect the pressure input line to the engine 1 oil pressure transmitter.

7. Open the ENG 1 OIL PRESS circuit breaker and close the ENG 2 OIL PRESS circuit breaker.

8. Repeat Step 1. through Step 3. for the engine 2 oil pressure transmitter.

9. Take readings from the engine No. 2 ENGINE OIL temperature and pressure indicator.

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Transmission Oil Pressure Indicator System - Description

The transmission oil pressure transmitter sends a signal to the transmission oil temperature and pressure indicator, which is given in PSIG.

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Transmission Oil Pressure Indicator System - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

1. Close the XMSN OIL PRESS circuit breaker.

2. Check that the XMSN OIL temperature and pressure indicator reads 0 ±5 PSIG.

NOTE

A pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 150 PSIG (1034 kPa) must be used when you do the following checks.

3. Disconnect the pressure input line from the transmission oil pressure transmitter. Attach the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) and apply pressure to the transmitter input, but do not exceed 110 PSIG (758 kPa). Check that the XMSN OIL indicator reads 100 PSIG when the input pressure is 100 ±10 PSIG (689 ±69 kPa).

4. Relieve the pressure and disconnect the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F). Reconnect the pressure input line to the transmission oil pressure transmitter.

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Hydraulic Pressure Indicator System - Description

In either hydraulic pressure indicator system, the hydraulic pressure transmitter sends a signal to the hydraulic pressure indicator, which is given in PSIG.

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Hydraulic Pressure Indicator System - Operational Check

Procedure

1. Close the NO. 1 HYD PRESS circuit breaker. Apply external hydraulic pressure to the No. 1 hydraulic system. Check that, with external hydraulic pressures of 1000 and 1500 PSIG (6895 and 10,342 kPa), the HYDR SYS 1 hydraulic pressure and temperature indicator reads within 50 PSIG of each pressure setting.

2. Use the No. 2 HYD PRESS circuit breaker and the HYDR SYS 1 hydraulic pressure and temperature indicator and repeat Step 1. for the HYDR SYS 2 hydraulic pressure and temperature indicator,

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Combining Gearbox (C Box) Oil Pressure Indicator Systems - Description

The C BOX oil pressure transmitter sends a signal to the combining gearbox oil pressure and temperature indicator, which is given in PSIG.

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Combining Gearbox (C Box) Oil Pressure Indicator Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

1. Close the C BOX OIL PRESS circuit breaker. Remove the pressure transmitter from the C BOX.

NOTE

A pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 150 PSIG (1034 kPa) must be used when you do the following checks.

2. Attach the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) and apply pressure to the transmitter input, but do not exceed 100 ±10 PSIG (689 ±69 kPa). Check that the gear box oil pressure and temperature indicator reads 100 PSIG when the input pressure is 100 ±10 PSIG (689 ±69 kPa).

3. Disconnect the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F). Install the pressure transmitter to the C BOX.

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Fuel Pressure Indicator Systems - Description

In either fuel pressure indicator system, the fuel pressure transmitter sends a signal to the fuel pressure indicator, which displays in PSIG.

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Fuel Pressure Indicator Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

1. Close the ENG 1 FUEL PRESS circuit breaker. Check that the engine 1 FUEL PRESS indicator reads 0 ±3 PSIG.

NOTE

A pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 150 PSIG (1034 kPa) must be used when you do the following checks.

2. Disconnect the pressure input line from the engine 1 fuel pressure transmitter on the engine No. 1. Attach the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) and apply pressure to the transmitter input line. Do not exceed 57 PSIG (393 kPa). Check that the engine 1 FUEL PRESS indicator reads 50 PSIG when the input pressure is 50 ±7 PSIG (345 ±48 kPa).

3. Disconnect the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F). Connect the pressure line to the engine 1 fuel pressure transmitter.

4. Close the ENG 2 FUEL PRESS circuit breaker. Repeat Step 1. through Step 3. for the engine 2 fuel pressure transmitter on the engine No. 2.

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Attitude Director Indicator Systems - Description

Applicable: SN: 33001-33213 or 36001-36247 or 36248-36999:

The attitude director indicators display flight attitude relative to the earth. The pilot and copilot attitude indicators use data from their respective three-axis reference sensor and rate gyro or the AHRS.

The applicable three-axis reference sensor or AHRS provide gyroscopic pitch, and roll attitude data to the pilot and copilot attitude director indicator.

Applicable: SN: 33001-33213 or 36001-36247 or 36248-36999:

The pilot and copilot rate gyros or AHRS provide data to the rate-of-turn pointers that, together with inclinometers, function as turn and slip portions of the respective indicators.

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Attitude Director Indicator Systems - Operational Check

Procedure

1. Close the TURN SLIP PLT circuit breaker. Check that the associated rate gyro begins operating. Check that the RT flag, FD flag, and horizontal and vertical command bars move out of sight and the ATT flag remains in view.
2. Close the PILOT ATT SYS circuit breaker. Check that the ATT flag moves out of view, attitude sphere erects to within 1° in both pitch and roll within 3 minutes, and operation is stable.
3. Rotate the roll trim knob on the pilot attitude indicator to the stop in a clockwise direction. Check that the roll indication is 15° minimum right bank.
4. Rotate the roll trim knob on the pilot attitude indicator to the stop in a counterclockwise direction. Check that the roll indication is 15° minimum left bank.
5. Rotate the pitch trim knob to the stop in a clockwise direction. Check that the pitch indication is 10° minimum dive.
6. Rotate the pitch trim knob to the stop in a counterclockwise direction. Check that the pitch indication is 15° minimum climb.
7. Open the TURN SLIP PLT and PILOT ATT SYS circuit breaker. Check that all the flags come into view.
8. Repeat Step 1. through Step 7. for the copilot attitude director indicator system.

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Rate Gyro - Maintenance

Procedure

1.

NOTE

The maintenance procedures are applicable for pilot and copilot rate-of-turn gyros; refer 412-A-96-11-00-07A-520A-A / 01961, 412-A-96-11-00-07A-280A-A / 01962, 412-A-96-11-00-07A-251A-A / 01963, 412-A-96-11-00-07A-685A-A / 01964 and 412-A-96-11-00-07A-720A-A / 01965, therefore, typical procedures are given.

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Rate Gyro - Removal

Procedure

NOTE

The following removal procedures are applicable for pilot and copilot rate-of-turn gyros.

1. Loosen and disengage the electrical connector from unit.

2. Protect the ends of the connector with caps or covers.

3. Remove the mounting screws and the washers and carefully remove the unit from the shelf.

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Rate Gyro - Inspection

Procedure

NOTE

The following inspection procedures are applicable for pilot and copilot rate-of-turn gyros.

1. Inspect the unit for a cracked or damaged case.

2. Inspect the unit for proper bonding and mounting.

3. Inspect the unit for broken connector pins or cracked connector inserts.

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Rate Gyro - Cleaning

Procedure

NOTE

The following cleaning procedures are applicable for pilot and copilot rate-of-turn gyros.

1. Remove grease, fungus, and ground-in dirt from the case with a soft cloth dampened with approved cleaning compound.

2. Remove dirt from the electrical connectors with a bristle brush and remove moisture with a dry, clean cloth.

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Rate Gyro - Repair or Replacement

Procedure

NOTE

The following repair or replacement procedures are applicable for pilot and copilot rate-of-turn gyros.

1. Replace the unit if the case is cracked or damaged.

2. Tighten, replace, or adjust the mounting hardware as necessary.

3. Repair or replace a damaged airframe wiring connector.

4. Replace a defective or inoperative unit.

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Rate Gyro - Installation

Procedure

NOTE

The following installation procedures are applicable for pilot and copilot rate-of-turn gyros.

1. Carefully position the unit in place on the shelf and install the mounting screws and washers.

2. Remove the protective caps or covers from the ends of the electrical connectors and connect to the unit.

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Triple Tachometer Indicator Systems - Description

Each of the power turbine and rotor tach generators sends a frequency signal, in percent of RPM, to each of the triple tach indicators.

NOTE

Triple tachometer and gas producer tachometer indicator systems are AC self-generating and do not require AC power from the helicopter inverters. In the event of loss of DC or AC electrical power from the helicopter bus, the AC self-generating equipment will continue to operate.

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Triple Tachometer Indicator Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Engine Test Set	Howell H337	1	or Equivalent

Procedure

1. Disconnect the connector (P35/1G7P1) from the main rotor tach generator and connect to the matching connector on the test set. Energize the tach generator in the test set. Check that the rotor tach pointer on the pilot triple tach indicator (and the copilot triple tach indicator, if installed) reads upscale and indicates the approximate speed at which the test set tach generator turns.

2. Disconnect the connector from the test set and reconnect to the rotor tach generator. Check that the connection is tight and properly mated.

3. Disconnect the connector (P111/1G1P1) from the engine No. 1 power turbine tach generator and connect to the test set. Energize the tach generator in the test set. Check that the power turbine No. 1 pointer on the pilot triple tach indicator (and the copilot indicator, if installed) reads upscale and indicates the approximate speed at which the test set tach generator turns.

4. Disconnect the connector from the test set and reconnect to the power turbine tach generator on the engine No. 1. Check that the connection is tight and properly mated.

5. Disconnect the connector (P112/1G2P1) from the engine No. 2 power turbine tach generator and connect to the test set. Energize the tach generator in the test set. Check that the power turbine No. 2 pointer on the pilot triple tach indicator (and the copilot indicator, if installed) reads upscale and indicates the approximate speed at which the test set tach generator turns.

6. Disconnect the connector from the test set and reconnect to the engine No. 2 power turbine tach generator. Check that the connection is tight and properly mated.

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Gas Producer Tachometer Indicator Systems - Description

The gas producer tach generator on the engine No. 1 sends a frequency signal to the No. 1 engine gas producer indicator proportional to the engine No. 1 gas producer speed. The gas producer tach generator on the engine No. 2 sends a frequency signal to the No. 2 engine gas producer indicator proportional to the engine No. 2 gas producer speed. The gas producer speed indication is given in percentage of RPM.

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Gas Producer Tachometer Indicator Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Engine Test Set	Howell H337	1	or Equivalent

Procedure

1. Disconnect the connector (P89/1G3P1) from the engine No. 1 gas producer tach generator and connect to the test set. Energize the tach generator in the test set. Check that the engine No. 1 gas producer tach indicator reads upscale and indicates the approximate speed at which the test set tach generator turns.

2. Disconnect the connector from the test set and reconnect to the engine No. 1 gas producer tach generator. Check that the connection is tight and properly mated.

3. Disconnect the connector (P90/1G4P1) from the engine No. 2 gas producer tach generator and connect to the test set. Energize the test set tach generator. Check that the engine No. 2 gas producer tach indicator reads upscale and indicates the approximate speed at which the test set tach generator turns.

4. Disconnect the connector from the test set and reconnect to the engine No. 2 gas producer tach generator. Check that the connection is tight and properly mated.

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Gas Producer Tachometer Indicator Systems - Maintenance

Procedure

1. For maintenance of the gas producer generators, refer to (412-A-71-01-00-01A-042A-A / 01663).

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Fuel Quantity Indicator System - Description

NOTE

SN: 33108-33213 or 36001-36999:

Refer to 412-A-96-12-00-11A-042A-A / 02005.

AC power is supplied to the fuel quantity indicating system through the FUEL QTY circuit breaker on the overhead console. Each fuel transmitter sends a signal to the fuel quantity indicator where it is summed and displayed in total pounds of fuel. A switch is provided, on the pilot instrument panel, for checking fuel quantity in the left or right fuel cells.

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Tank Unit And System Wiring Installation - Insulation Resistance Test

Procedure

1. Refer to the Simmonds Precision Maintenance Manual for Installation No. 380807-001, -002, -003, Chapter 28.

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Fuel Quantity Indicator System - Calibration Preferred Method (No Fuel In Tanks)

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Engine Test Set	Simmonds No. 387991-0031	1	or Equivalent

Procedure

NOTE

Make sure that the indicator remains connected when you do the following procedure.

1. Make sure that the helicopter is defueled.

2. Remove the fuel quantity indicator from the instrument panel.

3. Close the FUEL QTY circuit breaker.

4. Adjust the EMPTY control on the back of the indicator, if required, until the indicator reads exactly zero.

5. Verify that the indicator pointer is stable (does not oscillate).

6. Open the FUEL QTY circuit breaker.

7. Connect the field calibration unit to the system with the appropriate cabling (Figure 1 and Figure 2).

8. Set the FUEL QTY SEL switch to TOTAL.

9. Set the tank unit capacitance decade on the calibration unit for the values shown in Figure 1 and Figure 2.

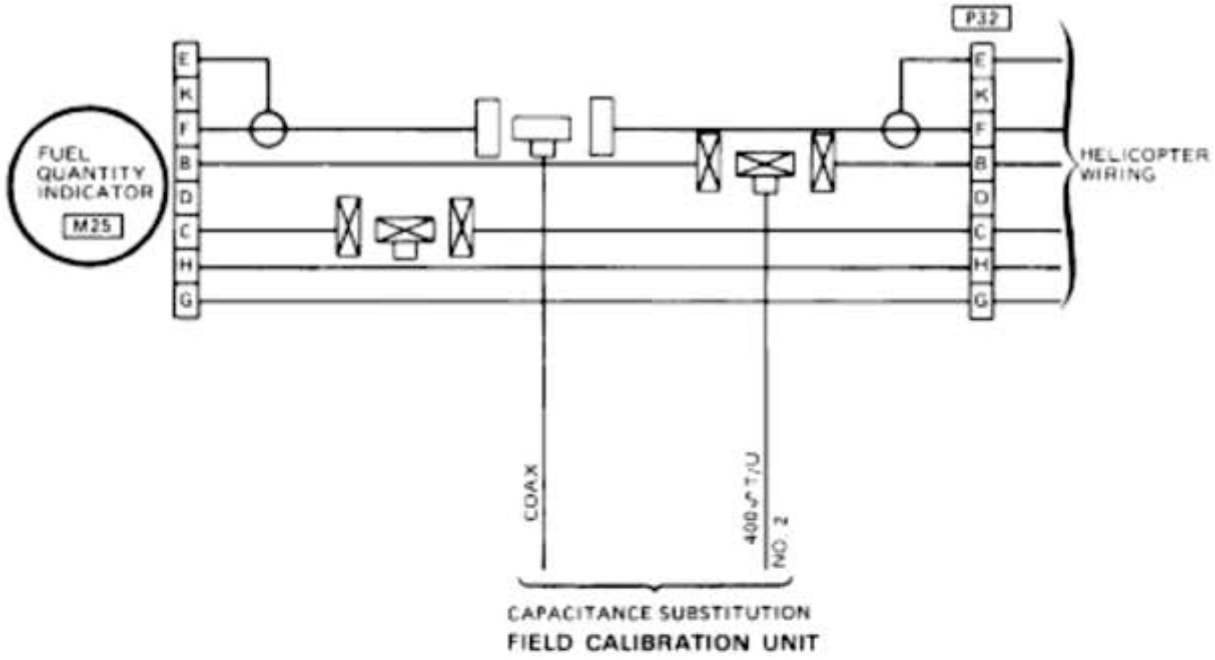
10. Adjust the FULL control on the back of the indicator until the indicator reads the correct full indication, and is stable.

11. Disconnect the Field Calibration Unit.

12. Return the fuel system to its original configuration.

13. Verify that all the connector are secure and tight.

14. Verify that the indicator reads zero.

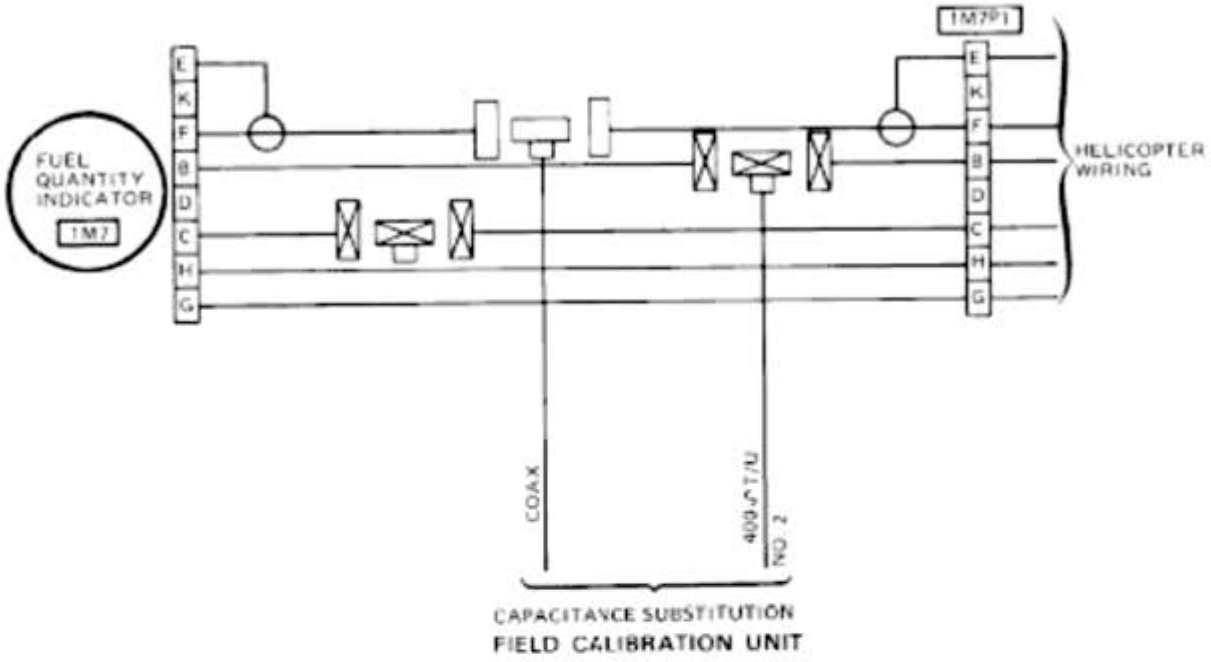


SYSTEM EMPTY AND FULL CAPACITANCE VALUES

SYSTEM	INDICATOR	CAPACITANCE (pF)			FULL IND READING
		EMPTY	ADDED	FULL	
BASIC	393008-040	112.3	136.6	248.9	1575
KIT No. 1	393008-046	112.3	136.6	248.9	1900
KIT No. 2	393008-047	207.8	208.3	416.1	2500

ICN-412-A-961100-B-97499-00001-A-002-01

Figure 1. Test Setup for Fuel Quantity Indicator Calibration (S/N 33001 through 33100) - Dry Tanks (Sheet 1 of 1)



SYSTEM EMPTY AND FULL CAPACITANCE VALUES

SYSTEM	INDICATOR	CAPACITANCE (pF)			FULL IND READING
		EMPTY	ADDED	FULL	
BASIC	393008-040	112.3	136.6	248.9	1575
KIT No. 1	393008-046	112.3	136.6	248.9	1900
KIT No. 2	393008-047	207.8	208.3	416.1	2500

ICN-412-A-961100-B-97499-00002-A-002-01

Figure 2. Test Setup for Fuel Quantity Indicator Calibration (S/N 33101 through 33107) - Dry Tanks (Sheet 1 of 1)

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Fuel Quantity Indicator System - Calibration Alternate Method (Fuel In Tanks)

Procedure

1. Refer to the Simmonds Precision Maintenance Manual for Installation No. 380807-001, -002, -003, Chapter 28.

NOTE

For information covering the auxiliary fuel tanks, refer to BHT-412-SI-4.

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Fuel Quantity Coupler CP1/1CP1 - Maintenance

Procedure

1. Disconnect and protect the connectors.
2. Remove the mounting hardware.
3. Remove the coupler from the compartment.

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Fuel Quantity Coupler CP1/1CP1 - Inspection

Procedure

1. Inspect the coupler case for cracks, dents, and corrosion.
2. Inspect the connector for loose, bent, or missing contacts.
3. Inspect the connector for dirt, grease, or corrosion.

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Fuel Quantity Coupler CP1/1CP1 - Repair or Replacement

Procedure

1. Replace the coupler if the inspection requirements are not met.

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Fuel Quantity Coupler CP1/1CP1 - Installation

Procedure

1. Install the coupler in the compartment and secure with the hardware.
2. Remove the protection from the connectors.
3. Install the connector on the coupler.

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DC Indicator Systems - Description

The DC indicator systems include the mast torque measurement system (Helicopters S/N 36020 through 36999, and helicopters modified by 412-570-001-103 or Post BHT-412-SI-74, Service Installation Instructions Manual (412SP to 412HP Upgrade)), engine torque pressure indicating system (412-A-98-00-00-46A-051A-A / 02506), standby attitude indicator (412-A-98-00-00-48B-051A-A / 02608), engine oil temperature indicator, transmission oil temperature indicator, combining gearbox oil temperature indicator (412-A-98-00-00-38A-051A-A / 02498, or 412-A-98-00-00-39A-051A-A / 02499), engine hourmeter (412-A-98-00-00-50A-051A-A / 02510, or 412-A-98-00-00-51A-051A-A / 02511, or 412-A-98-00-00-52A-051A-A / 02512), turn and slip indicator, interturbine temperature indicator (412-A-98-00-00-42A-051A-A / 02502, or 412-A-98-00-00-43A-051A-A / 02503), hydraulic oil temperature indicator systems (412-A-98-00-00-48A-051A-A / 02508, or 412-A-98-00-00-49A-051A-A / 02509), and fuel quantity indicator system (412-A-98-00-00-57A-051A-A / 02517, or 412-A-98-00-00-58A-051A-A / 02518, or 412-A-98-00-00-59A-051A-A / 02519). Refer for the appropriate system wiring diagram.

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Mast Torque Measurement System (412 Classic/SP to 412 HP Upgrade) - Description

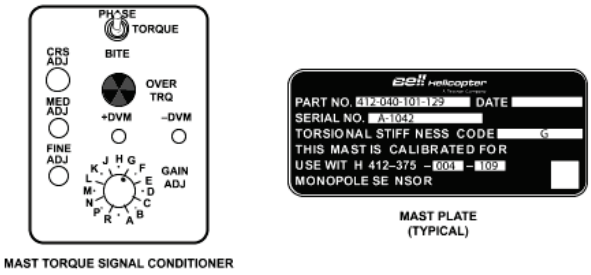
The mast torque measurement is accomplished by a sensor located on the bottom of the transmission and extending up into the bottom of the mast. The sensor consists of two monopole pickups. These pickups are located inside two sets of mast splines to detect the torsional displacement between them. The bottom sensor is solid in the mast and is the reference sensor. The upper pickup is located on the lower end of the torque tube and is the signal sensor. The signal sensor is connected to the top of the mast and is displaced in relation to the reference sensor when the mast is under a torque condition. During manufacture of the mast, the twist rate of the mast is determined, and the mast data plate is stamped with a letter from A to R indicating the stiffness of the mast. The electrical signals from the reference and signal pickups are in phase at zero torque and 145.4° out of phase at 100% torque, which corresponds to 4.8° of mechanical twist for an H graded mast. The torque signal conditioner has a cover plate on one end covering a 15 position switch marked A through R. This switch must be set to the letter corresponding to the letter on the mast data plate. The mast torque signal conditioner then provides an output to the triple torque indicators. All torque is indicated in percentages. Engine 1 and engine 2 torque is indicated by separate meter movements on the right side of the indicator and mast torque is indicated on the left side.

OVER TORQ warning lights for the pilot and copilot are mounted on the instrument panel. A self-test function is initiated by pressing the pilot or copilot OVER TORQ warning light or selecting BITE on the signal conditioner (Figure 1). The self-test function will cause the mast torque output signals to be 105 ±1.0% (as indicated on the triple torque indicators) and over torque output signal to illuminate the OVER TORQ waning lights.

A latching over torque indicator is provided to indicate when the output torque exceeds a preset torque level. The indicator trips within 0.5 seconds of the over torque occurrence. The indicator, located beneath the cover plate, is reset mechanically. The trip point is 108 ±0.5% for -105, and 110 ± -0.5% for -107 and -109. The indicator is visible through the cover plate.

A test function is provided to determine the phase difference during a zero torque calibration procedure. The function is initiated by selecting PHASE on the switch on the signal conditioner control panel. A DC voltage (phase voltage) is measurable between the +DVM and -DVM terminals.

A temperature sensor mounted on the transmission top case provides temperature compensation above and below 77°F (25°C) for changes in mast elasticity.



ICN-412-A-961200-B-97499-00001-A-001-01

Figure 1. Signal Conditioner Control Panel (Sheet 1 of 1)

Mast Torque - Indication Check

Procedure

1. Open the MAST TORQUE circuit breaker (1CB50).
2. Disconnect the connector (1A7P1) from the signal conditioner (1A7).

NOTE

The resistance measurement in Step 3. depends on the ambient air temperature. For a temperature of 77°F (25°C), the resistance should be 99 ±2 ohms.

3. Select the 200 ohm resistance scale on the multimeter.
4. Connect the leads to pins F and G of the connector (1A7P1).
5. The multimeter should indicate 80 to 110 ohms for the ambient temperature of -22°F to 131°F (-30°C to 55°C). Refer to Table 1 for additional values.
6. Connect the connector (1A7P1) to the signal conditioner (1A7).
7. Close the MAST TORQUE circuit breaker (1CB50).
8. Set the BATTERY BUS 2 switch to ON.

Table 1. Mast Torque Temperature Sensor Resistance Values

TEMPERATURE (°C)	RESISTANCE (OHMS)
-40	77.39 ±0.40
-30	80.56 ±0.40
-20	83.77 ±0.40
-10	87.04 ±0.40
0	90.38 ±0.40
10	93.80 ±0.40
20	97.31 ±0.40
30	100.91 ±0.40
40	104.60 ±0.40
50	108.39 ±0.40

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Mast Torque System - Operational Check

Procedure

1. Press the pilot OVER TORQ warning light and hold for approximately 10 seconds to initiate the mast torque systems Built-In-Test (BIT) function.
2. Verify that the pilot and copilot torque indicators read 105 ±1%.
3. Verify that the pilots and copilots OVER TORQ warning lights illuminate.
4. Repeat Step 1. through Step 3. except press the copilot OVER TORQ warning light.

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Mast Torque System - Calibration Check

Procedure

1. A mast torque system zero calibration is required for the following events (412-A-96-12-00-01A-273B-A / 01984):

- Replacement of the monopole torque sensor
- Replacement of the mast torque signal conditioner
- Replacement of the mast torque filter
- Replacement of the mast assembly or mast torque tube assembly.
- Alignment of mast torque tube assembly to mast assembly has changed

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Mast Torque System - Zero Torque Calibration

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Digital Multimeter	Fluke Model 85	1	or Equivalent

Procedure

1. Remove the blades from the main rotor hub (412-A-62-10-00-00A-520A-A / 00935).

2. Remove the cover plate from the signal conditioner control panel (Figure 1).

3. Connect the digital multimeter (Fluke Model 85) to the signal conditioner terminals marked +DVM and -DVM.

4. Select a scale on the digital multimeter (Fluke Model 85) of approximately 20 VDC.

5. Make sure that the GAIN adjustment rotary switch on the signal conditioner is set to the letter that corresponds to the torsional stiffness code letter stamped on the plate located on the mast.

6. Start either engine in accordance with the BHT-412-FM, Flight Manual.



WITHOUT MAIN ROTOR BLADES INSTALLED, THE ENGINE WILL REACT QUICKER TO THROTTLE MOVEMENTS. TO AVOID POSSIBLE OVERSPEED, SET RPM INCR/DECR SWITCH TO FULL DECR PRIOR TO INCREASING RPM AND MOVE THROTTLES SLOWLY.

7. Run the engine at 95 to 100% N_R.

8. Verify the standpipe alignment at zero torque as follows:

NOTE

The following Step 8.1. through Step 8.4. do not apply to -105 signal conditioners.

8.1. Rotate the coarse adjustment multiposition switch to the full clockwise position.

8.2. Hold the toggle switch in the PHASE position.

8.3. Measure and record the voltage the between the -DVM and +DVM terminals.

8.4. Verify that the voltage is between -2.43 and -3.68 VDC.

8.5. Release the toggle switch from the PHASE position.

8.6. If the voltage is not within range, the standpipe requires realignment.

9. Calibrate the signal conditioner to obtain the proper signal at zero torque (0.000 ±0.008 volts across -DVM and +DVM) as follows:



EXCESSIVE PRESSURE ON POTENTIOMETER ADJUST SHAFTS WILL CAUSE DAMAGE.

9.1. Make sure that the CRS ADJ switch is still in the fully clockwise position and then move two positions counterclockwise.

9.2. Measure and record the voltage between the -DVM and +DVM terminals.

NOTE

If the voltage does not change when the coarse adjustment switch is repositioned, stop the calibration and correct the wiring or the failed system component.

9.3. The voltage between the -DVM and +DVM terminal should be -0.6 to +0.6 VDC. Move the CRS ADJ switch clockwise or counterclockwise as needed to get the required voltage.

9.4. Rotate the MED ADJ counterclockwise or clockwise until the voltage between -DVM and +DVM is 0.000 ±0.030 volts.

9.5. Rotate the FINE ADJ counterclockwise or clockwise until the voltage between -DVM and +DVM is 0.000 ±0.008 volts.

NOTE

Better accuracy can be maintained by selecting a lower DC voltage range on the DVM after the signal has been reduced below 0.1 VDC.

It is normal for voltage at the DVM terminals to oscillate by as much as ±0.008 VDC. The average should be set as close to zero as possible.

9.6. If zero cannot be achieved, reposition the MED ADJ and FINE ADJ potentiometer at the center of rotation range as follows:

9.6.1. Rotate both potentiometer shafts 30 turns counterclockwise.

9.6.2. Rotate both potentiometer shafts 12 turns clockwise.

9.7. Repeat Step 9.2. through Step 9.6. as many times as required to get the results in Step 8.4.. Each time, reposition either the CRS ADJ switch one position either clockwise or counterclockwise from the previous position.



DO NOT ENGAGE ROTOR BRAKE WITHOUT MAIN ROTOR BLADES INSTALLED.

10. Shut down the engine.

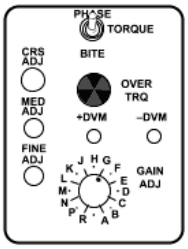
11. Reset the overtorque indicator, if required.

12. Install the cover plate on the signal conditioner.

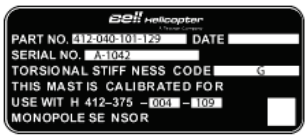
13. Install the blades on the main rotor hub (412-A-62-10-00-00A-720A-A / 00946).

14. A mast torque system flight check is necessary for the following events (412-A-96-12-00-01A-280B-A / 01985):

- Replacement of the monopole torque sensor
- Replacement of the mast torque signal conditioner
- Replacement of the mast assembly or mast torque tube assembly
- Alignment of mast torque tube assembly to mast assembly has changed.



MAST TORQUE SIGNAL CONDITIONER



MAST PLATE
(TYPICAL)

ICN-412-A-961200-B-97499-00001-A-001-01

Figure 1. Signal Conditioner Control Panel (Sheet 1 of 1)

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Mast Torque System - Flight Check

Procedure

1. Load the helicopter as near to maximum gross weight as practical in a symmetrical manner, which ensures a centered CG.

2. Do a ground run torque check as follows:

2.1. Start both engines in accordance with the appropriate Flight Manual.

2.2. Set NR to 100%.

2.3. Center the cyclic and directional controls.

2.4. Set the collective control so the rotor is at flat pitch.

2.5. The pilot and the copilot torque indicators should each indicate approximately 10% engine torque and 20% mast torque.

3. Do a normal take off and power climb within the following conditions:

- Less than 8000 feet altitude
- Straight attitude (cyclic centered) with no directional pedal movement and autopilot off
- Speed of 85 to 90 knots
- Pull sufficient collective to obtain engine torque readings greater than 45%

4. Record the engines and mast torques indicated on the pilot and the copilot torque indicators.

5. Readings from the pilot and the copilot torque indicators should be within 1% of each other.

6. The sum of the engine No. 1 and the engine No. 2 torque readings should be 2 to 9% higher than the mast torque reading.

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Engine Torque Pressure Indicating System - Description

The engine torque pressure indicator system simultaneously displays torque output of both engines on the right side of the pilot and the copilot, if installed, triple torque indicator. Pointer No. 1 indicates the engine No. 1 torque pressure transmitter output and pointer No. 2 indicates the engine No. 2 torque pressure transmitter output. The mast torque is indicated on the left side. Indications are in percent. DC power is supplied through the ENGINE NO 1 TORQUE and ENGINE NO 2 TORQUE circuit breakers. The pilot and the copilot indicators are independent but function identically.

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Engine Torque Pressure Indicating System - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Deadweight Tester	2311F	1	or Equivalent
Pressure Gauge	302-00102	1	

Procedure

NOTE

Refer to (412-A-76-10-00-00A-271B-A / 01728) for engine torque transmitter adjustment and functional check.

1. Open the engine cowling and reduction gearbox access doors.

2. Locate the combining gearbox data plate.

3. Record the LH valves letter code and the LH alpha value on the data plate (Figure 1).

4. Disconnect the hose at the engine No. 1 torque transmitter (1MT1) pressure port (Figure 2).



MAKE SURE PRESSURE TESTER IS SERVICED WITH SAME OIL AS USED IN COMBINING GEARBOX.

5. Attach the pressure tester to the torque transmitter pressure port.

6. Close the ENG 1 TORQUE circuit breaker.

7. Set the BATTERY BUS 1 and the BATTERY BUS 2 switches to ON.

8. Set the INV 1 and the INV 2 switches to ON.

NOTE

When you do the following, the pressure tester and gauge must be at same height as reduction gearbox torque meter (approximate level of oil cooler blower center line).

9. Apply a test pressure of 35.4 ±0.25 PSIG (244 ±1.7 kPa) to the transmitter input.

10. Record the reading on the pilot engine No. 1 torque indicator.

11. Calculate the engine torque % corresponding to the test pressure 35.4 PSIG (244 kPa) with following equation:

EQUATION:

Alpha value recorded in Step 3.÷ 2 = Engine torque percent.

EXAMPLE CALCULATION:

Engine No. 1 alpha value of 86.9 ÷ 2 = 43.45, which rounds off to 43.5 engine torque percent.

Engine No. 2 alpha value of 85.4 ÷ 2 = 42.7 engine torque percent.

12. The pilot engine No. 1 torque indicator should match the engine torque % calculated in Step 11..

13. The copilot torque indicator should read within 1% of pilot torque indicator.

14. If the torque indication does not match the calculated value, refer to (412-A-76-10-00-00A-271B-A / 01728) for the engine torque transmitter adjustment and functional check.

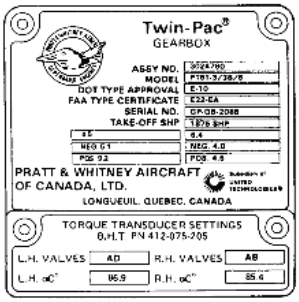
15. The engine No. 2 torque indicator for both the pilot and the copilot should indicate zero torque.

16. Connect the hose to the engine No. 1 torque transmitter pressure port.

17. Set the INV 1 and the INV 2 switches to OFF.

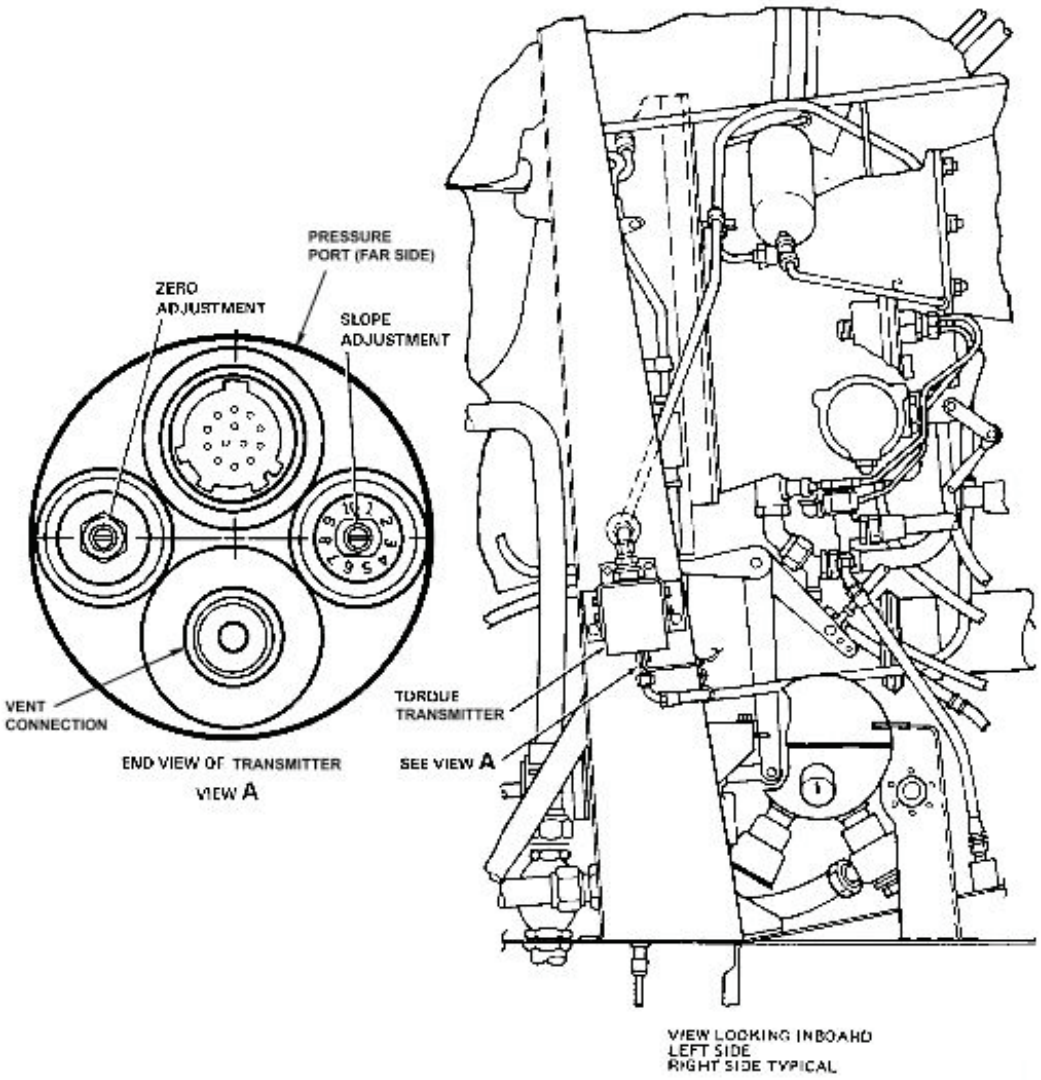
18. Set the BATTERY BUS 1 and the BATTERY BUS 2 switches to OFF.

19. Repeat Step 3. through Step 18. for the engine No. 2, except the references to engine No. 1 should be replaced with engine No. 2 and references to LH should be replaced with RH.



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Figure 1. Engine Combining Gearbox Data Plate (Sheet 1 of 1)



ICN-412-A-961200-B-97499-00002-A-001-01

Figure 2. Torque Transmitter (Sheet 1 of 1)

Standby Attitude Indicator - Description

The standby attitude indicator system is designed to provide attitude indication in the event of a No. 2 emergency bus being disabled. This system includes an independent battery for a power supply.



SET STBY ATTD TEST SWITCH TO OFF WHEN HELICOPTER BUSES ARE NOT ENERGIZED, THIS WILL PREVENT THE STANDBY ATTITUDE INDICATOR BATTERY CHARGE FROM BEING DEPLETED. THE STANDBY ATTITUDE INDICATOR BATTERY CHARGE WILL LAST FOR 1.5 HOURS WHILE OPERATING AS AN INDEPENDENT POWER SOURCE.



DO NOT ALLOW THE STANDBY ATTITUDE INDICATOR BATTERY CHARGE TO DROP BELOW 18 VOLTS. THIS COULD CAUSE CELL REVERSAL TO OCCUR, WHICH MAY PERMANENTLY DAMAGE THE BATTERY.

With the STBY ATTD TEST switch in the OFF position, the standby attitude indicator is inoperative for kit 412-706-117-XXX (BHT-412-SI-11) and the battery is not being charged. For kit 412-706-121-XXX (BHT-412-SI-35), the battery is being kept in a fully charged state if the No. 2 emergency bus is energized. With the switch in the TEST position, the indicator is disconnected from the No. 2 emergency bus and is being powered by the battery. With the switch in the ON position, the indicator is operating normally and the battery is being charged.

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Standby Attitude Indicator - Operational Check

Procedure

1. Connect the external power and energize the No. 2 emergency bus.

2. Close the STDY ATTD circuit breaker.

3. Set the STBY ATTD TEST switch to ON.

4. Verify that the standby attitude indicator is operating.

5. Set the STBY ATTD TEST switch to TEST.

6. Open the STDY ATTD circuit breaker.

7. Verify that the standby attitude indicator is operating.

8. Set the STBY ATTD TEST switch to OFF.

9. Verify that the standby attitude indicator is no longer operating.

10. De-energize the No. 2 emergency bus.

11. Disconnect the external power.

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Standby Battery - Maintenance/Testing

Procedure

1. Battery Removal.

1.1. Make sure that all electrical power is off.

1.2. Gain access to the standby battery (Figure 1) on the underside of the upper avionics shelf in the nose compartment.

1.3. Tag and remove the wires from the battery.

1.4. Remove the battery from the shelf and retain the hardware.

2. Battery Installation.

2.1. Install the battery on the underside of the upper avionics shelf in the nose compartment.

2.2. Connect the wires to the battery and remove the tags.

2.3. Do an operational check of the standby indicator system.

3. Annual Battery Test/Troubleshooting.

3.1. Remove the battery.

3.2. Electrical leakage check:

3.2.1. *Determine the extent of electrical leakage between the battery case and the cells by a breakdown of the insulation. Leakage greater than 25 milliamperes between the battery case and either the positive or negative terminals of the battery is cause to reject the battery as unserviceable.*

NOTE

Do not use the voltage reading between the battery case and the terminals as a criterion for rejection; current flow is the determining factor.

3.2.2. Set the range selector of a multimeter to the 500 milliampere range or higher. Place the meter positive lead on the positive terminal of the battery and momentarily touch the meter negative lead to any exposed metal on the battery case or the screws used to secure the cover.

3.2.3. If the reading is within the meter limits, connect the negative lead to the battery case. Decrease the range until the current flow, if any exists, is readable. Record this current value.

3.2.4. Repeat the previous steps, connecting the meter negative lead to the negative terminal of the battery and the positive lead to any exposed metal on the battery case.

3.2.5. *If either of the previous current measurements exceeds 25 milliamperes, reject the battery.*

3.3. Do a capacity test of the battery as follows:

3.3.1. Charge the battery for 16 hours at 190 milliamperes.



DURING THE FOLLOWING TEST, MONITOR BATTERY CLOSELY; DO NOT ALLOW BATTERY VOLTAGE TO DROP BELOW 17.5 VOLTS. IF VOLTAGE DROPS BELOW 17.5 VOLTS, CELL REVERSAL MAY OCCUR AND PERMANENTLY DAMAGE THE BATTERY.

NOTE

During the following test, verify that the 9 cell voltage remains 1/2 of the 18 cell voltage ±0.5 volts.

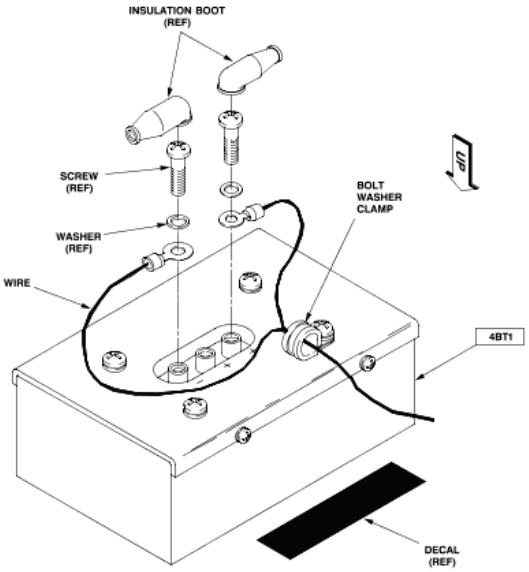
3.3.2. With a 5-amp resistive load, record the time required to discharge the battery to 18 volts.

3.3.3. Multiply the minutes required to discharge the battery to 18 volts by 5 (amp load); then divide by 60. The resultant is the amp-hour capacity of the battery.

3.3.4. *If the amp-hour capacity is 1.2 or greater, the battery is serviceable. Repeat* Step 3.3.1.

3.3.5. *If the amp-hour capacity is less than 1.2, repeat* Step 3.3.1. *through* Step 3.3.3. *and increase the charging time to 24 hours.*

3.3.6. *If the amp-hour capacity is still less than 1.2, the battery is not serviceable and shall be replaced.*



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Figure 1. Standby Attitude Indicator Battery (Sheet 1 of 1)

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Engine Oil Temperature Indicator System - Description

DC power is supplied through the ENGINE NO. 1 OIL TEMP circuit breaker and the ENGINE NO. 2 OIL TEMP circuit breaker. The engine 1 oil temperature bulb controls the engine No. 1 ENGINE OIL temperature and pressure indicator. The engine 2 oil temperature bulb controls the engine No. 2 ENGINE OIL temperature and pressure indicator. Each indicator also displays its respective engine oil pressure.

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Engine Oil Temperature Indicator System - Operational Check

Procedure

NOTE

Ambient temperature checks are not valid when ambient air is below -5.0°C (23°F).

1. Close the ENGINE NO. 1 OIL TEMP circuit breaker.

2. Check that the engine No. 1 ENGINE OIL temperature and pressure indicator indicates the approximate ambient temperature with the engine cooled to the ambient temperature.

3. Close the ENGINE NO. 2 OIL TEMP circuit breaker.

4. Verify that the engine No. 2 ENGINE OIL temperature and pressure indicator indicates the approximate ambient temperature with the engine cooled to the ambient temperature.

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Transmission Oil Temperature Indicator System - Description

DC power is supplied through the XMSN OIL TEMP circuit breaker. The transmission oil temperature and pressure bulb controls the transmission oil temperature indicator. The indicator also displays its respective transmission oil pressure.

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Transmission Oil Temperature Indicator System - Operational Check

Procedure

NOTE

Ambient temperature checks are not valid when ambient air is below -5.0°C (23°F).

1. Close the XMSN OIL TEMP circuit breaker.

2. Verify that the XMSN OIL temperature and pressure indicator indicates the approximate ambient temperature with the transmission cooled to the ambient temperature.

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Combining Gearbox (C Box) Oil Temperature Indicator System - Description

DC power is supplied through the C BOX OIL TEMP circuit breaker. The combining gearbox oil temperature bulb controls the combining gearbox oil temperature and pressure indicator. The indicator also displays its respective gearbox oil pressure.

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Combining Gearbox (C Box) Oil Temperature Indicator System - Operational Check

Procedure

NOTE

Ambient temperature checks are not valid when ambient air is below -5.0°C (23°F).

1. Close the C BOX OIL TEMP circuit breaker.

2. Verify that the GEAR BOX oil temperature and pressure indicator indicates the approximate ambient temperature with the combining gearbox cooled to the ambient temperature.

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Engine Hourmeter System - Description

With the HOURMETER, ENG 1 RPM, and ENG 2 RPM circuit breakers closed (rotor RPM indicator above 92.6 ±2%), the hourmeter starts to register by receiving 28 VDC power from the DC bus through the HOURMETER circuit breaker, de-energized hourmeter relay, C BOX oil pressure switch, engine No. 2 aft compartment connector, and to ground. When the engine speed decreases (rotor RPM below 92%), the hourmeter relay receives 28 VDC signal from the RPM limit detector thus energizing the relay and stopping the hourmeter.

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Engine Hourmeter System - Operational Check

Procedure

1. Disconnect the connector (8S8P1) from the C box oil pressure switch.
2. *Open the HOURMETER circuit breaker.*
3. Install a temporary short between pins B and C of the connector (8S8P1).
4. Close the HOURMETER circuit breaker.
5. *Make sure that the hourmeter operates.*
6. Open the HOURMETER circuit breaker.
7. Remove the temporary short installed in Step 3..
8. Reconnect the connector (8S8P1) to the C box oil pressure switch.
9. *Make sure that the connector is tight and engaged correctly.*

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Turn and Slip (Part of Attitude Director Indicator) - Description

The turn and slip indicator and the rate gyro (S/N 33001 through 33213 and 36001 through 36247) are DC powered. The pilot and copilot rate gyros and AHRS (S/N 36248 through 36999) provide data to the rate-of-turn pointers, which, together with the inclinometers, function as turn and slip portions of the respective ADIs.

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Turn and Slip - Operational Check

Procedure

1. For the operational check of the turn and slip function of ADI, refer to 412-A-96-11-00-07A-320A-A / 01959.

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Interturbine Temperature (ITT) Indicator Systems - Description

The voltage generated by the thermocouples located in the interturbine area of the engine No. 1 is transmitted through the internal engine harness, bias compensator, to the engine No. 1 ITT indicator. The engine No. 2 interturbine temperature voltage signal is transmitted through the internal engine harness, bias compensator, to the engine No. 2 ITT indicator.

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Interturbine Temperature (ITT) Indicator Systems - Functional Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Turbine Temperature Indicating Test Set	Barfield TT1000A	1	or Equivalent

Procedure

NOTE

The following procedure is for the connector (1Z11P2) for engine 1. The procedure is the same for the connector (1Z12P2) for engine 2.

NOTE

The tolerances stated do not include the test equipment tolerances.

1. Disconnect the connector (1Z11P2) from the ITT compensator unit for engine 1.

2. Disconnect the alumel and chromel leads from the terminals on the gas generator case of each engine.

NOTE

The smaller diameter lead of the gas generator is chromel, and the larger diameter lead is alumel.

3. Connect the turbine temperature indicating test set (Barfield TT1000A) to the leads disconnected from the terminals on the gas generator case on engine 1.

4. Check the connections for proper polarity.

5. Close the ENGINE NO 1 ITT COMP circuit breaker.

6. Adjust the test set for an input of 810°C and verify that the ITT indicator reads 810°C ±2.5°C.

7. Record the ITT TRIM value given in degrees Celsius from the PERFORMANCE REFERENCE DATA plate on the power section of the engine.

8. Subtract the trim value from 810°C.

9. Connect the ITT compensator 1 unit connector.

10. Verify that the connection is tight and secure.

11. Verify that the ITT indicator indication is ±5°C of the value calculated in Step 7. and Step 8.. Refer to (412-A-76-10-00-00A-280A-A / 01726) for the adjustment procedure.

12. Check the insulation resistance of the internal thermocouple harness with the test set.

13. The resistance to ground from either the chromel or alumel terminal shall be not less than 5000 ohms.

14. Measure the loop resistance between the chromel and alumel terminals on the terminal block.

15. The resistance shall not be less than 1.8 ohms or more than 2.6 ohms.

16. Connect the chromel and alumel leads to the gas generator case terminals.

17. Repeat Step 2. through Step 16. for the engine 2 (references to engine No. 1 shall become references to engine No. 2).

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Hydraulic Oil Temperature Indicator System - Description

DC power is supplied from the DC bus No. 1 and DC bus No. 2 through the hydraulic temperature circuit breakers to their respective dual hydraulic pressure/temperature indicator. AC power is used in the pressure indicating portion of the indicator. Hydraulic temperature bulbs control the indicators, which display fluid temperature in degrees Celsius.

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Hydraulic Oil Temperature Indicator System - Operational Check

Procedure

NOTE

Ambient temperature checks are not valid when the ambient air is below -5.0°C.

1. Check the No. 1 system as follows:

1.1. Close the HYD NO 1 TEMP circuit breaker.

1.2. Check that the HYDR SYS 1 indicator indicates the approximate ambient temperature with the engine cooled to the ambient temperature.

2. Check the No. 2 system as follows:

2.1. Close the HYD NO 2 TEMP circuit breaker.

2.2. Check that the HYDR SYS 2 indicator indicates the approximate ambient temperature with the engine cooled to the ambient temperature.

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Fuel Quantity Indicator System - Description

The fuel quantity indicator system is a capacitance type system that consists chiefly of eight fuel quantity probes, two signal conditioners, and a dual pointer fuel quantity indicator with digital totalizer. It is a 28 VDC system powered by both emergency buses through the ENGINE NO 1 FUEL QTY and the ENGINE NO 2 FUEL QTY circuit breakers.

Each cell, except the upper center cells, contains a fuel quantity probe. The quantities sensed by the probes are summed by the signal conditioners No. 1 (left) and the signal conditioner No. 2 (right) and transmitted to the FUEL QTY indicator for display. The fuel quantity select switch mounted on the instrument panel to the left of the indicator, enables the lower forward and lower middle cell quantities to be displayed separately. The DIGITS TEST switch, adjacent to the fuel quantity select switch, enables the indicator digital totalizer to be tested.

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Fuel Quantity Indicator System - Testing and Calibration

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
412 SP Adapter Module	101-00807	1	
Fuel Quantity Test Set	Barfield DC 400 Model 412	1	
Fuel Quantity Test Set	Simmonds Precision PSD412-1 and PSD30-2	1	

Procedure

NOTE

This is the preferred testing and calibration method based on empty fuel tanks and dry probes. Refer to the appropriate test equipment manual for an alternate procedure when it is required to do the calibration procedure with fuel in the tanks.

NOTE

The following test procedures are for the system 1 (left fuel quantity pointer and totalizer, signal conditioner No. 1, and fuel cells and their components on the left side of the helicopter) and to the signal conditioner No. 2 (same as above except for right side). To test the left side (system 1), connect the test set to the signal conditioner No. 1 and read and interpret results for outputs applicable to system 1. To test right side (system 2), connect test set to signal conditioner No. 2 and read and interpret results for outputs applicable to system 2.

1. Remove the connectors (1A5P1/1A6P1) from the signal conditioner No. 1 and No. 2.

2. Connect the fuel quantity test set in a tee connection with the respective signal conditioner and connectors.

3. Close the ENGINE NO 1 FUEL QTY, ENGINE NO 2 FUEL QTY, ENGINE NO 1 FUEL TRANS, and ENGINE NO 2 FUEL TRANS circuit breakers.

4. Position the following switches to the positions indicated:

BATTERY BUS 1 (2S3)	ON
BATTERY BUS 2 (2S4)	ON
NON-ESNTL BUS (2S7)	MANUAL

5. On the test set, do as follows:
- TEST FUNCTION switch to PROBE
- RANGE switch to 200 pF
- ON/OFF switch to ON
- ADAPTER FUNCTION switch to MEAS
- MONITOR switch to IND-VDC
- TANK PROBE switch to UPPER OUTBD

6. Press the PRESS-TO-READ-CAP switch.

7. Read and record the test set digital display.

8. The indication shall be 0.0 ±0.5 pF.

9. With the PRESS-TO-READ-CAP switch pressed, set the TANK PROBE switch to the FWD, AFT, MID, and AUX positions.

10. Read and record the digital display at each position.

11. The indication shall be 0.0 ±0.5 pF.

12. Disconnect the connector from the fuel quantity indicator.

13. Set the ADAPTER FUNCTION switch to CAL and the TEST FUNCTION switch to CAP-SIM-CAL.

14. Press the PRESS-TO-READ-CAP switch, and with the test set CAP SIM (pF) control, set the digital display to indicate the total probe value recorded in Step 8. and Step 11..

15. Set the TEST FUNCTION switch to IND AMP.

16. Adjust the E adjustment on the respective signal conditioner for an indication of -0.03 VDC on the digital display.

17. Set the TEST FUNCTION switch to CAP-SIM-CAL.

18. Press the PRESS-TO-READ-CAP switch, and with the CAP SIM control, set the digital display to indicate 99.9 pF.

19. Set the TEST FUNCTION switch to IND AMP.

20. Adjust the F adjustment on the respective signal conditioner for an indication of 7.50 VDC on the digital display.

21. Set the TEST FUNCTION switch to CAP-SIM-CAL.

22. Press the PRESS-TO-READ-CAP switch and with the CAP SIM control, set the digital display to read 54.3 pF.

23. Connect the connector to the fuel quantity indicator and set the TEST FUNCTION switch to IND AMP.

24. The digital display shall indicate 4.09 ± 0.02 VDC, and the fuel quantity indicator shall indicate 1200 ±40 pounds (both pointer and digital displays).

25. On the test set, set the LIQUID LEVEL switch to NORM and the TANK SIM switches 1, 2, and 6 to DRY.

26. Verify that the FUEL LOW caution panel segment on the master caution panel illuminates.

27. Set the LIQUID LEVEL switch to SIM.

28. The FUEL LOW caution panel segment shall remain illuminated.

29. Set the TANK SIM 6 switch to WET.

30. The FUEL LOW caution panel segment shall extinguish.

31. Set the TANK SIM 2 switch to WET.

32. The respective FUEL TRANS caution panel segment on the master caution panel shall illuminate.

33. Set the TANK SIM 1 switch to WET and the TANK SIM 2 switch to DRY.

34. The respective FUEL TRANS caution panel segment shall remain illuminated.

35. On the engine and fuel control panel (1A1), set the respective FUEL TRANS switch to ON.

36. Verify proper pump operation.

37. On the test set, set the TANK SIM 1 switch to DRY and record the time it takes for the transfer pump to de-energize.

38. The time shall be 30 ±10 seconds.

39. On the engine and fuel control panel (1A1), set the respective FUEL TRANS switch to OFF.

40. Pull the ENGINE 1 FUEL QTY (or ENGINE 2 FUEL QTY) circuit breaker. Observe that the respective FUEL TRANS caution light illuminates.

41. Push the circuit breaker back in.

42. Set the LIQUID LEVEL switch to NORM.

43. Set the MONITOR switch to THERM 1, 2, and 6.

44. Read and record the digital display voltage indications.

45. The indications shall be 1.8 ±0.3 VDC at 25°C.

Fuel Quantity Indicator System - Thermistor Check

Procedure

NOTE

The following is provided as an aid to troubleshooting when a thermistor problem is suspected.

1. Set the ohmmeter to RX100.

NOTE

Not all probes have thermistors.

2. Connect the red lead to pin 4 of the probe and the black lead to pin 6 of the probe. The resistance should be 1000 ohms ±20 ohms at 25°C.

3. Move the black lead to pin 5 of the probe. The resistance should be 400 ohms ±20 ohms at 25°C.

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Interior Lighting System - Description

The interior lighting systems include the cockpit lights (412-A-98-00-00-91A-051A-A / 02551 or 412-A-98-00-00-92A-051A-A / 02552), approach plate and map lights, dome lights (412-A-98-00-00-93A-051A-A / 02553 or 412-A-98-00-00-94A-051A-A / 02554), instrument panel lights (412-A-98-00-00-70A-051A-A / 02530 or 412-A-98-00-00-71A-051A-A / 02531, 412-A-98-00-00-72A-051A-A / 02532 or 412-A-98-00-00-73A-051A-A / 02533), engine instrument panel lights (412-A-98-00-00-74A-051A-A / 02534 or 412-A-98-00-00-75A-051A-A / 02535), instrument secondary lights (412-A-98-00-00-86A-051A-A / 02546 or 412-A-98-00-00-87A-051A-A / 02547, 412-A-98-00-00-88A-051A-A / 02548), pedestal lights (412-A-98-00-00-84A-051A-A / 02544 or 412-A-98-00-00-85A-051A-A / 02545), overhead console lights (412-A-98-00-00-89A-051A-A / 02549 or 412-A-98-00-00-90A-051A-A / 02550), and baggage compartment lights systems (412-A-98-00-00-03B-051A-A / 02563 or 412-A-98-00-00-04B-051A-A / 02564). Refer to for the applicable wiring diagrams.

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Cockpit Lights System - Description

The cockpit lights are multipurpose utility lights designed to selectively provide either red or white illumination using a narrow spotlight beam or a wide floodlight beam. Electrical protection is provided by the LIGHTING MAP PILOT and the LIGHTING MAP CPLT circuit breakers. The controls necessary to obtain the operational modes of ON-OFF, DIM-BRIGHT, SPOT-FLOOD, and red or white illumination are incorporated in the lamp body.

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Cockpit Lights System - Operational Check

Procedure

1. Close the MAP PILOT circuit breaker.
2. Check that the pilot map light is operational in each mode (ON/OFF, DIM/BRIGHT, and SPOT-FLOOD on both red and white).
3. Check that the pilot map light illuminates.
4. Close the MAP CPLT circuit breaker.
5. Check that the copilot map light is operational in each mode (ON/OFF, DIM/BRIGHT, and SPOT/FLOOD on both red and white).
6. Check that the copilot map light illuminates.

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Applicable to: ALL
ECCN EAR99

Approach Plate and Map Lights System - Description

The approach plate and map lights system consists of the pilot and copilot approach plate and the pilot and copilot map light resistor controls mounted on the pilot instrument panel lower right, and the copilot instrument panel lower left. Rotate the control to vary the intensity of light.

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Approach Plate and Map Lights System - Operational Check

Procedure

1. Close the MAP PILOT or the MAP CPLT circuit breaker.
2. Rotate the pilot or copilot approach plate and the map light control.
3. Check that the approach plate and map light illuminates from dim to bright as the control is rotated clockwise.
4. Rotate the approach plate and map light control counterclockwise.
5. Check that the light decreases in brightness and is extinguished at an extreme counterclockwise direction.

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Dome Lights System - Description

The aft dome lights circuit is protected by the CABIN DOME circuit breaker on the overhead console breaker panel. To illuminate the aft dome lights to red or white, place the AFT DOME LIGHTS switch to the RED or the WHITE position. The brightness level of the aft dome lights is determined by the AFT DOME LIGHTS rheostat.

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Dome Lights System - Operational Check

Procedure

1. Position the NON-ESNTL BUS switch to MANUAL.
2. Close the LT DOME circuit breaker.
3. Position the AFT DOME LIGHTS switch to RED.
4. Rotate the AFT DOME LIGHTS control clockwise.
5. Check that the red dome lights increase in brightness.
6. Rotate the AFT DOME LIGHTS control counterclockwise.
7. Check that the red dome lights decrease in brightness.
8. Position the AFT DOME LIGHTS switch to WHITE.
9. Repeat Step 1. through Step 7. to check the white dome lights.

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Instrument Panel Lights System - Description

The pilot instrument lighting circuit is protected by the INST PILOT circuit breaker. The rotation of the PILOT INSTR LT control clockwise closes the attached switch, which applies 28 VDC from the essential bus through the INST PILOT circuit breaker to the instrument lights power supply. The power supply in turn provides an output voltage of 5 VDC required to operate the 5 volt lamp instrument lights. The output voltage of the power supply is controlled by the PILOT INSTR LT control, which determines the brightness level of the 5 volt lamps. The copilot instrument lighting circuit is protected by the INST CPLT circuit breaker. The copilot instrument lights circuit and operation is identical to the pilot instrument lights circuit. The COPLT INSTR LT control determines the brightness level of the 5 volt lamps.

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Pilot Instruments and Passenger Warning Lighting Check - Operational Check

Procedure

1. Set the PILOT INSTR LT control to OFF.

2. Close the INST PILOT circuit breaker.

3. Rotate the PILOT INSTR LT control slightly clockwise out of detent.

4. Check that the instrument lights on the pilot instrument panel are dimly illuminated.

5. Check that the passenger warning lights are dimly illuminated.

6. Rotate the PILOT INSTR LT control clockwise.

7. Check that the instrument lights on the pilot instrument panel and the passenger warning lights increase in brightness and are brightest when the control is at the extreme clockwise position.

8. Rotate the PILOT INSTR LT control counterclockwise.

9. Check that the pilot instrument lights and the passenger warning lights decrease in brightness.

10. Rotate the PILOT INSTR LT control to OFF.

11. Check that the pilot instrument lights and the passenger warning lights extinguish.

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Copilot Lighting Check - Operational Check

Procedure

1. Set the CPLT INSTR LT control to OFF.
2. Close the INST CPLT circuit breaker.
3. Rotate the CPLT INSTR LT control slightly clockwise out of detent.
4. Check that the copilot instrument lights are dimly illuminated.
5. Rotate the CPLT INSTR LT control clockwise.
6. Check that the copilot instrument lights increase in brightness and are brightest at the extreme clockwise position.
7. Rotate the CPLT INSTR LT control counterclockwise.
8. Check that the copilot instrument lights decrease in brightness and are extinguished at the extreme counterclockwise position.
9. Rotate the CPLT INSTR LT control to OFF.
10. Check that the copilot instrument lights extinguish.

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Engine Instrument Lights System - Description

This circuit is protected by the ENG INST circuit breaker. The engine instrument lights circuit and operation are identical to the pilot and copilot instrument lights circuits. The ENG INSTR LT rheostat determines the brightness level of the 5 volt lamps.

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Engine Instrument Lights System - Operational Check

Procedure

1. Set the ENG INSTR LT control to OFF.
2. Close the ENG INST circuit breaker.
3. Rotate the ENG INSTR LT control slightly clockwise until the switch just closes.
4. Check that the engine instrument lights are dimly illuminated.
5. Rotate the ENG INSTR LT control clockwise.
6. Check that the engine instrument lights increase in brightness and are brightest at the extreme clockwise position.
7. Rotate the control counterclockwise.
8. Check that the engine instrument lights decrease in brightness and are extinguished at the extreme counterclockwise position.

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Instrument Secondary Lights System - Description

This circuit is protected by the INST SEC circuit breaker. The rotation of the SEC INSTR LT control clockwise closes its attached switch, which applies power to instrument secondary lights through the instrument secondary lights control transistor. The SEC INSTR LT control determines the brightness level of the instrument secondary lights by varying the base voltage of the 28 VDC secondary instrument lights transistor.

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Instrument Secondary Lights Systems - Operational Check

Procedure

1. Set the SEC INSTR LT control to OFF.
2. Close the INST SEC circuit breaker.
3. Rotate the SEC INSTR LT control clockwise until the switch just closes.
4. Check that the instrument secondary lights are dimly illuminated.
5. Rotate the SEC INSTR LT control clockwise.
6. Check that the instrument secondary lights increase in brightness and are brightest at the extreme clockwise position.
7. Rotate the control counterclockwise.
8. Check that the instrument secondary lights decrease in brightness and are extinguished at the extreme counterclockwise position.

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Pedestal Lights System - Description

The pedestal lights circuit and operation are identical to the instrument secondary and overhead console lights circuit. the PED LT control determines the brightness level of the pedestal lights by varying the base voltage of 28 VDC pedestal lights transistor.

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Pedestal Lights System - Operational Check

Procedure

1. Set the PED LT control to OFF.
2. Close the PED circuit breaker.
3. Rotate the PED LT control slightly clockwise until the switch just closes.
4. Check that the pedestal lights, panel lights on the pilot collective stick, and outlet valve lights on the right doorpost are dimly illuminated.
5. Rotate the PED LT control clockwise.
6. Check that the lights that were illuminated in Step 4. increase in brightness and are brightest at the extreme clockwise position.
7. Rotate the PED LT control counterclockwise.
8. Check that the lights illuminated in Step 4. decrease in brightness and are extinguished at the extreme counterclockwise position.

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Overhead Console Lights System - Description

This circuit is protected by the OVHD CSL circuit breaker. The overhead console lights circuit and operation are identical to the instrument secondary lights circuit. The CONSOLE LT control determines the brightness level of the overhead console lights by varying the base voltage of the 28 VDC console lights transistor.

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Overhead Console Lights System - Operational Check

Procedure

1. Set the CONSOLE LT control to OFF.
2. Close the OVHD CSL circuit breaker.
3. Rotate the CONSOLE LT control slightly clockwise until the switch just closes.
4. Check that the panel lights on the overhead console are dimly illuminated.
5. Rotate the CONSOLE LT control clockwise.
6. Check that the panel lights on the overhead console increase in brightness and are brightest at the extreme clockwise position.
7. Rotate the CONSOLE LT control counterclockwise.
8. Check that the panel lights on the overhead console decrease in brightness and are extinguished at the extreme counterclockwise position.

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Baggage Compartment Lights System - Description

The baggage compartment lights are protected by the the BAG COMPT circuit breaker. They illuminate the compartment when the door is opened. Part of the baggage compartment door switch energizes the caution panel segment in case the baggage compartment door is left open.

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Baggage Compartment Lights System - Operational Check

Procedure

1. Close the BAG COMPT circuit breaker.
2. Open the baggage compartment door.
3. Check that the baggage compartment lights are illuminated.
4. Manually, press the plunger on the baggage compartment door switch.
5. Check that the baggage compartment lights extinguish.

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Baggage Compartment Lights System - Maintenance

Procedure

1. For adjustment procedures of the baggage compartment door switch, refer to 412-A-96-03-01-14A-271A-A / 02061.

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Exterior Lighting Systems - Description

The exterior lighting systems include the utility, position lights (412-A-98-00-00-99A-051A-A / 02559 or 412-A-98-00-00-00B-051A-A / 02560), anti-collision lights (412-A-98-00-00-97A-051A-A / 02557 or 412-A-98-00-00-98A-051A-A / 02558 or 412-A-98-00-00-97A-051A-A / 02557 or 412-A-98-00-00-98A-051A-A / 02558), searchlight (412-A-98-00-00-95A-051A-A / 02555 or 412-A-98-00-00-96A-051A-A / 02556), and landing light systems (412-A-98-00-00-01B-051A-A / 02561 or 412-A-98-00-00-02B-051A-A / 02562). Refer for the applicable wiring diagrams.

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Utility Lights - Description

The utility light circuit is protected by the UTIL 5 amp circuit breaker and controlled by the UTILITY LIGHT switch. The left and right utility lights illuminate simultaneously.

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Utility Light System - Operational Check

Procedure

1. Apply electrical power to the helicopter electrical system.
2. Set the UTILITY LIGHT switch to ON. Verify that the eft and right utility lights illuminate.
3. Set the UTILITY LIGHT switch to OFF. Verify that both utility lights extinguish.
4. Turn the electrical power off.

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Position Lights System - Description

The position lights system gives constant navigation signals during flight. The position lights, fuselage and tail, are powered from the 28 VDC non-essential bus No. 1, and are protected by the LIGHTING POSN/POS circuit breaker. The position lights are controlled by the POSITION LIGHT switch on the overhead console. When the POSITION LIGHT switch is in the ON position the lights illuminate at full brilliance.

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Position Lights System - Operational Check

Procedure

1. Close the LIGHTING POSN/POS circuit breaker.
2. Set the POSITION switch to ON.
3. Verify that the right and left upper position lights illuminate.
4. Verify that the right and left lower position lights Illuminate.
5. Verify that the left and right tail lights illuminate.
6. Set the POSITION switch to OFF.
7. Verify that all the position lights extinguish.

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Position Lights - Inspection

Procedure

1. During the preflight exterior inspection (BHT-412-FM), inspect each lens for security , scratching, discoloration, or cracking. Replace the lens if it is excessively scratched, discolored, or cracked.

2. The LED tail position light assemblies are designed with six LEDs. If any one LED fails, repair or replace the unit.

3. The LED left-hand and right-hand position light assemblies are designed with three LEDs. If any one LED fails, repair or replace the nut.

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Anti-collision Lights System - Description

The anti-collision lights system gives constant navigation signals during flight. The lights are powered from the 28 VDC non-essential bus No. 2 and are protected by the ANTI COLL LT circuit breaker and controlled by the ANTI-COLL LT switch. When the switch is set to ON, the anti-collision strobe light assemblies flash at a rate of approximately 80 flashes per minute. One light is mounted on the aft pylon fairing and one light is mounted on the bottom of the fuselage.

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Anti-collision Light - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-136	AR	

Procedure

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR THE ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

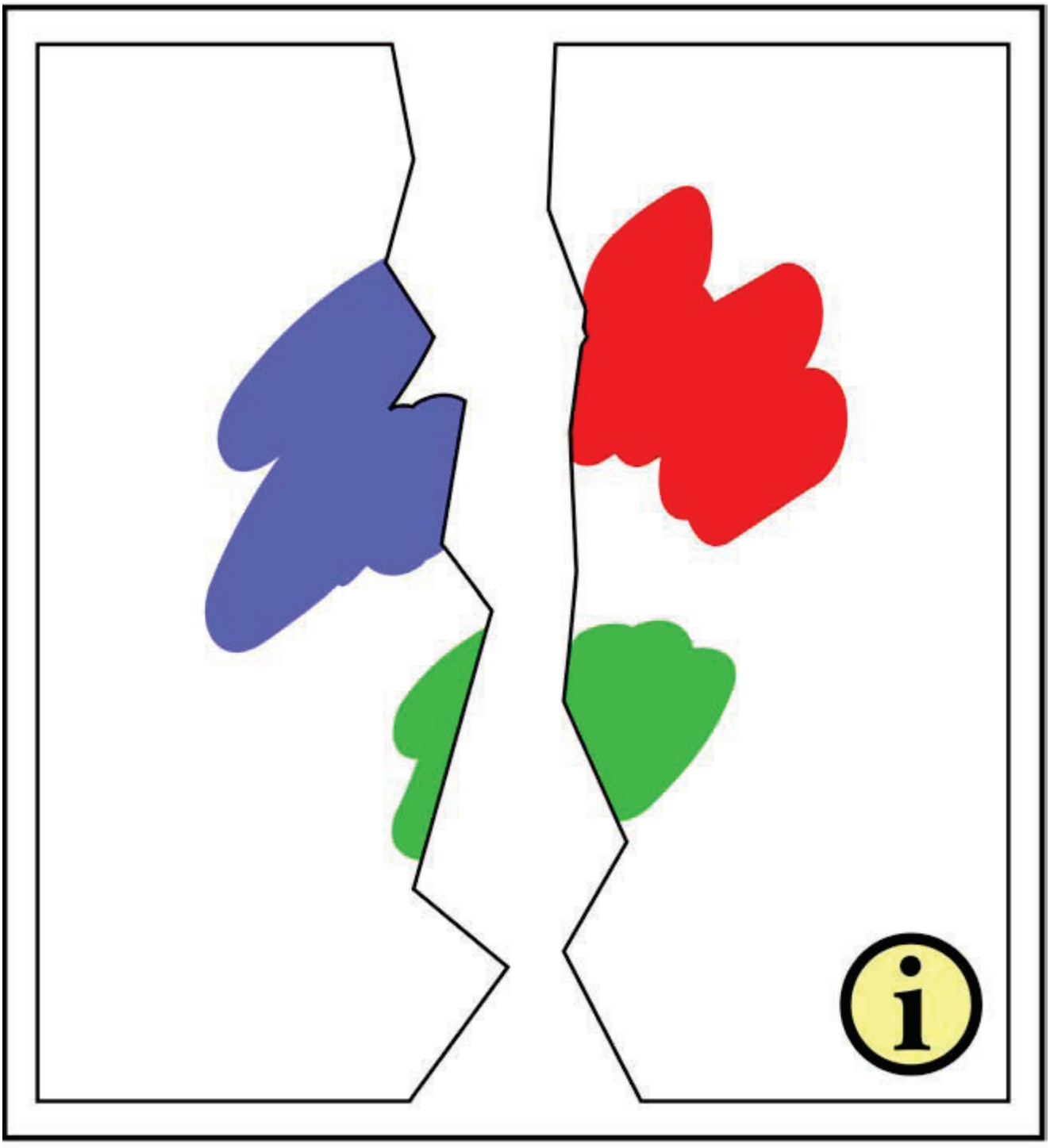
WARNING

THE ANTI-COLLISION LIGHT (DS8) (1, Figure 1) IS A HIGH VOLTAGE DEVICE. DO NOT TOUCH OR REMOVE THESE UNITS WHILE IN OPERATION. BEFORE YOU START MAINTENANCE ON THE LIGHTING SYSTEM, WAIT A MINIMUM OF 10 MINUTES AFTER YOU STOP THE UNIT. INJURY TO PERSONNEL CAN OCCUR.

NOTE

The procedure that follows is for the upper anti-collision light (DS8) (1). The procedure is the same for the lower anti-collision light (DS7).

1. Disconnect the battery and external DC power from the helicopter.
2. Remove screws (2) and washers (3).
3. Lift the anti-collision light (DS8) (1) sufficiently to get access to the electrical connector (P116) (4).
4. Disconnect the electrical connector (P116) (4) and carefully remove the anti-collision light (DS8) (1).
5. Put protective caps or plugs (C-136) on the electrical connector (P116) (4) and receptacle.



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Figure 1. Anti-collision Light - Removal (Sheet 1 of 1)

Anti-collision Light - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-136	AR	

Procedure

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR THE ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

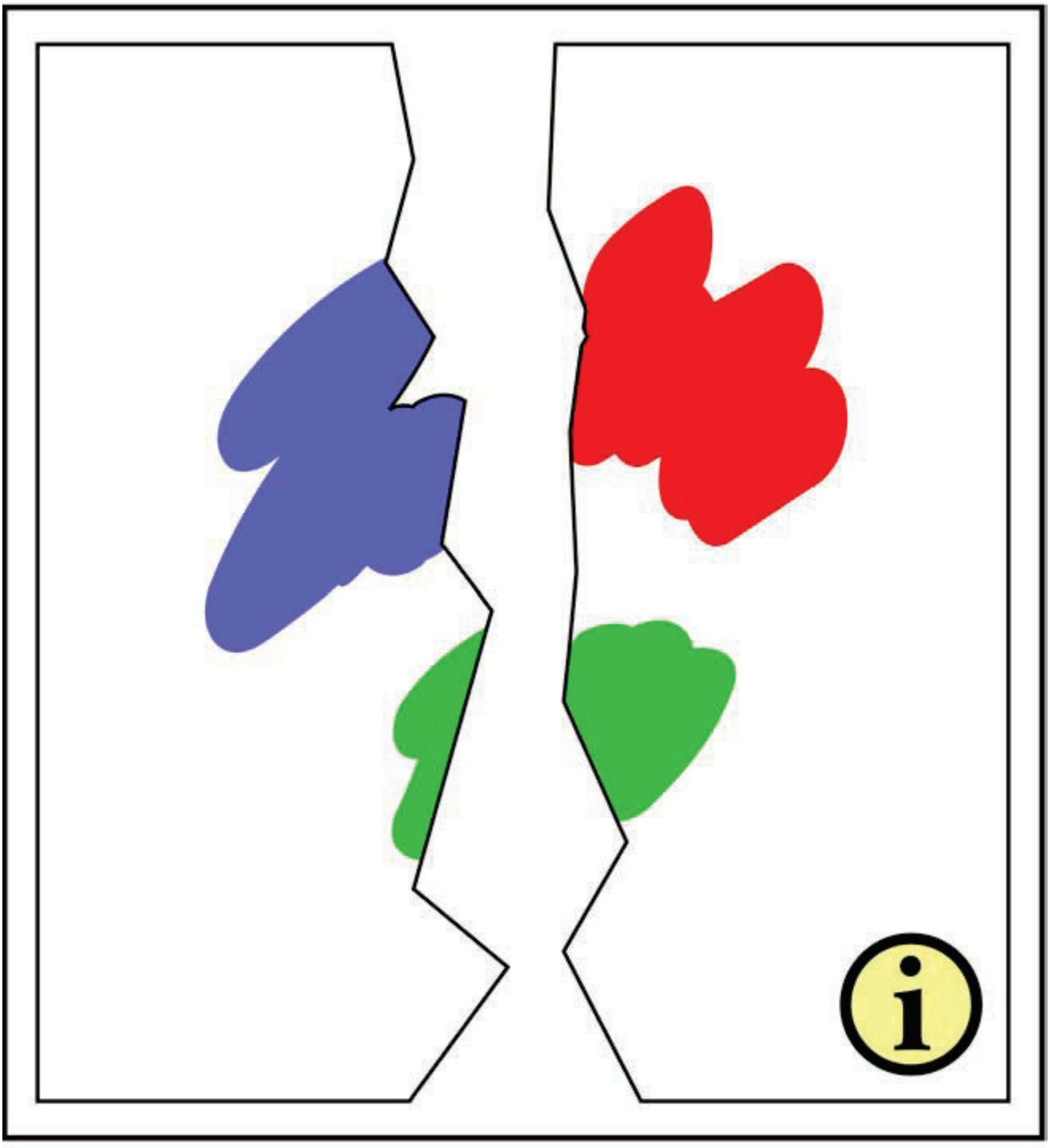
WARNING

THE ANTI-COLLISION LIGHT (8DS8) (1, Figure 1) IS A HIGH VOLTAGE DEVICE. DO NOT TOUCH OR REMOVE THESE UNITS WHILE IN OPERATION. BEFORE YOU START MAINTENANCE ON THE LIGHTING SYSTEM, WAIT A MINIMUM OF 10 MINUTES AFTER YOU STOP THE UNIT. INJURY TO PERSONNEL CAN OCCUR.

NOTE

The procedure that follows is for the upper anti-collision light (8DS8) (1). The procedure is the same for the lower anti-collision light (8DS3).

1. Disconnect the battery and external DC power from the helicopter.
2. Remove screws (2) and washers (3).
3. Lift the anti-collision light (8DS8) (1) sufficiently to get access to the electrical connector (8DS8P1) (4).
4. Disconnect the electrical connector (8DS8P1) (4) and carefully remove the anti-collision light (8DS8) (1).
5. Put protective caps or plugs (C-136) on the electrical connector (8DS8P1) (4) and receptacle.



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Figure 1. Anti-collision Light - Removal (Sheet 1 of 1)

Anti-collision Light - Inspection

Procedure

1. During the Preflight Exterior Inspection (BHT-412-FM, Section 2), examine each lens for correct installation, scratches, discoloration, or cracks. Replace the lens, if there are scratches, discoloration, or cracks.

2. The LED anti-collision light assembly has eight vertical columns with four LEDs each. If there is a failure in one or more LED or the vertical column, repair or replace the unit ((412-A-96-02-00-03A-520A-A / 02035.2 and 412-A-96-02-00-03A-720A-A / 02035.5) or (412-A-96-02-00-03B-520A-A / 02035.1 and 412-A-96-02-00-03B-720A-A / 02035.4)).

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Anti-collision Light - Installation

Procedure

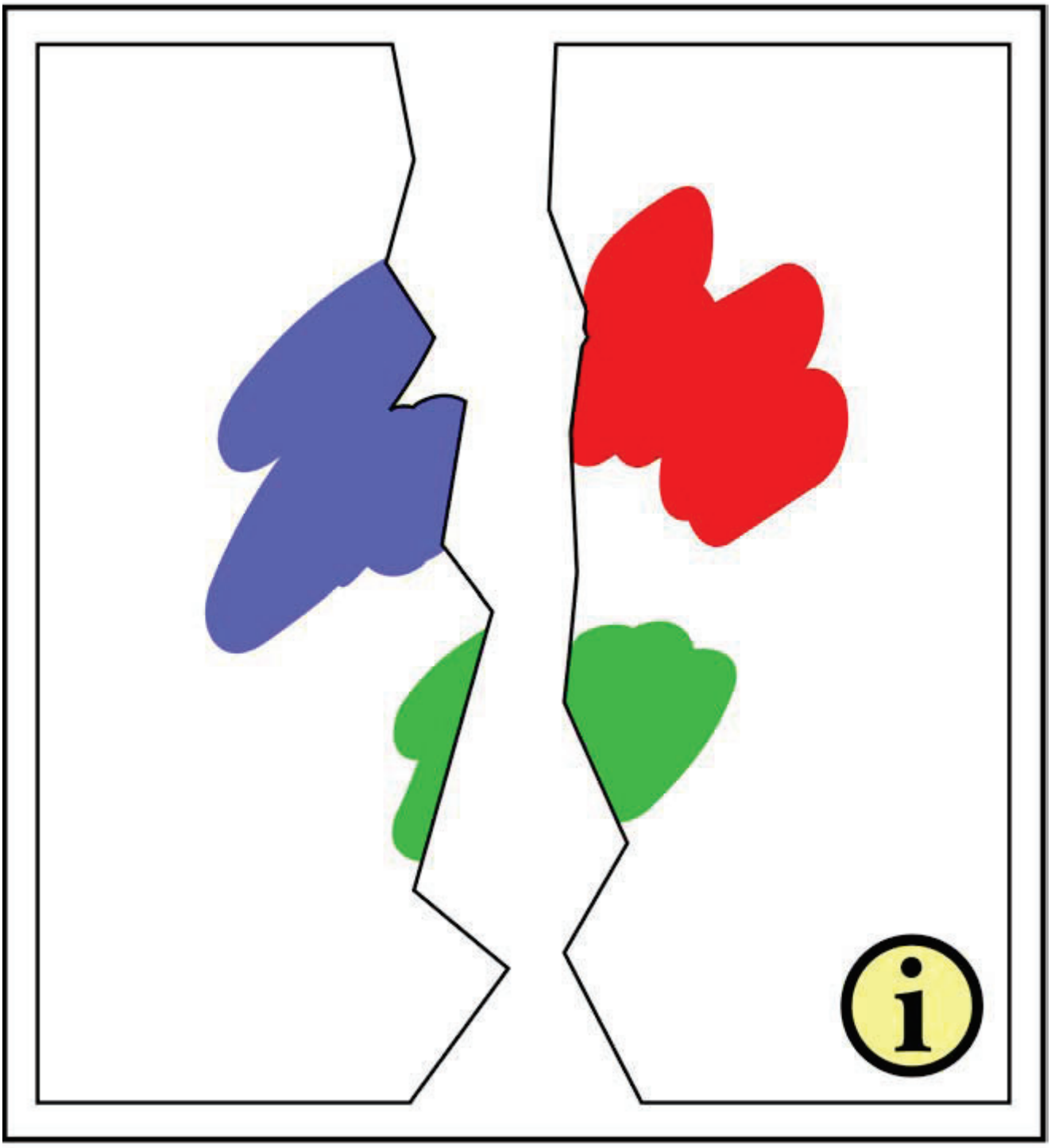


OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR THE ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The procedure that follows is for the upper anti-collision light (DS8) (1, Figure 1). The procedure is the same for the lower anti-collision light (DS7).

1. Remove protective caps and/or plugs from the electrical connector (P116) (4) and receptacle.
2. Connect the electrical connector (P116) (4) to anti-collision light (DS8) (1).
3. Install the anti-collision light (DS8) (1) with screws (2) and washers (3).
4. Do an operational check of the anti-collision light (DS8) (1) (412-A-96-02-00-03A-320A-A / 02036).



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Figure 1. Anti-collision light - Installation (Sheet 1 of 1)

Anti-collision Light - Installation

Procedure

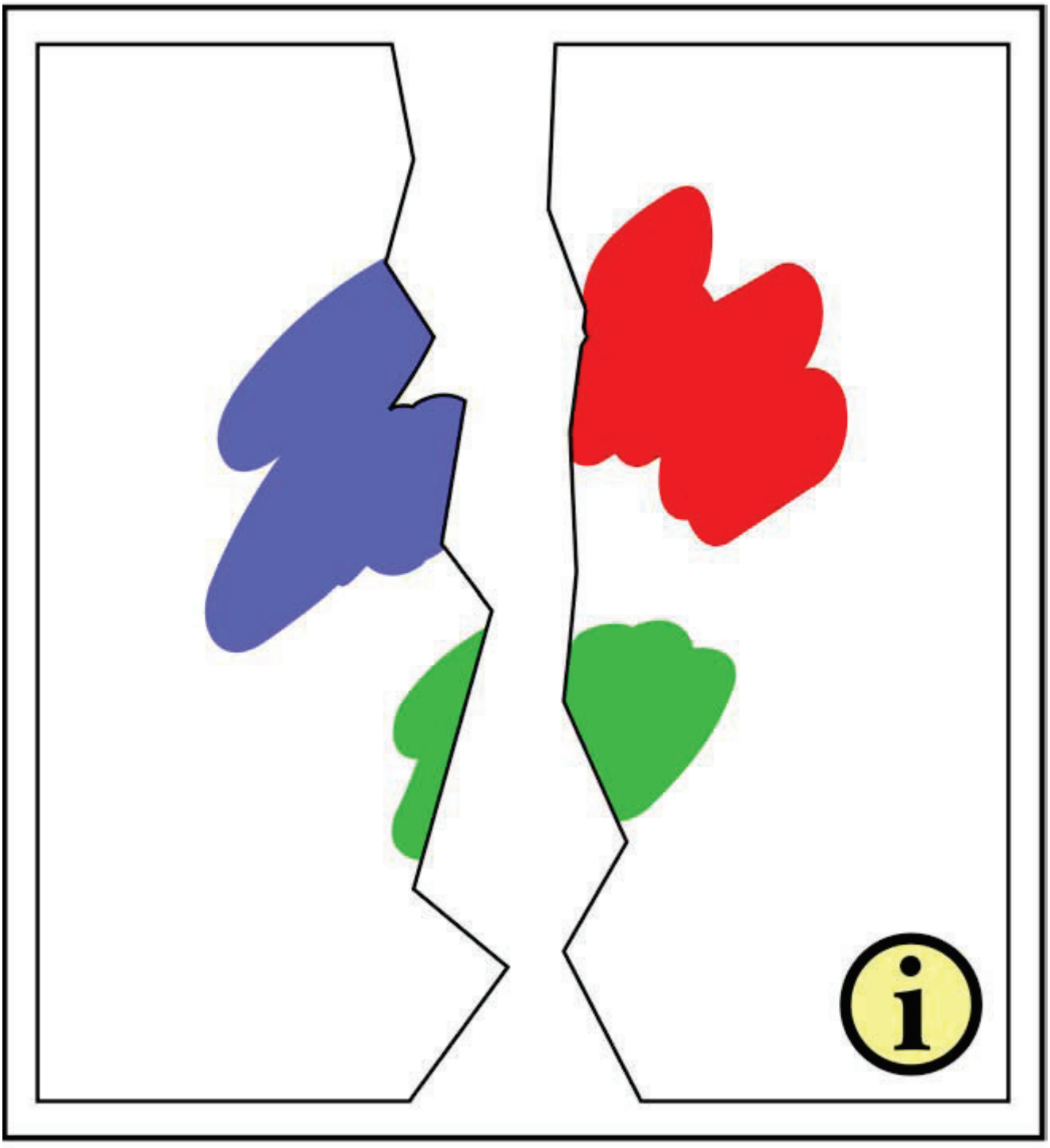


OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR THE ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The procedure that follows is for the upper anti-collision light (8DS8) (1, Figure 1). The procedure is the same for the lower anti-collision light (8DS3).

1. Remove protective caps and/or plugs from the electrical connector (8DS8P1) (4) and receptacle.
2. Connect the electrical connector (8DS8P1) (4) to anti-collision light (8DS8) (1).
3. Install the anti-collision light (8DS8) (1) with screws (2) and washers (3).
4. Do an operational check of the anti-collision light (8DS8) (1) (412-A-96-02-00-03A-320A-A / 02036).



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Figure 1. Anti-collision Light - Installation (Sheet 1 of 1)

Anti-collision Lights System - Operational Check

Procedure

1. Close the ANTI COLL LT circuit breaker.
2. Place the ANTI-COLL LT switch to ON.
3. Check that the upper and lower anti-collision lights are illuminated and flashing approximately 80 to 90 times per minute.
4. Set the ANTI-COLL LT switch to OFF.
5. Check that the anti-collision lights extinguish.

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Searchlight System - Description

The searchlight system provides close range illumination of the terrain and obstacles to aid in low level flight and landing. The searchlight is powered from the 28 VDC emergency bus No. 2, protected by the SCH LT PWR and SCH LT CNTL circuit breakers, and controlled by the SEARCH LIGHT NORM/OFF switch. The searchlight assembly includes a sealed beam-type light, rotation motor, extend-retract motor, control relays, and limit switches. The SRCH LT ON/OFF switch controls the searchlight lamp. The four-way SRCH LT EXT/L/R/RETR switch controls the extend/retract and rotate right/left motion of the searchlight.

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Searchlight System - Operational Check

Procedure

1. Close the SCHLT PWR and SCHLT CONTR circuit breakers.

2. Set the SRCH LT ON/OFF switch on the pilot collective stick to ON.

3. Verify that the searchlight illuminates.

4. Set the four-way SRCH LT switch to EXT.

5. Check that the searchlight extends approximately 120°.

6. Set the four-way SRCH LT switch to RETR.

7. Check that the searchlight retracts.

8. With the searchlight partially extended do as follows:

8.1. Set the four-way SRCH LT switch to L.

8.2. Check that the searchlight rotates to the left.

8.3. Set the four-way SRCH LT switch to R.

8.4. Check that the searchlight rotates to the right.

8.5. Use the four-way SRCH LT switch to rotate the searchlight to point forward.

8.6. Set the SRCH LT ON/OFF switch to OFF.

8.7. Set the four-way SRCH LT switch to RETR.

8.8. Check that the searchlight retracts.

9. Open the SCHLT PWR and SCHLT CONTR circuit breakers.

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Landing Light System - Description

The landing light system illuminates the landing area during the final approach to landing. The landing light is powered from the 28 VDC essential bus No. 1, protected by the LDN LT PWR and LND LT CNTL circuit breakers, and controlled by the LNDG LT ON/OFF and LNDG EXT/RET switches. The landing light assembly includes one sealed beam-type light, an extend-retract motor, a control relay, and limit switches. The LDG LT ON/OFF switch controls the ON and OFF functions. The LDG LT EXT/RETR switch controls the extend and retract motion of the landing light.

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Landing Light System - Operational Check

Procedure

1. Close both the LDG PWR and LDG CONTR circuit breakers.

2. Place and hold the LDG LT EXT/RETR switch on the pilot collective stick to EXT.

3. Check that the landing light extends and stops at approximately 90°.

4. Set the LDG LT ON/OFF switch to ON.

5. Check that the landing light is illuminated.

6. Open the LDG PWR circuit breaker.

7. Check that the landing light is extinguished.

8. Close the LDG PWR circuit breaker and open the LDG CONTR circuit breaker.

9. Set the LDG LT EXT/RETR switch to RETR.

10. Check that the LDG LT EXT/RETR switch cannot operate the landing light.

11. With the LDG CONTR circuit breaker closed and the landing light illuminated do as follows:

11.1. Set the LDG LT EXT/RETR switch to RETR and hold.

11.2. Check that the landing light retracts completely.

11.3. Check that the landing light remains illuminated.

11.4. Set the LDG LT ON/OFF switch to OFF.

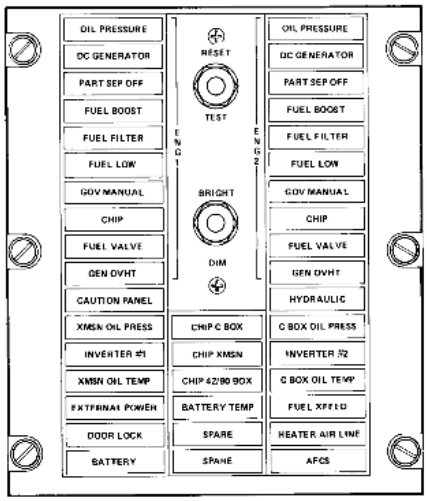
11.5. Check that the landing light is extinguished.

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Caution and Warning System - Description

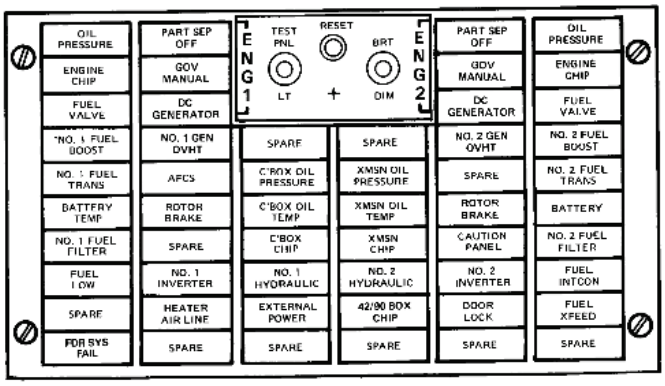
The caution and warning system includes the master caution panel, pilot MASTER CAUTION light, copilot MASTER CAUTION light, pilot RPM warning light, copilot RPM warning light, ENG 1 OUT light, ENG 2 OUT light, switches, diodes, relays, and interconnected wiring.

The master caution panel (Figure 1, Figure 2, and Figure 3) is a multicapsulated (each capsule includes an indicator light) panel that is protected by the MASTER CAUTION circuit breaker. Each indicator light is in series with a respective caution light system that contains switches and associated components. The MASTER CAUTION lights mounted on the instrument panel illuminate simultaneously when one or more capsule indicator(s) illuminates. The caution light and warning circuits include items as listed in this section. Refer to 412-A-98-00-00-77A-051A-A / 02537, 412-A-98-00-00-78A-051A-A / 02538, 412-A-98-00-00-79A-051A-A / 02539, 412-A-98-00-00-80A-051A-A / 02540, 412-A-98-00-00-25A-051A-A / 02485 or 412-A-98-00-00-26A-051A-A / 02486 for the appropriate system wiring diagram.



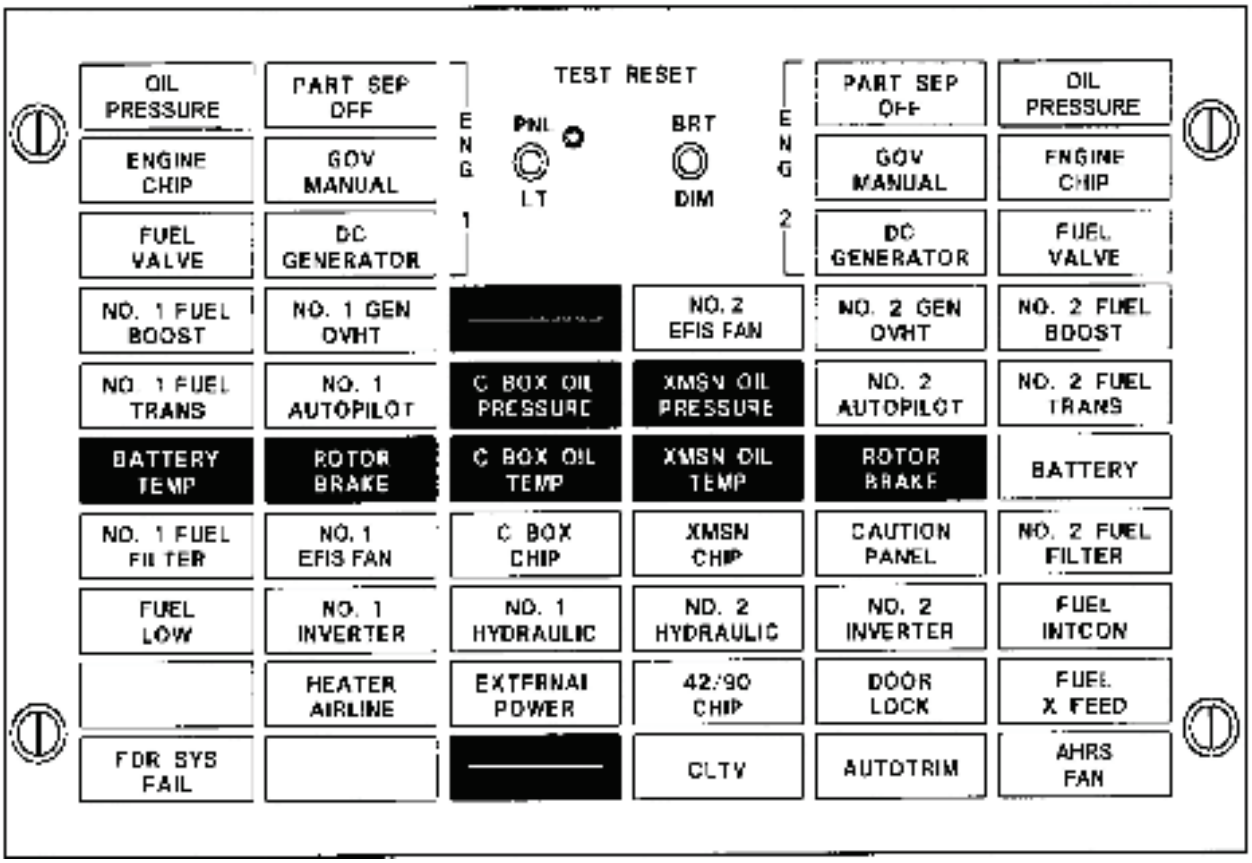
ICN-412-A-960300-B-97499-00001-A-001-01

Figure 1. Master Caution Panel (S/N 33001 through 33107) (Sheet 1 of 1)



ICN-412-A-960300-B-97499-00002-A-001-01

Figure 2. Master Caution Panel (S/N 33108 through 36086) (Sheet 1 of 1)



ICN-412-A-960300-B-97499-00003-A-001-01

Figure 3. Master Caution Panel (S/N 36087 through 36999) (Sheet 1 of 1)

Caution and Warning System - Operation

The MASTER CAUTION lights on the pilot and copilot instrument panel illuminate in conjunction with one or more indicator panel lights to announce a fault or malfunction in a monitored system. The MASTER CAUTION lights can be reset in readiness for the next fault by momentarily depressing the TEST-RESET switch to RESET. The individual indicator lights on the caution panels remain illuminated as long as the fault condition exists. Momentarily pressing the TEST switch illuminates all the indicator lights on the master caution panel (MASTER CAUTION lights, the RPM warning lights, ENG 1 OUT light, and ENG 2 OUT light).

The caution and warning lights are designed so that after each initial application of power, all indicator lights will illuminate bright. The manual selection of a bright or dimmed condition is provided by a BRIGHT-DIM switch on the master caution panel. Momentarily placing the BRIGHT-DIM switch to DIM will momentarily dim the master caution panel indicator lights (MASTER CAUTION light, RPM ENG 1 OUT, and ENG 2 OUT). The lights will dim and remain dim if the pilot INSTR LTS rheostat (R1/8R1) is rotated clockwise from OFF. The lights can be reset to bright by momentarily placing the BRIGHT-DIM switch to BRIGHT or by rotating pilot INSTR LTS rheostat counterclockwise to OFF.

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Master Caution Light System - Operational Test

Procedure

1. During system operational checks, the MASTER CAUTION lights on the pilot instrument panel should illuminate each time a caution panel segment illuminates, and shall be reset each time in readiness for another fault indication.

1.1. Close the MASTER CAUTION and CAUTION FAIL circuit breakers.

1.2. Check that the MASTER CAUTION lights illuminate and each caution light illuminates in accordance with following listing.

1.3. Reset the MASTER CAUTION lights by placing the RESET/TEST switch to RESET and releasing.

1.4. Check that the MASTER CAUTION lights extinguish and the individual caution lights remain either on or off in accordance with the following lists.

Applicable: SN: 33001-33107:

1.4.1.

CAUTION LIGHTS	ON/OFF CONDITION
ENG 1 DC GENERATOR	ON
ENG 2 DC GENERATOR	ON
ENG 1 OIL PRESSURE	ON
ENG 2 OIL PRESSURE	ON
ENGINE 1 PART SEP	ON
ENG 2 PART SEP	ON
ENG 1 FUEL BOOST	ON
ENG 2 FUEL BOOST	ON
ENG 1 FUEL FILTER	OFF
ENG 2 FUEL FILTER	OFF
ENG 1 FUEL LOW	ON (1)
ENG 2 FUEL LOW	ON (1)
ENG 1 GOV MANUAL	OFF
ENG 2 GOV MANUAL	OFF
ENG 1 CHIP	OFF
ENG 2 CHIP	OFF
ENG 1 FUEL VALVE	OFF
ENG 2 FUEL VALVE	OFF
ENG 1 GEN OVHT	OFF
ENG 2 GEN OVHT	OFF
CAUTION PANEL	OFF
HYDRAULIC	ON
XMSN OIL	ON
PRESSURE CHIP C BOX	ON
C BOX OIL PRESS	ON
INVERTER #1	ON
CHIP XMSN	OFF
INVERTER #2	ON
XMSN OIL TEMP	OFF
C BOX OIL TEMP	OFF
EXTERNAL POWER	ON

CAUTION LIGHTS	ON/OFF CONDITION
CHIP 42/90 BOX	OFF
FUEL XFEED	OFF
DOOR LOCK	ON
HEATER AIR LINE	OFF
BATTERY	ON
AFCS	OFF
BATTERY TEMP	OFF

NOTE:

[1] Light in ON state only when fuel quantity is less than 10% of tank capacity. Quantity is less than 10% of tank capacity.

Applicable: SN: 33108-36086:

1.4.2.

CAUTION LIGHTS	ON/OFF CONDITION
ENG 1 OIL PRESSURE	ON
ENG 1 PART SEP OFF	ON
ENG 2 PART SEP OFF	ON
ENG 2 OIL PRESSURE	ON
ENG 1 ENGINE CHIP	OFF
ENG 1 GOV MANUAL	OFF
ENG 2 GOV MANUAL	OFF
ENG 2 ENGINE CHIP	OFF
ENG 1 FUEL VALVE	OFF
ENG 1 DC GENERATOR	ON
ENG 2 DC GENERATOR	ON
ENG 2 FUEL VALVE	OFF
NO. 1 FUEL BOOST	ON
NO. 1 GEN OVHT	OFF
NO. 2 GEN OVHT	OFF
NO. 2 FUEL BOOST	ON
NO. 1 FUEL TRANS	ON
AFCS	ON
C BOX OIL PRESSURE	ON
XMSN OIL PRESSURE	ON
NO. 2 FUEL TRANS	ON
BATTERY TEMP	OFF
ROTOR BRAKE	OFF
C BOX OIL TEMP	OFF
XMSN OIL TEMP	OFF
ROTOR BRAKE	OFF
BATTERY	ON
NO. 1 FUEL FILTER	OFF
C BOX CHIP	OFF
XMSN CHIP	OFF

CAUTION LIGHTS	ON/OFF CONDITION
CAUTION PANEL	OFF (2)
NO. 2 FUEL FILTER	OFF
FUEL LOW	ON or OFF (1)
NO. 1 INVERTER	ON
NO. 1 HYDRAULIC	ON
NO. 2 HYDRAULIC	ON
NO. 2 INVERTER	ON
FUEL INTCON	OFF
HEATER AIR LINE	OFF
EXTERNAL POWER	ON
42/90 BOX CHIP	OFF
DOOR LOCK	ON
FUEL XFEED	OFF

NOTE:

1 Light in ON state only when fuel quantity is less than 10% of tank capacity.

2 Dimming not required.

Applicable: SN: 36087-36999:

1.4.3.

CAUTION LIGHTS	ON/OFF CONDITION
ENG 1 OIL PRESSURE	ON
ENG 1 PART SEP OFF	ON
ENG 2 PART SET OFF	ON
ENG 2 OIL PRESSURE	ON
ENG 1 ENGINE CHIP	OFF
ENG 1 GOV MANUAL	OFF
ENG 2 GOV MANUAL	OFF
ENG 2 ENGINE CHIP	OFF
ENG 1 FUEL VALVE	OFF
ENG 1 DC GENERATOR	ON
ENG 2 DC GENERATOR	ON
ENG 2 FUEL VALVE	OFF
NO. 1 FUEL BOOST	ON
NO. 1 GEN OVHT	OFF (3)
NO. 2 EFIS FAN	ON/OFF (4)
NO. 2 GEN OVHT	OFF (3)
NO. 2 FUEL BOOST	ON
NO. 1 FUEL TRANS	ON
NO. 1 AUTOPILOT	ON
C BOX OIL PRESSURE	ON
XMSN OIL PRESSURE	ON
NO. 2 AUTOPILOT	ON
NO. 2 FUEL TRANS	ON
BATTERY TEMP	OFF

CAUTION LIGHTS	ON/OFF CONDITION
ROTOR BRAKE	OFF
C BOX OIL TEMP	OFF
XMSN OIL TEMP	OFF
ROTOR BRAKE	OFF
BATTERY	ON
NO. 1 FUEL FILTER	ON/OFF
NO. 1 EFIS FAN	ON/OFF (4)
C BOX CHIP	OFF
XMSN CHIP	OFF
CAUTION PANEL	OFF (2)
NO. 2 FUEL FILTER	OFF
FUEL LOW	ON or OFF (1)
NO. 1 INVERTER	ON
NO. 1 HYDRAULIC	ON
NO. 2 HYDRAULIC	ON
NO. 2 INVERTER	ON
FUEL INTCON	OFF
HEATER AIR LINE	OFF
EXTERNAL POWER	ON
42/90 BOX CHIP	OFF
DOOR LOCK	ON
FUEL XFEED	OFF
FDR SYS FAIL	OFF
CLTV	ON/OFF (4)
AUTOTRIM	OFF
AHRS FAN	OFF (5)

NOTE:

1 Light in ON state only when fuel quantity is less than 10% of tank capacity.

2 Dimming not required.

3 Unless CAA VFR Kit (412-705-016-117/-119) is installed.

4 Four-axis configuration only.

5

Caution Panel Light System - Operational Check

Procedure

1. To test the caution lights, set the TEST PNL/LT switch to TEST and release.

2. Check that the MASTER CAUTION lights momentarily illuminate brightly and then extinguish.

3. Check that all the caution lights momentarily illuminate brightly and return to the former condition.

4. Set the BRT/DIM switch to DIM and release.

5. Check that the caution lights do not dim.

6. Rotate the PILOT INSTR LT rheostat clockwise from OFF.

7. Actuate the BRT/DIM switch again.

8. Check that the lights dim and hold.

9. Rotate the PILOT INSTR LT rheostat counterclockwise to OFF and check that the lights return to bright.

10. Make sure that the ENGINE 1 GOV and ENGINE 2 GOV switches are not set to the MANUAL position.

11. Check that the following occurs while the TEST switch on master caution panel is set to PNL.

11.1. The CAUTION PANEL segment illuminates in bright mode.

11.2. All other segments are extinguished.

11.3. The ENG 1 OUT, ENG 2 OUT lights are illuminated in bright mode.

11.4. The MASTER CAUTION light(s) are illuminated in bright mode.

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DC Generator Caution Light Systems - Operational Check

Procedure

1. The DC GENERATOR caution lights are checked as part of the generator system circuitry (412-A-96-61-00-04A-320A-A / 01923 and 412-A-96-61-00-04A-320B-A / 01924).

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Generator No. 1 and No. 2 Overheat Caution Light Systems - Operational Check

Procedure

1. Install a temporary jumper between the No. 1 starter-generator temperature sensor terminals.
2. Check that the ENG 1 GEN OVHT caution light illuminates.
3. Remove the temporary jumper.
4. Check that the ENG 1 GEN OVHT caution light extinguishes.
5. Install a temporary jumper between the No. 2 starter-generator temperature sensor terminals.
6. Check that the ENG 2 GEN OVHT caution light illuminates.
7. Remove the temporary jumper.
8. Check that the ENG 2 GEN OVHT caution light extinguishes.

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Inverter No. 1 and No. 2 Caution Light Systems - Operational Check

Procedure

1. The inverter No. 1 and No. 2 caution light systems are checked as a portion of the inverter circuitry (412-A-96-62-00-01A-320A-A / 01939).

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Engine Oil Pressure Caution Light Systems - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

NOTE

When you do the following checks, a pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 50 PSIG (345 kPa) must be used.

1. Connect a pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) to the engine 1 oil pressure switch (S73/8S5) and apply pressure.
2. Check that the ENG 1 OIL PRESSURE caution light extinguishes at an increasing pressure of 40 PSIG (276 kPa).
3. Relieve the pressure on the engine 1 oil pressure switch.
4. Check that the ENG 1 OIL PRESSURE caution light illuminates at a decreasing pressure of 31 ±1.5 PSIG (214 ±10 kPa).
5. Connect the pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) to the engine 2 oil pressure switch (8S6) and apply pressure.
6. Check that the ENG 2 OIL PRESSURE caution light extinguishes at an increasing pressure of 40 PSIG (276 kPa).
7. Relieve the pressure on the engine 2 oil pressure switch.
8. Check that the ENG 2 OIL PRESSURE caution light illuminates at a decreasing pressure of 31 ±1.5 PSIG (214 ±10 kPa).

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Transmission Oil Pressure Caution Light System - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

NOTE

When you do the following checks, a pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 50 PSIG (345 kPa) must be used.

1. Connect a pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) to the transmission oil pressure switch (8S2).

2. Check that the XMSN OIL PRESS caution light extinguishes at an increasing pressure of 38 PSIG (262 kPa) maximum.

3. Relieve the pressure on the transmission oil pressure switch.

4. Check that the XMSN OIL PRESS caution light illuminates at 30 ±3 PSIG (207 ±20 kPa) decreasing pressure.

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Transmission Oil Temperature Caution Light Systems - Operational Check

Procedure

1. Connect a temporary short between the stud on top of the transmission oil temperature switch (located on transmission) and ground.
2. Check that the XMSN OIL TEMP caution light illuminates.
3. Remove the ground from the transmission oil temperature switch.
4. Check that the XMSN OIL TEMP caution light extinguishes.

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Combining Gearbox Oil Pressure Caution Light System - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Pressure Tester, 0 to 150 PSI (0 to 1034 kPa)	Barfield Model 2311F	1	or Equivalent

Procedure

NOTE

When you do the following checks, a pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) capable of measuring at least 50 PSIG (345 kPa) must be used.

1. Connect a pressure tester, 0 to 150 psi (0 to 1034 kpa) (Barfield Model 2311F) to the gearbox oil pressure switch (S90/8S8) and apply pressure.

2. Check that the C BOX OIL PRESS caution light extinguishes at an increasing pressure of 40 PSIG (276 kPa).

3. Relieve the pressure on the gearbox oil pressure switch.

4. Check that the C BOX OIL PRESS caution light illuminates at a decreasing pressure of 31 ±1.5 PSIG (214 ±10 kPa).

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Combining Gearbox Oil Temperature Caution Light Systems - Operational Check

Procedure

1. Disconnect the connector (8S11P1) from the gearbox oil temperature switch.
2. Connect a temporary short between pins A and B.
3. Check that the C BOX OIL TEMP caution light illuminates.
4. Remove the temporary short and connect the connector.
5. Check that the C BOX OIL TEMP caution light extinguishes.

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Particle Separator Caution Light System - Operational Check

Procedure

1. The PART SEP OFF caution light is checked as part of the particle separator operational check (412-A-96-04-00-11A-320A-A / 02129).

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Governor Manual Control Caution Light Systems - Operational Check

Procedure

1. The GOV MANUAL caution lights are checked as portions of the fuel control valve test (412-A-96-04-00-05A-320B-A / 02114 and 412-A-96-04-00-05A-320C-A / 02115).

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Engine No. 1 Chip Detector Caution Light Circuitry - Operational Check

Procedure

1. Disconnect the connector (8E3P1) from the engine No. 1 forward chip detector.
2. Connect a temporary short between pins A and B.
3. Check that the ENG 1 CHIP caution light illuminates.
4. Remove the temporary short and connect the connector to chip detector.
5. Disconnect the connector (8E5P1) from the engine No. 1 aft chip detector.
6. Connect a temporary short between pins A and B.
7. Check that the ENG 1 CHIP caution light illuminates.
8. Remove the temporary short and connect the connector to the chip detector.

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Engine No. 2 Chip Detector Caution Light Circuitry - Operational Check

Procedure

1. Disconnect the connector (8E4P1) from the engine No. 2 forward chip detector.
2. Connect a temporary short between pins A and B.
3. Check that the ENG 2 CHIP caution light illuminates.
4. Remove the temporary short and connect the connector to the chip detector.
5. Disconnect the connector (8E6P1) from the engine No. 2 aft chip detector.
6. Connect a temporary short between pins A and B.
7. Check that the ENG 2 CHIP caution light illuminates.
8. Remove the temporary short and connect the connector to the chip detector.

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Combining Gearbox Chip Detector Caution Light Systems - Operational Check

Procedure

1. Disconnect the connector (8E2P1) from the gearbox chip detector.
2. Connect a temporary short between pins A and B.
3. Check that the CHIP C BOX caution light illuminates.
4. Remove the temporary short and connect the connector to the chip detector.

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Transmission Chip Detector Caution Light System - Operational Check

Procedure

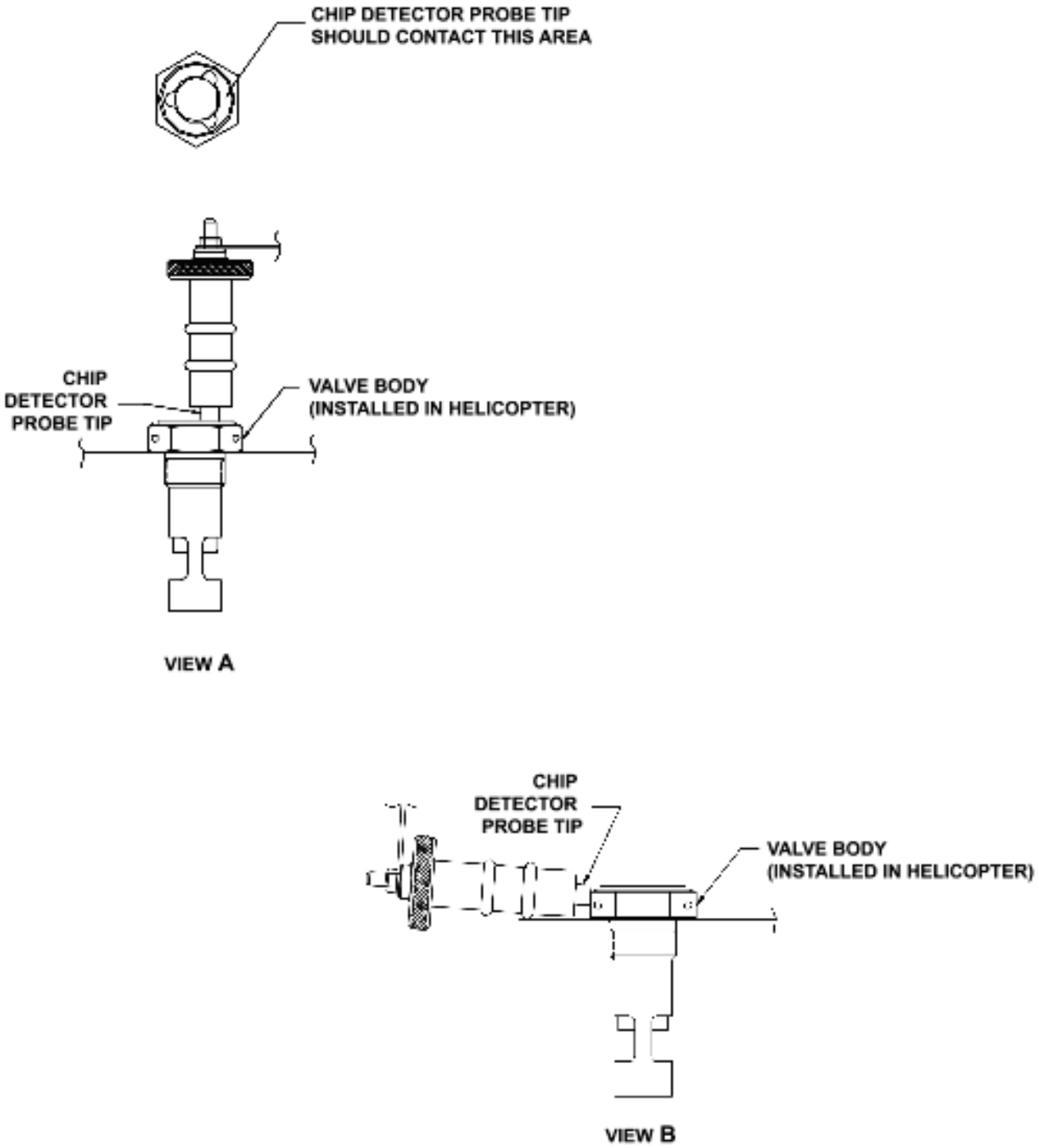
1. Disconnect the connector (8E11P1) from the transmission oil monitor/detector.
2. Connect a temporary short between pins A and B.
3. Check that the CHIP XMSN caution light illuminates.
4. Verify that the XMSN CHIP IND SUMP/OIL MONITOR indicator on the right side of the pedestal shows three white segments.
5. Remove the temporary short and the connect connector (8E11P1).
6. Reset the indicator (all black).
7. Connect a temporary short from the terminal of the transmission planetary chip detector to the ground.
8. Check that the CHIP XMSN caution light illuminates.
9. Verify that the XMSN CHIP IND PLNTY indicator on the right side of the pedestal shows three white segments.
10. Remove the temporary short.
11. Reset the indicator (all black).
12. Connect a temporary short from the terminal of the transmission upper mast chip detector to the ground.
13. Check that the CHIP XMSN caution light illuminates.
14. Verify that the XMSN CHIP IND UPPER MAST indicator on the right side of the pedestal shows three white segments.
15. Remove the temporary short.
16. Reset the indicator (all black).

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42 degree/90 degree Gearbox Chip Detector Caution Light Systems - Operational Check

Procedure

1. Connect a temporary short between the terminal (not ground terminal) on the 42° gearbox chip detector and the ground.
2. Check that the 42/90 CHIP caution light illuminates.
3. If the 42/90 CHIP caution light does not illuminate, determine the cause.
4. Remove the temporary short from the terminal on the 42° gearbox chip detector and the ground.
5. Check that the 42/90 CHIP caution light extinguishes.
6. Remove the 42° gearbox chip detector probe from the valve body.
7. Hold the chip detector probe tip firmly against the steel insert of the valve body, as shown in Figure 1, View A.
8. Check that the 42/90 CHIP caution light illuminates.
9. If the 42/90 CHIP caution light does not illuminate, hold the chip detector probe tip firmly against the side of the valve body, as shown in Figure 1, View B.
10. If the 42/90 CHIP caution light illuminates, replace the chip detector assembly. If the 42/90 CHIP caution light does not illuminate, check the helicopter wiring. For more information, refer to ASB 412-01-108.
11. Install the 42° gearbox chip detector probe into the valve body.
12. Connect a temporary short between terminal (not ground terminal) on the 90° gearbox chip detector and the ground.
13. Check that the 42/90 CHIP caution light illuminates.
14. If the 42/90 CHIP caution light does not illuminate, determine the cause.
15. Remove the temporary short from the terminal on the 90° gearbox chip detector and the ground.
16. Check that the 42/90 CHIP caution light extinguishes.
17. Remove the 90° gearbox chip detector probe from the valve body.
18. Hold the chip detector probe tip firmly against the steel insert of the valve body, as shown in Figure 1, View A.
19. Check that the 42/90 CHIP caution light illuminates.
20. If the 42/90 CHIP caution light does not illuminate, hold the chip detector probe tip firmly against the side of the valve body, as shown in Figure 1, View B.
21. If the 42/90 CHIP caution light illuminates, replace the chip detector assembly. If the 42/90 CHIP caution light does not illuminate, check the helicopter wiring. For more information, refer to ASB 412-01-108.
22. Install the 90° gearbox chip detector probe into the valve body.



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Figure 1. Chip Detector Serviceability Check (Sheet 1 of 1)

Door Lock Caution Light System - Operational Check

Procedure

1. Close and lock the baggage compartment door and the two passenger compartment doors.
2. Check that the DOOR LOCK caution light is extinguished.
3. Unlock, open, then close each of the three doors in turn.
4. Check that the DOOR LOCK caution light is illuminated when any one of three doors is unlocked and open.
5. Check that the DOOR LOCK caution light is extinguished when all doors are closed and locked.

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Door Lock Switches/Switch Levers - Adjustment Procedure

Procedure

1. Gain access to the switches (8S3, 8S4, and 8S7).

2. Slowly close either door, one at a time, and make sure that the switch is being activated by the switch lever.

NOTE

If the switch is activated by the switch lever and the DOOR LOCK caution light is illuminated, do an electrical continuity check of the system. Repair the wire or replace and adjust the door lock switch.

3. If the switch is not activated by the switch lever, adjust the switch bracket (with door in closed position) for a 0.12-inch (3.05-mm) nominal switch lever travel.

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Fuel Boost Caution Light System - Operational Check

Procedure

1. Disconnect the wire Q67A22 from the terminal 3 (TB11).
2. Verify that the ENG 1 FUEL BOOST light extinguishes.
3. Connect wire Q67A22 to the terminal 3 (TB11).
4. Verify that the ENG 1 FUEL BOOST light illuminates.
5. Disconnect wire Q68A22 from the terminal 3 (TB10).
6. Verify that the ENG 2 FUEL BOOST light extinguishes.
7. Connect wire Q68A22 to the terminal 3 (TB10).
8. Verify that the ENG 2 FUEL BOOST light illuminates.

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Fuel Boost Caution Light System - Operational Check

Procedure

1. Disconnect wire Q67C22 from the terminal 3 (1Z15TB1).
2. Verify that the ENG 1 FUEL BOOST light extinguishes.
3. Connect wire Q67C22 to the terminal 3 (1Z15TB1).
4. Verify that the ENG 1 FUEL BOOST light illuminates.
5. Disconnect wire Q68A22 from the terminal 3 (1Z14TB1).
6. Verify that the ENG 2 FUEL BOOST light extinguishes.
7. Connect wire Q68A22 to the terminal 3 (1Z14TB1).
8. Verify that the ENG 2 FUEL BOOST light illuminates.

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Fuel Boost Caution Light System - Operational Check

Procedure

1. Disconnect the connector (1S5P1) from the No. 1 fuel pressure switch (1S5).
2. Check that the NO. 1 FUEL BOOST caution light extinguishes.
3. Connect the connector (1S5P1) to the No. 1 fuel pressure switch.
4. Check that the NO. 1 FUEL BOOST caution light illuminates.
5. Disconnect the connector (1S6P1) from the No. 2 fuel pressure switch (1S6).
6. Check that the NO. 2 FUEL BOOST caution light extinguishes.
7. Connect the connector (1S6P1) to the No. 2 fuel pressure switch.
8. Check that the NO. 2 FUEL BOOST caution light illuminates.

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Fuel Filter Caution Light System - Operational Check

Procedure

1. Disconnect the connector (P182/8S9P1) from the engine No. 1 fuel filter pressure switch (S100/8S9).
2. Connect a temporary short between pins B and C of the connector.
3. Check that the ENG 1 FUEL FILTER caution light illuminates.
4. Remove the temporary short. The ENG 1 FUEL FILTER caution light should extinguish.
5. Connect the connector (P182/8S9P1) to the switch.
6. Disconnect the connector (P183/8S10P1) from the engine No. 2 fuel filter pressure switch (S101/8S10).
7. Connect a temporary short between pins C and B of the connector.
8. Check that the ENG 2 FUEL FILTER caution light illuminates.
9. Remove the temporary short.
10. The ENG 2 FUEL FILTER caution light should extinguish.
11. Connect the connector (P183/8S10P1) to the switch.

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Procedure for Low Fuel Level (10% or Less) - Operational Check

Procedure

NOTE

If there is little or no fuel in the tanks, the low level fuel switches will be closed and the FUEL LOW caution lights will be illuminated.

1. Disconnect wire E39A22 from the terminal 4 (TB11).

2. Verify that the ENG 1 FUEL LOW caution light extinguishes.

3. Connect wire E39A22 to the terminal 4 (TB11).

4. Verify that the ENG 1 FUEL LOW caution light illuminates.

5. Disconnect wire E40A22 from the terminal 4 (TB10).

6. Verify that the ENG 2 FUEL LOW caution light extinguishes.

7. Connect wire E40A22 to the terminal 4 (TB10).

8. Verify that the ENG 2 FUEL LOW caution light illuminates.

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Procedure for Fuel in Tanks (10% or More) - Operational Check

Procedure

NOTE

If there is more than 10% fuel in the tanks, the low level fuel switches will be open and the FUEL LOW caution lights will be extinguished.

1. Disconnect wire E39A22 from the terminal 4 (TB11).

2. Temporarily ground wire E39A22.

3. Verify that the ENG 1 FUEL LOW caution light illuminates.

4. Connect wire E39A22 to the terminal 4 (TB11).

5. Verify that the ENG 1 FUEL LOW caution light extinguishes.

6. Disconnect wire E40A22 from the terminal 4 (TB10).

7. Temporarily ground wire E40A22.

8. Verify that the ENG 2 FUEL LOW caution light illuminates.

9. Connect wire E40A22 to the terminal 4 (TB10).

10. Verify that the ENG 2 FUEL LOW caution light extinguishes.

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Procedure for Low Fuel Level (10% or Less) - Operational Check

Procedure

NOTE

If there is little or no fuel in the tanks, the low level fuel switches will be closed and the FUEL LOW caution lights will be illuminated.

1. Disconnect wire E39B22 from the terminal 4 (1Z15TB1).

2. Verify that the ENG 1 FUEL LOW caution light extinguishes.

3. Connect wire E39B22 to the terminal 4 (1Z15TB1).

4. Verify that the ENG 1 FUEL LOW caution light illuminates.

5. Disconnect wire E40A22 from the terminal 4 (1Z14TB1).

6. Verify that the ENG 2 FUEL LOW caution light extinguishes.

7. Connect wire E40A22 to the terminal 4 (1Z1TB1).

8. Verify that the ENG 2 FUEL LOW caution light illuminates.

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Procedure for Fuel in Tanks (10% or More) - Operational Check

Procedure

NOTE

If there is more than 10% fuel in the tanks, the low level fuel switches will be open and the FUEL LOW caution lights will be extinguished.

1. Disconnect wire E39B22 from the terminal 4 (1Z15TB1).

2. Temporarily ground wire E39B22.

3. Verify that the ENG 1 FUEL LOW caution light illuminates.

4. Connect wire E39B22 to the terminal 4 (1Z15TB1).

5. Verify that the ENG 1 FUEL LOW caution light extinguishes.

6. Disconnect wire E40A22 from the terminal 4 (1Z14TB1).

7. Temporarily ground wire E40A22.

8. Verify that the ENG 2 FUEL LOW caution light illuminates.

9. Connect wire E40A22 to the terminal 4 (1Z14TB1).

10. Verify that the ENG 2 FUEL LOW caution light extinguishes.

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Fuel Low Caution Light Systems - Operational Check

Procedure

1. The FUEL LOW caution light is checked as a portion of the fuel quantity indicator circuitry (412-A-96-12-00-11A-273A-A / 02006).

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Engine No. 1 and No. 2 Fuel Valve Caution Light Systems - Operational Check

Procedure

1. Close the No. 1 FUEL VALVE circuit breaker.

2. Set the ENGINE 1 FUEL switch to ON.

3. Check that the ENG 1 FUEL VALVE caution light illuminates momentarily and then extinguishes.

4. Set the ENGINE 1 FUEL switch to OFF.

5. Check that the ENG 1 VALVE caution light illuminates momentarily and then extinguishes.

6. Close the No. 2 FUEL VALVE circuit breaker.

7. Set the ENGINE 2 FUEL switch to ON.

8. Check that the ENG 2 FUEL VALVE caution light illuminates momentarily and then extinguishes.

9. Set the ENGINE 2 FUEL switch to OFF.

10. Check that the ENG 2 FUEL VALVE caution light illuminates momentarily and then extinguishes.

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No. 1 and No. 2 Fuel Transfer Caution Light Systems - Operational Check

Procedure

1. The NO. 1 and NO. 2 FUEL TRANS caution light systems are checked as a portion of the fuel quantity indicator circuitry (412-A-96-12-00-11A-273A-A / 02006).

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Fuel Interconnect Caution Light Systems - Operational Check

Procedure

1. The FUEL INTCON caution light is checked as a portion of the fuel interconnect valve circuitry (412-A-96-04-00-06A-320A-A / 02117).

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Fuel Crossfeed Valve Caution Light System - Operational Check

Procedure

1. Close the MASTER CAUTION circuit breaker.

2. Verify that the FUEL XFEED caution light illuminates.

3. Close both the FUEL XFEED circuit breakers.

4. Verify that the fuel crossover valve runs to the open position and stops.

5. Verify that the FUEL XFEED caution light is extinguished.

6. Press the FUEL XFEED test switch to TEST BUS 1 and then TEST BUS 2.

7. Verify that the FUEL XFEED caution light remains extinguished.

8. Open the ENGINE NO. 1 FUEL XFEED circuit breaker.

9. Press the FUEL XFEED test switch to TEST BUS 1.

10. Verify that the FUEL XFEED caution light illuminates.

11. Press the FUEL XFEED test switch to TEST BUS 2.

12. Verify that the FUEL XFEED caution light extinguishes.

13. Close the ENGINE NO. 1 FUEL XFEED circuit breaker.

14. Open the ENGINE NO. 2 FUEL XFEED circuit breaker.

15. Press the FUEL XFEED test switch to TEST BUS 2.

16. Verify that the FUEL XFEED caution light illuminates.

17. Press the FUEL XFEED test switch to TEST BUS 1.

18. Verify that the FUEL XFEED caution light extinguishes.

NOTE

The FUEL XFEED caution light shall remain extinguished in the following steps unless otherwise specified.

19. Close both the FUEL XFEED circuit breakers.

20. Verify that the fuel crossfeed valve runs to the open position and stops.

21. Place the FUEL XFEED switch to OVRD CLOSE.

22. Verify that the fuel crossfeed valve runs to the closed position and stops.

23. Place the FUEL XFEED switch to NORM.

24. Verify that the fuel crossfeed valve runs to the open position and stops.

25. Temporarily disconnect wire Q29A20N from the terminal 5 (TB11 or 1Z15TB1).

26. Verify that the fuel crossover valve remains open.

27. Temporarily disconnect wire Q37A20N from the terminal 5 (TB10 or 1Z14TB1).

28. Verify that the fuel crossfeed valve closes.

29. Reconnect wire Q29A20N to the terminal 5 (TB11 or 1Z15TB1).

30. Verify that the fuel crossfeed valve returns to open.

31. Reconnect wire Q29A20N to the terminal 5 (TB11 or 1Z15TB1).

32. Verify that the fuel crossfeed valve remains open.

33. Verify that all the wiring connections are tight and secure.

Fuel Crossfeed Valve Caution Light System - Operational Check

Procedure

1. The FUEL XFEED caution light is checked as a portion of the fuel crossfeed valve circuitry (412-A-96-04-00-07A-320A-A / 02119).

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Hydraulic Caution Light System - Operational Check

Procedure

1. The hydraulic caution light(s) is checked as part of the hydraulic control circuitry (412-A-96-05-00-02A-320A-A / 02134 and 412-A-96-05-00-02A-320C-A / 02136).

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External Power Caution Light System - Operational Check

Procedure

1. Set the BATTERY BUS 1 or BUS 2 switch to ON.
2. Open the external power access door.
3. Check that the EXTERNAL POWER caution light illuminates.
4. Close the external power access door.
5. Check that the EXTERNAL POWER caution light extinguishes.

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Battery Sensor Temperature Caution Light System - Operational Check

Procedure

1. Close the MASTER CAUTION circuit breaker.
2. Disconnect the connector (P281/8Z10P1) from the battery temperature sensor.
3. Jumper the pin B to C.
4. Check that the BATTERY TEMP caution light illuminates.
5. Remove the jumper wire.
6. Open the MASTER CAUTION LIGHT circuit breaker.
7. Connect the connector to the battery temperature sensor.

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Battery Sensor Temperature Caution Light System - Test Procedure

Procedure

NOTE

Test procedures are recommended at battery removal interval or a maximum interval of 500 hours in service.

1. Remove the sensor from helicopter.

2. Connect the sensor to the test harness, as shown in Figure 1.

3. Submerge the sensor in a temperature controlled liquid bath.

4. Increase and monitor the liquid temperature.

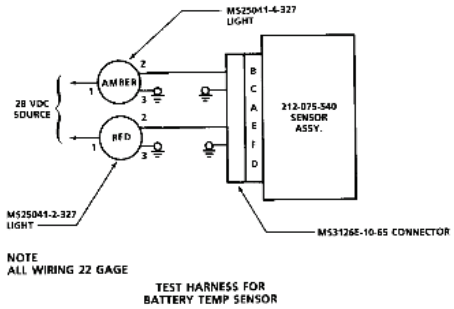
5. Verify that the sensor operation is as follows:

TEMPERATURE	INDICATION
130 ±3°F (54 ±2°C)	Amber Light ON
130 ±3°F (54 ±2°C)	Red Light ON

6. Check that both lights illuminate, then allow the lights to cool.

7. Verify that the lights extinguish within 6°F (3.5°C) of the temperature at which the lights illuminated.

8. If the unit under test fails the above check, replace the sensor.



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Figure 1. Battery Temp Sensor Test Harness (only) Wiring Diagram (Sheet 1 of 1)

Battery Sensor Temperature Caution Light System - Operational Check

Procedure

1. Close the MASTER CAUTION circuit breaker.
2. Disconnect the connector (8Z10P1) from the battery temperature sensor.
3. Jumper the pin B to C.
4. Check that the BATTERY TEMP annunciator light illuminates.
5. Remove the jumper wire.
6. Jumper the pin E to F.
7. Check that the BATTERY TEMP annunciator light illuminates.
8. Remove the jumper wire.
9. Open the MASTER CAUTION LIGHT circuit breaker.
10. Connect the connector to the battery temperature sensor.

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Battery Caution Light System - Operational Check

Procedure

1. The BATTERY caution light system is checked as part of the battery system (412-A-96-61-00-01A-320A-A / 01907).

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Heater Air Line Caution Light System - Operational Check

Procedure

1. Close the CABIN HTR circuit breaker.
2. Connect a temporary short between wire H26A22 on the heater air line overheat switch (S187/10S4) and the helicopter structure.
3. Check that the HEATER AIR LINE caution light illuminates.
4. Remove the short installed in Step 2..
5. Check that the HEATER AIR LINE caution light extinguishes.

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Heater Air Line Caution Light System - Operational Check

Procedure

1. Close the CABIN HTR circuit breaker.
2. Connect a temporary short between wire W112C22 on the heater air line overheat switch (10S2) and the helicopter structure.
3. Check that the HEATER AIR LINE caution light illuminates.
4. Remove the short installed in Step 2..
5. Check that the HEATER AIR LINE caution light extinguishes.

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AFCS Caution Light Systems - Operational Check

Procedure

1. Set the INV 1 and INV 2 switches to ON.
2. Verify that the AFCS caution light is extinguished.
3. Press the HP1 pushbutton on the Helipilot controller.
4. Verify that the HP1 legend illuminates.
5. Verify that the AFCS caution light illuminates.
6. Press the HP1 pushbutton on the Helipilot controller.
7. Verify that the HP1 legend extinguishes.
8. Verify that the AFCS caution light extinguishes.
9. Press the HP2 pushbutton on the Helipilot controller.
10. Verify that the HP2 legend illuminates.
11. Verify that the AFCS caution light illuminates.
12. Press the HP1 pushbutton on the Helipilot controller.
13. Verify that the HP1 and HP2 legend illuminates.
14. Verify that the AFCS caution light extinguishes.
15. Open the PILOT ATT SYS circuit breaker.
16. Verify that the AFCS caution light illuminates.
17. Close the PILOT ATT SYS circuit breaker.
18. Verify that the AFCS caution light extinguishes.
19. Open the CPLT ATT SYS circuit breaker.
20. Verify that the AFCS caution light illuminates.
21. Close the CPLT ATT SYS circuit breaker.
22. Verify that the AFCS caution light extinguishes.

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No. 1 Autopilot and No. 2 Autopilot Caution Light Systems - Operational Check

Procedure

1. The NO. 1 AUTOPILOT and NO. 2 AUTOPILOT caution lights are checked as part of the DAFCS functional test (412-A-96-10-00-15A-340J-A / 02189).

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Autotrim Caution Light System - Operational Check

Procedure

1. The AUTOTRIM caution light is checked as part of the DAFCS functional test (412-A-96-10-00-15A-340J-A / 02189).

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Rotor Brake Caution Light Systems - Operational Check

Procedure

1. Pull down on the ROTOR BRAKE handle until both ROTOR BRAKE caution lights on the master caution panel are illuminated.

2. Return the ROTOR BRAKE handle to its original retracted position. Observe that both ROTOR BRAKE caution lights extinguish.

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Caution Panel Caution Light Systems - Operational Check

Procedure

1. Close the MASTER CAUTION and CAUTION FAIL circuit breakers.
2. Momentarily set the MASTER CAUTION TEST switch to TEST/PNL.
3. Verify that the CAUTION PANEL caution light illuminates brightly.
4. Release the MASTER CAUTION TEST switch.
5. Verify that the CAUTION PANEL caution light extinguishes.

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Flight Data Recorder Caution Light System - Description

The FDR SYS FAIL caution light illuminates with Flight Data Acquisition Unit (FDAU) failure, Flight Data Recorder (FDR) failure, loss of 28 VDC to either unit, or if either unit is not installed. In normal operation, the circuits from the FDR and FDAU actuate the relay 27K1 to remove the ground from the FDR SYS FAIL caution light.

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Flight Data Recorder Caution Systems - Operational Check

Procedure

1. Open the FDR circuit breaker.
2. Verify that the FDR SYS FAIL caution light illuminates.
3. Close the FDR circuit breaker.
4. Verify that the FDR SYS FAIL caution light extinguishes.

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RPM Limit Warning Systems - Description

The RPM limit warning system consists of the RPM limit detector unit located on the shelf in the nose compartment, RPM caution light assembly on both the pilot and copilot instrument panels, rotor tach generator mounted on the transmission, and two gas producer tachometer generators, one mounted on each engine. The speed of each engine and rotor speed are electronically monitored by RPM limit detector unit. When a rotor overspeed condition exists, the RPM limit detector will cause the RPM caution lights on the instrument panel to flash, warning of an overspeed condition. When a rotor underspeed condition exists, the RPM limit detector will illuminate the RPM caution lights and create an audible signal for the pilot and copilot. When an engine underspeed condition exists, the RPM limit detector will illuminate the respective ENG 1/2 OUT caution light on the instrument panel.

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RPM Limit Warning System - Operational Check (Without Engines Operating)

Procedure

1. Apply external electrical power to the helicopter.
2. Close the ENG 1 RPM and ENG 2 RPM circuit breakers.
3. Check that the ENG 1 OUT and ENG 2 OUT warning lights on the instrument panel are illuminated.
4. Open the ENG 1 RPM and ENG 2 RPM circuit breakers.
5. Check that the ENG 1 OUT and ENG 2 OUT warning lights extinguish.
6. Close the ENG 1 RPM and ENG 2 RPM circuit breakers.
7. Check that the ENG 1 OUT and ENG 2 OUT warning lights illuminate.
8. Open the ENG 1 RPM and ENG 2 RPM circuit breakers.
9. Remove the external electrical power.

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RPM Limit Warning System - Operational Check and Alignment (With Engines Operating)

Procedure

NOTE

In addition to a check of warning lights, this check includes a check of audio warning signal.

1. Remove the RPM limit detector unit cover (Figure 1).

2. Operate both engines in accordance with the applicable operational instructions in the BHT-412-FM.

3. With both engines sharing the load, do as follows:

3.1. Adjust for a rotor indication of 98%.

3.2. Observe that the RPM caution lights are extinguished and no audio warning signal is audible in the headsets.

3.3. Decrease the rotor speed very slowly to point where the RPM caution lights and the limit detector unit light illuminate, with an audible warning signal heard in the headsets.

4. If the RPM caution lights and the limit detector unit light did not illuminate at the rotor speed of 95%, proceed with following:

4.1. Rotate the RTR LOW (R33) control slowly clockwise, while at a rotor speed of 95%, until the RPM caution lights just illuminate.

4.2. Vary the rotor speed slowly between 94 and 98% while observing the RPM caution lights.

4.3. Verify that the warning occurs at 95%.

5. Repeat Step 3.. through Step 4., if required.

6. If the RPM caution lights and the limit detector unit light illuminated above 95% in Step 4.3., proceed with following:

6.1. While at a rotor speed of 95%, rotate the RTR LOW (R33) control slowly counterclockwise until the RPM caution lights just extinguish, and then very slowly clockwise until the RPM caution lights just illuminate.

6.2. Vary the rotor speed slowly between 94 and 98% while observing the RPM caution lights.

6.3. Verify that the warning occurs at 95%.

6.4. Repeat Step 6.1. through Step 6.3., if required.

7. If audio was not heard at 95%, do the following:

7.1. Adjust the limit detector unit AUDIO control while at a rotor speed of 95% until audio is heard in the headsets.

7.2. Vary the rotor speed to 98% to automatically reset the audio signal circuit.

8. Position the ROTOR RPM AUDIO switch to OFF. Verify that the audio warning signal ceases.

9. Do a rotor high RPM warning check and alignment as follows:

9.1. Position the ROTOR RPM AUDIO switch to AUDIO.



DO NOT EXCEED ROTOR OR ENGINE RPM AS ESTABLISHED IN THE BHT-412-FM.

9.2. With the engine at idle and the rotor at flat pitch, set the ENGINE GOV switch to MANUAL.

9.3. Slowly increase the throttle for that engine until the RPM caution lights and the detector ROTOR light illuminate.

9.4. Observe that the RPM caution lights illuminate at an indication of 103.5% and no audio warning signal is audible in the headsets.

10. If the RPM caution lights illuminated before reaching a rotor speed of 103.5%, do as follows:

10.1. Maintain a rotor speed of 103.5%.

10.2. Rotate the RTR HIGH (R18) control clockwise until the warning occurs.

10.3. Rotate the control counterclockwise until the RPM CAUTION lights just illuminate.

11. If the RPM caution lights did not illuminate, do as follows:

11.1. Maintain a rotor speed of 103.5%.

11.2. Rotate the RTR HIGH control very slowly counterclockwise until the RPM caution lights just illuminate.

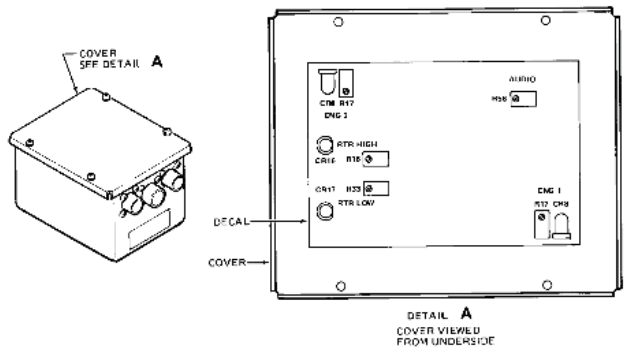
12. Vary the rotor speed slowly between 103.5% and 100.5% while observing the RPM caution lights.

13. Verify that the warning occurs at a rotor speed of 103.5% and the warning extinguishes between 103.5% and 100.5% as the rotor speed is decreased.

14. Repeat Step 10. through Step 13., if required.

15. Decrease the rotor speed.

16. With the engine at idle, set the ENGINE GOV switch to AUTOMATIC.



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Figure 1. Limit Detector Unit (Sheet 1 of 1)

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Engine Low RPM Warning Check and Alignment - Operational Check

Procedure

NOTE

The following procedure is for engine 1. The procedure is the same for engine 2, but use the engine 2 callouts.

1. Adjust for an N1 (gas producer) indication of 56%.

2. Verify that the RPM caution lights and the limit detector unit ENG 1 OUT light are extinguished.

3. Decrease the engine 1 speed slowly to 52.5%.

4. Verify that the ENG 1 OUT caution light illuminates at N1 indication of 53 ±2%.

5. If the ENG 1 OUT caution light did not illuminate, do as follows:

5.1. Maintain an engine RPM of 53%.

5.2. Rotate the limit detector unit ENG 1 (R17) control slowly clockwise until the ENG 1 OUT caution light just illuminates.

6. If the ENG 1 OUT caution light illuminated above 55%, do as follows:

6.1. Maintain an engine RPM of 53%.

6.2. Rotate the limit detector unit ENG 1 control slowly counterclockwise until the light just extinguishes and then rotate very slowly clockwise until the ENG 1 OUT caution light just illuminates.

7. Vary the engine 1 speed slowly between 50% and above 58% while observing the ENG 1 OUT caution light.

8. Verify that the warning occurs at an engine speed between 51 and 55% and that the lights extinguish within 3% higher than the percentage where the lights illuminated.

9. Repeat Step 1. through Step 8., if required.

10. Shut down both engines in accordance with the operational instructions in the BHT-412-FM.

11. Install the RPM limit detector unit cover.

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RPM Limit Detector Repair and Bench Check - Operational Check

Procedure

1. For the repair and bench check procedures, refer to TB 412-92-105.

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Engine No. 1 Fire Detection Warning - Operational Check

Procedure

1. Close the ENG 1 FIRE DETR circuit breaker.
2. Press the PRESS TO TEST fire detector test switch on the instrument panel.
3. Check that the FIRE 1 PULL warning light on the left emergency handle is illuminated.
4. Release the test switch.
5. Check that the FIRE 1 PULL warning light on the left emergency handle is extinguished.

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Engine No. 2 Fire Detection Warning - Operational Check

Procedure

1. Close the ENG 2 FIRE DETR circuit breaker.
2. Press the PRESS TO TEST fire detector test switch.
3. Check that the FIRE 2 PULL warning light on the right emergency handle is illuminated.
4. Release the test switch.
5. Check that the FIRE 2 PULL warning light on the right emergency handle is extinguished.

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Baggage Compartment Fire Detector System - Operational Check

Procedure

1. Close the BAG COMPT FIRE circuit breaker.
2. Press the BAGGAGE FIRE TEST switch on the instrument panel.

NOTE

If BAGGAGE FIRE warning light does not illuminate during test, verify serviceability of beacon and test lamps located in baggage compartment smoke detector. Replace defective lamp(s).

3. Check that the BAGGAGE FIRE warning light on the instrument panel is illuminated flashing.
4. Release the test switch.
5. Check that the BAGGAGE FIRE warning light is extinguished.
6. Introduce smoke into the baggage compartment near the baggage compartment smoke detector unit.
7. Check that the BAGGAGE FIRE warning light illuminates.
8. Remove the smoke or allow it to dissipate.
9. Check that the BAGGAGE FIRE warning light extinguishes.

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Hydraulic Filter Clogged Warning System - Description

The hydraulic filter clogged warning indicator is located on the right beam forward of the pedestal and is viewed by looking inboard through the pilot chin bubble. If the filters are not clogged, the hydraulic filter clog indicator will display a green dot in the viewing port. If the filters are clogged, the indicator will display a red dot in the viewing port.

If any of the filters becomes partially clogged, causing a hydraulic differential pressure exceeding 70 PSI (483 kPa), the contacts of either the hydraulic No. 1 or the hydraulic No. 2 switch close. This applies voltage from the HYD NO. 2 TEMP circuit breaker to the hydraulic filter clog indicator, which shows the red dot at the viewing sight. The dot in the viewing sight remains red until manually reset at the hydraulic filter clog indicator.

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Hydraulic Filter Clogged Warning System - Operational Check

Procedure

Applicable: SN: 33001-33107:

1. Refer to 412-A-96-05-00-02A-320A-A / 02134.

Applicable: SN: 33108-33213 or 36001-36999:

2. Refer to 412-A-96-05-00-02A-320C-A / 02136.

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Engine Control and Accessories Systems - Description

The engine controls and accessories systems include the fuel control heaters (412-A-98-00-00-60A-051A-A / 02520 or 412-A-98-00-00-61A-051A-A / 02521 or 412-A-98-00-00-62A-051A-A / 02522 or 412-A-98-00-00-63A-051A-A / 02523 or 412-A-98-00-00-64A-051A-A / 02524 or 412-A-98-00-00-65A-051A-A / 02525), fuel boost pumps, fuel transfer pumps, fuel shutoff valves, fuel control valves, fuel interconnect valves (412-A-98-00-00-69A-051A-A / 02529) and crossfeed valves (412-A-98-00-00-66A-051A-A / 02526 or 412-A-98-00-00-66A-051A-A / 02526 or 412-A-98-00-00-67A-051A-A / 02527 or 412-A-98-00-00-68A-051A-A / 02528), governor RPM actuator (412-A-98-00-00-29A-051A-A / 02489 or 412-A-98-00-00-30A-051A-A / 02490 or 412-A-98-00-00-31A-051A-A / 02491), sump drain valves, idle stop release solenoid (412-A-98-00-00-23A-051A-A / 02483 or 412-A-98-00-00-24A-051A-A / 02484), and engine particle separator systems. Refer for the applicable wiring diagrams.

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Fuel Control Heater - Description

The fuel control sense line heaters prevent accumulation of ice in the engine governing system. The fuel control No. 1 heater receives power from the No. 1 essential DC bus through the ENGINE NO 1 FUEL CONTR HTR circuit breaker and is controlled by the ENGINE 1 FUEL switch located on the engine and fuel control panel. The fuel control No. 2 heater is supplied power from the No. 2 essential bus through the ENGINE NO 2 FUEL CONTR HTR circuit breaker and is controlled by the ENGINE 2 FUEL switch on the engine and fuel control panel.

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Fuel Control Heater - Operational Check

Procedure

1. Close the ENGINE NO 1 FUEL CONTR HTR circuit breaker.

2. Set the ENGINE 1 FUEL switch to ON.

3. Verify that the fuel control No. 1 heater (1HR1) begins to heat.

4. Set the ENGINE 1 FUEL switch to OFF.

5. Verify that the fuel control No. 1 heater begins to cool.

6. Open the ENGINE NO 1 FUEL CONTR HTR circuit breaker.

7. Close the ENGINE NO 2 FUEL CONTR HTR circuit breaker.

8. Set the ENGINE 2 FUEL switch to ON.

9. Verify that the fuel control No. 2 heater (1HR2) begins to heat.

10. Set the ENGINE 2 FUEL switch to OFF.

11. Verify that the fuel control No. 2 heater begins to cool.

12. Open the ENGINE NO 2 FUEL CONTR HTR circuit breaker.

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Fuel Boost Pump System - Description

Each lower main cell sump assembly is equipped with a fuel boost pump that provides a flow of fuel to the respective engine (crossfeed operation enables a single boost pump to supply fuel to both engines). The No. 1 DC essential bus powers the No. 1 (left) boost pump, which is protected by the ENGINE NO 1 FUEL BOOST circuit breaker. The ENGINE 1 BOOST PUMP switch on the engine and fuel control panel controls the pump. Similarly, the No. 2 (right) boost pump is powered by the No. 2 DC essential bus through the ENGINE NO 2 FUEL BOOST circuit breaker and is controlled by the ENGINE 2 BOOST PUMP switch on the engine and fuel control panel.

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Engine 1 - Operational Check

Procedure

1. Close the ENGINE NO 1 FUEL BOOST circuit breaker.
2. Set the ENGINE 1 BOOST PUMP switch to ON.
3. Verify that the No. 1 (left) boost pump is operating.
4. Set the ENGINE 1 BOOST PUMP switch to OFF.
5. Verify that the No. 1 (left) boost pump stops operating.
6. Open the ENGINE NO 1 FUEL BOOST circuit breaker.

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Engine 2 - Operational Check

Procedure

1. Close the ENGINE NO 2 FUEL BOOST circuit breaker.
2. Set the ENGINE 2 BOOST PUMP switch to ON.
3. Verify that the No. 2 (right) boost pump is operating.
4. Set the ENGINE 2 BOOST PUMP switch to OFF.
5. Verify that the No. 2 (right) boost pump stops operating.
6. Open the ENGINE NO 2 FUEL BOOST circuit breaker.

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Fuel Transfer Pump Systems - Description

The sump assemblies in both lower forward cells are equipped with transfer pumps that pump fuel from the lower forward cells to the lower main cells through ejector pumps mounted in the lower middle cells. Fuel flowing from the transfer pumps provides motive force for the ejector pumps to empty fuel from the lower middle cells into the lower main cells. In normal operations, fuel level sensors (thermistors) in the forward fuel cells trigger the fuel signal conditioners to shut off the transfer pumps 30 seconds after the forward fuel cells empty.

The left lower forward cell transfer pump is powered by the No. 1 DC essential bus through the ENGINE NO 1 FUEL TRANS circuit breaker and in addition to the automatic shutoff feature, is controlled by the ENGINE 1 FUEL TRANS switch located on the engine and fuel control panel.

Similarly, the right lower forward cell transfer pump is powered by a 28 VDC emergency bus No. 2 through the ENGINE NO 2 FUEL TRANS circuit breaker and is controlled by the ENGINE 2 FUEL TRANS switch on the engine and fuel control panel.

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Engine 1 - Operational Check

Procedure

1. Close the ENGINE NO 1 FUEL TRANS circuit breaker.
2. Set the ENGINE 1 FUEL TRANS switch to ON.
3. Verify that the left lower forward transfer pump (1B17) is operating.
4. Set the ENGINE 1 FUEL TRANS switch to OFF.
5. Verify that the left lower forward transfer pump stops operating.
6. Open the ENGINE NO 1 FUEL TRANS circuit breaker.

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Engine 2 - Operational Check

Procedure

1. Close the ENGINE NO 2 FUEL TRANS circuit breaker.
2. Set the ENGINE 2 FUEL TRANS switch to ON.
3. Verify that the right lower forward transfer pump (1B18) is operating.
4. Set the ENGINE 2 FUEL TRANS switch to OFF.
5. Verify that the right lower forward transfer pump stops operating.
6. Open the ENGINE NO 2 FUEL TRANS circuit breaker.

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Fuel Shutoff Valve Systems - Description

Motor-operated fuel shutoff valves open or close to control the flow of fuel to the engines. The valves receive power from the DC emergency buses through the ENGINE NO 1 FUEL VALVE and ENGINE NO 2 FUEL VALVE circuit breakers.

During normal operation the No. 1 fuel shutoff valve is controlled by the ENGINE 1 FUEL switch located on the engine and fuel control panel. When pulled during an engine fire emergency, the FIRE 1 PULL handle switch, located on the instrument panel, causes the valve to close, stopping the flow of fuel to the engine No. 1. The No. 2 fuel shutoff valve is similarly controlled by the ENGINE 2 FUEL switch on the engine and fuel control panel and the instrument panel-mounted FIRE 2 PULL handle switch.

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Engine 1 - Operational Check

Procedure



FIRE EXTINGUISHER CONTAINERS ARE PRESSURIZED AND EQUIPPED WITH PYROTECHNIC CARTRIDGES. PRIOR TO ACTIVATING FIRE PULL HANDLES, FIRE EXTINGUISHER SQUIB ELECTRICAL CONNECTORS SHALL BE DISCONNECTED AND JUMPER WIRES INSTALLED ON SQUIB.

NOTE

Due to presence of diodes (1A1CR7 and 1A1CR8) on ENGINE 1 FUEL and ENGINE 2 FUEL switches, FUEL VALVE caution segment on master caution panel will illuminate when engine fuel switch position does not agree with fuel shutoff valve position after a fire pull handle is pulled. Also, illumination will occur during opening or closing of fuel shutoff valve provided an engine fuel switch is positioned to ON or OFF or when a fire pull handle is actuated.

1. Close the ENGINE NO 1 FUEL VALVE circuit breaker.

2. Set the ENGINE 1 FUEL switch to ON.

3. Verify that the No. 1 fuel shutoff valve opens.

4. Pull the FIRE 1 PULL handle switch.

5. Verify that the No. 1 fuel shutoff valve closes.

6. Push the FIRE 1 PULL handle switch in.

7. Verify that the No. 1 fuel shutoff valve opens.

8. Set the ENGINE 1 FUEL switch to OFF.

9. Verify that the No. 1 fuel shutoff valve closes.

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Engine 2 - Operational Check

Procedure



FIRE EXTINGUISHER CONTAINERS ARE PRESSURIZED AND EQUIPPED WITH PYROTECHNIC CARTRIDGES. PRIOR TO ACTIVATING FIRE PULL HANDLES, FIRE EXTINGUISHER SQUIB ELECTRICAL CONNECTORS SHALL BE DISCONNECTED AND JUMPER WIRES INSTALLED ON SQUIB.

NOTE

Due to presence of diodes (1A1CR7 and 1A1CR8) on ENGINE 1 FUEL and ENGINE 2 FUEL switches, FUEL VALVE caution segment on master caution panel will illuminate when engine fuel switch position does not agree with fuel shutoff valve position after a fire pull handle is pulled. Also, illumination will occur during opening or closing of fuel shutoff valve provided an engine fuel switch is positioned to ON or OFF or when a fire pull handle is actuated.

1. Close the ENGINE NO 2 FUEL VALVE circuit breaker.

2. Set the ENGINE 2 FUEL switch to ON.

3. Verify that the No. 2 fuel shutoff valve opens.

4. Pull the FIRE 2 PULL handle switch.

5. Verify that the No. 2 fuel shutoff valve closes.

6. Push the FIRE 2 PULL handle switch in.

7. Verify that the No. 2 fuel shutoff valve opens.

8. Set the ENGINE 2 FUEL switch to OFF.

9. Verify that the No. 2 fuel shutoff valve closes.

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Fuel Control Valve System - Description

Each engine fuel control contains a solenoid valve that provides a method of bypassing the engine fuel control governors. The fuel control valve No. 1 solenoid is powered by the DC essential bus No. 1 through the ENGINE NO 1 FUEL CONTR circuit breaker and is controlled by the ENGINE 1 GOV switch, located on engine and fuel control panel. Similarly, the fuel control valve No. 2 solenoid is powered by the DC essential bus No. 2 through the ENGINE NO 2 FUEL CONTR circuit breaker and is controlled by the ENGINE 2 GOV switch on the engine and fuel control panel.

When the switches ENG 1 GOV and ENG 2 GOV are positioned to AUTO, the solenoids are de-energized, twist grip throttle controls are manually set full-on, and speed of each engine is controlled by the governor. When the switches ENG 1 GOV and ENG 2 GOV are positioned to MANUAL, the solenoids are energized, governors are bypassed, and engine fuel control is by manual twist grip operation.

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Engine 1 - Operational Check

Procedure

1. Close the ENGINE NO 1 FUEL CONTR circuit breaker.
2. Set the ENGINE 1 GOV switch to MANUAL.
3. Verify that the fuel control valve No. 1 solenoid opens to the bypass position.
4. Verify that the ENG 1 GOV MANUAL caution light illuminates.
5. Set the ENGINE 1 GOV switch to AUTO.
6. Verify that the fuel control valve No. 1 closes.
7. Verify that the ENG 1 GOV MANUAL caution panel segment extinguishes.

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Engine 2 - Operational Check

Procedure

1. Close the ENGINE NO 2 FUEL CONTR circuit breaker.
2. Set the ENGINE 2 GOV switch to MANUAL.
3. Verify that the fuel control valve No. 2 opens to the bypass position.
4. Verify that the ENG 2 GOV MANUAL caution light illuminates.
5. Set the ENGINE 2 GOV switch to AUTO.
6. Verify that the fuel control valve No. 2 closes.
7. Verify that the ENG 2 GOV MANUAL caution light extinguishes.

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Fuel Interconnect Valve System - Description

The lower main cells are connected by a fuel line that contains a fuel interconnect valve. This motor-operated fuel interconnect valve enables interconnection or isolation of the lower main cells, as desired. Powered by DC emergency buses through the ENGINE NO 1 FUEL INTCON and ENGINE NO 2 FUEL INTCON circuit breakers, the valve is controlled by the FUEL INTCON switch, located on the engine and fuel control panel and by associated fuel interconnect relays.

The fuel interconnect valve is closed when the FUEL INTCON switch is in the normal (NORM) operating position, thus isolating the lower main cells from each other and, therefore, isolating all the lower right cells from all the lower left cells. The valve opens when the FUEL INTCON switch is positioned to OPEN, and it opens automatically when the FUEL LOW caution light on the master caution panel illuminates. Positioning the FUEL INTCON switch to OVRD CLOSE, closes the valve after it has opened automatically.

The FUEL INTCON switch is mounted on the engine and fuel control panel, and permits testing of the system on the ground. Verification of proper operation is possible by monitoring the FUEL INTCON caution light on the master caution panel. The caution panel segment should be illuminated only during valve transit.

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Fuel Interconnect Valve System - Operational Check

Procedure

NOTE

Verify that sufficient fuel is on board to cause the FUEL LOW caution panel segment to remain extinguished.

1. Close the following circuit breakers:
- ENGINE NO 1 FUEL QTY
- ENGINE NO 2 FUEL QTY
- ENGINE NO 1 FUEL INTCON
- ENGINE NO 2 FUEL INTCON
- MASTER CAUTION

2. Set the FUEL INTCON and FUEL XFEED/INTCON switches to NORM.

3. Verify that the FUEL INTCON caution panel segment is extinguished.

4. Set the FUEL INTCON switch to OPEN.

5. Verify that the FUEL INTCON caution light illuminates and extinguishes, which indicates that the valve has opened.

6. Return the FUEL INTCON switch to NORM.

7. Verify that the FUEL INTCON caution light illuminates and extinguishes, which indicates that the valve has closed.

8. Set the FUEL XFEED/INTCON switch to TEST BUS 1, then to TEST BUS 2.

9. Verify that the FUEL INTCON caution light remains extinguished, which indicates that the valve remained closed.

NOTE

In the following step, monitor the FUEL LOW and FUEL INTCON caution panel segments on the master caution panel. With the helicopter level, the FUEL INTCON segment should illuminate and then extinguish at about the moment the FUEL LOW segment illuminates. If fuel level in the upper right and upper left cells is unequal, the FUEL LOW segment may illuminate while the FUEL INTCON segment remains extinguished. If this condition occurs, additional fuel must be drained to cause the AFT fuel interconnect valve to move from closed to open.

10. Refer to 412-A-12-10-01-00A-221A-A / 00131 and drain fuel from the helicopter until the FUEL LOW caution panel segment illuminates.

11. Verify that the FUEL INTCON caution panel segment illuminates and then extinguishes, which indicates that the valve opened automatically in response to the low fuel level condition.

12. Set the FUEL INTCON switch to OVRD CLOSE.

13. Verify that the FUEL INTCON caution panel segment illuminates and then extinguishes, which indicates that the valve has closed.

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Fuel Crossfeed Valve System - Description

The motor-operated fuel crossfeed valve enables interconnection, or isolation, of the engine No. 1 and engine No. 2 fuel feed lines. The system is powered from both DC essential buses through the ENGINE NO 1 FUEL XFEED and ENGINE NO 2 FUEL XFEED circuit breakers. The valve is controlled by the FUEL XFEED switch on the engine and fuel control panel and is controlled automatically (opened) when a loss of fuel pressure in either engine feed line is sensed by the No. 1 fuel pressure switch (left) or the No. 2 fuel pressure switch (right).

The fuel crossfeed valve is closed when the FUEL XFEED switch is in the normal (NORM) operating position. Loss of fuel pressure in either engine feed line causes the fuel crossfeed relay, located in the engine and fuel control panel, to actuate, thus opening the valve and allowing a single boost pump to provide fuel to both engines. Positioning the FUEL XFEED switch to OVRD CLOSE closes the valve after it has been opened automatically.

The FUEL XFEED switch permits testing the system on the ground. Verification of proper operation of the fuel crossfeed valve and associated circuitry is possible by monitoring the FUEL XFEED caution panel segment on the master caution panel. The caution panel segment should be illuminated only during valve transit.

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Fuel Crossfeed Valve System - Operational Check

Procedure

1. Close the following circuit breakers:
- MASTER CAUTION
- ENGINE NO 1 FUEL XFEED
- ENGINE NO 2 FUEL XFEED
- ENGINE NO 1 FUEL BOOST
- ENGINE NO 2 FUEL BOOST
- ENG 1 FUEL PRESS
- ENG 2 FUEL PRESS

2. Verify that the NO. 1 FUEL BOOST and NO. 2 FUEL BOOST caution panel segments are illuminated.

3. Verify that the No. 1 FUEL PRESS and No. 2 FUEL PRESS indicators indicate zero or the low end of the dial.

4. Set the FUEL XFEED switch to NORM.

5. Set the ENGINE 1 BOOST PUMP and ENGINE 2 BOOST PUMP switches to ON.

6. Verify that the NO. 1 FUEL BOOST and NO. 2 FUEL BOOST caution panel segments extinguish.

7. Verify that both the fuel pressure indicators indicate normal fuel pressure.

8. Set the FUEL XFEED switch to TEST BUS 1 and then to TEST BUS 2.

9. Verify that there is no change in the caution panel segment indications or the fuel pressure indication.

10. Open the ENGINE NO 1 FUEL BOOST circuit breaker.

11. Verify that the No. 1 FUEL PRESS indicator continues to indicate normal fuel pressure, which indicates that the fuel crossfeed valve is open.

12. Position the FUEL XFEED switch to OVRD CLOSE.

13. Verify that the No. 1 FUEL PRESS indicator indicates zero or the low end of the dial.

14. Verify that the No. 2 FUEL PRESS indicator indicates normal fuel pressure.

15. Close the ENGINE NO 1 FUEL BOOST circuit breaker.

16. Verify that both the fuel pressure indicators indicate normal fuel pressure.

17. Open the ENGINE NO 2 FUEL BOOST circuit breaker.

18. Verify that the No. 2 FUEL PRESS indicator indicates zero or the low end of the dial.

19. Verify that the No. 1 FUEL PRESS indicator indicates normal fuel pressure.

20. Set the FUEL XFEED switch to NORM.

21. Verify that both the fuel pressure indicators indicate normal fuel pressure, which indicates that the fuel crossfeed valve is open.

22. Close the ENGINE NO 2 FUEL BOOST circuit breaker.

23. Verify that both the fuel pressure indicators indicate normal fuel pressure.

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Governor RPM Actuator System - Description

The governor RPM actuator allows engine RPM of both engines to be fine tuned over a small range centered around a value determined by the engine governors. The actuator has no effect when in MANUAL control mode. The system consists of a two spring loaded center off, RPM loop switches, pilot and copilot, and governor RPM actuator. The circuit is powered through the 5-amp GOV CONT circuit breaker. Whenever the pilot or copilot switch is positioned to INCR, the actuator will retract, and the governor RPM will increase. Positioning the switch to DECR will cause the actuator to extend and the governor RPM to decrease.

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Governor RPM Actuator System - Operational Check

Procedure

1. Close the GOV/CONTR circuit breaker.
2. Set the RPM INCR/DECR switch on the pilot collective stick to INCR.
3. Verify that the governor RPM actuator retracts.
4. Set the RPM INCR/DECR switch on the pilot collective stick to DECR.
5. Verify that the governor RPM actuator extends.
6. If a copilot collective is installed, repeat Step 2. through Step 5. with the copilot switch.

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Governor RPM Actuator System - Description

The governor RPM actuator allows the RPM of both engines to be fine tuned over a small range centered around a value determined by the engine governors through the use of the INCR/DECR switch located on the pilot and copilot collective. The engine 2 RPM actuator allows only the engine 2 to be fine tuned by using the +2/-2 switch, located on the pilot collective.

This system consists of the spring loaded, center-off, pilot RPM loop beep switch for increasing or decreasing the engine RPM on both the engines simultaneously or the RPM loop beep switch for increasing or decreasing the engine 2 RPM only. The copilot collective contains a spring loaded, center-off, RPM loop beep switch for increasing or decreasing the engine RPM on both engines simultaneously.

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Governor RPM Actuator System - Operational Check

Procedure

1. Close the GOV/CONTR circuit breaker.
2. Set the RPM INCR/DECR switch on the pilot collective stick to INCR.
3. Verify that the governor RPM actuator retracts.
4. Set the RPM INCR/DECR switch on the pilot collective stick to DECR.
5. Verify that the governor RPM actuator extends.
6. Set the +2/-2 switch on the pilot collective stick to +2.
7. Verify that the engine 2 governor actuator retracts.
8. Set the +2/-2 switch on the pilot collective stick to -2.
9. Verify that the engine 2 governor actuator extends.
10. Repeat Step 1. through Step 5. for the RPM INCR/DECR switch on the copilot collective stick.
11. Repeat Step 6., except set the copilot RPM INCR/DECR switch to INCR.
12. Verify that the governor RPM actuator extends.

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Sump Drain System - Description

The sump drain valve systems enable fuel to be drained from the lower main cell sumps by depressing the remote switches located on either side of the fuselage.

The lower left main cell sump drain valve solenoid, located on the lower left main cell sump assembly, is powered from the No. 1 DC emergency bus through the ENGINE NO 1 FUEL VALVE circuit breaker and is controlled by the left SUMP DRAIN VALVE 1 switch located on the fuselage skin.

The lower right main cell sump drain valve solenoid, located on the lower right main cell sump assembly, is powered from the No. 2 DC emergency bus through the ENGINE NO 2 FUEL VALVE circuit breaker and is controlled by the right SUMP DRAIN VALVE 2 switch located on the fuselage skin.

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Sump Drain System - Operational Check

Procedure

1. Set the ENGINE 1 FUEL switch to OFF.
2. Close the ENGINE NO 1 FUEL VALVE circuit breaker.
3. Press the left SUMP DRAIN VALVE 1 switch.
4. Verify that the lower left main cell sump drain valve solenoid opens.
5. Release the left SUMP DRAIN VALVE 1 switch.
6. Verify that the lower left main cell dump drain valve closes.
7. Set the ENGINE 1 FUEL switch to ON.
8. Press the left SUMP DRAIN VALVE 1 switch.
9. Verify that the lower left main cell sump drain valve solenoid does not open.
10. Set the ENGINE 2 FUEL switch to OFF.
11. Close the ENGINE NO 2 FUEL VALVE circuit breaker.
12. Press the right SUMP DRAIN VALVE 2 switch.
13. Verify that the lower right main cell sump drain valve solenoid opens.
14. Release the right SUMP DRAIN VALVE 2 switch.
15. Verify that the lower right main cell sump drain valve solenoid closes.
16. Set the ENGINE 2 FUEL switch to ON.
17. Press the right SUMP DRAIN VALVE 2 switch.
18. Verify that the lower right main cell sump drain valve solenoid does not open.

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Idle Stop Release Solenoid System - Description

The idle stop circuits limit the minimum throttle setting for each engine to flight idle during normal or ground run operations. The solenoids receive power from the 28 VDC essential bus No. 1 through the IDLE STOP circuit breaker. When the pilot IDLE STOP switch is placed in the ENG 1 or ENG 2 position, the engine No. 1 or engine No. 2 idle stop solenoid causes the stop to retract. This allows the throttle to be actuated further toward the OFF position, thus slowing or stopping the engine No. 1 or engine No. 2, respectively. Time delay relays keep the solenoid energized for 5 seconds after the switch is released.

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Idle Stop Release Solenoid System - Operational Check

Procedure

1. Close the IDLE STOP circuit breaker.
2. Press the IDLE STOP switch on the pilot collective stick to ENG 1.
3. Check that the idle stop 1 solenoid retracts, then after approximately 5 seconds, extends again.
4. Press the IDLE STOP switch on the pilot collective stick to ENG 2.
5. Check that the idle stop 2 solenoid retracts, then after approximately 5 seconds, extends again.
6. Repeat Step 1. through Step 5. for the IDLE STOP switch on the copilot collective.

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Applicable to: ALL
ECCN EAR99

Engine Particle Separator System - Description

The engine particle separator system filters the intake air to prevent foreign objects from entering the engines. The system consists of the engine No. 1 and No. 2 particle separator, engine No. 1 and No. 2 particle separator actuators, particle separator 1 and 2 relays, and ENG 1 PART SEP OFF and ENG 2 PART SEP OFF caution lights. The particle separator actuators are energized from the essential DC buses through the ENGINE NO 1 PART SEP and ENGINE NO 2 PART SEP circuit breakers. Control of the particle separator actuators is done by signals from the RPM LIMIT DETECTOR, which extend the engine No. 1 and engine No. 2 particle separator actuators and illuminate the ENG 1 PART SEP OFF and ENG 2 PART SEP OFF caution lights. When the engines are operating at flight idle speed or above, the particle separator control relays are de-energized, enabling the engine No. 1 and No. 2 particle separator actuators to retract and extinguish the ENG 1 PART SEP OFF and ENG 2 PART SEP OFF caution lights. The FIRE 1 PULL handle switch and the FIRE 2 PULL handle switch override the above control when energized. Pulling either fire pull handle extends the corresponding particle separator actuator and illuminates the corresponding caution light.

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Engine Particle Separator System - Operational Check

Procedure



FIRE EXTINGUISHER CONTAINERS ARE PRESSURIZED AND EQUIPPED WITH PYROTECHNIC CARTRIDGES. PRIOR TO ACTIVATING FIRE PULL HANDLES, FIRE EXTINGUISHER SQUIB ELECTRICAL CONNECTORS SHALL BE DISCONNECTED AND JUMPER WIRES INSTALLED ON SQUIB (Figure 1).

1. Check the engine No. 1 particle separator as follows:

1.1. Open the ENG 1 RPM circuit breaker.

1.2. Close the MASTER CAUTION and ENGINE NO 1 PART SEP circuit breakers.

1.3. Check that the engine No. 1 air particle separator opens.

1.4. Check that the ENG 1 PART SEP OFF caution light extinguishes.

1.5. Pull the FIRE 1 PULL handle switch.

1.6. Check that the engine No. 1 air particle separator closes.

1.7. Check that the ENG 1 PART SEP OFF caution light illuminates.

1.8. Push the FIRE 1 PULL handle switch in.

1.9. Check that the engine No. 1 air particle separator opens.

1.10. Check that the ENG 1 PART SEP OFF caution light extinguishes.

1.11. Close the ENG 1 RPM circuit breaker.

1.12. Check that the No. 1 air particle separator closes.

1.13. Check that the ENG 1 PART SEP OFF caution light illuminates.

1.14. Set the ENGINE 1 PART SEP switch to OVRD ON.

1.15. Check that the air particle separator opens.

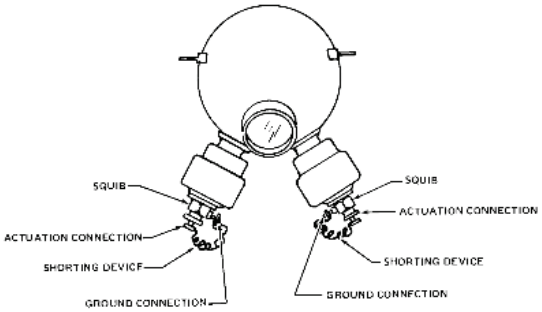
1.16. Check that the ENG 1 PART SEP OFF caution light extinguishes.

1.17. Set the ENGINE 1 PART SEP switch to NORM.

1.18. Check that the No. 1 air particle separator closes.

1.19. Check that the ENG 1 PART SEP OFF caution light illuminates.

2. Check the engine No. 2 particle separator by repeating the engine No. 1 particle separator tests, except use engine 2 controls in lieu of engine 1.



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Figure 1. Shorting Device - Installation (Sheet 1 of 1)

Flight Control Systems - Description

The flight control systems include the force trim (412-A-98-00-00-12A-051A-A / 02472 or 412-A-98-00-00-13A-051A-A / 02473 or 412-A-98-00-00-14A-051A-A / 02474) and the hydraulic control solenoid systems (412-A-98-00-00-06B-051A-A / 02566 or 412-A-98-00-00-07B-051A-A / 02567 or 412-A-98-00-00-08B-051A-A / 02568). Refer for the applicable wiring diagrams.

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Force Trim System - Description

Applicable: SN: 33001-33100:

The magnetic brake, pitch trim actuator, and roll trim actuator are wired in parallel to the 28 VDC ESS BUS NO. 2 through the control switches and the FORCE TRIM circuit breakers.

Applicable: SN: 33101-33213 or 36001-36999:

The magnetic brake, pitch trim actuator, and roll trim actuator are wired in parallel to the 28 VDC EMERGENCY BUS NO. 2 through the control switches and the FORCE TRIM circuit breakers.

Control switching is done through the series wired FORCE TRIM switch located on the MISC control panel, and the normally closed pilot and copilot pushbutton cyclic stick force trim switches. The force trim system provides spring forces to maintain the control system linkage position and thus relieve the pilot/copilot from the strain of holding the controls in a particular position. The switches allow the magnetic brakes and the trim actuators to be disengaged and repositioned as required. The force trim system is de-energized when the FORCE TRIM switch is set to OFF.

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Force Trim System - Operational Check

Procedure

NOTE

If the copilot stick is not installed, a cable assembly 412-075-037-119 must be installed in the copilot cyclic stick receptacle.

1. Close the FORCE TRIM circuit breaker.
2. Set the FORCE TRIM switch on the MISC control panel to ON.
3. Verify that the cyclic stick has holding force in the forward, aft, and lateral directions.
4. Verify that the control pedals have holding force in both directions.
5. Verify that the FT OFF light, on pilot instrument panel, is extinguished.
6. Press the FORCE TRIM RELEASE switch on the pilot cyclic stick.
7. Verify that the forces in Step 3. and Step 4. are no longer present.
8. Release the FORCE TRIM RELEASE switch on the pilot cyclic stick.
9. Verify that the forces in Step 3. and Step 4. are now present.
10. Repeat Step 3. through Step 9. with the copilot cyclic stick FORCE TRIM RELEASE switch.
11. Set the FORCE TRIM switch on the MISC control panel to OFF.
12. Open the FORCE TRIM circuit breaker.

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Hydraulic Control Systems - Description

The hydraulic bypass solenoid systems receive electrical power from the 28 VDC essential buses through the NO. 1 HYD SYST and NO. 2 HYD SYST circuit breakers. The hydraulic bypass solenoid valves are controlled by the switches on the miscellaneous control panel. The valves are normally de-energized (open) when the switches are in the ON position and energized (closed) when the switches are placed to OFF.

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Hydraulic Control Systems - Operational Check

Procedure

NOTE

An electrical interlock prevents both hydraulic systems from being switched off at the same time. If one system is off and the second system is switched off, the second system will remain on.

1. Close the following circuit breakers:
- NO. 1 HYD SYST
- NO. 2 HYD SYST
- NO. 1 HYD TEMP
- NO. 2 HYD TEMP
- MASTER CAUTION

2. Place the NO. 1 and NO. 2 HYD SYS switches ON.

3. Check that the HYDRAULIC caution lights illuminate.

4. Check that the No. 1 and No. 2 hydraulic temperature indicators indicate ambient temperature.

5. Disconnect the hydraulic 1 filter switch connector (P174/9S1P1).

6. Place a jumper between pins C and B.

7. Check that the hydraulic filter clog indicator (located in the right side of the nose below and forward of pilot pedals) displays a red indication.

8. Remove the jumper.

9. Press the reset plunger on the hydraulic filter clog indicator.

10. Check that the indicator displays a green indication.

11. Repeat Step 5. through Step 10. for the hydraulic 2 filter switch connector.

12. Connect and secure the connectors to the hydraulic 1 and 2 filter switches.

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Hydraulic Control Systems - Systems Check

Procedure

NOTE

Maintain proper hydraulic pressure during these tests. Failure to do so may result in unsatisfactory test results.

1. Apply external hydraulic pressure, 750 PSIG (5171 kPa) minimum, to the helicopter hydraulic system.

2. Check that the HYDRAULIC caution light extinguishes.

3. Verify that the hydraulic filter clog indicator displays green indication.

4. Operate the cyclic stick, collective stick, and pedal controls.

5. Check that only normal force is required to operate the controls.

6. Set the NO. 1 HYD SYS switch to OFF.

7. Check that the HYDRAULIC caution light illuminates.

8. Check that only normal force is required to operate the cyclic and collective controls.

9. Check that pedal controls are harder.

10. Set the NO. 2 HYD SYS switch to OFF.

11. Check for the same results as in Step 8. and Step 9., and the hydraulic caution light remains illuminated (No. 2 hydraulic does not turn off).

12. Set the NO. 1 HYD SYS switch to ON.

13. Check that the HYDRAULIC caution light remains illuminated.

14. Verify that only normal force is required to operate the cyclic, collective, and pedal controls.

15. Set the NO. 2 HYD SYS switch to ON.

16. Check that the HYDRAULIC caution light extinguishes.

17. Disconnect the No. 1 hydraulic temperature switch connector (P106/8S13P1) from the No. 1 hydraulic temperature switch.

18. Place a jumper between pins C and B.

19. Check that the HYDRAULIC caution light illuminates.

20. Remove the jumper.

21. Disconnect the No. 2 hydraulic temperature switch connector (P105/8S14P1) from the No. 2 hydraulic temperature switch.

22. Place a jumper between pins C and B.

23. Check that the HYDRAULIC caution light illuminates.

24. Remove the jumper.

25. Connect and secure the connectors to the hydraulic temperature switches.

26. Check that the HYDRAULIC caution light is extinguished.

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Hydraulic Control Systems - Operational Check

Procedure

1.

NOTE

An electrical interlock prevents both hydraulic systems from being switched off at the same time. If one system is off and the second system is switched off, the second system will remain on.

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Hydraulic Control Systems - Clogged Filter Check

Procedure

1. Close the following circuit breakers:
   - NO. 1 HYD SYST
   - NO. 2 HYD SYST
   - NO. 1 HYD TEMP
   - NO. 2 HYD TEMP
   - MASTER CAUTION
2. Place the NO. 1 and NO. 2 HYD SYS switches ON.
3. Check that the NO. 1 HYDRAULIC and NO. 2 HYDRAULIC caution lights illuminate.
4. Check that the No. 1 and No. 2 hydraulic temperature indicators indicate ambient temperature.
5. Disconnect the hydraulic 1 filter switch connector (9S1P1).
6. Place a jumper between pins C and B.
7. Check that the hydraulic filter clog indicator (located in the right side of the nose below and forward of the pilot pedals) displays a red indication.
8. Remove the jumper.
9. Press the reset plunger on the hydraulic filter clog indicator.
10. Check that the indicator displays a green indication.
11. Repeat Step 5. through Step 10. for the hydraulic 2 filter switch connector.
12. Connect and secure the connectors to the hydraulic 1 and 2 filter switches.

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Hydraulic Control Systems - Systems Check

Procedure

NOTE

Maintain proper hydraulic pressure during these tests. Failure to do so may result in unsatisfactory test results.

1. Apply external hydraulic pressure, 750 PSIG (5171 kPa) minimum, to the helicopter hydraulic system.

2. Check that the HYDRAULIC caution lights extinguish.

3. Verify that the hydraulic filter clog indicator displays green indication.

4. Operate the cyclic stick, collective stick, and pedal controls.

5. Check that only normal force is required to operate the controls.

6. Set the NO. 1 HYD SYS switch to OFF.

7. Check that the NO. 1 HYDRAULIC caution light illuminates.

8. Check that only normal force is required to operate the cyclic and collective controls.

9. Check that the pedal controls are harder.

10. Set the NO. 2 HYD SYS switch to OFF.

11. Check for the same results as in Step 8. and Step 9., and the hydraulic caution light remains illuminated (No. 2 hydraulic does not turn off).

12. Set the NO. 1 HYD SYS switch to ON.

13. Check that the NO. 1 HYDRAULIC caution light extinguishes.

14. Check that the NO. 2 HYDRAULIC caution light illuminates.

15. Verify that only normal force is required to operate the cyclic, collective, and rudder controls.

16. Set the NO. 2 HYD SYS switch to ON.

17. Check that the NO. 2 HYDRAULIC caution light extinguishes.

18. Disconnect the No. 1 hydraulic temperature switch connector (8S13P1) from the No. 1 hydraulic temperature switch (8S13).

19. Place a jumper between pins C and B.

20. Check that the NO. 1 HYDRAULIC caution light illuminates.

21. Disconnect the No. 2 hydraulic temperature switch connector (8S14P1) from the No. 2 hydraulic temperature switch (8S14).

22. Place a jumper between pins C and B.

23. Check that the NO. 2 HYDRAULIC caution light illuminates.

24. Remove the jumpers.

25. Connect and secure the connectors to the hydraulic temperature switches.

26. Check that the NO. 1 HYDRAULIC and NO. 2 HYDRAULIC caution lights extinguish.

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Heating Systems - Description

Heating systems (electrical portion) include the cabin heater (412-A-98-00-00-33B-051A-A / 02593, 412-A-98-00-00-34B-051A-A / 02594, 412-A-98-00-00-35B-051A-A / 02595, 412-A-98-00-00-36B-051A-A / 02596) and vent blower systems (412-A-98-00-00-24B-051A-A / 02584). Refer for the applicable wiring diagrams.

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Applicable to: ALL
ECCN EAR99

Cabin Heater System - Description

The bleed air heating system mixes engine bleed air with ambient air and distributes mixed air in the cabin area. The system has two electrical control circuits, outlet air control, and temperature control. The outlet air control controls amount of heated air needed for cabin heating and defrosting. The temperature control combines engine bleed air with outside air for distribution in heating system. Overheat switches and relays are provided as protective devices to prevent excessively hot air from entering distribution system. Particle separator relays are provided as safety devices to cause bleed air flow to be shut off when rpm at either engine is below normal operational limits.

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Cabin Heater System - Operational Check

Procedure

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874). SQUIB CARTRIDGES ARE PYROTECHNIC DEVICES. INADVERTENT DETONATION CAN RESULT IN SERIOUS INJURY. BEFORE YOU DO ANY OPERATIONAL CHECK, THE SQUIB ELECTRICAL TERMINAL GROUND CONNECTION AND ACTUATION CONNECTION ON EACH OF THE FOUR SQUIBS MUST BE SHORTED TOGETHER (Figure 1).

1. Disconnect the battery (2BT1).

2. Open the following circuit breakers on the overhead console:
- MAIN FIRE EXT
- RES FIRE EXTG

3. Verify that all electrical power is removed from the helicopter and the helicopter is properly grounded.

4. To gain access to the fire extinguisher bottles, remove the access plates on the right and left engine cowl (412-A-52-00-00-00A-028A-A / 00775).

5. Tag and remove all the actuating and ground wires from both the main and the reserve fire extinguisher bottles.

6. Install the shorting devices between the ground and the actuation connections on all four squibs (Figure 2).

7. Open the ROTOR RPM, ENG 1 and the ENG 2 RPM CAUTION/WARNING circuit breakers.

8. Close the CABIN HTR circuit breaker.

9. Set the AFT OUTLET switch to ON.

9.1. Result

- The aft outlet valve and the door post outlet valve open.

10. Move the heating bleed air control lever, located on the right side of the pedestal, below the instrument panel, to the full ON position.

10.1. Result

- The aft outlet valve and the door post outlet valve return to the closed position.

11. Move the heating bleed air control lever to the full OFF position.

11.1. Result

- The aft outlet valve and the door post outlet valve return to the maximum open position.

12. Set the aft outlet switch to OFF.

12.1. Result

- The aft outlet valve and the door post outlet valve close.

13. Set the HEATER switch to ON.

13.1. Result

- Both bleed air valves energize.

Applicable: SN: 33001-33100:

14. Connect a temporary short between wire H10B22 on the heater overheat switch and the helicopter structure.

Applicable: SN: 33101-33213 or 36001-36999:

15. Connect a temporary short between wire H10C22 on the heater overheat switch and the helicopter structure.

15.1. Result

- After a 30-second time delay, both bleed air valves de-energize and the CABIN HTR circuit breaker trips.

16. Remove the short installed in Step 6. and reset the CABIN HTR circuit breaker.

16.1. Result

- Both bleed air valves energize.

17. Pull out the FIRE 1 PULL handle switch.

17.1. Result

- Both bleed air valves de-energize.

18. Push in the FIRE 1 PULL handle switch.

18.1. Result

- Both bleed air valves energize.

19. Pull out the FIRE 2 PULL handle switch.

19.1. Result

- Both bleed air valves de-energize.

20. Push in the FIRE 2 PULL handle switch.

20.1. Result

- Both bleed air valves are energized.

21. Close the ENG 1 RPM, ENG 2 RPM, and ROTOR RPM CAUTION/WARNING circuit breakers.

21.1. Result

- Both bleed air valves de-energize.

22. Open the ENG 1 RPM, ENG 2 RPM, and ROTOR RPM CAUTION/WARNING circuit breakers.

22.1. Result

- Both bleed air valves are energized.

23. Rotate the cabin temperature controller from the HOT to COLD position.

23.1. Result

- The cabin temperature motor shaft rotates counterclockwise (viewed looking at the bleed air duct).

24. Repeat Step 23., and rotate from the COLD to HOT position and check for a clockwise rotation.

24.1. Result

- The cabin temperature motor shaft rotates clockwise (viewed looking at the bleed air duct).

25. Do an operational check of the bleed air pressure valve with both engines running as follows:

25.1. Start both engines in accordance with the Flight Manual procedures and set the SYSTEM SELECTOR switch to HEATER.

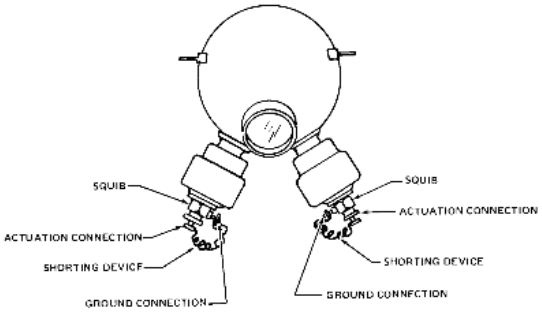
25.2. Shut down engine No. 1 after reaching the operating RPM. The cabin heater will automatically cease to function even if the bleed air valve in system 2 is operating normally.

25.3. A faulty valve will permit the cabin heater to continue to function.

25.4. Start engine No. 1 and shut down engine No. 2. The cabin heater will automatically cease to function even if the bleed air valve in system 1 is operating normally.

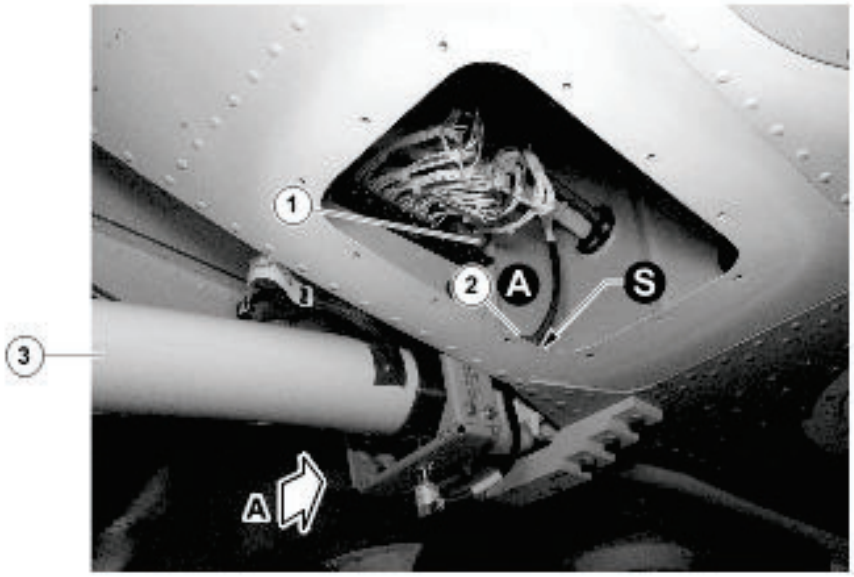
25.5. Shut down the engine.

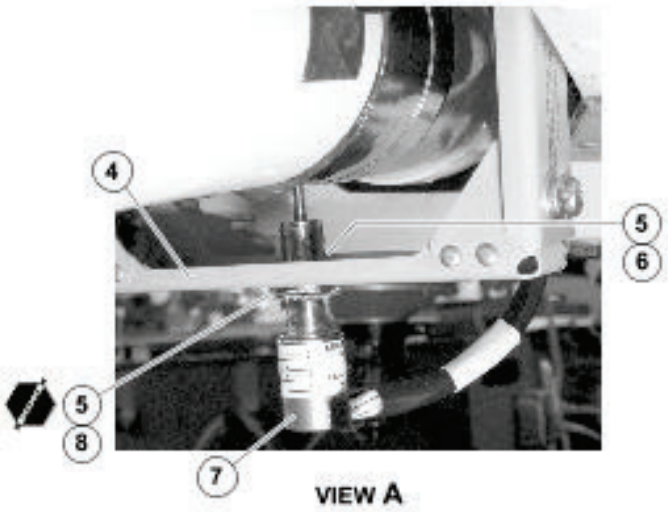
26. Turn the electrical power off.



ICN-412-A-960400-B-97499-00001-A-001-01

Figure 1. Shorting Device - Installation (Sheet 1 of 1)





1. Terminal board module 9TB3-U
2. Grommet
3. Forward crosstube
4. Support bracket
5. Jam nut
6. Lockwasher
7. Weight-on-gear (WOG) switch
8. Key washer

A ADHESIVE (C-325)

LOCKWIRE (C-405)

S SEALANT (C-251)

ICN-412-A-960600-B-97499-00001-A-001-01

Figure 2. Weight-On-Gear (WOG) Switch - Maintenance Practices (Sheet 1 of 1)

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Cabin Heater System - Operational Check

Procedure

WARNING

OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874). SQUIB CARTRIDGES ARE PYROTECHNIC DEVICES. INADVERTENT DETONATION CAN RESULT IN SERIOUS INJURY. BEFORE YOU DO ANY OPERATIONAL CHECK, THE SQUIB ELECTRICAL TERMINAL GROUND CONNECTION AND ACTUATION CONNECTION ON EACH OF THE FOUR SQUIBS MUST BE SHORTED TOGETHER (Figure 1).

1. Disconnect the battery (2BT1).

2. Open the following circuit breakers on the overhead console:
- MAIN FIRE EXT (2620CB1)
- RESERVE FIRE EXT (2629CB2)
- BUS MON (9250CB1)

3. Verify that all electrical power is removed from the helicopter and the helicopter is properly grounded.

4. To gain access to the fire extinguisher bottles, remove the access plates on the right and left engine cowl (412-A-52-00-00-00A-028A-A / 00775).

5. Tag and remove all the actuating and ground wires from both the main and the reserve fire extinguisher bottles.

6. Install the shorting devices between the ground and the actuation connections on all four squibs (Figure 2).

7. Close the following circuit breakers on the overhead console:
- CABIN HTR (2140CB1)
- MAIN FIRE EXT (2620CB1)
- RESERVE FIRE EXT (2629CB2)
- BUS MON (9250CB1)

NOTE

Use an external power unit, if available, to prevent the battery from being drained of its stored electrical power.

8. Connect the battery (2BT1) and/or apply the external ground power.

9. On the overhead console, set the AFT OUTLET switch (10S7) to ON.

9.1. Result

- Aft outlet valve (10L5) and door post outlet valve (10L2) open.

10. Move the heating bleed air control lever, located on the right side of the pedestal, below the instrument panel, to full ON position.

10.1. Result

- Aft outlet valve (10L5) and door post outlet valve (10L2) return to the closed position.

11. Move the heating bleed air control lever to the full OFF position.

11.1. Result

- Aft outlet valve (10L5) and door post outlet valve (10L2) return to the maximum open position.

12. Set the AFT OUTLET switch (10S7) to OFF.

12.1. Result

- Aft outlet valve (10L5) and door post outlet valve (10L2) close.

13. On the overhead console, set the HEATER switch (10S8) to ON.

13.1. Result

- Both bleed air valves (10L3 and 10L4) open.

14. Connect a jumper between wire H10C22 on heater overheat switch (aft) (10S3) and the helicopter structure.

14.1. Result

- After 30 second time delay, both bleed air valves (10L3 and 10L4) close.

15. Remove the jumper installed in Step 14..

15.1. Result

- Both bleed air valves (10L3 and 10L4) open.

16. Connect a jumper between wire W112C22 on the heater airline overheat switch (10S2) and the helicopter structure.

16.1. Result:

- CABIN HEAT HOT caution is displayed in the CAS message window.

17. Remove the jumper installed in Step 16..

17.1. Result:

- CABIN HEAT HOT caution is not displayed in the CAS message window.

18. Push the FIRE ARM ENG 1 switch (2610DS1) to select.

18.1. Result

- Both bleed air valves (10L3 and 10L4) close.

19. Push the FIRE ARM ENG 1 switch (2610DS1) to unselect.

19.1. Result

- Both bleed air valves (10L3 and 10L4) open.

20. Push the FIRE ARM ENG 2 switch (2610DS2) to select.

20.1. Result

- Both bleed air valves (10L3 and 10L4) close.

21. Push the FIRE ARM ENG 2 switch (2610DS2) to unselect.

21.1. Result

- Both bleed air valves (10L3 and 10L4) open.

22. Rotate the cabin temperature controller from HOT to COLD.

22.1. Result

- Cabin temperature motor shaft rotates counterclockwise (viewed looking at bleed air duct).

23. Repeat Step 22., rotating from COLD to HOT position and checking for a clockwise rotation.

24. Set the HEATER switch (10S8) to OFF.

25. Remove the external electrical power and disconnect the battery (2BT1).

26. Open the following circuit breakers on the overhead console:
- MAIN FIRE EXT (2620CB1)
- RESERVE FIRE EXT (2629CB2)
- BUS MON (9250CB1)

27. Verify all electrical power is removed from the helicopter and the helicopter is properly grounded.

28. Gain access to the fire extinguisher bottles by removing the access plates on the right and left engine cowl (412-A-52-00-00-00A-028A-A / 00775).

29. Remove the shorting devices between the ground and the actuation connections on all four squibs (Figure 2).



FAILURE TO PROPERLY CONNECT WIRING MAY RESULT IN FIRE EXTINGUISHING SYSTEM FAILURE OR DISCHARGE OF WRONG FIRE EXTINGUISHER BOTTLE.

30. Remove the tags and connect the wires to the main and reserve fire extinguisher bottles.

31. Check that the connections are properly mated, tight, and secure.

32. Close the following circuit breakers on the overhead console:

- MAIN FIRE EXT (2620CB1)
- RESERVE FIRE EXT (2629CB2)
- BUS MON (9250CB1)

33. Install the access plates on the right and left engine cowl.



A QUALIFIED PERSON MUST BE AT THE HELICOPTER CONTROLS DURING THE FOLLOWING PROCEDURE.

34. Perform an operational check of the bleed air pressure valve with both engines running as follows:

34.1. Start both engines (BHT-412-FM-5).

34.2. On the overhead console, set the HEATER switch (10S8) to ON.

34.2.1. Result:

- HEATER switch is electromagnetically held in the ON position.
- Cabin heater is operational.

34.3. Shut down Engine No. 1 after reaching the operating RPM.

34.3.1. Result

- Engine No. 1 out condition is set. EAPS No.1 Relay (1K5) is energized resulting in loss of power to the HEATER switch (10S8). Switch is electromagnetically held in the ON position. Loss of power releases the switch to the OFF position.
- Cabin heater will automatically cease to function even if the bleed air valve in system 2 is operating normally.

NOTE

A faulty valve will permit the cabin heater to continue to function.

34.4. Restart Engine No. 1.

34.5. Set the HEATER switch (10S8) to ON.

34.5.1. Result:

- HEATER switch is electromagnetically held in the ON position.
- Cabin heater is operational.

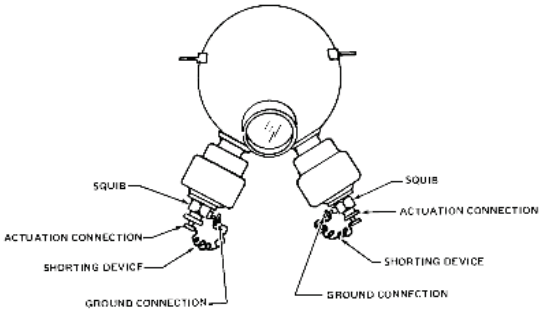
34.6. Shut down Engine No. 2 after reaching the operating RPM.

34.6.1. Result

- Engine No. 2 out condition is set. EAPS No.2 Relay (1K6) is energized resulting in loss of power to the HEATER switch (10S8). Switch is electromagnetically held in the ON position. Loss of power releases the switch to the OFF position.

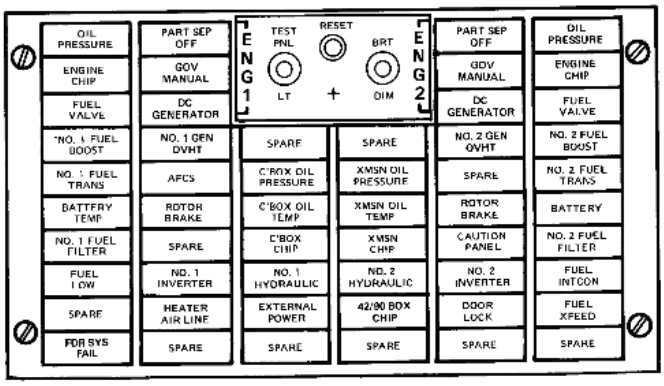
- Cabin heater will automatically cease to function even if the bleed air valve in system 1 is operating normally.

34.7. Shut down the engine (BHT-412-FM-5).



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Figure 1. Shorting Device - Installation (Sheet 1 of 1)



ICN-412-A-960300-B-97499-00002-A-001-01

Figure 2. Master Caution Panel (S/N 33108 through 36086) (Sheet 1 of 1)

Cabin Temperature Control - Rigging

Procedure

1.

NOTE

For the removal, installation, and rigging procedures, refer to 412-A-21-20-00-00A-028A-A / 00190.

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Cabin Temperature Control - Operational Check

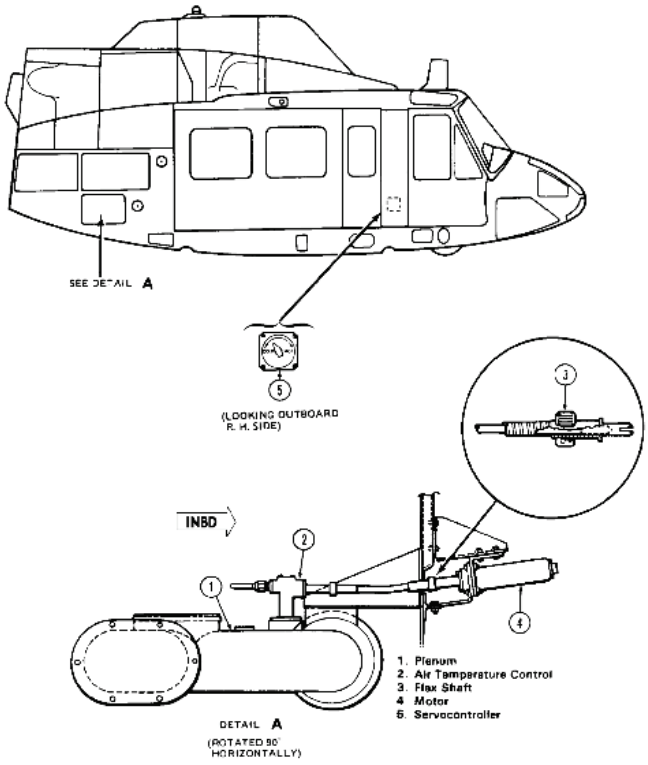
Procedure

1. Close the CABIN HTR circuit breaker.
2. Set the HEATER switch to ON.
3. Check that the bleed air valves actuate.
4. Rotate the servocontroller (5, Figure 1) through its entire range, from COLD position to HOT position.
5. The motor (4) travel shall correspond with the control position within 3%.

NOTE

The direction of motor (4) travel shall be the same as the direction of the servocontroller (5) knob rotation when viewed from the end of the motor, opposite the shaft. The motor shall develop a minimum torque of 5 inch-pounds (0.56 Nm) throughout total travel. The time required to travel through 270° of arc shall not exceed 45 seconds. The full load current should not exceed 1.0 amp with 28 VDC input voltage applied.

6. Rotate the servocontroller (5) knob from the HOT to COLD position.
7. Check that the motor travel corresponds with the control position within 3%.
8. Check that the motor travel is in the same direction as the direction of knob rotation.
9. The time required to travel through 270° of arc shall not exceed 45 seconds.



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Figure 1. Air Temperature Control Group (Sheet 1 of 1)

Vent Blower System - Description

A right and left vent blower is provided. The electrical circuit includes two 28 VDC blower motors, control switch, and VENT BLO circuit breaker. Both blowers are energized with the VENT BLOWER switch on the overhead console.

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Vent Blower System - Operational Check

Procedure

1. Set the NON-ESNTL BUS switch to MANUAL.
2. Close the VENT BLO circuit breaker.
3. Set the VENT BLOWER switch to ON.
4. Check that the right and left vent blowers operate.
5. Set the VENT BLOWER switch to OFF.
6. Check that the right and left vent blowers are not operating.

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Miscellaneous Systems - Description

The miscellaneous systems include the windshield wiper systems (412-A-98-00-00-40B-051A-A / 02600 or 412-A-98-00-00-41B-051A-A / 02601), engine fire extinguisher systems (412-A-98-00-00-33A-051A-A / 02493 or 412-A-98-00-00-34A-051A-A / 02494), weight-on-gear system (412-A-98-00-00-59B-051A-A / 02619), powered passenger step system (412-A-98-00-00-26B-051A-A / 02586 or 412-A-98-00-00-27B-051A-A / 02587 or 412-A-98-00-00-28B-051A-A / 02588 or 412-A-98-00-00-29B-051A-A / 02589), and emergency float system (412-A-98-00-00-42B-051A-A / 02602 or 412-A-98-00-00-43B-051A-A / 02603 or 412-A-98-00-00-44B-051A-A / 02604). Refer for the applicable wiring diagrams.

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Windshield Wiper Systems - Description

Windshield wipers may be operated individually by positioning the pilot or copilot switch to low, medium, high, and de-energized in the off and park position. Park position is a spring-loaded position (returns to OFF) for stowing the wiper blades to a location where vision is not obstructed.

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Windshield Wiper Systems - Operational Check

Procedure



DO NOT OPERATE WIPERS ON DRY WINDSHIELD. SUITABLY PROTECT WINDSHIELD AGAINST SCRATCHING BY WIPER BLADES.

1. Close the WIND SHIELD WIPER PILOT and CPLT circuit breakers.
2. Set the WIPERS PILOT switch to the LO, MED, HI, OFF, and PK positions respectively.
3. Check that only the pilot wiper operates in accordance with the WIPERS PILOT switch position.
4. Set the WIPERS COPILOT switch to the LO, MED, HI, OFF, and PK positions respectively.
5. Check that only the copilot wiper operates in accordance with the WIPERS COPILOT switch position.
6. Open the WIND SHIELD WIPER PILOT and CPLT circuit breakers.

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Engine Fire Extinguisher Systems - Description

If a fire occurs in the engine compartment No. 1, the No. 1 fire detector element and fire detector amplifier causes the FIRE 1 PULL handle switch to illuminate. In the event fire extinguishing action is required, the actuating FIRE 1 PULL handle switch will arm the main and reserve the fire extinguisher circuit for engine 1. The pull handle will also close the particle separator door, shut off fuel to engine 1, and de-energize both the engine 1 and engine 2 bleed air valves. Actuating the FIRE EXT switch to MAIN will discharge the main fire extinguisher bottle into engine 1. If the above action does not extinguish the fire, a reserve fire extinguisher bottle can be discharged into the engine compartment No. 1 by placing the FIRE EXT switch to RESERVE. If fire occurs in the engine compartment No. 2, the No. 2 fire detector element and fire detector amplifier causes the FIRE 2 PULL handle switch to illuminate. In the event fire extinguishing action is required, actuating the FIRE 2 PULL handle switch will arm the main and reserve fire extinguisher circuit for engine 2. The pull handle will also close the particle separator door, shut off fuel to engine 2, and de-energize both the engine 2 and engine 1 bleed air valves. Actuating the FIRE EXT switch to MAIN will discharge the main fire extinguisher bottle into engine 2. If the above action does not extinguish the fire, a reserve fire extinguisher bottle can be discharged into the engine compartment No. 2 by placing the FIRE EXT switch to RESERVE.

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Engine Fire Extinguisher Systems - Operational Check

Procedure



A SHORTING DEVICE MUST BE INSTALLED ON THE VACANT TERMINAL OF THE SQUIB CARTRIDGE BEFORE DOING ANY MAINTENANCE. A CARTRIDGE IS AN EXPLOSIVE DEVICE. INADVERTENT DETONATION OF THE CARTRIDGE MAY CAUSE INJURY.



DO NOT USE A BATTERY POWERED CONTINUITY LIGHT, OR ANY SIMILAR DEVICE IN AN ATTEMPT TO TEST SQUIB ASSEMBLY. THE USE OF TEST DEVICES OR OHMMETER THAT PASS MORE THAN 30 MA MIGHT DETONATE SQUIB ASSEMBLY AND CAN AFFECT ITS USEFUL LIFE.

1. Make sure that the external power is removed and the BATTERY BUS 1 and BUS 2 switches are set to OFF.
2. Open the following circuit breakers:
   - ENG 1 FIRE DET
   - ENG 2 FIRE DET
   - INST PILOT
   - MAIN FIRE EXT
   - RES FIRE EXT
3. To gain access to the fire extinguisher bottles, remove the access plates on the right and left engine cowl.
4. Remove all the actuating and ground wires from both the main and reserve fire extinguisher bottles.
5. Restrain the loose wires from contact with the airframe and remaining wires.
6. Install a shorting device (wire) between the actuate and ground connection of each squib (Figure 1).
7. Apply electrical power to the helicopter buses.
8. Close the MAIN FIRE EXT circuit breaker.
9. Pull the FIRE 1 PULL handle.
10. Set the FIRE EXT selector switch to MAIN.
11. Check that the squib firing voltage (28 VDC) is present at wire W3D20.
12. Check that there is no squib firing voltage on the remaining fire extinguisher wires.
13. Set the FIRE EXT selector switch to RESERVE.
14. Check for zero voltage on all disconnected wires.
15. Set the FIRE EXT selector switch to OFF.
16. Check for zero voltage on all disconnected wires.
17. Reset the FIRE 1 PULL handle.
18. Pull the FIRE 2 PULL handle.
19. Position the FIRE EXT selector switch to MAIN.
20. Check that squib firing voltage is present at the end of wire W6D20.
21. Check that there is no squib firing voltage on the remaining fire extinguisher wires.
22. Set the FIRE EXT selector switch to RESERVE.
23. Check for zero voltage on all disconnected wires.

24. Set the FIRE EXT selector switch to OFF.

25. Check for zero voltage on all disconnected wires.

26. Reset the FIRE 2 PULL handle.

27. Open the MAIN FIRE EXT circuit breaker.

28. Close the RES FIRE EXT circuit breaker.

29. Pull the FIRE 1 PULL handle.

30. Set the FIRE EXT selector switch to MAIN.

31. Check for zero voltage on all disconnected wires.

32. Set the FIRE EXT selector switch to RESERVE.

33. Check that squib firing voltage is present on wire W2C20.

34. Check that there is no squib firing voltage on all the remaining fire extinguisher wires.

35. Set the FIRE EXT switch to OFF.

36. Check for zero voltage on all disconnected wires.

37. Reset the FIRE 1 PULL handle.

38. Pull the FIRE 2 PULL handle switch.

39. Set the FIRE EXT switch to MAIN.

40. Check for zero voltage on all disconnected wires.

41. Set the FIRE EXT switch to RESERVE.

42. Check that squib firing voltage is present on wire W5C20.

43. Check that there is no squib firing voltage on all the remaining fire extinguisher wires.

44. Set the FIRE EXT switch to OFF.

45. Check for zero voltage on all disconnected wires.

46. Reset the FIRE 2 PULL handle.

47. Check for zero voltage on all disconnected wires.

48. Open the RES FIRE EXTG circuit breaker.

49. Close the ENG 1 FIRE DET circuit breaker.

50. Hold the FIRE EXT PRESS TO TEST switch in the TEST position.

51. Check that the FIRE 1 PULL handle switch illuminates.

52. Close the ENG 2 FIRE DETR circuit breaker.

53. Hold the FIRE EXT PRESS TO TEST switch in the TEST position.

54. Check that the FIRE 2 PULL handle switch illuminates.

55. Open the ENG 1 FIRE DET and ENG 2 FIRE DETR circuit breakers.

56. Remove the electrical power from the helicopter buses.

57. Set the BATTERY BUS 1 and BUS 2 switches to OFF.

58. Remove the shorting device on the squibs (Figure 1).



FAILURE TO PROPERLY CONNECT WIRING MAY RESULT IN FIRE EXTINGUISHING SYSTEM FAILURE OR DISCHARGE OF WRONG FIRE EXTINGUISHER BOTTLE.

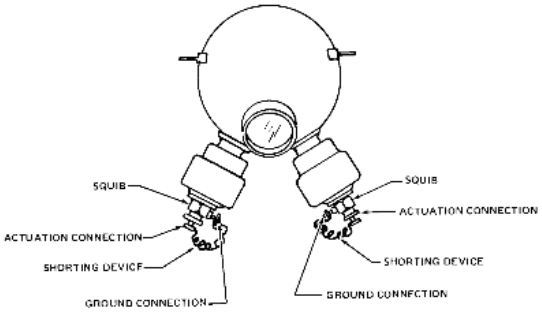
59. Connect the wires to the fire extinguisher bottles.

60. Check that the connections are properly mated, tight, and secure.

61. Close the following circuit breakers:

- ENG 1 FIRE DET
- ENG 2 FIRE DETR
- INST PILOT
- MAIN FIRE EXT
- RES FIRE EXTG

62. Install the access plates on the right and left engine cowl.



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Figure 1. Shorting Device - Installation (Sheet 1 of 1)

Weight-On-Gear Switch - Description

NOTE

The weight-on-gear switch is installed as basic equipment on helicopters S/N 36342 through 36999. It is also applicable to helicopters S/N 36248 through 36461 with WOG Switch and Annunciator Retrofit kits (BHT-412-SI-84) installed. Refer to the Service Instruction for the wiring diagram for helicopters with this kit installed.

The Weight-On-Gear (WOG) system consists of a WOG switch 3420S1 mounted on the forward crosstube and WOG override switch 3420S2 mounted on the hourmeter panel on the right side of the pedestal.

NOTE

Weight-On-Wheels (WOW) terminology may be called out in conjunction with weather radar system.

With the helicopter weight on the skid and the WOG override switch 3420S2 in the NORM position, the WOG switch provides a ground to the MST-67A mode S transponder system (if installed) AIR/GND ATCRBS inhibit circuit. It also provides a ground to the weather radar system (if installed) indicator and controller Forced Standby (FSB) WOW circuits. This prevents operation of the weather radar when the helicopter is on the ground.

Placing the guarded two position (AIR, NORM) WOG override switch 3420S2 in the AIR position removes the ground from both the transponder and weather radar. This allows testing of both systems with the helicopter on the ground.

NOTE

SN: 36248-36527:

The WOG switch also affects AHRS system on helicopters with WOG Switch and Annunciator Retrofit kits (BHT-412-SI-84) installed.

Applicable: SN: 36528-36999:

The WOG switch also provides a ground to the Attitude and Heading Reference System (AHRS) (412-A-97-22-00-00A-042A-A / 02319) ground/air relay 3407K1. With the ground applied to the relay, the relay energizes and the ground/air discretes are removed from AHRU No. 1 and AHRU No. 2. With the discretes removed, the AHRS system can be tested.

Placing the guarded two-position (AIR, NORM) WOG override switch 3420S2 in the AIR position removes the ground from ground/air relay 3407K1. This reapplies ground/air discretes to AHRU No. 1 and AHRU No. 2. With the discretes applied, testing is inhibited.

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Weight-On-Gear Annunciator - Description

NOTE

The weight-on-gear switch is installed as basic equipment on helicopters S/N 36342 through 36999. It is also applicable to helicopters S/N 36248 through 36461 with WOG Switch and Annunciator Retrofit kits installed. Refer to the Service Instruction for the wiring diagram for helicopters with this kit (BHT-412-SI-84) installed.

The Weight-on Gear (WOG) annunciator provides a visual indication to the pilot that the helicopter weight is on the skid gear and the WOG switch is closed.

Power for the WOG annunciator comes from the caution light control panel on the instrument panel. It is controlled with the BRT/DIM switch on the caution light panel. If the PILOT INSTR LT rheostat on the overhead console is rotated clockwise from OFF and the BRT/DIM switch on the caution light panel is set to DIM, the WOG annunciator will dim. Setting either the PILOT INSTR LT rheostat to OFF or the BRT/DIM switch on the caution control panel to BRT will reset the WOG annunciator to full brightness.

The annunciator can also be manually dimmed by rotating the annunciator bezel. This opens and closes an internal shutter.

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Weight-On-Gear Switch - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Insertion/Removal Tool	M81969/14-10	1	

Procedure

1. Remove the access panel on the underside of the helicopter, aft of the left side of the forward crosstube (3, Figure 1).

2. Cut and discard the lockwire from the Weight-on-Gear (WOG) switch (7).

3. Loosen and remove the two jam nuts (5), lockwasher (6), and key washer (8). Remove the switch from the support bracket (4).

4. Remove the sealant from the grommet (2) where the Weight-on-Gear (WOG) switch (7) electrical wires go through the structure.

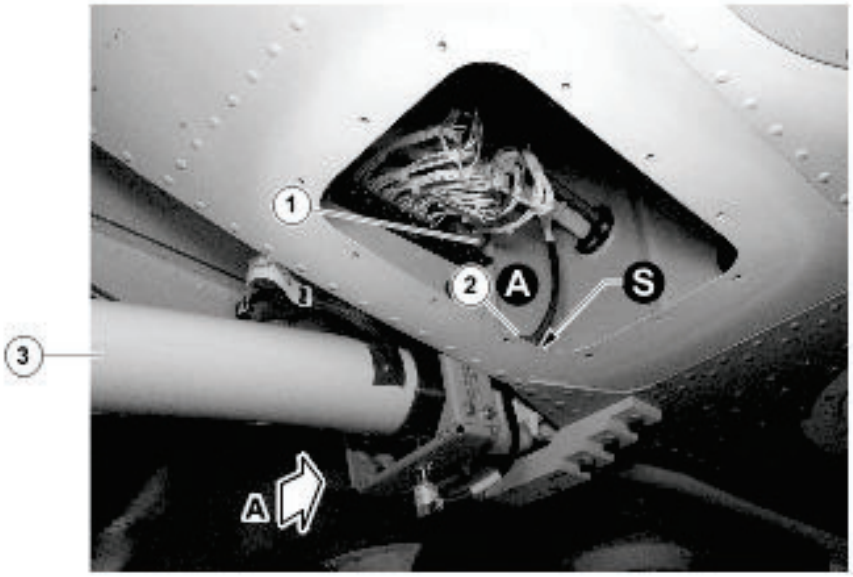
5. Locate and tag the wires that follow. With the insertion/removal tool (M81969/14-10), disconnect them from the terminal board module 9TB3-U (1).

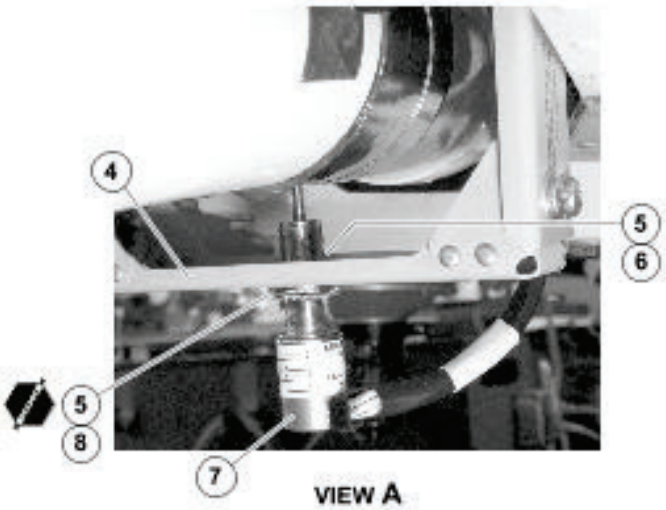
WIRE NO.	DISCONNECT FROM 9TB3
1-20	UA
2-20	UD
4-20 (1)	UG
5-20 (1)	UE

NOTE:

1 With WOG Switch and Annunciator Retrofit kits (BHT-412-SI-84) installed.

6. Locate and unstow the remaining WOG switch wires from the electrical harness. Remove the WOG switch from the helicopter.





1. Terminal board module 9TB3-U
2. Grommet
3. Forward crosstube
4. Support bracket
5. Jam nut
6. Lockwasher
7. Weight-on-gear (WOG) switch
8. Key washer

A ADHESIVE (C-325)

LOCKWIRE (C-405)

S SEALANT (C-251)

ICN-412-A-960600-B-97499-00001-A-001-01

Figure 1. Weight-On-Gear (WOG) Switch - Maintenance Practices (Sheet 1 of 1)

Weight-On-Gear - Installation

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Insertion/Removal Tool	M81969/14-10	1	

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Sealant	C-251	AR	
Adhesive	C-325	AR	

Procedure

NOTE

The lockwasher (6, Figure 1) is installed on the top side of the support bracket (4) and the key washer (8) is installed on the lower side of the bracket.

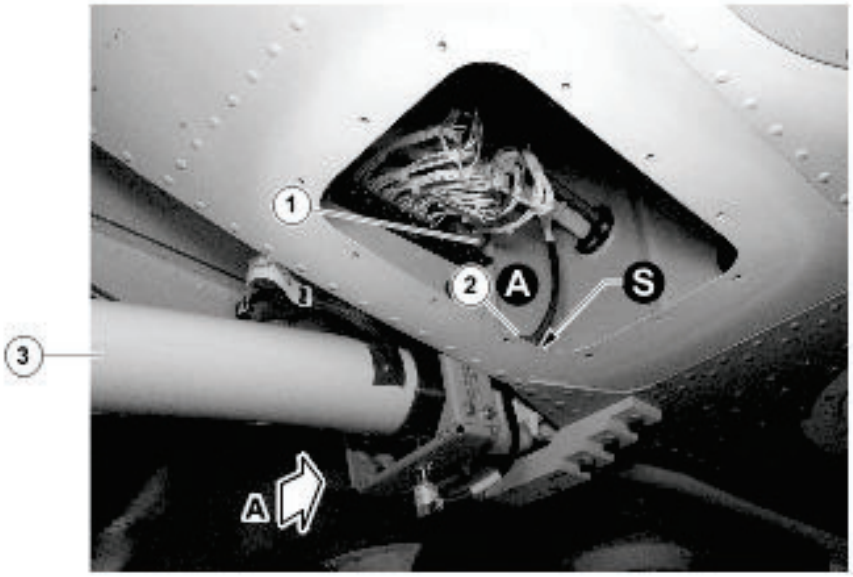
1. Install the Weight-on-Gear (WOG) switch (7) on the support bracket (4) with two jam nuts (5), lockwasher (6), and key washer (8). Tighten the jam nuts fingertight only.
2. Inspect the grommet (2) for condition. If it is loose, reattach to the structure with adhesive (C-325).
3. Route the Weight-on-Gear (WOG) switch (7) wires through the grommet (2) to the terminal board module 9TB3-U (1).
4. If the Weight-on-Gear (WOG) switch (7) is new, cut the wires to length and install new contacts (BHT-ELEC-SPM, Chapter 4, Electrical Standard Practices Manual).
5. Remove the tags and install the wires that follow in the terminal board module 9TB3-U (1) with the insertion/removal tool (M81969/14-10).

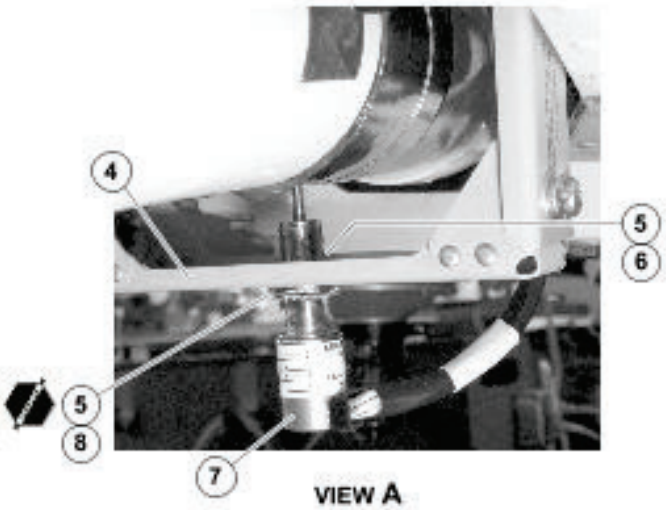
WIRE NO.	CONNECT TO 3408DS2P1
1-20	UA
2-20	UD
4-20 (1)	UG
5-20 (1)	UE

NOTE:

1 With WOG Switch and Annunciator Retrofit kits (BHT-412-SI-84).

6. Stow the remaining wires with the existing harness (BHT-ELEC-SPM, Chapter 4, Electrical Standard Practices Manual).
7. Apply sealant (C-251) to both sides of the grommet (2).
8. Install the access panel on the underside of the helicopter, aft of the left side of the forward crosstube (3).
9. Do the weight-on-gear switch adjustment procedure (412-A-96-07-00-03A-271A-A / 02156).





VIEW A

1. Terminal board module 9TB3-U
2. Grommet
3. Forward crosstube
4. Support bracket
5. Jam nut
6. Lockwasher
7. Weight-on-gear (WOG) switch
8. Key washer

A ADHESIVE (C-325)

LOCKWIRE (C-405)

S SEALANT (C-251)

ICN-412-A-960600-B-97499-00001-A-001-01

Figure 1. Weight-On-Gear (WOG) Switch - Maintenance Practices (Sheet 1 of 1)

Weight-On-Gear Switch - Adjustment

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. With the helicopter weight on the skid tube, make sure the Weight-on-Gear (WOG) switch (7, Figure 1) plunger does not contact the forward crosstube (3). If it does, do the steps that follow:

1.1. Cut and discard the lockwire from the jam nuts (5).

1.2. Loosen the jam nuts (5) and lower the switch until the plunger does not make contact. Tighten the jam nuts fingertight.

2. On the right side of the pedestal, make sure that the WOG override switch 3420S2 is in the NORM position.

Applicable: SN: 36248-36527 BHT-412-SI-87: Not installed:

3. Do the steps that follow:

3.1. If installed, remove the MST 67A transponder (412-A-97-23-00-04A-520B-A / 02352).

3.2. In the transponder rack, connect an ohmmeter from pin 79 of receptacle 3410TR1P1A to ground.

3.3. Make sure that the ohmmeter shows a closed circuit or approximately zero ohms.

Applicable: SN: 36528-36999 or (SN: 36248-36527 BHT-412-SI-87: Installed):

4. Apply electrical power. Make sure that the WOG annunciator on the pilot instrument panel is illuminated.

5. On the forward crosstube (3), cut and discard the lockwire on the jam nuts (5). Loosen the jam nuts.

6. Jack the helicopter until there is no weight on the skid gear (412-A-07-10-00-00A-172A-A / 00080).

7. With the two jam nuts (5), adjust the Weight-on-Gear (WOG) switch (7) upward until the plunger contacts the bottom of the forward crosstube (3). Continue adjusting upward until the applicable condition that follows is met:

Applicable: SN: 36248-36527 BHT-412-SI-87: Not installed:

7.1. The resistance on the ohmmeter shows an open circuit.

Applicable: SN: 36528-36999 or (SN: 36248-36527 BHT-412-SI-87: Installed):

7.2. The WOG annunciator light goes out.

7.3. With a fine permanent marker, draw an index line on the face of the bottom jam nut (5). Using the index mark as a reference, unscrew the bottom jam nut two full turns. Tighten the upper jam nut against the support bracket (4).

8. Lower the helicopter until there is weight on the skid gear (412-A-07-10-00-00A-172A-A / 00080). Make sure that the applicable condition that follows is met:

Applicable: SN: 36248-36527 BHT-412-SI-87: Not installed:

8.1. The resistance on the multimeter shows a closed circuit (approximately zero ohms).

Applicable: SN: 36528-36999 or (SN: 36248-36527 BHT-412-SI-87: Installed):

8.2. The WOG annunciator light illuminates.

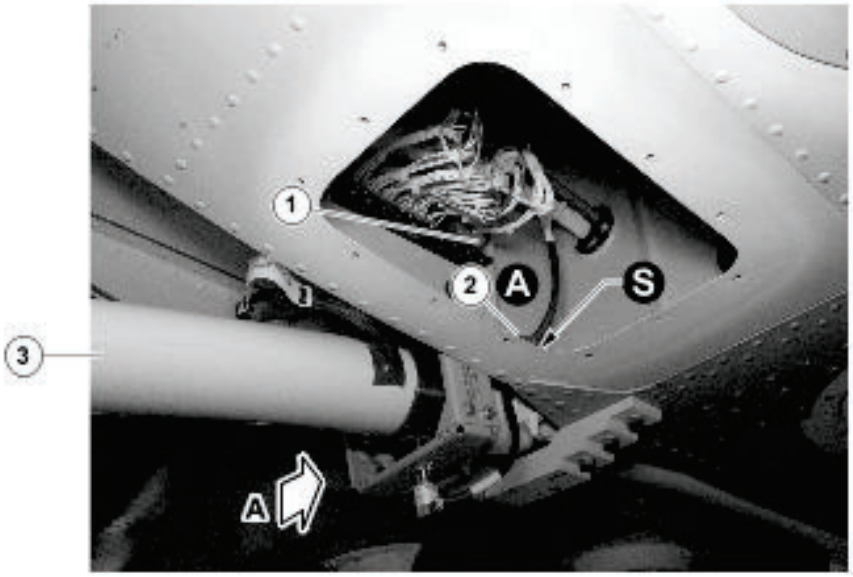
Applicable: SN: 36528-36999 or (SN: 36248-36527 BHT-412-SI-87: Installed):

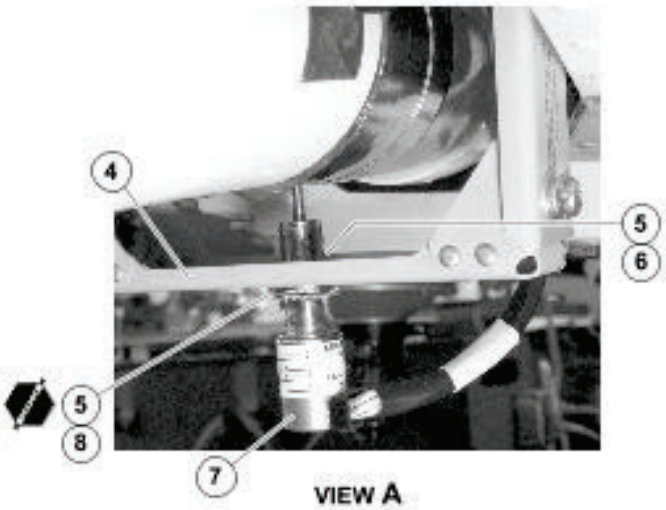
9. Remove electrical power.

Applicable: SN: 36248-36527 BHT-412-SI-87: Not installed:

10. Remove the ohmmeter. Reinstall the MST 67A transponder (412-A-97-23-00-04A-720B-A / 02356).

11. Safety the jam nuts (5) with lockwire (C-405).





1. Terminal board module 9TB3-U
2. Grommet
3. Forward crosstube
4. Support bracket
5. Jam nut
6. Lockwasher
7. Weight-on-gear (WOG) switch
8. Key washer

A ADHESIVE (C-325)

LOCKWIRE (C-405)

S SEALANT (C-251)

ICN-412-A-960600-B-97499-00001-A-001-01

Figure 1. Weight-On-Gear (WOG) Switch - Maintenance Practices (Sheet 1 of 1)

Powered Passenger Step System (Kit) - Description

The powered passenger step system is powered by 28 VDC from the bus No. 2 through the STEP circuit breaker, STEP switch, and right and left step actuators. When the STEP switch is set to RAISE, the actuator attached to the respective step retracts to raise the step and then stops. When the STEP switch is set to STOW, the actuator attached to the respective step extends to lower the step and then stops.

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Powered Passenger Step System - Operational Check

Procedure



TO PREVENT DAMAGE TO ACTUATOR MOTORS, NEVER REPEAT Step 1. AND Step 2. MORE THAN ONE TIME IN SUCCESSION.

1. Make sure that 28 VDC power is applied to the helicopter.
2. Close the STEP circuit breaker.
3. Set the STEP switch to RAISE.
4. Verify that both steps raise freely (without binding) and stop when fully extended.
5. Set the STEP switch to STOW.
6. Verify that both steps lower freely (without binding) to each skid gear.
7. Set the STEP switch to OFF.
8. Verify that the steps remain stowed.

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Emergency Float (Kit) - Description

The emergency float system is powered from the 28 VDC EMERGENCY BUS 2 through the EMER FLOATS circuit breaker and the STEP circuit breaker. The system is controlled by the FLOATS switch on the pilot collective stick.

Applicable: SN: 33001-33100:

When the FLOATS switch is set to ARMED, the float arm relay is energized and completes the circuit to automatically stow the powered passenger steps.

Applicable: SN: 33101-33213 or 36001-36999:

When the STEP switch is set to RAISE, the STEP EXTEND light on the instrument panel illuminates.

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Emergency Float - Operational Check

Procedure

1. Disconnect the electrical connector from the squib valve.
2. Make sure that 28 VDC is applied to the EMERG BUS 2.
3. Close the ENERG FLOATS and the STEP circuit breakers.
4. Set the STEP switch to RAISE.

Applicable: SN: 33101-33213 or 36001-36999:

5. Verify that the STEP EXTEND light illuminates.
6. Set the FLOATS switch to ARMED.
7. Verify that the FLOATS ARMED light on the instrument panel illuminates.

Applicable: SN: 33001-33100:

8. Verify that steps stow.
9. Actuate a pair of float switches with a container of water.
10. Verify that 28 VDC is present between A(+) and B(-) of the connector.
11. Verify that 28 VDC is present between C(+) and D(-) of the connector.
12. Repeat Step 9. through Step 11. for the remaining pair of float switches.

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Applicable to: ALL
ECCN EAR99

Heated Windshield Kit (412-706-010/-138) - Description

The optional heated windshield kit has two controller units that are installed in the cabin roof area.

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Controller Unit - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-136	AR	

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

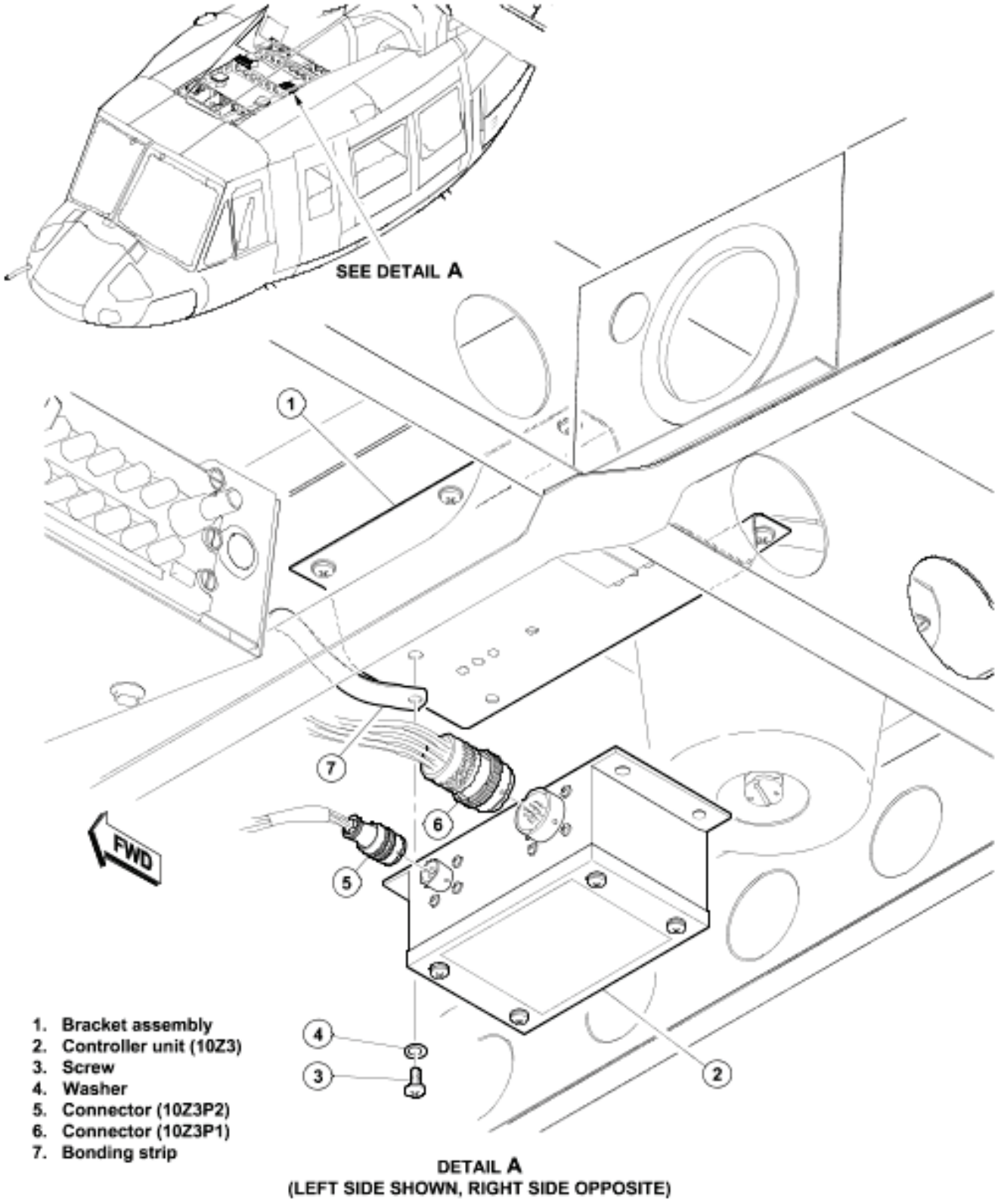
NOTE

The following procedure is for the controller unit (10Z3) (2, Figure 1) on the left side. The procedure is the same for the controller unit (10Z2) on the right side.

NOTE

Remove the components to the extent necessary to do maintenance.

1. Open the nose doors and/or panels, as applicable (412-A-52-40-00-01A-520A-A / 00832).
2. Disconnect the battery and/or the external power from the helicopter.
3. Disconnect the connectors (10Z3P1 and 10Z3P2) (6 and 5) from the controller unit (10Z3) (2).
4. Install caps or plugs (C-136) on the connectors.
5. Remove the screws (3), washers (4), and controller unit (10Z3) (2) from the bracket assembly (1) and the bonding strip (7).



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Figure 1. Controller Unit - Removal and Installation (Sheet 1 of 1)

Controller Unit - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Isopropyl alcohol	C-385	AR	
Clean cloth	C-516	AR	

Procedure

1. Remove all moisture and loose dirt with a dry, clean cloth (C-516).
2. Remove grease, fungus, and ground-in dirt with a clean cloth (C-516) moistened with isopropyl alcohol (C-385).
3. Remove dirt from the electrical connector with a soft bristle brush.

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Controller Unit - Inspection

Procedure

NOTE

The following inspection criteria are primarily intended when the part or component has been removed from the helicopter. To accommodate all inspection requirements, applicable criteria may also be used when the part or component is installed on the helicopter.

1. Examine the electrical connectors for corrosion, damage, and bent or burnt contacts.

2. Examine the case for dents, cracks, wear, mechanical damage, and corrosion damage.

3. Visually examine the attaching hardware for cracks, wear, mechanical damage, and corrosion. Replace when necessary.

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Controller Unit - Installation

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON OR NEAR ELECTRICAL/ELECTRONIC EQUIPMENT (412-A-96-00-01-02A-042A-A / 01874).

NOTE

The following procedure is for the controller unit (10Z3) (2, Figure 1) on the left side. The procedure is the same for the controller unit (10Z2) on the right side.

1. Make sure that the mating surfaces of the bracket assembly (1), bonding strip (7), and controller unit (10Z3) (2) are clean and show no signs of corrosion. Otherwise, refer to the BHT-ELEC-SPM, Chapter 8, Electrical Standard Practices Manual for surface preparation.

2. Install the controller unit (10Z3) (2) on the bracket assembly (1) and the bonding strip (7) with the screws (3) and the washers (4). Tighten the screws.

3. If the mating surfaces of the components were clean and showed no signs of corrosion at the time of installation, Step 4. is not required.

4. Make sure that the bond resistance between the bracket assembly (1), bonding strip (7), and controller unit (10Z3) (2) meets Class R-I electrical bonding requirements (BHT-ELEC-SPM, Chapter 8, Electrical Standard Practices Manual).

5. Remove the protective caps and/or plugs from all the connectors.

6. Connect the connectors (10Z3P1 and 10Z3P2) (6 and 5) to the controller unit (10Z3) (5).

7. Connect the battery and/or the external power to the helicopter.

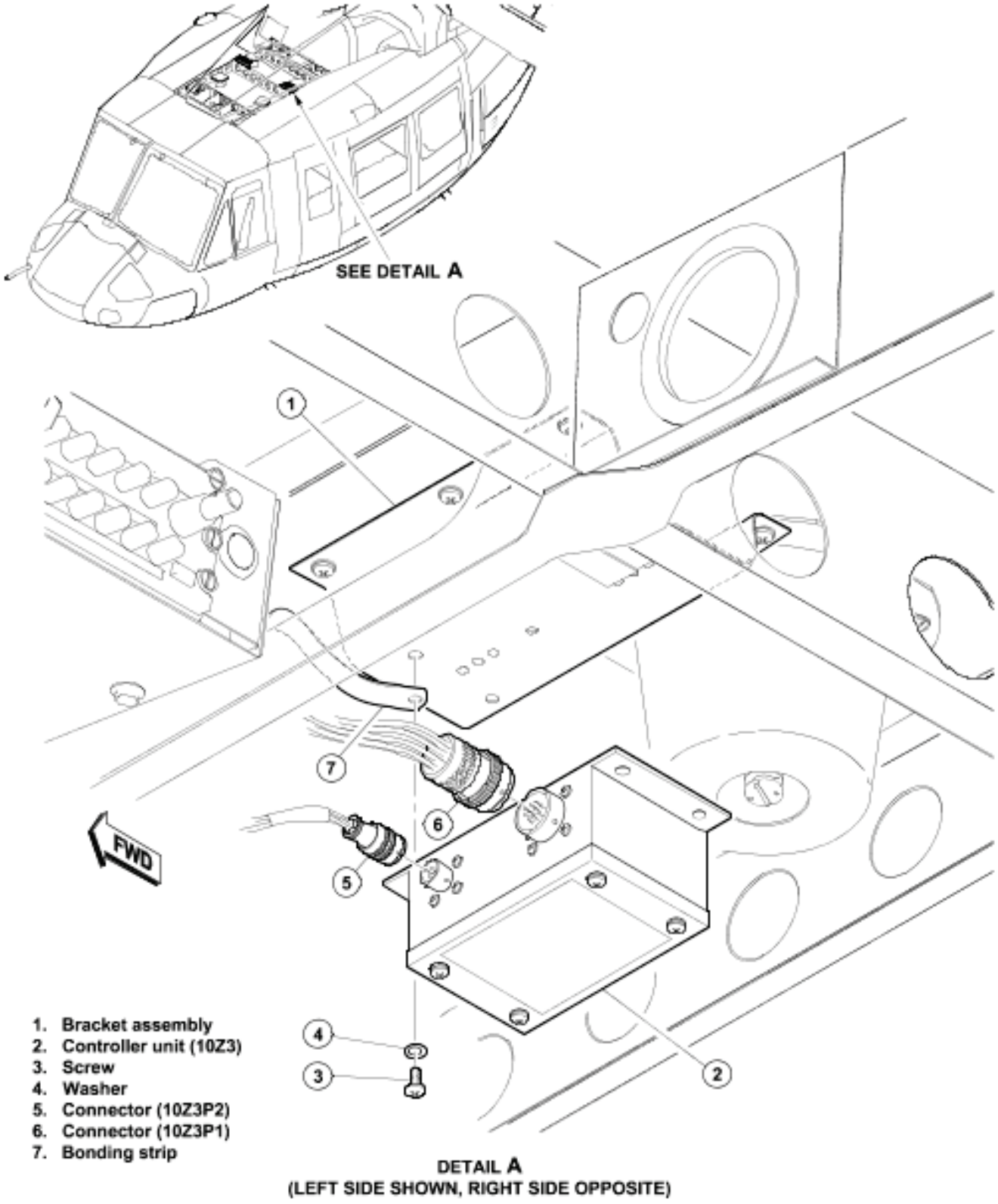
8. Close the nose doors and/or panels, as applicable (412-A-52-40-00-01A-720A-A / 00835).

Applicable: SN: 36087-36999:

9. Do an operational check of the heated windshield (BHT-412-SI-49, Chapter 2, Service Installation Instructions Manual).

Applicable: SN: 37002-37999:

10. Do an operational check of the heated windshield (BHT-412-MMS-EPI, Chapter 96, Maintenance Manual Supplement).



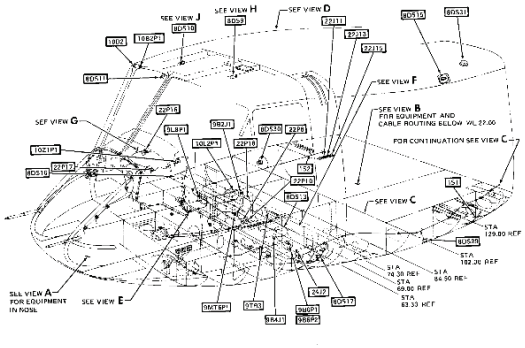
ICN-412-A-960700-B-97499-00001-A-001-01

Figure 1. Controller Unit - Removal and Installation (Sheet 1 of 1)

Voice/Flight Data Recorder System - Description

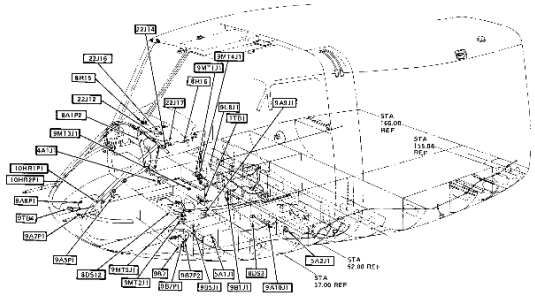
The voice and flight data recorder system consists of a flight data recorder, cockpit voice recorder, and their associated hardware and wiring. Refer to Figure 1 for equipment location and (412-A-98-00-00-15B-051A-A / 02575, or 412-A-98-00-00-16B-051A-A / 02576) for the applicable wiring diagrams.

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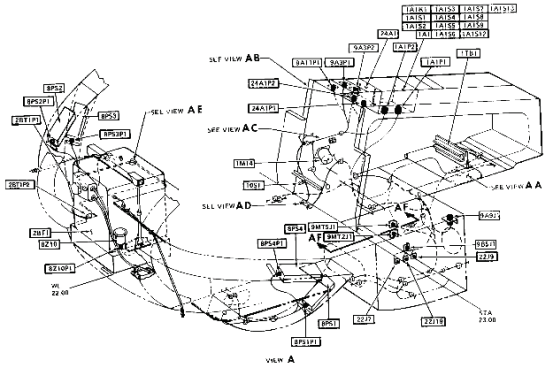
ICN-412-A-960002-B-97499-00024-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 1 of 11)



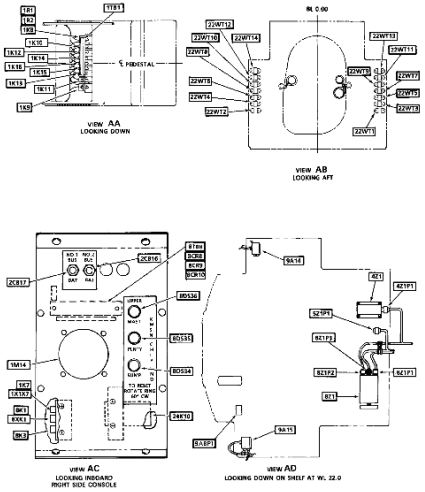
ICN-412-A-960002-B-97499-00025-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 2 of 11)



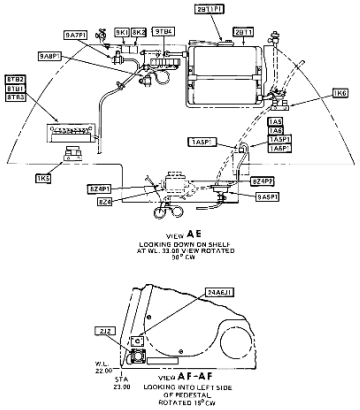
ICN-412-A-960002-B-97499-00026-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 3 of 11)



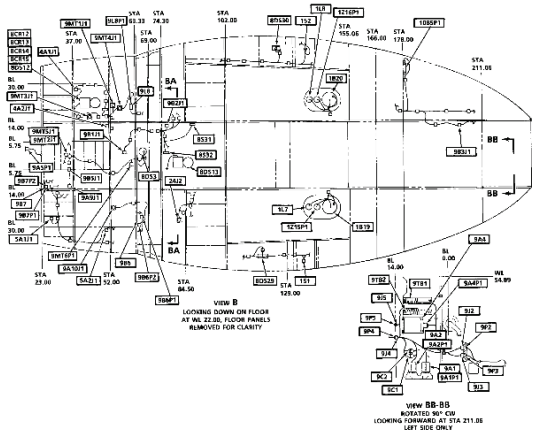
ICN-412-A-960002-B-97499-00027-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 4 of 11)



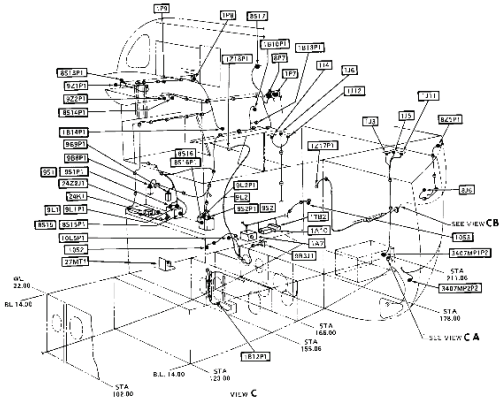
ICN-412-A-960002-B-97499-00028-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 5 of 11)



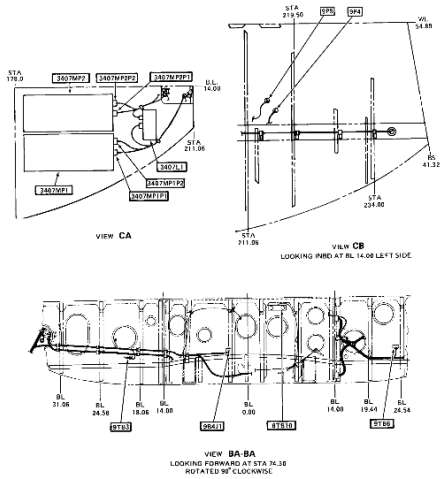
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Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 6 of 11)



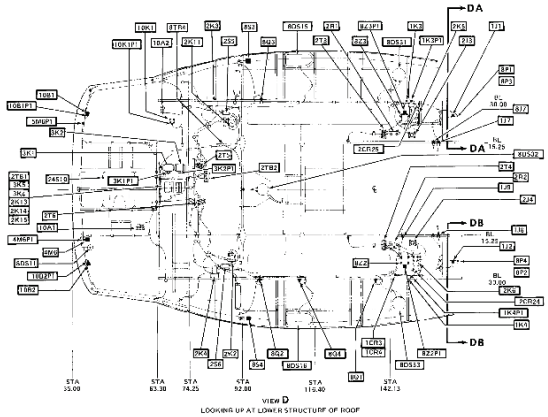
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Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 7 of 11)



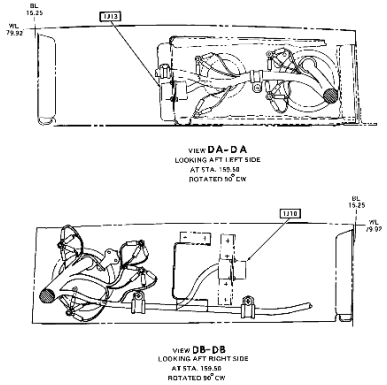
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Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 8 of 11)



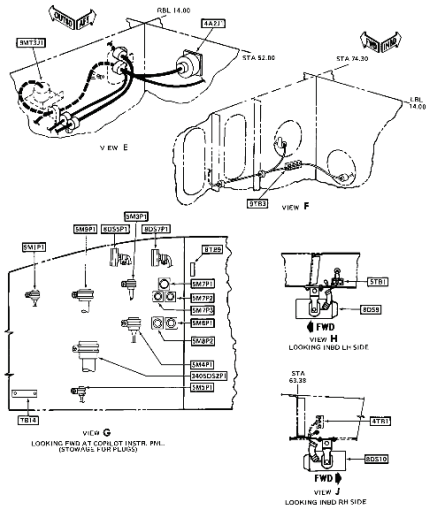
ICN-412-A-960002-B-97499-00032-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 9 of 11)



ICN-412-A-960002-B-97499-00033-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 10 of 11)



ICN-412-A-960002-B-97499-00034-A-001-01

Figure 1. Fuselage Equipment Location (S/N 33108 through 33213 and 36001 through 36086) (Sheet 11 of 11)

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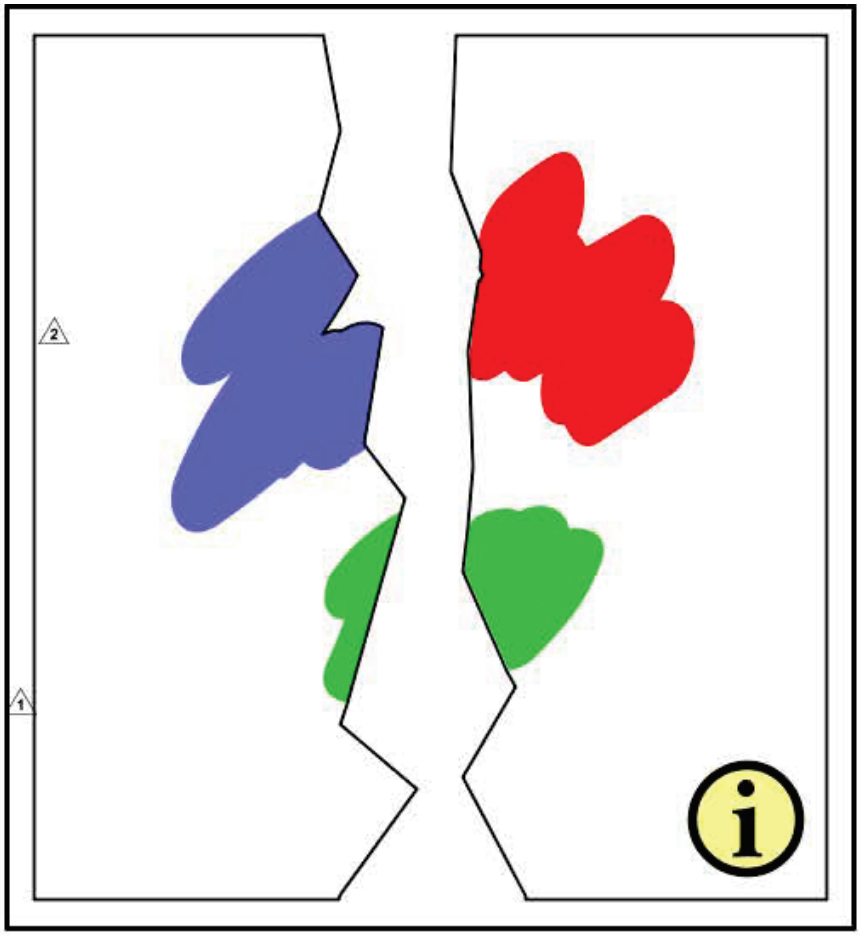
Flight Data Recorder (FDR) System - Description

The flight data recorder system (Figure 1) consists of a Flight Data Acquisition Unit (FDAU) , Flight Data Recorder (FDR) , accelerometer, Outside Air Temperature (OAT) probe, Air Data Computer (ADC) , and system fail relay.

The flight data acquisition unit gathers data from the air data computer and various instruments and sensors. It also passes data to the flight data recorder and clock signals to the Cockpit Voice Recorder (CVR) . DC power is required for the flight data acquisition unit and both AC and DC power are required for the flight data recorder unit. The FDR SYS FAIL caution light on the master caution panel illuminates with a FDAU failure, FDR failure, loss of power to either unit, or if either unit is not installed. In addition, the flight data acquisition unit contains two Light Emitting Diode (LED) s on its front cover that are independently activated by a fault signal for either the flight data acquisition unit or the flight data recorder.

The FDR SYS FAIL caution light and the flight data recorder (LED) extinguish automatically once the fault is cleared. But, in order to extinguish the LED caution indicator on the flight data acquisition unit, activated by a FDAU fault, the fault must first be cleared, and then the FDAU circuit breaker must be reset.

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NOTES

1. S/N 36248 through 36333.
2. S/N 36037 through 36333.

ICN-412-A-960800-B-97499-00001-A-001-01

Figure 1. Simplified Block Diagram Voice/Flight Data Recorder System (S/N 36037 through 36333) (Sheet 1 of 1)

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Flight Data Recorder (FDR) System - Retrieval of Data

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Portable Analysis Unit (PAU)	17TES0010	1	
Portable Printer Unit (PPU)	17TES0020	1	

Procedure

1. The portable analysis unit and the portable printer unit are used to extract, analyze, monitor, or copy recorded data from the flight data recorder. For additional information, refer to the LORAL Fairchild component Maintenance Manual, Publication No. 165-E0514-00.

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Cockpit Voice Recorder (CVR) - Description

The Cockpit Voice Recorder (CVR) system consists of a voice recorder, voice recorder control panel, and inertia (impact) switch.

This system is DC powered and protected by the CVR circuit breaker. The CVR, through the ICS panels, records the microphone signals, keyed and unkeyed, and the headphone audio. It also records the audio signals from the area microphone, and time correlation signals from the flight data acquisition unit. The record time is a minimum of 30 minutes on a 1/4-inch preconditioned continuous loop magnetic tape. In the event of a crash, the inertia switch disables power to the CVR to prevent erasure of audio data.

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Cockpit Voice Recorder (CVR) - Operational Check

Procedure

NOTE

The CVR has a TEST interlock through the pilot and copilot Intercommunication System (ICS) control units to ensure ICS control units 212-077-200-105 are installed. If the meter fails to indicate during the test, an earlier model (lower dash number) ICS control unit may have been installed.

1. Close the CVR circuit breaker.

2. Press the test button on the CVR control panel, and hold for 3 seconds.

3. Verify that the meter indication reads GOOD.

NOTE

The meter needle will normally oscillate slightly. A channel failure will result in the needle dropping from the green section of the meter scale or returning to zero, either of which indicate a definite NO GO condition.

4. Cover the area microphone.

NOTE

The CVR control panel requires a 600 ohm headset with a standard phone plug to monitor CVR audio.

5. Monitor the CVR audio.

6. Using the pilot microphones, announce the date, time, and helicopter serial number.

7. Verify that the CVR audio is clear, intelligible, and delayed by approximately 0.5 seconds.

NOTE

A continual BEEP from the FDAU is normal.

8. Repeat Step 4. through Step 7. using the copilot microphone.

9. Set the ICS volume control to midrange and cover the area microphone.

10. Monitor the CVR audio.

11. Using the pilot microphones, key and announce the date, time, and helicopter serial number.

12. Verify that the CVR audio is clear, intelligible, and delayed by approximately 0.5 seconds.

13. Monitor the CVR audio.

14. Using the copilot microphones, key and announce the date, time, and helicopter serial number.

15. Verify that the CVR audio is clear, intelligible, and delayed by approximately 0.5 seconds.

16. Uncover the area microphone.

17. Disconnect the pilot and copilot headsets.

18. Monitor the CVR audio.

19. From the pilot or copilot seat, announce the date, time, and helicopter serial number.

20. Verify that the CVR audio is clear, intelligible, and delayed by approximately 0.5 seconds.

21. Remove all power from the CVR system.

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Integrated Flight Control System (IFCS) - Description

NOTE

The Integrated Flight Control System (IFCS) is manufactured by Sperry Flight Systems, Avionic Division, Phoenix AZ. The system consists of an Automatic Flight Control System (AFCS) , a Flight Director System (FDS) , and is integrated with the Model 412 helicopter electrical and hydraulic systems. For additional information on the AFCS and FDS, refer to Sperry Flight Systems SHZ-412 Integrated Flight Control System Manual 22-15-00. For the electrical interface wiring diagram of the AFCS, refer to (412-A-98-00-00-09B-051A-A / 02569, or 412-A-98-00-00-10B-051A-A / 02570).

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Digital Automatic Flight Control System (DAFCS) - Description

The Digital Automatic Flight Control System (DAFCS) (Figure 1) includes the following major components: copilot and pilot digital flight control computers, copilot and pilot flight director mode selectors, autopilot controller, air data command display (altitude preselect indicator), and AFCS Actuator Position Indicator (API) panel.

NOTE

For standardization, the left side of the helicopter has been designated No. 1 and the right side No. 2 respectively when seated in the cockpit, regardless of the pilot in command position. This applies to all avionics systems, including DAFCS, gyros and radios.

The DAFCS provides fully coupled autopilot and flight director functions in the pitch, roll, and yaw axis. The autopilot and flight director functions are done by two identical, interchangeable digital flight control computers.

The air data sensor uses static and pitot pressure inputs to produce DC voltage outputs proportional to static pressure and differential pressure for conversion to altitude and airspeed voltages respectively. It also provides a 28 volt DC output to warn of an equipment failure.

Applicable: SN: 36087-36247:

The yaw rate gyro provides a reference signal to the DAFCS flight control computer and is used to compute yaw actuator displacement for helicopter speeds above 60 knots.

Applicable: SN: 36248-36999:

The AHRS provides a reference signal to the DAFCS flight control computer and is used to compute yaw actuator displacement for helicopter speeds above 60 knots.

The mode selector enables the pilot or copilot to select and annunciate the various flight modes of the helicopter flight director system. The bezel and panel display characteristics consist of illuminating pushbutton switches for selecting flight director standby, roll axis, or pitch axis modes of operation. When a mode is selected, that mode is annunciated by a bright light level. The light level is controlled by external cockpit light dimming circuitry. The SBY switch, when held, tests all lamps in the mode selector and autopilot controller.

The digital flight control computer uses a combination of digital and analog computational techniques to implement various automatic flight control system functions. The computer processes information about actual attitude versus a desired attitude as a function of selected flight mode to produce autopilot pitch, roll, and yaw outputs and flight director pitch and roll steering commands.

The DAFCS relay control box assembly contains relays associated with the DAFCS. This box may be replaced as a complete unit or individual relays may be isolated for replacement.

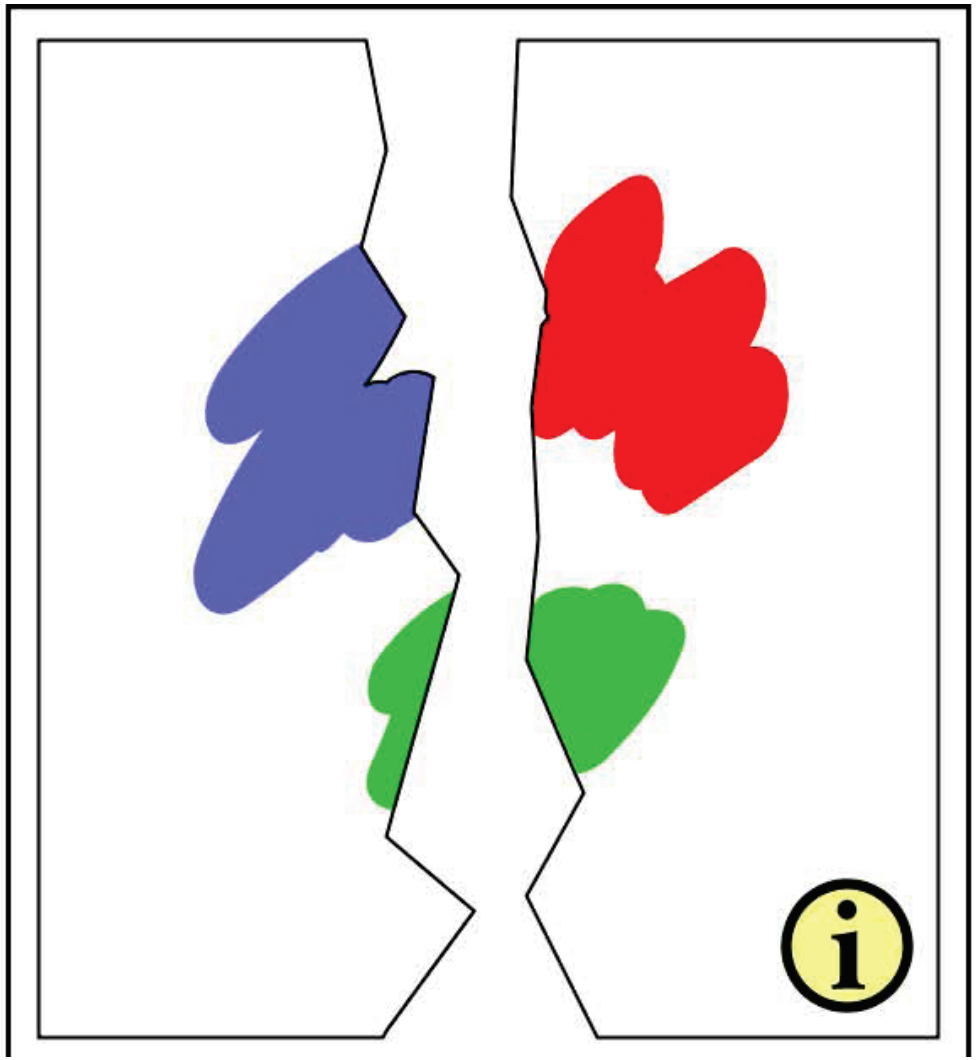
The autopilot controller enables the pilot to engage the autopilot system and to select the desired mode of operation. The three basic modes are stability augmentation (SAS), attitude retention (ATT), and automatic flight path control (CPL). The controller also contains a flight director select switch (selects system 1 or system 2), and a system test switch. The panel display characteristics consist of illuminating pushbutton switches for engaging the autopilot system and selecting the desired mode. When a mode is selected, that mode is annunciated by a bright light level. The light level is controlled by external cockpit light dimming circuitry. A lamp test function is controlled by the flight director mode selector SBY switch.

The air data command display is used to display air data parameter references. It is also used to display fault codes during the DAFCS test cycle. The display receives 8-bit parallel digital data from the digital flight control computer and decodes this data to illuminate a five digit, seven segment incandescent display. At power-up, the display will indicate all dashes (——).

The Actuator Position Indicator (API) panel indicates the position of the pitch, roll, and yaw series actuators during operation.

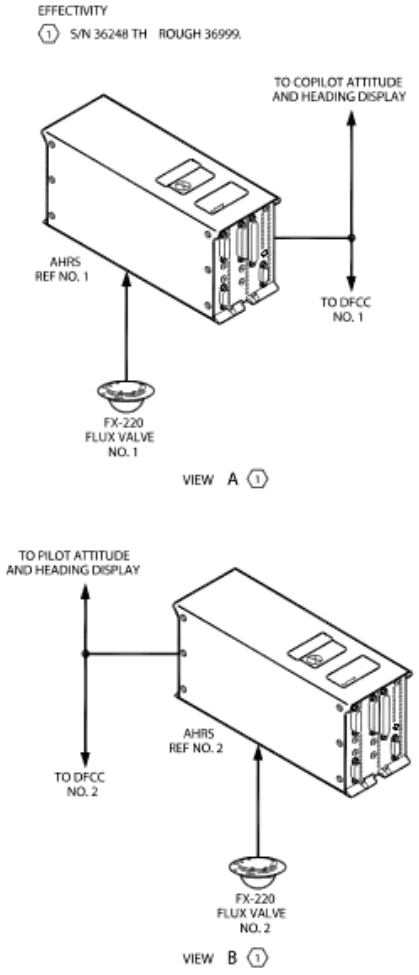
The DAFCS improves helicopter stability and control through two types of operations, the attitude retention mode and the stability augmentation system mode.

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ICN-412-A-961000-B-97499-00001-A-001-01

Figure 1. 3-Axis System with Dual Autopilot/Flight Director (S/N 36068 through 36999) (Sheet 1 of 2)



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Figure 1. 3-Axis System with Dual Autopilot/Flight Director (S/N 36068 through 36999) (Sheet 2 of 2)

Attitude (ATT) Retention Mode - Description

The ATT mode provides long-term stabilization for hands-off flying (or coupling to the flight director if installed). Helicopter pitch and roll attitude is held using both the series and trim actuators, while the yaw axis provides yaw damping. The pilot may change the attitude reference by pushing the force trim button on his cyclic grip or by actuating the trim beeper switch.

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Stability Augmentation System (SAS) Mode - Description

The SAS is a hands-on pitch and roll mode in which short-term rate damping is applied to the series actuators only. The SAS mode provides short-term rate damping to external disturbances and enhances helicopter handling qualities during hands-on flight. The SAS also provides yaw damping and automatic turn coordination. The SAS is intended for use where extensive helicopter maneuvering is required and the pilot prefers to be hands-on without attitude retention.

NOTE

When operating in SAS, the cyclic beep switches are inoperative.

The Digital Automatic Flight Control System (DAFCS) also has an automatic preflight TEST mode that checks the operational readiness of the system.

Actuator drive capability, sensor reasonableness, and system monitors are also checked.

The primary component of this is the digital flight control computer. This computer provides the capability to automatically control pitch and roll axes.

SN: 36087-36247: The digital flight control computer interfaces to the vertical gyros and a yaw rate gyro or to the AHRS and provides the drive signals for the pitch, roll, and yaw series actuators. The computer also provides the pitch and roll trim driver outputs, in the form of 28 volt pulse signals, and can drive a variety of existing DC trim actuators.

SN: 36248-36999: The digital flight control computer interfaces to the vertical gyros and a yaw rate gyro or to the AHRS and provides the drive signals for the pitch, roll, and yaw series actuators. The computer also provides the pitch and roll trim driver outputs, in the form of 28 volt pulse signals, and can drive a variety of existing DC trim actuators.

All control inputs are applied to the series actuators and the trim system is driven in a proportional manner to keep the series actuators operating near their center positions. The pilot must apply pedal inputs as needed to satisfy long-term requirements in yaw.

Series actuator stroke lengths are selected to meet certification failure criteria. The trim system is monitored to prevent failures from causing a runaway trim condition.

Components that make up the three-axis system with dual autopilot/dual flight director are listed in Table 1.

The Digital Automatic Flight Control System (DAFCS) (Figure 1), which includes pitch and roll flight director coupling, has the following modes available.

- Altitude Hold (ALT-ON)
- Indicated Airspeed Hold (IAS-ON)
- Vertical Speed Hold (VS-ON)
- Heading Select (HDG-ON)
- VOR/LOC Capture and Tracking (NAV-ARM/CAP)
- Glideslope Capture and Tracking (ILS-ARM/CAP)
- Back Course Capture and Tracking (BC-ARM/CAP)
- VOR Approach (VOR APR-ARM/CAP)
- Go-Around (GA-ON)
- Standby (SBY-ON).

Table 1. 3-Axis System with Autopilot/Flight Director Components (S/N 36087 through 36247)

COMPONENTS	QTY
DAFCS COMPONENTS	
Digital Flight Control Computer	2
Autopilot Controller (3-Axis)	1
Air Data Command Display	1
Yaw Rate Gyro	2
Pitch Control Rod Assembly	1
Roll Control Rod Assembly	1
Yaw Control Rod Assembly	1
AFCS Indicator Panel	1

Table 1. 3-Axis System with Autopilot/Flight Director Components (S/N 36087 through 36247) (continued)

COMPONENTS	QTY
FORCE GRADIENT ASSEMBLIES	
Pitch	1
Roll	1
Yaw	1
CONTROL MOTION TRANSDUCERS	
Pitch	2
Roll	2
Yaw	2
Cltv	2
TRIM ACTUATOR/MAGNETIC BRAKES	
Pitch Actuator	1
Roll Actuator	1
Yaw Actuator	1
ATTITUDE AND HEADING COMPONENTS	
TARSYN Three-Axis Reference	2
Flux Valve	2
Dual Remote Compensator	1
FLIGHT DIRECTOR COMPONENTS	
FD Mode Selector (3-Axis)	2
Air Data Sensor	2

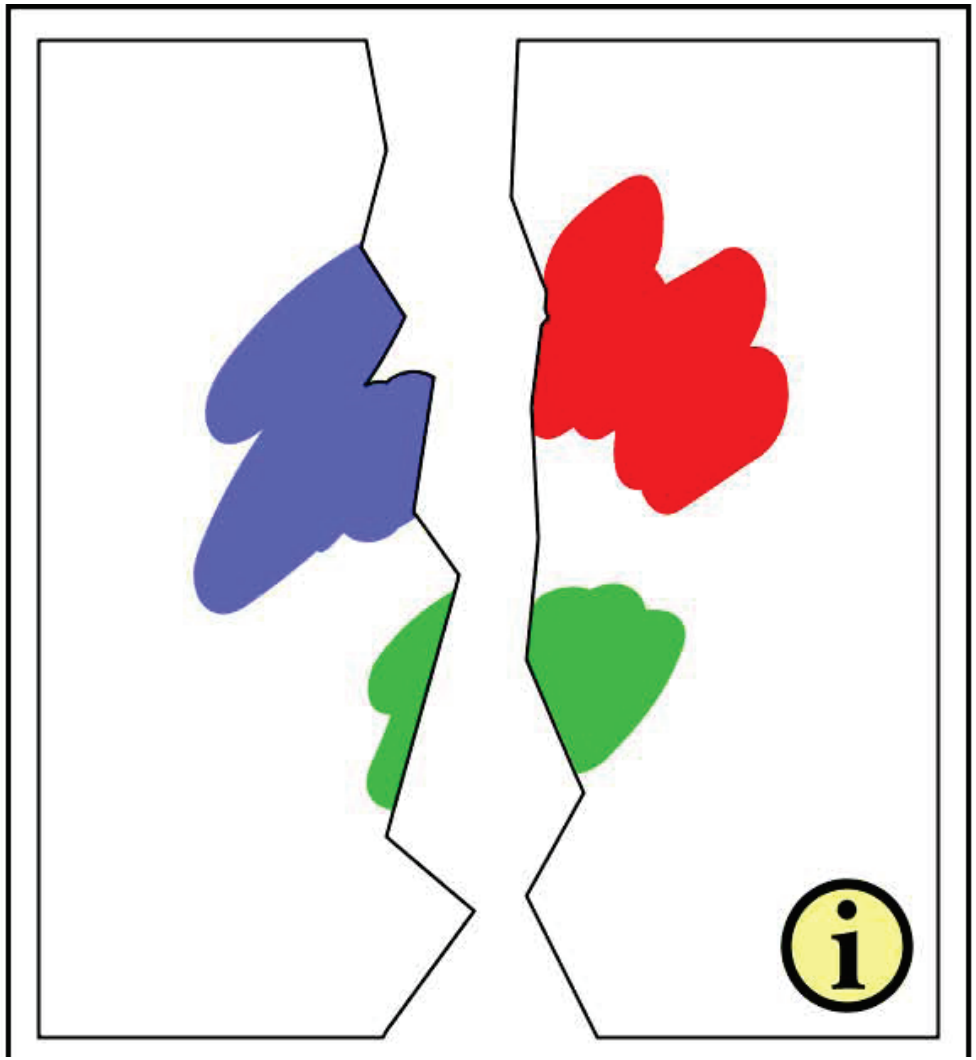
Table 2. Attitude Heading Reference System with Autopilot/Flight Director Components (S/N 36248 through 36999)

COMPONENTS	QTY
DAFCS COMPONENTS	
Digital Flight Control Computer	2
Autopilot Controller (3-Axis)	1
Air Data Command Display	1
Pitch Control Rod Assembly	1
Roll Control Rod Assembly	1
Yaw Control Rod Assembly	1
AFCS Indicator Panel	1
FORCE GRADIENT ASSEMBLIES	
Pitch	1
Roll	1
Yaw	1
CONTROL MOTION TRANSDUCERS	
Pitch	2
Roll	2
Yaw	2
Cltv	2
TRIM ACTUATOR/MAGNETIC BRAKES	
Pitch Actuator	1
Roll Actuator	1

Table 2. Attitude Heading Reference System with Autopilot/Flight Director Components (S/N 36248 through 36999) (continued)

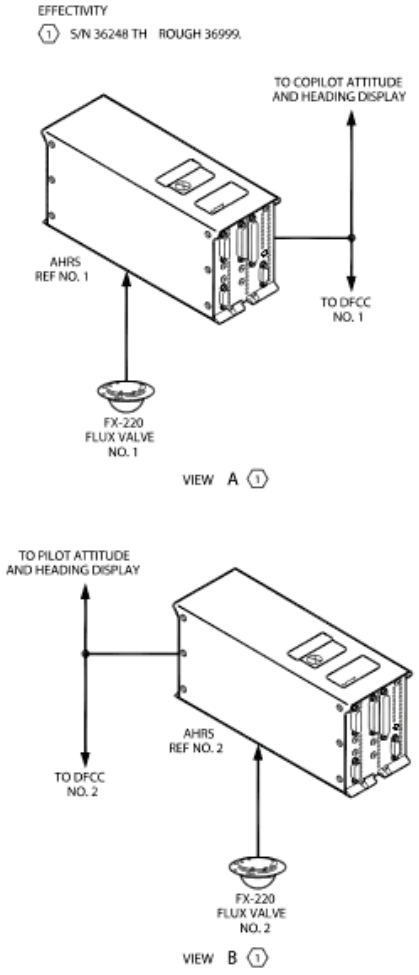
COMPONENTS	QTY
Yaw Actuator	1
ATTITUDE AND HEADING COMPONENTS	
Attitude Heading and Reference Unit (AHRU)	2
Flux Valve	2
FLIGHT DIRECTOR COMPONENTS	
FD Mode Selector (3-Axis)	2
Air Data Sensor	2

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Figure 1. 3-Axis System with Dual Autopilot/Flight Director (S/N 36068 through 36999) (Sheet 1 of 2)



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Figure 1. 3-Axis System with Dual Autopilot/Flight Director (S/N 36068 through 36999) (Sheet 2 of 2)

Digital Flight Control Computer - Description

The digital flight control computer is an autopilot and a flight director computer integrated into a single unit.

The digital flight control computer is a microprocessor based digital computer that uses a combination of digital and analog computational techniques to implement automatic flight control system functions. This computer accepts both digital and analog inputs, performs digital and analog computations, and supplies both digital and analog outputs. It processes information about actual aircraft attitude versus a desired attitude as a function of selected flight mode to produce autopilot pitch roll, and yaw control outputs and flight director pitch and roll steering command outputs. Operating modes for the digital flight control computer are classified into outer loop modes, generally associated with flight director functions, and inner loop modes, generally associated with autopilot functions.

In addition to providing autopilot and flight director functions, the digital flight control computer also performs the following tasks:

- Manages incoming sensor data (validation and selection)
- Monitors autopilot and yaw damper operation
- Computes selected vertical speed, altitude hold and air speed hold references, and selected heading

Internally, the digital flight control computer is designed to prevent a single fault within this computer from causing a hazard to the helicopter. Self-test for latent faults is built-in, so that exposure to latent faults is minimized.

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Autopilot Controller - Description

The autopilot controller (Figure 1) provides the pilot with a convenient method of engaging various autopilot flight control functions and system preflight test. It is used to select the active flight director, and control decoupling/recoupling of the autopilots to flight director commands.

The panel display controls consist of illuminating button switches for engaging the autopilot system and selecting the desired mode. When a switch is engaged or a mode selected, it is annunciated by a light. This light is controlled by external light dimming circuitry. A lamp test function is controlled by the flight director mode selector SBY switch. The annunciator lamps in the button switches are line replaceable.

1.1. AP1/AP2 (Autopilot Engage) Buttons

Actuation of either AP button engages that autopilot in the ATT mode and automatically engages its yaw axis. Autopilot engagement is annunciated by the ON legend on the button. Each autopilot may be individually turned off by again pushing its respective button. Autopilot disengagement is indicated by a blank button, and illumination of the corresponding AP caution annunciator.

1.2. SAS/ATT (Stability Augmentation System/Attitude) Button

The SAS/ATT button allows selection of the SAS or ATT mode of autopilot operation. In normal operation, both autopilots are on in the ATT (attitude retention) mode. The ATT mode is selected automatically upon engagement of either autopilot and is annunciated by the ATT legend on the button. The ATT mode must be selected for coupling the autopilots to flight director commands.

NOTE

The autopilot(s) will engage in SAS if the cyclic stick FORCE TRIM switch is off.

1.3. TEST Button

The TEST button controls engagement of the preflight test mode.

1.4. TRIM/FD (Trim/Flight Director) Select Button

The TRIM/FD select button is mainly used to select which flight director, FD1 or FD2, will supply steering commands to both autopilots. The flight director selected is referred to as the active flight director and is indicated by a 1 or 2 legend on the button. In normal operation, the system automatically selects FD2 upon power-up.

NOTE

Pushing the TRIM/FD button places the newly selected FD into standby (SBY) automatically to prevent coupling to possibly old or invalid information.

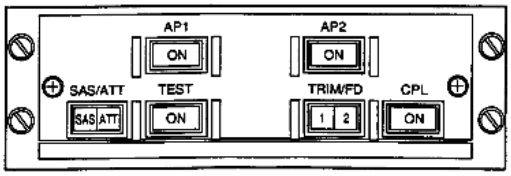
The autopilot (AP) on the active flight director side controls the parallel trim actuators and the Air Data Command Display. Thus, if one AP actuator or the air data command display outputs fail, the TRIM/FD button may be used to switch these functions to the other AP.

NOTE

If a single AP is engaged, its FD automatically becomes the active flight director, to ensure that the single AP controls the parallel trim actuators. Pressing the TRIM/FD button has no effect in this case.

1.5. CPL (Couple) Button

When both autopilots are engaged in the ATT mode, steering commands from the active flight director are automatically coupled to the autopilots upon engagement of a flight director mode. Hands-off automatic flightpath steering is then provided. The CPL button is used to uncouple commands from the autopilots, allowing the pilot to manually satisfy the cues on the ADI. Coupling is indicated by the ON legend on the button. To uncouple, the button is pushed to blank the legend. Turning CPL off will light the DCPL annunciator. Assuming a flight director mode is engaged, coupling can be restored by pressing the CPL button to indicate on or by cycling either autopilot off and then on.



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Figure 1. Autopilot Controller (S/N 36087 through 36999) (Sheet 1 of 1)

Flight Director Mode Selector - Description

Dual three-axis flight director mode selectors (Figure 1) are installed in the instrument panel. Each is used to engage modes of its respective flight director, FD1 (left side) or FD2 (right side). The active FD is selected with the TRIM/FD button on the autopilot controller.

Flight director modes can be engaged and disengaged by pushing the button. Mode engagement is annunciated by an illuminated legend on the button face. In normal operation (both autopilots on in the ATT mode), engagement of any single flight director mode will result in steering commands to both autopilots.

The light level is controlled by external cockpit light dimming circuitry. The SBY switch, when held, tests all lamps in the mode selector and autopilot controller.

1.1. Modes

Altitude (ALT) Hold Mode. The ALT mode maintains the altitude present at mode engagement. Engagement is annunciated by the ON legend.

Indicated Airspeed (IAS) Hold Mode. The IAS mode maintains the airspeed present at mode engagement. Engagement is annunciated by the ON legend on the button.

Vertical Speed (VS) Hold Mode. The VS mode maintains the vertical speed present at the moment of engagement. Engagement is annunciated by the ON legend on the button.

Heading (HDG) Select Mode. The HDG mode maintains the heading selected on the heading bug. Engagement is annunciated by the ON legend on the button.

Navigation (NAV) Tracking Mode. The ARM and CAP legends on the button indicate armed or captured status of the desired navigation course to be flown (e.g., VOR, localizer, long range navigation, etc.).

Instrument Landing System (ILS) Mode. Used in conjunction with an ILS approach, the ARM and GS legends on the button indicate armed or captured status of the ILS glideslope.

Back Course (BC) Mode. Used in conjunction with a localizer back course approach, the ARM and CAP legends indicate armed or captured status of the localizer.

VOR Approach (VOR APR) Mode. Same as NAV except mode performance is optimized for VOR instrument approach procedures.

Go-Around (GA) Mode. Intended for aborting an approach to the runway, engagement is annunciated by the ON legend on the button. Upon engagement, all other flight director modes are canceled; the aircraft will automatically achieve wings level and smoothly enter a climb of approximately 75 feet per minute. The collective axis will automatically maintain airspeed at approximately 75 knots or more (Vy).

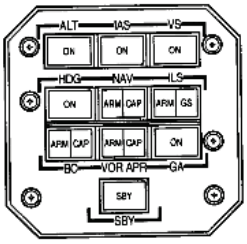
Standby (SBY) Mode. Pushing this button cancels all flight director modes and returns the flight director to the standby (ready) condition. The SBY button is also used to sequence through power-up, continuous, and preflight test error codes displayed on the air data command display. In normal operation, the system powers up in standby as shown by the ON legend on the button.

NOTE

An invalid flight director will result in a visible FD warning flag on the appropriate ADI and may result in a loss of SBY ON annunciation on the button.

A remote SBY button is located on each cyclic grip and is identical in operation to the flight director mode selector SBY button.

Pushing the pilot or copilot SBY or remote SBY button for more than 1/4 second will activate a lamp test of all lamps associated with the pilot or copilot flight control computer. Holding the button for approximately 5 seconds will cause all associated lamps to dim and will sound the altitude alert horn if installed.



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Figure 1. Flight Director Mode Selector (S/N 36087 through 36999) (Sheet 1 of 1)

Air Data Command Display - Description

The air data command display (Figure 1) is used to display the command of reference values for the following modes: IAS, VS, and altitude alert. It is also used to display fault codes during system power-up tests, continuous tests, and preflight test. The active flight director (as selected on the TRIM/ FD button) is responsible for air data command display operation.

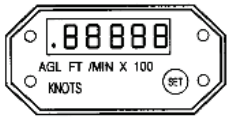
The display receives 8-bit parallel digital data from the digital flight control computer and decodes this data to illuminate a five-digit display. Annunciators below the display indicate the parameter being displayed.

Upon coupled engagement of the IAS or VS modes, the air data command display will display the current reference value for that mode. Backlit legends on the face of the air data command display simultaneously annunciate the appropriate units of the displayed reference (for example, KNOTS or FT/MIN × 100).

To eliminate potentially mistaking the display for DME or other information presented on similar cockpit instruments, the display reverts to dashes 5 seconds after mode engagement (except as described as follows). The display will be restored while making adjustments to the reference using the FTR buttons or beep switches.

The SET knob is used to select an altitude for the altitude alert. Rotating the knob activates the altitude reference display, lights the Feet (FT) legend on the face, and automatically arms altitude alert. The MACH annunciation, while visible during lamp test, is not used in this application.

When the flight director SBY button is pushed for more than 1/4 second, the air data command display will display all eights (88888) and illuminate all the face legends as a lamp test. After completion of system power-up, the display will show all dashes (——).



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Figure 1. Air Data Command Display (S/N 36087 through 36999) (Sheet 1 of 1)

Rate Gyro and AHRS - Description

The applicable yaw rate gyro or AHRS provides yaw rate information to the digital flight control computer for yaw axis stabilization. This rate-of-turn signal is proportional to the rate of angular displacement about an axis perpendicular to the mounting surface. The yaw rate signal is summed against the rate command output by the digital flight control computer to drive the yaw series actuator. The yaw rate signal is also input by the digital flight control computer for yaw rate gain reduction calculations.

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Control Rod Assemblies - Description

The control rod assemblies (Figure 1) are extendable links in the primary flight controls (pitch, roll, and yaw axis) of the helicopter. The control rods position the control surfaces (rotor swashplate and tail rotor pitch control) of the helicopter in response to the input of the stability augmentation system. Overall rod length is varied by a motor-driven actuator. The actuator is installed in the control rod linkages in a series arrangement.

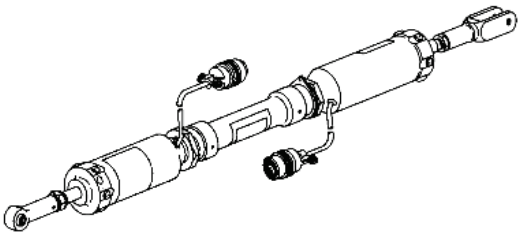


DO NOT APPLY TORQUE BETWEEN THE LINEAR ACTUATOR OUTPUT ROD AND STRUCTURAL COVER. APPLICATION OF TORQUE DURING REMOVAL OR INSTALLATION CAN COMPROMISE THE DESIGN INTEGRITY OF THE CONTROL ROD ASSEMBLY ACTUATOR.

DO NOT REMOVE THE LINEAR ACTUATORS FROM THE FLIGHT CONTROL ROD ASSEMBLIES. THESE ASSEMBLIES MUST BE MAINTAINED IN ACCORDANCE WITH THE MANUFACTURER SPECIFICATIONS TO SAFEGUARD DESIGN INTEGRITY. THE CONTROL RODS REQUIRE SPECIFIC LENGTH, NUT TORQUE, AND CLEVIS OPERATION.

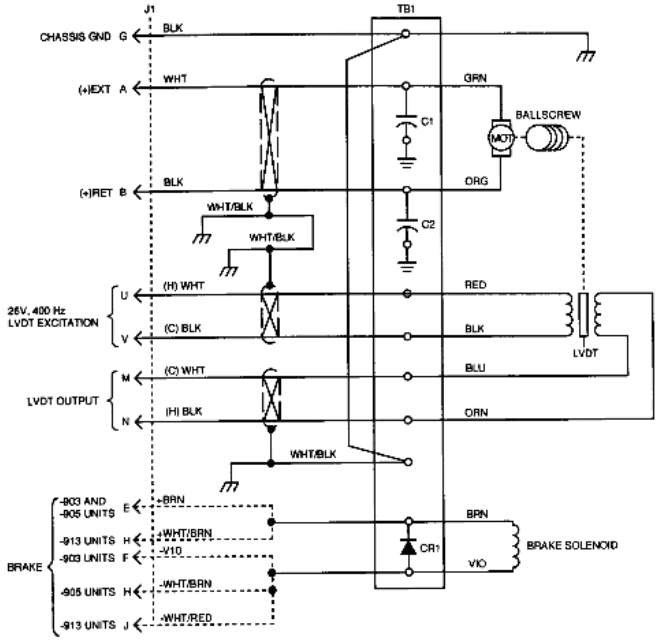
IF AN ACTUATOR OR CONTROL ROD IS SUSPECTED OR KNOWN TO BE DEFECTIVE, THE ENTIRE CONTROL ASSEMBLY MUST BE REMOVED FROM THE HELICOPTER. REFER TO (412-A-67-11-00-14A-520A-A / 01372 AND 412-A-67-11-00-14A-720A-A / 01374) FOR THE REMOVAL AND INSTALLATION PROCEDURES.

The actuator (Figure 2) is comprised of four major assemblies: a DC torque motor, Linear Variable Differential Transformer (LVDT) , adjustable mechanical stops, and brake solenoid. Electrical (DC) commands from the digital flight control computer drive the motor of the actuator to extend or retract the control rod linkages. Position feedback to the digital flight control computer servo loop is provided by the LVDT. The brake solenoid releases the motor for operation when power is applied at autopilot engagement.



ICN-412-A-961000-B-97499-00006-A-001-01

Figure 1. Control Rod Assembly (Typical) (S/N 36087 through 36999) (Sheet 1 of 1)



ICN-412-A-961000-B-97499-00007-A-001-01

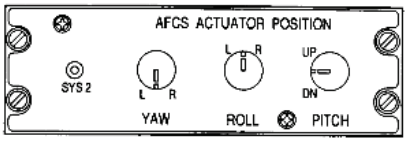
Figure 2. Linear Actuator Block Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

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AFCS Indicator Panel - Description

The position of the pitch, roll, or yaw linear (series) actuators may be checked at any time during autopilot operation by observing the AFCS indicator panel (Figure 1). Pushing the System 2 (SYS 2) button replaces the AP1 actuator display with AP2 actuator positions.

During normal operation in the ATT mode, the actuator positions should remain fairly close to center for optimum authority. If autotrim is not functioning, the pilot can manually retrim the controls by pushing FTR on the cyclic and moving the control in the appropriate labeled direction to recenter the actuator(s). When approximately recentered, the switches are released.



ICN-412-A-961000-B-97499-00008-A-001-01

Figure 1. AFCS Indicator Panel (S/N 36087 through 36999) (Sheet 1 of 1)

Air Data Sensor - Description

The air data sensor uses static and pitot pressure inputs to produce DC voltage outputs proportional to static pressure and differential pressure for conversion to altitude and airspeed voltages respectively. It also provides a 28 volt DC output to warn of an equipment failure.

Degraded performance may result if the air data sensor is not allowed sufficient warmup. In ambient temperature above 32°F (0°C) a minimum 10-minute warmup is recommended.

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System Operation - Description

The system is divided into three subsystems: Autopilot, Flight Director, and Compass Systems. Table 1 lists the system limits.

Table 1. System Performance/Operating Limits (S/N S/N 36087 through 36999)

MODE	CONTROL OR SENSOR	PARAMETER	VALUE
Basic Autopilot	Heading Hold	Roll Angle Limit	Less than 6° and no roll mode selected
Heading Select	HDG knob on ADI	Roll angle limit	±20°
		Pitch rate limit	±5°/second
VOR or VOR APR	Course Knob, NAV Receiver and DME Receiver	Capture	
		Beam Angle Intercept (HDG SEL)	Up to ±90°
		Roll Angle Limit	±22°
		Course Cut Limit at Capture	30° VOR or VOR APR 45° LNAV or RNAV
		Capture Point	Function of beam, beam rate, course error, and DME distance. Maximum trip point is 190 mV. Minimum trip point is 15 mV.
		On Course	
		Roll Angle Limit	±13° of roll
		Roll Rate Limit	1°/second VOR 5°/second VOR APR
		Crosswind Correction	Up to ±45° Course Error
		Over Station	
		Course Change	Up to ±90°
		Roll Angle Limit	±17°
LOC or ILS or BC	Course Knob and Nav Receiver	LOC Capture	
		Beam Intercept	Up to ±90°
		Roll Angle Limit	±22°
		Roll Rate Limit	±5°/second
		Capture Point	Function of beam, beam rate, and course error. Maximum trip point 190 mV. Minimum trip point is 60 mV.
		NAV On-Course	
	Radio Altimeter and Marker Beacon Receiver	Roll Angle Limit	±13° of roll
		Crosswind Correction Limit	Function of True Airspeed (TAS)
		Gain Programming	Function of time and TAS starts at 1200 feet radio altitude, gain reduction = 1 to 0.5
		Glideslope Capture	
	GS Receiver and Air Data Sensor	Beam Capture	Function of beam and beam rate. Maximum trip point to 100 mV. Minimum trip point is 20 mV.

Table 1. System Performance/Operating Limits (S/N S/N 36087 through 36999) (continued)

MODE	CONTROL OR SENSOR	PARAMETER	VALUE
		Pitch Command Limit	20° up 3° down
		Glideslope Damping	Vertical velocity
		Pitch Rate limit	Function of TAS, 3° maximum
		Gain Programming	Function of time and TAS, starts at 1200 feet radio altitude, gain reduction = 1 to 0.16
			Function of Radio Alt, starts at 1200 feet, gain reduction = 0.16 at 100 feet to 0 at 50 feet
GA	Pushbutton Switch on Mode Selector or Pushbutton Switch on Collective	Power Command	Pitch up and collective command for > 74 knots and 750 fpm climb
ALT Hold	Air Data Sensor	ALT Hold Engage Range	0 to 20,000 feet
		ALT Hold Engage Error	±50 feet
		Pitch Limit	±20°
		Pitch Rate Limit	Function of TAS
VS Hold	Air Data Sensor	VS Engage Range	0 to ±2100 feet/minute
		VS Beep Range	±1000 feet/minute
		VS Beep Rate	±300 feet/minute/second
		VS Hold Engage Error	±50 feet/minute
		Pitch Limit	±20°
		Pitch Rate Limit	Function of TAS
IAS Hold	Air Data Sensor	IAS Engage Range	0 to 200 knots
		IAS Hold Engage Error	±5 knots
		IAS Beep Range	0 to 140 knots
		IAS Beep Rate	5 knots/sec
		Pitch Limit	±12°
		Pitch Rate Limit	Function of TAS

Self-Test - Description

The self-test features of the integrated flight control system are controlled by the two digital flight control computers. There are three types of self-test. The first two types, the power-up test and continuous monitoring, each consist of basic routines, which the computers perform automatically in the background. Pilot interaction is not required for these tests and, unless a failure occurs, performance of these tests is undetectable. The third type, the preflight test, requires pilot interaction and the control system is visibly moved in order to verify proper operation of the servos.

1.1. Power-Up Test

Upon initial power-up of the helicopter, the digital flight control computers do a self-initiated power-up test. Failure to pass this test successfully usually indicates an internal failure of the computer.

Briefly, the following areas of the digital flight control computers are tested:

- RAM Read/Write Test - A test to check that all bits of each scratch pad location can be set (reset) and read
- CPU Register Read/Write Test - Checks that all bits of each of the CPU registers used by the operational program can be set (reset) and read
- Power Supply Monitor Test - A check is made to verify that the power supply monitor is operational

The power-up test is approximately 2 seconds in duration. If the test is not successfully completed within 30 seconds, the helicopter should be shut down and a maintenance check done. If necessary, flight may still be conducted in VFR conditions with autopilots off.

Failure to pass this test will result in the following:

- The FD warning flag on the ADI will appear and the SBY button on the FD mode selector will be extinguished
- The AUTOPILOT NO. 1 and/or AUTOPILOT NO. 2 fail light on the caution panel will illuminate
- Power-up error code(s) for the selected flight director will be displayed on the air data command display.

1.2. Continuous Monitoring

The continuous monitors begin to function as soon as the power-up test has been successfully completed. Each digital flight control computer does continuous checks of the following major areas at least once per second:

- Program Memory Sum Check - The contents of each memory location is summed and compared against the known value for this sum.
- Analog Loop Test - Dedicated analog input and output is checked using a 12-bit test word, confirming that the test is received at the analog input, complementing the test word and repeating the loop check.
- CPU Monitor - The function of this monitor is to assure that the CPU is executing its real time programs. The monitor is designed to automatically disconnect the autopilot and drop the flight director valid in the event of a CPU failure.
- Trim Monitor - When the autopilot is engaged and the trim switch is on, the computers monitor autotrim operation. This monitor compares control stick position and motion versus command. That is, if a control moves without: a) an accompanying command, b) the forced gradient spring cartridge being out-of-detent or c) FTR being actuated, it is considered a trim failure and the digital flight control computer will disable trim by removing power from the trim servo.

1.3. Preflight Test

The preflight test is a test of both No. 1 and No. 2 flight control computers (autopilots) that checks the hardware and software considered to be airworthiness-critical. In addition, the preflight test checks those system functions, which can be automatically checked prior to dispatch.

The level 1 test sequence is executing when LE 1 is displayed on the air data command display. Level 1 does checks of all essential parameters for autopilot and flight director mode engagement in both computers simultaneously. Failure of a computer to pass level 1 completely will not inhibit engagement of the computer. When level 1 tests are completed, the air data command display will display END 1. The pilot may exit the Preflight Test, or continue with level 2.

Level 2 is typically used for more thorough troubleshooting after failing a level 1 test. The air data command display will display LE 2 when this test sequence is executing in both computers simultaneously.

NOTE

Pilot performance of preflight test level 1 prior to dispatch is dictated by the approved Flight Manual or supplement.

The TRIM/FD button indicates which flight control computer is responsible for preflight test performance in both computers and readout of error codes on the air data command display.

The preflight test is inhibited while in flight through AOG logic. That is, the test can only be engaged when the following three conditions are satisfied; airspeed is less than 60 knots, radar altitude is less than 20 feet, and collective position is less than 25% up travel. If any one of these parameters is exceeded, the helicopter is assumed to be airborne and test engagement will be inhibited. If the air data sensor is invalid, AOG will be based on conditions 2 and 3, as stated previously. If radar altitude is invalid, condition 1 and 3 will define AOG. If both air data and radar altitude are invalid, AOG will be determined solely by collective position.

Preflight test is done with the following initial conditions; helicopter parked on a reasonably level surface with throttles set for flight, engine torques matched, helicopter DC and AC power operating normally, all gyros valid (i.e. attitude and heading flags out of view) and autopilot(s) OFF, cyclic and pedals approximately centered, collective down, and miscellaneous control panel FORCE TRIM switch set to ON.

NOTE

Since the preflight test checks for proper command response of cyclic, hands should be off the controls to prevent nuisance error codes.

When the above conditions are met, the TEST button on the autopilot controller is pressed and the TEST button will annunciate ON. While TEST is ON, the pilot should be attentive to the flashing buttons on the autopilot controller and flight director mode selectors. The flashing indicates that the pilot must press that button for the test to continue. For example, several seconds after TEST is turned ON, the AP1 button will begin flashing. The pilot then presses the button for the test to continue. This will be repeated for the AP2 button. THE PREFLIGHT TEST WILL NOT CONTINUE UNTIL THE FLASHING BUTTON IS PRESSED. The pilot may dispatch at any time after the level 1 test sequence has been completed by pushing the TEST button on the autopilot controller.

Upon completion of each level of the preflight test, any errors detected will be displayed as error codes on the air data command display and the SBY button will flash. To proceed, the SBY or cyclic NAV STBY button is pressed to display the next error code in sequence. Error codes should be recorded and reported to the maintenance technicians.

A lamp and switch test is also included in the preflight test for all buttons on the autopilot controller and flight director mode selector. The pilot may press each flashing button individually or may elect to bypass this portion by pressing SBY or the cyclic NAV STBY switch twice.

When both level 1 and level 2 of the preflight test have been completed, END is displayed, and the TEST button on the autopilot controller will flash. TEST is then turned OFF by pressing the TEST button.

Flight Director Control Functions - Description

The following paragraphs discuss the conditions, functions and signal flow of each flight path mode of the flight director section of the computer. Figure 1 through Figure 5 are simplified diagrams that show the unit signal path and interconnect wiring for the applicable selected mode. These diagrams are used in conjunction with Figure 6, which is a simplified diagram of the mode selection logic.

1.1. Lateral Beam Sensor (LBS)

When flying to intercept the VOR or LOC beam, the LBS will determine the capture point. The capture point is determined from beam rate and distance from the station.

When the LBS trips, the flight director commands a turn toward the desired VOR radial or runway at the optimum point for a smooth capture of the beam. If the intercept angle to the beam center is very shallow, the LBS will not trip until the helicopter is almost on beam center. For this reason, an override on the LBS occurs when the beam deviation reaches a specified minimum. The minimum beam sensor trip point for VOR or VOR APR mode is ±15 mV. In the LOC mode, the minimum trip point is ±60 mV. In all modes, the maximum LBS trip point is ±190 mV.

1.2. NAV On-Course (NOC)

The NOC is defined as: LBS tripped, course error less than 22° and bank angle less than 6°. In VOR based modes, the course error and bank angle must exist simultaneously for 8 seconds for NOC to latch up. In the LOC mode, the course error and bank angle conditions must exist simultaneously for 4 seconds for NOC to latch up. When in the NOC submode, the maximum bank angle commanded is 13 ±2°. NOC starts the crosswind correction function in the computer.

1.3. Over Station Sensor (OSS)

The OSS senses the beam rate variation in the zone of confusion over the VOR station. In the VOR or VOR APR modes if the DME is valid and less than 5.5 nautical miles the OSS trips. If the DME is invalid and if the beam deviation rate reaches a predetermined level (8 mV of deviation per second with deviation greater than 75 mV), the OSS also trips. OSS removes the radio beam deviation from the flight director roll command output. In the VOR mode, the beam deviation is removed until the deviation has decreased below the trip point for 20 seconds. In the VOR APR mode, the deviation is removed until the beam deviation has decreased below the trip point for 4 seconds.

1.4. After Over Station Sensor (AOSS)

AOSS is the period between the loss of OSS and the reoccurrence of NOC.

1.5. Vertical Beam Sensor (VBS)

The vertical beam sensor is armed when the nav radio is tuned to a localizer (LOC) frequency, The LOC receiver is valid, and the LBS has tripped. The VBS trips as a function of vertical speed, TAS, and glideslope deviation. The VBS will trip at 21 mV of glideslope deviation. This will reset the previously selected pitch mode and capture the glideslope beam.

When capturing from above the beam, the helicopter must be descending at a rate that will create a suitable intercept angle.

In the Back Course (BC) mode, the VBS is locked out.

1.6. Localizer Gain Programming (LOC II)

Localizer gain programming (LOC II) is required due to the directional properties of the localizer antenna, which causes localizer beam convergence as the helicopter approaches the localizer transmitter. Signal gain is changed to maintain the helicopter within the localizer beam envelope.

For front course ILS approaches, this submode is armed after the LBS has tripped and NOC is achieved. With a valid radio altimeter, LOC II programming starts when the radio altitude is approximately 1200 feet. If the radio altimeter is not valid, LOC II occurs at GS capture.

For back course approaches, this submode is armed after LBS has tripped and NOC has been established. With a valid radio altimeter, operation is as described for front course approach. If the radio altimeter is not valid, LOC II starts when NOC has been established.

1.7. Glideslope Gain Programming

Glideslope gain programming is required due to the directional properties of the glideslope antenna, which causes glideslope beam convergence as the helicopter approaches the glideslope transmitter. Glideslope signal gain is changed to maintain the helicopter within the glideslope beam envelope.

Gain programming starts after the VBS trips, when LOC II is established. The gain is programmed as a function of radio altitude. If the radio altimeter is invalid, gain programming occurs at GS capture and the gain is reduced as a function of barometric altitude rate until the middle marker is received. The middle marker, if installed, slews the gain to a predetermined value for 250 feet.

1.8. True Airspeed (TAS) Function Generator

An approximation of TAS is calculated as a function of altitude and indicated airspeed. TAS gain programming is done to achieve approximately the same response over the helicopters altitude and airspeed envelope. This function is used to program heading select error, course select error, air data commands, vertical beam sensor pitch rate limit, and ILS gain programmers. If the air data sensor is invalid, the TAS programmer reverts to a fixed airspeed reference.

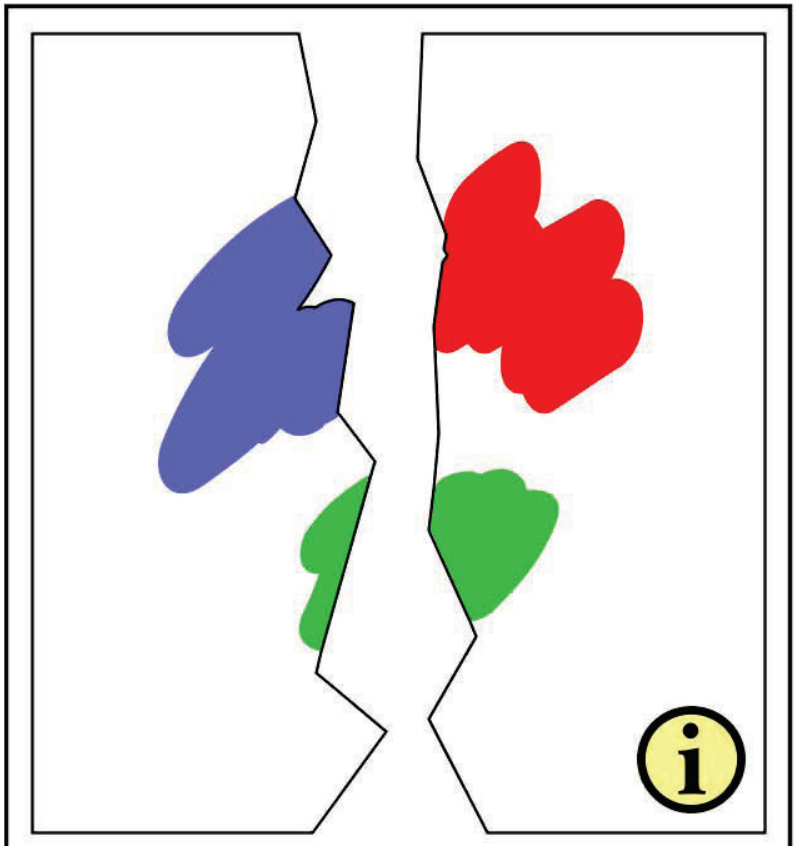
1.9. Lateral Gain Programmer

This function optimizes VOR and VOR APR operation by changing the gain of the radio signal as a function of DME distance to the station. As the helicopter approaches the station, signal strength is increasing, so gain is decreased to maintain proper helicopter response. The programmer reverts to a fixed gain level and remains there if the DME is invalid, DME is placed in the DME HOLD mode, or the RNAV mode is selected.

1.10. ADI Command Bars

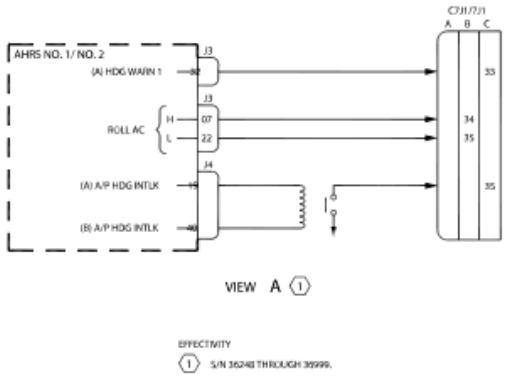
When a command signal is applied to the bar input, the bar will move left or right (roll) or up or down (pitch). This provides the required visual cue to allow the pilot to maneuver the helicopter in the proper direction to reach the desired flight path.

If the information required to fly the desired flight path becomes invalid, the command bar is biased from view.



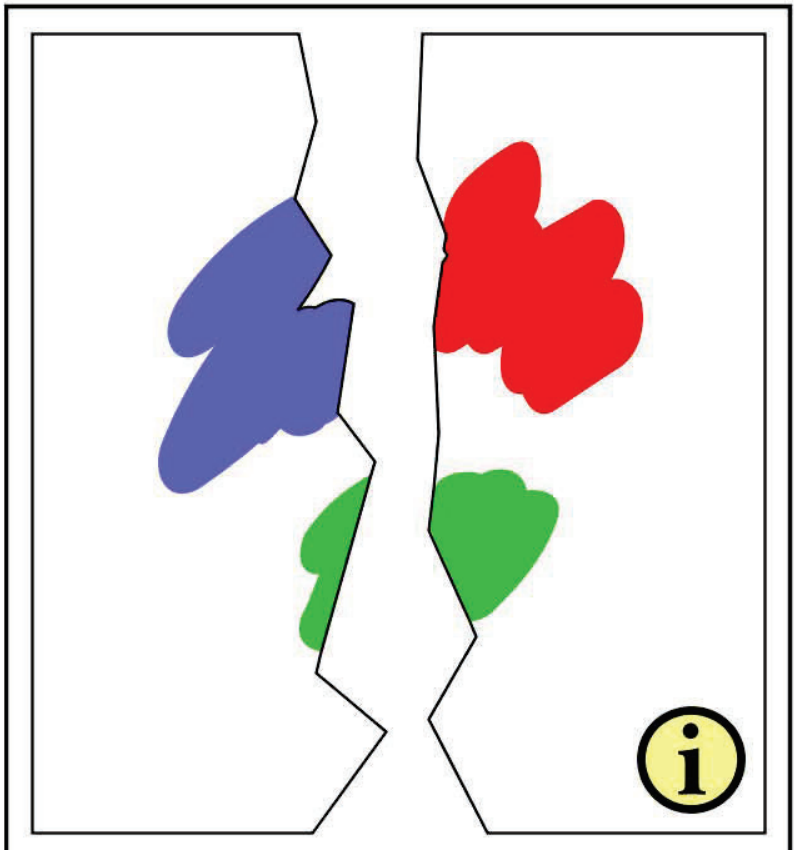
ICN-412-A-961000-B-97499-00030-A-001-01

Figure 1. Heading Select (HDG) and Go-Around (GA) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



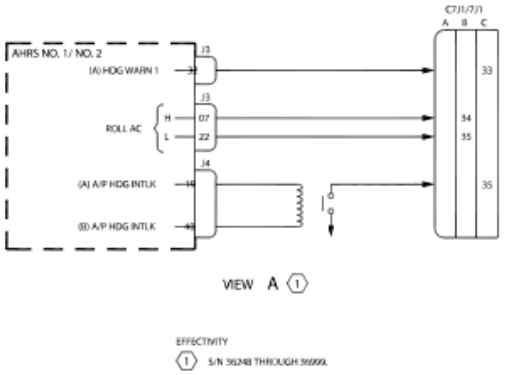
ICN-412-A-961000-B-97499-00031-A-001-01

Figure 1. Heading Select (HDG) and Go-Around (GA) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



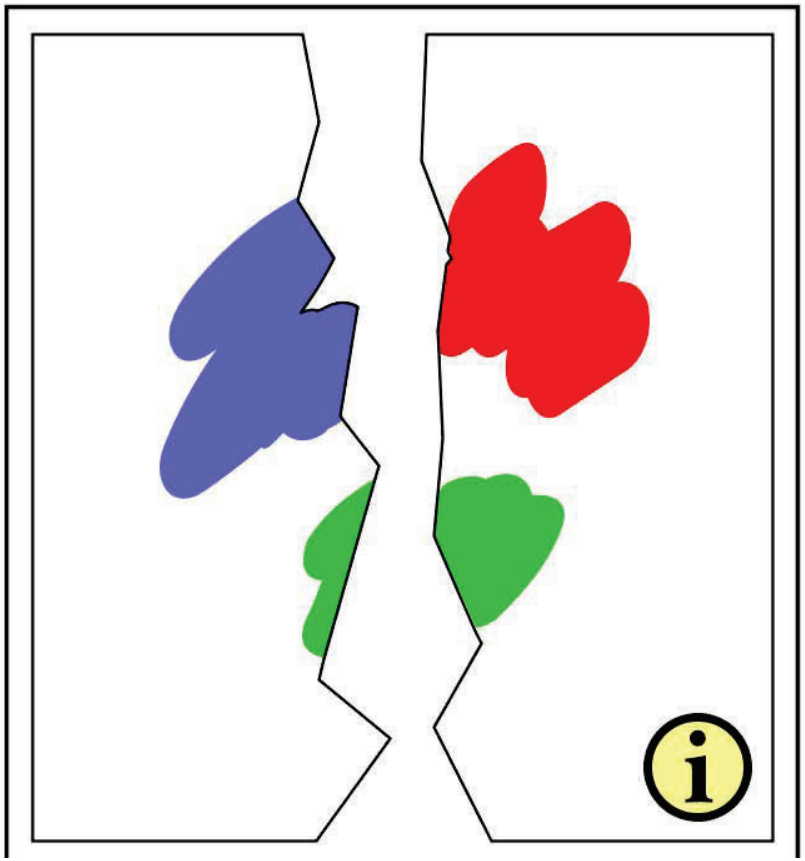
ICN-412-A-961000-B-97499-00032-A-001-01

Figure 2. VOR, VOR APR, RNAV, and LNAV Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



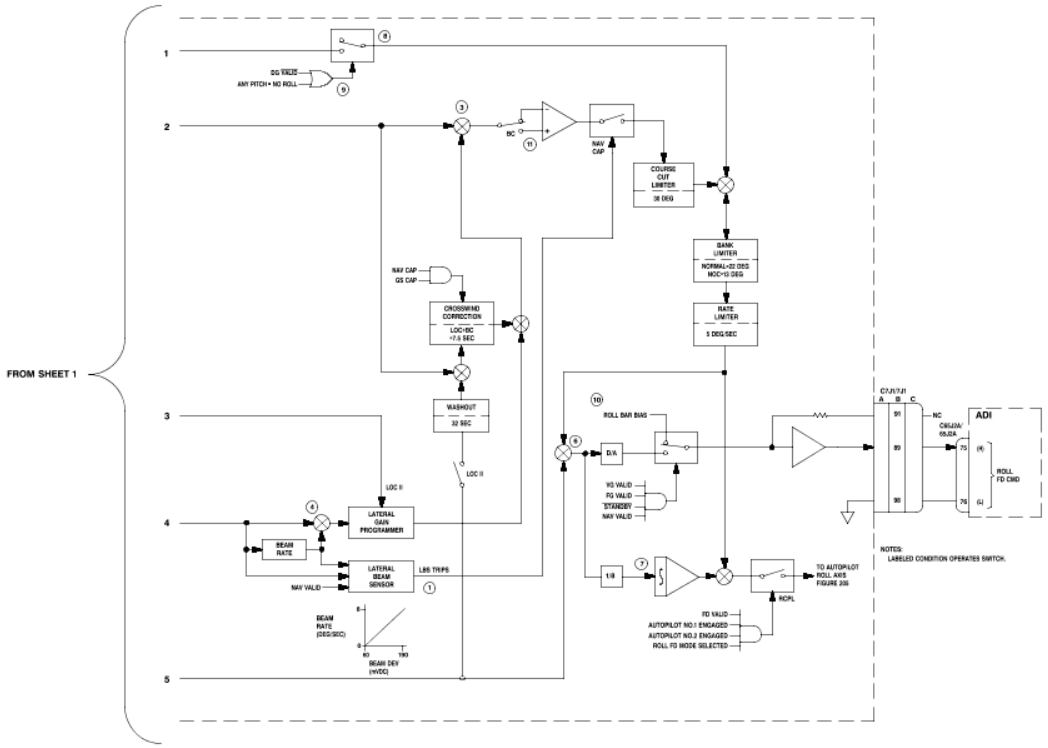
ICN-412-A-961000-B-97499-00033-A-001-01

Figure 2. VOR, VOR APR, RNAV, and LNAV Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



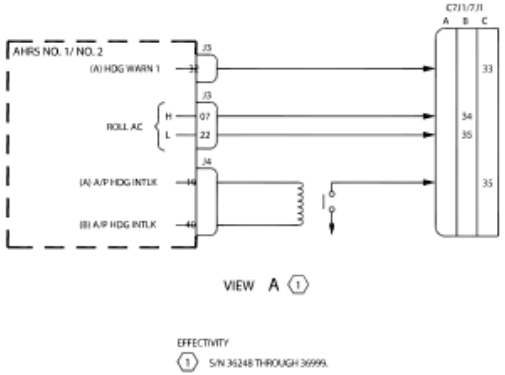
ICN-412-A-961000-B-97499-00034-A-001-01

Figure 3. Localizer and Back Course (BC) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 3)



ICN-412-A-961000-B-97499-00035-A-001-01

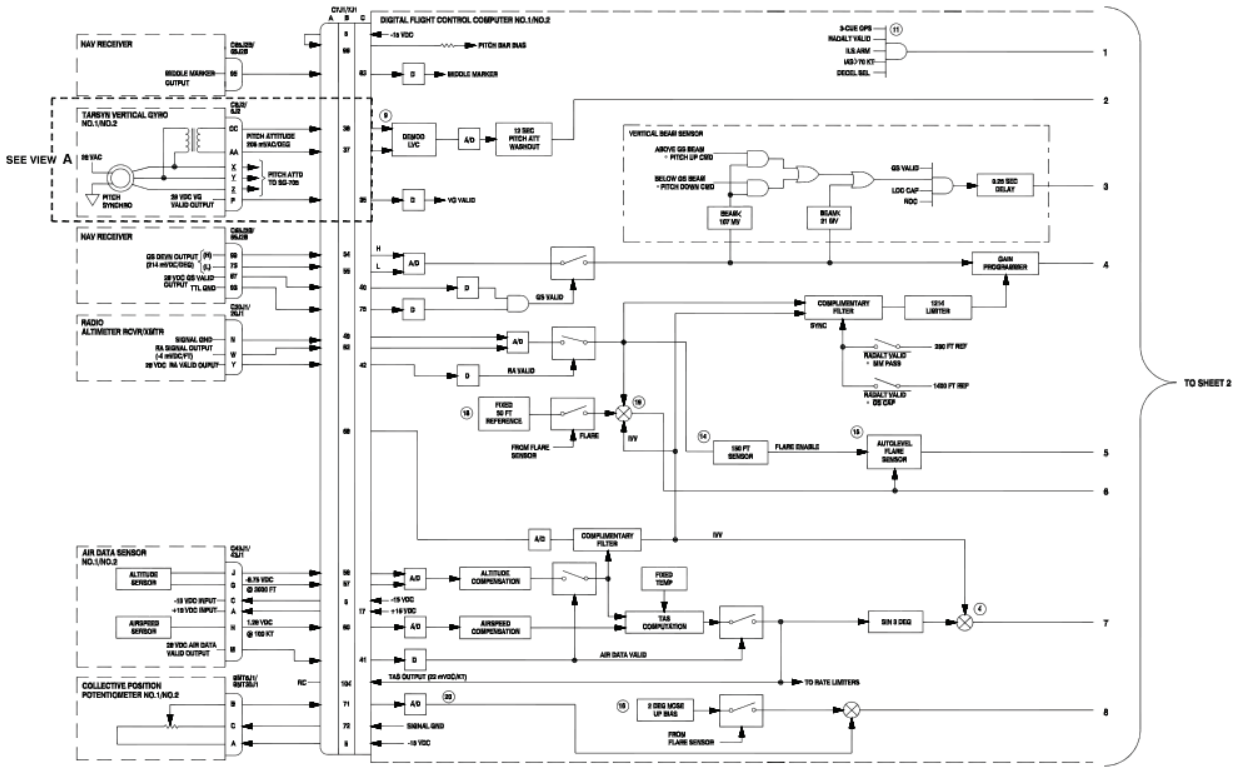
Figure 3. Localizer and Back Course (BC) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 3)



ICN-412-A-961000-B-97499-00036-A-001-01

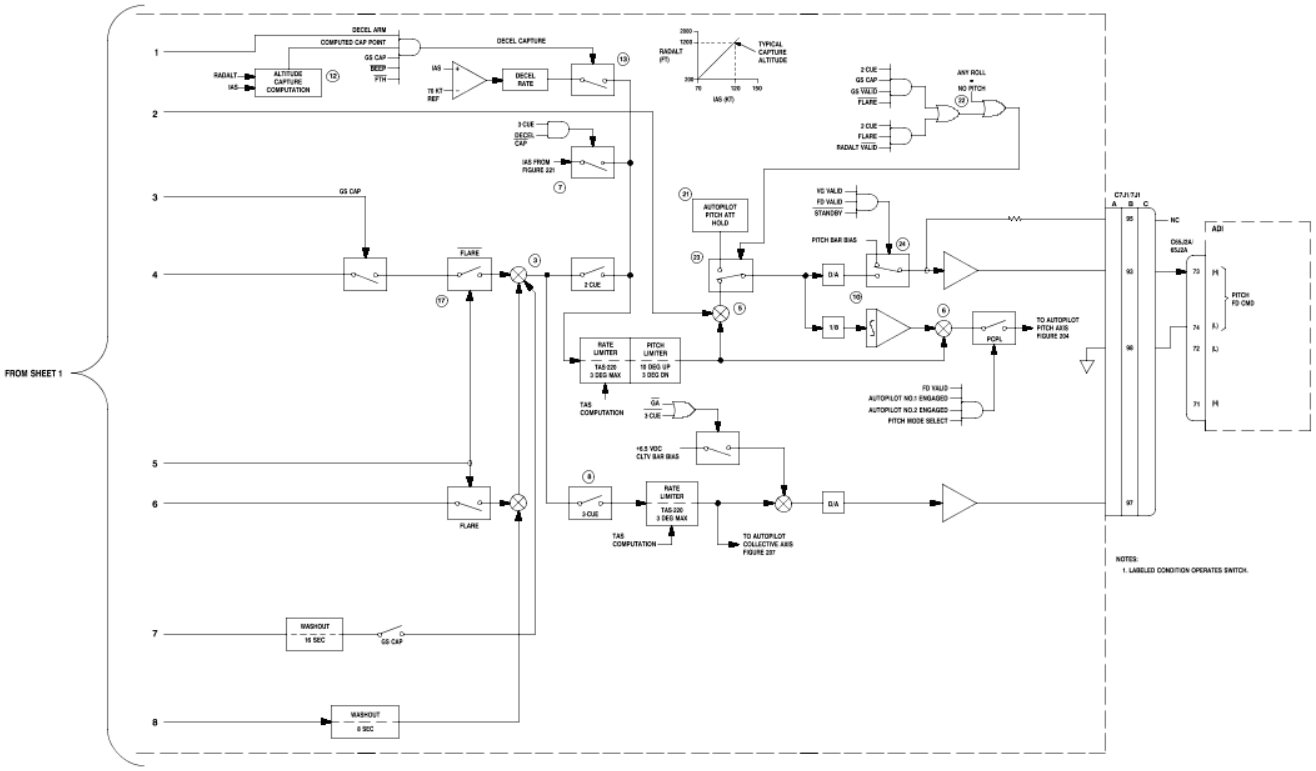
Figure 3. Localizer and Back Course (BC) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 3 of 3)

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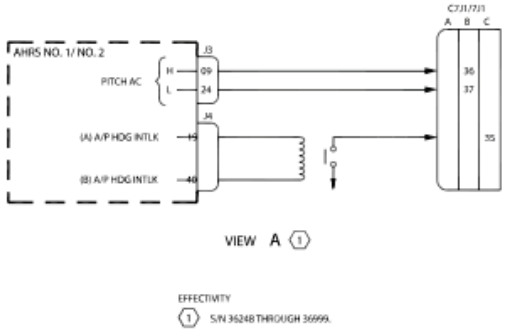
ICN-412-A-961000-B-97499-00037-A-001-01

Figure 4. Glideslope and Autolevel Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 3)



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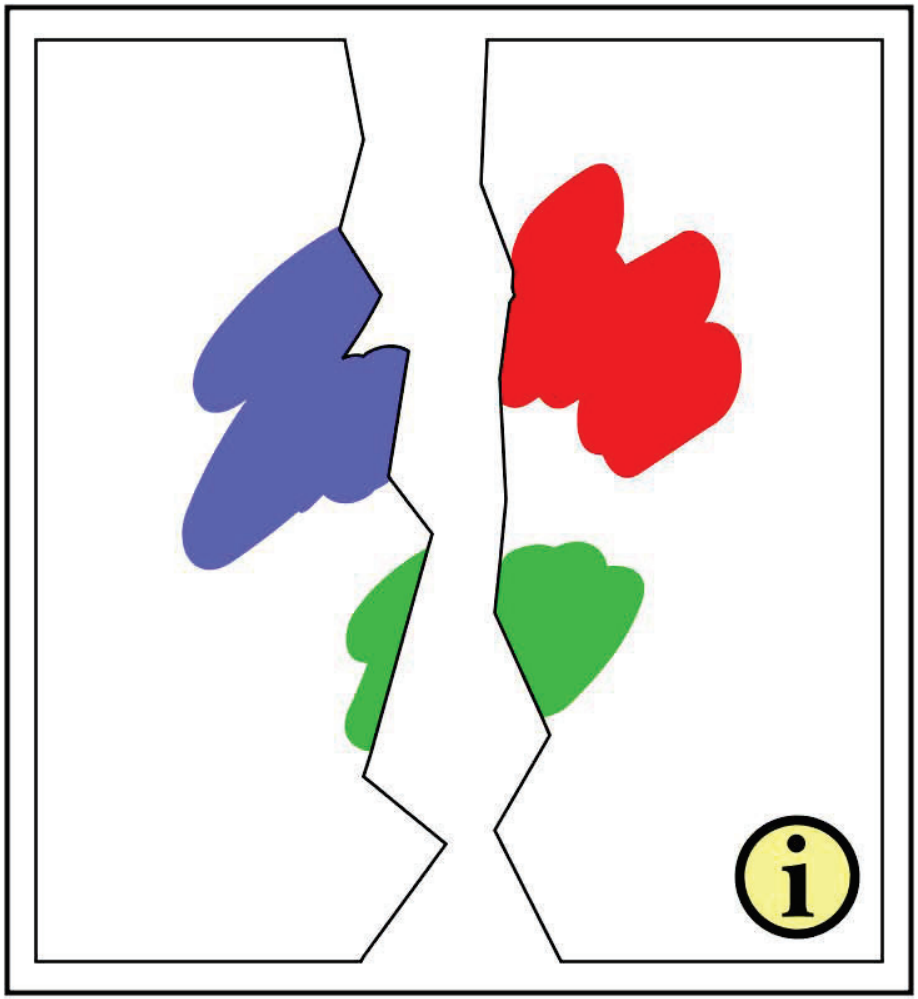
Figure 4. Glideslope and Autolevel Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 3)



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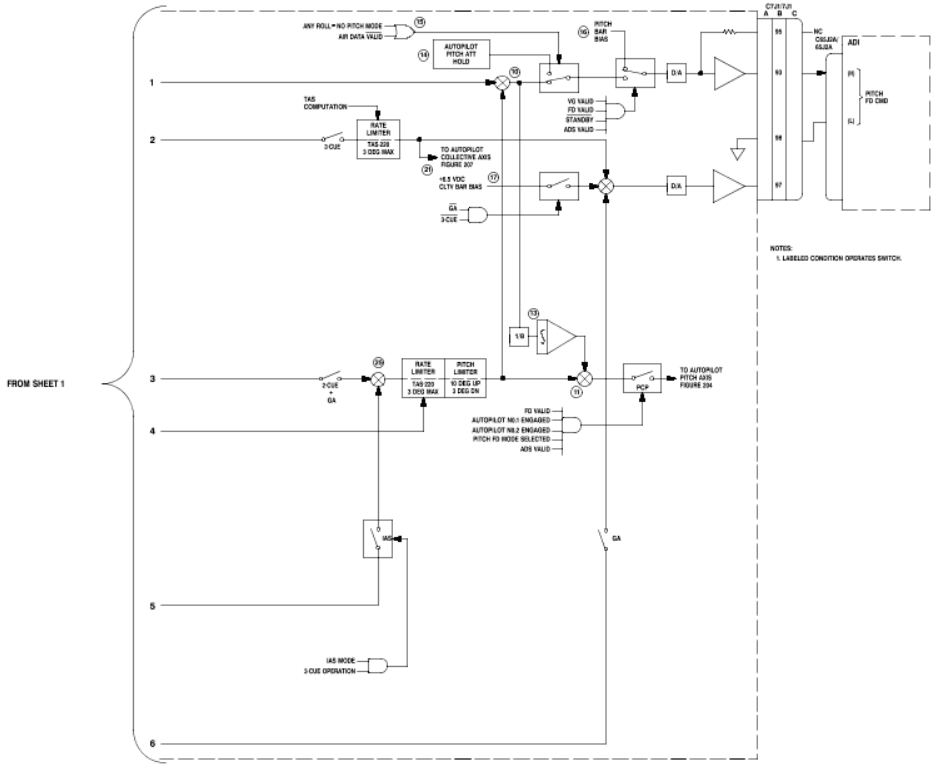
Figure 4. Glideslope and Autolevel Mode Flow Diagram (S/N 36087 through 36999) (Sheet 3 of 3)

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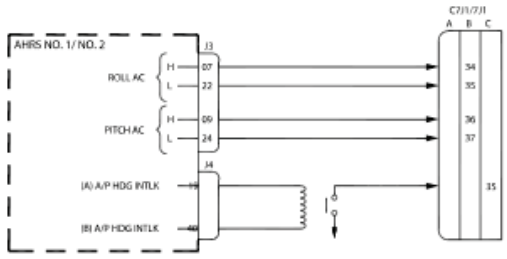
ICN-412-A-961000-B-97499-00040-A-001-01

Figure 5. ALT, VS, IAS, and Go-Around Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 3)



ICN-412-A-961000-B-97499-00041-A-001-01

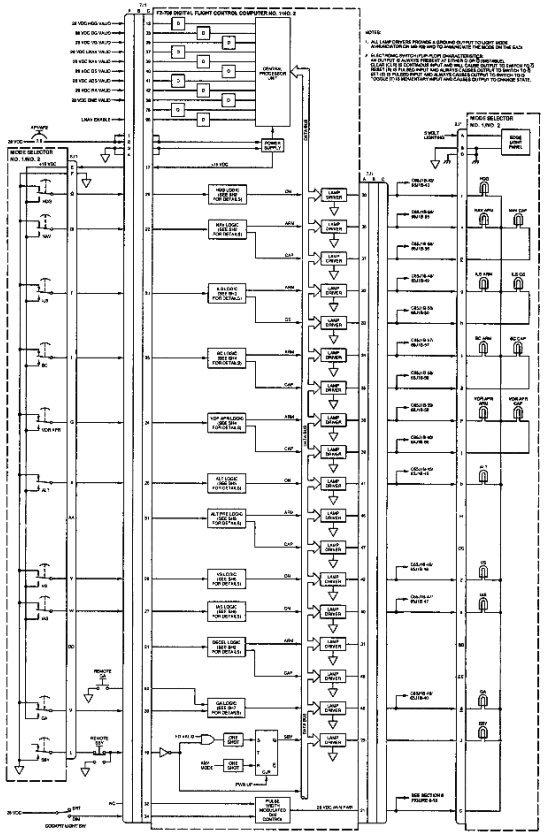
Figure 5. ALT, VS, IAS, and Go-Around Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 3)



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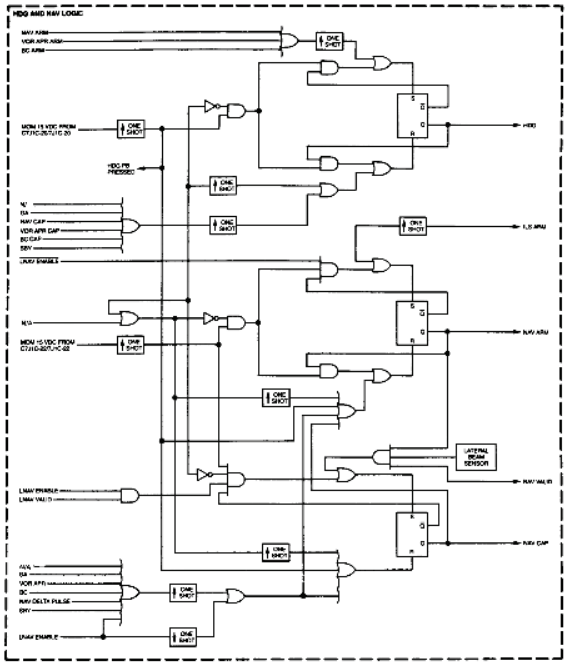
Figure 5. ALT, VS, IAS, and Go-Around Mode Flow Diagram (S/N 36087 through 36999) (Sheet 3 of 3)

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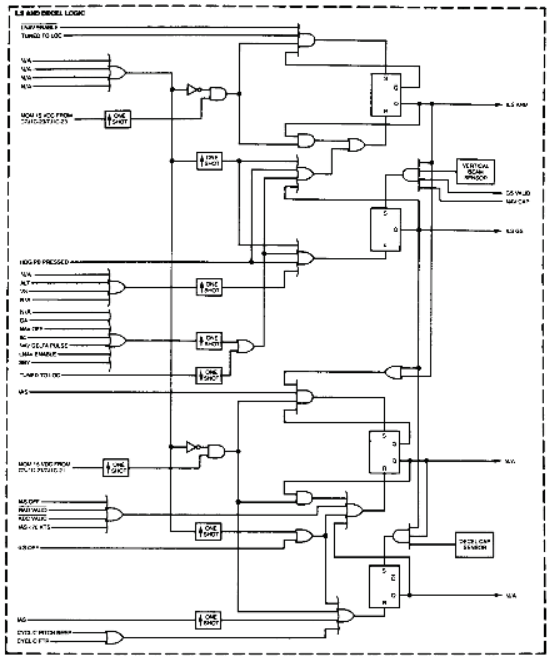
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Figure 6. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 1 of 7)



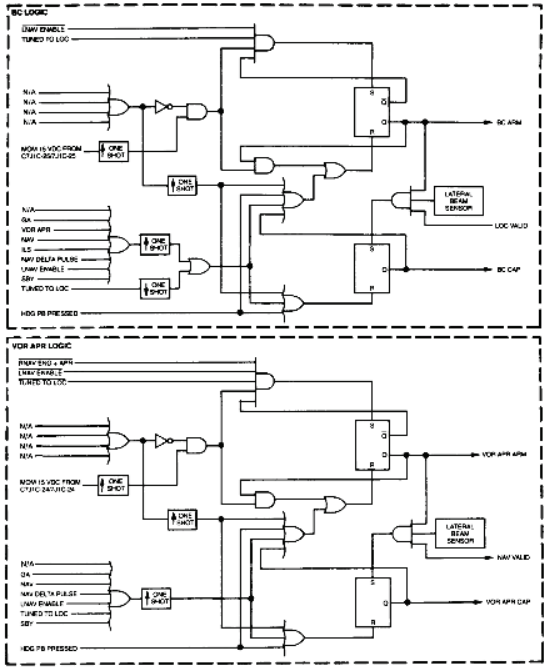
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Figure 6. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 2 of 7)



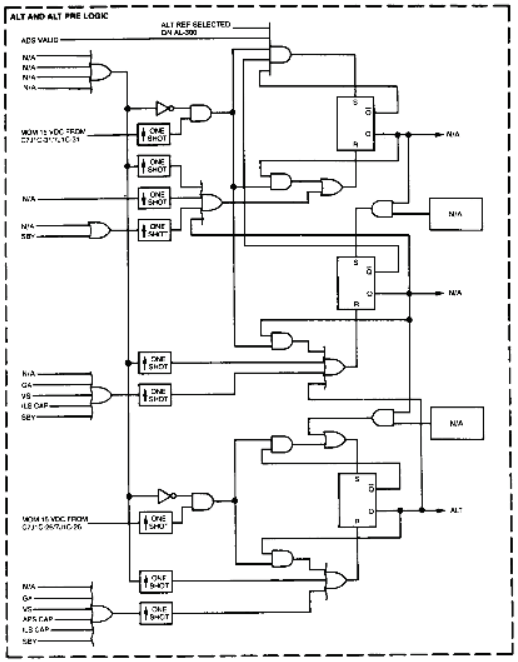
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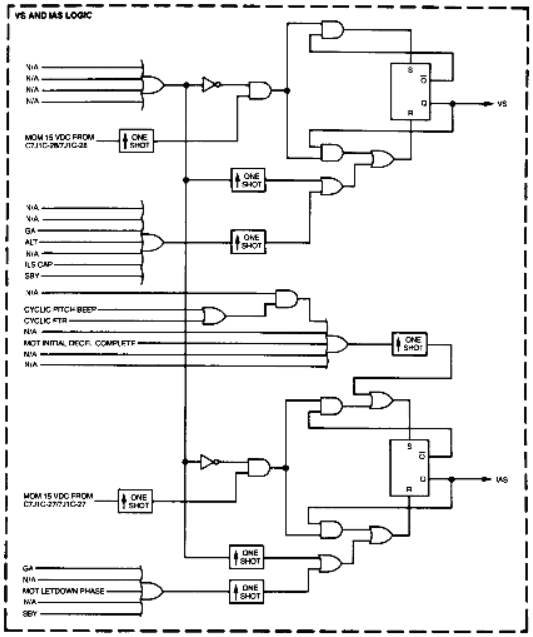
Figure 6. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 3 of 7)



ICN-412-A-961000-B-97499-00026-A-001-01

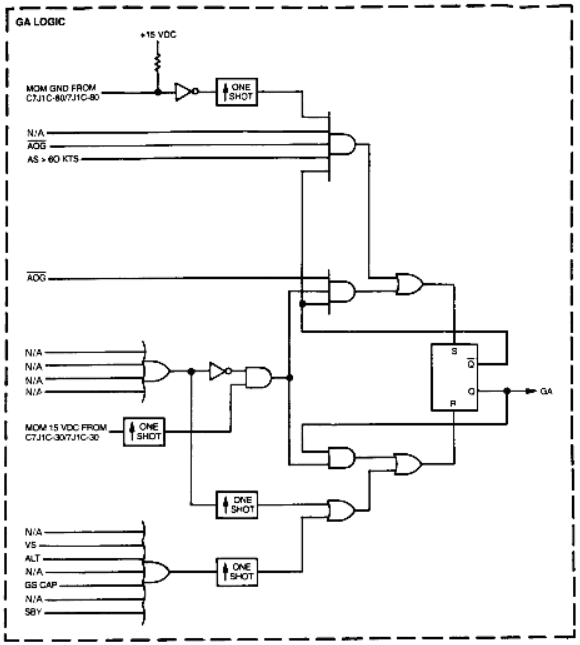
Figure 6. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 4 of 7)


ALT AND ALT PRE LOGIC
ALT REF SELECTED
DH AL-200
ADS VALID
N/A
N/A
N/A
N/A
ONE SHOT
ONE SHOT
ONE SHOT
ONE SHOT
N/A
N/A
SBY
N/A
N/A
N/A
N/A



ICN-412-A-961000-B-97499-00028-A-001-01

Figure 6. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 6 of 7)



ICN-412-A-961000-B-97499-00029-A-001-01

Figure 6. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 7 of 7)

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Flight Director Mode Flow - Description

The following modes of operation are used with the flight director:

- Standby (SBY)
- Lateral (Roll) Axis Modes
  - Heading Select (HDG)
  - Navigation (NAV)
  - VOR Approach (VOR APR)
  - Localizer (ILS)
  - Back Course (BC)
  - Go-Around (GA)
- Longitudinal (Pitch) Axis Modes
  - Glideslope and Autolevel (ILS)
  - Altitude Hold (ALT)
  - Indicated Airspeed (IAS) Hold
  - Vertical Speed (VS) Hold
  - Go-Around (GA)

1.1. Standby (SBY)

Pushing the SBY button on the mode selector, or the remote NAV STBY button on either cyclic stick, will cancel all modes and place the flight director in standby. Holding the button for more than 0.50 second will light all mode selector and autopilot controller buttons as a lamp test and bring the FD FAIL annunciators into view on the ADIs. This lamp test will also cause the air data command display to display all eights (88888) and light all face legends. Holding the button for more than 5 seconds will dim all lamps and test the altitude alert horn if installed. Upon release, all buttons except SBY and TRIM/FD will go out and the FD FAIL annunciators will be removed from view.

The SBY button is also used to clear any error codes displayed on the air data command display. If any flight director modes were engaged, the first push of the button will cancel all modes. Subsequent pushes will cancel the code currently being displayed.

1.2. Heading Select (HDG) and Go-Around (GA) Mode

The heading select mode (Figure 1) is used to intercept and maintain a magnetic heading. The mode is engaged by pressing the HDG button on the flight director mode selector. The annunciator will illuminate ON. The flight control computer will generate the proper roll command to bank the helicopter to intercept and maintain the heading selected by the pilot. The heading cursor (bug) on the HSI is positioned around the compass card to the heading the pilot desires to intercept. When the heading select mode is engaged, a heading error signal composed of the difference between the actual helicopter heading and the selected heading is routed to the flight control computer (1).

In the computer, the heading error signal is demodulated and converted from analog to digital form. Next, the heading error signal is True Airspeed (TAS) gain programmed (2). TAS gain programming is done on the heading error signal to achieve approximately the same response, regardless of the helicopters airspeed and altitude. The TAS gain is computed as airspeed divided by 125 knots. The TAS computation is derived from airspeed and barometric altitude information provided by the air data sensor (3). This information is combined with a fixed temperature reference to produce a TAS signal. The TAS signal runs the gain programmer and is also brought outside the flight control computer to optional use by other systems. A fixed 110-knot bias is used for TAS gain programming if the air data sensor is invalid.

After the heading error signal is TAS gain programmed, it goes through a ±20° bank limiter and a 5° per second rate limiter. From here the signal is sent to the roll command cue on the ADI and also to the autopilot roll axis for automatic flightpath steering (4).

SN: 36087-36247: As the helicopter banks in response to the heading error signal, the roll axis of the vertical gyro detects the change in helicopter roll attitude. The applicable compass system provides a signal to the flight control computer that is opposite in polarity to the heading error signal (5). The vertical gyro signal as applicable, is demodulated, line voltage compensated and converted from analog to digital form and summed with the heading error signal (1). When these signals match each other, the flight director command is satisfied and the roll command cue on the ADI is centered.

SN: 36248-36999: As the helicopter banks in response to the heading error signal, the roll axis of the AHRS detects the change in helicopter roll attitude. The applicable compass system provides a signal to the flight control computer that is opposite in polarity to the heading error signal (5). The AHRS signal as applicable, is demodulated, line voltage

compensated and converted from analog to digital form and summed with the heading error signal (1). When these signals match each other, the flight director command is satisfied and the roll command cue on the ADI is centered.

If the heading error signal and the signals from the vertical gyro or AHRS, as applicable, do not match exactly, the difference between the two will be seen at the integrator circuit (7). This will cause the integrator to develop an output to increase the steering command going to the autopilot and eliminate standoffs. Note that the integration input signal goes through a gain block, indicating that the integral output is only as large as the steering command at the summation point.

SN: 36087-36247: Selecting a pitch axis flight director mode without an accompanying roll mode, or if the directional gyro is invalid, will cause OR gate (8) to operate switch (9). This allows autopilot roll attitude hold to be displayed on the roll command cue on the ADI.

SN: 36248-36999: Selecting a pitch axis flight director mode without an accompanying roll mode, or if the AHRS is invalid, will cause OR gate (8) to operate switch (9). This allows autopilot roll attitude hold to be displayed on the roll command cue on the ADI.

With the OR gate not activating the switch, the roll command cue will present flight director steering information. With the OR gate activating the switch, the roll command cue will present autopilot attitude reference information.

Roll bar bias (10) is a fixed voltage level used to bias the ADI roll command cue out of view whenever an invalid condition exists. This is done to prevent the pilot from following an invalid steering command. The roll command cue is also biased from view with the flight director in standby.

The go-around mode (Figure 1) is used to transition from an approach to a climb out condition when a missed approach has occurred. The pilot selects the mode by pressing the go-around button on his collective control. When go-around has been selected, the GA annunciator on the flight director mode selector will illuminate and any previously engaged lateral and vertical flight director modes are canceled.

With go-around selected (11), either a heading hold or roll attitude hold signal is presented on the ADI roll command cue. This signal commands the wings level condition for the climbout.

1.3. Navigation (NAV) Mode

The NAV mode (Figure 2) provides for automatic intercept, capture, and tracking of a selected VOR radial. To use the mode, the pilot must do the following.

- Tune the navigation receiver to the desired VOR frequency
- If the station is listed on the navigation chart as being VORTAC (combination of VOR and TACAN), then tune the DME receiver to the station frequency to obtain distance to the station information
- Set the course pointer on the HSI for the desired course to be flown toward or away from the station

The helicopter must be flown initially on a heading that will intercept the selected course. The heading select mode is normally used to establish the intercept. With the helicopter outside the normal capture range of the VOR signal (between one and two dots on the HSI), pressing the NAV button on the flight director mode selector will:

- Illuminate ARM on the NAV button
- Illuminate Heading ON

The helicopter will fly the heading select mode as previously discussed. While the helicopter approaches the selected course intercept, the lateral beam sensor (1) monitors the VOR signal for beam deviation, beam rate, and validity. At the proper time, the lateral beam sensor will trip and capture the selected VOR course. At this time, the flight control computer switches from the heading select mode to the NAV mode of steering. On the flight director mode selector the:

- HDG ON light goes out
- NAV ARM light goes out
- NAV CAP light comes on

The flight control computer will then generate the proper roll command to bank the helicopter to capture and track the course selected by the pilot.

With the VOR mode captured, a course select error signal composed of the difference between actual helicopter heading and selected course is routed to the flight control computer.

In the computer, the course error signal is demodulated and the line voltage compensated and converted from analog to digital form at (2). Line voltage compensation corrects the signal amplitude ±10% due to fluctuations in the helicopters power supply voltage, to maintain signal gradients. Next, the course error signal is TAS gained programmed. RAS gain programming is done on the course select error signal to achieve approximately the same

response, regardless of the helicopters airspeed and altitude. The TAS gain change is computed as airspeed divided by 125 knots.

The TAS computation is derived from airspeed and barometric altitude information provided by the air data sensor. This information is combined with a fixed temperature reference to produce a TAS signal. The TAS signal runs the gain programmer and is also brought outside the flight control computer, for optional use by other systems. Should the air data sensor be programmer and is also brought outside the flight control computer, for optional use by other systems. Should the air data sensor be invalid, TAS gain is set as a function of a fixed 110-knot bias. From the TAS gain programmer, the course select error signal is routed to summation point (3). When the lateral beam sensor trips, radio deviation is mixed with course error in the flight control computer to provide the required steering command. Radio deviation from the nav receiver is brought to the flight control computer, changed from analog to digital form, then applied to a filter and beam rate circuit. The filter removes spurious noise spikes from the radio signal while the rate circuit provides a damping function. Radio deviation and rate are added at summation point (4). From summation point (4) the radio signal is lateral gain programmed to compensate for beam convergence and signal strength as the helicopter approaches the VOR station. The gain programming is done as a function of DME and is computed as distance to station, divided by 25 knots. If the DME receiver becomes invalid, then programming is done as a function of a fixed bias. From the lateral gain programmer, the radio signal is routed to summation point (3) where it is added to the course error signal. From (3), the steering command is routed to the course cut limiter.

When the helicopter approaches the desired course at an intercept angle that exceeds 45° and at a high speed, the course cut limiter functions to help prevent beam overshoots. When NAV CAP occurs for this condition, the helicopter will make an initial heading change to reduce the intercept angle, roll level, and then make a second heading change to smoothly get on course.

SN: 36087-36247: From the course cut limiter, the steering command is bank limited, rate limited, and then sent to the roll command cue on the ADI and also to the autopilot. As the helicopter banks in response to the steering command, the roll axis of the vertical gyro detects the change in roll attitude and provides a signal to the flight control computer (5) that is opposite in polarity to the steering command. The signal from the vertical gyro signal is demodulated, line voltage compensated and changed from analog to digital form. The signal is then summed with the steering command at summation point (6). When these signals match, the flight director command is satisfied and the ADI roll command cue is centered.

SN: 36248-36999: From the course cut limiter, the steering command is bank limited, rate limited, and then sent to the roll command cue on the ADI and also to the autopilot. As the helicopter banks in response to the steering command, the roll axis of the AHRS detects the change in roll attitude and provides a signal to the flight control computer (5) that is opposite in polarity to the steering command. The signal from the AHRS signal is demodulated, line voltage compensated and changed from analog to digital form. The signal is then summed with the steering command at summation point (6). When these signals match, the flight director command is satisfied and the ADI roll command cue is centered.

If the steering command and vertical gyro signal do not exactly match, the difference between the two will be seen at the integrator input (7). This will cause the integrator to develop an output, which will increase the steering command going to the autopilot and eliminate standoffs. Note that the input to the integrator goes through a -gain block, indicating that the integral signal is only as large as the steering command at summation point (6).

As the helicopter levels out to track the selected VOR course, the flight director enters a sub mode called Nav On Course (NOC) . NOC will occur when the course error is less than 22° and the helicopters bank angle is less than 6° simultaneously for 8 seconds. Two things will occur as a function of NOC. The bank limit is reduced from 22 to 13° and the rate limit is reduced from 3 to 1° per second. The bank limit and rate limit are reduced, because as the helicopter gets closer to beam center, less control is required to maintain the helicopter on its desired track. Also, as a function of NOC, crosswind correction (8) will start. If the helicopter is not on beam center due to a crosswind, radio deviation will try to turn the helicopter onto beam center. The resultant course error that is generated is washed out, and the helicopter will track the center of the beam. As the helicopter approaches the VOR station, it will enter an area of unstable signal (the zone of confusion). The zone of confusion radiates upwards in the shape of a cone around the VOR station. In this area, the radio signal becomes highly erratic, and it is undesirable to have the flight director follow the radio steering command. To monitor for this condition, the flight director has an Over Station Sensing (OSS) circuit (1). When the OSS detects that the helicopter has entered the zone of confusion, by monitoring beam deviation and beam rate, the radio path in the flight director is opened (10).

During OSS, the NAV mode is only responding to crosswind corrected course error. As the helicopter passes through the zone, the beam rate drops back down to a usable level and a 20-second timer starts. After 20 seconds, the radio path is closed again and radio deviation is again mixed with course error in the flight director. The 20-second delay in using the radio signal ensures that the radio signal is stable and the helicopter is out of the zone of confusion.

After passing through the zone of confusion, the VOR mode goes into the submode of After Over Station Sensing (AOSS) for 4 seconds, prior to returning to NOC. AOSS allows bank limits of 17° and rate limits of 3° per second. This

is allowed so that if the helicopter drifted too far off the desired radial while passing through the zone of confusion, the larger bank angle will quickly fly the helicopter back onto the selected course.

SN: 36087-36247: Selecting a pitch axis flight director mode without an accompanying roll mode, or if directional gyro is not valid, will cause switch (11) to operate. This allows autopilot roll attitude hold, or heading hold to be displayed on the roll command cue on the ADI. With the OR gate (12) not activating switch (11), the lateral command cue will display flight director steering information. With the OR gate activating the switch, the lateral command cue will display autopilot attitude reference information.

SN: 36248-36999: Selecting a pitch axis flight director mode without an accompanying roll mode, or if the AHRS is not valid, will cause switch (11) to operate. This allows autopilot roll attitude hold, or heading hold to be displayed on the roll command cue on the ADI. With the OR gate (12) not activating switch (11), the lateral command cue will display flight director steering information. With the OR gate activating the switch, the lateral command cue will display autopilot attitude reference information.

Roll bar bias (13) is a fixed voltage level used to bias the ADI roll command cue out of view whenever an invalid condition exists. This is done so the pilot will not follow an invalid lateral steering command. Roll bar bias will occur if:

- **SN: 36087-36247:**
  Vertical gyro is invalid
- **SN: 36248-36999:**
  AHRS is invalid
- Flight director is invalid
- Nav receiver is invalid
- Flight director is in standby

1.4. VOR Approach (VOR APR) Mode

The VOR APR mode (Figure 2) provides for intercept, capture, and tracking of a selected VOR radial when the helicopter is less than 10 DME miles from the station and is using the VOR as an approach to land. The VOR APR mode is set up and flown exactly like the NAV mode with the following differences:

- On the flight director mode selector, press the VOR APR button instead of the NAV button
- Crosswind correction time constant changes from 64 to 16 seconds
- Rate limit changes from 3 to 1° per second
- The OSS timer changes from 20 to 4 seconds
- DME fixed bias gains change as called out on the mode flow diagram

The VOR APR gains are adjusted to optimize flight director performance in this mode of operation.

1.5. RNAV Enroute Function

The RNAV function (Figure 2) provides for the automatic intercept, capture, and tracking of a selected ground track to the VORTAC station, or to an offset waypoint using the VORTAC station as a reference. To fly RNAV information, the navigation problem is established like the NAV mode, with the following differences:

- Tune the RNAV and DME receivers to the frequency of the VORTAC reference
- Have the RNAV selected as an input to the flight director
- Establish the navigation problem in the RNAV in accordance with the operating procedures of the RNAVs manufacturer
- Lateral gain programming of the RNAV signal is done in the RNAV receiver and therefore not needed in the flight director

The RNAV signal is brought into the flight director and treated as a radio input. The flight director will process this information and generate a steering command, as in the VOR mode of operation.

1.6. RNAV Approach Function

The RNAV approach function (Figure 2) provides for automatic intercept, capture, and tracking of a selected ground track when using a VORTAC or offset waypoint as an approach to land. By selecting the approach mode on the RNAV controller, a discrete input into the flight director establishes the proper gains to fly RNAV approach. The mode is established just like the RNAV enroute function, and the flight director will provide a steering command just like the VOR APR mode.

1.7. LNAV Function

The LNAV or lateral navigation function (Figure 2) mode is used to fly a composite steering command from a long range navigation unit such as an INS or LORAN. The LNAV function is established and flown like the NAV mode, with the following differences.

- The course select pointer on the HSI becomes a desired track pointer and course error information is not used by the flight director
- As a function of an LNAV valid and LNAV select ground, the flight director is set up to receive course angle and cross-track error (composite) steering information
- With LNAV selected, pressing the NAV button on the flight director mode selector will cause it to illuminate CAP
- Bank limits are ±32° and the rate limits are 5° per second

The LNAV signal is a composite signal that bypasses the radio path in the flight director. With the signal already gain programmed in the LNAV unit, the composite steering command can be brought into the flight director, which will then provide a steering command to fly the desired ground track as called for by the LNAV unit.

1.8. Localizer Function

The localizer function (Figure 3) provides for automatic intercept, capture, and tracking of the front course localizer beam, to line up on the centerline of the runway in use. To use this function, the pilot must do the following.

- Tune the navigation receiver to the published front course localizer frequency for the runway in use
- Set the course pointer on the HSI for the inbound runway heading
- Set the heading cursor to the desired heading to intercept the selected course

The HSI now displays pictorially the relative position of the helicopter to the center of the localizer beam and the desired inbound course. With the heading cursor set to intercept the selected course, the heading select mode will be used to establish the intercept. Outside the normal capture range of the localizer signal (between one and two dots on the HSI), pressing the NAV button on the flight director mode selector will:

- Illuminate ARM on the NAV button
- Illuminate heading ON

The helicopter will now fly the heading select mode. With the helicopter approaching the selected course intercept, the lateral beam sensor (1) monitors the localizer signal for beam deviation, beam rate, and validity. At the computed time, the lateral beam sensor will trip and capture the localizer beam. At this time, the flight control computer switches from the heading select, to the NAV mode of operation. On the flight director mode selector the:

- HDG ON light goes out
- NAV ARM light goes out
- NAV CAP light comes on

The flight control computer will now generate the proper roll command to bank the helicopter to capture and track the localizer signal. With the NAV mode captured, a course select error signal composed of the difference between actual helicopter heading and selected course is routed to the flight control computer from the HSI.

In the computer, the course error signal is demodulated, line voltage compensated, and converted from analog to digital form (2). Line voltage compensation corrects the signal amplitude ±10% due to fluctuations in the helicopters power supply voltage, to maintain proper gradient levels. Next, the course error signal is TAS gain programmed. TAS gain programming is done to achieve approximately the same response, regardless of the helicopters airspeed and altitude. The TAS gain change is computed as airspeed divided by 110 knots.

The TAS computation is derived from airspeed and barometric altitude information provided by the air data sensor. This information is combined with a fixed temperature reference to produce a TAS signal. The TAS signal runs the gain programmer and is also brought outside the flight control computer for optional use by other systems. Should the air data sensor not be valid, the TAS gain is set as a function of a fixed 110 knot bias. From the TAS gain programmer, the course select error signal is felt at summation point (3).

The localizer radio signal is sent to the flight control computer from the NAV receiver. In the computer, the signal is changed from analog to digital form, filtered, and applied to a beam rate circuit. The filter removes spurious noise from the radio signal, while the rate circuit provides a damping function. Beam deviation and beam rate are added at summation point (4). From (4) the localizer signal is lateral gain programmed to compensate for signal strength and beam convergence as the helicopter gets closer to the localizer transmitter.

Lateral gain programming is accomplished primarily as a function of radio altitude and will start at 1200 feet AGL. If the radio altimeter is not valid, then lateral gain programming will be a function of barometric altitude and will start with localizer and glideslope capture. When lateral gain programming begins, this is known as LOC II and the gain of

the localizer signal is reduced from 1.0 to 0.5 over a period of time which varies according to radio altitude. After being gain programmed, the localizer signal is added to the course error signal at summation point (3). From (3), the computed steering command goes to the course cut limiter.

When the helicopter approaches the desired course at an intercept angle that exceeds 45° and at a high speed, the course cut limiter functions to help prevent beam overshoots. When NAV CAP occurs for this condition, the helicopter will make an initial heading change to reduce the intercept angle, roll level, and then make a second heading change to smoothly get on course.

SN: 36087-36247: From the course cut limiter, the steering command is bank and rate limited and sent to the roll command cue on the ADI and to the autopilot. As the helicopter banks in response to the steering command, the roll axis of the vertical gyro detects the change in the helicopter roll attitude and provides a signal to the flight control computer (5) that is opposite in polarity to the steering command. The change of attitude signal is demodulated, line voltage compensated and changed from analog to digital form. Then, it is summed with the steering command at summation point (6). When these signals match, the flight director command is satisfied and the ADI roll command cue is centered.

SN: 36248-36999: From the course cut limiter, the steering command is bank and rate limited and sent to the roll command cue on the ADI and to the autopilot. As the helicopter banks in response to the steering command, the roll axis of the AHRS detects the change in the helicopter roll attitude and provides a signal to the flight control computer (5) that is opposite in polarity to the steering command. The change of attitude signal is demodulated, line voltage compensated and changed from analog to digital form. Then, it is summed with the steering command at summation point (6). When these signals match, the flight director command is satisfied and the ADI roll command cue is centered.

If the steering command and change of attitude signals from the vertical gyro or AHRS, as applicable, do not exactly match, the difference between the two will be seen at the integrator input (7). This will cause the integrator to develop an output to increase the steering command going to the autopilot to eliminate standoffs. Note that the input to the integrator goes through a -gain block, indicating that the integral signal is only as large as the steering command at summation point (6).

SN: 36087-36247: As the helicopter aligns itself on localizer beam center, the localizer mode enters a submode called Nav On Course (NOC) . NOC will occur when the course error is less than 22°and the helicopters bank angle is less than 6° simultaneously for 4 seconds. As a function of NOC, two things will occur; the bank limit is reduced, because as the helicopter gets closer to beam center, less control is required to maintain it there. If the helicopter is not on localizer beam center due to a crosswind, radio deviation will try to turn the helicopter onto beam center. The resultant course error that is generated is washed out, and the helicopter will now track the center of the localizer beam. Notice that as a function of LOC II, vertical gyro roll attitude signal is summed with course error through a 32-second washout. This is done to enhance lateral stability and minimize track divergence.

SN: 36248-36999: As the helicopter aligns itself on localizer beam center, the localizer mode enters a submode called Nav On Course (NOC) . NOC will occur when the course error is less than 22°and the helicopters bank angle is less than 6° simultaneously for 4 seconds. As a function of NOC, two things will occur; the bank limit is reduced, because as the helicopter gets closer to beam center, less control is required to maintain it there. If the helicopter is not on localizer beam center due to a crosswind, radio deviation will try to turn the helicopter onto beam center. The resultant course error that is generated is washed out, and the helicopter will now track the center of the localizer beam. Notice that as a function of LOC II, the AHRS roll attitude signal is summed with course error through a 32-second washout. This is done to enhance lateral stability and minimize track divergence.

SN: 36087-36247: Selecting a pitch axis flight director mode without an accompanying roll mode, or if the directional gyro goes invalid, will cause switch (8) to operate. This allows autopilot roll attitude hold or heading hold to be displayed on the ADI roll command cue. With the OR gate (9) not operating switch (8), the roll command cue will display flight director steering information. With the OR gate operating the switch, the roll command cue will display autopilot attitude reference information.

SN: 36248-36999: Selecting a pitch axis flight director mode without an accompanying roll mode, or if the AHRS goes invalid, will cause switch (8) to operate. This allows autopilot roll attitude hold or heading hold to be displayed on the ADI roll command cue. With the OR gate (9) not operating switch (8), the roll command cue will display flight director steering information. With the OR gate operating the switch, the roll command cue will display autopilot attitude reference information.

Roll bar bias (10) is a fixed voltage level used to bias the ADI roll command cue out of view whenever an invalid condition exists. This is done so that the pilot will not follow an invalid lateral steering command. Roll bar bias will occur if:

- **SN: 36087-36247:**
  Vertical gyro is invalid
- **SN: 36248-36999:**

AHRS is invalid
- Flight director is invalid
- Nav receiver is invalid
- Flight director is in standby

1.9. Back Course (BC) Mode

The back course mode (Figure 3) provides for automatic intercept, capture, and tracking of the back course localizer signal. The mode is selected by pressing the BC button on the flight director mode selector. The mode is set up just like the front course localizer mode, with the course error and radio deviation signals internally reversed (11) to make the back course localizer approach. When BC is selected and outside the lateral beam sensor trip point, the flight director mode selector will illuminate:

- BC ARM
- HDG ON

At capture, the mode selector will illuminate:

- BC CAP

The back course mode has a gain change take place in the flight control computer (12), to compensate for closer proximity of the localizer transmitter. Capture of the glideslope signal is inhibited in the flight control computer while in the BC mode.

1.10. ILS Mode

The ILS mode (Figure 3) is to automatically capture and track the localizer and glideslope signals. The navigation receiver must be tuned to the standard ILS frequencies. The glideslope signal is used to guide the helicopter down to the runway threshold in a linear descent. Typical glideslope angles vary between 2 and 3°. The helicopter can intercept and capture the glideslope from either above or below the beam; however glideslope capture is inhibited without first capturing the localizer signal and is inhibited for back course localizer approaches.

After capturing and tracking the localizer beam, but outside the normal capture point of the vertical beam sensor, the flight director mode selector will illuminate the following:

- NAV CAP
- ILS ARM

Capture will normally occur when the helicopter is within 107 Mv (1-1/3 dot) of beam center and proper polarity of pitch command versus beam deviation exists. Capture will occur immediately if the helicopter is within 21 Mv (1/3 dot) of beam center. At capture, the flight director mode selector will illuminate the following:

- NAV CAP
- ILS GS
- Any other previously selected vertical mode goes out except IAS

Glideslope deviation is routed from the NAV receiver to the flight control computer, where it is changed from analog to digital form and applied to the vertical beam sensor. Next, the glideslope signal is gain programmed. This is done to compensate for beam convergence as the helicopter gets closer to the glideslope transmitter.

Glideslope gain programming (Figure 4) is done by the radar altimeter starting at 1200 feet AGL. Gain is reduced as a function of radio altitude from 1.0 to 0.08. Should the radio altimeter become invalid, gain programming will start at glideslope capture and will be done as a function of barometric altitude. From the gain programmer, the glideslope signal goes to summation point (3).

At GS capture, a combination of Instantaneous Vertical Velocity (IVV) and True Airspeed (TAS) is used to control how quickly the helicopter noses over (4). IVV is a combination of barometric altitude rate and vertical acceleration. The TAS signal, multiplied by the sine of 3°, provides a descent rate proportional to airspeed.

The TAS computation is derived from airspeed and barometric altitude information provided by the air data sensor. This information is combined with a fixed temperature reference to produce an approximate TAS signal. The TAS signal runs the gain programmer and is also brought outside the computer for optional use by other systems.

From summation point (3), the glideslope signal is routed for either 2-cue or 3-cue operation. In 2-cue operation (standard pitch and roll commands with the collective cue biased out of view), the glideslope signal is rate limited, pitch limited, and appears at summation points (5) and (6). From summation point (5), the signal is sent to the pitch command cue on the ADI, and from summation point (6), the signal is applied to the pitch axis of the autopilot. In 3-cue operation, IAS (7) is commanded on the pitch cue of the ADI and the pitch axis of the autopilot, while the glideslope command is presented on the collective command cue (8) and drives the autopilot collective axis.

SN: 36087-36247: As the helicopter noses over to follow the glideslope signal, the pitch axis of the vertical gyro detects the change in helicopter attitude and provides change of pitch signal to the flight control computer (9) that is opposite in polarity to the glideslope signal. The change of pitch signal is demodulated, line voltage compensated, and changed from analog to digital form. Line voltage compensation corrects signal amplitude ±10%, due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. After being changed to digital form, the change of pitch signal is summed at (5) through a 12-second washout circuit.

SN: 36248-36999: As the helicopter noses over to follow the glideslope signal, the pitch axis of the AHRS detects the change in helicopter attitude and provides change of pitch signal to the flight control computer (9) that is opposite in polarity to the glideslope signal. The change of pitch signal is demodulated, line voltage compensated, and changed from analog to digital form. Line voltage compensation corrects signal amplitude ±10%, due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. After being changed to digital form, the change of pitch signal is summed at (5) through a 12-second washout circuit.

Once the helicopter is lined up on the glideslope beam (2-cue), or maintaining desired IAS (3-cue), the charge of pitch signal is washed out so that it does not fight the steering command to control the helicopter.

If a gust causes the attitude to change, the vertical change of pitch signal will cause immediate control response to reestablish the chosen flight path attitude.

With the pitch command and change of pitch signals exactly matched at summation point (5), the pitch command cue on the ADI will be centered. If these two signals do not exactly match, the difference between the two will be seen at the integrator circuit (10). The integrator will develop an output to increase the vertical axis command to eliminate standoffs. Note that the integrator input signal goes through a -gain block, indicating that the integral signal is only as large as the pitch command at summation point (6).

1.11. Autolevel

The autolevel (Figure 4) feature is a submode of ILS and provides a smooth level-off, or flare, to terminate descent of the glideslope. Autopilot causes the helicopter to level off and fly 50 feet above the runway while following localizer guidance. Glideslope tracking is automatically disconnected. If the radar altimeter is invalid, autolevel is disabled and glideslope tracking will continue.

With a valid radio altimeter, as the helicopter passes through 150 feet absolute altitude, the 150-foot sensor (14) arms the flare sensor (15). At approximately 90 feet radio altitude, the flare sensor will trip. When the flare sensor trips, a 2° noseup bias (16) is injected at summing point (3) through an 8-second washout circuit. This is done to arrest the descent to approximately level attitude. The glideslope signal path is open at (17). A fixed 50-foot reference voltage (18) is applied to summation point (19), where it is compared to actual radio altitude. Any difference between these two signals will be measured at summation point (3), and from there will command a return to 50-foot radio altitude. The IVV signal at (19) is a damping term used to control helicopter response.

In either 2-cue or 3-cue operation, a change in collective stick position will tend to change the helicopters flight path. To minimize helicopter flight path deviations caused by collective stick movement, collective to pitch axis crossfeed is done by means of a collective position potentiometer signal (20).

Autopilot pitch attitude hold (21) will normally occur when a lateral flight director mode is selected without an accompanying vertical mode. The flight director and autopilot pitch axis will also revert to pitch attitude hold attitude hold in 2-cue operation automatically, for the following conditions:

- Glideslope is captured, the helicopter has not flared, and glideslope valid is lost or
- The helicopter has flared and the radio altimeter valid is lost

If OR gate (22) has not operated switch (23), the pitch command cue on the ADI will present flight director information. If OR gate (22) has operated switch (23), the pitch command cue on the ADI will present autopitch attitude reference information.

Pitch bar bias (24) is a fixed voltage level used to bias the ADI pitch command cue out of view whenever an invalid condition exists. This is done to prevent the pilot from following an invalid longitudinal flight director command. Pitch bar bias will occur if:

- Vertical gyro is invalid
- Flight director is invalid
- Flight director is in standby

1.12. Altitude Hold (ALT) Mode

The ALT mode (Figure 5) is a vertical axis flight director mode used to maintain a barometric altitude reference. In standard 2-cue operation (pitch and roll commands on the ADI), the vertical command for altitude hold is displayed on

the pitch command cue. In 3-cue operation (pitch, roll and collective commands on the ADI), the vertical command for altitude hold is displayed on the collective command cue. To fly altitude hold, the pilot would:

- Fly to desired altitude
- Press the ALT button on the flight director mode selector

NOTE

Pressing and releasing cyclic FTR during 2-cue operation will resynchronize the altitude reference while the mode is engaged. During 3-cue operation, collective FTR use will resynchronize the altitude reference.

At the time engagement, any previously selected vertical flight director mode except IAS is canceled, and the mode selector will illuminate ALT ON. The vertical axis of the flight director will synchronize to barometric altitude at the time of mode engagement and will generate a command to maintain that altitude.

Prior to mode engagement, pressure altitude information provided by the air data sensor is present at summation point (1). It is then corrected by the setting of the baro set potentiometer on the barometric encoding altimeter. The signal is then routed to summation point (2) and with the mode not engaged, from there to the input of the altitude hold synchronizer. The synchronizer develops an output equal in amplitude but opposite in polarity to its input.

When the altitude hold mode is engaged, switch (3) opens and clamps the synchronizer output as the reference for the mode. With the helicopter at the desired altitude, the barometric altitude and synchronizer signals at summation point (2) match each other and no error signal results. If the helicopter departs from the desired altitude, the synchronizer output remains unchanged as a reference. The change in altitude is detected by the air data sensor and presented to summation point (2).

The difference in signals at summation point (2) generates a displacement error signal to fly the helicopter back to the selected altitude. The displacement signal is TAS gain programmed and routed to summation point (4).

TAS gain programming is done on the altitude displacement signal to achieve approximately the same response, regardless of the helicopters airspeed and altitude. The TAS computation is derived from airspeed and barometric altitude information provided by the air data sensor. This information is combined with a fixed temperature reference to produce an approximate TAS signal. The TAS signal runs the gain programmer and rate limiter.

As the helicopter departs the desired altitude, the change is applied to the rate circuit (5) and from there, to summation point (6). At summation point (6) the rate of altitude change signal is summed with vertical acceleration to produce Instantaneous Vertical Velocity (IVV) . The vertical acceleration signal is compensated by the roll axis of the vertical gyro at (7). This is necessary to prevent the accelerometer from giving an increased output due to roll attitude changes. From summation point (6), the IVV signal is filtered, inverted, and added to the displacement signal at summation point (4). The IVV signal at summation point (4) is a control term that helps to stop the helicopter from departing the desired altitude, and as it comes back to the desired altitude, the IVV term controls how quickly the helicopter comes back (reduces overshoot).

The altitude error signal is summed at (8) with a collective position potentiometer signal. In either 2-cue or 3-cue operation, a change in collective stick position will change the helicopters flight path. To counteract this, collective-to-pitch crossfeed is done by means of a collective potentiometer signal (9). Pitch changes slightly to maintain the selected altitude.

In 3-cue operation, the altitude error signal is routed to the autopilot collective axis (21) and to the collective command cue on the ADI. In 2-cue operation, the altitude error signal is rate and pitch limited and summed at points (10) and (11). From summation point (10), the signal is displayed on the ADI pitch command cue. From summation point (11), the signal is routed to the autopilot pitch axis.

SN: 36087-36247: As the helicopter responds to the altitude error signal, the change in helicopter attitude is detected by the vertical gyro pitch axis. The gyro provides a change of pitch signal (12) that is opposite in polarity to the altitude error signal. The change of pitch signal is demodulated, line voltage compensated and changed from analog to digital form. Line voltage compensation corrects signal amplitude ±10%, due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. The change of pitch signal is felt at summation point (10) through a 12-second washout circuit.

SN: 36248-36999: As the helicopter responds to the altitude error signal, the change in helicopter attitude is detected by the AHRS. The gyro provides a change of pitch signal (12) that is opposite in polarity to the altitude error signal. The change of pitch signal is demodulated, line voltage compensated and changed from analog to digital form. Line voltage compensation corrects signal amplitude ±10%, due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. The change of pitch signal is felt at summation point (10) through a 12-second washout circuit.

Since the helicopter normally flies at some pitch attitude other than 0°, it is necessary to washout the long-term signal from the vertical gyro or AHRS, as applicable, so it does not always command a nose level attitude. The 12-second washout time is long enough to allow the autopilot to respond to short term attitude changes.

SN: 36087-36247: Attitude changes due to wind gusts will be detected and corrective control applied to bring the helicopter back to its reference attitude. With the altitude error and vertical gyro signals matched at summation point (10) the ADI pitch command cue will be centered. If these two signals do not exactly match, the difference between the two will appear at the integrator circuit (13).

SN: 36248-36999: Attitude changes due to wind gusts will be detected and corrective control applied to bring the helicopter back to its reference attitude. With the altitude error and AHRS signals matched at summation point (10) the ADI pitch command cue will be centered. If these two signals do not exactly match, the difference between the two will appear at the integrator circuit (13).

The integrator will develop an output to sum with the signal at (11) to increase the pitch command and eliminate standoffs. The signal to the integrator goes through a -gain block, indicating that the signal is as large as the vertical axis command signal.

Autopilot pitch attitude hold (14) will occur if a roll flight director mode is selected without an accompanying pitch mode, or if the air data sensor is not valid. If OR gate (15) has not operated the logic switch, then flight director commands are displayed on the pitch axis command cue. If OR gate (15) has operated the logic switch, then autopilot pitch attitude is displayed on the command cue.

Pitch bar bias (16) is a fixed voltage level used to bias the pitch command cue out of view whenever an invalid condition exists. This prevents the pilot from following an invalid longitudinal flight director command. Pitch bar bias will occur if:

- **SN: 36087-36247:**
  Vertical gyro is invalid
- **SN: 36248-36999:**
  AHRS is invalid
- Flight director is invalid
- Air data sensor is invalid
- Flight director is in standby.

1.13. Vertical Speed Hold (VS) Mode

The VS mode (Figure 5) is used to automatically maintain the helicopter at a pilot selected vertical speed reference. In standard 2-cue operation (pitch and roll commands on the ADI), the command for vertical speed is displayed on the pitch command cue. In 3-cue operation (pitch, roll, and collective commands on the ADI), the vertical speed command is displayed on the collective command cue. The vertical speed reference may be changed in either 2-cue or 3-cue operation by use of the cyclic or collective beep switches. The new VS reference will be displayed for 5 seconds on the air data command display. To fly vertical speed hold, the pilot would:

- Establish a rate of climb or descent
- Press the VS button on the flight director mode selector

At the time of engagement, the mode selector will illuminate VS ON. The flight director will synchronize the vertical speed at the time of mode engagement and display it on the air data command display. Any deviation from this reference will generate an error signal to fly the helicopter back to the selected vertical speed reference.

Prior to mode engagement, barometric altitude information provided by the air data sensor is present at summation point (1). It is then corrected by the setting of the baro set knob on the barometric encoding altimeter. The altitude signal is applied to the rate circuit (5), and then routed to summation point (6). Here, vertical acceleration and pitch attitude are added to produce an IVV signal. The accelerometer signal is compensated by the roll axis signal at (7). This is necessary to prevent increased accelerometer outputs due to roll attitude changes.

The IVV signal is filtered and routed to summation point (18). With the VS mode not engaged, the signal is sent through switch (19) to the VS synchronizer, which develops an output equal in amplitude, but opposite in polarity to its input. The output of summation point (18) is then zero. When the VS mode is engaged, switch (19) opens. This clamps the synchronizer output as the reference for the mode. With the helicopter flying the desired vertical speed, the IVV and integrator signals match at summation point (18), and no error signal is developed. Should the helicopter depart from the desired vertical speed, the IVV signal would reflect the change, while the synchronizer output is held constant. The resultant error signal at (18) appears at summation point (20), and from there to summation point (8).

The VS error signal is summed at (8) with a collective position potentiometer signal (9). In either 2-cue or 3-cue operation a change in collective position will change the helicopter flight path. To counteract this effect,

collective-to-pitch crossfeed is developed through the collective potentiometer signal (9). In 3-cue operation, the VS error signal is routed to the autopilot collective axis (21), and the collective command cue on the ADI. In 2-cue operation, the VS error signal is rate limited, displacement limited, and sent to summation points (10) and (11). From (10), the signal is displayed on the ADI pitch command cue. From (11), the signal is routed to the autopilot pitch axis.

SN: 36087-36247: As the helicopter responds to the VS error signal, the change in aircraft attitude is detected by the vertical gyro pitch axis and a change of pitch signal is generated that is opposite in polarity to the VS error signal. The charge of pitch signal is then demodulated, line voltage compensated and changed from analog to digital form. Line voltage compensation corrects signal amplitude ±10% due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. The change of pitch signal appears at summation point (10) through a 4-second washout circuit. Since the helicopter normally flies at some pitch attitude other than 0°, it is necessary to washout the long term vertical gyro long term signal, so it does not always command a nose level attitude. The 4-second washout time is long enough to allow the autopilot to respond to short term changes.

SN: 36248-36999: As the helicopter responds to the VS error signal, the change in aircraft attitude is detected by the AHRS and a change of pitch signal is generated that is opposite in polarity to the VS error signal. The charge of pitch signal is then demodulated, line voltage compensated and changed from analog to digital form. Line voltage compensation corrects signal amplitude ±10% due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. The change of pitch signal appears at summation point (10) through a 4-second washout circuit. Since the helicopter normally flies at some pitch attitude other than 0°, it is necessary to washout the long term AHRS long term signal, so it does not always command a nose level attitude. The 4-second washout time is long enough to allow the autopilot to respond to short term changes.

With the VS error and change of pitch signals matched at summation point (10), the ADI pitch command cue is centered. Should the two signals not exactly match, the difference between them will be measured at the integrator circuit (13). The integrator will develop an output to sum with the signal at (11) to increase the pitch command to eliminate standoffs. Note that the integrator input signal goes through a -gain block, indicating that the integral output signal is only as large as the pitch axis command signal.

The reference vertical speed clamped by the synchronizer at the time of mode engagement can be changed through the use of the cyclic or collective beep switches. Activating either of these switches will cause a change in the vertical speed reference of 300 feet per minute per second (22).

Autopilot pitch attitude hold (14) will occur if a roll flight director mode is selected without an accompanying pitch mode, or if the air data sensor is not valid. If OR gate (15) has not operated the logic switch, then flight director commands are displayed on the pitch axis command cue. If OR gate (15) has operated the logic switch, then autopilot pitch attitude reference is displayed on the command cue.

Pitch bar bias (16) is a fixed voltage level used to bias the pitch command cue out of view whenever an invalid condition exists. This is done to prevent the pilot from following an invalid longitudinal flight director command. Pitch bar bias will occur if:

- **SN: 36087-36247:**
  Vertical gyro is invalid
- **SN: 36248-36999:**
  AHRS is defective
- Flight director is invalid
- Air data sensor is invalid
- Flight director is in standby

1.14. Indicated Airspeed Hold (IAS) Mode

The IAS mode (Figure 5) is used to automatically maintain the helicopter at a pilot selected airspeed reference. In standard operation (pitch and roll commands on the ADI) the IAS command is always displayed on the pitch command bar. The IAS reference may be changed by use of the cyclic beep switch. The new IAS reference will be displayed for 5 seconds on the air data command display. To fly IAS hold, the pilot shall:

- Establish the desired airspeed
- Press the IAS button on the flight director mode selector

At the time of engagement, any previously selected vertical flight director mode would be automatically transferred to the collective axis and IAS hold would be maintained by the pitch axis. The flight director mode selector will illuminate IAS ON and whichever previously selected vertical mode that was in use. The pitch axis of the flight director will synchronize to the indicated airspeed at the time of mode engagement and display the airspeed on the air data command display. Any deviation from the reference airspeed will generate a displacement error signal to fly the helicopter back to the pilot selected IAS reference.

Prior to mode engagement, airspeed information from the air data sensor is routed to summation point (23). From (23), the signal is routed through switch (24) to the IAS synchronizer. The synchronizer develops an output equal in amplitude, but opposite in polarity to its input. The output of summing point (23) is then zero. When the IAS mode is engaged, switch (24) opens and clamps the IAS synchronizer output as the reference for the mode. With the helicopter flying the pilot selected IAS, the airspeed signal and synchronizer output match each other at summation point (23), and no displacement error signal is developed. If the helicopter departs from its reference airspeed, the air data sensor will detect the change, while the synchronizer output is held constant. The resultant displacement error signal at (23) is routed to summation point (8) in 2-cue operation, or summation point (25) in 3-cue operation. In either 2-cue or 3-cue operation, the IAS displacement error signal is rate and pitch limited and sent to summing points (10) and (11).

SN: 36087-36247: As the helicopter responds to the IAS error signal, the change in attitude is detected by the vertical gyro pitch axis and a change of pitch signal is generated that is opposite in polarity to the IAS error signal. The change of pitch signal is demodulated, line voltage compensated, and changed from analog to digital form. Line voltage compensation corrects signal amplitude 10% due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. The change of pitch signal appears at summation point (10), through a 12-second washout circuit. Since the helicopter maintains airspeed as a function of pitch attitude, it is necessary to washout the long-term vertical gyro signal reference signal so it does not always command a nose level attitude. The 12-second washout time is long enough to allow the autopilot to respond to short term attitude changes.

SN: 36248-36999: As the helicopter responds to the IAS error signal, the change in attitude is detected by the AHRS and a change of pitch signal is generated that is opposite in polarity to the IAS error signal. The change of pitch signal is demodulated, line voltage compensated, and changed from analog to digital form. Line voltage compensation corrects signal amplitude 10% due to fluctuations in the helicopters power supply voltage. This is to maintain proper gradient levels. The change of pitch signal appears at summation point (10), through a 12-second washout circuit. Since the helicopter maintains airspeed as a function of pitch attitude, it is necessary to washout the long-term AHRS reference signal so it does not always command a nose level attitude. The 12-second washout time is long enough to allow the autopilot to respond to short term attitude changes.

With the IAS error and change of pitch signals matched at summation point (10), the ADI pitch command cue is centered. If the two signals do not exactly match, the difference between them will be measured at the integrator circuit (13). The integrator will develop an output to sum with the signal at (11) to increase the pitch command to eliminate standoffs. The integrator input signal goes through a -gain block, indicating the integral signal is only as large as the pitch command signal.

The airspeed reference clamped by the synchronizer at the time of mode engagement can be changed through the use of the cyclic beep switches. Activating the cyclic beep switch will change the reference airspeed 5 knots for every second the beep switch is held. The airspeed reference may also be changed by momentarily pushing the FTR button on the cyclic, flying to the new reference and then releasing the FTR button.

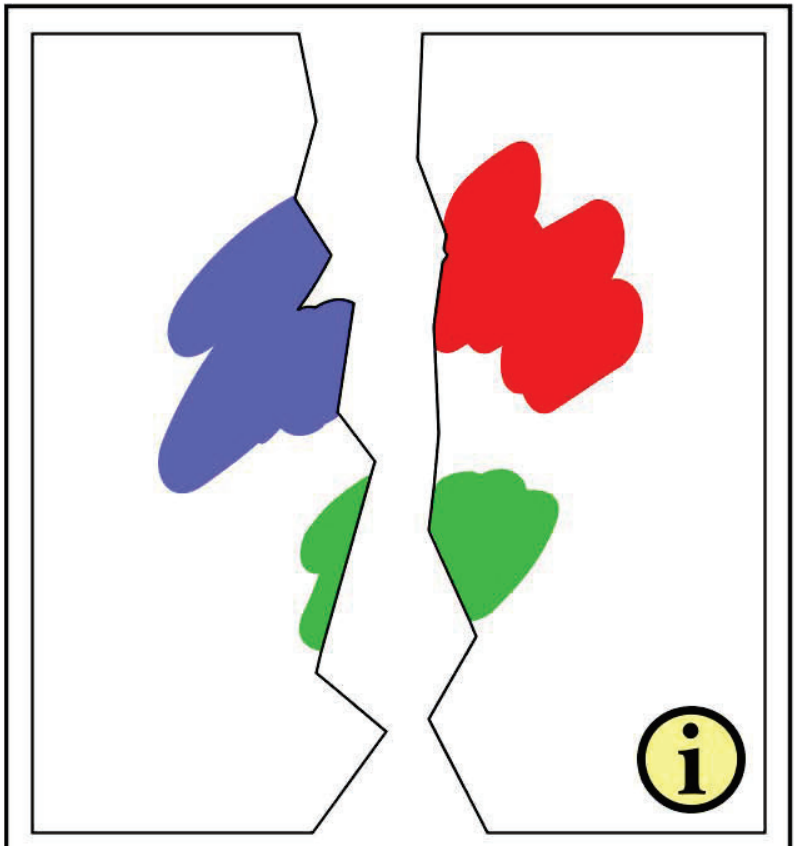
Automatic pitch attitude hold (14) will occur if a roll axis flight director mode is selected without an accompanying pitch mode, or if the air data sensor is not valid. If OR gate (15) has not operated the logic switch, then flight director commands are displayed on the pitch axis command cue. If OR gate (15) has operated the logic switch, then autopilot pitch attitude reference is displayed on the command cue.

1.15. Go-Around (GA) Mode

The go-around mode (Figure 5) is used to transition from an approach to a climb out condition, when a missed approach is required. The pilot selects the mode by pressing the go-around button on his collective control or the mode selector. The GA annunciator on the flight director mode selector will illuminate and all other previously selected modes are canceled. Laterally, go-around commands wings level. The go-around programmer will command a 700 Feet Per Minute (FPM) climb rate to be displayed on the pitch command cue.

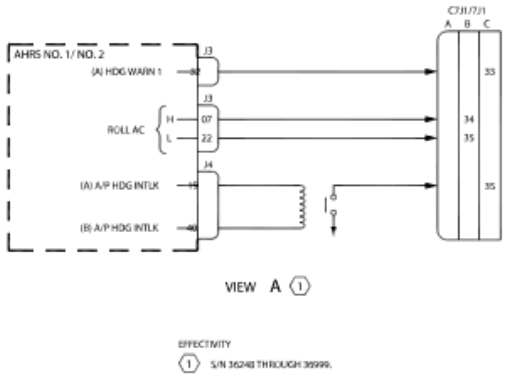
Pitch bar bias (16) is a fixed voltage level used to bias the pitch command cue out of view whenever an invalid condition exists. This is done to prevent the pilot from following an invalid longitudinal flight director command. Pitch bar bias will occur if:

- **SN: 36087-36247:**
  Vertical gyro is invalid
- **SN: 36248-36999:**
  AHRS is invalid
- Flight director is invalid
- Flight director is in standby



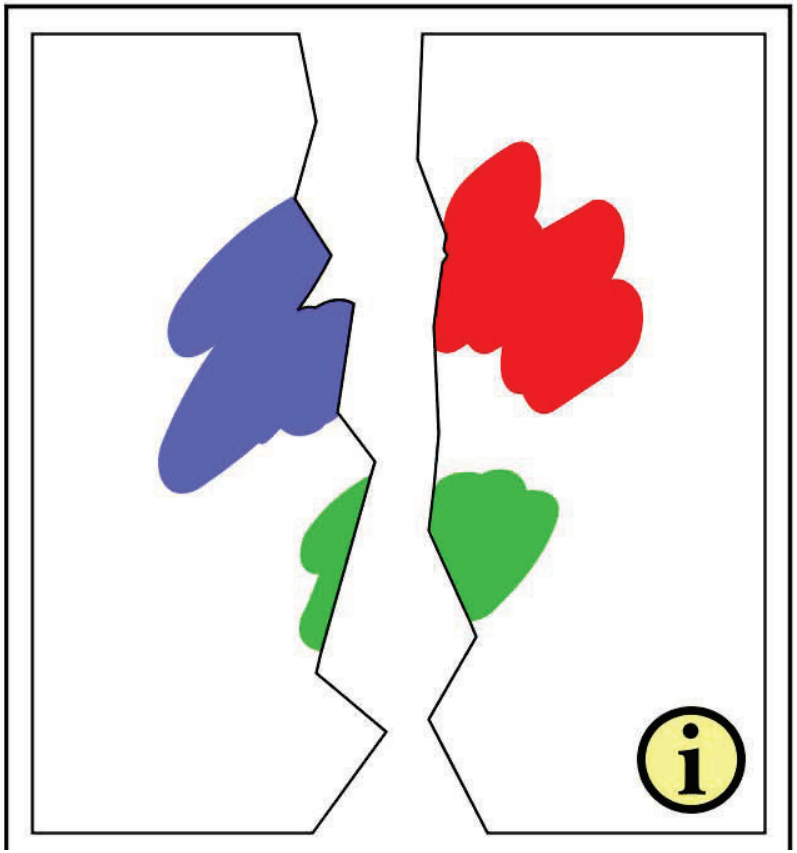
ICN-412-A-961000-B-97499-00030-A-001-01

Figure 1. Heading Select (HDG) and Go-Around (GA) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



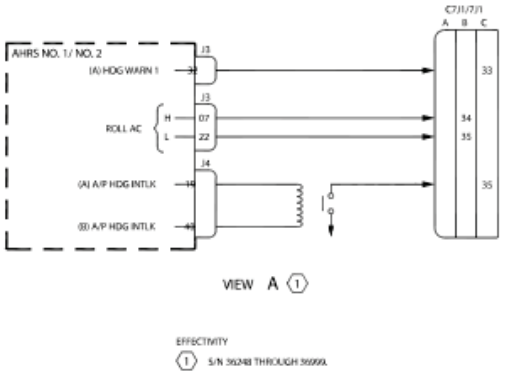
ICN-412-A-961000-B-97499-00031-A-001-01

Figure 1. Heading Select (HDG) and Go-Around (GA) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



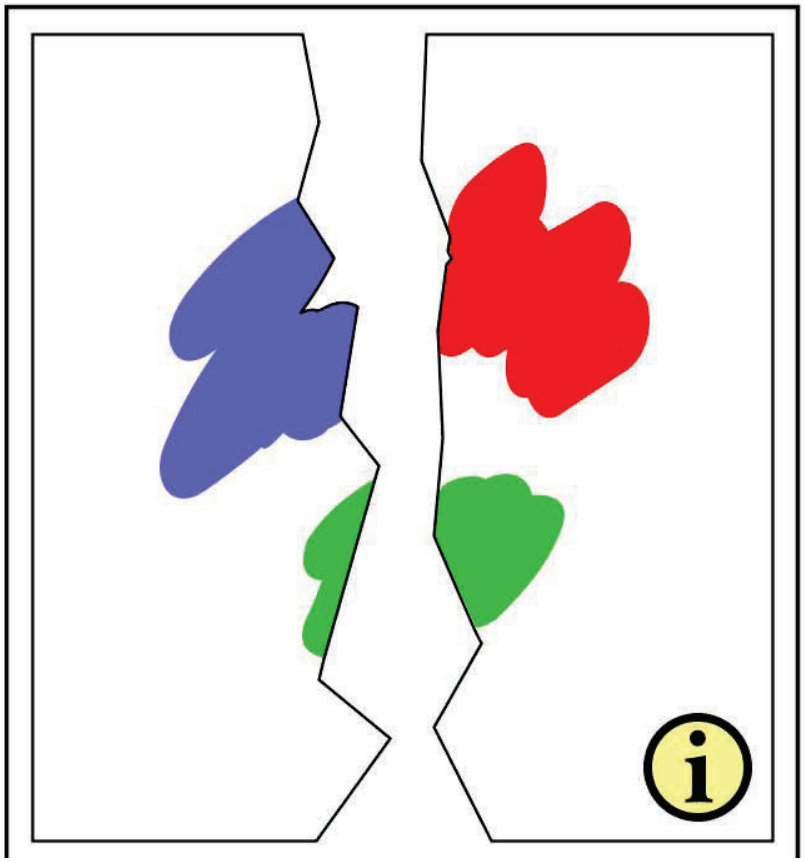
ICN-412-A-961000-B-97499-00032-A-001-01

Figure 2. VOR, VOR APR, RNAV, and LNAV Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



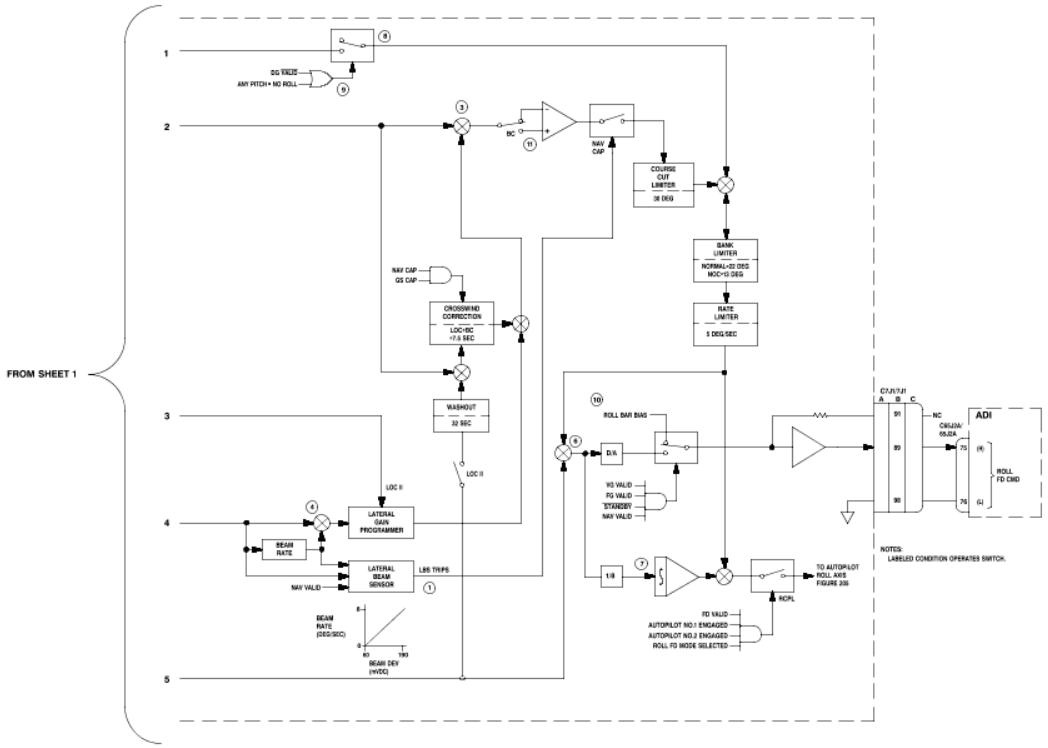
ICN-412-A-961000-B-97499-00033-A-001-01

Figure 2. VOR, VOR APR, RNAV, and LNAV Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



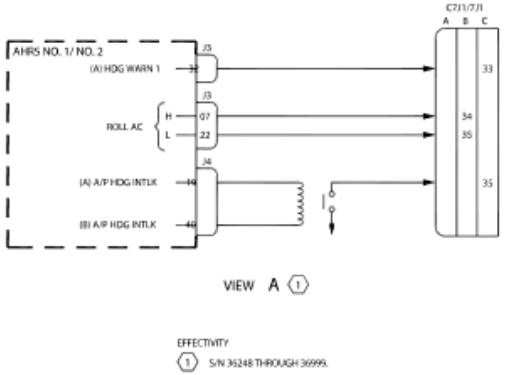
ICN-412-A-961000-B-97499-00034-A-001-01

Figure 3. Localizer and Back Course (BC) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 3)



ICN-412-A-961000-B-97499-00035-A-001-01

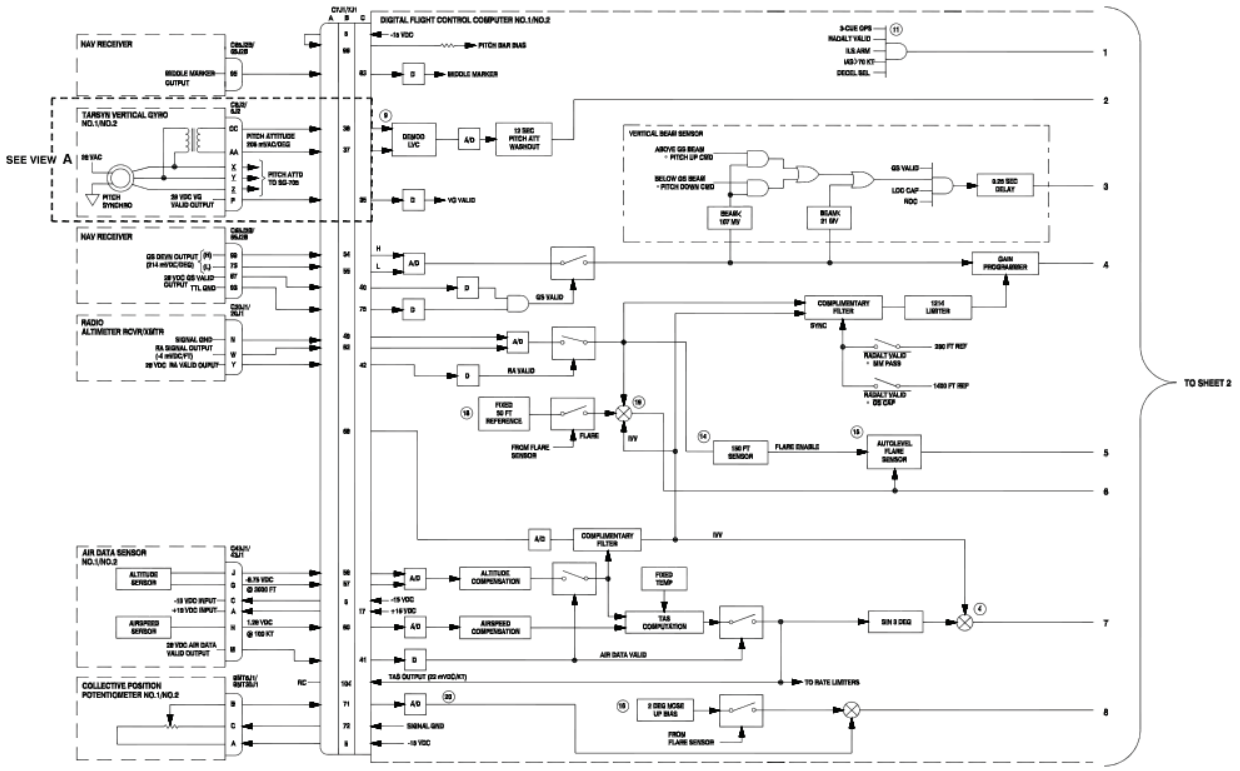
Figure 3. Localizer and Back Course (BC) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 3)



ICN-412-A-961000-B-97499-00036-A-001-01

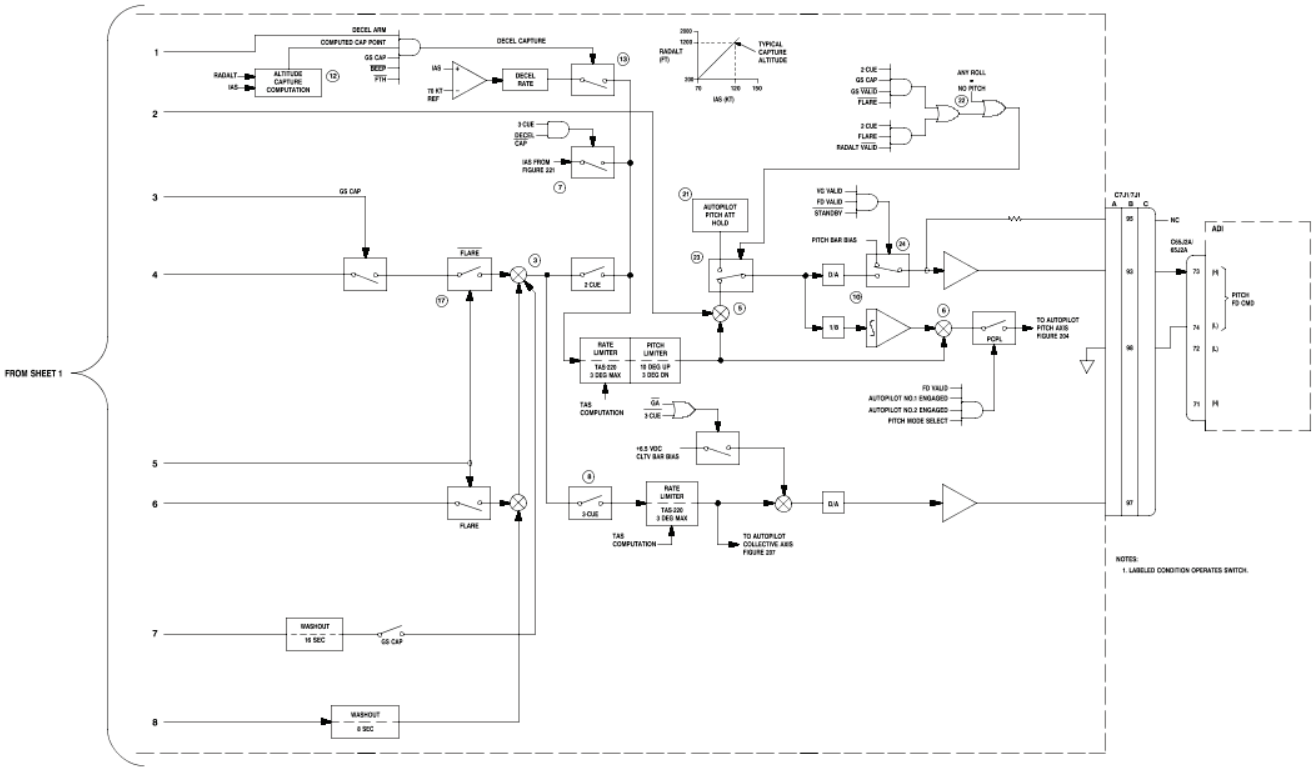
Figure 3. Localizer and Back Course (BC) Mode Flow Diagram (S/N 36087 through 36999) (Sheet 3 of 3)

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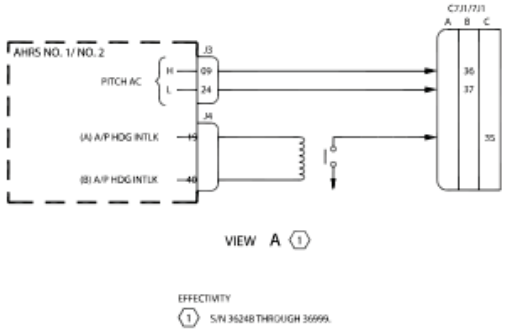
ICN-412-A-961000-B-97499-00037-A-001-01

Figure 4. Glideslope and Autolevel Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 3)



ICN-412-A-961000-B-97499-00038-A-001-01

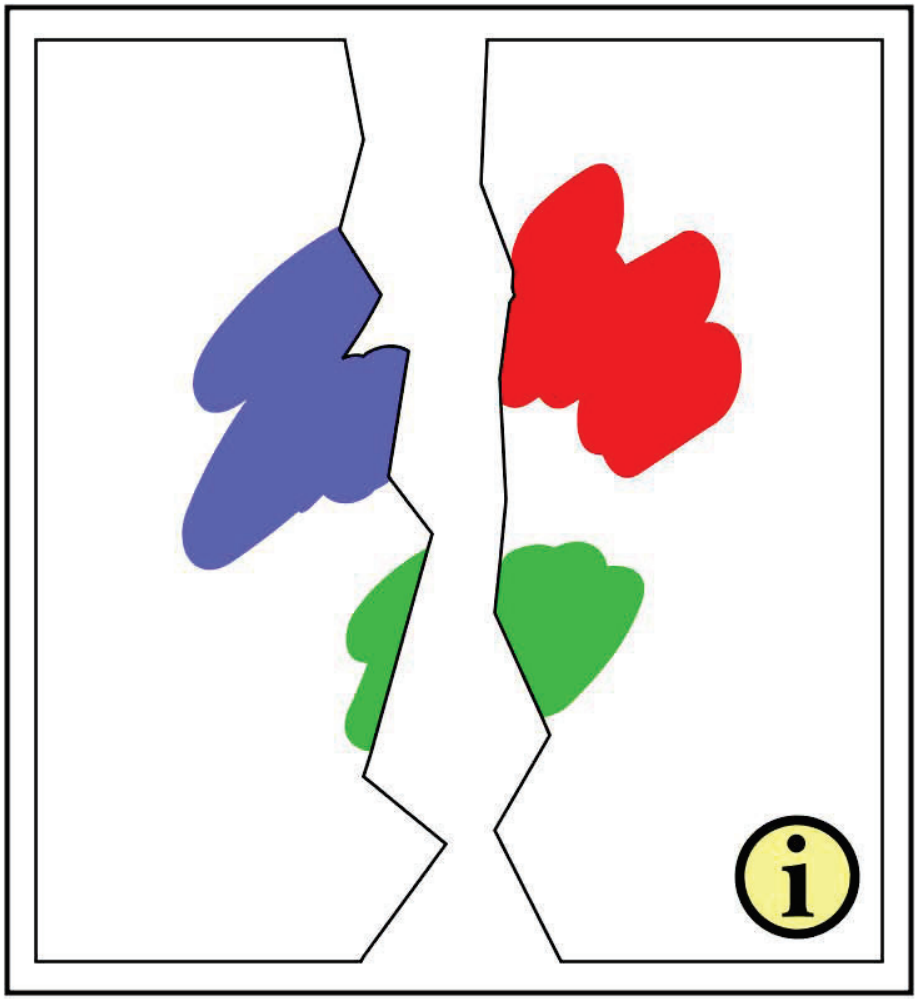
Figure 4. Glideslope and Autolevel Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 3)



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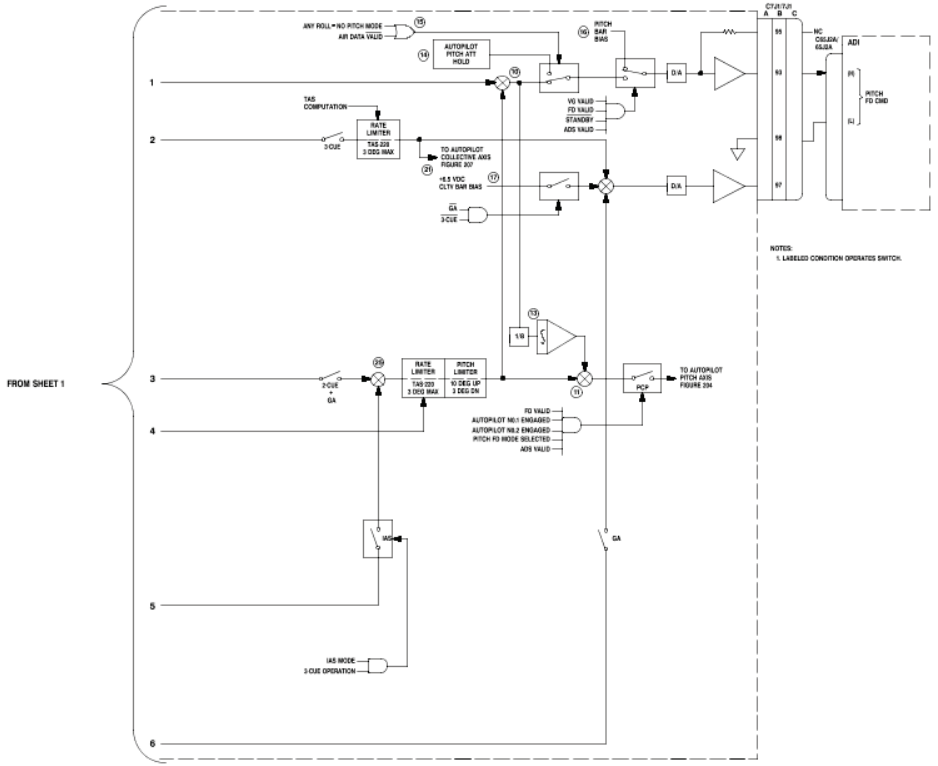
Figure 4. Glideslope and Autolevel Mode Flow Diagram (S/N 36087 through 36999) (Sheet 3 of 3)

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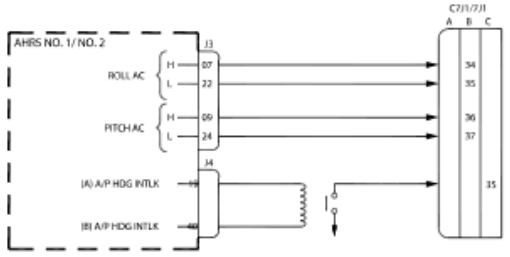
ICN-412-A-961000-B-97499-00040-A-001-01

Figure 5. ALT, VS, IAS, and Go-Around Mode Flow Diagram (S/N 36087 through 36999) (Sheet 1 of 3)



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Figure 5. ALT, VS, IAS, and Go-Around Mode Flow Diagram (S/N 36087 through 36999) (Sheet 2 of 3)



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1 S/N 36248 THROUGH 36999.

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Figure 5. ALT, VS, IAS, and Go-Around Mode Flow Diagram (S/N 36087 through 36999) (Sheet 3 of 3)

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Autopilot - Operational Description

The following paragraphs discuss the flight control system, modes of operation, self test, and signal flow through the autopilot section of the computer for each helicopter axis.

Figure 1 and Figure 2 are simplified diagrams of the flight controls. Figure 3 through Figure 8 are simplified diagrams that show signal path and interconnect wiring for the applicable axis. They are used in conjunction with Figure 9, which is a simplified diagram of the autopilot engage/control logic.

1.1. Functional Operation Autopilot System

The AFCS provides pitch, roll, and yaw axis stabilization. The system is composed of dual attitude sensors, dual digital flight control computers, and dual control actuators in each axis. Motion detectors connected to the cyclic control stick, anti-torque pedals, and the collective control stick provide position information to the computers for specific operating modes. Flight path guidance commands can also be coupled to the computers for automatic flight path steering.

The autopilot system has two basic modes of operation: stability augmentation and attitude retention. For automatic flightpath steering, the flight director commands are coupled to the autopilots, but may be decoupled at the pilots discretion.

1.2. Stability Augmentation System (SAS) Mode

The SAS mode provides short-term attitude and attitude rate stabilization for use in hands-on flying. It is referred to as a stability augmentation system that stabilizes the helicopter against outside disturbances, and thereby augments pilot effort. The SAS mode has been calibrated so that pilot control motions are enhanced while helicopter motions caused by outside disturbances are counteracted. This type of operation provides a stabilized mode wherein the basic helicopter handling qualities are improved.

SN: 36087-36247: When helicopter motion is detected by the gyros (i.e., wind gust), a stabilizing control proportional to the amplitude and rate of the motion is computed in the autopilot computers and applied to the appropriate actuators.

SN: 36248-36999: When helicopter motion is detected by the AHRS (i.e., wind gust), a stabilizing control proportional to the amplitude and rate of the motion is computed in the autopilot computers and applied to the appropriate actuators.

1.3. Attitude Retention (ATT) Mode

In the attitude retention mode, the autopilot pitch and roll channels are engaged (autopilots engaged but not coupled to the flight director) to maintain indefinitely the attitude existing at the moment of engagement. The pilot can fly completely hands-off. If a change in attitude is desired, the pilot simply pushes the beep trim switch on the cyclic control to obtain a small attitude change or the Force Trim Release (FTR) switch on the cyclic control and then moves the cyclic to obtain the desired new attitude. When the pilot releases the selected switch, the system again resumes the attitude retention function. When collective (power) changes are made, it is generally necessary to press the FTR switch and establish new cyclic positions and thereby recenter the actuators to operate with full authority about the neutral position. As the new trim condition is established, releasing the FTR switch will cause the system to maintain the new control position. The yaw channel continues to perform yaw damping as in the SAS mode.

In the ATT mode, the flight director is not coupled to the autopilot as long as the flight director mode selector is in SBY.

1.4. Automatic Flight Path Steering (CPL)

Automatic flight path steering allows flight director commands to be coupled to the autopilots pitch and roll channels. This is accomplished by having the autopilots engaged in the ATT mode and selecting any flight director mode. These lateral and vertical commands to the autopilot maneuver the helicopter to keep the ADI flight director command bars centered just as the human pilot does.

When operating in the coupled mode, retrimming the cyclic control is normally done automatically through autotrim. When both systems are engaged, autotrim starts when both actuators reach 10% of their travel from center for a period of 0.25 second, provided both actuators are commanding the same direction. Trimming stops when the actuators are essentially recentered. If only one autopilot is engaged, the trim operation is identical except that the operating actuator is required to reach 25% of its travel for 3.5 seconds to start the trimming.

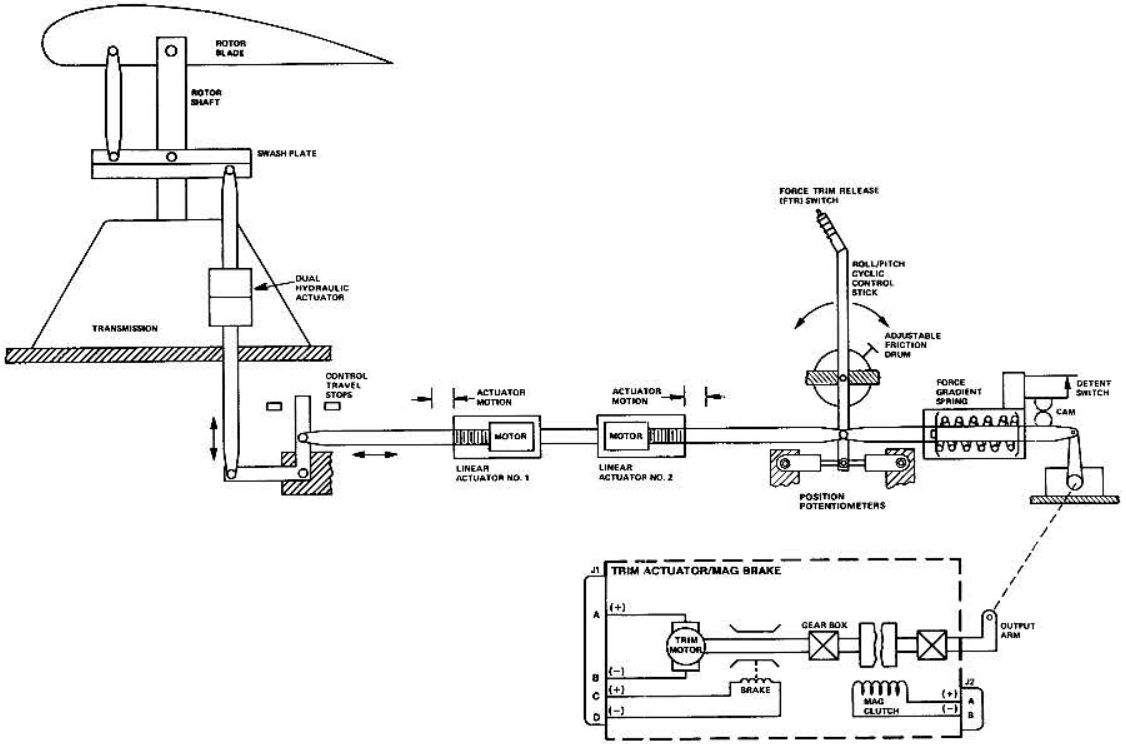
NOTE

When below 70 KIAS, single system trim operation will occur upon 25% actuator travel for 0.25 second.

In the event of a trim servo failure, manual retrim is done by momentarily pushing the cyclic FTR switch and adjusting the cyclic control to center the API needle on the AFCS indicator panel. When an invalid signal occurs, as evidenced by complete retraction of one or both flight director command bars, the affected autopilot control axis reverts to attitude retention mode.

NOTE

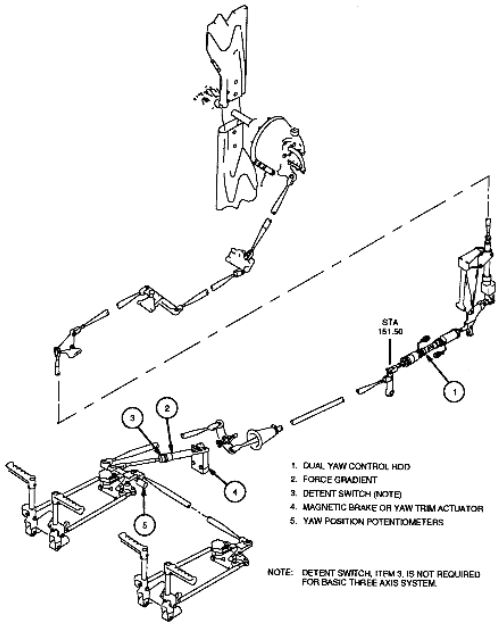
Coupling of a single autopilot to a flight director is inhibited; however, steering commands are still displayed on the appropriate ADI.



ICN-412-A-961000-B-97499-00009-A-001-01

Figure 1. Cyclic (Pitch or Roll) Control Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

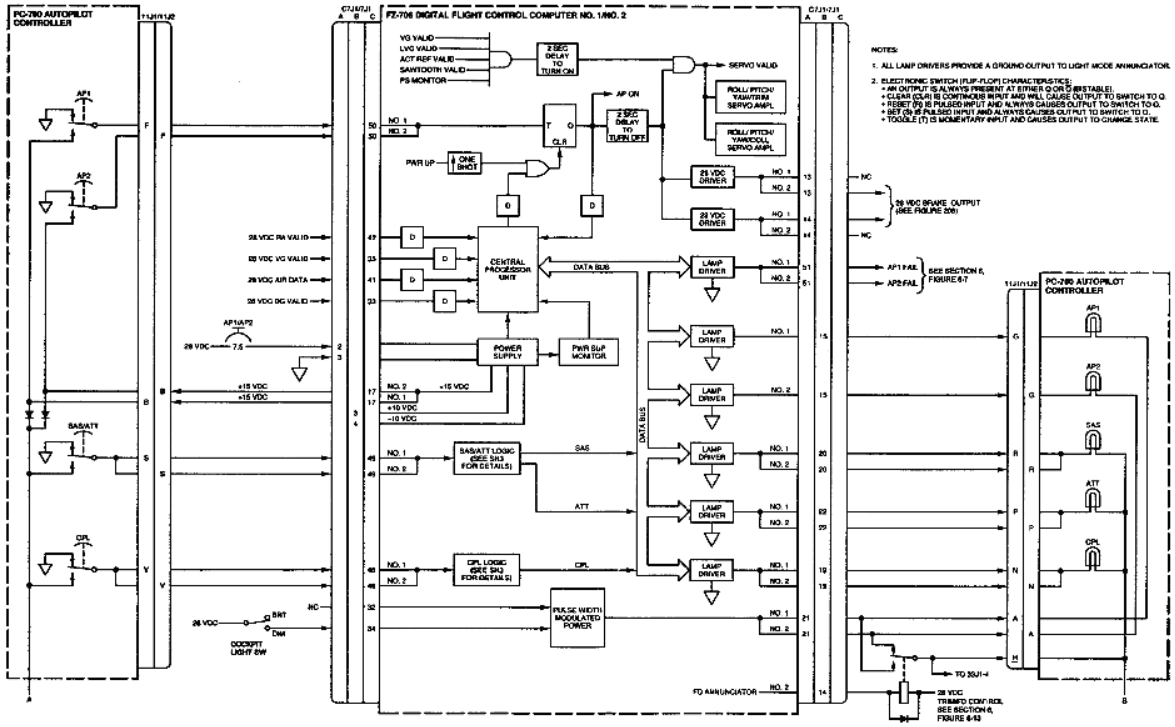
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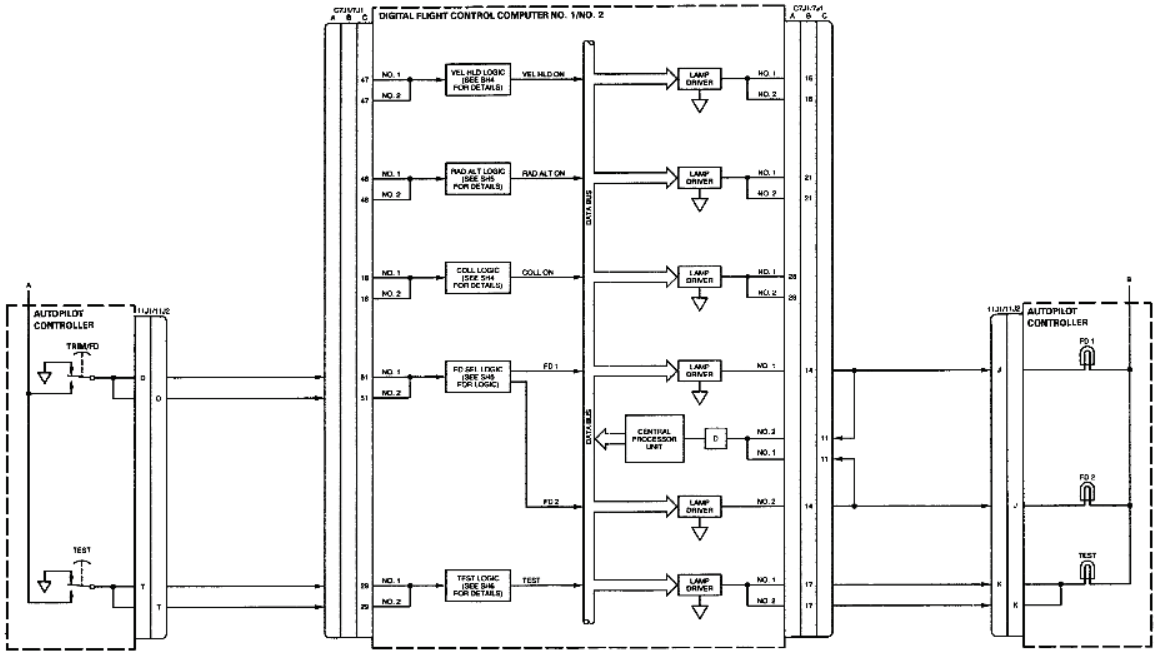
Figure 2. Pedal Control (Yaw) Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

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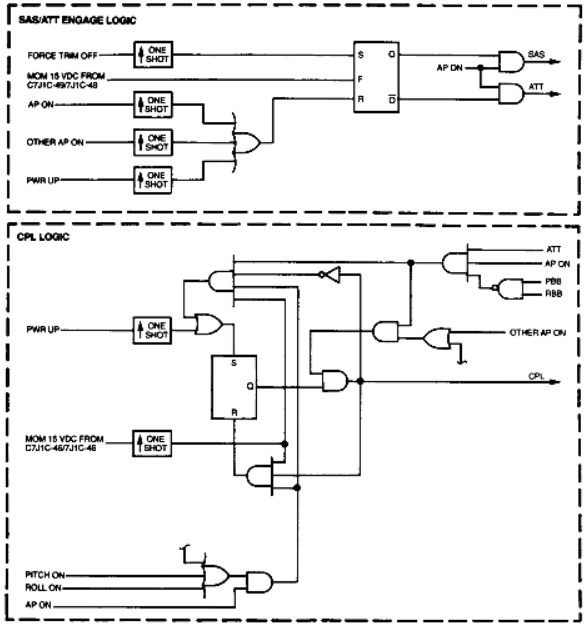
ICN-412-A-961000-B-97499-00011-A-001-01

Figure 3. DFCS Engage/Control Logic Diagram (S/N 36087 through 36999) (Sheet 1 of 4)



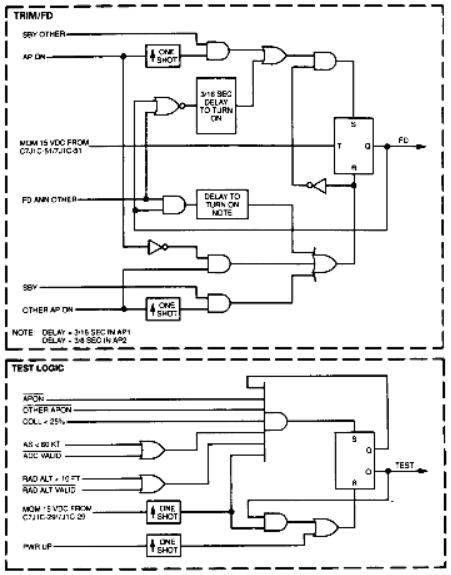
ICN-412-A-961000-B-97499-00012-A-001-01

Figure 3. DFCS Engage/Control Logic Diagram (S/N 36087 through 36999) (Sheet 2 of 4)



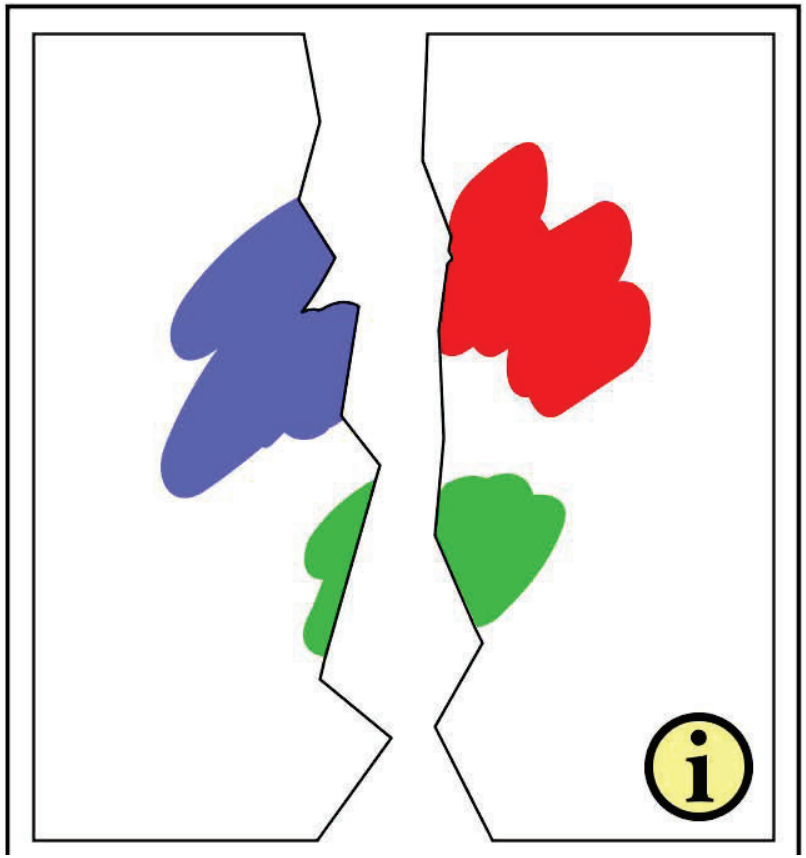
ICN-412-A-961000-B-97499-00013-A-001-01

Figure 3. DFCS Engage/Control Logic Diagram (S/N 36087 through 36999) (Sheet 3 of 4)



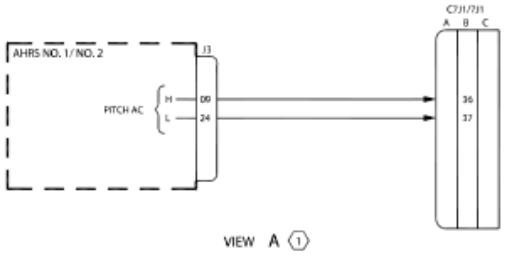
ICN-412-A-961000-B-97499-00014-A-001-01

Figure 3. DFCS Engage/Control Logic Diagram (S/N 36087 through 36999) (Sheet 4 of 4)



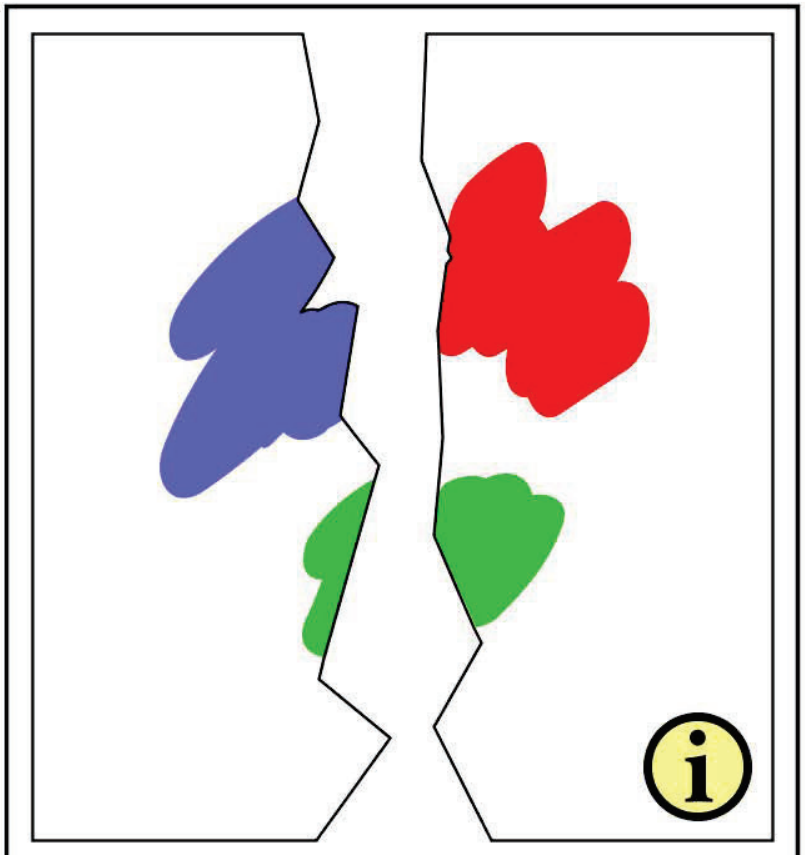
ICN-412-A-961000-B-97499-00015-A-001-01

Figure 4. Autopilot Pitch Axis Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



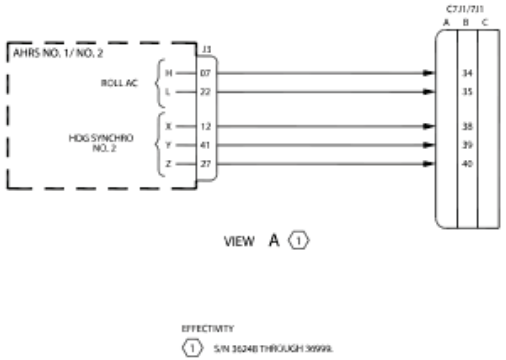
ICN-412-A-961000-B-97499-00016-A-001-01

Figure 4. Autopilot Pitch Axis Functional Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



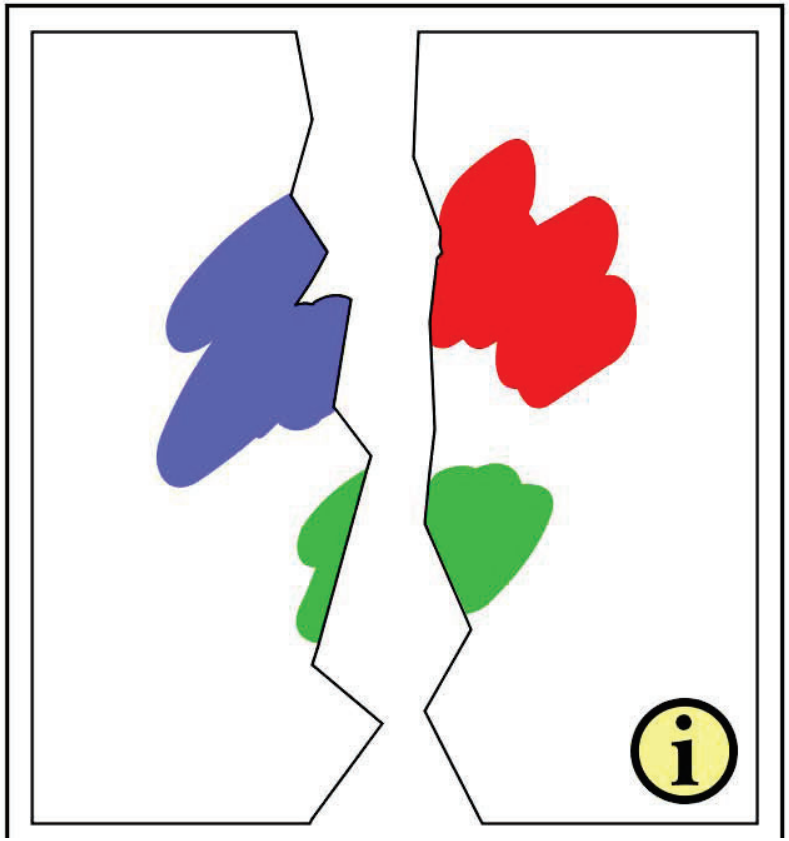
ICN-412-A-961000-B-97499-00017-A-001-01

Figure 5. Autopilot Roll Axis Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



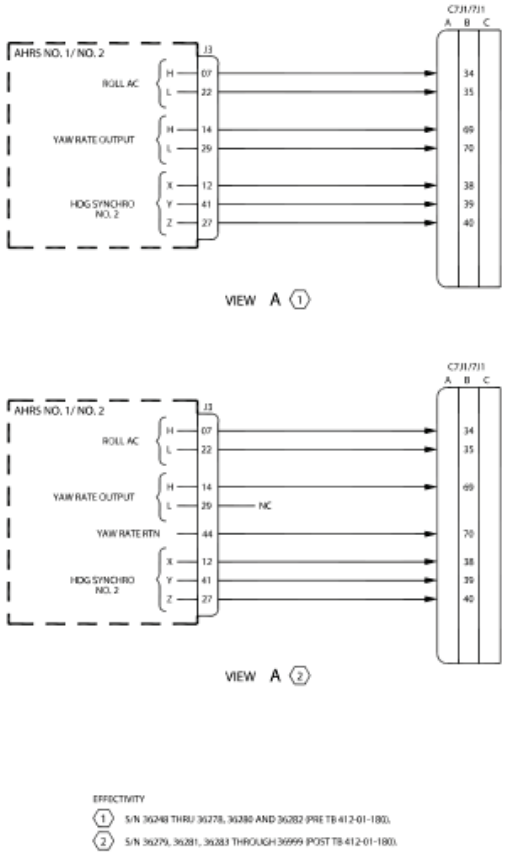
ICN-412-A-961000-B-97499-00018-A-001-01

Figure 5. Autopilot Roll Axis Functional Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



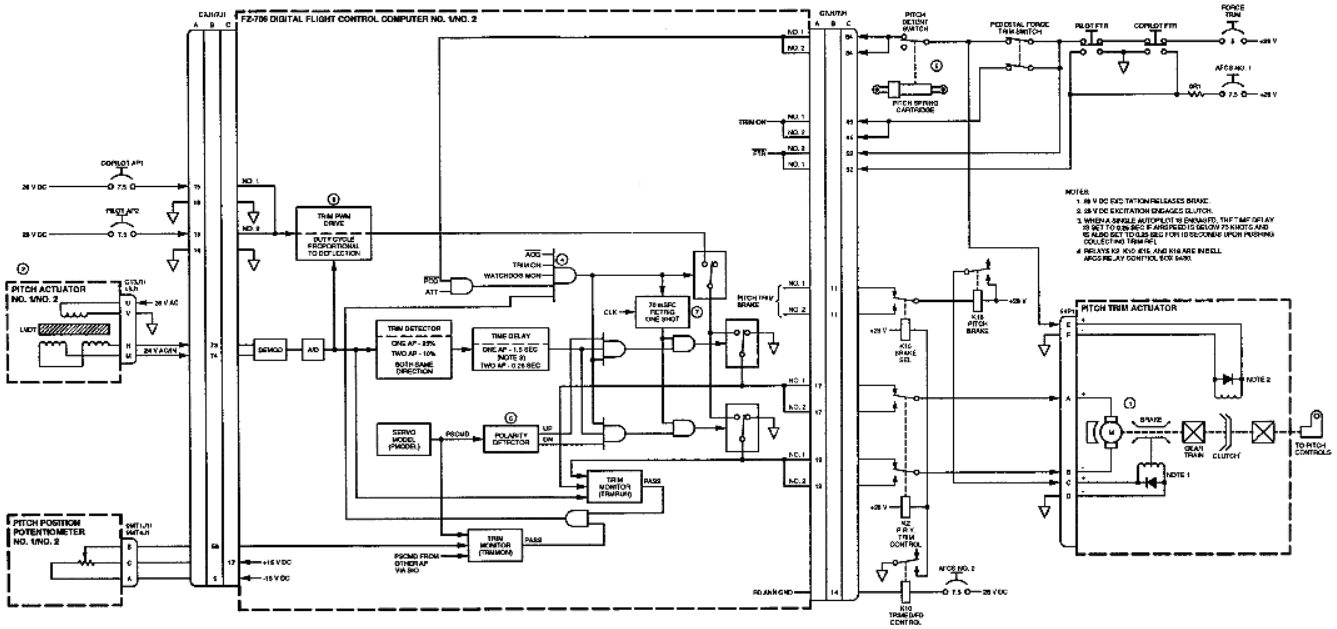
ICN-412-A-961000-B-97499-00019-A-001-01

Figure 6. Autopilot Yaw Axis Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



ICN-412-A-961000-B-97499-00020-A-001-01

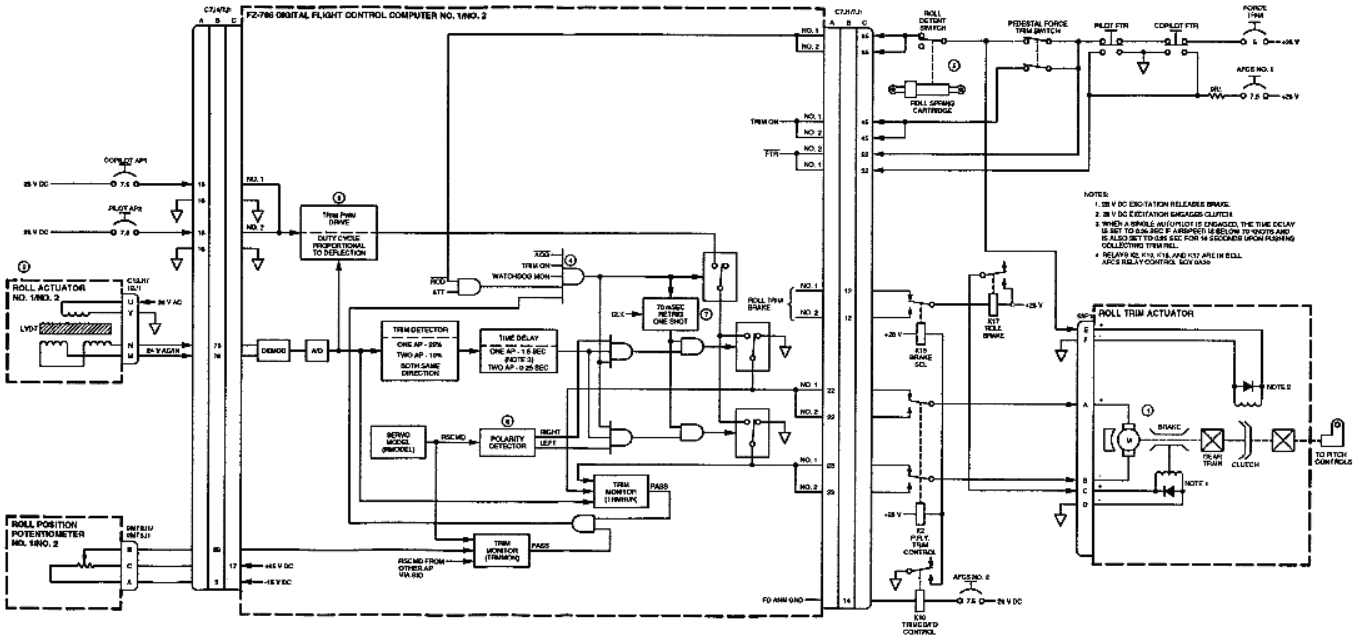
Figure 6. Autopilot Yaw Axis Functional Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



ICN-412-A-961000-B-97499-00021-A-001-01

Figure 7. Autopilot Pitch Trim Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

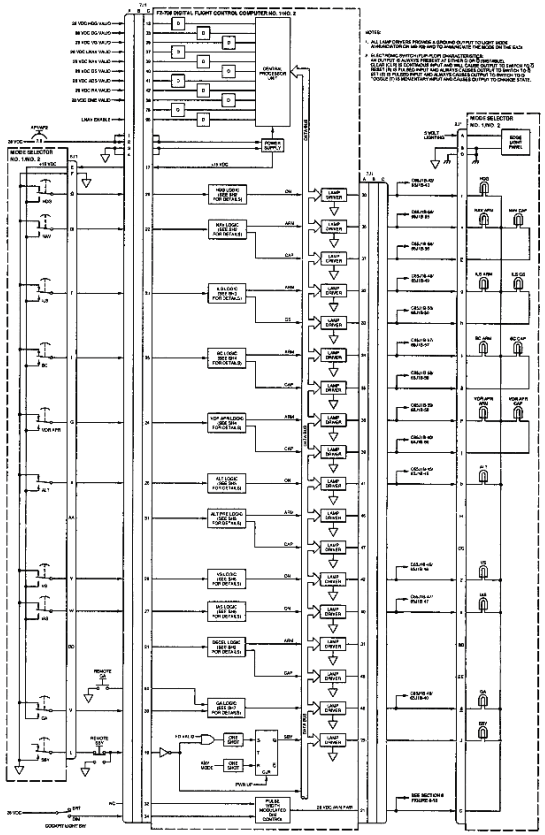
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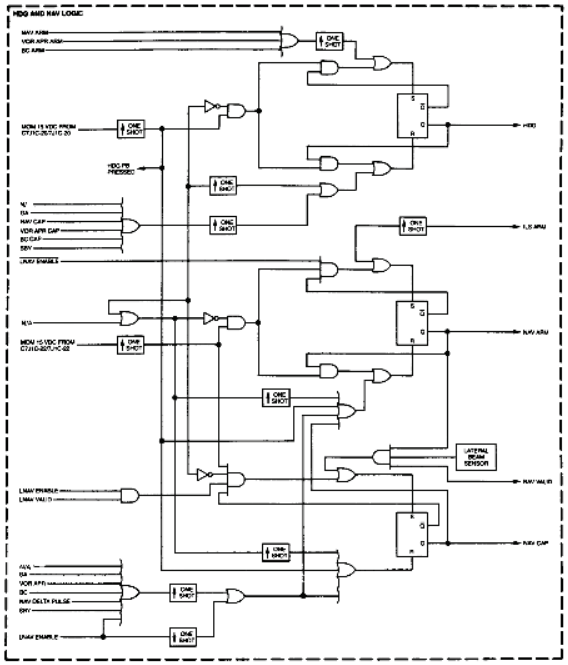
Figure 8. Autopilot Roll Trim Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

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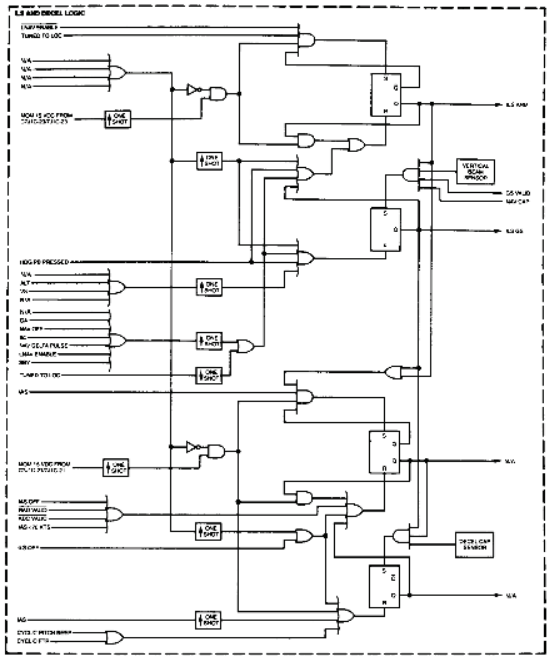
ICN-412-A-961000-B-97499-00023-A-001-01

Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 1 of 7)



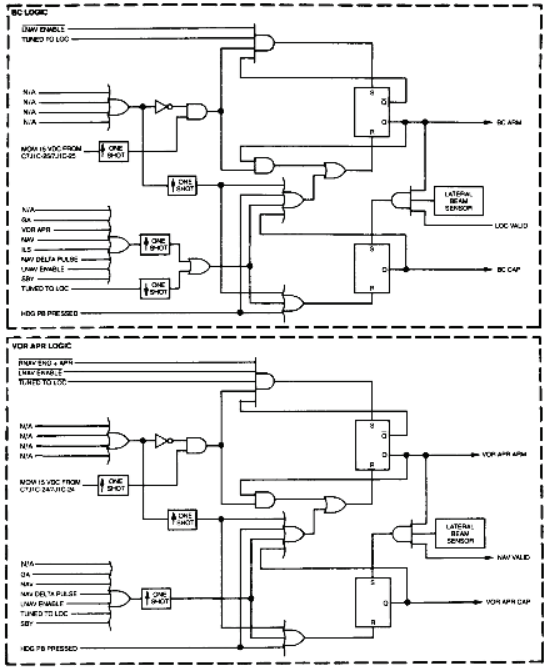
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Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 2 of 7)



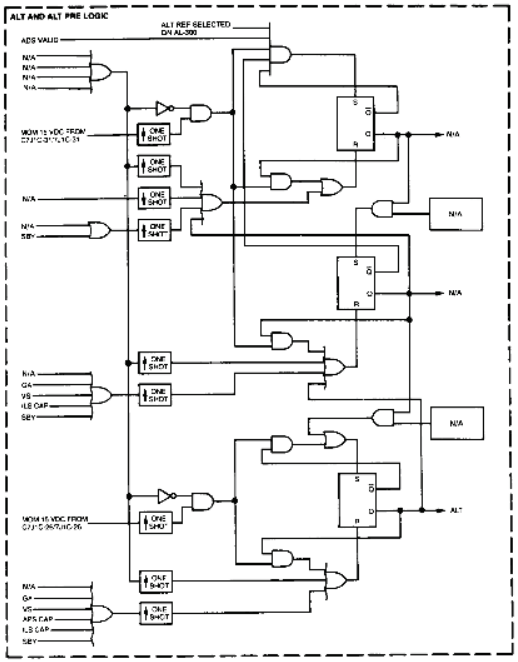
ICN-412-A-961000-B-97499-00025-A-001-01

Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 3 of 7)



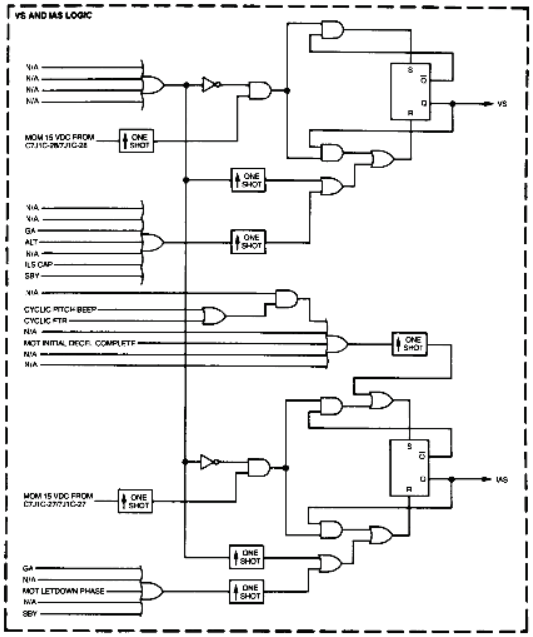
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Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 4 of 7)



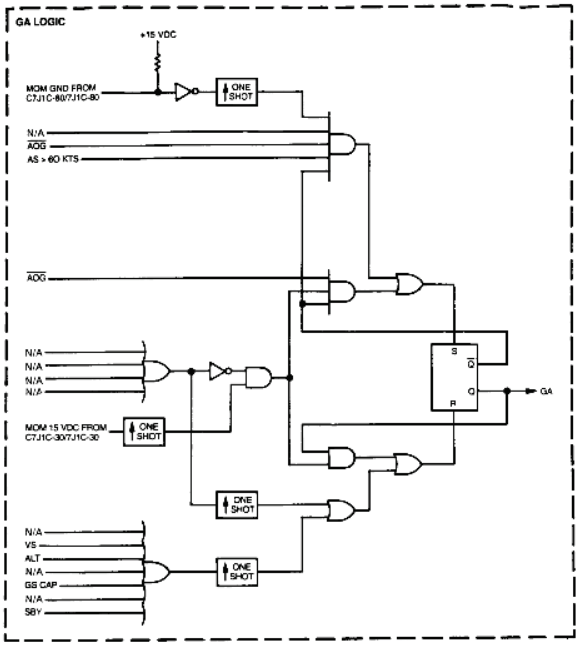
ICN-412-A-961000-B-97499-00027-A-001-01

Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 5 of 7)



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Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 6 of 7)



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Figure 9. Flight Director Mode Select Logic Diagram (S/N 36087 through 36999) (Sheet 7 of 7)

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Autopilot - Functional Description

NOTE

The system consists of autopilot and flight director functions. The circuitry for both functions is contained in the digital flight control computer. In the following text when autopilot or flight director is specified, it refers to that function of the digital flight control computer.

1.1. Pitch Axis Operation

The pitch axis (Figure 1) of the helicopter is stabilized by dual linear actuators, each driven by one digital flight control computer. The actuators are mounted in series in the helicopters pitch controls and extend or retract to achieve the desired control position.

Each of the dual autopilots has a complete set of sensor and control components and is capable of controlling the helicopter independently.

The pitch axis operates in three primary modes: Attitude Retention (ATT), Stability Augmentation (SAS), and Coupled (CPL).

1.2. SAS Mode

The SAS mode is used during low and slow maneuvering where the pilot may be making constant attitude changes in preparation for a landing. The system provides rate damping of upset forces, thereby augmenting his control efforts. Additionally, the helicopter response is shaped by the AFCS for a given cyclic control input. This provides the pilot with additional or boosted control to enhance the handling qualities of the helicopter. This is a hands-on control mode, therefore, the AFCS may be operated with force trim on or off.

A pitch position potentiometer attached to the pitch cyclic control provides an electrical signal to the autopilot at (7) and is routed through a lag circuit and SAS mode switch to the attitude wash-out and follow-up loop. The attitude and stick washout in this mode are programmed functions to cause the best combination of positive and smooth control for widely varying attitudes and airspeeds. The stick position signal causes an initial actuator response to aid in getting the helicopter attitude to change. As the pitch attitude changes, the vertical gyro signal increases until it matches the stick potentiometer signal. At this point, the stick command is satisfied and the actuator stops.

A roll position potentiometer attached to the roll cyclic control provides an electrical signal to the autopilot at (7) (Figure 2) and is routed through a lag circuit and SAS mode switch to the attitude wash-out and follow-up loop. The attitude and stick washouts in this mode are programmed functions to cause the best combination of positive and smooth control for widely varying attitudes and airspeeds. The stick position signal causes an initial actuator response to aid in getting the helicopter attitude to change.

SN: 36087-36247: As the roll attitude changes, the vertical gyro provide a change of attitude signal that increases until it matches the stick potentiometer signal.

SN: 36248-36999: As the roll attitude changes, the AHRS provide a change of attitude signal that increases until it matches the stick potentiometer signal.

At this point, the stick command is satisfied and the actuator stops.

It can be seen that the autopilot control never opposes what the pilot has commanded. The pilot centers the cyclic to maintain his new attitude. The stick signal washes out and the change of attitude signal is washed out leaving zero signal to the actuator. This then, is a steady state condition awaiting the next control input.

If an upset transient is experienced while the pilot is maneuvering, the change of attitude signal will cause actuator control to damp the rate of upset. The amount of control applied will be proportional to the magnitude and the rate of the upset.

1.3. Coupled (CPL) Mode

The CPL mode is used when navigating with the flight director. It is engaged when both autopilots are ON in the ATT mode, and any flight director mode is selected on the mode selector. In the CPL mode, the autopilot system stabilizes the helicopter in the normal manner and automatically follows computed steering commands from the flight director to steer the helicopter.

Both autopilots must be engaged and both will be coupled simultaneously to a single flight director. The number two (pilots) flight director is chosen as primary and coupling will occur automatically when flight director modes are selected. The coupling may be switched to the number one flight director if a second pilot is available to operate the flight director controls.

The selected flight director (8) (Figure 1) supplies a signal through the PCPL switch and connects into the autopilot circuits at (4) where it is summed with the attitude signal. The actuator response will cause the helicopter to follow the flight director commands. The steering signal is also displayed on the ADI pitch command bar. Normal autopilot response will be indicated when the command bar appears to stay centered. Note that the attitude reference is maintained at zero while coupled. This causes the system to roll the helicopter level if roll steering data is lost for any reason.

1.4. Roll Axis Operation

The roll axis of the helicopter (Figure 2) is stabilized by dual linear actuators, each driven by one digital flight control computer. The actuators are mounted in series in the helicopters roll controls and extend or retract to achieve the desired control position.

Each of the dual autopilots has a complete set of sensor and control components and is capable of controlling the helicopter independently.

The roll axis operates in three primary modes: Attitude Retention (ATT), Stability Augmentation (SAS), and Coupled (CPL).

1.5. Attitude Retention (ATT) Mode

SN: 36087-36247: In the ATT mode, the vertical gyro provide helicopter data that are processed by the computer and supplied to the linear actuator for control motion as needed.

SN: 36248-36999: In the ATT mode, the AHRS provide helicopter data that are processed by the computer and supplied to the linear actuator for control motion as needed.

The attitude signal enters a demodulator (1) for conversion from AC to DC voltage and then is voltage regulated by the line voltage compensator. This compensation corrects varying helicopter power supply levels that would affect precise gain control. Note a direct dedicated analog circuit path from (1) to (2) to the servoamplifier. This is special feature needed to make sure that no digital computer fault can cause a multiple axis hardover control input.

Also, along this path is a derived rate circuit that measures how fast the signal is changing. This rate signal is summed with the attitude displacement signal, which boosts control in turbulence to stop the upset and then reduces control as the ship nears its reference attitude in order to prevent overshoot or oscillation.

At autopilot engagement, logic signals are supplied by the autopilot controller to the computer logic circuits that apply enabling power to the servoamplifier (3). This amplifier is a pulse-width modulated-type that can supply ample motor power with minimum continuous current. It provides voltage of either polarity to drive the DC actuator motor in either direction. Motor speed will be proportional to the voltage supplied, which in turn, is proportional to the error signal entering the amplifier. The servoamplifier and motor are protected from sustained high currents (such as a stalled motor) by a device that will reduce the amplifier duty cycle to 25% after 3 seconds.

A gain changer operates to provide full gain with one autopilot engaged. When both autopilots are engaged, the gain is reduced to 50% and each system then supplies half of the required control motion.

An easy-on/easy-off circuit (14) drives the actuator gently back to center at AFCS engagement or disengagement. This helps prevent unexpected transients in the controls at autopilot engagement. A 2 second delay (12) holds power on the roll actuator brake for 2 seconds after autopilot disengagement. This allows time for the easy-on/easy-off circuit to drive the actuator to its center position. At the end of 2 seconds the 28-volt brake excitation is removed, causing the brake to engage. This locks the actuator at its center position.

The roll attitude signal is routed in a digital path from (1) through an analog-to-digital converter (5), to an integrator, to a 10° limiter, and then summed with the rate signal, which is also converted to digital format. The combined signal then is reconverted to analog format and routed to an adaptive rate limiter circuit. The limiter protects against rapid signal changes in the digital circuitry, which might drive the controls at excessive rates. The digital path from (5) to (2) is in parallel with the analog path from (1) to (2), and is of opposite polarity, thereby summing to zero at (2).

SN: 36087-36247: The autopilots may be engaged in any normal flight attitude. However, when not level, the signal from the vertical gyro is proportional and must be reduced to zero to prevent transients in the controls at engagement.

SN: 36248-36999: The autopilots may be engaged in any normal flight attitude. However, when not level, the signal from the AHRS is proportional and must be reduced to zero to prevent transients in the controls at engagement.

This synchronization process is accomplished in the attitude error loop through the attitude washout circuit, which provides a controlled delay. At the moment of engagement a zero signal is output at (5). Any change in helicopter attitude after engagement then produces a change of attitude signal at (5). The difference between this new signal level and the reference level becomes an error signal to provide control to drive the helicopter back to the selected

attitude. Whenever the roll axis is coupled to the flight director, or the bank angle is less than 6° and roll beep is not in use, the synchronizer input is set to zero. This essentially keeps the helicopter at a wings level attitude reference. The voltage summed with the rate signal appears at the pulse-width modulated servoamplifier. This device is a power amplifier, which provides drive control to the linear actuator motor. The motor is excited by a permanent magnet. As the motor drives, a position feedback LVDT follows. The LVDT provides a signal voltage proportional to displacement. The signal is routed to the computer and summed with the attitude error signal as it enters the servoamplifier. A derived servo rate signal serves to stabilize the high-gain servo loop.

When the servo position signal equals the attitude error, the results sum to zero and the motor stops. Now the controls are displaced and the helicopter will start to change attitude back toward the reference level.

SN: 36087-36247: As the attitude changes, the vertical gyro detects it and the error signal starts to decrease.

SN: 36248-36999: As the attitude changes, the AHRS detects it and the error signal starts to decrease.

The attitude error/actuator position sum shows a predominant position signal. This process continues until the helicopter returns to its reference attitude and the attitude error signal is reduced to zero and a state of equilibrium is restored. Thus, the autopilot maintains a selected attitude.

If the pilot chooses to make a roll attitude change he can activate the four-way beep switch on the cyclic control left or right. This causes a fixed value to be applied to the attitude reference integrator. The reference will change at approximately 3° per second. When the new desired attitude is reached, the switch is released. For larger corrections, the pilot may choose to depress the Force Trim Release (FTR) switch and manually place the helicopter in a new attitude with the cyclic control. During this maneuver, the attitude reference synchronizer rapidly follows and establishes the new reference level when the FTR switch is released.

An added feature of the attitude retention mode is the heading hold function. This feature is used to maintain the heading selected by the pilot without being coupled to the flight director. The heading hold function is provided by the roll axis for airspeeds above 60 knots.

SN: 36087-36247: In the heading hold function, the heading sensing is accomplished by the directional gyro. The three wire data from the heading synchro is routed to a Scott Tee transformer, which converts the format to sine/cosine

SN: 36248-36999: In the heading hold function, the heading sensing is accomplished by the AHRS module. The three wire data from the AHRS is routed to a Scott Tee transformer, which converts the format to sine/cosine

It is then demodulated and routed through an analog-to-digital converter and then to a heading synchronizer circuit. Before engagement or any time the roll out-of-detent logic is present, the synchronizer reduces any standing voltage to zero. The synchronizer will also go to zero if the FTR switch is pressed, the cyclic beep switch is used, or roll is greater than 6°. The heading value existing at engagement (logic switch open) becomes the synchronizer heading reference. Any heading change will then be compared with this reference and the difference becomes an error signal. The error signal is then TAS programmed, which converts the heading error signal to an attitude error signal. This signal is routed through the AS >60 KT switch and the ATT/RCPL switch to (4), where it is summed with the attitude error signal. The sum of these signals becomes an error signal to drive the roll controls for correction.

1.6. Yaw Axis Operation

The autopilot controls the helicopter yaw axis (Figure 3) to provide yaw damping.

Dual linear actuators mounted in the tail rotor control rods provide the motive power. Each actuator is controlled by one of the autopilots.

SN: 36087-36247: During yaw damping the sensor is the yaw rate gyro, which provide a signal (1) that is processed through an analog to digital converter, a washout circuit with variable rates, a gain changer responsive to airspeed, and finally to a central summation point (2). A limiter prevents computer signals from exceeding the actuators 100% control voltage.

SN: 36248-36999: During yaw damping the sensor is the AHRS, which provide a signal (1) that is processed through an analog to digital converter, a washout circuit with variable rates, a gain changer responsive to airspeed, and finally to a central summation point (2). A limiter prevents computer signals from exceeding the actuators 100% control voltage.

An easy-on/easy-off circuit at (3) will cause the actuators to drive to center at a gentle rate at both engagement and disengagement of the autopilots. A 2-second time delay (15) holds power on the yaw actuator brake for 2 seconds after autopilot disengagement. This allows time for the easy-on/easy-off circuit to drive the actuator to its center position. At the end of 2 seconds, the 28V brake excitation is removed, causing the brake to engage. This locks the actuator at its center position.

At autopilot engagement, logic signals are supplied by the autopilot controller to the computer logic circuits which apply enabling power to the servoamplifier (3). This amplifier is a pulsewidth-modulated type that can supply ample motor power with minimum continuous current. It provides voltage of either polarity to drive the DC actuator motor in either direction. Motor speed will be proportional to the voltage supplied, which in turn, is proportional to the error signal entering the amplifier. The servoamplifier and motor are protected from sustained high currents (such as a stalled motor) by a device that will reduce the amplifier duty cycle to 25% after 3 seconds.

A gain changer operates to provide full gain with one autopilot engaged. When both autopilots are engaged, the gain is reduced to 50% and each system then supplies half of the required control motion.

An easy-on/easy-off circuit (14) drives the actuator gently back to center at AFCS engagement or disengagement. This helps prevent unexpected transients in the controls at autopilot engagement. A 2-second delay (12) holds power on the yaw actuator brake for 2 seconds after autopilot disengagement. This allows time for the easy-on/easy-off circuit to drive the actuator to its center position. At the end of 2 seconds the 28-volt brake excitation is removed, causing the brake to engage. This locks the actuator at its center position.

SN: 36087-36247: Turn coordination is provided at speeds above 60 knots by the yaw axis. Calculated turn rate is a function of bank angle and airspeed. This signal is compared with actual turn and rate signal measured by the rate gyro. The difference between these signals represents the correction required by the yaw axis for proper turn coordination.

SN: 36248-36999: Turn coordination is provided at speeds above 60 knots by the yaw axis. Calculated turn rate is a function of bank angle and airspeed. This signal is compared with actual turn and rate signal measured by the AHRS (5). The difference between these signals represents the correction required by the yaw axis for proper turn coordination.

All sensor signals have a direct control path at (2) and an integral path at (7). The integrator causes small signals to build up until the controls respond, thus preventing standoff conditions. The integrator is deactivated and reset when the helicopter is on the ground (AOG), the trim is not enabled, or the pedals are pressed (YOD).

The gain switch at (6) operates at approximately 60 knots and reduces the yaw axis gain to compensate for the gain in effectiveness of the tail fin with increased airspeed.

1.7. Pitch Trim Operation

The automatic trim (autotrim) (Figure 4) system is used to keep the series linear actuators near their center-of-travel, thereby maintaining maximum authority in anticipation of control demands imposed by helicopter attitude changes. Autotrim is functional when either or both autopilots are engaged in any mode except SAS, and the cyclic force trim switch is on.

The trim actuator (1) is a servo unit that contains a force trim clutch, trim motor, and motor brake.

The linear actuator position LVDTs (2) provide a signal voltage to the computers that, after control logic is satisfied, applies servoamplifier drive power to the rotary trim actuator motor.

Control logic required to permit trim operation is as follows:

- FORCE TRIM switch (3) ON
- Computer watchdog monitor (4) is satisfied
- Autopilot engaged in any mode except SAS
-

NOTE

The following conditions define helicopter on ground state: RADALT less than 20 feet, airspeed less than 60 knots, collective less than 25% up travel.

Helicopter not on ground (RADALT Valid)

- Trim actuator detent switch (5) closed (no force on cyclic control)
- Linear servoamplifier polarity detector (6) is tripped indicating servo command direction
- A retriggerable one-shot device (7) fires every 70 milliseconds and will close the logic circuit thereby verifying that trim is still needed

Threshold LVDT voltage representing 10% of linear actuator stroke length when both autopilots are engaged (25% for single autopilot) and sustained for 0.25 second (1.5 seconds for single autopilot) will start the trim motor. Also, when one autopilot is engaged the 0.25-second delay time is enabled for 10 seconds whenever the pilot depresses collective FTR while making a power change. Once started, the trim motor runs until the linear actuator position signal decreases below 6%

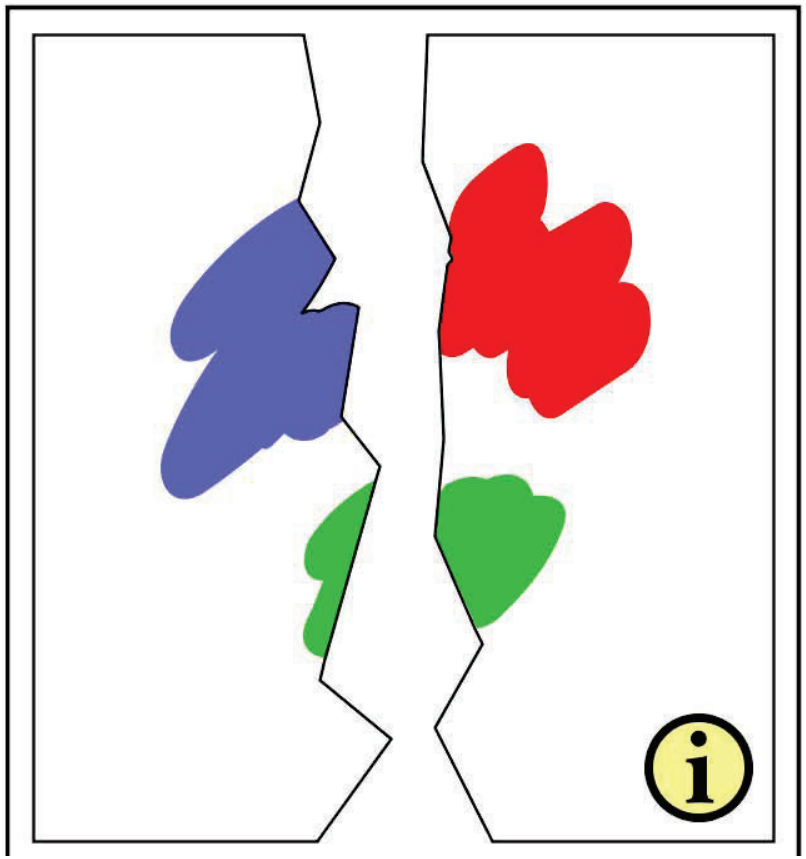
Trim motor speed is variable and controlled by the pulse-width modulated servoamplifier (8). In turn, the amplifier duty cycle is proportional to linear actuator displacement-maximum speed of trim motor for maximum deflection of linear actuator.

Note that the trim drive is from the active flight director as indicated on the TRIM/FD button on the autopilot controller.

1.8. Roll Trim Operation

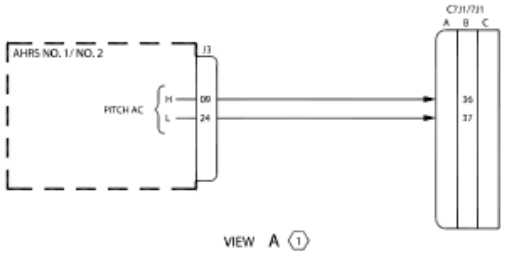
The roll channel auto trim (Figure 5) works in a manner identical to that described for the pitch channel. It uses the roll channel linear actuators and rotary trim actuator.

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ICN-412-A-961000-B-97499-00015-A-001-01

Figure 1. Autopilot Pitch Axis Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 2)

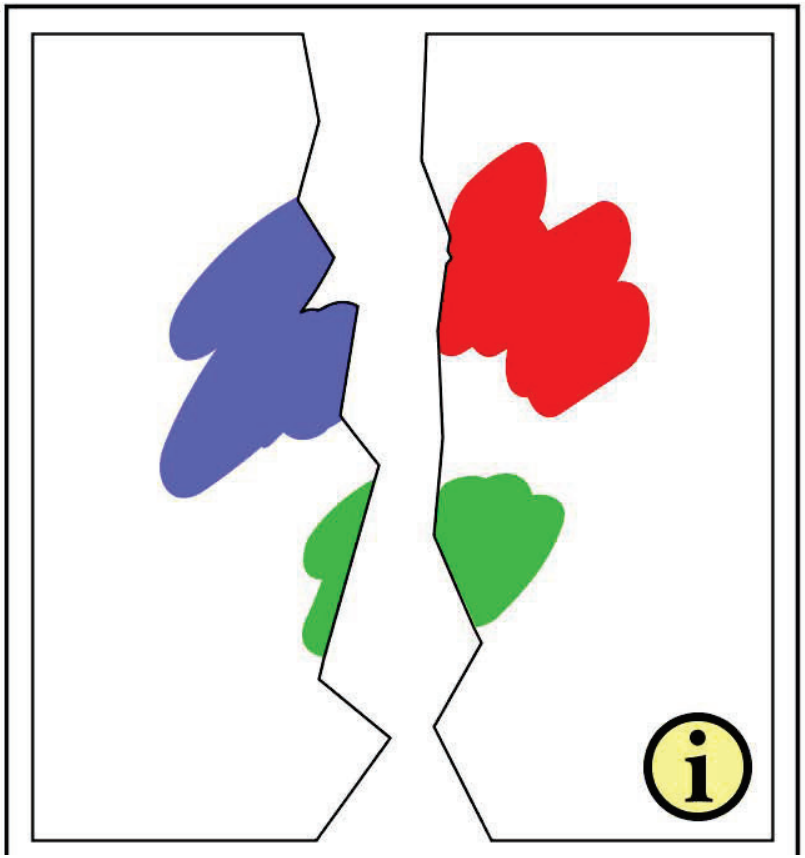


EFFECTIVITY

1 S/N 36248 THROUGH 36999.

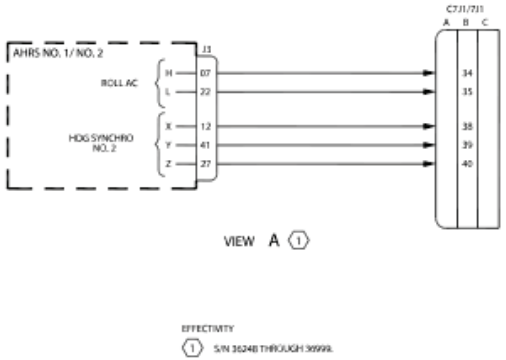
ICN-412-A-961000-B-97499-00016-A-001-01

Figure 1. Autopilot Pitch Axis Functional Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



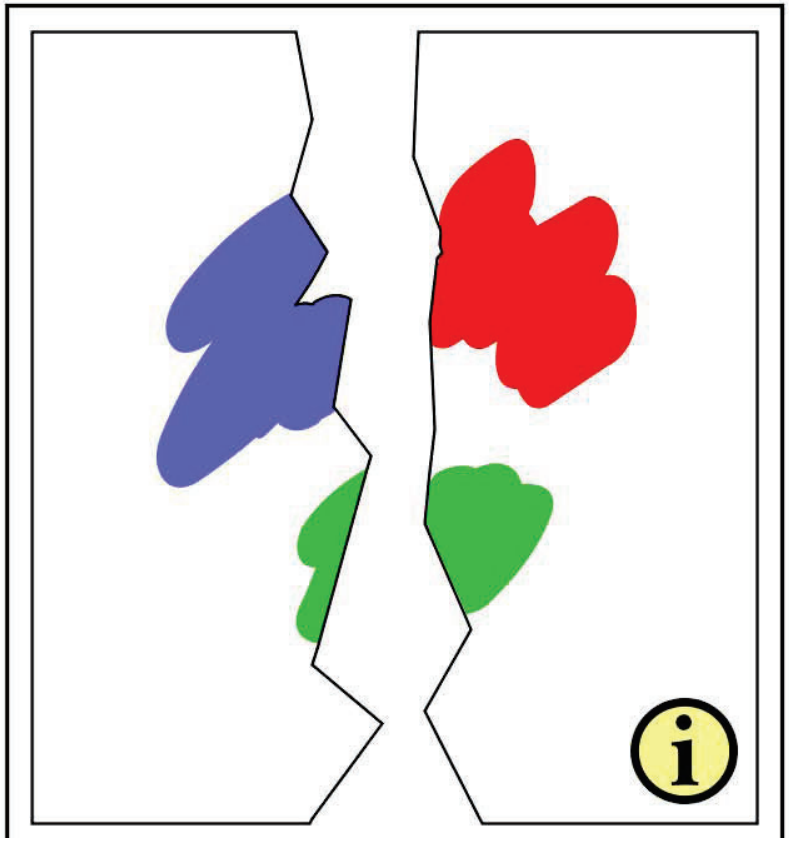
ICN-412-A-961000-B-97499-00017-A-001-01

Figure 2. Autopilot Roll Axis Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



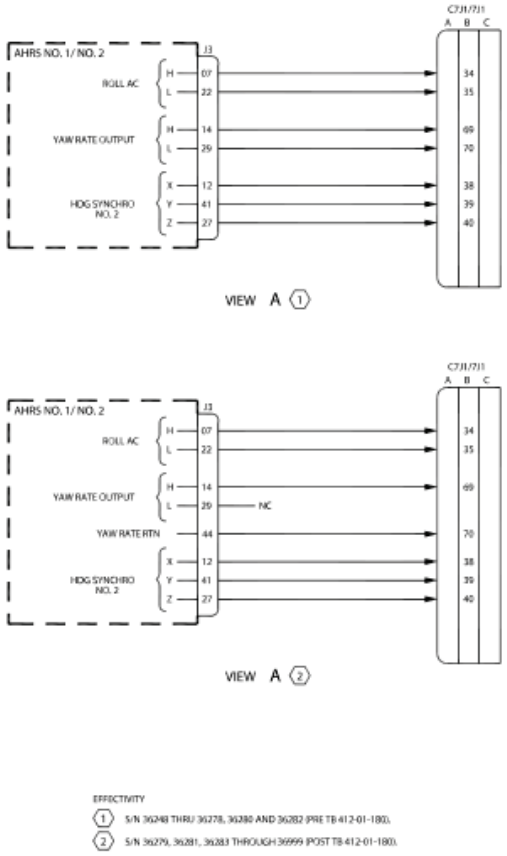
ICN-412-A-961000-B-97499-00018-A-001-01

Figure 2. Autopilot Roll Axis Functional Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



ICN-412-A-961000-B-97499-00019-A-001-01

Figure 3. Autopilot Yaw Axis Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 2)



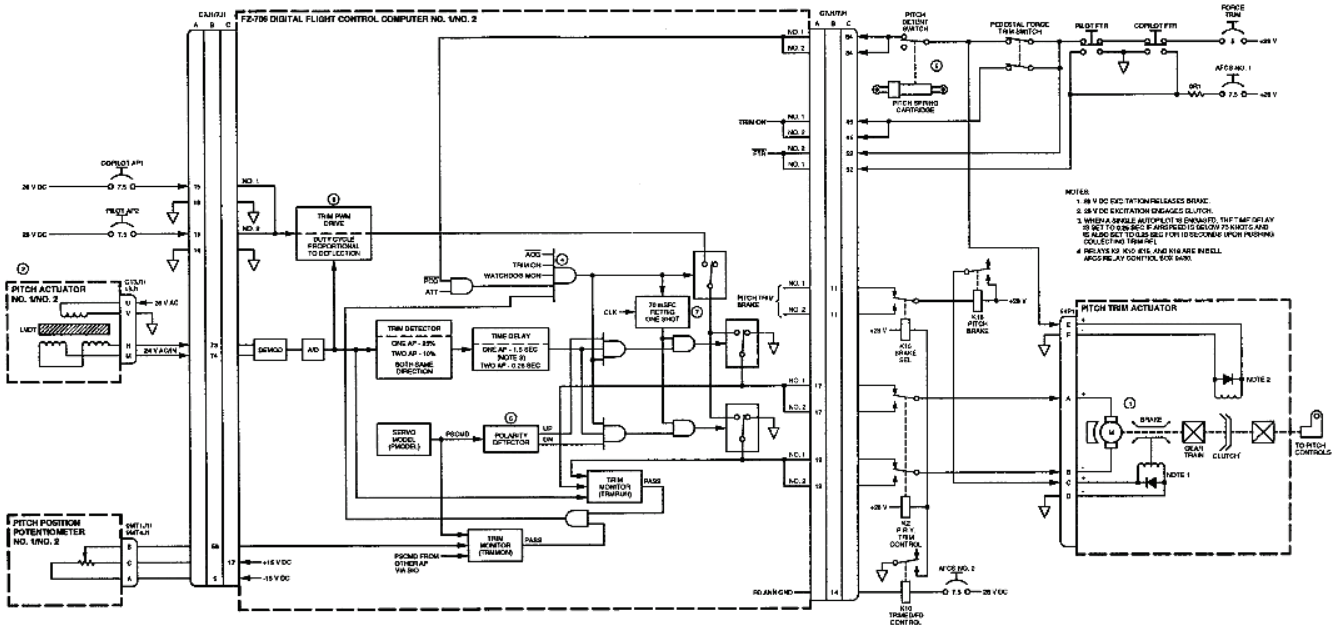
EFFECTIVITY

1 S/N 36248 THRU 36278, 36280 AND 36282 (PRE TB 412-01-180).

2 S/N 36279, 36281, 36283 THROUGH 36999 (POST TB 412-01-180).

ICN-412-A-961000-B-97499-00020-A-001-01

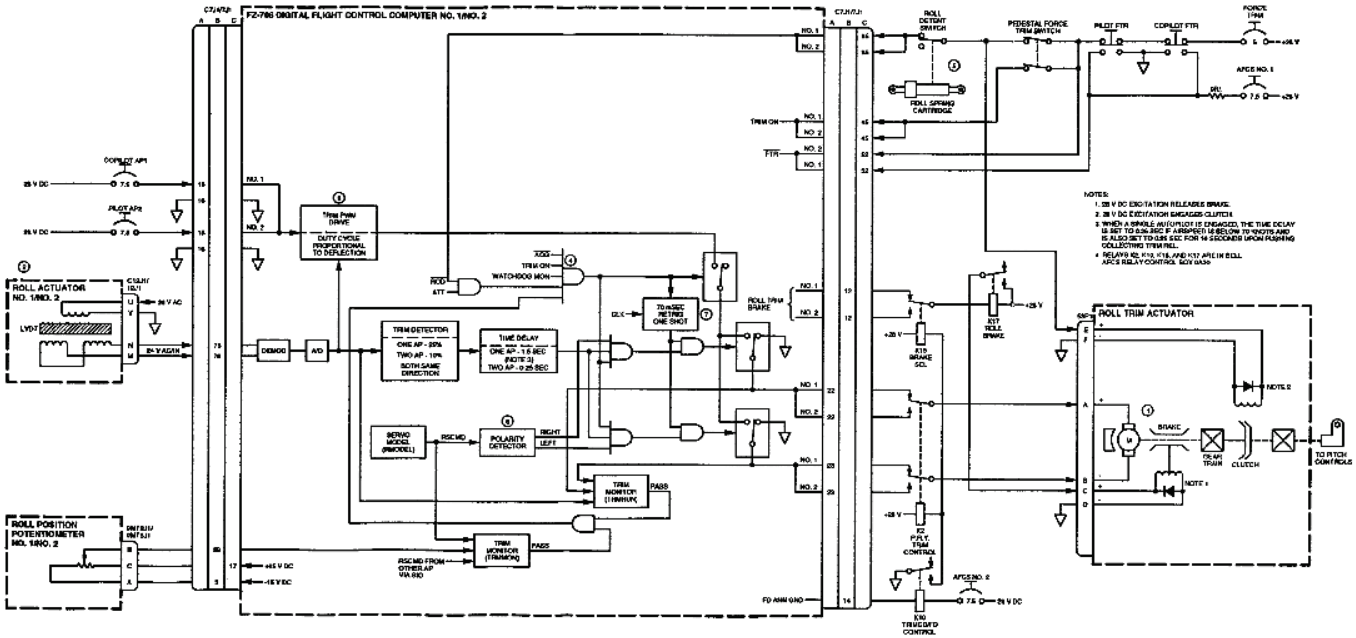
Figure 3. Autopilot Yaw Axis Functional Diagram (S/N 36087 through 36999) (Sheet 2 of 2)



ICN-412-A-961000-B-97499-00021-A-001-01

Figure 4. Autopilot Pitch Trim Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

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ICN-412-A-961000-B-97499-00022-A-001-01

Figure 5. Autopilot Roll Trim Functional Diagram (S/N 36087 through 36999) (Sheet 1 of 1)

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DAFCS - Functional Test

Procedure

1.

NOTE

Before you do a functional test on the DAFCS, verify that the attitude system, compass system, radio altimeter system, and navigation radios are fully operational.

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Preliminary Test Procedures - Functional Test

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
XPNDR/DME Test Set	ATC-600A	1	
DC Power Cart	MH28-200RA2M	1	
Pitot Static Tester	1811G-345	1	
VOR/ILS Test Set	966Y-2	1	
Hydraulic Mule	9780-120	1	

Procedure

1. Visually inspect each component to make sure that it is properly installed and the electrical connectors are properly mated and tight.

2. Connect the external power.

3. Connect and apply hydraulic power to the No. 1 hydraulic system.

4. Energize the power buses.

5. Select the following initial switch positions:

SWITCH	POSITION
AP1	-OFF
AP2	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-OFF
FD 1/2	-2

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Lamp Test - Functional Test

Procedure

1. Press and hold the pilot DAFCS lamp test pushbutton (SBY on pilot mode selector).
2. Verify that all the lamps in the pilot flight director mode selector, air data command display, and autopilot controller (except AP1-ON and FD 1) illuminate brightly.
3. Press and hold the SBY switch for more than 5 seconds.
4. Verify that all lamps dim.
5. Release the SBY switch.

NOTE

Pilot instrument light variable control must be out of the OFF position.

6. Set the BRT/DIM switch on the caution panel to DIM.
7. Press and hold the pilot cyclic decouple switch (remote SBY).
8. Verify that all the lamps are on and dim.
9. Release the SBY switch and set the caution panel BRT/DIM switch to BRT.
10. Select FD1 and repeat Step 1. through Step 9. with the copilot SBY switch.

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Force Trim Test - Functional Test

Procedure

1. Move the cyclic stick and pedals.
2. Verify that there is no spring force and minimum friction on the controls.
3. Set the force trim switch to ON.
4. Verify that the cyclic stick has spring force when moved from the detent position.
5. Press and hold the pilot cyclic FTR switch.
6. Verify that there is no spring force and minimum friction on the cyclic and pedals.
7. Release the pilot cyclic FTR switch.
8. Repeat Step 5. through Step 7. with the copilot cyclic FTR switch.

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Autopilot Engage Test - Functional Test

Procedure

1. Press the AP1 pushbutton on the autopilot controller.

NOTE

If the ATT flag in the copilot ADI has retracted, the AP1 ON and ATT legends will illuminate and remain engaged.

2. Verify that the AP1 ON and ATT legends illuminate and remain engaged.
3. Set the force trim switch to OFF.
4. Verify that the ATT legend extinguishes and the SAS legend illuminates.
5. Press the AP1 pushbutton.
6. Verify that the AP1 ON and SAS legends extinguish.
7. Press the AP2 pushbutton.

NOTE

If the ATT flag in the pilot ADI has retracted, the AP2 ON and SAS legends will illuminate.

8. Verify that the AP2 ON and SAS legend illuminate.
9. Set the force trim switch to ON and press the SAS/ATT button.
10. Verify that the AP2 reverts to the ATT mode.
11. Press the AP1 pushbutton.
12. The AP1 ON legend shall illuminate.
13. Open the AFCS NO. 1 28 VDC circuit breaker.
14. Verify that the AP1 ON legend extinguishes and the AP2 ON and ATT legends remain illuminated.
15. Verify that the NO. 1 AUTOPILOT warning annunciator on the caution panel illuminates and the MASTER CAUTION RESET switch is also illuminated.
16. Close the AFCS NO. 1 28 VDC circuit breaker.
17. Press the AP1 pushbutton.
18. Verify that the NO. 1 AUTOPILOT warning annunciator extinguishes and the AP1 ON legend illuminates.
19. Open the AFCS NO. 1 26 VAC circuit breaker.
20. Verify that the AP1 ON legend extinguishes and the AP2 ON and ATT legends remain illuminated.
21. Verify that the NO. 1 AUTOPILOT warning annunciator on the caution panel illuminates and the MASTER CAUTION RESET switch is also illuminated.
22. Close the AFCS NO. 1 26 VAC circuit breaker.
23. Press the AP1 pushbutton.
24. Verify that the AP1 ON legend illuminates and the AP1 warning annunciator extinguishes.
25. Repeat Step 13. through Step 24. with the AFCS NO. 2 circuit breakers.

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ATT Gain Polarity Test - Functional Test

Procedure

NOTE

SN: 36087-36247: Helicopters are equipped with two TARSYN three-axis reference systems having two independent yaw rate gyros.

SN: 36248-36999: Helicopters are equipped with two Attitude Heading and Reference Systems (AHRS) modules that do the same basic functions.

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-OFF
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON

2. Position the collective stick above the AOG position (25%).

3. Move the cyclic stick Pitch Out of Detent (POD) .

4. Unfasten and tilt the copilot VG or AHRS, as applicable, to simulate a 5° pitch up attitude.

5. Verify that the No. 1 pitch Actuator Position Indicator (API) indicates a maximum forward deflection.

6. Level the VG or AHRS, as applicable.

7. Return the cyclic stick to the detent position.

8. Momentarily press the cyclic FTR switch.

9. Verify that the pitch API returns to center.

10. Move the cyclic stick POD.

11. Tilt the copilot VG or AHRS, as applicable, to simulate a 5° pitch down attitude.

12. Verify that the No. 1 pitch API indicates a maximum aft deflection.

13. Return the cyclic stick to the detent position.

14. Momentarily press the cyclic FTR switch.

15. Verify that the pitch API returns to center.

16. Move the cyclic stick Roll Out of Detent (ROD) .

17. Tilt the copilot VG or AHRS, as applicable, to simulate a 5° roll right attitude.

18. Verify that the No. 1 roll API indicates a maximum left deflection.

19. Level the copilot VG or AHRS, as applicable.

20. Return the cyclic stick to the detent position.

21. Momentarily press the cyclic FTR switch.

22. Verify that the roll API returns to center.

23. Move the cyclic stick ROD.

24. Tilt the copilot VG or AHRS, as applicable, to simulate a 5° roll left attitude.

25. Verify that the No. 1 roll API indicates a maximum right deflection.

26. Level the copilot VG or AHRS module, as applicable.

27. Return the cyclic stick to the detent position.

28. Momentarily press the cyclic FTR switch.

29. Verify that the roll API returns to center.

30. Disengage AP1.

31. Engage AP2.

32. Repeat Step 3. through Step 29. with the pilot VG or AHRS, as applicable, and the No. 2 pitch and roll API.

ATT Mode Test - Functional Test

Procedure

NOTE

When running these tests, trim may run and cause the actuators to deflect toward center on the APIs.

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-ON
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON

2. Position the collective stick above the AOG position.

3. With both autopilots engaged, momentarily press cyclic FTR to center the actuators.

4. Beep forward on the pilots cyclic beep switch while monitoring the No. 2 pitch API. The pitch actuator shall command pitch forward and shall stop at 50% deflection on the API.

5. Repeat Step 3. and Step 4. while monitoring the No. 1 pitch API.

6. Disengage AP1 and momentarily press cyclic FTR.

7. Beep forward on the copilot beep switch. The No. 2 pitch API shall move to a full forward deflection.

8. Repeat Step 6. and Step 7. for AP1 ON, AP2 OFF, and monitoring No. 1 pitch API.

9. With both autopilots on, momentarily press cyclic FTR.

10. Beep aft on the pilot cyclic beep switch. Both pitch APIs shall move to 50% pitch aft indication.

11. Disengage AP1 and momentarily press cyclic FTR.

12. Beep aft on the copilot beep switch. The No. 2 pitch API shall move to a full aft deflection.

13. Repeat Step 11. and Step 12. for AP1 ON, AP2 OFF and monitoring No. 1 pitch API.

14. With both autopilots engaged, momentarily press cyclic FTR to center the actuators.

15. Beep right on the pilot cyclic beep switch. Both roll actuators shall command roll right and shall stop at 50% deflection on the APIs.

16. Disengage AP1 and momentarily press cyclic FTR.

17. Beep right on the copilot beep switch. The No. 2 roll API shall move to a full right deflection.

18. Repeat Step 16. and Step 17. for AP2 OFF, AP1 ON and monitoring No. 1 roll API.

19. With both autopilots on, momentarily press cyclic FTR.

20. Beep left on the pilots cyclic beep switch. Both roll APIs shall move to 50% roll left indication.

21. Disengage AP1 and momentarily press cyclic FTR.

22. Beep left on the copilot beep switch. The No. 2 roll API shall move to a full left deflection.

23. Repeat Step 21. and Step 22. for AP1 ON, AP2 OFF, and monitoring No. 1 roll API.

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Yaw Axis Test - Functional Test

Procedure

NOTE

SN: 36087-36247: Helicopters are equipped with two TARSYN three-axis reference systems having two independent yaw rate gyros.

SN: 36248-36999: Helicopters are equipped with two Attitude Heading and Reference Systems (AHRS) modules that do the same basic functions.

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-OFF
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON

2. Connect a pitot-static test set to the helicopter (412-A-95-11-05-00A-320A-A / 01808).

3. Make sure that both air data sensors are connected to the test set.

4. Simulate a 100-knot airspeed on both the pilot and copilot airspeed indicators.

5. Rotate the copilot Directional Gyro (DG) or AHRS as applicable, 10° counter clockwise as viewed from the top. The No. 1 roll API shall deflect right.

6. Return the DG or AHRS, as applicable, to its original position

7. Momentarily press the cyclic FTR switch. The No. 1 roll API shall center.

8. Displace the copilot yaw rate gyro or AHRS, as applicable, to the right, and notice the No. 1 yaw API will initially move left.

9. Return the copilot yaw rate gyro or AHRS, as applicable, to its original position.

10. Momentarily press the FTR switch. The No. 1 yaw API shall center.

11. Disengage AP1.

12. Engage AP2.

13. Repeat Step 2. through Step 10. with the pilot DG and yaw rate gyro or AHRS, as applicable.

14. Disengage AP2.

15. Release the pitot-static test set pressure (412-A-95-11-05-00A-320A-A / 01808).

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SAS Mode Test - Functional Test

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-OFF
ATT/SAS	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-ON

2. Press the ATT/SAS switch to engage the SAS mode.
3. Move the cyclic approximately 1 inch (25 mm) forward. The No. 1 pitch API shall deflect maximum forward.
4. Momentarily press the cyclic FTR switch.
5. Move the cyclic approximately 1 inch (25 mm) aft. The No. 1 pitch API shall deflect maximum aft.
6. Momentarily press the cyclic FTR switch.
7. Move the cyclic approximately 1 inch (25 mm) left. The No. 1 roll API shall deflect maximum left.
8. Momentarily press the cyclic FTR switch.
9. Disengage AP1.
10. Engage AP2.
11. Press the ATT/SAS switch to engage the SAS mode.
12. Repeat Step 2. through Step 9. monitoring the No. 2 pitch and roll APIs.

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Flight Director-Logic Test - Operational Check

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
DME Test Set	ATC-600A	1	
DC Power Cart	MH28-200RA2M	1	
Pitot Static Tester	1811G-345	1	
VOR/ILS Test Set	966Y-2	1	
Hydraulic Mule	9780-120	1	

Procedure

1. Prior to receiving gyro valids, engage HDG and VS modes on both the pilot and copilot mode selector. The HDG and VS annunciators shall illuminate; the ADI pitch and roll command bars shall be out of view.

2. After 3 minutes following power application, the pilot and copilot ADI/HSI flags shall indicate the following:

- HSI - HDG flag out of view
- ADI - ATT and FD flags out of view and pitch and roll command bars shall come into view
- No. 1 autopilot - annunciator on (caution panel)
- No. 2 autopilot - annunciator on (caution panel)
- DCPL - annunciators off (instrument panel)

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Heading Mode Test - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-OFF
AP2	-OFF
ATT/SAS	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Set the heading bug on the HSI to the helicopter heading.
3. Engage the HDG mode on the pilot mode selector. The HDG annunciator shall light and the roll command bar on the ADI shall center.
4. Set the heading bug to 10° right of the helicopter heading. The ADI roll command bar shall deflect right.
5. Select FD1 and repeat Step 2. through Step 4. for the copilot side.

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VOR Mode Test - Operational Check

Procedure

NOTE

Display the pilot Nav information on the pilot HSI and the copilot Nav information on the copilot HSI.

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-OFF
AP2	-OFF
ATT/SAS	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Connect a DME test set with a simulated range of 50 miles.

3. Set the course pointer on the pilot HSI to helicopter heading.

4. With a VOR/ILS test set, transmit a VOR radial positioning the helicopter 10° right of the selected course.

5. Turn on the pilot navigation radio and tune to the transmitted signal. The HSI deviation bar shall displace full-scale left and the TO-FROM pointers shall display TO. In addition, check that the HSI NAV flag is retracted.

6. Engage the NAV mode on the pilot mode selector. The NAV ARM and HDG annunciators shall illuminate and the pilot ADI roll command bar shall still respond to the heading bug.

7. Slowly reduce the Nav deviation to zero. (Change the transmitted radial to the present helicopter heading.) The course deviation bar shall be centered, the pilot mode selector HDG and NAV ARM annunciators shall be extinguished, and the NAV CAP legend shall be illuminated. Note the ADI roll command bar shall be near center.

8. Transmit a VOR radial positioning the helicopter 4° left of the selected course.

9. Note the position of the ADI roll command bar, then reduce the DME to 5 miles and adjust the VOR radial to helicopter heading.

10. Disengage NAV mode. After 30 seconds, re-engage NAV mode. The course deviation bar shall be centered, the mode selector HDG and NAV ARM annunciators shall be extinguished and the NAV CAP legend shall be illuminated.

11. Transmit a VOR radial positioning helicopter 4° left of the selected course. The roll command bar shall be much closer to center than noted in Step 9..

12. Press VOR APR on the pilot mode selector.

13. Note that NAV CAP extinguishes and the VOR APR ARM light illuminates.

14. Slowly reduce deviation and make sure that VOR APR CAP illuminates.

15. Select FD1 and repeat Step 7. through Step 14. for the copilot side.

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ILS Mode Test - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-OFF
AP2	-OFF
ATT/SAS	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Set the VOR/ILS test set to a localizer frequency and adjust deviation for full-scale left (greater than two dots) deflection (helicopter right of desired course).

3. Open the 28 VDC RAD ALT circuit breaker(s).

4. Tune the Nav radio to the localizer frequency.

5. Engage the ILS mode on the pilot mode selector. The HDG, NAV ARM, and ILS ARM annunciators shall be illuminated. Observe that the glideslope flag and the roll command bar are in view.

6. Reduce deviation on the VOR/ILS test set to zero. The pilot mode selector HDG and NAV ARM annunciators shall be extinguished and NAV CAP shall be illuminated. The pilot ADI roll command bar shall be centered.

7. Rotate the HSI course pointer clockwise 30°. The pilot ADI roll command bar shall move to the right.

8. Engage SBY mode on the pilot mode selector.

9. Set the course pointer back to the original heading.

10. Set the VOR/ILS test set localizer deviation to zero and glideslope deviation to maximum up (aircraft below beam). The glideslope pointer on the HSI shall indicate full-scale up and the GS flag on the HSI shall be out of view.

11. Engage the BC mode on the pilot mode selector. The BC CAP annunciator shall illuminate and the GS pointer shall be removed from display.

12. Engage the ILS and ALT modes on the pilot mode selector. The NAV CAP, ILS ARM, and ALT annunciators shall illuminate and the pilot ADI pitch and roll command bars shall be centered.

13. Reduce the glideslope deviation on the VOR/ILS test set to zero. The pilot mode selector ALT and ILS ARM annunciators shall extinguish and the ILS GS annunciator shall illuminate.

14. Set the VOR/ILS test set glideslope deviation to maximum up (aircraft below beam). The pilot ADI pitch command bar shall move up.

15. Set the VOR/ILS test set glideslope deviation for zero deviation.

16. LOC programmer/flare command. Set the simulated airspeed to 100 knots with pitot-static test set (412-A-95-11-05-00A-320A-A / 01808). Engage the 28 VDC RAD ALT circuit breaker(s).

17. About 2 seconds after engaging the RAD ALT circuit breakers, the pilot ADI pitch bar shall move up indicating a climb command.

18. Open the 28 VDC RAD ALT circuit breaker; the ADI pitch bar shall bias from view in about 5 seconds.

19. Press SBY.

20. Select FD1 and repeat Step 10. through Step 19. for the copilot side.

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VS Select Mode Test - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-ON
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Engage VS on the pilot mode selector. The pilot ADI pitch command bar shall come into view and center. The air data command display shall indicate 0 FT/MIN for 5 seconds and then revert to dashes.

3. Beep aft on the pilot cyclic beep switch to achieve 500 FPM on the air data command display. The pilot ADI pitch command bar shall deflect up.

4. Beep forward on the pilot cyclic beep switch to achieve -500 FPM on the air data command display. The pilot ADI pitch command bar shall deflect down.

5. Select FD1 and repeat Step 1. through Step 4. for the copilot side.

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IAS Hold Mode Test - Operational Check

Procedure



THIS PROCEDURE MUST BE DONE (ON THE APPROPRIATE SIDE) WHENEVER AN AIR DATA SENSOR OR AIRSPEED INDICATOR IS REPLACED OR RECALIBRATED.

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-ON
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Simulate an airspeed of 100 knots with a calibrated pitot-static test set.

3. Verify that the helicopter airspeed indicating system is within tolerance.

4. Select FD2 and engage the IAS mode on the pilot mode selector. The pilot ADI pitch command bar shall center. The air data command display shall indicate 100 ±5 knots for 5 seconds and then revert to dashes.

5. Beep forward on the pilot cyclic beep switch to achieve a command of 130 knots on the air data command display. The pilot ADI pitch command bar shall move down indicating a descend command.

6. Increase the simulated airspeed to 130 knots. The pilot ADI pitch command bar shall center.

7. Beep aft on the pilot cyclic beep switch to achieve a command of 70 knots on the air data command display. The pilot ADI pitch command bar shall move up indicating a climb command. The air data command display shall revert to dashes after 5 seconds.

8. Select FD1 and repeat Step 1. through Step 7. for copilot side.

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ALT Hold Test - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-ON
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Simulate an altitude of 4000 feet with the pitot-static test set.

3. Engage the ALT hold mode on the pilot mode selector. The pilot ADI pitch command bar shall come into view and center. The air data command display shall remain dashes.

4. Increase the simulated altitude to 4500 feet. The pilot ADI pitch command bar shall move down indicating a descend command.

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Go Around Mode Test - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-ON
AP2	-ON
ATT/SAS	-ATT
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Move the collective to its full down position.
3. Simulate an airspeed to 70 knots.
4. Engage GA on the pilot mode selector. The GA annunciator on the pilot mode selector shall illuminate. The pilot ADI pitch command bar shall indicate a climb command and the roll command bar shall center.
5. Press SBY. The pilot ADI pitch command bar shall bias from view.
6. Re-engage the GA mode with the remote GA switch on the pilot collective. Note that GA engages.
7. Press SBY.
8. Move the collective to its full down position.
9. Simulate an airspeed of 0 knots.
10. Engage the GA mode with the switch on the pilot mode selector. The GA annunciator on the pilot mode selector shall not illuminate, indicating that the GA engagement is inhibited.
11. Select FD1 and repeat Step 1. through Step 10. for the copilot side.

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Flight Director Coupling Test (Pitch and Roll) - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-OFF
AP2	-OFF
ATT/SAS	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Engage AP1 and AP2 in the ATT mode.
3. Make sure that the FD 2 annunciator is illuminated.
4. Align the pilot heading select bug with the actual helicopter heading.
5. Engage the VS and HDG modes on the pilot mode selector.
6. Make sure that the CPL annunciator on the autopilot controller illuminates and the pilot pitch and roll command bars are in view and centered.
7. Rotate the pilot heading select bug 45° left. The roll APIs and pilot ADI roll command bar shall deflect left.
8. Beep forward on the cyclic beep switch to achieve a vertical speed descent command of -400 FPM. The pitch APIs and pilot ADI pitch command bar shall deflect forward and down, respectively.
9. Decouple FD 2 by pressing the CPL pushbutton. The CPL-ON annunciator on the autopilot controller shall extinguish and the DCPL annunciator shall illuminate. Pitch and roll APIs shall return to center. The pilot ADI command bars shall remain deflected.
10. Recouple FD 2 by pressing the CPL pushbutton. The CPL-ON annunciator shall illuminate and the DCPL annunciator shall extinguish.
11. Select FD1 and repeat Step 1. through Step 10. for the copilot.

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Autotrim Test - Operational Check

Procedure

1. Select the following initial switch positions:

SWITCH	POSITION
AP1	-OFF
AP2	-OFF
ATT/SAS	-OFF
MODE SELECTORS	-SBY
FORCE TRIM	-ON
FD 1/2	-2

2. Apply external hydraulic power to the No. 1 hydraulic system.
3. Connect an pitot-static test set to the pitot-static system (412-A-95-11-05-00A-320A-A / 01808).
4. Position the collective at 50% travel.
5. Simulate 100 knots airspeed.
6. Engage AP1.
7. Press cyclic FTR and position the cyclic stick aft of center.
8. Beep forward on cyclic beep switch. The cyclic shall begin trimming forward approximately 1 to 2 seconds after beeping starts.

NOTE

Break detent while trim is running.

9. Move the cyclic stick about 5 centimeters (2.00 in.) aft (out of detent); the autotrim shall stop.
10. Allow the stick to return to the detent position. The autotrim shall start again.
11. Press the cyclic FTR switch for 1 second. Beep the cyclic beep switch aft. The cyclic shall begin trimming aft approximately 1 to 2 seconds after beeping starts.
12. Press the cyclic FTR switch for 1 second.
13. Beep the cyclic beep switch right. Continue holding beep switch right until Step 13. through Step 15. are completed. Cyclic shall begin trimming right approximately 1 to 2 seconds after beeping starts.

NOTE

Break detent while trim is running.

14. Move the cyclic stick about 5 centimeters (2.00 in.) left (out of detent); the autotrim shall stop.
15. Allow the stick to return to the detent position. The autotrim shall start again.
16. Press the cyclic FTR switch for 1 second.
17. Beep and hold the cyclic beep switch left. The cyclic shall begin trimming left approximately 1 to 2 seconds after beeping starts.
18. Release the beep switch and then briefly press the cyclic FTR switch.
19. Disengage AP1.
20. Repeat Step 3. through Step 18. with AP2.
21. Engage both autopilots.
22. Beep the cyclic switch forward. The cyclic shall begin trimming forward less than 1 second (0.25 second) after beeping starts.

23. Press the cyclic FTR switch for 1 second.

24. Beep the cyclic beep switch right. The cyclic shall begin trimming right less than 1 second (0.25 second) after beeping starts.

25. Press the cyclic FTR switch for 1 second.

26. Disengage AP2.

27. Reduce simulated airspeed to 0 knots.

28. Repeat Step 21. through Step 25. with AP1 only (single system below 70 knots).

NOTE

SN: 36087-36247: Helicopters are equipped with two TARSYN three-axis reference systems having two independent yaw rate gyros.

SN: 36248-36999: Helicopters are equipped with two Attitude Heading and Reference Systems (AHRS) modules that do the same basic functions.

29. Position collective at full down position (AOG).

30. Unfasten and tilt the copilot VG or AHRS, as applicable, right and forward to obtain at least 50% deflection on the No. 1 pitch and roll APIs. The cyclic shall not trim.

31. Position collective above the AOG position.

32. Use the cyclic FTR switch, and position the cyclic stick full forward.

33. Unfasten and tilt the copilot VG or AHRS, as applicable, to simulate a steady 20° pitch down attitude. The time for the cyclic to travel full aft shall be 20 to 27 seconds.

34. Repeat Step 31. and Step 32. except tilt the pilot VG or AHRS, as applicable, 20° pitch up and do the test for full aft to full forward direction.

35. Use the cyclic FTR switch, and position the cyclic stick full left.

36. Tilt the copilot VG or AHRS, as applicable, to simulate steady 30° roll left attitude. The time for the cyclic stick to travel full right shall be 20 to 27 seconds.

37. Repeat Step 34. and Step 35. except tilt the copilot VG or AHRS, as applicable, 30° roll right and do the test for full right to full left direction.

Airspeed System - Adjustment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
DC Power Cart	MH28-200RA2M	1	
Pitot Static Tester	1811G-345	1	

Procedure

NOTE

Matching of the Air Data Sensor (ADS) and/or air data command display to the pilot and copilot airspeed indicator is required whenever an ADS or airspeed indicator is replaced or calibrated.

1. With a calibrated pitot-static test set, plumb static source to pilot and copilot static systems so that identical static pressure is seen by both systems.

2. Make sure that all static ports are sealed.

3. Connect the pitot output of the test set to both pitot tubes.

4. Adjust altitude on the test set to 1500 feet at 29.92 IN HG.

5. Adjust the airspeed on the test set to output 60 knots. (Make sure that test set correction chart is properly applied.)

6. After airspeed has stabilized (15 to 20 seconds), engage IAS mode on the pilot mode selector and record the airspeed reference shown on the air data command display and the airspeed shown on the pilots airspeed indicator. Refer to Table 1 and Table 2.

INSTRUCTIONS

6.1. Record the airspeed indicator readings (KIAS) and air data command display readings in specified columns.

6.2. Determine BASIC SPLIT between KIAS and air data command display. No negative numbers.

6.3. Add offsets of -5, +5, and +10 to air data command display and place on the left of the diagonal line. This is the new air data command display value.

6.4. Determine the compensated split by taking the difference between KIAS and new air data command display value from the Step 6.3. and place on the right of the diagonal line. No negative numbers.

6.5. Total columns as shown, then divide by 5 to obtain the average split.

6.6. Determine the lowest average split and record where shown. This determines the best offset.

6.7. Proceed to Table 3 to determine the proper offset wiring, if required, and wire accordingly.

7. Select FD1 on the AP controller and repeat Step 6. for the copilot side.

8. Repeat Step 6. and Step 7. as necessary to complete Table 1 and Table 2.

9. Complete Table 1 and Table 2 offset calculations per instructions and adjust the wiring on each digital flight control computer to provide proper offset per Table 3.

NOTE

Table 3 is used in conjunction with Table 1 and Table 2 to determine proper airspeed offset wiring on digital flight control computer.

Table 1. Copilot OFFSET/SPLIT (S/N 36087 through 36999)

AIRSPEED	KIAS	FD1 AL-300	BASIC SPLIT	-5/SPLIT	+5/SPLIT	+10/SPLIT
60				/	/	/
80				/	/	/

Table 1. Copilot OFFSET/SPLIT (S/N 36087 through 36999) (continued)

AIRSPEED	KIAS	FD1 AL-300	BASIC SPLIT	-5/SPLIT	+5/SPLIT	+10/SPLIT
100				/	/	/
120				/	/	/
140				/	/	/
Total Split			_____	_____	_____	_____
Total Split/5 =			_____	_____	_____	_____
Lowest Average Split (Pick from Above) = _____						

Table 2. Pilot OFFSET/SPLIT (S/N 36087 through 36999)

AIRSPEED	KIAS	FD1 AL-300	BASIC SPLIT	-5/SPLIT	='PLUS'>5/ SPLIT	+10/SPLIT
60				/	/	/
80				/	/	/
100				/	/	/
120				/	/	/
140				/	/	/
Total Split			_____	_____	_____	_____
Total Split/5 =			_____	_____	_____	_____
Lowest Average Split (Pick from Above) = _____						

Table 3. Airspeed Offset Wiring (S/N 36087 through 36999)

REQUIRED ADS OFFSET (KNOTS)	PIN C72	PIN C71
0	OPEN	OPEN
-5	OPEN	SIGNAL GND
+5	SIGNAL GND	OPEN
+10	SIGNAL GND	SIGNAL GND

EXAMPLE

If lowest average split is from the +5 column on Table 1 or Table 2, Pin C72 should be wired to signal ground while Pin C71 should remain open. Refer to helicopter wiring diagrams to determine wiring location.

DAFCS - Troubleshooting

Procedure

1. Make sure that the appropriate circuit breakers are closed and the interconnecting cabling is in normal condition.

1.1. Procedure

1.1.1. Table 1 is used to isolate a faulty component of the system. If an error code is displayed on the air data command display immediately at power-up or as different modes are engaged, Table 1 is used to determine the possible cause of the problem along with the mode flow or mode logic diagrams in SYSTEM OPERATION, that are associated with the error code.

1.1.2. When the pilot reports a faulty condition, the technician should do the preflight test to verify the reported fault condition. If a fault is verified, refer to the applicable error code in Table 1 to determine the most probable cause of the failure. If the reported faulty condition cannot be verified by doing the preflight test, then the GROUND CHECK procedure associated with the reported fault should be done.

1.1.3. To verify digital automatic flight control system integrity and assist troubleshooting, test and monitoring functions are built into the system. These functions include automatic power-up and continuous monitoring tests, and a pilot performed preflight test.

1.2. Power-Up Test

1.2.1. Each time power is applied to the digital flight control computer, an automatic power-up test is done. Failure of power-up test typically indicates a computer failure and autopilot and flight director engagement is inhibited.

1.2.2. A power-up test failure will also result in the following:

- The FD flag will remain in view on the ADI (FD FAIL).
- The AP1 and/or AP2 failure annunciator will light on the DAFCS caution panel. Actual annunciator legends are NO 1 AUTOPILOT and NO 2 AUTOPILOT.
- Any power-up error codes will be displayed on the air data command display for 2 seconds and then automatically sequence to the next code.
- If power-up error codes do not self-clear within 30 seconds, do as follows:
  - Note error code on air data command display.
  - Press SBY repeatedly until all codes are removed and the air data command display has returned to dashes.

NOTE

Electrical power fluctuations during helicopter start may cause erroneous failure of the power-up test. Allow electrical power to stabilize, then reset each digital flight control computer with the AFCS DC circuit breakers. If codes persist, recommended procedure is to shut down helicopter and notify maintenance personnel.

1.2.3. It is recommended that the helicopter be shut down and the problem be reported (with error codes) to maintenance personnel. System error codes are presented in Table 1.

1.3. Continuous Tests and Monitoring

1.3.1. When not in power-up or preflight tests, the digital automatic flight control system does its own continuous tests and monitoring to verify continued system integrity. System response to a test failure or tripped monitor is dependent on the mode in use when the problem occurred. A failure of this type may result in autopilot disengagement or simply degraded performance. However, an error code will always be displayed on the air data command display (provided that 28 VDC is still available to at least one digital flight control computer). If more than one code is present, each will be displayed with automatic sequencing as with power-up test.

1.3.2. If a continuous error code does not self-clear within 30 seconds, do as follows:

- Note all error codes on air data command display
- Press SBY repeatedly until all codes are removed and the air data command display has returned to dashes

1.3.3. If an error code is displayed as a result of a software monitor trip, an additional fail annunciation will be activated (i.e., NO. 1 AUTOPILOT, NO. 2 AUTOPILOT, or AUTOTRIM). If it is suspected that the monitor has falsely tripped, these annunciators can be cleared by resetting the software monitor, which can be done by disengaging and re-engaging the failed autopilot(s).

1.3.4. If code(s) occurred during flight, continued flight is at the discretion of the pilot. Problem and error codes should be reported to maintenance personnel as soon as practical. System error codes are presented in Table 1.

NOTE

Loss of an autopilot, or other equipment, may impose operational limitations. Consult the approved Flight Manual or appropriate supplements for further information.

Electrical power fluctuations during and following a bus failure and/or bus tie switching may cause erroneous failure of the continuous test. Allow electrical power to stabilize, then reset each digital flight control computer separately using the AFCS DC circuit breakers. If code(s) return(s) after resetting, continue flight at pilots discretion. After landing, shut down the helicopter and notify maintenance personnel.

1.4. Preflight Test

1.4.1. Preflight test in the Model 412 helicopter is a two-level test of both digital flight control computers and peripheral system components. Level 1 checks hardware and software considered to be airworthiness critical.

1.4.2. Level 2 tests those non-critical hardware and software system functions that can be checked automatically prior to dispatch. Level 2 is optional, and is intended to allow more thorough troubleshooting of system problems.

1.4.3. Preflight test is inhibited in flight by Aircraft-On-Ground (AOG) logic within each digital flight control computer. Preflight test may only be engaged when both autopilots are off, and the following three conditions are satisfied:

- Airspeed less than 60 knots
- Radar altitude less than 20 feet
- Collective position less than 25% up travel

1.4.4. If any one of these parameters is exceeded, the helicopter is assumed to be airborne and test engagement will be inhibited. If the air data sensor is invalid, AOG will be based on conditions 2 and 3 as previously stated. If radar altitude is invalid, conditions 1 and 3 will define AOG. If both air data and radar altitude are invalid, AOG will be determined solely by collective position.



DURING PREFLIGHT TEST, THE MAIN ROTOR TIP PATH PLANE MAY BE DEFLECTED SLIGHTLY. USE CAUTION NOT TO ENDANGER PERSONS OR OBJECTS NEAR THE HELICOPTER ROTOR DISC.

1.4.5. Successful performance of the preflight test also requires the following initial conditions:

- Helicopter parked on a reasonably level surface
- Rotor RPM above 77%
- Helicopter DC and AC operating normally
- **SN: 36087-36247:**
  All gyros valid (flags out of view)
- **SN: 36248-36999:**
  All AHRS valid (flags out of view)
- Autopilots OFF
- Cyclic and pedals approximately centered, collective down
- Hands and feet off of the controls

NOTE

To prevent nuisance failures, hands must remain off the controls.

1.4.6. Preflight test is started by pressing the TEST button. The digital automatic flight control system will then flash certain buttons on the autopilot controller in a specific sequence. The pilot must then press that button in order for the test to continue. Test will not continue until the flashing button is pressed. Without interruptions, level 1 will take approximately 20 to 25 seconds. Failures detected will be displayed as error codes on the air data command display. Refer to Table 1 for error listings.

1.5. Preflight Test Sequence

1.5.1. Level 1 Performance

Press the TEST button.

1.5.2. Verify that the NO. 1 AUTOPILOT and NO. 2 AUTOPILOT failure annunciators extinguish.

1.5.3. Air data command display illuminates LE 1 indicating activation of level 1.

1.5.4. AP1 annunciator button illuminates flashing.

NOTE

If TEST will not engage, refer to Table 1.

1.5.5. Press the AP1 button when flashing.

1.5.6. Verify that the AP1 engages and the annunciator button shows steady ON.

1.5.7. Verify that during testing, the cyclic, pedals, and collective will move slightly to check for proper command and feedback as well as binding or interference in the controls.

1.5.8. Verify that upon completion of AP1 testing, the AP2 button illuminates flashing.

1.5.9. Press the AP2 button when flashing.

1.5.10. Verify that the AP2 engages and the annunciator button shows steady ON.

1.5.11. Verify that the cyclic, pedals, and collective will move slightly doing similar checks as previously listed.

1.5.12. Verify that upon completion of AP2 testing, the AP1 and AP2 buttons automatically extinguish.

1.5.13. If no error codes are present, the air data command display will display END 1.

1.5.14. Proceed to the next step (Press TEST button to end), or proceed to level 2 performance if desired.

1.5.15. Press the TEST button to end. The TEST button will turn off and the SBY buttons will display ON.

1.5.16. Air data command display will revert to dashes. If desired, the autopilots may be engaged.

1.5.17. If error codes are displayed do as follows:

1.5.17.1. Press the SBY (or NAV STBY switch on either cyclic grip) once to view each code.

1.5.17.2. When all level 1 codes have been displayed, the air data command display will display END 1.

1.5.18. Proceed with level 2 testing, as follows, to obtain further information, which may explain the cause of level 1 errors.

1.5.19. All level 1 and 2 error codes displayed should be recorded and used with Table 1 to aid in troubleshooting.

1.5.19.1. Reset each digital flight control computer using the DFCS DC circuit breakers and repeat preflight test.

1.5.19.2. If codes persist, refer to Table 1.

NOTE

Failures of preflight test do not inhibit autopilot or flight director engagement. Refer to the approved Flight Manual to determine the course of action taken following a test failure.

Error codes are stored in non-volatile computer memory for retrieval by maintenance personnel. However, it is recommended that pilots write down error codes to prevent confusion with any codes that were previously stored, but possibly not erased from memory.

1.5.20. Level 2 Performance

NOTE

Level 2 may be entered upon completion of level 1 when the air data command displays END 1 and the SBY buttons are flashing.

Press either SBY annunciator button when flashing and END 1 is displayed.

1.5.21. Verify that the air data command display illuminates LE 2 indicating that the system has entered level 2.

1.5.22. Verify that the DCPL annunciator and three annunciators in the caution panel (NO. 1 AUTOPILOT, NO. 2 AUTOPILOT and AUTO TRIM) flash.

1.5.23. Verify that all annunciator lamps are working.

NOTE

The NAV STBY button on either cyclic may be used in place of the SBY button on the mode selectors.

1.5.24. Press either Master Caution annunciator button above the ADI.

1.5.25. Verify that the DCPL annunciator and the three annunciators in the caution panel extinguish.

NOTE

The lamp and button test may be bypassed by pressing either SBY button or cyclic NAV STBY button twice.

1.5.26. Verify that the lamp and button test is then activated;

1.5.26.1. Check that all the buttons on both flight director mode selectors flash.

1.5.26.2. Check that all the buttons on the autopilot controller also flash, except; AP1, AP2, TRIM/ FD, and TEST.

1.5.26.3. Press each flashing button individually to extinguish it.

1.5.26.4. After all the buttons have been pressed and no error codes are present, the air data command display will display END and the TEST button will flash.

1.5.27. If error codes are displayed, do as follows:

1.5.27.1. Press SBY (or cyclic NAV STBY button) once to view each code.

1.5.27.2. Error codes displayed should be recorded and used with Table 1 for troubleshooting.

1.5.27.3. Reset each digital flight control computer using the DFCS DC circuit breakers and repeat preflight test, if required.

1.5.28. Upon completion of the error code readout, the air data command display will display END and the TEST annunciator button will flash.

1.5.29. Press the TEST annunciator button when flashing.

1.5.30. Verify that the TEST annunciator button extinguishes.

1.5.31. Verify that the SBY annunciator buttons display ON.

1.5.32. Verify that the air data command display will revert to dashes.

1.5.33. Verify that the NO. 1 AUTOPILOT and NO. 2 AUTOPILOT failure annunciators in the caution panel illuminate.

1.5.34. Engage the autopilots to extinguish AP fail annunciators, if required.

1.6. TEST Will Not Engage

1.6.1. The following is a troubleshooting aid of the preflight test.

1.6.1.1. Verify that both autopilots are OFF.

1.6.1.2. Verify that the helicopter electrical power operates normally.

1.6.1.3. Verify that the collective position is less than 25% up travel. With the rotor turning, it is unlikely that the collective position inhibits the test; however, a defective collective position sensor could be at fault.

NOTE

For the purpose of engaging TEST, only the master computer (as selected with the TRIM/FD button on autopilot controller) monitors the TEST pushbutton and determines whether the required conditions for engaging preflight test are met (i.e., helicopter on ground and both APs off).

1.6.1.4. If unable to engage TEST when the autopilot 2 is selected, select the autopilot 1 as the master.

1.6.1.5. Attempt to engage the preflight test.

1.6.1.6. If TEST now engages, do level 1 and level 2 testing of the autopilot 2.

1.6.1.7. If TEST still does not engage, or if switching master computer was necessary to engage it, reference error codes to aid in additional investigation into the source of the problem.

1.7. Error Codes

1.7.1. Power-up test, continuous monitoring, and preflight test error codes are displayed on the air data command display according to the following format:

NE###

where: N - is the computer side (i.e., 1 or 2).

1 = AP1, copilot side.

2 = AP2, pilot side

E - error code follows.

- is the error code number. Where the first digit may be blank, or a 1 or 2 indicating the test level.

1.7.2. The error code listing does these functions.

- Defines the failure associated with each code for troubleshooting purposes
- Defines the effect of the code on system performance

1.8. Error Code Memory

1.8.1. All error codes, regardless of power-up, continuous test, or preflight test, are repetitively stored in non-volatile memory (EEPROM) for later retrieval by maintenance personnel.

1.8.2. A guarded toggle switch is used to read stored error codes and a momentary button switch is used in conjunction with the toggle switch to erase stored error codes. These switches are located on the right sidewall of the pedestal.

1.8.3. This feature is most useful for verifying pilot reported error codes on a flight by flight basis or verifying system integrity on a regular basis. This regular basis may be in conjunction with normal periodic helicopter inspections. Since a particular error code may be stored repeatedly, trends may be spotted in system and/or component performance. These trends may prove helpful in diagnosing chronic problems. After system or component repair has been done, the memory should be erased and then read again after the next flight to help ensure that the problem has been corrected.

1.8.3.1. Reading Stored Error Codes.

1.8.3.1.1. To read the error codes stored in memory, do as follows:

- Make sure AOG conditions are met (collective fully down)
- Turn all helicopter electrical power OFF
- Place the READ switch in READ position
-

NOTE

For non flight director equipped helicopters, press and hold the CPL switch on the autopilot control panel.

Turn helicopter electrical power ON

1.8.3.1.2. In normal operation, the system automatically selects System 2 upon power-up. Stored error codes from System 2 will then be displayed automatically in reverse sequence (i.e., last in, first out) for 5 seconds each, until all codes have been displayed. After all error codes have been displayed, the air data command display will revert to dashes. To read the other system (System 1) codes press the TRIM/FD (FDSEL, if 4-Axis kit is installed) button on the autopilot controller. To view the error codes again, repeat the previous retrieval sequence.

- Return the READ switch to the OFF position when finished or erase as required

1.8.3.2. Erasing Stored Error Codes.

1.8.3.2.1. To erase stored error codes from memory, do as follows

NOTE

For the following procedure, make sure that the System 2 is automatically selected upon power-up.

- Make sure that the AOG conditions are met (collective fully down).
- Turn all helicopter electrical power OFF
- Place the READ switch in READ position

NOTE

For non flight director equipped helicopters, press and hold the CPL switch on the autopilot control panel.

- Press and hold the ERASE button switch and turn the helicopter electrical power ON. After power application, continue to hold the ERASE button for an additional 5 seconds
- Release the ERASE switch. The air data command display will read 2ErAS while erasing

NOTE

Erasing time depends upon the number of codes stored. If no codes are in memory the AL-300 will display dashes immediately.

- When erasure of System 2 is complete, the air data command display will revert to dashes
- To erase the other system (System 1), press the TRIM/FD or FDSEL button on the autopilot controller to select FD1
- The air data command display will read 1ErAS while System 1 is erasing.

1.8.3.2.2. Upon completion of erasure, the air data command display will revert to dashes and the SBY annunciator on the mode selector will light, indicating the system is ready to operate.

- Return READ switch to the OFF position when erasure is complete

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E0 Lost processor	Defective digital flight control computer	Replace digital flight control computer
E1 Ram partition fail	Defective digital flight control computer	Replace digital flight control computer
E2 Instruction set fail	Defective digital flight control computer	Replace digital flight control computer
E3 Register test fail	Defective digital flight control computer	Replace digital flight control computer
E4 RAM RD/WR fail	Defective digital flight control computer	Replace digital flight control computer
E5 EPROM checksum fail	Defective digital flight control computer	Replace digital flight control computer
E6 CPU fail	Defective digital flight control computer	Replace digital flight control computer
E7 D/A-A/D fail	Defective digital flight control computer	Replace digital flight control computer
E8 Interrupt reg fail	Defective digital flight control computer	Replace digital flight control computer
E9 +15 V monitor fail	Defective digital flight control computer	Replace digital flight control computer
E10 -15 V monitor fail	Defective digital flight control computer	Replace digital flight control computer
E11 +5 V monitor fail	Defective digital flight control computer	Replace digital flight control computer
E12 +10 V monitor fail	Defective digital flight control computer	Replace digital flight control computer
E13 LVC monitor fail	Defective digital flight control computer	Replace digital flight control computer
E14 Sawtooth monitor fail	Defective digital flight control computer	Replace digital flight control computer
E15 Actuator reference monitor fail	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E16 Watchdog monitor fail	Defective digital flight control computer	Replace digital flight control computer
E17 Ticket failure	Defective digital flight control computer	Replace digital flight control computer
E18 LVC fail	26 VAC circuit breaker not set	Set circuit breaker
	26 VAC bus failure	Troubleshoot electrical wiring
	Defective digital flight control computer	Replace digital flight control computer
E19 Sawtooth fail	Defective digital flight control computer	Replace digital flight control computer
E20 Actuator reference fail	26 VAC circuit breaker not set.	Set circuit breaker
	26 VAC bus failure	Troubleshoot electrical wiring
	Defective digital flight control computer	Replace digital flight control computer
E21 Watchdog fail	Defective digital flight control computer	Replace digital flight control computer
E25 Trim end around fail	Helicopter trim switching relay(s) shorted	Replace relay
	Trim motor shorted	Replace trim motor
	Wire to trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E26 Trim monitor checksum fail	Defective digital flight control computer	Replace digital flight control computer
E28 No roll trim	Wire to roll trim motor broken or shorted.	Repair/replace wire
	Roll trim motor inoperative	Replace roll trim motor
	Roll trim clutch disengaged	
	Defective digital flight control computer	Replace digital flight control computer
E29 No pitch trim	Wire to pitch trim motor broken or shorted	Repair/replace wire
	Pitch trim motor inoperative	Replace pitch trim motor
	Pitch trim clutch disengaged.	
	Defective digital flight control computer	Replace digital flight control computer
E31 Roll trim fail	Wire to roll trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E32 Pitch trim fail	Wire to pitch trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E34 Roll trim runaway	Wire to roll trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E35 Pitch trim runaway	Wire to pitch trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E36 Trim run checksum fail	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E38 Yaw series servo fail	Wire to yaw actuator broken or shorted	Repair/replace wire
	Yaw actuator motor failed	Replace yaw actuator motor
	Yaw actuator LVDT failed	Replace yaw actuator LVDT
	Defective digital flight control computer	Replace digital flight control computer
E39 Roll series servo fail	Wire to roll actuator broken or shorted	Repair/replace wire
	Roll actuator motor failed	Replace roll actuator motor
	Roll actuator LVDT failed	Replace roll actuator LVDT
	Defective digital flight control computer	Replace digital flight control computer
E40 Pitch series servo fail	Wire to pitch actuator broken or shorted	Repair/replace wire
	Pitch actuator motor failed	Replace pitch actuator motor
	Pitch actuator LVDT failed	Replace pitch actuator LVDT
	Defective digital flight control computer	Replace digital flight control computer
E42 Vertical gyro miscompare	SN: 36087-36247: Wire from either vertical gyro shorted or open	Repair/replace wire
	SN: 36248-36999: Wire from either AHRS shorted or open	Repair/replace wire
	SN: 36087-36247: Unflagged failure of either vertical gyro	Replace vertical gyro
	SN: 36248-36999: Unflagged failure of either AHRS module	Replace AHRS module
	Defective digital flight control computer	Replace digital flight control computer
E51 Collective CMT fail	Wire to/from either collective position pot broken or shorted	Repair/replace wire
	Either collective position pot failed	Replace collective position pot
	Defective digital flight control computer	Replace digital flight control computer
E55 Error code storage nearly full		Erase stored error codes
E56 Error code storage full		Erase stored error codes
E101 Pitch adaptive rate limiter test fail	POD, ROD, or FTR	See error codes E125, E126, or E129
	Defective digital flight control computer	Replace digital flight control computer
E102 Roll adaptive rate limiter test fail	POD, ROD, or FTR	See error codes E125, E126, or E129
	Defective digital flight control computer	Replace digital flight control computer
E103 Pitch actuator moved during test	Defective digital flight control computer	Replace digital flight control computer
E104 Roll actuator moved during test	Defective digital flight control computer	Replace digital flight control computer
E105 Pitch trim actuator moved	Defective digital flight control computer	Replace digital flight control computer
E106 Roll trim actuator moved	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E110 No roll trim	Wire to roll trim motor or controls position sensor broken or shorted.	Repair/replace wire
	Roll trim motor inoperative	Replace roll trim motor
	Roll trim clutch disengaged	
	Defective digital flight control computer	Replace digital flight control computer
E111 No pitch trim	Wire to pitch trim motor or controls position sensor broken or shorted	Repair/replace wire
	Pitch trim motor inoperative	Replace pitch trim motor
	Pitch trim clutch disengaged	
	Defective digital flight control computer	Replace digital flight control computer
E113 Roll trim backwards	Wires to roll trim motor reversed	Reverse wires
	Excitation to roll position pot reversed	Reverse wires
	Defective digital flight control computer	Replace digital flight control computer
E114 Pitch trim backwards	Wires to pitch trim motor reversed	Reverse wires
	Excitation to pitch position pot reversed	Reverse wires
	Defective digital flight control computer	Replace digital flight control computer
E115 Servo valid delay (2 seconds) fail	Defective digital flight control computer	Replace digital flight control computer
E116 Watchdog cannot disengage AP	Defective digital flight control computer	Replace digital flight control computer
E117 Yaw actuator/servo amp fail	Wire to yaw actuator broken or shorted	Repair/replace wire
	Yaw actuator fail	Replace yaw actuator
	Defective digital flight control computer	Replace digital flight control computer
E118 Roll actuator/servo amp fail	Wire to roll actuator broken or shorted	Repair/replace wire
	Roll actuator fail	Replace roll actuator
	Defective digital flight control computer	Replace digital flight control computer
E119 Pitch actuator/servo amp fail	Wire to pitch actuator broken or shorted	Repair/replace wire
	Pitch actuator failure	Replace pitch actuator
	Defective digital flight control computer	Replace digital flight control computer
E121 FD ANN fail	Either wire from on-side FD ANN to cross-side FD ANN OTHER broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
F124 FD ANN fail other	Either wire from on-side FD ANN to cross-side FD ANN OTHER broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E125 POD during test	Wire to pitch detent switch broken or shorted	Repair/replace wire
	Controls held during preflight test	
	Cyclic friction too high	
	Defective digital flight control computer	Replace digital flight control computer
E126 ROD during test	Wire to roll detent switch broken or shorted	Repair/replace wire
	Cyclic held during preflight test	
	Cyclic friction too high	
	Defective digital flight control computer	Replace digital flight control computer
E129 FTR during test	Force trim pedestal switch off or circuit breaker open	Turn force trim switch on or close circuit breaker
	Cyclic grip FTR button(s) pressed	
	Wire to force trim circuit breaker, switch or button(s) broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E131 Trim switch OFF during test	Force trim pedestal switch or circuit breaker open	Close force trim switch or circuit breaker
	Wire to force trim circuit breaker or switch broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E133 Pitch trim actuator moved with drive off	Wire to pitch trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E134 Roll trim actuator moved with drive off	Wire to roll trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E136 Pitch trim actuator moved with AP off	Defective digital flight control computer	Replace digital flight control computer
E137 Roll trim actuator moved with AP off	Defective digital flight control computer	Replace digital flight control computer
E139 Yaw series actuator moved	Defective digital flight control computer	Replace digital flight control computer
E141 Pitch trim (H) end around ON	Defective digital flight control computer	Replace digital flight control computer
E142 Pitch trim (L) end around ON	Defective digital flight control computer	Replace digital flight control computer
E143 Roll trim (H) end around ON	Defective digital flight control computer	Replace digital flight control computer
E144 Roll trim (L) end around ON	Defective digital flight control computer	Replace digital flight control computer
E147 Trim enable fail	Defective digital flight control computer	Replace digital flight control computer
E148 Trim one-shot fail	Defective digital flight control computer	Replace digital flight control computer
E149 Pitch trim (H) end around OFF	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E150 Pitch trim (L) end around OFF	Defective digital flight control computer	Replace digital flight control computer
E151 Roll trim (H) end around OFF	Defective digital flight control computer	Replace digital flight control computer
E152 Roll trim (L) end around OFF	Defective digital flight control computer	Replace digital flight control computer
E155 Servo valid fail	Defective digital flight control computer	Replace digital flight control computer
E156 Trim end around fail	Wire to any trim motor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E157 APON other input fail	Either wire from on-side AP engage to cross-side AP engage other broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E201 Air data display knob motion	Air data display failure	Replace air data display
	Wire to air data display shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E202 Airspeed more than 60 knots	Air data sensor not warmed up	
	Air data sensor failure	Replace air data sensor
	Wire to air data sensor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E203 Primary radio altimeter more than 20 feet	Radalt indication more than 20 feet.	
	Wire to radalt shorted	Repair/replace wire
	Defective digital flight control computer.	Replace digital flight control computer
E204 Analog ground fail	Defective digital flight control computer	Replace digital flight control computer
E208 Roll attitude more than 10°	Roll attitude indication greater than 10°	
	SN: 36087-36247: Wire to vertical gyro shorted	Repair/replace wire
	SN: 36248-36999: Wire to AHRS module shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E209 Pitch attitude more than 10°	Pitch attitude indication greater than 10°	
	SN: 36087-36247: Wire to vertical gyro shorted	Repair/replace wire
	SN: 36087-36247: Wire to AHRS shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E210 Yaw rate gyro more than 10° per second	SN: 36087-36247: Wire to yaw rate gyro shorted	Repair/replace wire
	SN: 36248-36999: Wire to AHRS shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E211 Directional gyro fail.	SN: 36087-36247: DG circuit breaker open	Close circuit breaker
	SN: 36248-36999: AHRS circuit breaker open	Close circuit breaker
	SN: 36087-36247: Wire to directional gyro broken or shorted	Repair/replace wire
	SN: 36248-36999: Wire to AHRS broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E212 Vertical gyro not valid.	SN: 36087-36247: VG circuit breaker open	Close circuit breaker
	SN: 36248-36999: AHRS circuit breaker open	Close circuit breaker
	SN: 36087-36247: Wire to vertical gyro broken or shorted	Repair/replace wire
	SN: 36248-36999: Wire to AHRS broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E213 Directional gyro not valid	SN: 36087-36247: DG circuit breaker open	Close circuit breaker
	ATTD SYS circuit breaker open	Close circuit breaker
	SN: 36087-36247: Wire to directional gyro broken or shorted	Repair/replace wire
	SN: 36248-36999: Wire to AHRS broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer.
E214 Primary radio altimeter not valid	Radalt circuit breaker open	Close circuit breaker
	Wire to radalt broken or shorted.	Repair/replace wire.
	Defective digital flight control computer	Replace digital flight control computer
E215 Air data sensor not valid	ADC circuit breaker open	Close circuit breaker
	Air data sensor not warmed up	
	Air data sensor failure	Replace air data sensor
	Wire to air data sensor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E216 Serial I/O not valid	Only one digital flight control computer installed	
	Serial I/O wire(s) broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E217 Other autopilot not in test	Only one digital flight control computer installed	
	Either wire from on-side AP engage to cross-side AP engage other broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E218 Mode selector button(s) pressed	Any button on the mode selector pressed	
	Wire to mode selector shorted	Repair/replace wire
	Mode selector failure	Replace mode selector
	Defective digital flight control computer	Replace digital flight control computer
E219 AP controller button(s) pressed	Any button on the autopilot controller pressed	
	Wire to autopilot controller shorted	Repair/replace wire
	Autopilot controller failure	Replace autopilot controller
	Defective digital flight control computer	Replace digital flight control computer
E221 Hardware not valid	26 VAC circuit breaker not set	Close circuit breaker
	26 VAC bus failure	Troubleshoot electrical wiring
	Defective digital flight control computer	Replace digital flight control computer
E222 Four-way beep switch pressed	Wire to either cyclic or either collective beep switch broken or shorted	Repair/replace wire
	Either cyclic or either collective beep switch stuck or failed	Replace switch
	Defective digital flight control computer	Replace digital flight control computer
E224 Barometric altimeter <-600 or >12,000 feet	Air data sensor not warmed up	
	Air data sensor failure	Replace air data sensor
	Wire to air data sensor shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E225 POD during test	Refer to possible causes for E125	
E226 ROD during test	Refer to possible causes for E126	
E229 FTR during test	Refer to possible causes for E129	
E231 Trim switch OFF during test	Refer to possible causes for E131	
E236 Airspeed input open	Wire from air data sensor broken or shorted	Repair/replace wire
	Air data sensor failure	Replace air data sensor
	Defective digital flight control computer	Replace digital flight control computer
E237 Heading error not valid	SN: 36087-36247: DG circuit breaker open	Close circuit breaker
	SN: 36087-36247: Wire to Tarsyn broken or shorted	Repair/replace wire
	SN: 36248-36999: Wire to AHRS broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer

Table 1. DAFCS (S/N 36087 through 36999) - Troubleshooting (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
E241 Caution advisory RESET button pressed	Caution advisory reset button stuck or failed	Replace caution panel
	Wire from caution advisory reset button broken or shorted	Repair/replace wire
	Defective digital flight control computer	Replace digital flight control computer
E243 Cyclic beep active during test	Wire to second pole of either cyclic beep switch broken or shorted	Repair/replace wire
	Either cyclic beep switch stuck or failed	Replace beep switch
	Wire to remote trim enable relay broken or shorted	Repair/replace wire
	Remote trim enable relay failed	Replace relay
	Defective digital flight control computer	Replace digital flight control computer
E244 Erase EEPROM during test	Wire to EEPROM switch shorted	Repair/replace wire
	Erase EEPROM switch failure	Replace switch
	Defective digital flight control computer	Replace digital flight control computer
E245 Read EEPROM during test	Read EEPROM switch left on during test	
	Wire to read EEPROM switch shorted	Repair/replace wire
	Read EEPROM switch failure	Replace switch
	Defective digital flight control computer	Replace digital flight control computer

Autopilot Controller - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the controller in the pedestal.
2. Lift the panel far enough out of the pedestal assembly to disconnect the electrical connector.
3. Lift the controller free of the pedestal.

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Air Data Command Display (Altitude Preselector) - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Refer to the removal procedures, panel mounted instruments, (412-A-95-00-01-00A-520A-A / 01791).

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Actuator Position Indicator Panel - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Loosen the spring lock fasteners that secure the panel in the pedestal.
2. Lift the panel far enough out of the pedestal to disconnect the electrical connector.
3. Lift the panel free of the pedestal.

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Digital Flight Control Computer (No. 1 or No. 2) - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Loosen the knurled nuts on the front of the computer mounting rack.
2. Slide the computer out of its mounting rack.

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Flight Director Mode Selector (No. 1 or No. 2) - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Refer to the removal procedures, panel mounted instruments, (412-A-95-00-01-00A-520A-A / 01791).

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Air Data Sensor (No. 1 or No. 2) - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Disconnect the electrical connector and the pitot/static lines.
2. Protect the connector and the lines with caps or tape.
3. Remove and retain the mounting hardware.
4. Remove the air data sensor from the shelf.

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Rate Gyro (No. 1 or No. 2) - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.



A GYRO MAY BE DAMAGED BY MOVING IT AFTER ELECTRICAL POWER IS REMOVED BUT BEFORE THE GYRO ROTOR STOPS. THE GYRO ROTOR WILL NOT COME TO A STOP FOR APPROXIMATELY 15 MINUTES AFTER ELECTRICAL POWER IS REMOVED. WHEN REMOVING OR INSTALLING A GYRO, DO SO GENTLY WITH NO QUICK MOTION.

1. Remove the hardware that secures the lid on the yaw rate gyro assembly.
2. Remove the lid.
3. Disconnect the electrical connector.
4. Remove the mounting hardware from the gyro.
5. Remove the gyro from the yaw rate gyro assembly box.

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Relay Box - Removal

Procedure



MAKE SURE THAT ELECTRICAL POWER IS OFF.

1. Disconnect the electrical connectors.
2. Protect the connectors with cap or tape.
3. Remove the mounting hardware.
4. Remove the relay box.

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Relay Box Assembly - Repair

Procedure

1. Gain access to the relay box assembly in the aft left avionics bay.
2. Remove the hardware that secures the lid on the relay box assembly.
3. Remove the lid.
4. Remove the hardware that secures the defective relay to be removed.
5. Remove the relay from the socket.
6. Install the replacement relay in the socket.
7. Secure the relay with the hardware.
8. Position the lid on the relay box assembly and secure with the attaching hardware.

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Autopilot Controller - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the controller.
2. Put the controller in its position on the pedestal.
3. Attach the controller to the pedestal with the spring lock fasteners.

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Air Data Command Display (Altitude Preselector) - Installation Procedure

1. For the installation procedure of Air Data Command Display (Altitude Preselector), refer to 412-A-95-00-01-00A-720A-A / 01792.

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Actuator Position Indicator Panel - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Connect the electrical connector to the panel.
2. Put the panel in its position on the pedestal.
3. Attach the panel to the pedestal with the spring lock fasteners.

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Digital Flight Control Computer (No.1 or No.2) - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Move the computer into the mounting rack.
2. Lift the holddown clamps above the lug on the front of the computer.
3. Tighten the knurled nuts to safety the computer.

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Flight Director Mode Selector (No. 1 or No. 2) - Installation

Procedure

1. For the installation procedure of Flight Director Mode Selector (No. 1 or No. 2), refer to 412-A-95-00-01-00A-720A-A / 01792.

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Air Data Sensor (No. 1 or No. 2) - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Put the air data sensor in its position on the shelf.
2. Attach with the mounting hardware.

NOTE

Torque the pitot static lines 10 to 20 inch-pounds (1.13 to 2.26 Nm).

3. Connect and tighten the pitot static lines.
4. Connect the electrical connector.
5. Do the pitot static system leak check (412-A-95-11-05-00A-364A-A / 01807).

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Rate Gyro (No. 1 or No. 2) - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.



EXTREME CARE MUST BE USED WHEN HANDLING A GYRO. MAKE SURE POWER HAS BEEN REMOVED FROM UNIT FOR A MINIMUM OF 15 MINUTES. WHEN REMOVING OR INSTALLING A GYRO, DO SO GENTLY WITH NO QUICK MOTION.

1. Put the gyro in its position in the assembly box.
2. Attach with the mounting hardware.
3. Connect the electrical connector.
4. Attach the lid on the assembly box with the attaching hardware.

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Relay Box - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Put the relay box in its position on the structure.
2. Attach with the mounting hardware.
3. Remove the protective cap or tape from the electrical connector.
4. Connect the electrical connectors.

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Digital Automatic Flight Control System (DAFCS) - Inspection Procedures

Procedure

1. Inspect the DAFCS for completeness, proper installation, cleanliness, and secure mounting of each component.
2. Inspect the switches and controls on DAFCS system components for proper mechanical action.
3. Inspect the components to ensure the panel lights illuminate with the electrical power on.
4. Inspect the electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

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Digital Automatic Flight Control System (DAFCS) - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a clean, soft cloth.



CLEANING COMPOUND FUMES ARE FLAMMABLE AND TOXIC.

2. Remove grease, fungus, and ground-in dirt with a soft cloth dampened with approved cleaning compound.
3. Remove dirt from the connector with a brush and remove moisture with a dry cloth.

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Digital Automatic Flight Control System (DAFCS) - Repair or Replacement

Procedure

1. Repair of the DAFCS components is limited to replacement of the relays in the relay box assembly and the replacement of the lamps in the annunciator switches.

2. Replace any defective or inoperative units.

3. Further repairs, other than the removal and installation, must be conducted by an authorized repair facility.

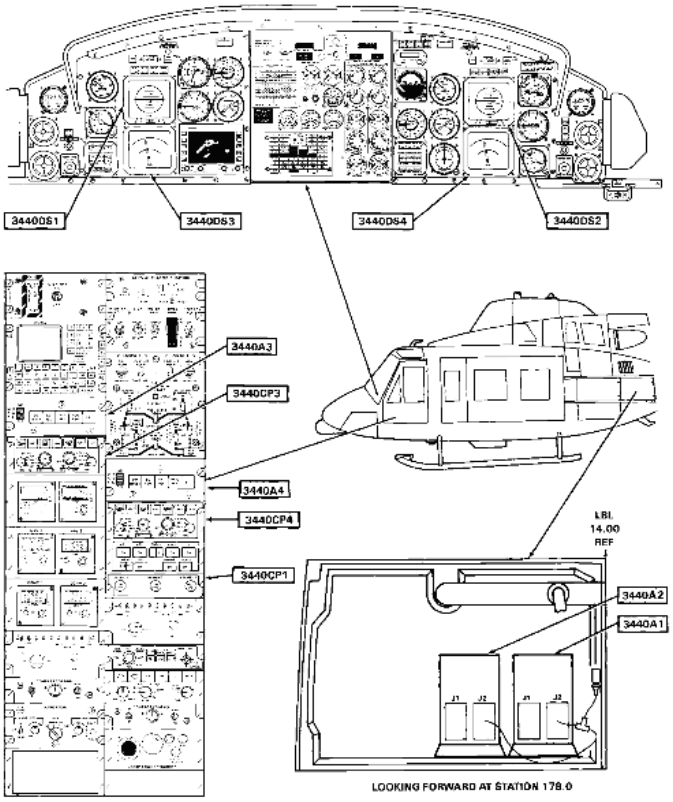
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Electronic Flight INstrument System (EFIS) - Description

The Electronic Flight Instrument System (EFIS) is comprised of the following components: copilot and pilot display controllers (3440CP3 and 3440CP4), copilot and pilot Electrical Attitude Director Indicators (EADI) (3440DS1 and 3440DS2), copilot and pilot Electronic Horizontal Situation Indicators (EHSI) (3440DS3 and 3440DS4), copilot and pilot symbol generators (3440A1 and 3440A2), and remote instrument controller (3440CP1) (Figure 1).

The EFIS is a dual system that provides attitude and heading information to the pilot and copilot. The components of each system are identical and interchangeable. The EFIS displays pitch and roll attitude, heading, course orientation, weather presentations, source annunciations, and doppler velocities.

The major component of the EFIS is the symbol generator. The symbol generator receives heading, attitude, radio altitude, rate of turn, doppler/velocities, short and long range navigation, and weather inputs. All inputs are processed and transmitted to the electronics displays as a function of the selections made on the display controller and the remote instrument controller. If one symbol generator fails, backup is provided by the other symbol generator. The symbol generators communicate over the Avionics Standard Communications Bus (ASCB) to do comparison monitoring.



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Figure 1. Electronic Flight Instrument System - Component Location (Sheet 1 of 1)

EFIS - Operational Check

Procedure

1. Visually inspect the equipment and correct any faulty mechanical mountings or electrical connections. Connect the external power, adjusted to 27.5 VDC, to the helicopter power receptacle.

2. Energize the power buses per 412-A-96-00-01-01A-042A-A / 01873.

3. Apply external hydraulic pressure (412-A-29-10-00-08A-042A-A / 00654).

4. Close the EFIS No. 1 circuit breakers ADI (3440CB1), HSI (3440CB3), SG/DL (3440CB5), EFIS 26V NO.1 (3440CB7), BLWR (3440CB9), and EFIS No. 2 circuit breakers ADI (3440CB2), HSI (3440CB4), SG/DL (3440CB6), EFIS 26V NO.2 (3440CB8), and BLWR (3440CB10).

5. Close the circuit breakers AFCS 26V NO. 1 (9CB4), RATE GYRO NO. 1 (9CB31), AFCS NO. 1 PWR (9CB31), AFCS NO. 1 ADD/CAUT (9CB32), and FORCE TRIM (9CB3).

6. Close the circuit breakers AFCS 26V NO. 2 (9CB7), RATE GYRO NO. 2 (9CB30), AFCS NO. 2 PWR (9CB2), AFCS NO. 2 CAUT (9CB33), and ENC ALT (3410CB1).

7. Set the EFIS MASTER switch to ON.

8. Do the operational checkout according to the Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024.

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EFIS - Troubleshooting

Procedure

1. Make sure that all the appropriate circuit breakers are closed, interconnecting cabling is in normal operating condition, and power is applied to the system. (Refer to the appropriate wiring diagram.) Refer to the Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024 for troubleshooting.

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EFIS - Removal

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Auxiliary Switch Panel

Refer to 412-A-96-00-02-00A-520A-A / 01888 for the removal of the pedestal mounted component.



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

2. Display Controller

Refer to 412-A-96-00-02-00A-520A-A / 01888 for the removal of the pedestal mounted component.



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

3. Instrument Controller

Refer to 412-A-96-00-02-00A-520A-A / 01888 for the removal of the pedestal mounted component.



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

4. Symbol Generator

4.1. Loosen the mounting tray knob at the front of the symbol generator.

4.2. Slide the symbol generator out of its mounting rack.

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EFIS - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. Auxiliary Switch Panel

Refer to 412-A-96-00-02-00A-720A-A / 01891 for the installation of the pedestal mounted component.



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

2. Display Controller

Refer to 412-A-96-00-02-00A-720A-A / 01891 for the installation of the pedestal mounted component.



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

3. Instrument Controller

Refer to 412-A-96-00-02-00A-720A-A / 01891 for the installation of the pedestal mounted component.



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

4. Symbol Generator



DO NOT FORCE FIT. IF MATING IS DIFFICULT, REMOVE THE UNIT AND CHECK FOR CONNECTOR PINS THAT MAY BE BENT OR OUT OF ALIGNMENT. ALSO CHECK THE ALIGNMENT OF THE RECEPTACLE IN THE MOUNTING TRAY.

4.1. Slide the symbol generator into its mounting rack.

4.2. Tighten the mounting tray knob at the front of the symbol generator.

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EFIS - Inspection Procedures

Procedure

1. Inspect the electronic flight instrument system for completeness, proper installation, cleanliness, and secure mounting of each component.

2. Inspect the switches and the controls on the electronic flight instrument system components for proper mechanical action.

3. Inspect the components to make sure that the panel lights illuminate with the electrical power on.

4. Inspect the electrical connections for corroded or bent pins and proper mating, interconnecting cable for frayed or broken insulation, and circuit breakers for proper operation.

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EFIS - Cleaning Procedures

Procedure

1. Remove moisture and loose dirt with a soft, clean cloth.
2. Remove dust and dirt from the system components with a soft, clean cloth.
3. Remove grease, fungus, and ground-in dirt with a soft, clean cloth dampened with approved cleaning compound.
4. Remove dirt from the connectors with a soft bristle brush. Remove moisture with a clean, dry cloth.

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EFIS - Repair or Replacement

Procedure

1. Tighten or replace loose or cracked control knobs or system components.
2. Replace defective panel light bulbs.
3. Replace defective or inoperative units.
4. Further repairs, other than removal or replacement, must be conducted at an authorized repair facility.

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EFIS Cooling System - Description

One external cooling source is provided for each EADI/EHSI pair of electronic displays to provide adequate cooling and proper display operation. Each fan supplies a minimum of 120 cubic feet per minute free air delivery at sea level.

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EFIS Cooling System - Operational Check

Procedure

1. The EFIS cooling system operates when the EFIS system is operating. Refer to the Honeywell, SPZ-7600 Integrated Flight Control System Bell 412 System Maintenance Manual, Publication Number A09-1169-024.

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EFIS Cooling System - Removal

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. **EFIS Fan**

1.1. Disconnect the electrical connector.

1.2. Remove and retain the mounting hardware.

1.3. Remove the fan.

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EFIS Cooling System - Installation

Procedure



MAKE SURE THAT THE ELECTRICAL POWER IS OFF.

1. EFIS Fan

1.1. Position the fan in the mounting bracket.

1.2. Install the mounting hardware.

1.3. Connect the electrical connector.

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Electrical Fault Analysis - Description

Electrical fault analysis is provided for DC power, AC power, exterior lighting, interior lighting, master caution panel, and bleed air heating systems.

Applicable: SN: 33001-33100:

Refer to Table 1.

Applicable: SN: 33101-33107:

Refer to Table 2 or,

Applicable: SN: 33108-33213 or 36001-36019:

Refer to Table 3.

DEFINITIONS OF TERMS USED:

- OPEN - Infinite Resistance.
- SHORT - 0 to minimum resistance.
- RESULT OF FAULT - Circuit condition after fault
- INDICATION TO CREW - Clues given to crew; i.e., visual, audible, etc.
- INDICATION ON AIRCRAFT - Physical damage or effect on helicopter/system.
- POSSIBLE CAUSE OF FAULT - Wire, equipment, or area where fault may have occurred.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
			DC POWER SYSTEM			
Battery	Power wire from battery connector to battery bus	Open circuit	No battery power to battery bus	Zero voltage reading on voltmeters	No battery power available	Wire P2A1 (P4 + to bus bar) loose, disconnected, or broken.
		Short to ground	Battery switch opens	Battery switch in OFF position	No battery power on buses	Wire P2A1 broken, touching structure or ground.
	Ground wire from battery connector to helicopter structure	Open circuit	No battery power to battery bus	Zero voltage reading on voltmeters	No battery power available	Wire P1A1N (P4 - to ground) loose, disconnected, or broken.
		Short to ground	None	None	None	Wire P1A1N broken, touching structure or ground.
	Wire from terminal A1 of No. 1 BUS battery relay to terminal A1 of No. 1 generator relay	Open circuit	No battery power on No. 1 buses	None	No battery power to No. 1 buses. Engine No. 1 cannot be started.	Wire P5A1 (K9-A1 to K3-A1) BATT bus No. 1 relay to No. 1 relay loose, disconnected, or broken.
		Short to ground	No battery power on battery bus	Zero voltage reading on voltmeters	No battery power available	Wires P5A1, P5B1 (K3-A1 to K39-A1) No. 1 non-essential bus relay, P5C1 (K39-A1 to main DC bus No. 1) and wire P45A22 (K9-A1 to P11-E) No. 1 DC control unit connector, broken, touching structure or ground, shorted to ground at disconnect.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
	Wire from BUS 1 BATT switch circuit breaker to terminal 3 of BATTERY switch No. 1	Open circuit	Battery relay No. 1 will not close	None, if battery switch No. 2 is closed (normal operation)	Battery cannot be connected in No. 1 DC buses by battery switch No. 1. Cannot start engine No. 1.	Wires P224A12 (K9-A2 to 2CB1) EMERG BUS 1 circuit breaker, P114A12 (2CB1 to CB5) BUS 1 BAT switch, P138A20 (CB5 to S46-3) BAT switch No. 1, broken, loose or disconnected. BUS 1 BAT SWITCH CB open.
		Short to ground	BUS 1 BATTERY circuit breaker trips	BUS 1 BATTERY circuit breaker tripped	Battery cannot be connected to No. 1 buses by battery switch No. 1. Cannot start engine No. 1.	Wires P224A12, P114A12, P216A12 (2CB1 to 2S1-12) EMERG LOADS switch, P216B22 (2S1-12 to 2TB1-2C), P242A22 (2TB1-2H to 2XK5-5) emergency bus relay socket, P212C22 (2XK5-4 to splice), P212A22 (splice to 2XK3-6), P212B22 (splice to 2XK3-20), P242B22 (2TB1-G to 2XK4-19) emergency bus relay socket, P138A20 (CB5 to S46-3), P124A20 (CB5 to K33) non-essential bus relay, broken, touching structure or ground.
	Wire from terminal A2 of No. 2 bus battery relay to terminal A2 to No. 2 generator relay	Open circuit	No battery power on No. 2 buses	None	No battery power on No. 2 buses. Engine No. 2 cannot be started from battery.	Wire P30A1 (K10-A2 to K4-A2) No. 2 generator relay loose or disconnected.
		Short to ground	No. 2 generator trips off line	ENG 2 GEN caution light	Loss of generator No. 2	Wires P30A1, P30B1 (K4-A2 to K2-A2) No. 2 non-essential bus relay, P30C1 (K2-A2 to main DC bus No. 2) and P29A22 (K4-A2 to P10-E) No. 2 DC control unit broken, touching structure or ground, or shorted to ground at disconnect.
	Wire from BUS 2 BATT switch circuit breaker to terminal 3 of battery switch	Open circuit	Battery relay No. 2 will not close	BATTERY caution light on unless BATT switch No. 1 is closed	Battery cannot be connected to No. 2 DC buses by battery switch No. 2. Cannot start No. 2 engine from battery.	Wires P118A12 (CB5, BUS 2 BAT switch CB to CB2) EMERG BUS 2 CB, P139A20 (CB5 to S202-3) BAT switch No. 2, broken, loose, or disconnected. BUS 2 BAT switch open.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
		Short to ground	Bus 2 battery circuit breaker trips	BUS 2 BATTERY circuit breaker tripped	Battery cannot be connected to No. 2 DC buses by battery switch No. 2. Cannot start No. 2 engine from battery	Wires P139A20 (S202-3 to CB5), P118A12 (CB5 to 2CB2) EMERG BUS 2 CB, P227A12 (2CB2 to 2S1-9) EMERG LOADS switch, P227B22 (2S1-9 to 2TB1-2B), P242B22 (2TB1-2G to 2XK4-19) emergency bus interlock relay socket, P242A22 (2TB1-H to 2XK5-5) emergency bus interlock relay socket, P212C22 (2SK5-4 to splice), P212B22 (splice to 2XK3-20) emergency bus interlock relay socket, P212A22 (splice to 2XK3-6) broken, touching ground or structure.
Generator No. 1	Terminal A	Open circuit	No generator No. 1 output	ENG 1 GENERATOR caution light	Generator No. 1 fails to supply power	Wires P54A16 (G1-A to P59-D), P54B16 (J59-C to K7-C1) No. 1 start relay, P50A16 (K7-C1 to P11-B) No. 1 DC control unit, broken, loose, or disconnected.
		Short to ground	Generator No. 1 drops off line	ENG 1 GENERATOR caution light	Generator No. 1 fails to supply power	Wires P54A16, P54B16, and P50A16 broken, touching structure or ground, shorted to ground at disconnect, or grounded with shield.
	Terminal B	Open circuit	No generator No. 1 output	ENG 1 GENERATOR caution light	Loss of power from generator No. 1	Wires P72A1 (G1-B to P61-B), P72B1, (J61-B to K5-B2) No. 1 P71A1 (K5-A1 to K3-A2) No. 1 generator relay, P46A16 (K3-A2 to P11-D) No. 1 DC control panel connector, loose, disconnected, or broken. Bus bar at K5 missing.
		Short to ground	Generator drops off line	ENG 1 GENERATOR caution	Loss of power from generator No. 1	Wires P72A1, P72B1, P71A1, P46A16 (K3-A2 to P11-D), No. 1 DC control unit, P128A16 (K3-A2 to CB15) No. 1 GEN FIELD CB, P129A16 (CB15) No. 1 GEN FIELD CB to (K7-D3) No. 1 start relay, P49A16 (K7-D2 to P11-A), P38A22 (K7-D2 to P11-S) broken, touching structure or ground, shorted to ground at disconnect, or grounded with shield.
	Terminal C	Open circuit	Starter No. 1 inoperative	None	Starter inoperative	Wires P95A1 (G1-C to P61-D), P95B1 (J61-D to K5-A2) No. 1 start relay P71A1 (K5-A1 to K3-A2) No. 1 generator relay, P46A16 (K3-A2 to P11-D) No. 1 DC control panel connector loose, broken, or disconnected. K5 contacts A1 to A2 bent or broken.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
		Short to ground	Generator drops off line	ENG 1 GENERATOR caution light	Loss of power from generator No. 1	Wires P95A1, P95B1, P71A1 (K5-A1 to K3-A2), P46A16 (K3-A2 to P11-D), P128A16 (K3-A2 to CB15) No. 1 GEN FIELD CB, P129A16 (CB15 to K7-D3) No. 1 start control relay broken, touching structure or ground.
	Terminal D	Open circuit	Generator paralleling circuitry inoperative	None	Generators not paralleled	Wires P51B16 (G1-D to P59-A), P51A22 (J59-A to P11-G) broken, loose, or disconnected.
		Short to ground	No immediate discernible effect, but paralleling circuit is affected	None	Generator paralleling affected	Wires P51B16, P51A22, P51C16 broken, touching structure or ground, or shorted to ground at disconnects.
	Terminal E	Open circuit	Generator No. 1 inoperative	ENG 1 GENERATOR caution light	Loss of power from generator No. 1	Wires P73A1 (G1-E to P61-F), P73B1 (J61-F to R9) No. 1 shunt P74A1N (No. 1 shunt) to structure ground, loose or disconnected. No. 1 shunt burned, disconnected, or broken.
		Short to ground	No effect	None	None	Wires P73A1, P73B1 broken, touching structure or ground. Shorted to ground at disconnects or grounded with shield.
Generator No. 2	Terminal A	Open circuit	No generator No. 1 output	ENG 2 GENERATOR caution light	Generator No. 2 fails to supply power	Wire P57A16 (G2-A to P60-C), P57B16 (J60-C to K8-C1) No. 2 start control relay, P34A16 (K8-C1 to P10-B) No. 2 DC control unit, broken, loose, or disconnected.
		Short to ground	Generator No. 2 drops off line	ENG 2 GENERATOR caution light	Generator No. 2 fails to supply power	Wires P57A16, P57B16 P34A16 broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
	Terminal B	Open circuit	No generator No. 2 output	ENG 2 GENERATOR caution light	Loss of power from generator No. 2	Wires P85A1 (G2-B to P62-B) generator disconnect, P85B1 (J62-B to K6-B2) No. 2 start relay, P84A1 (K6-A1 to K4-A1), No. 2 generator relay. Bus bar on K6 missing. Relay K4 not energized. Wires P91A22N (K4X1 to ground) P90A22 (K4-X2 to K8-A2), P22A22 (K8-A3 to K4-11) P22B22 (K4-11 to K6-12), P88A22 (K6-13 to P10-H) wire P30A1 (K4-A2 to K1-A1) external power relay, wires either broken, loose, or disconnected.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
		Short to ground	Generator No. 2 drops off line	ENG 2 GENERATOR caution light	Loss of power from generator No. 2	Wires P85A1, P85B1, P84A1, P30B1 (K4-A2 to K2-A2) No. 2 non-essential bus relay P30C1 (K2-A2 to overhead No. 2 main DC bus) P154A16 (K4-A1 to GEN 2 FIELD CB) P84C16 (K4-A1 to P10-D) broken, touching structure or ground, shorted to ground at disconnectors, or grounded with shields.
	Terminal C	Open circuit	Starter No. 2 circuit inoperative	None	Starter No. 2 inoperative	Wire P98A1 (G2-C to P62-D), P98B1 (J62-D to K6-A2) loose or disconnected.
		Short to ground	Generator No. 2 drops off line	ENG 2 GENERATOR caution light	Loss of power from generator No. 2	Wires P98A1, P98B1 broken and touching structure or ground.
	Terminal D	Open circuit	Generator paralleling circuitry inoperative	None	Generators not paralleled	Wires P35B16 (G2-D to P60-A), P35A22 (J60-A to P10-G), and P35C16 (K8-C2 to splice), broken, loose, or disconnected.
		Short to ground	No immediate discernible effect, but paralleling circuit is affected	None	Generator paralleling affected	Wires P35B16, P35A22, and P35C16 broken, touching structure or ground, or shorted to ground at disconnects.
	Terminal E	Open circuit	Generator No. 2 inoperative	ENG 2 GENERATOR caution light	Loss of power from generator No. 2	Wires P86A1 (G2-E to P62-F) P86B1 (J62-F to No. 2 shunt) R10, P87A1N (No. 2 shunt to ground) loose or disconnected. No. 1 shunt burned, disconnected, or broken.
		Short to ground	No effect	None	None	Wires P86A1, P86B1, broken, touching structure or ground, shorted to ground at disconnects, or grounded with shields.
No. 1 Main DC bus	Feeder wire at bus	Open circuit	No 28 VDC power on No. 1 main DC bus	Caution panel caution light no voltage indicated on voltmeter	No 28 VDC power supplied to buses by generator No. 1	Wires P5A1 (K9-A1 to K3-A1) P5B1 (K3-A1 to K39-A1) No. 1 non-essential bus relay, P5C1 (K39-A1 to overhead main DC bus No. 1) loose or disconnected. K9 contacts A1 to A2 broken.
		Short to ground	Generator No. 1 drops off line	ENG 1 GENERATOR caution light	Loss of power from generator No. 1	Wires P5A1, P5B1, P5C1, P45A22 to P11-E) broken and touching structure or ground.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
No. 2 Main DC bus	Feeder wire at bus	Open circuit	No 28 VDC power on No. 2 main DC bus	No voltage on indication voltmeter	No 28 VDC power supplies to buses by generator No. 2	Wires P30A1 (K1-A1 to K4-A2), P30B1(K4-A2 to K2-A2) No. 2 non-essential bus relay, P30C1 (K2-A2 to overhead main DC bus No. 2) loose or disconnected.
		Short to ground	Generator No. 2 drops off line	ENG 2 GENERATOR caution light	Loss of power from generator No. 2	Wires P30A1, P30B1 and P30C1, P29A2 broken and touching structure or ground.
No. 1 Essential DC bus		Short to ground	No. 1 feeder bus circuit breakers trip (after 1 or 2 seconds when BAT BUS switch No. 2 is ON, BAT BUS switch No. 1 is OFF; immediately when BAT BUS switch No. 2 is OFF, and BAT BUS switch No. 1 is ON).	No. 1 feeder circuit breakers trip	No. power on No.1 essential bus from generator No. 1. Generator No. 2 supplies power to No. 1 essential bus. Heavy bus loads may also cause No. 2 feeder circuit breakers to trip. Inverters are not affected.	Wires P247A16, P136A16 broken, touching structure or ground.
		Short to ground	No. 2 feeder bus circuit breakers trip (immediately if BAT BUS switch No. 2 is ON and BAT BUS switch No. 1 is OFF; after several seconds if BAT BUS switch No. 1 is ON and BAT BUS No. 2 is OFF.)	No. 2 feeder circuit breakers trip	No power on No. 1 essential bus from generator No. 2, generator No. 2 supplies power to No. 1 essential bus. Heavy bus loads may cause No. 1 feeder breakers to trip.	Wires P15A10 and P16A10 broken, touching structure or ground, or diodes shorted to ground.
No. 2 Essential DC bus		Short to ground	No. 2 feeder bus circuit breakers trip (Immediately when BAT BUS switch No. 2 is ON, after several seconds when BAT BUS switch No. 1 is ON, and BAT BUS switch No. 2 is OFF).	No. 2 feeder circuit breakers tripped	No power on No. 2 essential bus from generator No. 2. Generator No. 1 supplies power to No. 2 essential bus (heavy bus loads may cause No. 1 feeder breakers to trip). Inverters are not affected.	Wires P241A12 and P246A16, P234A20, P244A16 broken, touching structure or ground, or diodes shorted to ground.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
Non-essential DC bus	No. 1 non-essential bus feeder wire at TB 52	Open circuit	No DC power on non-essential bus from generator No. 1	None	Non-essential bus receives no power from generator No. 1. Generator. No. 2 generator continues to supply power to non-essential bus (heavy load on non-essential bus will cause No. 2 overload sensor to trip and remove all power from non-essential bus).	Wires P174A6 (TB52-2 to diodes), P148A6 (diodes to S185-L2) non-essential 1 sensor, P147A6 (S185-L1 to K39-A2). Non-essential bus 1 sensor loose or disconnected, diodes open, sensor opened, or non-essential 1 sensor de-energized.
	No. 2 non-essential bus feeder wire at TB 52	Open circuit	No DC power on non-essential bus from generator No. 2	None	Non-essential bus receives no power from generator No. 2. Generator No. 1 generator continues to supply power to non-essential bus (heavy load on non-essential bus will cause No. 1 overload relay to trip and remove all power from non-essential bus).	Wires P175A6 (tB52-2 to diodes), P11A6 (diodes to S186-L2) Non-essential bus 2 sensor, and P12A6 (S186-L1 to K2A1) non-essential bus 2 sensor loose or disconnected, diodes open, sensor opened, or non-essential sensor de-energized.
	No. 1 and No. 2 Non-essential bus feeders at TB 52	Short to ground	Non-essential bus trips off	Non-essential equipment inoperative	No power on non-essential bus	Wires P174A6, P175A6 broken, touching structure or ground.
			AC POWER SYSTEM			
Inverter No. 1	Pins A and B	Open circuit	Inverter No. 1 inoperative. Inverter No. 2 automatically supplies all AC buses.	INVERTER #1 caution light	None Inverter No. 2 supplies all AC buses	Wires X21A16N (3PS1P1-B to ground) V3A12N (3PS1P1-A to ground) broken, loose, or disconnected.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
	Pin C	Open circuit	Inverter No. 1 inoperative. Inverter No. 2 automatically supplies all AC buses.	INVERTER #1 caution light	None Inverter No. 2 supplies all AC buses	Wires V15A16 (3PS1P1-C to 3S1-5), V14A16N (3S1-6 to ground) broken, loose, or disconnected. Contacts 5 and 6 of 3S1 open.
		Short to ground	Inverter will not turn off with INV 1 switch, 3S1	None	None	Wire V15A16 or contacts 5 and 6 of 3S1 shorted to ground.
	Pin D	Open circuit	No 28 VDC power to inverter No. 1	INVERTER #1 caution light	None Inverter No. 2 supplies all AC buses	Wire V1B12 (3PS1P1-D to INV 1 PWR circuit breaker 3CB7) broken, loose, or disconnected. Circuit breaker 3CB7 inoperative or wire P187A12 broken, loose, or disconnected.
		Short to ground	No. 1 inverter inoperative, INV 1 PWR circuit breaker opens	INV 1 PWR circuit breaker (3CB7) open inverter #1 caution light	None. Inverter No. 2 can supply all AC buses	Wire V1B12 (3PS1P1-D to INV 1 PWR circuit breaker) shorted to structure or ground.
	Pin E	Open circuit	26 VAC not supplied to 26 VAC buses No. 1 and No. 2 by inverter No. 1	Equipment requiring 26 VAC power becomes inoperative. Instrument flags come in view.	No 26 VAC power is available on 26 VAC buses No. 1 and No. 2.	Wire X6A18 (3PS1P1-E to 3K3-B1) broken, loose, or disconnected.
		Short to ground	Inverter No. 1 drops off line	INVERTER #1 caution light	None Inverter No. 2 can supply power to No. 1 and No. 2 buses	Wires X6A18, X6B18, or X6C18, 3K3-B1, 3K2P1-B touching structure or ground.
	Pin G	Open circuit	115 VAC not supplied to 115 VAC buses No. 1 and No. 2	INVERTER #1 caution light	None Inverter No. 2 can supply No. 1, No. 2, and No. 3 115 VAC buses.	Wire X2A18 (3PS1P1-G to splice), X2C22 (splice to 3K1P1-C), X2B22 (splice to 3K1P1-A) broken, loose, or disconnected. Relay 3K1 contacts C and D open, wire X4A18 (3K1P1-D to splice) broken, loose, or disconnected.
		Short to ground	Inverter No. 1 drops off line immediately. Inverter No. 2 automatically switched to supply all AC buses.	INVERTER #1 caution light	None inverter No. 2 supplies power to all AC buses.	Wire X2A18, X2B22, X2C22, X4A18, X4C18, X4B22, X4D18, X4E18 touching structure or ground.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE
Inverter No. 2	Pins A and B	Open circuit	Inverter No. 2 inoperative. Inverter No. 1 automatically supplies bus 3, if required.	INVERTER #2 caution light	None Inverter No. 1 supplies all AC buses.	Wires X22A16N (3PS2P1-B to ground), V11A12N (3PS2P1 to ground) broken, loose, or disconnected.
	Pin C	Open circuit	Inverter No. 2 inoperative. Inverter No. 1 automatically supplies all AC buses.	INVERTER #2 caution light	None. Inverter No. 1 can supply all AC buses.	Wires V13A16 (3PS1P1-C to 3S2-5) V12A16N (3S2-6 to ground) broken, loose, or disconnected. Contacts 5 and 6 of 3S2 open.
		Short to ground	Inverter will not turn off with INV 2 switch, 3S2	None	None	Wire V13A16 or contacts 5 and 6 of 3S2 shorted to ground.
	Pin D	Open circuit	No. 28 VDC power to inverter No. 2	INVERTER #2 caution light	None Inverter No. 1 will supply all AC buses.	Wire V9B12 (3PS2P1-D to INV 2 PWR circuit breaker (3CB8) broken, loose, or disconnected. Circuit breaker (3CB8) inoperative. Wire P188A12 broken, loose, or disconnected.
		Short to ground	Inverter No. 2 inoperative INV 1 PWR circuit breaker opens	INV 1 PWR circuit breaker (3CB8) open INVERTER #2 caution light	None Inverter No. 1 will supply all AC buses.	Wire V9B12 (3PS2P1-D to INV 2 PWR circuit breaker) shorted to structure or ground.
	Pin E	Open circuit	26 VAC not supplied to 26 VAC Bus 3 by inverter 2	Equipment requiring power from 26 VAC bus 3 inoperative	No. 2 VAC power is available on 26 VAC Bus 3 (AFCS)	Wires X10A18 (3PS2P1 to splice), X10C18 (splice to 3CB10) broken, loose, or disconnected.
		Short to ground	Inverter No. 2 drops off line	INVERTER #2 caution light	None Inverter No. 1 can supply power to 26 VAC bus 3, as well as 26 VAC buses 1 and 2.	Wires X10A18, X10B18, X10C18, X11A18, X11B18 shorted to structure or ground.
	Pin G	Open circuit	115 VAC not supplied to 115 VAC buses No. 1 and No. 2	INVERTER #2 caution light	None Inverter No. 1 can supply power to 115 VAC buses No. 1 and No. 2	Wires X8A18 (3PS2P1-G to splice), X8C18 (splice to 3CB9), X8D18 (splice to 3K3-C3) broken, loose, or disconnected.
		Short to ground	Inverter No. 2 drops off line immediately. Inverter No. 1 automatically switches to supply all AC buses.	INVERTER #2 caution light	None Inverter No. 1 supplies power to all AC buses.	Wires X8A18, X8B22, X8C18, X8D18, X8E18 broken, and touching structure or ground.

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						WIRE NO.	FROM	TO
			EXTERIOR LIGHTING					
Search-light	Power wire from circuit breaker to searchlight	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L77A12	CB25	DS12-3
							CB25	Inopera-tive
		Short to ground	Circuit breaker trips	Circuit breaker trips	Searchlight inoperative	L77A12	CB25	DS12
	Ground wire. (Searchlight to ground)	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L106A12 N	DS12-1	Ground
	Control wire	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L73A22	CB5	J48-Z
						L73B22	P48-Z	S21-2
							S21-2	S25-G
							CB5	Inopera-tive
		Short to ground	Searchlight inoperative	Circuit breaker trips	Searchlight inoperative	L73A22	CB5	J48-Z
						L73B22	P48-Z	S21-2
							S21-2	S25-G
	Left rotation circuit	Open circuit	Loss of left control	None	Loss of left control	L64A22	S25-F	P48-W
						L64B22	J48-W	DS12-5
	Right rotation circuit		Loss of right control		Loss of right control	L63A22	S25-D	P48-X
						L63B22	J48-X	DS12-4
	Left rotation circuit	Short to ground	Control breaker trips	Circuit breaker trips	Searchlight inoperative	L64A22	S25-F	P48-W
						L64B22	J48-W	DS12-5
	Right rotation circuit					L63A22	S25-D	P48-X
						L63B22	J48-X	DS12-4
	Extend circuit	Open circuit	Loss of extend control	None	Loss of extend control	L65A22	S25-C	P48-V
						L65B22	J48-V	DS12-6
	Retract circuit		Loss of retract control	Circuit breaker trips	Loss of retract control	L76A22	S25-E	P48-Y
						L76C22	J48-Y	DS12-7
	Power wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L69A12	CB25	P225-H
							CB25	Inopera-tive
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L69A12	CB25	P255-H
	Ground wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L70A12N	P225-1	Ground

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	Control wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L78A22	CB5	J48-c
						L78B22	P48-c	S22-2
							S20-2	S22-2
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L105A22	S22-3	P48-e
						L105B22	J48-e	P225-C
	Extend wire	Open circuit	Unable to extend	None	Loss of extend control	L82A22	S20-3	P48-b
						L82B22	J48-b	P225-A
		Short to ground	Unable to retract	Circuit breaker trips	Loss of retract control	L83A22	S20-1	P48-d
						L83B22	J48-d	P225-B
Lower anti-collision lights	Power wire	Open circuit	Lower anti-collision light inoperative	None	Lower anti-collision light inoperative	L11A20	S62-3	Splice
						L11A20	Splice	P50-B
						L11B20	Splice	Splice
						L11C20	Splice	P50-A
	Ground wire					L12A20N	P50-C	Ground
Upper anti-collision lights	Power wire		Upper anti-collision light inoperative	None	Upper anti-collision light inoperative	L8A20	S62-3	J59-E
						L8B20	P59-E	J355-A
						L8C20	P355-A	Splice.
						L8D20	Splice	P116-B
						L8E20	Splice	P116-A
	Ground wire					L9A20	P116-C	P355-B
						L8B20	J355-B	P59-F
						L9C20N	J59-F	Ground.
Position lights	Power wire	Open circuit	Light/lights inoperative	None	Light/lights inoperative	L1A20	CB7	S59-2
						L5A20	S59-3	TB9-6
			Lower left position light inoperative		Lower left position light inoperative	L5C20	DS15	TB9-6
							Lower	
			Upper left position light inoperative		Upper left position light inoperative	L2C20	DS15	TB9-5
							Upper	
			Upper right position light inoperative		Upper right position light inoperative	L2B20	DS16	TB9-5
							Upper	
			Lower right position light inoperative		Lower right position light inoperative	L5B20	DS16	TB9-6
							Lower	
			Right tail position light inoperative		Right tail position light inoperative	L5D20	TB9-6	P117-A
						L5E20	DS 17	J117-A
						RED	Right	

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
			Left tail position light inoperative		Left tail position light inoperative	L2D20	TB9-5	P117-C
						L2E20	DS17	J117-C
						RED	Left	
	Ground wire		Left tail position light inoperative		Left tail position light inoperative	L3B20N	P117D	Ground.
						L3A20	DS17	J117-D
						BLU	Left	
			Right tail position light inoperative		Right tail position light inoperative	L6B20N	P117-B	Ground.
						L6A20	DS17	J117-B
						BLU	Right	
LH and RH utility lights	Power wire circuit breaker to switch	Open circuit	Utility lights inoperative	None	Utility lights inoperative	L124A22	CB5	S96-2
						L125A22	S96-2	S96-3
							S96-3	Splice
L utility light			L utility light inoperative		L Utility light inoperative	L125C22	Splice	DS29
R utility light			R utility light inoperative		R utility light inoperative	L125B22	Splice	DS30
R or L utility light	Ground wire	Open circuit	L or R Utility light inoperative	None	L or R Utility light inoperative	Chassis	Ground	
Landing light	Power wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L69A12	8CB25	22P11-E
						L69B12	22J11-E	8DS13 P1-H
							8CB25	Inopera-tive
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	L69A12	8CB25	22P11-E
						L69B12	22J11-E	8DS13 P1-H
	Ground wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L70A12N	8DS13 P1-I	Ground
	Control wire	Open circuit	Landing light inoperative	None	Landing light inoperative	L78A22	8CB27	22P15-95
						L78B22	22J15-95	22J7-36
						L78C22	22P7-36	4A2J1-c
							S22-2	S20-2

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to ground	Circuit breaker trips	Circuit breaker trips	Landing light inoperative	78A22	8CB27	22P15-95
						L78B22	22J15-95	22J7-36
						L78C22	4A2J1-c	22P7-36
						L105B22	S22-2	S20-2
						L105C22	S22-2	S22-3
							4A2J21-e	22P7-35
							22J7-35	8DS13 P1-C
	Extend wire	Open circuit	Unable to extend	None	Loss of extend control	L82C22	8DS13 P1-A	22J7-33
						L82B22	22P7-33	4A2J1-b
							S20-3	4A2 P1-b
		Short to ground	Unable to retract	Circuit breaker trips	Loss of retract control.	L83B22	S20-1	4A2 P1-d
						L83C22	4A2 J1-d	22P7-34
							22J7-34	8DS13 P1-B
Lower anti-collision lights	Power wire	Open circuit	Lower anti-collision lights inoperative	None	Lower anti-collision lights inoperative	L11B20	Splice	22J13-d
						L11A20	22P13-d	8S22-3
	Ground wire					L12A20N	8DS3 P1-C	Ground
Upper anti-collision lights	Power wire		Upper anti-collision light inoperative	None.	Upper anti-collision light inoperative	L8A20	8S22-3	1J1-E
						L8B20	1P1-E	8J5-A
						L8C20	8P5-A	Splice
	Ground wire					L9A20	8DS8 P1-C	8P5-B
						L9B20	8J5-B	1P1-F
						L9C20N	1J1-F	Ground
Position lights	Power wire	Open circuit	Light/lights inoperative	None	Light/lights inoperative	L1A20	8CB23	8S21-2
						8S21-2	8S21-3	8TB4-6
						L5A20	8S21-3	
			Lower left position light inoperative		Lower left position light inoperative	L5E20	8TB4-6	22P13-C
						L5J20	22J13-C	8DS17
			Upper left position light inoperative		Upper left position light inoperative	L2C20	8TB4-5	8DS15

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
			Upper right position light inoperative		Upper right position light inoperative	L2B20	8TB4-5	8DS14
			Lower right position light inoperative		Lower right position light inoperative	L5B20 L5C20 L5D20	8TB4-6 22J12-B 22J8-T	22P12-B 22P8-T 8DS16
			Right tail position light inoperative		Right tail position light inoperative	L5F20 L5G20 L5H20 (WHT)	8TB4-6 22J13-B 8J6-A	22P13-B 8P6-A 8DS28
			Left tail position light inoperative		Left tail position light inoperative	L2D20 L2E20 L2F20	8TB4-5 22J13-A 8J6-C	22P13-A 8P6-C 8DS27
	Ground wire		Left tail position light inoperative		Left tail position light inoperative	L3B20N L2F20 (BLU)	8P6-d 8J6-d	Ground. 8DS27
			Right tail position light inoperative.		Right tail position light inoperative.	L6B20N L5H20 9BLU)	8P6-B 8J6-B	Ground. 8DS28
LH and RH utility lights	Power wire-circuit breaker to switch	Open circuit	Utility lights inoperative	None	Utility lights inoperative	L124A22 L125A22 125B22	8CB18 8S23-2 8S23-3 22J15-93	8S23-2 8S23-3 22P15-93 Splice
L utility light			L utility light inoperative		L utility light inoperative	L125C22	Splice	8DS29
R utility light			Utility light inoperative		Utility light inoperative	L125D22 L125E22 L125F22	Splice 22P7-8 22J10-4	22J7-8 22P10-4 8DS30
R or L utility light	Ground wire	Open circuit	L or R utility light inoperative	None	L or R utility light inoperative	Chassis	Ground	
Search-light	Power wire from circuit breaker to searchlight	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L77A12	8CB24 8CB24	8DS12-3 Inopera-tive
		Short to ground	Circuit breaker trips	Circuit breaker trips	Searchlight inoperative	L77A12	8CB24	8DS12

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	Ground wire. (Searchlight to ground)	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L106A12 N	8DS12	Ground.
	Control wire	Open circuit	Searchlight inoperative	None	Searchlight inoperative	L73A22	8CB26	22P14-7 9
						L73B22	22J14-7 9	4A2J1-Z
							4A2 P1-Z	S21-2
							S21-2	S25-G
							8CB26	Inopera-tive
		Short to ground	Searchlight inoperative	Circuit breaker trips	Searchlight inoperative	L73A22	8CB26	22P14-7 9
						L73B22	22J14-7 9	4A2J1-Z
							4A2 P1-Z	S21-2
							S21-2	S25-G
	Left rotation circuit	Open circuit	Loss of left control	None	Loss of left control	L64A22	S25-F	4A2 P1-W
						L64B22	4A2 J1-W	22P10-1 0
						L64C22	22J10-1 0	8DS1 2-5
	Right rotation circuit		Loss of right control		Loss of right control	L63A22	S25-D	4A2 P1-X
						L63B22	4A2 J1-X	22P10-9
						L63C22	22J10-9	8DS1 2-4
	Left rotation circuit	Short to ground	Control breaker trips	Circuit breaker trips	Searchlight inoperative	L64A22	S-25F	4A2 P1-W
						L64B22	4A2 J1-W	22P10-1 0
						L64C22	22J10-1 0	8DS1 2-5
	Right rotation circuit					L63A22	S-25D	4A2 P1-X
						L63B22	22J10-9	22P10-9
						L63C22	22J10-1 1	8DS1 2-4

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	Extend circuit	Open circuit	Loss of extend control	None	Loss of extend control	L65A22 L65B22 L65C22	S25-C 4A2 J1-V 22J10-11	4A2 P1-V 22P10-11 8DS1 2-6
	Retract circuit		Loss of retract control	Circuit breaker trips	Loss of retract control	L76A22 L76C22 L76D22	S25-E 4A2 J1-Y 22J10-12	4A2 P1-Y 22P10-12 8DS1 2-7
INTERIOR LIGHTING 5-VOLT INSTRUMENT LIGHTS								
a) Pilot instrument lights	(1) Wire from circuit breaker to control switch	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a1) L47A22 (b1) L54A22 (c1) L42A22	CB3 CB3 CB3	R1-2 R2-2 R3-2
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of instrument lights			
b) Copilot instrument lights	(2) Wire from control switch to power supply	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a2) L136A22 (b2) L55B22	R1-1 R2-1	198-A 196-A
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of instrument lights	(c2) L43B22	R3-1	Splice
c) Engine instrument lights		Short to power.	Lights on all the time	Lights on all the time	Cannot extinguish lights	L43B22 L43A22	Splice Splice	P197-A P294-A
	(3) Wires from power supply to rheostat	Open circuit.	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a3) L49A22 (b3) L52A22	P198-D 196-D	R1-4 R2-4
		Short to ground	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(c3) L44A22 L44B22 L44A22	P197-D P294-D Splice	Splice Splice R3-4

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to power	Lights on bright all the time	Lights on bright all the time	Lights on bright all the time			
	(4) Power supply ground return	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a4) L114A20N (b4) L112A20N	P198-B P196-B	Ground Ground
		Short to power	Loss of circuit supplying power	Circuit breaker trips	Loss of circuit supplying power	(c4) L113A20N L141A2N	P197-B P294-B	Ground Ground
	(5) Wire from rheostat to ground	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a5) L50A22 (b5) L57A22	R1-5 R2-5	TB3-5 TB2-6
		Short to power	Loss of circuit supply power	Circuit breaker trips	Loss of circuit	(c5) L45A22	R3-5	TB3-5
	(6) Power supply output wire to terminal block	Open circuit	Loss of instrument lights	Loss of instrument lights	Loss of instrument lights	(a6) L51C20 (b6) L58A20	P198-C 196-C	TB27-12 TB27-2
		Short to ground	Loss of instrument lights, circuit breaker trips	Loss of instrument lights, circuit breaker trip.	Loss of instrument lights	(c6) L46A20 L140A20	P197-C P294-C	TB27-8 TB27-10
		Short to power	Lights on bright all the time	Lights burn bright or bulbs burn out	Loss of instrument light control or loss of lights			
	(7) Power supply output wire from TB27 to instruments	Open circuit	Loss of individual instrument light.	Loss of individual instrument lights	Loss of individual instrument light	(a7) L5120	TB27-13	P14-g Pilot Attitude Indicator
		Short to ground	Lights dim all the time	Lights dim all the time	Lights dim all the time	L51L20	TB27-12	P186 Pilot Standby Com-pass
		Short to power	Lights bright all the time	Lights bright all the time	Lights bright all the time	L51G20	TB27-14	P188-2 Pilot Vertical Speed Indicator

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L51H20	TB27-14	P108-2 Pilot Clock
						L51K20	L27-14	P26-E Pilot Triple Tachom-eter
						L51J20	TB27-13	P191-A Pilot Air-speed Indicator
						L51E20	TB27-13	P190-A Pilot Al-timeter
						L51R20	TB27-13	TB15-13
						L51X20	TB15-13	P237-G Pilot Dual Torque Indicator
						L51W20	TB15-13	P28-G Dual Torque Indicator
						(b7)	TB27-2	TB15-7
						L58N20	TB15-7	P238-G
						L58Q20	TB15-7	P29-G
						L58P20		Copilot Dual Torque Indicator
						L143A20	TB27-2	P295-2 Copilot Clock
						L143B20	TB27-2	P296 Copilot Standby Com-pass
						L58H20	TB27-3	P189-A Copilot Altime-ter

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L58K20	TB27-4	P15-g Copilot Attitude Indicator
						L58C20	TB27-4	P27-E Copilot Triple Tachom-eter
						L58G20	TB27-4	P193-2 Copilot Rate of Climb Indicator
						L58J20	TB27-4	P192-A Copi-lot Air-speed Indicator
						(c7)	TB27-6	P210-A
						L46X20		No. 1 Hy-draulic Pres-sure and Temper-ature In-dicator
						L46C20	TB27-5	P160-A Dual Volt-meter 2
						L46H20	TB27-5	P16-A Engine 2 Fuel Pres-sure Indicator
						L46M20	TB27-5	P32-A Fuel Quantity Indicator
						L46N20	TB27-10	J33-b

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
						L46P20	P33-b	J206
								Engine Instrument Panel
						L46T20	TB27-10	P42-b
								Master Caution Panel
						L46E20	TB27-9	P158-E
								Dual Ammeter
						L46G20	TB27-9	P17-A
								Engine 1 Fuel Pressure Indicator
						L46D20	TB27-8	P150-A
								Engine 1 Interturbine Temperature Indicator
						L46Q20	TB27-8	P31-A
								Combining Gearbox Oil Pressure and Temperature Indicator
						L46V20	TB27-8	P161-A
								Dual Voltmeter 1
						L46R20	TB27-7	P22-D
								Engine 2 Gas Producer Tachometer

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
						L46F20	TB27-7	P151-A Engine 2 Interturbine Temperature Indicator
						L46K20	TB27-7	P18-A Engine 2 Oil Pressure and Temperature Indicator
						L46W20	TB27-7	P211-A No. 2 Hydraulic Pressure and Temperature Indicator
						L46J20	TB27-6	P19-A Engine 1 Oil Temperature and Pressure Indicator
						L46L20	TB27-6	P30-A Transmission Oil Temperature and Pressure Indicator
						L46S20	TB27-6	P23-D Engine 1 Gas Producer Tachometer
	(8) Individual ground wire from instrument	Open circuit	Loss of individual light	Loss of individual light	Loss of individual light	(a8) L52A20	P14-h	TB28-12 Pilot Attitude Indicator

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
		Short to power	Loss of circuit supplying power	Circuit breaker trips	Loss of circuit supplying power	L95D20	P108-1	TB28-14
						95C20	P27F	PilotClo-ckTB2
								8-14 Pi-lot Triple Tachom-eter
						L95F20	P188-1	TB28-14
								Pilot Vertical Speed Indicator
						L95L20	P28-H	TB15-14
						L95M20	P237-H	TB15-14
						L95G20	TB15-14	TB28-13
								Pilot Dual Torque Indicator
						L95J20	P191-B	TB28-13
								Pilot Air-speed Indicator
						L95K20	P190-B	TB28-13
								Pilot Al-timeter
						(b8)	P238-H	TB15-8
						L97N20	P29-H	TB15-8
						L97M20	TB15-8	TB28-2
						P97L20		Copilot Dual Torque Indicator
						L98A20	P15-h	TB28-3
								Copilot Attitude Indicator
						L97E20	P27-F	TB28-4
								Copilot Triple Tachom-eter

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L97C20	P193-1	TB28-4
								Copilot Rate of Climb Indicator
						L97D20	P192-B	TB28-4
								Copi-lot Air-speed Indicator
						L97H20	P189-B	TB28-3
								Copilot Altime-ter
						L142A20	P295-1	TB28-2
								Copilot Clock
						(c8)	TB28-6	P210-E
						L96R20		No. 1 Hy-draulic Pres-sure and Temper-ature In-dicator
						L96D20	TB28-5	P160-J
								Dual Volt-meter 2
						L96N20	TB28-5	P16-E
								Engine 2 Fuel Pres-sure In-dicator
						L96M20	TB28-5	P32-J
								Fuel Quantity Indicator
						L96L20	TB28-6	P23-E
								Engine 1 Gas Pro-ducer Tachom-eter

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
						L96K20	TB28-6	P30-E
								Trans-mission Oil Tem-pera-ture and Pres-sure In-dicator
						L96J20	TB28-6	P19-E
								Engine 1 Oil Tem-pera-ture and Pres-sure In-dicator
						L96Q20	TB28-7	P211-E
								No. 2 Hy-draulic Pres-sure and Temper-ature In-dicator
						L96E20	TB28-7	P18-E
								Engine 2 Oil Tem-pera-ture and Pres-sure In-dicator
						L96T20N	TB28-7	P151-J
								Engine 2 Inter-turbine Temper-ature In-dicator
						L96F20	TB28-7	P22-E
								Engine 2 Gas Pro-ducer Tachom-eter
						L96P20	TB28-8	P161-J
								Dual Volt-meter 1

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELICOPTER	POSSIBLE CAUSE		
						L96C20	TB28-8	P31-E Combining Gearbox Oil Pressure and Temperature Indicator
						L96S20	TB28-8	P150-J Engine 1 Interturbine Temperature Indicator
						L96G20	TB28-9	P17-E Engine 1 Fuel Pressure Indicator
						L96H20	TB28-9	P158-F Dual Ammeter
	(9) Wire from pilot switch to caution panel	Open circuit	None	None	Cannot dim caution panel	(a9) L107A22	P42-a	R-7
INTERIOR LIGHTING 28-VOLT INSTRUMENT LIGHTS								
a) Overhead console lights	(1) Wire from circuit breaker to switch	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a1) [1]	CB5	R5-2
						L32A22	CB5	Inoperative
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of lights	(b1) [1] L111A22	CB5	R6-2
b) Pedestal lights						(c1) L59A22	CB3	R4-2
c) Secondary instruments lights	(2) Wire from switch to transistor collector	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a2) [1]	R5-1	Q5-C
						L33B22 L33C22	Q5-C	Q7-C
		Short to ground	Circuit breaker trips	Circuit breaker trips	Loss of lights	(b2) [1] L38B22	R6-4	Q6-C
		Short to power	Lighting circuit energized continuously	None	None	(c2) [1] L60B22	R4-1	Q4-C

Table 1. Electrical Fault Analysis (S/N 33001 through 33100) (continued)

CIRCUIT	LOCATION	TYPE	RESULT ON CIRCUIT	INDICATION TO CREW	EFFECT ON HELI-COPTER	POSSIBLE CAUSE		
	(3) Wire from switch to transistor base	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a3) (1)	R5-4	Q5-B
		Short to ground				L34A22	Q5-B	Q7-B
						L34B22	R6-1	Q6-B
						(b3) (1)	R4-4	Q4-B
						L39A22		
						(c3)		
						L61A22		
	(4) Wire from switch to ground	Open circuit	None	None	Loss of circuit supplying power	(a4) (1)	R5-5	TB3-6
						L35A22		
		Short to ground	None	None	Loss of dimming control	(b4) (1)	R6-5	TB3-6
						L40A22	TB3-6	Ground
						L36B22N		
						(C4)	R4	TB2-6
						L62A22	TB2-6	Ground
						L57C22N		
	(5) Wire from transistor emitter to terminal board	Open circuit	Loss of lights	Loss of lights	Loss of lights	(a5) (1)	Q5-E	TB3-3
		Short to ground				L162A22	Q7-E	TB2-3
						L163A22	Q6-E	TB12-5
						L41A22		
		Short to power	None	None	Lights bright all the time			
						(c5) (1)	Q4-E	J33-M
						L36A22	P33-M	TB14-2
						L36B22		
	(6) Individual wires from terminal board to lights	Open circuit	Loss of individual light	Loss of individual light	Loss of individual light	(a6) (1)	TB3-1	J200
						L123C22	TB3-1	J202
						L123D22		
		Short to ground	Loss of lights	Loss of lights	Loss of lights	L123G22	TB3-1	J213
						L154A22	TB3-2	J3035
						L123G22	TB3-1	
						L154A22	TB3-2	
		Short to power	None	None	Lights on bright all the time	L123E22	TB2-1	J201
						L123F22	TB2-1	J203
						L146A22	TB2-1	S201
						L134A22	TB2-2	J231